Linear Algebra and Its Applications

Printed Chapters 1–8: collected worked solutions

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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1.1 Systems of linear equations

A solution is one ordered list of values satisfying *every* equation. The solution set is the set of all such lists. A consistent system has at least one solution; an inconsistent system has none. Two systems are equivalent when their solution sets are equal.

The three permitted operations are swapping two equations, multiplying an equation by a nonzero scalar, and adding a multiple of one equation to another. In matrix notation $R_i \leftarrow R_i + cR_j$ means to replace the *whole* row, including its last entry. We use a vertical bar to separate the coefficient columns from the constant column. A zero row means $0 = 0$; a row $[0 \ \cdots \ 0 \mid d]$ with $d \neq 0$ means an impossible equation.

Practice problem 1.1.P1: Choose the next elimination step

(a) Use the simpler coefficient to make the next pivot.

Swap equations 3 and 4, so the coefficient of x_3 in the third equation is 1 instead of 5. Next one would replace equation 4 by equation 4 minus five times equation 3. This eliminates x_3 from the last equation. Do not restart elimination on x_2: the forward phase first establishes successive pivots below the previous ones.

(b) Begin the backward phase at the last pivot.

The fourth equation is $x_4 = 1$. Add twice equation 4 to equation 1 to remove its term $-2x_4$. The other two equations already have no x_4 term. After that, divide equation 3 by 2 and use it to clear the x_3 terms above it. Each operation preserves all solutions.

Practice problem 1.1.P2: Consistency of a triangular system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 5 & 0 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{4})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 5 & 0 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 0 & -6 \\ 0 & 1 & -\frac{7}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{7}{4})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 5 & 0 & -6 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (5)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{17}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -\frac{17}{2} \\ x_2 = \frac{1}{2} \\ x_3 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Practice problem 1.1.P3: Test a proposed solution in every equation

Substitute $(x_1, x_2, x_3) = (3, 4, -2)$ into the three left sides:

$$5(3) - 4 + 2(-2) = 7, \qquad -2(3) + 6(4) + 9(-2) = 0,$$

$$-7(3) + 5(4) - 3(-2) = 5.$$

The first two constants match, but the third required constant is -7, not 5. One failed equation is enough to reject the proposed solution.

Conclusion. $(3, 4, -2)$ is not a solution.

Practice problem 1.1.P4: A consistency condition on two constants

The second left side is -3 times the first. Replace equation 2 by equation 2 plus three times equation 1. The resulting last equation is $0 = k + 3h$. It can hold precisely when $k = -3h$. In that case choose any $x_2 = t$ and put $x_1 = (h + t)/2$; these values satisfy both equations.

Conclusion. Consistency holds if and only if $k = -3h$.

Exercise 1.1.1: A two-equation elimination

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ -2 & -7 & -5 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 3 & 9 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 5 & 7 \\ 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (5)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -8 \\ 0 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -8 \\ x_2 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.2: A second two-equation elimination

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 3 & 6 & -3 \\ 5 & 7 & 10 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{3})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 5 & 7 & 10 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (5)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & -3 & 15 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -5 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & -5 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 9 \\ x_2 = -5 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.3: The intersection of two lines

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 1 & -1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -3 & -3 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 2 \\ x_2 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.4: A second line intersection

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 3 & -2 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 0 & -8 & 40 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{8})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -13 \\ 0 & 1 & -5 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & -5 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -3 \\ x_2 = -5 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.5: Begin backward elimination

The last two equations already give $x_3 = 2$ and $x_4 = -5$. To begin clearing above the x_3 pivot, subtract four times row 3 from row 2, then add three times row 3 to row 1:

$$R_2 \leftarrow R_2 - 4R_3, \qquad R_1 \leftarrow R_1 + 3R_3.$$

The two changed equations become $x_2 = -2$ and $x_1 - 4x_2 = 13$. Both operations act on all entries, including the constants.

Conclusion. Subtract four times row 3 from row 2; add three times row 3 to row 1.

Exercise 1.1.6: Create the last pivot

Use the coefficient 1 in row 3 to remove the coefficient 4 below it:

$$R_4 \leftarrow R_4 - 4R_3 \quad \Longrightarrow \quad [0\ 0\ 0\ -7 \mid 14].$$

Now multiply the last row by $-1/7$, giving $[0\ 0\ 0\ 1 \mid -2]$. These steps create the next pivot without disturbing the earlier zeros.

Conclusion. Subtract four times row 3 from row 4, then divide row 4 by -7.

Exercise 1.1.7: A contradiction row cannot be repaired

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_3 \leftrightarrow R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-4)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_4$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 7 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (7)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.8: Four variables fixed at zero

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{3})R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right].$$

Use $R_4 \leftarrow (\frac{1}{2})R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-5)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 0 \\ x_4 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.9: Back substitution through four pivots

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & -2 & 0 & -7 \\ 0 & 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & -2 & 0 & -7 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 0 & 21 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 16 \\ 0 & 1 & 0 & 0 & 21 \\ 0 & 0 & 1 & 0 & 14 \\ 0 & 0 & 0 & 1 & 4 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 16 \\ x_2 = 21 \\ x_3 = 14 \\ x_4 = 4 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.10: A triangular four-variable system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -7 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & -11 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & -11 \\ 0 & 1 & 0 & 0 & 12 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -47 \\ 0 & 1 & 0 & 0 & 12 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = -47 \\ x_2 = 12 \\ x_3 = 2 \\ x_4 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.11: A system with a zero leading coefficient

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 0 & 1 & 5 & -4 \\ 1 & 4 & 3 & -2 \\ 2 & 7 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftrightarrow R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 2 & 7 & 1 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & -1 & -5 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & -2 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 5 & -4 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-4)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -17 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.12: A three-variable contradiction

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 2 & -7 & 3 & -2 \\ -2 & 1 & 7 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 3 & -5 & 4 \\ -2 & 1 & 7 & -1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 3 & -5 & 4 \\ 0 & -9 & 15 & -7 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & -9 & 15 & -7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-9)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 5 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & -3 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & 0 \\ 0 & 1 & -\frac{5}{3} & \frac{4}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{4}{3})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -5 & 4 & 0 \\ 0 & 1 & -\frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-5)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & -\frac{13}{3} & 0 \\ 0 & 1 & -\frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.13: A unique three-variable solution

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 2 & 15 & -9 \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 1 & 5 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & -\frac{5}{2} & \frac{5}{2} \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{2}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & 8 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & \frac{15}{2} & -\frac{9}{2} \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{15}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 5 \\ x_2 = 3 \\ x_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.14: Solve after scaling a leading row

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 2 & 0 & -6 & -8 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 6 & 7 & 8 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (6)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -5 & -10 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 2 \\ x_2 = -1 \\ x_3 = 2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.15: Detect inconsistency without solving

The augmented rows are

$$\begin{bmatrix} 1 & -6 & 0 & 0 & 5 \\ 0 & 1 & -4 & 1 & 0 \\ -1 & 6 & 1 & 5 & 3 \\ 0 & -1 & 5 & 4 & 0 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 + R_1$ and $R_4 \leftarrow R_4 + R_2$. Rows 3 and 4 become $[0\ 0\ 1\ 5 \mid 8]$ and $[0\ 0\ 1\ 5 \mid 0]$. Then $R_4 \leftarrow R_4 - R_3$ gives $[0\ 0\ 0\ 0 \mid -8]$. Its equation $0 = -8$ is impossible.

Conclusion. The system is inconsistent. No further solving is needed.

Exercise 1.1.16: Detect consistency without solving

Divide row 1 by 2, then use $R_4 \leftarrow R_4 + 3R_1$:

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -5 \\ 0 & 3 & 3 & 0 & 0 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 2 & 3 & -5 & -10 \end{bmatrix}.$$

Now $R_4 \leftarrow R_4 - \frac{2}{3}R_2$ gives $[0\ 0\ 1\ -5 \mid -10]$, and $R_4 \leftarrow R_4 - R_3$ gives $[0\ 0\ 0\ -9 \mid -9]$. There is a nonzero pivot in each variable column and no contradiction row. Back substitution is therefore possible from the last equation upward.

Conclusion. The system is consistent. The exercise does not require its values.

Exercise 1.1.17: Do three lines meet at one point?

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 2 & 3 & -1 \\ 6 & 5 & 0 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 6 & 5 & 0 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (6)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -4 & 3 \\ 2 & -5 & 7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -4 & 3 \\ 0 & -8 & 8 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{4})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & -8 & 8 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-8)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{3}{4})R_3$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{3}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{3}{2})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.1.18: Do three planes have a common point?

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & 4 \\ 0 & 1 & -2 & -2 \\ 2 & 3 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{2})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 2 & 3 & 0 & 0 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & -1 & -4 & -4 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -6 & -6 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{6})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Exercise 1.1.19: A forbidden value of the parameter

Replace row 2 by row 2 minus three times row 1:

$$\begin{bmatrix} 1 & h & 4 \\ 0 & 6-3h & -4 \end{bmatrix}.$$

If $h = 2$, the last equation is $0 = -4$, so there is no solution. If $h \neq 2$, the second variable is $-4/(6-3h)$ and the first is then determined by row 1. Thus a solution does exist in every other case.

Conclusion. $h \neq 2$.

Exercise 1.1.20: A second forbidden parameter value

Use $R_2 \leftarrow R_2 - 2R_1$ to obtain $[0\ -8-2h \mid 16]$. If $h = -4$, this asserts $0 = 16$. For $h \neq -4$ the nonzero coefficient determines x_2, then row 1 determines x_1.

Conclusion. Consistency holds precisely for $h \neq -4$.

Exercise 1.1.21: A parameter that never destroys consistency

After $R_2 \leftarrow R_2 - 3R_1$, the last row is $[0\ h-12 \mid 0]$. If $h \neq 12$ it says $x_2 = 0$ and then $x_1 = -2$. If $h = 12$ it says $0 = 0$, leaving x_2 free and $x_1 = -2 - 4x_2$. Neither case produces a contradiction.

Conclusion. Every real h gives a consistent system.

Exercise 1.1.22: Match a multiple equation

Use $R_1 \leftarrow R_1 + 2R_2$. Its result is $[0\ 0 \mid h-6]$. Thus consistency requires $h = 6$. Conversely, if $h = 6$, the first equation is -2 times the second, so solving $2x_1 - 6x_2 = -3$ supplies solutions of both.

Conclusion. $h = 6$.

Exercise 1.1.23: Definitions and reversible operations

(a) True. Swapping twice restores the original rows. Scaling by $c \neq 0$ is reversed by scaling by $1/c$. Adding cR_j to R_i is reversed by adding $-cR_j$ to R_i.

(b) False. In an $m \times n$ matrix, m counts rows and n counts columns. A 5×6 matrix has five rows and six columns.

(c) False. The described list is *one solution*. The solution *set* contains all lists satisfying the system, and may be empty, a singleton, or infinite.

(d) True. Existence asks whether at least one solution exists; uniqueness asks whether exactly one exists. These are separate questions.

Exercise 1.1.24: Row equivalence and consistency

(a) False. Equal row counts do not imply row equivalence. For example, [0 0] cannot be row reduced to [1 0]: all allowed operations keep a zero row zero.

(b) True. Each row operation replaces equations by reversible consequences, so it preserves the common solution set.

(c) False. Having identical solution sets is the definition of equivalent systems; different sets would contradict that definition.

(d) True. "Consistent" means that at least one solution exists.

Exercise 1.1.25: Compatibility of dependent rows

The coefficient rows satisfy $r_3 = -2r_1 - r_2$. Make this relation explicit by $R_3 \leftarrow R_3 + 2R_1$ and then $R_3 \leftarrow R_3 + R_2$. The augmented last row becomes $[0\ 0\ 0 \mid k + 2g + h]$. Therefore consistency requires $k + 2g + h = 0$. If it holds, the remaining first two rows have nonzero pivots in columns 1 and 2, so they can be solved with x_3 free.

Conclusion. $2g + h + k = 0$ is necessary and sufficient.

Exercise 1.1.26: Consistency for every right side

Replace row 2 by row 2 minus $c/2$ times row 1. This gives

$$[0\ d - 2c \mid g - cf/2].$$

If $d = 2c$, choose $f = 0, g = 1$. The resulting row asserts $0 = 1$, contradicting the assumption of consistency for *every* right side. Thus $d \neq 2c$. Conversely, that inequality supplies a second pivot for any f, g, so it also suffices.

Conclusion. $d \neq 2c$.

Exercise 1.1.27: The condition on a general two-by-two system

Because $a \neq 0$, use $R_2 \leftarrow R_2 - (c/a)R_1$. The new row is

$$[0\ d - cb/a \mid g - cf/a].$$

Its coefficient cannot be zero: otherwise $f = 0, g = 1$ would make the system inconsistent. Multiplying $d - cb/a \neq 0$ by the nonzero a gives $ad - bc \neq 0$. Conversely, this condition makes the second pivot nonzero for every right side.

Conclusion. $ad - bc \neq 0$. The condition states exactly that the two equations have independent left sides.

Exercise 1.1.28: Construct three equivalent augmented matrices

Start with the equations $x_1 = 3$, $x_2 = -2$, $x_3 = -1$. Three possible augmented matrices are

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \end{array}\right], \qquad \left[\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & 3 & 0 & -6 \\ 0 & 0 & 4 & -4 \end{array}\right],$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

The second merely rescales equations by nonzero numbers. The third comes from adding row 2 to row 1 and then the original row 3 to row 2. Alternatively back substitution in the third gives $x_3 = -1$, $x_2 = -2$, $x_1 = 3$. Thus all three have exactly the requested single solution.

Exercise 1.1.29: Reverse a row interchange

The first and third rows exchange places; the middle row stays fixed.

$$\text{Forward: } R_1 \leftrightarrow R_3, \qquad \text{reverse: } R_1 \leftrightarrow R_3.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.30: Reverse a row scaling

Each entry in $[0, -5, 10]$ is multiplied by $-1/5$, giving $[0, 1, -2]$.

$$\text{Forward: } R_3 \leftarrow -\tfrac{1}{5}R_3, \qquad \text{reverse: } R_3 \leftarrow -5R_3.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.31: Reverse a multiple-row subtraction

The subtraction is $[4, -1, 3, -6] - 4[1, -2, 1, 0] = [0, 7, -1, -6]$.

$$\text{Forward: } R_3 \leftarrow R_3 - 4R_1, \qquad \text{reverse: } R_3 \leftarrow R_3 + 4R_1.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.32: Reverse a second multiple-row subtraction

The subtraction is $[0, 4, -12, 7] - 4[0, 1, -3, -2] = [0, 0, 0, 15]$.

$$\text{Forward: } R_3 \leftarrow R_3 - 4R_2, \qquad \text{reverse: } R_3 \leftarrow R_3 + 4R_2.$$

The reverse operation cancels exactly the change made by the forward one; unchanged rows supply the same entries in both directions.

Exercise 1.1.33: Set up the four-node temperature model

At each node, multiply "temperature equals the average of its four neighbors" by 4. The two fixed neighboring boundary values contribute to the constant; the two interior neighbors stay as unknowns. Reading the labels in the figure gives

$$\begin{aligned} 4T_1 - T_2 - T_4 &= 10 + 20 = 30, \\ -T_1 + 4T_2 - T_3 &= 20 + 40 = 60, \\ -T_2 + 4T_3 - T_4 &= 40 + 30 = 70, \\ -T_1 - T_3 + 4T_4 &= 10 + 30 = 40. \end{aligned}$$

Conclusion. These four equations are the required equilibrium model. The order of nodes around the square is $1, 2, 3, 4$, not a row-wise ordering.

Exercise 1.1.34: Solve the temperature model

Use the equations from Exercise 33. Adding the first and third gives $4(T_1 + T_3) - 2(T_2 + T_4) = 100$. Adding the second and fourth gives $-2(T_1 + T_3) + 4(T_2 + T_4) = 100$. Let $S = T_1 + T_3$ and $U = T_2 + T_4$. Dividing both equations by 2 gives $2S - U = 50$ and $-S + 2U = 50$. Thus $S = U = 50$. Now substitute these pair sums into the original equations:

$$T_1 = (30 + U)/4 = 20, \quad T_2 = (60 + S)/4 = 55/2,$$

$$T_3 = (70 + U)/4 = 30, \quad T_4 = (40 + S)/4 = 45/2.$$

For example, the first average is $(10 + 20 + 55/2 + 45/2)/4 = 20$; the other three check in the same way by the four displayed equations. Every solution must have these pair sums and hence these values, so uniqueness is also established.

Conclusion. $T_1 = 20$, $T_2 = 27.5$, $T_3 = 30$, $T_4 = 22.5$ degrees.

1.2 Row reduction and echelon forms

A pivot is a leading nonzero entry of an echelon row. In echelon form, zero rows are at the bottom, pivots move strictly to the right as one moves downward, and all entries below a pivot are zero. In reduced echelon form (RREF), every pivot is 1 and is the only nonzero entry of its column. RREF is unique even though the sequence of operations leading to it is not.

For a system, pivot variable columns correspond to basic variables, and nonpivot variable columns correspond to free variables. *First test consistency*: a pivot in the augmented column means no solution. Only a consistent system with a free variable has infinitely many solutions.

Practice problem 1.2.P1: Describe every solution with one parameter

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & -5 & 0 \\ 0 & 1 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 9 \\ 0 & 1 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 9 \\ x_2 = 3 - t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Practice problem 1.2.P2: Elimination with two free variables

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ -2 & 4 & 5 & -5 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 0 & 0 & -3 & -1 & 2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & -3 & -1 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 5 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & 0 & \frac{10}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.2.1: Distinguish echelon form from reduced form

(a) Echelon only. The first pivot is 10, not 1, so reduced form fails, although pivots move right and their columns are zero below them.

(b) Reduced echelon form. The pivots are 1 in columns 1, 2, 4, and these pivot columns have no other nonzero entries. The 1 in column 3 of the first row does not matter because column 3 is not a pivot column.

(c) Neither. Row 3 is zero but row 4 is nonzero; zero rows must be last.

(d) Echelon only. The successive pivots $1, 2, 3, 4$ move right, but three pivots are not 1, and pivot columns may have entries above them.

Exercise 1.2.2: Check all echelon requirements

(a) Reduced echelon form: pivots in columns 1 and 2 are 1 and are the only nonzero entries in their columns; the zero row is last.

(b) Echelon only: the middle pivot is 2, so it is not reduced.

(c) Neither: the zero row is above nonzero rows.

(d) Echelon only: pivots in columns 2, 3, 5 move right and are 1, but there are nonzero entries above the latter pivots. A pivot being 1 by itself does not make a matrix reduced.

Exercise 1.2.3: Locate two pivot positions

What we must show. Reduce the matrix itself; it is not being declared an augmented matrix.

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 3 & 6 & 9 & 12 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & -3 & -12 \end{bmatrix}.$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -3 & -12 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (-3)R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Conclusion. The pivot positions are $(1,1)$ and $(2,3)$ in both the final matrix and the original matrix; the pivot columns are columns 1 and 3.

The circled entries mark those positions explicitly:

$$\begin{bmatrix} \textcircled{1} & 2 & 4 & 8 \\ 2 & 4 & \textcircled{6} & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 2 & 0 & -8 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A pivot position refers to a location, not necessarily the same numerical entry before and after reduction.

Exercise 1.2.4: Locate three pivot positions

What we must show. Reduce the matrix itself; it is not being declared an augmented matrix.

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 2 & 4 & 5 & 4 \\ 4 & 5 & 4 & 2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -6 \\ 4 & 5 & 4 & 2 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (4)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -6 \\ 0 & -3 & -12 & -18 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & -3 & -12 & -18 \\ 0 & 0 & -3 & -6 \end{bmatrix}.$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & -3 & -6 \end{bmatrix}.$$

Use $R_3 \leftarrow (-\frac{1}{3})R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (4)R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Conclusion. The pivot positions are $(1,1)$, $(2,2)$, and $(3,3)$ in both the final matrix and the original matrix; the pivot columns are columns 1, 2, and 3.

The original and reduced pivot positions can be circled as follows:

$$\begin{bmatrix} \textcircled{1} & 2 & 4 & 5 \\ 2 & \textcircled{4} & 5 & 4 \\ 4 & 5 & \textcircled{4} & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & -2 \\ 0 & 0 & \textcircled{1} & 2 \end{bmatrix}.$$

Exercise 1.2.5: All nonzero two-by-two echelon patterns

Let ■ mean a nonzero pivot and $*$ an arbitrary entry. A nonzero 2×2 matrix has either one or two nonzero echelon rows. With one nonzero row, its pivot is in column 1 or column 2. With two, pivots must be in columns 1 then 2. Thus all possibilities are

$$\begin{bmatrix} ■ & * \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & ■ \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} ■ & * \\ 0 & ■ \end{bmatrix}.$$

There are no other choices: a pivot cannot occur to the left of the pivot above it, and zero rows cannot precede nonzero rows.

Exercise 1.2.6: All nonzero three-by-two echelon patterns

There can be at most two pivots because there are only two columns. As in Exercise 5, either the sole pivot is in column 1, the sole pivot is in column 2, or the two pivots occupy columns 1 and 2 in succession:

$$\begin{bmatrix} ■ & * \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & ■ \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} ■ & * \\ 0 & ■ \\ 0 & 0 \end{bmatrix}.$$

The third row must be zero in every case, and nonzeroness excludes the all-zero matrix. These exhaust the echelon possibilities.

Exercise 1.2.7: Two pivots and one free variable

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{5})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_2. Set $x_2 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = -3t_1 - 5 \\ x_2 = t_1 \\ x_3 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.8: A variable absent from both equations

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ -3 & 7 & 0 & 9 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ 0 & -2 & 0 & -6 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ 0 & 1 & 0 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 4 \\ x_2 = 3 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.9: Interchange before eliminating

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 0 & 1 & -2 & 3 \\ 1 & -3 & 4 & -6 \end{array}\right].$$

Use $R_1 \leftrightarrow R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 4 & -6 \\ 0 & 1 & -2 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 3 \\ 0 & 1 & -2 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 3 \\ x_2 = 2t_1 + 3 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.10: A nonzero second pivot after elimination

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ -2 & 4 & -5 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ 0 & 0 & -7 & 14 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{7})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_2. Set $x_2 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 2 \\ x_2 = t_1 \\ x_3 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.11: Repeated equations leave two free variables

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{3})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (9)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (6)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variables are x_2, x_3. Set $x_2 = t_1$, $x_3 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = \frac{2t_1}{3} - \frac{4t_2}{3} \\ x_2 = t_1 \\ x_3 = t_2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.12: An augmented-column pivot rules out solutions

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 4 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-7)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.2.13: Five variables and two free choices

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccccc|c} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccccc|c} 1 & -3 & 0 & 0 & 9 & 2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -3 & 5 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variables are x_3, x_5. Set $x_3 = t_1$, $x_5 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 3t_2 + 5 \\ x_2 = 4t_2 + 1 \\ x_3 = t_1 \\ x_4 = 4 - 9t_2 \\ x_5 = t_2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.14: A zero basic variable simplifies the parameters

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccccc|c} 1 & 0 & -5 & 0 & -8 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-8)R_3$ to obtain

$$\left[\begin{array}{cccccc|c} 1 & 0 & -5 & 0 & 0 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variables are x_3, x_4. Set $x_3 = t_1$, $x_4 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 5t_1 + 3 \\ x_2 = -4t_1 + t_2 + 6 \\ x_3 = t_1 \\ x_4 = t_2 \\ x_5 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.15: Read two symbolic echelon patterns

(a) Two pivots among three variable columns.

The last row is zero, and the augmented column is not a pivot column. Thus the system is consistent. Columns 1 and 2 are pivot columns; column 3 supplies one free variable. Every choice of that variable allows back substitution, so there are infinitely many solutions.

(b) Three pivots among four variable columns.

The pivot columns are 2, 3, 4; column 5 is the augmented column and has no pivot. Thus the system is consistent. The first variable is free, so the solution is not unique: there are infinitely many solutions.

Conclusion. Both systems are consistent and neither has a unique solution.

Exercise 1.2.16: Contrast variable pivots with a constant pivot

(a) A pivot in the constant column.

There are three columns total, so column 3 is augmented. The last row is $[0\ 0 \mid \blacksquare]$, where $\blacksquare \neq 0$. It asserts $0 = \blacksquare$, a contradiction. The system is inconsistent.

(b) No pivot in the constant column.

Column 5 is augmented and has no pivot; the pivots are in columns 1, 3, 4. The system is consistent, and variable 2 is free. It consequently has infinitely many solutions.

Conclusion. Part (a) has no solution; part (b) is consistent but not unique.

Exercise 1.2.17: A parameter in the constant column

Eliminate the first coefficient in row 2 by $R_2 \leftarrow R_2 + 2R_1$. The last row becomes $[0\ 1 \mid h + 8]$. Its variable coefficient is 1 for every h, so it never produces a contradiction. Set $x_2 = h + 8$ and then $x_1 = 4 + x_2 = h + 12$.

Conclusion. The system is consistent for every real h.

Exercise 1.2.18: A parameter that can remove one equation

Use $R_2 \leftarrow R_2 - hR_1$. Its result is $[0\ 6+3h \mid -2-h]$. If $h \neq -2$, the variable coefficient is nonzero and gives a solution by back substitution. If $h = -2$, both entries are zero, so the second row is merely $0 = 0$; solutions still exist.

Conclusion. All real h work. At $h = -2$ the solutions are not unique; at every other h there is a unique solution.

Exercise 1.2.19: Choose parameters for all three outcomes

Elimination gives

$$\begin{bmatrix} 1 & h & 2 \\ 0 & 8-4h & k-8 \end{bmatrix}.$$

If $h \neq 2$, the second coefficient is nonzero, giving a unique solution for every k. If $h = 2$, the last equation is $0 = k - 8$. It is impossible for $k \neq 8$ and imposes no constraint for $k = 8$, leaving x_2 free.

Conclusion. (a) No solution: $h = 2, k \neq 8$, for example $(2, 0)$. (b) Unique: $h \neq 2$, any k, for example $(0, 0)$. (c) Infinitely many: $h = 2, k = 8$.

Exercise 1.2.20: A second three-outcome parameter family

Replace row 2 by row 2 minus twice row 1. The last row is $[0\ h+6 \mid k-2]$. If $h \neq -6$ it determines x_2 uniquely. If $h = -6$, it is compatible precisely when $k = 2$, and then x_2 is unrestricted.

Conclusion. (a) No solution: $h = -6, k \neq 2$, for example $(-6, 0)$. (b) Unique: $h \neq -6$, any k, for example $(0, 0)$. (c) Infinitely many: $h = -6, k = 2$.

Exercise 1.2.21: Definitions of basic variables and solutions

(a) False. The uniqueness-of-RREF theorem says every matrix has exactly one reduced echelon form; different operation sequences give that same matrix.

(b) False. Row reduction applies to any rectangular matrix, whether or not its last column represents constants in a system.

(c) True. A basic variable corresponds by definition to a pivot column of the coefficient matrix.

(d) True. Solving a system means describing all its solutions. A parametric description does this while recording the free choices.

(e) False. In $[0\ 0\ 0\ 5 \mid 0]$ the nonzero entry is in a variable column: the equation is $5x_4 = 0$, which is satisfiable.

Exercise 1.2.22: Uniqueness of RREF and consistency cautions

(a) True. This is precisely the uniqueness theorem for RREF.

(b) False. A pivot in *every* augmented-matrix column includes a pivot in the final column, implying an impossible equation. For example [0 | 1] already exhibits the obstruction (and an identity augmented matrix gives examples with every column a pivot column).

(c) False. Pivot positions are determined from the unique RREF, not by the particular path of row operations.

(d) True. A general solution explicitly describes all solutions, including the parameters and their allowed values.

(e) False. Consistency is also required. The system $0x_1 + 0x_2 = 1$ has nonpivot variable columns but no solutions.

Exercise 1.2.23: Four coefficient pivots force uniqueness

There are four variable columns and all four are pivot columns. In echelon form, the four pivots occupy all four rows, so the final augmented column cannot contain an additional pivot. Hence there is no contradiction row and the system is consistent. Since all variables are basic, no free choice remains; back substitution determines exactly one value for each variable.

Conclusion. The system has a unique solution.

Exercise 1.2.24: No pivot in the last augmented column

The fifth column is the constant column. By hypothesis it is not a pivot column. Thus RREF has no row $[0\ 0\ 0\ 0 \mid d]$ with $d \neq 0$, which is exactly the consistency criterion.

Conclusion. Yes, the system is consistent. It cannot have a unique solution: four variable columns but only three rows leave at least one free variable.

Exercise 1.2.25: A pivot in every coefficient row

A pivot in every coefficient-matrix row uses all available rows. Appending a right side cannot change the coefficient entries or create an additional row in which all coefficients vanish but the constant is nonzero. More concretely, apply to the right side the same row operations that put the coefficient matrix in echelon form. Every resulting row has a variable pivot, so no resulting row can assert $0 = d \neq 0$.

Conclusion. The system is consistent for every possible right side.

Exercise 1.2.26: Three pivots in a three-by-five coefficient matrix

There are three rows and three pivot columns. Different pivots occupy different rows, so there is a pivot in every coefficient row. By the argument in Exercise 25, there can be no contradiction row for any right side. There are five variables but only three pivots, so two variables are free.

Conclusion. The system is consistent and has infinitely many solutions.

Exercise 1.2.27: Restate uniqueness in pivot language

The exact completion is: "every column of the coefficient matrix is a pivot column." A nonpivot variable column is equivalent to a free variable. In a consistent system a free variable gives infinitely many solutions; without any free variables, all variables are fixed.

Conclusion. For a consistent system, uniqueness is equivalent to a pivot in every variable column.

Exercise 1.2.28: Combine existence and uniqueness conditions

Two facts are needed. The last (augmented) column must *not* be a pivot column, ensuring consistency. Every preceding (coefficient) column must be a pivot column, ensuring that there are no free variables. Conversely those two facts give existence and uniqueness by Theorem 2.

Conclusion. Pivots in every coefficient column, and no pivot in the augmented column. Neither requirement alone is sufficient.

Exercise 1.2.29: Why underdetermined systems cannot be unique

If there are m equations and n unknowns with $m < n$, there can be at most m pivots. Thus at least $n - m > 0$ variable columns are nonpivot. If the system is consistent, these give free variables and infinitely many solutions. If it is inconsistent, it has no solutions. Neither case has exactly one solution.

Conclusion. No underdetermined linear system can have a unique solution.

Exercise 1.2.30: An inconsistent underdetermined example

For example,

$$x_1 + x_2 + x_3 = 0, \qquad x_1 + x_2 + x_3 = 1.$$

There are only two equations for three unknowns. Subtracting the first from the second gives $0 = 1$, so the system is inconsistent.

Conclusion. Underdetermined does not mean consistent.

Exercise 1.2.31: A consistent overdetermined example

Yes. Consider

$$x_1 = 1, \qquad x_2 = 2, \qquad x_1 + x_2 = 3.$$

The pair $(1, 2)$ satisfies every equation, so this three-equation, two-variable system is consistent. The third equation is redundant: it is the sum of the first two, rather than an incompatible additional constraint.

Conclusion. An overdetermined system can be consistent; equation counts alone do not decide consistency.

Exercise 1.2.32: Estimate the backward share of reduction cost

The numerical note estimates the forward cost by $F_n = 2n^3/3 + n^2/2 - 7n/6$ and bounds the backward cost by $B_n = n^2$. Using the leading forward term, the approximate backward fraction of *total* cost is

$$\frac{n^2}{(2/3)n^3 + n^2} = \frac{3}{2n+3}.$$

At $n = 20$ this is $3/43 \approx 0.06977$, about 7%; at $n = 200$ it is $3/403 \approx 0.007444$, about 0.74%. Using the full stated forward count instead gives $400/(5510 + 400) \approx 6.77\%$ and $40000/(5353100 + 40000) \approx 0.742\%$. The cost of the backward phase is stated as an upper bound, so these are cost-model estimates, not an exact count for every input matrix.

Conclusion. About 7% for $n = 20$ and 0.74% for $n = 200$.

Exercise 1.2.33: Interpolate three points with a quadratic

The coefficient vector is (a_0, a_1, a_2). Its augmented matrix is

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 4 & 15 \\ 1 & 3 & 9 & 28 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$ to get rows [0 1 3 | 9] and [0 2 8 | 22] below the first. Then $R_3 \leftarrow R_3 - 2R_2$ gives [0 0 2 | 4]. Thus $a_2 = 2$, $a_1 = 9 - 3(2) = 3$, and $a_0 = 6 - 3 - 2 = 1$.

$$p(t) = 1 + 3t + 2t^2.$$

Check: $p(1) = 6$, $p(2) = 15$, and $p(3) = 28$.

Conclusion. The unique interpolating quadratic is $1 + 3t + 2t^2$.

Exercise 1.2.34: Interpolate wind-tunnel measurements

What we must show. Interpolate all six measured data points and convert the requested velocity into the table's units.

Set up six equations for six coefficients.

For each listed pair (t, F), use $a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 = F$. The coefficient matrix has rows $[1, t, t^2, t^3, t^4, t^5]$ for $t = 0, 2, 4, 6, 8, 10$; the right side is $[0, 29/10, 74/5, 198/5, 743/10, 119]^T$. Treating the printed decimals as the given exact data, elimination gives

$$\begin{aligned} a_0 &= 0, & a_1 &= 137/80, & a_2 &= -1147/960, \\ a_3 &= 127/192, & a_4 &= -269/3840, & a_5 &= 1/384. \end{aligned}$$

Thus

$$p(t) = \frac{137t}{80} - \frac{1147t^2}{960} + \frac{127t^3}{192} - \frac{269t^4}{3840} + \frac{t^5}{384}.$$

Substitution at the six inputs gives exactly the six measured outputs, so the computed coefficients satisfy the full system. They are unique: a difference of two degree-at-most-five interpolants would have six distinct zeros and hence would be the zero polynomial.

Use the measurement units.

The velocity 750 ft/sec corresponds to $t = 7.5$, not $t = 750$. The force is tabulated in hundreds of pounds. Consequently

$$p(7.5) = \frac{132789}{2048} \approx 64.83838, \qquad \text{force} \approx 6483.84 \text{ lb}.$$

Explain why lowering the degree fails.

The unique degree-at-most-five interpolant has nonzero t^5 coefficient $1/384$. Thus no polynomial of degree less than five can pass through all six data points. For a direct cubic check, fitting the first four points gives

$$q(t) = \frac{13t^3}{160} + \frac{51t^2}{80} - \frac{3t}{20}.$$

But $q(8) = 81.2 \neq 74.3$ and $q(10) = 143.5 \neq 119$. The six-equation cubic system is inconsistent, rather than merely less accurate.

Conclusion. The displayed quintic interpolates the data; its prediction at 750 ft/sec is approximately 6484 lb.

1.3 Vector equations

A vector is an ordered list; its coordinates are added and multiplied by scalars coordinate by coordinate. We display vectors as columns unless explicitly discussing points. A linear combination $c_1v_1+\cdots+c_kv_k$ uses scalar weights c_i. The span is the *set of all* combinations, not merely the finite list of generating vectors.

To test whether b is in a span, put the generators into the columns of a matrix and put b in the augmented column. Consistency of that system means the needed weights exist. An inconsistent system means b is outside the span. Neither the weights nor a solution should be guessed from a picture when exact coordinates are supplied.

Practice problem 1.3.P1: Commutativity from the coordinate definition

Take arbitrary $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$. Their ith sum coordinate is u_i+v_i. Real-number commutativity gives $u_i + v_i = v_i + u_i$ for each i. Thus

$$u + v = (u_1 + v_1, \ldots, u_n + v_n) = (v_1 + u_1, \ldots, v_n + u_n) = v + u.$$

Conclusion. The vector identity holds because all corresponding coordinates are equal, not because the vector symbols can be rearranged without reason.

Practice problem 1.3.P2: Find the height that lies in a span

Seek weights x_1, x_2, x_3 and reduce

$$\left[\begin{array}{rrr|r} 1 & 5 & -3 & -4 \\ -1 & -4 & 1 & 3 \\ -2 & -7 & 0 & h \end{array}\right].$$

Use $R_2 \leftarrow R_2 + R_1$ and $R_3 \leftarrow R_3 + 2R_1$:

$$\left[\begin{array}{rrr|r} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 3 & -6 & h-8 \end{array}\right].$$

Now $R_3 \leftarrow R_3 - 3R_2$ gives $[0\ 0\ 0 \mid h - 5]$. Consequently consistency requires $h = 5$. If $h = 5$, choose $x_3 = 0$; row 2 gives $x_2 = -1$, and row 1 gives $x_1 = 1$. Indeed $v_1 - v_2 = (-4, 3, 5)^T$.

Conclusion. Exactly $h = 5$ works.

Exercise 1.3.1: Add vectors and subtract twice a vector

Add coordinates in the same positions, and first double v for the second expression:

$$u + v = \begin{bmatrix} -1 + (-3) \\ 2 + (-1) \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}, \qquad u - 2v = \begin{bmatrix} -1 - 2(-3) \\ 2 - 2(-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

Conclusion. $u + v = (-4, 1)^T$ and $u - 2v = (5, 4)^T$.

Exercise 1.3.2: A second coordinate calculation

Coordinatewise addition and scalar multiplication give

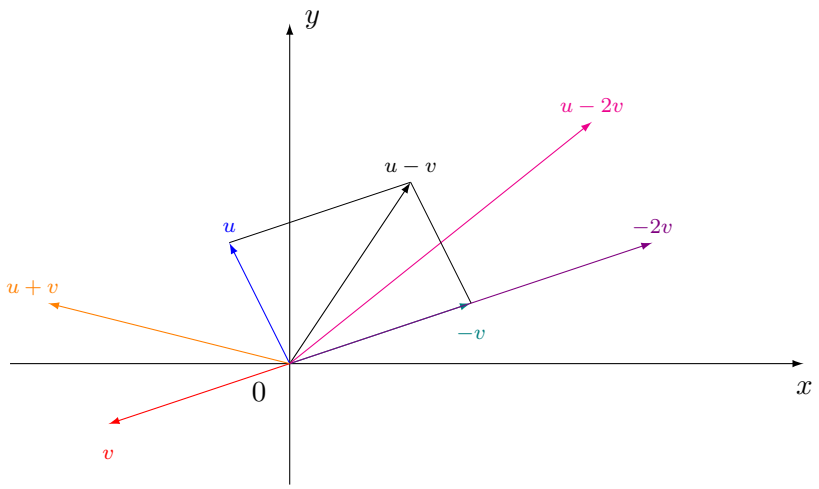
$$u+v=\begin{bmatrix}3+2\\2-1\end{bmatrix}=\begin{bmatrix}5\\1\end{bmatrix},\qquad u-2v=\begin{bmatrix}3-4\\2-2(-1)\end{bmatrix}=\begin{bmatrix}-1\\4\end{bmatrix}.$$

Conclusion. $u+v=(5,1)^T$ and $u-2v=(-1,4)^T$.

Exercise 1.3.3: Seven vectors and their parallelogram

Compute the coordinates before drawing. Every arrow below begins at the origin and ends at its vector coordinates.

vector	x	y
u	-1	2
v	-3	-1
$-v$	3	1
$-2v$	6	2
$u+v$	-4	1
$u-v$	2	3
$u-2v$	5	4

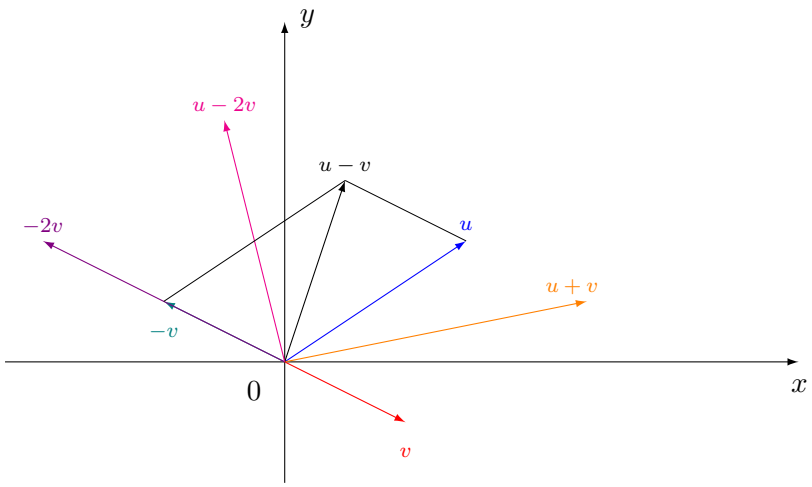


The thin extra sides join u to $u-v$ and $-v$ to $u-v$. They complete the parallelogram with vertices $0, u, u-v, -v$. The coordinate table fixes the exact endpoints even where arrows cross.

Exercise 1.3.4: A second arrow diagram

Compute the coordinates before drawing. Every arrow below begins at the origin and ends at its vector coordinates.

vector	x	y
u	3	2
v	2	-1
$-v$	-2	1
$-2v$	-4	2
$u+v$	5	1
$u-v$	1	3
$u-2v$	-1	4



The thin extra sides join u to $u-v$ and $-v$ to $u-v$. They complete the parallelogram with vertices $0, u, u-v, -v$. The coordinate table fixes the exact endpoints even where arrows cross.

Exercise 1.3.5: Read a vector equation coordinate by coordinate

The left side has coordinates $(3x_1+5x_2, -2x_1, 8x_1-9x_2)$. Two vectors are equal exactly when each corresponding coordinate is equal. Thus the equivalent system is

$$3x_1+5x_2=2, \qquad -2x_1=-3, \qquad 8x_1-9x_2=8.$$

Conclusion. These three scalar equations express precisely the given vector equation.

Exercise 1.3.6: Three weights but two coordinate equations

Multiply each column by its weight, add the two coordinates separately, and compare with the zero vector:

$$3x_1+7x_2-2x_3=0, \qquad -2x_1+3x_2+x_3=0.$$

Conclusion. The two scalar equations are equivalent to the vector equation; there are two equations even though there are three unknown weights.

Exercise 1.3.7: Read four vectors on an oblique grid

The grid consists of lines parallel to u and lines parallel to v. Following a line parallel to u changes only the u coefficient; following a line parallel to v changes only the v coefficient. Use the marked $u, v, -u, -v, -2v, 2v$ to fix unit steps. Counting those steps, including the halfway position of c, gives

$$a = u - 2v, \qquad b = 2u - 2v, \qquad c = 2u - \frac{7}{2}v, \qquad d = 3u - 4v.$$

For example, the line through a parallel to u meets the v-axis at $-2v$, and moving one u step from there reaches a. The vectors u, v point in distinct directions, so their two axes are not parallel. Through any plane point draw the corresponding two grid lines; they give one coefficient along each axis.

Conclusion. The displayed combinations describe a, b, c, d. Yes, every vector in $\mathbb{R}^2$ is a linear combination of these two nonparallel vectors.

Exercise 1.3.8: Read four more vectors on the grid

Count displacements on the same two oblique axes as in Exercise 7. The marked points give

$$w = -u + 2v, \qquad x = -2u + 2v, \qquad y = -2u + \frac{7}{2}v, \qquad z = -3u + 4v.$$

For instance, starting at $2v$ and moving one step in the $-u$ direction reaches w; moving one more step reaches x. The halfway placement of y in the v direction gives coefficient $7/2$, not 3 or 4. As the two generating directions are nonparallel, their grid extends across the whole plane.

Conclusion. The four combinations are as displayed, and the span is all of $\mathbb{R}^2$.

Exercise 1.3.9: Collect coefficients into three columns

Read down the coefficients of each variable, including its zero entry:

$$x_1 \begin{bmatrix} 0 \\ 4 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ -1 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The first coordinate reproduces $x_2 + 5x_3 = 0$, the second reproduces $4x_1 + 6x_2 - x_3 = 0$, and the third reproduces the last equation. Thus no equation or coefficient has been omitted.

Exercise 1.3.10: Encode a nonhomogeneous system as a vector equation

Collect the coefficients down the three equations:

$$x_1 \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ -7 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}.$$

Equality of each coordinate is exactly the corresponding original scalar equation, so the vector equation is equivalent to the system.

Exercise 1.3.11: A span test with a selected-answer discrepancy

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2 - 5t_1 \\ x_2 = 3 - 4t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

For a concrete combination take $x_3 = 0$, giving $b = 2a_1 + 3a_2$. Therefore b belongs to the span. The selected-answer entry for Exercise 11 in the supplied PDF says "No, but the explicit coordinate check $2(1, -2, 0)^T + 3(0, 1, 2)^T = (2, -1, 6)^T$ shows that entry is inconsistent with the printed data.

Exercise 1.3.12: A span test with unique weights

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 1 & -2 & 5 & 9 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 0 & 0 & 11 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 11 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{11})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & 11 \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-6)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & \frac{109}{11} \\ 0 & 1 & \frac{7}{3} & -\frac{5}{3} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (\frac{7}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & \frac{109}{11} \\ 0 & 1 & 0 & -\frac{41}{33} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{245}{33} \\ 0 & 1 & 0 & -\frac{41}{33} \\ 0 & 0 & 1 & -\frac{2}{11} \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} x_1 = \frac{245}{33} \\ x_2 = -\frac{41}{33} \\ x_3 = -\frac{2}{11} \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Thus $b = (245/33)a_1 - (41/33)a_2 - (2/11)a_3$, so it is a linear combination.

Exercise 1.3.13: An inconsistent column-span test

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ 0 & 0 & 0 & 3 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 3 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 3 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & \frac{5}{3} & -\frac{7}{3} \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{7}{3})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & 0 \\ 0 & 1 & \frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{26}{3} & 0 \\ 0 & 1 & \frac{5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

There are no weights producing b, so it is not a linear combination of the columns.

Exercise 1.3.14: An explicit combination with one free weight

What we must show. Decide whether weights exist, rather than whether b equals one of the columns. Form $[a_1\ a_2\ a_3 \mid b]$.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2 - 5t_1 \\ x_2 = 3 - 4t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

For a concrete combination take $x_3 = 0$, giving $b = 2a_1 + 3a_2$. Therefore b belongs to the span.

Exercise 1.3.15: Find the height in a generated plane

Write $x_1a_1 + x_2a_2 = b$ and reduce the augmented matrix:

$$\left[\begin{array}{rr|r} 1 & -5 & 3 \\ 3 & -8 & -5 \\ -1 & 2 & h \end{array}\right].$$

Use $R_2 \leftarrow R_2 - 3R_1$ and $R_3 \leftarrow R_3 + R_1$. The last two rows become $[0\ 7 \mid -14]$ and $[0\ -3 \mid h + 3]$. The former gives $x_2 = -2$, then the first row gives $x_1 = -7$. The last row requires $6 = h + 3$, or $h = 3$. Conversely these weights then give $-7a_1 - 2a_2 = (3, -5, 3)^T$ exactly.

Conclusion. $h = 3$, and only that height lies in the plane.

Exercise 1.3.16: Find a first coordinate in a generated plane

From the second coordinate of $x_1v_1 + x_2v_2 = y$, we immediately get $x_2 = -3$. The third coordinate is $-2x_1 + 7x_2 = -5$, so $-2x_1 - 21 = -5$ and $x_1 = -8$. The first coordinate then requires $h = x_1 - 2x_2 = -8 + 6 = -2$. Substitution verifies all three coordinates.

Conclusion. Exactly $h = -2$ works, with $y = -8v_1 - 3v_2$.

Exercise 1.3.17: Five combinations and their weights

Choose five different pairs of weights and compute their coordinates. The choice of weights itself proves membership in the span.

c_1	c_2	$c_1v_1 + c_2v_2$		
0	0	0	0	0
1	0	3	1	2
0	1	−4	0	1
1	1	−1	1	3
2	−1	10	2	3

Conclusion. These are five explicit vectors with their generating weights. A sketch is not needed or requested.

Exercise 1.3.18: Five combinations of a second generating pair

Choose five different pairs of weights and compute their coordinates. The choice of weights itself proves membership in the span.

c_1	c_2	$c_1v_1 + c_2v_2$		
0	0	0	0	0
1	0	1	1	−2
0	1	−2	3	0
1	1	−1	4	−2
2	−1	4	−1	−4

Conclusion. These are five explicit vectors with their generating weights. A sketch is not needed or requested.

Exercise 1.3.19: Dependent directions generate a line

We have $v_2 = (3/2)v_1$, since multiplying each coordinate of $(8, 2, -6)^T$ by $3/2$ gives $(12, 3, -9)^T$. Therefore $c_1v_1 + c_2v_2 = (c_1 + (3/2)c_2)v_1$. Every combination lies on the line through the origin in direction v_1, and every point on that line occurs by taking $c_2 = 0$.

Conclusion. The span is the line $\{t(8, 2, -6)^T : t \in \mathbb{R}\}$, not a plane.

Exercise 1.3.20: Two nonparallel vectors generate a plane

The vectors $(1, 1, -2)^T$ and $(-2, 3, 0)^T$ are not scalar multiples: a multiple of the first with third coordinate zero would have multiplier zero and hence be the zero vector. Two nonparallel directions

through the origin generate a plane. To identify it algebraically, observe that $n = (6, 4, 5)^T$ satisfies $n \cdot v_1 = 6 + 4 - 10 = 0$ and $n \cdot v_2 = -12 + 12 = 0$. Thus every combination obeys $6x + 4y + 5z = 0$. Conversely, given a point on this plane, choose $c_1 = -z/2$ and $c_2 = (y - c_1)/3$. Its second and third coordinates match, and the plane equation forces the first coordinate to match as well.

Conclusion. The span is the plane through the origin with equation $6x + 4y + 5z = 0$.

Exercise 1.3.21: Express every plane vector using two generators

Seek scalars a, b with $a(2, -1)^T + b(2, 1)^T = (h, k)^T$. The two equations are $2a + 2b = h$ and $-a + b = k$. The second gives $b = a + k$; substitute this into the first to get $4a + 2k = h$. Thus

$$a = \frac{h - 2k}{4}, \qquad b = \frac{h + 2k}{4}.$$

These real numbers are defined for every real h, k. Their substitution gives the desired coordinates exactly.

Conclusion. Every $(h, k)^T$ lies in the span, with the displayed weights.

Exercise 1.3.22: A nonzero-entry matrix with a proper span

Take

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

All entries of A are nonzero. Each linear combination of its columns has form $(t, t, t)^T$, where t is the sum of the three weights. But b does not have three equal coordinates. Therefore no combination can produce b.

Conclusion. The displayed A, b meet both requirements.

Exercise 1.3.23: Vector notation and span definitions

(a) False. A column vector has shape 2×1, whereas the displayed row has shape 1×2. Changing the orientation requires a transpose; the two matrix shapes are not interchangeable.

(b) False. If $(-5, 2)$ were a scalar multiple of $(-2, 5)$, the first coordinate would require multiplier $5/2$, but the second would then be $25/2$, not 2. Thus the two points do not lie on one line through zero.

(c) True. The expression $(1/2)v_1$ is $(1/2)v_1 + 0v_2$; a coefficient may be zero.

(d) True. Coordinatewise expansion of the vector equation gives exactly the equations encoded by $[a_1\ a_2\ a_3 \mid b]$.

(e) False. If $u = v \neq 0$ the span is a line, and if $u = v = 0$ it is just the origin. A plane requires two independent directions.

Exercise 1.3.24: What linear combinations do and do not require

(a) False. With $u = (1, 0)^T, v = (0, 1)^T$, the combination $u + v = (1, 1)^T$ lies on neither generating axis. The span contains combinations involving both directions.

(b) True. $\mathbb{R}^5$ is the set of ordered lists of five real coordinates.

(c) True. By definition, b is in the span exactly when suitable weights exist, which is the same as consistency of the augmented system.

(d) False. Adding $u - v$ to v yields $u - v + v = u$, not generally v. For example $u = 0$, $v \neq 0$ gives zero as the result.

(e) False. All zero weights are allowed and produce the zero vector. Consequently every span contains zero.

Exercise 1.3.25: A generating set is different from its span

(a) Distinguish the generating set from its span.

The three columns are $(1, 0, -2)^T$, $(0, 3, 6)^T$, and $(-4, -2, 3)^T$. None equals $b = (4, 1, -4)^T$. Thus b is not in the three-element set $\{a_1, a_2, a_3\}$, which contains exactly three distinct vectors.

(b) Produce actual weights.

Direct calculation gives

$$-4a_1 - a_2 - 2a_3 = (-4, 0, 8)^T + (0, -3, -6)^T + (8, 4, -6)^T = (4, 1, -4)^T = b.$$

So $b \in W$. Also W contains all multiples of the nonzero vector a_1; different real multiples give different vectors, so W is infinite.

(c) Use the definition without reducing a matrix.

We have $a_1 = 1a_1 + 0a_2 + 0a_3$, which explicitly represents a_1 as a combination of the generators.

Conclusion. The finite generating set does not contain b, but its infinite span does contain b and all three generating columns.

Exercise 1.3.26: A target outside the span but a column inside

(a) Check the three coordinate equations for compatible weights.

Writing the weights as c_1, c_2, c_3, the first coordinate gives $2c_1 + 6c_3 = 10$, or $c_1 = 5 - 3c_3$. The second is $-c_1 + 8c_2 + 5c_3 = 3$, so after substitution $c_2 + c_3 = 1$. The third would require

$$c_1 - 2c_2 + c_3 = 7.$$

Its left side, using the first two equations, is instead $(5 - 3c_3) - 2(1 - c_3) + c_3 = 3$, independent of c_3. Thus it cannot equal 7; no weights exist, and $b \notin W$.

(b) Exhibit the trivial combination for a generator.

The second column is $0a_1 + 1a_2 + 0a_3$, so it is certainly in W.

Conclusion. b is outside the column span, although each generating column itself is automatically inside it.

Exercise 1.3.27: Translate mine outputs into a vector equation

(a) Interpret the scale factor together with the units.

Five days of mine 1 produce $5v_1 = (150, 3000)^T$: 150 metric tons of copper and 3000 kilograms of silver.

(b) Express the target as the sum of the two outputs.

The vector equation is

$$x_1 \begin{bmatrix} 30 \\ 600 \end{bmatrix} + x_2 \begin{bmatrix} 40 \\ 380 \end{bmatrix} = \begin{bmatrix} 240 \\ 2824 \end{bmatrix}.$$

The unknowns count days, so physically they must be nonnegative.

(c) Solve and check.

Divide the copper equation by 10: $3x_1 + 4x_2 = 24$. Multiply it by 200 and subtract the silver equation to get $420x_2 = 4800 - 2824 = 1976$. Therefore

$$x_2 = \frac{494}{105}, \qquad x_1 = \frac{24 - 4(494/105)}{3} = \frac{544}{315}.$$

Both are positive. In the copper equation their contributions sum to $30(544/315) + 40(494/105) = 240$; in the silver equation they sum to $600(544/315) + 380(494/105) = 2824$.

Conclusion. Mine 1 runs $544/315 \approx 1.727$ days and mine 2 runs $494/105 \approx 4.705$ days.

Exercise 1.3.28: Check three coal-plant output coordinates

(a) Track just the heat coordinate.

Burning x_1 tons of A and x_2 tons of B gives $27.6x_1 + 30.2x_2$ million Btu of heat.

(b) Keep the three output types in a fixed order.

With coordinates heat (million Btu), sulfur dioxide (g), and particulates (g), the output vector is

$$x_1 \begin{bmatrix} 27.6 \\ 3100 \\ 250 \end{bmatrix} + x_2 \begin{bmatrix} 30.2 \\ 6400 \\ 360 \end{bmatrix}.$$

(c) Set this vector equal to the observed output.

Use right side $(162, 23610, 1623)^T$. The heat and particulate equations, clearing the heat decimal, are

$$276x_1 + 302x_2 = 1620, \qquad 250x_1 + 360x_2 = 1623.$$

Subtracting 250 times the first from 276 times the second gives $23860x_2 = 42948$, so $x_2 = 9/5 = 1.8$. Substitution in either equation gives $x_1 = 39/10 = 3.9$. Now check the remaining equation rather than ignoring the extra datum:

$$3100(3.9) + 6400(1.8) = 12090 + 11520 = 23610.$$

The other outputs are $107.64 + 54.36 = 162$ and $975 + 648 = 1623$.

Conclusion. The plant burned 3.9 tons of anthracite and 1.8 tons of bituminous coal, satisfying all three measurements.

Exercise 1.3.29: Compute a mass-weighted average

The total mass is $4 + 2 + 3 + 5 = 14$ grams. Form the weighted sum one coordinate at a time:

$$\begin{aligned} 4v_1 + 2v_2 + 3v_3 + 5v_4 &= (8, -8, 16) + (-8, 4, 6) \\ &\quad + (12, 0, -6) + (5, -30, 0) \\ &= (17, -34, 16). \end{aligned}$$

Dividing by the total mass gives

$$\overline{v} = \left(\frac{17}{14}, -\frac{17}{7}, \frac{8}{7}\right).$$

The division is by total *mass*, not by the number of points.

Conclusion. The center of mass is $(17/14, -17/7, 8/7)$.

Exercise 1.3.30: A center of mass is a linear combination

Assume the total mass m is nonzero, as required to define the center of mass. Rewrite its formula as

$$\overline{v} = \frac{m_1}{m}v_1 + \cdots + \frac{m_k}{m}v_k.$$

The coefficients m_j/m are real scalars. This is precisely a linear combination of the given points viewed as vectors, so it belongs to their span. For positive physical masses the coefficients also sum to one and are nonnegative, although those extra facts are not needed for span membership.

Conclusion. Yes; the displayed normalized masses are the required weights.

Exercise 1.3.31: Move the center of a triangular plate

(a) Average the three equal point masses.

The uniform triangular plate has the same center as three 1-gram masses at the vertices, so

$$\overline{v} = \frac{(0, 1) + (8, 1) + (2, 4)}{3} = \left(\frac{10}{3}, 2\right).$$

(b) Include the plate's existing mass in the new balance.

The total mass after adding 6 grams will be 9 grams. The plate already has first moment $3\overline{v} = (10, 6)$. Let w_i be the added grams at vertex v_i. The new center $(2, 2)$ requires total first moment $(18, 18)$, so

$$w_1 + w_2 + w_3 = 6, \qquad 8w_2 + 2w_3 = 18 - 10 = 8,$$

$$w_1 + w_2 + 4w_3 = 18 - 6 = 12.$$

Subtract the mass equation from the last equation to obtain $3w_3 = 6$, so $w_3 = 2$. The horizontal equation then gives $8w_2 = 4$ and $w_2 = 1/2$. Finally $w_1 = 6 - 2 - 1/2 = 7/2$. All added masses are nonnegative, as the physical request requires.

Conclusion. Add 3.5 g at $(0, 1)$, 0.5 g at $(8, 1)$, and 2 g at $(2, 4)$. Their contribution is $(8, 12)$, making the final first moment $(18, 18)$ and the desired center $(2, 2)$.

Exercise 1.3.32: A plane-spanning diagram with nonunique weights

The figure shows v_1, v_2 pointing in nonparallel directions. Their span is therefore the whole plane, so in particular there are weights a, b with $av_1 + bv_2 = b_{\text{target}}$. Taking the third weight to be zero already gives a solution. The same plane contains v_3, so write $v_3 = pv_1 + qv_2$. For every $t \in \mathbb{R}$,

$$(a - pt)v_1 + (b - qt)v_2 + tv_3 = av_1 + bv_2 = b_{\text{target}}.$$

These are distinct coefficient triples because their third entries are distinct. Negative weights are allowed; the target need not lie between the positive arrows.

Conclusion. A solution exists, and it is not unique: there are infinitely many coefficient triples.

Exercise 1.3.33: Associativity and distributivity coordinatewise

(a) Associativity.

For each coordinate i, real-number associativity gives $(u_i + v_i) + w_i = u_i + (v_i + w_i)$. The ith coordinates of $(u + v) + w$ and $u + (v + w)$ are therefore equal for every i, which proves equality of the vectors.

(b) Distributivity over vector addition.

For each i, $c(u_i + v_i) = cu_i + cv_i$ by real-number distributivity. The left side is the ith coordinate of $c(u + v)$; the right side is the ith coordinate of $cu + cv$. Equality coordinate by coordinate gives $c(u + v) = cu + cv$.

Conclusion. Both vector laws follow from the corresponding real-number laws.

Exercise 1.3.34: Additive inverses and scalar associativity

(a) The additive inverse.

The ith coordinate of $u + (-u)$ is $u_i + (-u_i) = 0$. Similarly the ith coordinate of $(-u) + u$ is $-u_i + u_i = 0$. Both sums are therefore the vector $(0, \ldots, 0)$.

(b) Two successive scalar multiplications.

The ith coordinate of $c(du)$ is $c(du_i)$. Real multiplication is associative, so $c(du_i) = (cd)u_i$, which is the ith coordinate of $(cd)u$. This holds for every coordinate and for all real c, d, including zero and negative values.

Conclusion. $u + (-u) = (-u) + u = 0$ and $c(du) = (cd)u$.

1.4 The matrix equation $Ax = b$

A matrix–vector product is a weighted sum of the matrix's columns. If $A = [a_1 \ \cdots \ a_n]$ and $x = (x_1, \ldots, x_n)^T$, then

$$Ax = x_1 a_1 + \cdots + x_n a_n.$$

Thus Ax is defined only when x has one entry for each column of A. The row rule gives the same product: entry i is $a_{i1}x_1 + \cdots + a_{in}x_n$. A vector b belongs to the span of the columns exactly when $Ax = b$ is consistent. Theorem 4 says the columns span the whole target $\mathbb{R}^m$ exactly when A has a pivot in every row. A pivot in an augmented column does not count as a pivot in A.

Practice problem 1.4.P1: Read coefficients from a known solution

The columns are $a_1 = (1, -3, 4)^T$, $a_2 = (5, 1, -8)^T$, $a_3 = (-2, 9, -1)^T$, and $a_4 = (0, -5, 7)^T$. The given solution has entries $(3, -2, 0, -4)^T$, so the definition gives

$$3a_1 - 2a_2 + 0a_3 - 4a_4 = (3, -9, 12)^T + (-10, -2, 16)^T + (0, 20, -28)^T = (-7, 9, 0)^T = b.$$

The coefficients are the entries of the solution, not row entries of A.

Practice problem 1.4.P2: Check additivity by both calculations

Here $A = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$, $u = (4, -1)^T$, and $v = (-3, 5)^T$. First $u + v = (1, 4)^T$, so

$$A(u + v) = (2 + 20, 3 + 4)^T = (22, 7)^T.$$

Separately $Au = (8 - 5, 12 - 1)^T = (3, 11)^T$ and $Av = (-6 + 25, -9 + 5)^T = (19, -4)^T$. Their sum is again $(22, 7)^T$. This checks $A(u + v) = Au + Av$ with the actual data.

Exercise 1.4.1: Check dimensions before multiplying

What we must show. Compute the indicated product, if possible, in both requested ways. The matrix $\begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix}$ has two columns, but $(3, -2, 7)^T$ has three entries. In the column definition there are only two columns for three proposed weights. In the row rule a row has two entries and cannot be paired with a three-entry vector.

Conclusion. The product is undefined; no numerical result can be assigned.

Exercise 1.4.2: A column matrix is not a row matrix

The matrix $(1, 3, 1)^T$ has size 3×1, whereas $(5, -1)^T$ has size 2×1. The inner sizes 1 and 2 differ. The column definition would need one scalar, and each row contains only one entry.

Conclusion. The product is undefined by either rule.

Exercise 1.4.3: Compute one product in two ways

(a) Weight the two columns.

$$-2\begin{bmatrix}1\\-3\\1\end{bmatrix}+3\begin{bmatrix}2\\1\\6\end{bmatrix}=\begin{bmatrix}-2+6\\6+3\\-2+18\end{bmatrix}=\begin{bmatrix}4\\9\\16\end{bmatrix}.$$

(b) Use the rows.

The three dot products are $1(-2)+2(3)=4$, $(-3)(-2)+1(3)=9$, and $1(-2)+6(3)=16$.

Conclusion. Both rules give $(4,9,16)^T$.

Exercise 1.4.4: Two rows give two output entries

(a) Use the column definition.

$$1\begin{bmatrix}1\\3\end{bmatrix}+2\begin{bmatrix}3\\2\end{bmatrix}+1\begin{bmatrix}-4\\1\end{bmatrix}=\begin{bmatrix}1+6-4\\3+4+1\end{bmatrix}=\begin{bmatrix}3\\8\end{bmatrix}.$$

(b) Use the row rule.

The first entry is $1(1)+3(2)-4(1)=3$; the second is $3(1)+2(2)+1(1)=8$. Both entries have now been checked independently.

Exercise 1.4.5: Expand the matrix equation into columns

Read each column vertically and use the corresponding entry of the multiplier vector as its weight:

$$2\begin{bmatrix}1\\-2\end{bmatrix}-\begin{bmatrix}2\\-3\end{bmatrix}+\begin{bmatrix}-3\\1\end{bmatrix}-\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}-4\\1\end{bmatrix}.$$

For a direct check, the first coordinate is $2-2-3-1=-4$ and the second is $-4+3+1+1=1$.

Exercise 1.4.6: A four-entry output from two columns

The equivalent vector equation is

$$-3\begin{bmatrix}2\\3\\8\\-2\end{bmatrix}+5\begin{bmatrix}-3\\2\\-5\\1\end{bmatrix}=\begin{bmatrix}-21\\1\\-49\\11\end{bmatrix}.$$

The coordinates are $-6-15=-21$, $-9+10=1$, $-24-25=-49$, and $6+5=11$.

Exercise 1.4.7: Assemble the coefficient vectors as columns

Place the coefficient of x_1 in column 1, that of x_2 in column 2, and that of x_3 in column 3. The result is

$$\begin{bmatrix} 4 & -5 & 7 \\ -1 & 3 & -8 \\ 7 & -5 & 0 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}.$$

The output has four entries because each original vector lies in $\mathbb{R}^4$.

Exercise 1.4.8: Four scalar weights and a two-dimensional target

The same column convention gives

$$\begin{bmatrix} 2 & -1 & -4 & 0 \\ -4 & 5 & 3 & 2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}.$$

There are four columns, one for each weight, and two rows, one for each coordinate equation.

Exercise 1.4.9: Keep the missing coefficient equal to zero

The coefficient of x_1 in the second equation is zero. Therefore

$$x_1 \begin{bmatrix} 5 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix},$$

and the matrix equation is

$$\begin{bmatrix} 5 & 1 & -3 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}.$$

Exercise 1.4.10: Three equations become three coordinates

The vector equation is

$$x_1 \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}.$$

Equivalently,

$$\begin{bmatrix} 4 & -1 \\ 5 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}.$$

No equation has been discarded in changing notation.

Exercise 1.4.11: Solve an augmented system and check its vector

Form and reduce the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 3 & -4 & -2 \\ 1 & 5 & 2 & 4 \\ -3 & -7 & 6 & 12 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3+3R_1]{R_2 \leftarrow R_2-R_1} \left[\begin{array}{ccc|c} 1 & 3 & -4 & -2 \\ 0 & 2 & 6 & 6 \\ 0 & 2 & -6 & 6 \end{array}\right].$$

Replace R_3 by $R_3 - R_2$ to obtain $-12x_3 = 0$. Thus $x_3 = 0$. The second row gives $2x_2 + 6x_3 = 6$, hence $x_2 = 3$. The first gives $x_1 + 9 = -2$, hence $x_1 = -11$.

Substitute into the original rows.

They give $-11 + 9 = -2$, $-11 + 15 = 4$, and $33 - 21 = 12$.

Conclusion. The solution vector is $x = (-11, 3, 0)^T$.

Exercise 1.4.12: A second exact augmented reduction

Start with

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ -3 & -4 & 2 & 2 \\ 5 & 2 & 3 & -3 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3-5R_1]{R_2 \leftarrow R_2+3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 2 & -1 & 5 \\ 0 & -8 & 8 & -8 \end{array}\right].$$

Now $R_3 \leftarrow R_3 + 4R_2$ gives $4x_3 = 12$, so $x_3 = 3$. Then $2x_2 - 3 = 5$ gives $x_2 = 4$, and $x_1 + 8 - 3 = 1$ gives $x_1 = -4$. The original products are $-4 + 8 - 3 = 1$, $12 - 16 + 6 = 2$, and $-20 + 8 + 9 = -3$.

Conclusion. $x = (-4, 4, 3)^T$.

Exercise 1.4.13: Membership in a plane means consistency

What we must show. Decide whether $u = (0, 4, 4)^T$ is a linear combination of $(3, -2, 1)^T$ and $(-5, 6, 1)^T$. The equations for the two weights s, t are $3s - 5t = 0$, $-2s + 6t = 4$, and $s + t = 4$. The last equation gives $s = 4 - t$. Substitution into the first gives $12 - 8t = 0$, so $t = 3/2$ and $s = 5/2$. The second equation checks: $-5 + 9 = 4$. Hence

$$u = \tfrac{5}{2}(3, -2, 1)^T + \tfrac{3}{2}(-5, 6, 1)^T.$$

Conclusion. Yes: the explicit weights show that u lies in the plane.

Exercise 1.4.14: A contradictory coordinate rules out membership

The proposed equation $Ax = u$ gives

$$2x_1 + 5x_2 - x_3 = 4, \qquad x_2 - x_3 = -1, \qquad x_1 + 2x_2 = 4.$$

From the last two equations $x_3 = x_2 + 1$ and $x_1 = 4 - 2x_2$. The left side of the first would then be $2(4 - 2x_2) + 5x_2 - (x_2 + 1) = 7$, not 4.

Conclusion. No. The system is inconsistent, so u is outside the column span.

Exercise 1.4.15: Describe every attainable right-hand side

For $A = \begin{bmatrix} 3 & -1 \\ -9 & 3 \end{bmatrix}$, the second row is -3 times the first. Eliminating it in $[A \mid b]$ gives the equation $0 = b_2 + 3b_1$. Thus consistency is equivalent to $b_2 = -3b_1$. This is also sufficient: the first column $(3, -9)^T$ alone reaches $(b_1, -3b_1)^T$ with weight $b_1/3$.

Conclusion. The attainable vectors are exactly $t(1, -3)^T$, $t \in \mathbb{R}$; for example $(0, 1)^T$ is not attainable, so not every b works.

Exercise 1.4.16: A condition on three coordinates of the right side

For the given matrix, reduce symbolically:

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ -2 & 2 & 0 & b_2 \\ 4 & -1 & 3 & b_3 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & -2 & -2 & b_2 + 2b_1 \\ 0 & 7 & 7 & b_3 - 4b_1 \end{array}\right].$$

These steps use $R_2 + 2R_1$ and $R_3 - 4R_1$. Replacing the last row by $2R_3 + 7R_2$ gives $0 = 6b_1 + 7b_2 + 2b_3$. If this condition holds, the remaining two equations have pivots and can be solved, with one free variable. If it fails, the last row is a contradiction.

Conclusion. The attainable right sides form the plane $\{b : 6b_1 + 7b_2 + 2b_3 = 0\}$, not all of $\mathbb{R}^3$.

Exercise 1.4.17: Count pivot rows of the specified matrix

For the matrix A used in Exercises 17 and 19, exact elimination gives

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are three pivot rows, in columns 1, 2, and 4. The fourth row of the coefficient matrix has no pivot. By Theorem 4, $Ax = b$ is solvable for every $b \in \mathbb{R}^4$ if and only if A has a pivot in every row.

Conclusion. Three rows contain pivots; some right sides have no solution.

Exercise 1.4.18: Distinguish the target space from the number of columns

For the matrix B used here and in Exercise 20,

$$B = \begin{bmatrix} 1 & 4 & 1 & 2 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 6 & 7 \\ 2 & 9 & 5 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -11 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are only three pivots and one zero row. Theorem 4 therefore shows that its columns do not span $\mathbb{R}^4$. They do not span $\mathbb{R}^3$ either: the columns themselves have four entries and belong to $\mathbb{R}^4$, so $\mathbb{R}^3$ is not their ambient space.

Conclusion. Not every vector of $\mathbb{R}^4$ is a combination of the columns; the proposed $\mathbb{R}^3$ claim also fails.

Exercise 1.4.19: Translate the pivot calculation into a span statement

The reduction displayed in Exercise 1.4.17 has a zero fourth row and only three pivots. Theorem 4 says this is equivalent to failure of the columns to span $\mathbb{R}^4$. A right side whose transformed fourth entry is nonzero gives an inconsistent augmented system.

Conclusion. No to both questions: not every vector is a column combination, and the column span is a proper subset of $\mathbb{R}^4$.

Exercise 1.4.20: The span and solvability questions are equivalent

Exercise 1.4.18 exhibits three pivot rows for B, not four. By Theorem 4, the following are equivalent: the columns span $\mathbb{R}^4$; $Bx = y$ is consistent for every $y \in \mathbb{R}^4$; and B has a pivot in every row. The last condition fails.

Conclusion. The columns do not span $\mathbb{R}^4$, and some y give no solution.

Exercise 1.4.21: Three columns cannot fill four target directions

A 4×3 matrix formed from these three vectors has at most three pivot positions, because each pivot requires a different column. It therefore cannot have a pivot in every one of its four rows. Theorem 4 excludes spanning $\mathbb{R}^4$. More concretely each vector's coordinate sum is zero; every linear combination also has coordinate sum zero, whereas $(1, 0, 0, 0)^T$ does not.

Conclusion. The three vectors do not span $\mathbb{R}^4$.

Exercise 1.4.22: Three suitable vectors do span three-dimensional space

Put the vectors into columns:

$$V = \begin{bmatrix} 0 & 0 & 4 \\ 0 & -3 & -2 \\ -3 & 9 & -6 \end{bmatrix}.$$

Swap rows 1 and 3. The resulting matrix has pivots $-3, -3, 4$ in successively later columns, so it has three pivot rows. Theorem 4 implies its columns span $\mathbb{R}^3$. One can also solve $Vc = b$ directly: $c_3 = b_1/4$, then $c_2 = -(b_2 + 2c_3)/3$, and finally $c_1 = (9c_2 - 6c_3 - b_3)/3$. These formulas work for every b.

Conclusion. Yes, the vectors span $\mathbb{R}^3$.

Exercise 1.4.23: Six true-or-false statements about the matrix equation

(a) **False.** $Ax = b$ is called a matrix equation. Its equivalent column-sum expression is a vector equation.

(b) **True.** By definition, Ax runs through precisely the linear combinations of the columns; existence of a suitable x is the claim.

(c) **False.** A pivot in every row of $[A \mid b]$ can include a pivot in the last column. For instance $[0 \mid 1]$ has a pivot in its only row but represents $0x = 1$, an inconsistent equation.

(d) **True.** The first entry is $a_{11}x_1 + \cdots + a_{1n}x_n$, a sum of scalar products.

(e) **True.** This is the column-span equivalence in Theorem 4.

(f) **True.** If A had a pivot in every row, Theorem 4 would make every right side attainable. The stated inconsistent right side contradicts that possibility.

Exercise 1.4.24: Six further equivalences and counterexamples

(a) **True.** Expanding Ax into columns changes notation, not the equations or their solutions.

(b) **True.** A solution gives $b = x_1a_1 + \cdots + x_na_n$, which is exactly membership in the column span.

(c) **True.** Place the vectors being combined into the columns of A and their weights into x.

(d) **False.** A pivot in every coefficient row guarantees *consistency* for every right side, not inconsistency.

(e) **True.** Both descriptions use the coefficient matrix $A = [a_1\ a_2\ a_3]$ and the same right side b.

(f) **False.** Failure to span means that *some* b lies outside the span and gives an inconsistent equation. The right side 0 is always consistent, since $x = 0$ solves $Ax = 0$.

Exercise 1.4.25: Read a supplied product without repeating elimination

The given multiplier vector is $(-3, -1, 2)^T$. The definition of matrix multiplication therefore already states

$$(-7, -3, 10)^T = -3(4, 5, -6)^T - (-3, -2, 2)^T + 2(1, 5, -3)^T.$$

Conclusion. $c_1 = -3$, $c_2 = -1$, and $c_3 = 2$.

Exercise 1.4.26: Rearrange a known linear relation

The supplied equality $2u - 3v - w = 0$ rearranges to $w = 2u - 3v$. The displayed matrix has columns u and v, so its product with $(2, -3)^T$ is w. Checking the entries gives $14 - 9 = 5$, $4 - 3 = 1$, and $10 - 9 = 1$.

Conclusion. $x_1 = 2$ and $x_2 = -3$; no row operations are needed.

Exercise 1.4.27: Name every vector and every coefficient

Define

$$v_1 = (-3, 5)^T, \quad v_2 = (5, 8)^T, \quad v_3 = (-4, 1)^T, \quad v_4 = (9, -2)^T, \quad v_5 = (7, -4)^T.$$

Let $(c_1, c_2, c_3, c_4, c_5) = (-3, 1, 2, -1, 2)$ and $b = (11, -11)^T$. The symbolic vector equation is

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 + c_5v_5 = b.$$

Its coordinate sums are $9 + 5 - 8 - 9 + 14 = 11$ and $-15 + 8 + 2 + 2 - 8 = -11$, checking the interpretation.

Exercise 1.4.28: Package a symbolic vector equation

Set $Q = [q_1\ q_2\ q_3]$, a 5×3 matrix, and $x = (x_1, x_2, x_3)^T$. The required matrix equation is $Qx = v$. Its five coordinates reproduce those of $x_1q_1 + x_2q_2 + x_3q_3 = v$, because a product is a weighted column sum.

Exercise 1.4.29: Construct a spanning matrix outside echelon form

Take

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The leading entry in row 2 lies left of that in row 1, so A is not in echelon form. Swapping rows 1 and 2 produces I_3, which has a pivot in every row. Theorem 4 therefore proves its columns span $\mathbb{R}^3$. Explicitly, $(b_1, b_2, b_3)^T$ is obtained with weights $(b_2, b_1, b_3)^T$.

Exercise 1.4.30: Construct a nonspanning matrix outside echelon form

Take

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

The first two leading entries move left, so it is not echelon. Every product is $(x_2, x_1, x_2)^T$; its first and third entries agree. The vector $(1, 0, 0)^T$ violates that condition and cannot be obtained.

Conclusion. This matrix meets both requirements.

Exercise 1.4.31: More rows than columns leave a missing pivot row

A 3×2 matrix has at most two pivots, since there are only two columns. It cannot have a pivot in each of three rows. Theorem 4 then gives some $b \in \mathbb{R}^3$ for which $Ax = b$ is inconsistent. More generally, an $m \times n$ matrix with $m > n$ has at most $n < m$ pivots and hence cannot reach every vector of $\mathbb{R}^m$.

Conclusion. At least one target direction is missing whenever $m > n$.

Exercise 1.4.32: Too few spanning vectors

Place the three proposed vectors in the columns of a 4×3 matrix. At most three of its four rows can contain pivots, so Theorem 4 rules out spanning $\mathbb{R}^4$. In general n vectors in $\mathbb{R}^m$ with $n < m$ give an $m \times n$ matrix with too few columns to pivot in every row.

Conclusion. Neither three vectors in $\mathbb{R}^4$ nor $n < m$ vectors in $\mathbb{R}^m$ can span the whole target space.

Exercise 1.4.33: A unique solution fixes the reduced form

Since the given equation is consistent and has only one solution, none of its three unknowns can be free. Each of the three coefficient columns therefore contains a pivot. In reduced echelon form the pivot columns are unit vectors, with pivots successively lower. Thus the reduced echelon form of A is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The fourth coefficient row must be zero because only three pivots are possible. Consistency ensures the corresponding transformed right-side entry is zero as well.

Exercise 1.4.34: Add solutions to reach the sum of their images

The additivity identity of Theorem 5 gives

$$A(u_1 + u_2) = Au_1 + Au_2 = v_1 + v_2 = w.$$

The vector $u_1 + u_2$ belongs to $\mathbb{R}^4$, the correct input space. It is an explicit solution to $Ax = w$, so the system is consistent.

Exercise 1.4.35: Scale a known input

The scalar-multiplication identity of Theorem 5 gives $A(5y) = 5Ay = 5z$. Thus $x = 5y$ is a solution to the new system. The factor multiplies the entire input vector, not just one entry.

Exercise 1.4.36: A unique square-system solution forces spanning

The unique solution means no unknown is free. All four coefficient columns have pivots, so the 4×4 matrix has four pivot rows. By Theorem 4 its columns span $\mathbb{R}^4$. Consistency for one special b alone would not be enough; uniqueness supplies the crucial four column pivots.

Exercise 1.4.37: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -\frac{93}{50} \\ 0 & 1 & 0 & \frac{53}{50} \\ 0 & 0 & 1 & -\frac{189}{50} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3; there are 3 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. The last row has no pivot, so some right sides are unattainable.

Conclusion. The columns do not span $\mathbb{R}^4$.

Exercise 1.4.38: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 4 & -5 & -1 & 8 \\ 3 & -7 & -4 & 2 \\ 5 & -6 & -1 & 4 \\ 9 & 1 & 10 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 4; there are 3 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. The last row has no pivot, so some right sides are unattainable.

Conclusion. The columns do not span $\mathbb{R}^4$.

Exercise 1.4.39: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 10 & -7 & 1 & 4 & 6 \\ -8 & 4 & -6 & -10 & -3 \\ -7 & 11 & -5 & -1 & -8 \\ 3 & -1 & 10 & 12 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5; there are 4 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. That condition holds, so every right side in $\mathbb{R}^4$ is attainable.

Conclusion. The columns do span $\mathbb{R}^4$.

Exercise 1.4.40: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 5 & 11 & -6 & -7 & 12 \\ -7 & -3 & -4 & 6 & -9 \\ 11 & 5 & 6 & -9 & -3 \\ -3 & 4 & -7 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4, 5; there are 4 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. That condition holds, so every right side in $\mathbb{R}^4$ is attainable.

Conclusion. The columns do span $\mathbb{R}^4$.

Exercise 1.4.41: Delete a redundant column while retaining the span

The pivot columns in the reduction of Exercise 1.4.39 are 1, 2, 3, and 5. These four original columns remain spanning after column 4 is deleted. The reduced form also displays the relation $a_4 = a_1+a_2+a_3$; row operations preserve a relation among columns because they multiply every column by the same invertible operation. Thus the deleted column adds no new span.

Conclusion. Delete column 4. The remaining four pivot columns span $\mathbb{R}^4$.

Exercise 1.4.42: One deletion is possible but two are not

Exercise 1.4.40 has pivot columns 1, 2, 4, and 5. Delete column 3; the remaining four original pivot columns still span $\mathbb{R}^4$. Indeed the reduced matrix shows $a_3 = a_1 - a_2$, so this column was redundant. Deleting two columns from the original five would leave only three vectors. Exercise 1.4.32 proves that three vectors cannot span $\mathbb{R}^4$.

Conclusion. Column 3 can be deleted; no pair of columns can be deleted while preserving the span.

1.5 Solution sets of linear systems

A homogeneous system has right side zero: $Ax = 0$. It is always consistent, because $x = 0$ works. A nontrivial solution means any solution other than this entire zero vector; some individual coordinates may still be zero. A homogeneous system has a nontrivial solution exactly when there is a free variable.

To write a solution set parametrically, solve for pivot variables in terms of the free variables, then separate the resulting vector into its constant part and its parameter-dependent parts. If p is one solution of $Ax = b$, all solutions are $p + v$, where $Av = 0$. Thus a consistent nonhomogeneous solution set is a translation of the homogeneous solution set.

Practice problem 1.5.P1: Intersect two planes by solving their simultaneous equations

The two equations must hold at the same time. Starting with $x_1+4x_2-5x_3 = 0$ and $2x_1-x_2+8x_3 = 9$, subtract twice the first from the second to get $-9x_2 + 18x_3 = 9$. Thus $x_2 = -1 + 2x_3$ and the first equation gives $x_1 = 4 - 3x_3$. Write $x_3 = t$. Then

$$x = (4, -1, 0)^T + t(-3, 2, 1)^T, \qquad t \in \mathbb{R}.$$

Substitution gives $4 - 4 + (-3 + 8 - 5)t = 0$ in the first equation and $8 + 1 + (-6 - 2 + 8)t = 9$ in the second. The intersection is therefore the displayed line, not an empty set or an entire plane.

Practice problem 1.5.P2: A translated plane

Solve $10x_1 - 3x_2 - 2x_3 = 7$ for its pivot variable: $x_1 = 7/10 + (3/10)x_2 + (1/5)x_3$. With $x_2 = s, x_3 = t$ this gives

$$x = (7/10, 0, 0)^T + s(3/10, 1, 0)^T + t(1/5, 0, 1)^T.$$

The two direction vectors are the same as for the homogeneous equation in Example 2. The constant vector translates that plane; it satisfies the new equation itself, since $10(7/10) = 7$.

Exercise 1.5.1: Find a free variable with two eliminations

Add row 1 to row 2 and subtract twice row 1 from row 3. The coefficient matrix becomes

$$\begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 0 & 12 & -9 \end{bmatrix}.$$

The last two rows are negatives. Their sum is zero, leaving only two pivot rows for three unknowns. Hence x_3 is free and a nontrivial solution exists. For an explicit check, $(-17, 6, 8)^T$ gives $-34-30+64 = 0$, $34 - 42 + 8 = 0$, and $-68 + 12 + 56 = 0$.

Conclusion. Yes, the system has nontrivial solutions.

Exercise 1.5.2: Three pivots force the zero solution

The operations $R_2 \leftarrow R_2 + 2R_1$ and $R_3 \leftarrow R_3 - 2R_1$ give

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & -7 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

All three columns have pivots. Back substitution gives $x_3 = 0$, then $x_2 = 0$, then $x_1 = 0$.

Conclusion. No: the only solution is the trivial solution.

Exercise 1.5.3: Fewer equations guarantee a free variable here

A two-row coefficient matrix can have at most two pivots. With three unknowns, at least one variable must be free. Since the system is homogeneous, it is consistent, and the free variable gives a nontrivial solution. One such solution is $(-8, -4, 1)^T$: $24 - 16 - 8 = 0$ and $16 - 20 + 4 = 0$.

Conclusion. Yes; the pivot-count argument needs no row operations.

Exercise 1.5.4: A second two-equation homogeneous system

There are three coefficient columns but only two rows. There cannot be a pivot in every column, so a free variable exists. Homogeneity ensures consistency, hence a nontrivial solution. For a numerical check, $(-2, 16, 29)^T$ gives $-10 - 48 + 58 = 0$ and $6 - 64 + 58 = 0$.

Conclusion. Yes, the system has nontrivial solutions.

Exercise 1.5.5: Separate the common free parameter

The second equation is -2 times the first, so it adds no restriction. The last equation gives $x_2 = -x_3$. Divide the first equation by 2 to obtain $x_1 + x_2 + 2x_3 = 0$, hence $x_1 = -x_3$. Writing $x_3 = t$ gives

$$x = (-t, -t, t)^T = t(-1, -1, 1)^T, \qquad t \in \mathbb{R}.$$

Substitution into each original row gives zero.

Conclusion. The solution set is a line through the origin.

Exercise 1.5.6: Three equal coordinates

The third equation gives $x_2 = x_1$. Substituting into the first produces $3x_1 - 3x_3 = 0$, so $x_1 = x_3$. The second equation then also holds, since $2x_3 + x_3 - 3x_3 = 0$. Thus, with $x_3 = t$,

$$x = t(1, 1, 1)^T, \qquad t \in \mathbb{R}.$$

Conclusion. This is another line through the origin.

Exercise 1.5.7: Two free variables in four coordinates

Row equivalence preserves solutions to the homogeneous equation. The second displayed row gives $x_2 = 4x_3 - 5x_4$; substitute into the first to obtain $x_1 = -9x_3 + 8x_4$. Let $x_3 = s$ and $x_4 = t$.

$$x = s(-9, 4, 1, 0)^T + t(8, -5, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

These expressions satisfy both displayed rows, and every solution arises because its free coordinates may be chosen arbitrarily.

Exercise 1.5.8: Solve the second pivot row first

The second row gives $x_2 = -2x_3 + 4x_4$. The first gives $x_1 = 3x_2 + 8x_3 - 5x_4 = 2x_3 + 7x_4$. Hence

$$x = s(2, -2, 1, 0)^T + t(7, 4, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

Here $s = x_3$ and $t = x_4$ are independent parameters; neither is silently set to zero.

Exercise 1.5.9: The third coordinate is a pivot variable

The equations are $3x_1 - 6x_2 + 6x_3 = 0$ and $-2x_1 + 4x_2 - 2x_3 = 0$. Twice the first plus three times the second gives $6x_3 = 0$. Thus $x_3 = 0$ and $x_1 = 2x_2$.

$$x = t(2, 1, 0)^T, \qquad t \in \mathbb{R}.$$

The parameter is x_2, not x_3: the latter has been forced to zero.

Exercise 1.5.10: A missing column gives a completely free coordinate

The equations are $-x_1 - 4x_2 - 4x_4 = 0$ and $2x_1 - 8x_2 + 8x_4 = 0$. Adding twice the first to the second gives $-16x_2 = 0$. Hence $x_2 = 0$ and $x_1 = -4x_4$. The variable x_3 never appears, so it is independently free.

$$x = s(0, 0, 1, 0)^T + t(-4, 0, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

Exercise 1.5.11: Three free coordinates in a six-coordinate vector

The third nonzero row gives $x_5 = 4x_6$ and the second gives $x_3 = x_6$. Substitution into the first yields

$$x_1 = 4x_2 + 2x_3 - 3x_5 + 5x_6 = 4x_2 - 5x_6.$$

The free variables are $x_2 = s$, $x_4 = t$, and $x_6 = u$. Therefore

$$x = s(4, 1, 0, 0, 0, 0)^T + t(0, 0, 0, 1, 0, 0)^T + u(-5, 0, 1, 0, 4, 1)^T.$$

All three parameters range independently over $\mathbb{R}$. The zero fourth row imposes no additional equation.

Exercise 1.5.12: A pivot in the last coefficient column is not inconsistency

The third row is $x_6 = 0$; this is a coefficient matrix for $Ax = 0$, not an augmented matrix with a nonzero right-side pivot. The second row gives $x_4 = -4x_5 + 6x_6 = -4x_5$. The first then gives $x_1 = 2x_2 - 3x_3 + 6x_4 - 5x_5 = 2x_2 - 3x_3 - 29x_5$. Thus

$$x = s(2,1,0,0,0,0)^T + t(-3,0,1,0,0,0)^T + u(-29,0,0,-4,1,0)^T, \qquad s,t,u \in \mathbb{R}.$$

The parameters are precisely the free coordinates x_2, x_3, x_5.

Exercise 1.5.13: Read the point and direction of a line

Put $x_3 = t$ in the given coordinate description:

$$x = (5 + 4t, -2 - 7t, t)^T = (5,-2,0)^T + t(4,-7,1)^T.$$

Thus the line passes through $(5,-2,0)^T$ and has nonzero direction $(4,-7,1)^T$. Every real t is allowed.

Exercise 1.5.14: A line can lie in four-dimensional space

With $x_4 = t$, the given formulas assemble into

$$x = (5t, 3 - 2t, 2 + 5t, t)^T = (0,3,2,0)^T + t(5,-2,5,1)^T.$$

There is just one free parameter and one nonzero direction vector, so the set is a line in $\mathbb{R}^4$. A line does not require its ambient space to have only two or three dimensions.

Exercise 1.5.15: Parallel planes from the same left side

Solve for x_1 and set $x_2 = s, x_3 = t$. The homogeneous equation gives

$$x = s(-5,1,0)^T + t(3,0,1)^T.$$

The equation with right side -2 gives

$$x = (-2,0,0)^T + s(-5,1,0)^T + t(3,0,1)^T.$$

The same two independent directions occur in both descriptions. The second plane is the first translated by $(-2,0,0)^T$. They are distinct and disjoint: the same point cannot make the left-hand expression equal both 0 and -2.

Exercise 1.5.16: Translate the homogeneous plane by a particular solution

Using $x_2 = s, x_3 = t$, the homogeneous solutions are

$$x = s(2,1,0)^T + t(-3,0,1)^T.$$

For right side 4 the solutions become

$$x = (4,0,0)^T + s(2,1,0)^T + t(-3,0,1)^T.$$

Both are planes with the same two directions. One passes through zero, while the translated plane does not, since the left side at zero is 0, not 4.

Exercise 1.5.17: Translate the line from Exercise 5

The last equation gives $x_2 = -4 - x_3$. Divide the first by 2: $x_1 + x_2 + 2x_3 = 4$, so $x_1 = 8 - x_3$. The second equation is again -2 times the first and adds nothing. With $x_3 = t$,

$$x = (8, -4, 0)^T + t(-1, -1, 1)^T, \qquad t \in \mathbb{R}.$$

The constant vector satisfies the original right side $(8, -16, 12)^T$. The direction agrees with Exercise 1.5.5, so the two solution lines are parallel; the current line is translated away from the origin.

Exercise 1.5.18: Translate the line from Exercise 6

The third equation gives $x_2 = x_1 - 8$. The first becomes $3x_1 - 16 - 3x_3 = 5$, so $x_1 = 7 + x_3$ and $x_2 = -1 + x_3$. The second equation checks: $2(7 + x_3) + (-1 + x_3) - 3x_3 = 13$. Thus

$$x = (7, -1, 0)^T + t(1, 1, 1)^T, \qquad t \in \mathbb{R}.$$

This line has the same direction as the homogeneous line in Exercise 1.5.6 and is its translation by $(7, -1, 0)^T$.

Exercise 1.5.19: A line through a point with a specified direction

A point on the required line is obtained by starting at a and moving any real multiple of b. Thus

$$x = (-2, 0)^T + t(-5, 3)^T, \qquad t \in \mathbb{R}.$$

At $t = 0$ the point is a; differences between two points are scalar multiples of the nonzero vector b, so the line is parallel to b.

Exercise 1.5.20: A line through a point with a specified direction

A point on the required line is obtained by starting at a and moving any real multiple of b. Thus

$$x = (3, -2)^T + t(-7, 6)^T, \qquad t \in \mathbb{R}.$$

At $t = 0$ the point is a; differences between two points are scalar multiples of the nonzero vector b, so the line is parallel to b.

Exercise 1.5.21: Use the difference of two points as direction

Subtract the starting point from the endpoint: $q - p = (1, 4)^T$. Hence the line is

$$x = (3, -3)^T + t(1, 4)^T, \qquad t \in \mathbb{R}.$$

The value $t = 0$ gives p and $t = 1$ gives q; all displacements are parallel to $q - p$.

Exercise 1.5.22: Use the difference of two points as direction

Subtract the starting point from the endpoint: $q - p = (3, -5)^T$. Hence the line is

$$x = (-3, 2)^T + t(3, -5)^T, \qquad t \in \mathbb{R}.$$

The value $t = 0$ gives p and $t = 1$ gives q; all displacements are parallel to $q - p$.

Exercise 1.5.23: Five statements about homogeneous and translated systems

(a) **True.** $x = 0$ always solves a homogeneous equation.

(b) **False.** $Ax = 0$ is an implicit condition. Parametric vector form is an explicit description in terms of free parameters.

(c) **False.** The trivial solution exists whether or not there is a free variable. A free variable is equivalent to existence of a *nontrivial* solution.

(d) **False.** $x = p + tv$ describes the line through p parallel to v, not in general a line through v parallel to p. For example $p = (1, 0)^T, v = (0, 1)^T$ give the vertical line $x_1 = 1$.

(e) **False as stated.** Such a representation requires p to be a particular solution and $Ax = b$ to be consistent. Without those hypotheses it need not hold. With them it is Theorem 6.

Exercise 1.5.24: Five further statements and their justifications

(a) **False.** A homogeneous system has the solution 0, so it cannot be inconsistent.

(b) **False.** Nontrivial means not *all* coordinates are zero. For example $(1, 0)^T$ is a nontrivial solution of $[0\ 1]x = 0$ even though its second entry is zero.

(c) **True.** The displacement from v to $v + p$ is $(v + p) - v = p$, so it is in the direction of p (with no movement when $p = 0$).

(d) **True.** If 0 solves $Ax = b$, then $b = A0 = 0$; that is exactly the homogeneous case.

(e) **True.** Consistency supplies a particular solution p; Theorem 6 states the entire solution set is $p + \{v : Av = 0\}$.

Exercise 1.5.25: Prove the translation theorem in both directions

What we must show. Assume $Ap = b$. Relate every solution to the homogeneous system.

(a) Add a homogeneous solution.

If $Av_h = 0$ and $w = p + v_h$, additivity of multiplication gives $Aw = Ap + Av_h = b + 0 = b$. Thus every proposed vector really is a solution.

(b) Recover the homogeneous part from any solution.

If $Aw = b$, let $v_h = w - p$. Then $Av_h = A(w - p) = Aw - Ap = b - b = 0$, and $w = p + v_h$.

Conclusion. Both inclusions hold, so the two described solution sets are exactly equal, not merely one contained in the other.

Exercise 1.5.26: The zero coefficient matrix imposes no restriction

For the 3×3 zero matrix, $Ax = (0,0,0)^T$ for every input $x \in \mathbb{R}^3$. Thus every vector satisfies the equation.

$$x = s(1,0,0)^T + t(0,1,0)^T + u(0,0,1)^T, \qquad s,t,u \in \mathbb{R}.$$

Conclusion. The solution set is all of $\mathbb{R}^3$.

Exercise 1.5.27: Uniqueness is equivalent to a trivial homogeneous kernel

Because a solution exists, fix one and call it p. By Theorem 6, all solutions are $p + v_h$ with $Av_h = 0$. If the only homogeneous solution is 0, the only translated solution is $p + 0 = p$, so uniqueness holds. Conversely, if there were a nonzero homogeneous v_h, the vector $p + v_h \neq p$ would give a second solution. This contradicts uniqueness.

Conclusion. The equivalence follows in both directions. Consistency is needed in order to fix p.

Exercise 1.5.28: Use the numbers of pivots, rows, and columns

There are 3 unknowns and 3 coefficient pivots. Consequently there are 0 free variables.

(a) Nontrivial homogeneous solutions.

Every variable is a pivot variable, so the only homogeneous solution is zero; there is no nontrivial solution.

(b) Consistency for every right side.

Every one of the 3 rows contains a pivot. Theorem 4 therefore guarantees consistency for every $b \in \mathbb{R}^3$.

Exercise 1.5.29: Use the numbers of pivots, rows, and columns

There are 4 unknowns and 3 coefficient pivots. Consequently there are 1 free variables.

(a) Nontrivial homogeneous solutions.

Since there is at least one free variable, the homogeneous system has nontrivial solutions.

(b) Consistency for every right side.

There are fewer pivots than rows. Theorem 4 therefore says that some right sides have no solution.

Exercise 1.5.30: Use the numbers of pivots, rows, and columns

There are 5 unknowns and 2 coefficient pivots. Consequently there are 3 free variables.

(a) Nontrivial homogeneous solutions.

Since there is at least one free variable, the homogeneous system has nontrivial solutions.

(b) Consistency for every right side.

Every one of the 2 rows contains a pivot. Theorem 4 therefore guarantees consistency for every $b \in \mathbb{R}^2$.

Exercise 1.5.31: Use the numbers of pivots, rows, and columns

There are 2 unknowns and 2 coefficient pivots. Consequently there are 0 free variables.

(a) Nontrivial homogeneous solutions.

Every variable is a pivot variable, so the only homogeneous solution is zero; there is no nontrivial solution.

(b) Consistency for every right side.

There are fewer pivots than rows. Theorem 4 therefore says that some right sides have no solution.

Exercise 1.5.32: A nonzero right side excludes the origin

If a solution set were a plane through the origin, it would contain $x = 0$. But $A0 = 0 \neq b$ when $b \neq 0$.

Conclusion. No. A nonzero right side makes the origin impossible as a solution, regardless of the dimension of the solution set.

Exercise 1.5.33: Arrange for row sums to vanish

Take

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}.$$

This is nonzero, and multiplying by $(1, 1, 1)^T$ gives row sums $1 - 1 + 0 = 0$, $0 + 1 - 1 = 0$, and $1 + 0 - 1 = 0$.

Conclusion. The displayed matrix has the required nonzero null vector.

Exercise 1.5.34: Make every row orthogonal to the prescribed vector

Take

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$$

The row products with $(2, -1, 1)^T$ are respectively $2 - 2 = 0$, $-1 + 1 = 0$, and $2 - 2 = 0$. Thus $A(2, -1, 1)^T = 0$, and A is plainly not the zero matrix.

Exercise 1.5.35: Cancel proportional columns by inspection

The second column is three times the first. Therefore $-3a_1 + a_2 = 0$. By the column definition this means $A(-3, 1)^T = 0$. The vector is nonzero, so it is nontrivial. No elimination is needed.

Exercise 1.5.36: Cancel a different proportionality

The columns satisfy $2a_1 + 3a_2 = 0$, since $2(3, -6, 12)^T + 3(-2, 4, -8)^T = (0, 0, 0)^T$. Thus $(2, 3)^T$ is a nontrivial solution of $Ax = 0$.

Exercise 1.5.37: An inconsistent right side is not a translated line

Choose

$$A = \begin{bmatrix} 1 & -4 \\ 0 & 0 \end{bmatrix}.$$

The homogeneous equation is $x_1 = 4x_2$, with solution line $x = t(4, 1)^T$, exactly the line through $(4, 1)^T$ and zero. Now choose $b = (0, 1)^T$. The second row of $Ax = b$ says $0 = 1$, so its solution set is empty, not a line parallel to the original one.

Conclusion. There is no contradiction: Theorem 6 assumes consistency, and this selected right side violates that assumption.

Exercise 1.5.38: Scalar multiples stay in the homogeneous solution set

The scalar-multiplication rule gives

$$A(cw) = c(Aw) = c0 = 0.$$

This holds for every real c, including $c = 0$ and negative values.

Conclusion. Every multiple cw is another homogeneous solution.

Exercise 1.5.39: Sums and linear combinations stay in the solution set

Additivity first gives $A(v+w) = Av+Aw = 0+0 = 0$. For any scalars c, d, combine both multiplication rules:

$$A(cv + dw) = A(cv) + A(dw) = cAv + dAw = c0 + d0 = 0.$$

Conclusion. Every such linear combination is a homogeneous solution.

Exercise 1.5.40: A missing right side rules out any unique solution

Suppose there were a right-side vector y for which $Ax = y$ had a unique solution. Uniqueness would mean all three coefficient columns have pivots. A 3×3 matrix would then have a pivot in every row. Theorem 4 would make $Ax = b$ consistent for every $b \in \mathbb{R}^3$, contradicting the specified missing right side. Equivalently, the missing right side implies at most two pivots, so at least one variable is free. Every consistent system with this coefficient matrix consequently has infinitely many solutions.

Conclusion. No vector on the right can produce a unique solution.

1.6 Applications of linear systems

Three modeling ideas organize this section.

- In an exchange economy, column j of the exchange table records where sector j's output goes. Each column sums to 1. If E is this table and p lists the values of total outputs, equilibrium means $Ep = p$, or $(E - I)p = 0$.
- In a chemical reaction, conserve each type of atom separately. Assign positive unknown coefficients to all compounds; write reactant atom counts minus product atom counts equal to zero. Choose the smallest positive integer solution.
- In a flow network, write incoming flow equals outgoing flow at every node. After solving the linear equations, apply the required nonnegativity inequalities. A free flow parameter is not automatically allowed to take negative values.

Practice problem 1.6.P1: Build the exchange table in the stated order

Use the order Agriculture, Mining, Manufacturing for both rows and columns. The three columns are respectively $(.65, .05, .30)^T$, $(.20, .10, .70)^T$, and $(.20, .30, .50)^T$:

$$E = \begin{bmatrix} .65 & .20 & .20 \\ .05 & .10 & .30 \\ .30 & .70 & .50 \end{bmatrix}.$$

For example Agriculture retains $1 - .05 - .30 = .65$, so that value belongs in the first entry of the first column. Each displayed column sums to 1; this checks that all output was allocated exactly once.

Practice problem 1.6.P2: Use flow inequalities after solving the equations

Example 2 gives $x_1 = 600 - x_5$, $x_2 = 200 + x_5$, $x_3 = 400$, $x_4 = 500 - x_5$. Nonnegative x_5 and x_4 imply $0 \le x_5 \le 500$. The remaining two inequalities are then automatic. Consequently $100 \le x_1 \le 600$ and $200 \le x_2 \le 700$. Both endpoints occur, at $x_5 = 500$ and $x_5 = 0$, respectively. The two flows vary together: as x_1 decreases, x_2 increases.

Exercise 1.6.1: Two-sector equilibrium is a ratio, not a unique price list

Use the order Goods, Services. The exchange matrix and equations are

$$E = \begin{bmatrix} .20 & .70 \\ .80 & .30 \end{bmatrix}, \qquad .20p_G + .70p_S = p_G, \quad .80p_G + .30p_S = p_S.$$

Both equations simplify to $8p_G = 7p_S$. Thus

$$p = t(7, 8)^T, \qquad t > 0.$$

For example output values 7 and 8 work: Goods spends $.20(7) + .70(8) = 7$, while Services spends $.80(7) + .30(8) = 8$.

Conclusion. Any common positive multiple of $(7, 8)^T$ is an equilibrium.

Exercise 1.6.2: A currency change rescales every output value

In the Coal–Electric–Steel order of Example 1, the exact equations are

$$p_C = .4p_E + .6p_S, \qquad .9p_E = .6p_C + .2p_S.$$

Insert the first into the second: $.9p_E = .24p_E + .56p_S$, so $.66p_E = .56p_S$ and $p_E = (28/33)p_S$. Then $p_C = (31/33)p_S$. Thus an exact alternative is $(p_C, p_E, p_S) = (62, 56, 66)$. Checking the third row gives $.4(62) + .5(56) + .2(66) = 66$. The example rounds its decimal answer; these rational ratios avoid that rounding error. If one dollar is $k > 0$ yen, every component is multiplied by k. Because $E(kp) = kEp = kp$, the exchange proportions and equilibrium relations do not change, only the numerical currency units do.

Exercise 1.6.3: A three-sector exchange economy

(a) Place each sector's sales in its column.

In the order Fuels and Power, Manufacturing, Services,

$$E = \begin{bmatrix} .10 & .10 & .20 \\ .80 & .10 & .40 \\ .10 & .80 & .40 \end{bmatrix}.$$

The retained amounts are $.10, .10, .40$, respectively. Each column sums to 1.

(b) Set the output value equal to the cost of purchased inputs.

Writing the values as f, m, s, the equations $Ep = p$ are

$$.10f + .10m + .20s = f, \quad .80f + .10m + .40s = m, \quad .10f + .80m + .40s = s.$$

Multiplying $(I - E)p = 0$ by 10 gives the augmented matrix

$$\left[\begin{array}{rrr|c} 9 & -1 & -2 & 0 \\ -8 & 9 & -4 & 0 \\ -1 & -8 & 6 & 0 \end{array}\right].$$

(c) Set the Services value to 100 and solve exactly.

The first equation gives $m = 9f - 2s$. Insert this in the second: $-8f + 9(9f - 2s) - 4s = 0$, hence $73f = 22s$. Thus $f = 22s/73$ and $m = 52s/73$. For $s = 100$,

$$p = (2200/73, 5200/73, 100)^T \approx (30.136986, 71.232877, 100)^T.$$

The third equation checks because $-22 - 8(52) + 6(73) = 0$ after factoring out $s/73$.

Exercise 1.6.4: An exact four-sector equilibrium

(a) Construct the exchange matrix.

Use Mining, Lumber, Energy, Transportation in that order:

$$E = \begin{bmatrix} .30 & .15 & .20 & .20 \\ .10 & .15 & .15 & .10 \\ .60 & .50 & .45 & .50 \\ 0 & .20 & .20 & .20 \end{bmatrix}.$$

For instance Mining retains $1-.10-.60=.30$, and the Energy column retains $1-.20-.15-.20=.45$.

(b) Solve $(I-E)p=0$.

Multiplying by 20 avoids decimals. The coefficient matrix is

$$20(I-E) = \begin{bmatrix} 14 & -3 & -4 & -4 \\ -2 & 17 & -3 & -2 \\ -12 & -10 & 11 & -10 \\ 0 & -4 & -4 & 16 \end{bmatrix}.$$

Its reduced equations are

$$p_M = \frac{193}{141}p_T, \qquad p_L = \frac{118}{141}p_T, \qquad p_E = \frac{446}{141}p_T.$$

For clarity one can obtain these ratios without decimal arithmetic: the last equation gives $p_L + p_E = 4p_T$. Substituting $p_E = 4p_T - p_L$ in the second gives $20p_L = 2p_M + 14p_T$; the first gives $14p_M + p_L = 20p_T$. Solving these two equations gives the displayed ratios. Choose $p_T = 141$. Then $p = (193, 118, 446, 141)^T$ is positive. The rows of $20(I-E)p$ are $2702 - 354 - 1784 - 564 = 0$, $-386 + 2006 - 1338 - 282 = 0$, $-2316 - 1180 + 4906 - 1410 = 0$, and $-472 - 1784 + 2256 = 0$.

Conclusion. One exact equilibrium is $(193, 118, 446, 141)^T$; every positive common multiple also works.

Exercise 1.6.5: Compare the equilibrium before and after a change in trade

(a) Build the original table.

Use Agriculture, Manufacturing, Services, Transportation:

$$E = \begin{bmatrix} .20 & .35 & .10 & .20 \\ .20 & .10 & .20 & .30 \\ .30 & .35 & .50 & .20 \\ .30 & .20 & .20 & .30 \end{bmatrix}.$$

Each column is the allocation of the sector at its top, including its retained share on the diagonal.

(b) Normalize Transportation to 10.

Writing a, m, s, t for the four output values, multiply $(I-E)p=0$ by 20. The equations are

$$\begin{aligned} 16a - 7m - 2s - 4t &= 0, & -4a + 18m - 4s - 6t &= 0, \\ -6a - 7m + 10s - 4t &= 0, & -6a - 4m - 4s + 14t &= 0. \end{aligned}$$

The second and fourth give $2m + 2s = 7t - 3a$ and $-2a + 9m - 2s = 3t$, so $11m = 10t - a$. The first equation together with $2m + 2s = 7t - 3a$ gives $19a - 5m = 11t$. Substitute $11m = 10t - a$ to

obtain $209a - 5(10t - a) = 121t$, or $214a = 171t$. Therefore $a = (171/214)t$, $m = (179/214)t$, and $s = (627/428)t$. For $t = 10$,

$$p = (855/107, 895/107, 3135/214, 10)^T \approx (7.990654, 8.364486, 14.649533, 10)^T.$$

(c) Change just the Agriculture column.

The new shares from Agriculture are $.20, .10, .40, .30$; the other columns stay the same. Thus

$$\widetilde{E} = \begin{bmatrix} .20 & .35 & .10 & .20 \\ .10 & .10 & .20 & .30 \\ .40 & .35 & .50 & .20 \\ .30 & .20 & .20 & .30 \end{bmatrix}.$$

(d) Solve and compare with the same normalization.

The new coefficient matrix is

$$20(I - \widetilde{E}) = \begin{bmatrix} 16 & -7 & -2 & -4 \\ -2 & 18 & -4 & -6 \\ -8 & -7 & 10 & -4 \\ -6 & -4 & -4 & 14 \end{bmatrix}.$$

Its reduced equations give $a = (57/73)t$, $m = (56/73)t$, and $s = (114/73)t$. For example the last row again gives $2m + 2s = 7t - 3a$, and the second becomes $11m = 10t - 2a$; the first still gives $19a - 5m = 11t$. Substitution now yields $209a - 5(10t - 2a) = 121t$, hence $219a = 171t$ and $a = (57/73)t$, from which the other two ratios follow. At $t = 10$ the exact solution is

$$\widetilde{p} = (570/73, 560/73, 1140/73, 10)^T \approx (7.808219, 7.671233, 15.616438, 10)^T.$$

Direct multiplication verifies both $Ep = p$ and $\widetilde{E}\widetilde{p} = \widetilde{p}$. With Transportation fixed, Agriculture and Manufacturing decrease by about 0.182435 and 0.693253, while Services increases by about 0.966905. These are changes in the values of total sector outputs under the model's convention, not independently determined retail prices of individual products.

Exercise 1.6.6: Balance aluminum oxide and carbon

Let the four coefficients, in the printed order, be a, b, c, d. Using atom order Al, O, C gives the vector equation

$$a(2, 3, 0)^T + b(0, 0, 1)^T = c(1, 0, 0)^T + d(0, 2, 1)^T.$$

Thus $2a = c$, $3a = 2d$, and $b = d$. Choose the smallest positive a making all coefficients integral: $a = 2$ gives $d = 3, b = 3, c = 4$.

Conclusion. $2\mathrm{Al_2O_3} + 3\mathrm{C} \longrightarrow 4\mathrm{Al} + 3\mathrm{CO_2}$. The atom totals are Al: $4 = 4$, O: $6 = 6$, C: $3 = 3$.

Exercise 1.6.7: Count every hydrogen in citric acid

Let the coefficients of $\mathrm{NaHCO_3}, \mathrm{H_3C_6H_5O_7}, \mathrm{Na_3C_6H_5O_7}, \mathrm{H_2O}, \mathrm{CO_2}$ be a, b, c, d, e. The citric acid molecule has $3 + 5 = 8$ hydrogen atoms. In atom order Na, H, C, O, the conservation equations are

$$a = 3c, \quad a + 8b = 5c + 2d, \quad a + 6b = 6c + e, \quad 3a + 7b = 7c + d + 2e.$$

This is the coordinate form of the vector equation obtained by listing those four atom counts for each compound. Set $c = 1$, so $a = 3$. The hydrogen equation gives $d = 4b - 1$; the carbon equation gives $e = 6b - 3$. The oxygen equation becomes $9 + 7b = 7 + (4b - 1) + 2(6b - 3) = 16b$, hence $b = 1$. Thus $d = e = 3$.

Conclusion. $3NaHCO_3 + H_3C_6H_5O_7 \longrightarrow Na_3C_6H_5O_7 + 3H_2O + 3CO_2$. Atom totals on both sides are Na: 3, H: 11, C: 9, O: 16.

Exercise 1.6.8: Balance limestone neutralization

Use coefficients a, b, c, d, e in the printed order, and atom order H, O, Ca, C. The vector equation gives

$$3a = 2c, \quad a + 3b = c + 2e, \quad b = d, \quad b = e.$$

Substitute $d = e = b$ in the oxygen equation to get $c = a + b$. Then $3a = 2(a + b)$ gives $a = 2b$. The smallest choice $b = 1$ therefore gives $(a, b, c, d, e) = (2, 1, 3, 1, 1)$.

Conclusion. $2H_3O + CaCO_3 \longrightarrow 3H_2O + Ca + CO_2$. The H, O, Ca, C totals are respectively $6, 5, 1, 1$ on each side.

Exercise 1.6.9: Conserve boron, sulfur, hydrogen, and oxygen

With coefficients a, b, c, d, the atom-count vector equation yields

$$2a = c, \qquad 3a = d, \qquad 2b = 3c + 2d, \qquad b = 3c.$$

The first, second, and fourth give $c = 2a, d = 3a, b = 6a$. The hydrogen equation checks: $2(6a) = 3(2a) + 2(3a)$. Taking $a = 1$ gives the primitive integer coefficients.

Conclusion. $B_2S_3 + 6H_2O \longrightarrow 2H_3BO_3 + 3H_2S$. The atom totals B, S, H, O are $2, 3, 12, 6$.

Exercise 1.6.10: Use exact fractions in a six-compound reaction

Let the coefficients of the six compounds, in printed order, be a, b, c, d, e, f. The vector equation in atom order Pb, N, Cr, Mn, O has coefficient matrix

$$\begin{bmatrix} 1 & 0 & -3 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 & 0 & 0 \\ 0 & 2 & 0 & 0 & -1 & 0 \\ 0 & 8 & -4 & -3 & -2 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} = 0.$$

The first four rows give $a = 3c$, $f = 6a = 18c$, $b = 2d$, and $e = 2b = 4d$. The last row says

$$8b = 4c + 3d + 2e + f \quad \Longrightarrow \quad 16d = 22c + 11d, \quad 5d = 22c.$$

To get positive integers choose $c = 5$, so $d = 22$, $a = 15, b = 44, e = 88, f = 90$. The entries have common divisor 1.

Conclusion. $15PbN_6 + 44CrMn_2O_8 \longrightarrow 5Pb_3O_4 + 22Cr_2O_3 + 88MnO_2 + 90NO$. The oxygen check is $44(8) = 5(4) + 22(3) + 88(2) + 90 = 352$. The other rows already check each remaining element.

Exercise 1.6.11: Balance seven compounds without decimal rounding

Use coefficients a, b, c, d, e, f, g for the seven printed compounds. Conserve Mn, S, As, Cr, H, O, respectively:

$$\begin{aligned} a &= d, & a + c &= 3f, & 2b &= e, \\ 10b &= f, & 2c &= d + 3e + 2g, & 35b + 4c &= 4d + 12f + g. \end{aligned}$$

These are the coordinates of a single atom-count vector equation. The first four give $d = a$, $e = 2b$, $f = 10b$, $c = 30b - a$. The hydrogen equation becomes $2g = 54b - 3a$. The oxygen equation becomes $g = 35b - 8a$. Equating the two formulas gives

$$54b - 3a = 70b - 16a, \qquad 13a = 16b.$$

Choose $b = 13, a = 16$. Then $c = 374, d = 16, e = 26, f = 130, g = 327$.

$$16\mathrm{MnS} + 13\mathrm{As_2Cr_{10}O_{35}} + 374\mathrm{H_2SO_4} \longrightarrow 16\mathrm{HMnO_4} + 26\mathrm{AsH_3} + 130\mathrm{CrS_3O_{12}} + 327\mathrm{H_2O}.$$

Check sulfur: $16 + 374 = 390 = 3(130)$; hydrogen: $2(374) = 748 = 16 + 3(26) + 2(327)$; oxygen: $35(13) + 4(374) = 1951 = 4(16) + 12(130) + 327$. Mn, As, and Cr match by the first equations.

Conclusion. The displayed positive integer coefficients balance every atom; their common divisor is 1.

Exercise 1.6.12: Use junction equations before imposing nonnegative flow

Read the arrows at the three nodes. Node A receives $x_1 + x_4$ and sends x_2; node B receives x_2 and sends $x_3 + 100$; node C receives $x_3 + 80$ and sends x_4. Thus

$$x_1 + x_4 = x_2, \qquad x_2 = x_3 + 100, \qquad x_4 = x_3 + 80.$$

The last two inserted in the first give $x_1 = 20$. Set $x_3 = t$. The general algebraic flow is

$$(x_1, x_2, x_3, x_4) = (20, 100 + t, t, 80 + t).$$

Nonnegative flows require $t \geq 0$; that condition makes every other component nonnegative too. Hence $x_4 = 80 + t \geq 80$, and $t = 0$ attains the lower bound.

Conclusion. The smallest possible x_4 is 80.

Exercise 1.6.13: A five-node network with two independent free flows

(a) Record all node balances.

Following the arrows in the supplied figure gives

$$\begin{aligned} A:&\quad 30 + x_2 = 80 + x_1, & E:&\quad 60 + x_1 = 20 + x_3, \\ B:&\quad x_3 + x_5 = x_2 + x_4, & C:&\quad 100 + x_6 = 40 + x_5, \\ D:&\quad 40 + x_4 = 90 + x_6. \end{aligned}$$

Let $s = x_1$ and $t = x_6$. The first, second, fourth, and fifth balances give

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (s, s + 50, s + 40, t + 50, t + 60, t).$$

The balance at B checks identically: $(s + 40) + (t + 60) = (s + 50) + (t + 50)$.

(b) Apply the arrow-direction restrictions.

Nonnegative x_1 and x_6 require $s, t \geq 0$, and these inequalities make all other flows nonnegative. Their individual minima are therefore

$$\min x_2 = 50, \quad \min x_3 = 40, \quad \min x_4 = 50, \quad \min x_5 = 60.$$

They can all be attained at once by taking $s = t = 0$.

Exercise 1.6.14: A road closure specializes a two-parameter traffic pattern

(a) Write flow in equals flow out at each intersection.

The four balances read

$$80 = x_1 + x_5, \quad 100 + x_1 + x_2 = x_4, \quad x_3 = x_2 + 90, \quad x_4 + x_5 = x_3 + 90.$$

The last is dependent on the first three; total external inflow $80 + 100$ equals total external outflow $90 + 90$. Put $s = x_2$ and $t = x_5$. Then

$$(x_1, x_2, x_3, x_4, x_5) = (80 - t, s, 90 + s, 180 + s - t, t).$$

For the indicated traffic directions require $s \geq 0$ and $0 \leq t \leq 80$; the remaining inequalities then follow.

(b) Close the road carrying x_5.

Setting $t = 0$ gives

$$(x_1, x_2, x_3, x_4, x_5) = (80, s, 90 + s, 180 + s, 0), \qquad s \geq 0.$$

This is the entire feasible family with the road closed, rather than one arbitrarily selected flow pattern.

(c) Minimize x_4.

The formula $x_4 = 180 + s$ with $s \geq 0$ gives the lower bound 180. It is attained when $s = 0$.

Conclusion. With $x_5 = 0$, the minimum flow in branch 4 is 180 cars/minute.

Exercise 1.6.15: Circulating flow around a one-way roundabout

Read the successive balances around A, B, C, D, E, F:

$$x_6 + 60 = x_1, \quad x_1 = x_2 + 70, \quad x_2 + 100 = x_3,$$

$$x_3 = x_4 + 90, \quad x_4 + 80 = x_5, \quad x_5 = x_6 + 80.$$

Choose $t = x_6$. Working around the circle yields

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (t + 60, t - 10, t + 90, t, t + 80, t).$$

The final equation checks because $(t + 80) - t = 80$. To respect every arrow all six flows must be nonnegative. The strongest restriction is $x_2 = t - 10 \geq 0$, namely $t \geq 10$; this also ensures each other component is nonnegative.

Conclusion. All feasible flows are the displayed family with $t \geq 10$, and the smallest possible x_6 is 10.

1.7 Linear independence

The main question. For vectors $v_1, \ldots, v_p$, write $c_1v_1+\cdots+c_pv_p = 0$. The list is *linearly independent* if its only solution is $c_1 = \cdots = c_p = 0$. It is dependent if a solution exists with at least one nonzero coefficient. Notice that the zero-coefficient solution always exists, so its existence proves nothing. Putting the vectors into the columns of A turns the question into $Ac = 0$. A pivot in every column means no free variable, hence independence. A free variable supplies a nontrivial dependence relation.

Practice problem 1.7.P1: Check every pair, not only the whole list

The four vectors are $u = (3, 2, -4)$, $v = (-6, 1, 7)$, $w = (0, -5, 2)$, and $z = (3, 7, -5)$. None is zero. For two nonzero vectors, dependence means one is a scalar multiple of the other. We can disprove this by looking just at their first two coordinates: the quantities $a_1b_2 - a_2b_1$ for $(u, v), (u, w), (u, z), (v, w), (v, z), (w, z)$ are respectively $15, -15, 15, 30, -45, 15$. None is zero. For example, if $v = tu$, then $3(1) - 2(-6)$ would have to be zero; it is 15.

Conclusion. All six pairs are linearly independent.

Practice problem 1.7.P2: Pairwise independence is weaker than independence

No. Checking pairs rules out a relation using only two vectors, but it does not rule out a relation using three or four. For instance, $e_1, e_2, e_1 + e_2$ are pairwise independent yet $e_1 + e_2 - (e_1 + e_2) = 0$ is a nontrivial relation among all three. The same logical distinction applies to the four given vectors.

Practice problem 1.7.P3: Why testing a selected vector can miss dependence

It is not a reliable test to ask only whether w is in the span of the other three. A dependence relation may have coefficient zero on w. For these data, solving with the actual coordinates gives $z = 3u + v$. Thus $3u + v - z = 0$ already establishes dependence without involving w. Moreover w is outside $\text{Span}\{u, v\}$: the equations $3a - 6b = 0$, $2a + b = -5$ force $b = -1$, $a = -2$, but then $-4a + 7b = 1 \neq 2$. So $w \notin \text{Span}\{u, v, z\}$ despite dependence of the whole list. Testing one chosen vector could therefore mislead us.

Practice problem 1.7.P4: More vectors than coordinates

There are four vectors in $\mathbb{R}^3$, so their 3×4 column matrix cannot have a pivot in all four columns. At least one variable is free in the homogeneous system, and the list is dependent. The relation $3u + v - z = 0$ from the preceding problem gives a direct check.

Exercise 1.7.1: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 5 & 7 & 9 \\ 0 & 2 & 4 \\ 0 & -6 & -8 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{5}R_1$. Then

$$\begin{bmatrix} 1 & \frac{7}{5} & \frac{9}{5} \\ 0 & 2 & 4 \\ 0 & -6 & -8 \end{bmatrix}.$$

Use $R_2 \leftarrow \frac{1}{2}R_2, \quad R_1 \leftarrow R_1 - (\frac{7}{5})R_2, \quad R_3 \leftarrow R_3 - (-6)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{4}R_3, \quad R_1 \leftarrow R_1 - (-1)R_3, \quad R_2 \leftarrow R_2 - (2)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.2: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 2 & 0 & 3 \\ 3 & -8 & 1 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_1 \leftarrow \frac{1}{2}R_1, \quad R_3 \leftarrow R_3 - (3)R_1$. Then

$$\begin{bmatrix} 1 & 0 & \frac{3}{2} \\ 0 & 0 & -1 \\ 0 & -8 & -\frac{7}{2} \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_2 \leftarrow -\frac{1}{8}R_2$. Then

$$\begin{bmatrix} 1 & 0 & \frac{3}{2} \\ 0 & 1 & \frac{7}{16} \\ 0 & 0 & -1 \end{bmatrix}.$$

Use $R_3 \leftarrow -1R_3, \quad R_1 \leftarrow R_1 - (\frac{3}{2})R_3, \quad R_2 \leftarrow R_2 - (\frac{7}{16})R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.3: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 2 & -4 \\ -3 & 6 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1, \quad R_2 \leftarrow R_2 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

Conclusion. The vectors are linearly dependent.

Exercise 1.7.4: Test the column equation

What we must show. Determine whether the given vectors are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} -1 & -3 \\ 3 & -9 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_1 \leftarrow \frac{1}{3}R_1, \quad R_2 \leftarrow R_2 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -3 \\ 0 & -6 \end{bmatrix}.$$

Use $R_2 \leftarrow -\frac{1}{6}R_2, \quad R_1 \leftarrow R_1 - (-3)R_2$. Then

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.5: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 0 & -3 & 9 \\ 2 & 1 & -7 \\ -1 & 4 & -5 \\ 1 & -4 & -2 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_4$, $R_2 \leftarrow R_2 - (2)R_1$, $R_3 \leftarrow R_3 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -4 & -2 \\ 0 & 9 & -3 \\ 0 & 0 & -7 \\ 0 & -3 & 9 \end{bmatrix}.$$

Use $R_2 \leftarrow \frac{1}{9}R_2$, $R_1 \leftarrow R_1 - (-4)R_2$, $R_4 \leftarrow R_4 - (-3)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -\frac{10}{3} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & -7 \\ 0 & 0 & 8 \end{bmatrix}.$$

Use $R_3 \leftrightarrow R_4$, $R_3 \leftarrow \frac{1}{8}R_3$, $R_1 \leftarrow R_1 - (-\frac{10}{3})R_3$, $R_2 \leftarrow R_2 - (-\frac{1}{3})R_3$, $R_4 \leftarrow R_4 - (-7)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.6: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} -4 & -3 & 0 \\ 0 & -1 & 5 \\ 1 & 1 & -5 \\ 2 & 1 & -10 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_3$, $R_3 \leftarrow R_3 - (-4)R_1$, $R_4 \leftarrow R_4 - (2)R_1$. Then

$$\begin{bmatrix} 1 & 1 & -5 \\ 0 & -1 & 5 \\ 0 & 1 & -20 \\ 0 & -1 & 0 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_1 \leftarrow R_1 - (1)R_2$, $R_3 \leftarrow R_3 - (-1)R_2$, $R_4 \leftarrow R_4 - (-1)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 15 \\ 0 & 1 & -20 \\ 0 & 0 & -15 \\ 0 & 0 & -20 \end{bmatrix}.$$

Use $R_3 \leftarrow -\frac{1}{15}R_3$, $R_1 \leftarrow R_1 - (15)R_3$, $R_2 \leftarrow R_2 - (-20)R_3$, $R_4 \leftarrow R_4 - (-20)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Every column has a pivot. The reduced equations force every coefficient in c to be zero, so there is no nontrivial dependence relation.

Conclusion. The vectors are linearly independent.

Exercise 1.7.7: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 1 & 4 & -3 & 0 \\ -2 & -7 & 5 & 1 \\ -4 & -5 & 7 & 5 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1$, $R_3 \leftarrow R_3 - (-4)R_1$. Then

$$\begin{bmatrix} 1 & 4 & -3 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 11 & -5 & 5 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_2$, $R_3 \leftarrow R_3 - (11)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 6 & -6 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{6}R_3$, $R_1 \leftarrow R_1 - (1)R_3$, $R_2 \leftarrow R_2 - (-1)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

Conclusion. The vectors are linearly dependent.

Exercise 1.7.8: Test the column equation

What we must show. Determine whether the given vectors (the columns of the matrix) are independent. Form $Ac = 0$; row operations preserve exactly its solutions.

$$A = \begin{bmatrix} 1 & -2 & 3 & 2 \\ -2 & 4 & -6 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 & 2 \\ 0 & 0 & 0 & 6 \\ 0 & 1 & -1 & 3 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_1 \leftarrow R_1 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 & 8 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix}.$$

Use $R_3 \leftarrow \frac{1}{6}R_3, \quad R_1 \leftarrow R_1 - (8)R_3, \quad R_2 \leftarrow R_2 - (3)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There is a free variable. For example, the nonzero coefficient vector $c = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ satisfies $Ac = 0$ (substitution checks every coordinate). This is a nontrivial relation among the columns.

Conclusion. The vectors are linearly dependent.

Exercise 1.7.9: A span question can differ from a dependence question

(a) Reduce the span of the first two vectors.

Here $v_1 = (1, -3, 2)$ and $v_2 = (-3, 9, -6) = -3v_1$. Their span is the line of multiples of v_1. For $v_3 = (5, -7, h)$ to lie on that line, its first coordinate would force the multiplier to be 5, but its second would then have to be -15, not -7. No choice of h fixes that contradiction.

(b) Test the three-vector list instead.

The relation $3v_1 + v_2 + 0v_3 = 0$ is nontrivial for every h.

Conclusion. (a) No real h. (b) Every real h. The first two vectors are already dependent, whether or not the third lies in their span.

Exercise 1.7.10: The same distinction with another line

Here $v_1 = (1, -3, -5)$, $v_2 = (-3, 9, 15) = -3v_1$, and $v_3 = (2, -5, h)$.

(a) Compare coordinates of a proposed multiple.

If $v_3 = tv_1$, its first coordinate gives $t = 2$, but then the second coordinate would be -6, not -5. Thus v_3 is outside $\text{Span}\{v_1, v_2\}$ for every h.

(b) Exhibit a relation that does not use v_3.

Again $3v_1 + v_2 + 0v_3 = 0$, with coefficients not all zero.

Conclusion. (a) No h. (b) All $h \in \mathbb{R}$.

Exercise 1.7.11: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 2 & 4 & -2 \\ -2 & -6 & 2 \\ 4 & 7 & h \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1$, $\quad R_2 \leftarrow R_2 - (-2)R_1$, $\quad R_3 \leftarrow R_3 - (4)R_1$. Then

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -1 & h+4 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $\quad R_2 \leftarrow -1R_2$, $\quad R_1 \leftarrow R_1 - (2)R_2$, $\quad R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2h+7 \\ 0 & 1 & -h-4 \\ 0 & 0 & -2h-8 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $-2\,(h+4) = 0$.

Conclusion. The required value is $h = -4$. At other values all three coefficients are forced to zero, giving independence.

Exercise 1.7.12: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 3 & -6 & 9 \\ -6 & 4 & h \\ 1 & -3 & 3 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_3$, $R_2 \leftarrow R_2 - (-6)R_1$, $R_3 \leftarrow R_3 - (3)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & -14 & h+18 \\ 0 & 3 & 0 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{1}{3}R_2$, $R_1 \leftarrow R_1 - (-3)R_2$, $R_3 \leftarrow R_3 - (-14)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & h+18 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $h + 18 = 0$.

Conclusion. The required value is $h = -18$. At other values all three coefficients are forced to zero, giving independence.

Exercise 1.7.13: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 5 & -9 & h \\ -3 & 6 & -9 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (5)R_1$, $R_3 \leftarrow R_3 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & h-15 \\ 0 & 0 & 0 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2h-27 \\ 0 & 1 & h-15 \\ 0 & 0 & 0 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $0 = 0$.

This condition holds identically. The reduced equations are solved nontrivially by $c_3 = 1$, $c_2 = 15 - h$, and $c_1 = 27 - 2h$.

Conclusion. The list is dependent for every real h.

Exercise 1.7.14: Locate the parameter value that removes a pivot

Plan. Keep h symbolic, but eliminate using only the first two columns. Those pivots are nonzero constants, so these operations do not assume anything about h.

$$A = \begin{bmatrix} 1 & -3 & 2 \\ -2 & 7 & 1 \\ -4 & 6 & h \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-2)R_1, \quad R_3 \leftarrow R_3 - (-4)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & 5 \\ 0 & -6 & h+8 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-3)R_2, \quad R_3 \leftarrow R_3 - (-6)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 17 \\ 0 & 1 & 5 \\ 0 & 0 & h+38 \end{bmatrix}.$$

The first two pivot columns are present for all h. The three columns are dependent precisely when the last row contributes no third pivot, that is, when $h + 38 = 0$.

Conclusion. The required value is $h = -38$. At other values all three coefficients are forced to zero, giving independence.

Exercise 1.7.15: Four vectors in a two-coordinate space

The list has four vectors in $\mathbb{R}^2$. Its 2×4 column matrix has at most two pivots, leaving at least two free variables in $Ac = 0$.

Conclusion. Dependent. More than two vectors in $\mathbb{R}^2$ cannot be independent.

Exercise 1.7.16: Spot a scalar multiple

The second vector equals $-\frac{3}{2}$ times the first: $(-3, 6, -12) = -\frac{3}{2}(2, -4, 8)$. Thus $3v_1 + 2v_2 = 0$ is a relation with nonzero coefficients.

Conclusion. Dependent.

Exercise 1.7.17: A zero vector is enough

The second vector is zero. The coefficients $(0, 1, 0)$ produce $0v_1 + 1v_2 + 0v_3 = 0$ without being all zero.

Conclusion. Dependent. The values of the other vectors do not matter.

Exercise 1.7.18: Count the coordinates first

There are four vectors in $\mathbb{R}^2$, so a 2×4 column matrix cannot have four pivot columns. Its homogeneous system has a free variable.

Conclusion. Dependent. Row reduction is unnecessary.

Exercise 1.7.19: Two vectors that are not multiples

For $u = (-8, 12, -4)$ and $v = (2, -3, -1)$, the first coordinate of $v = tu$ would force $t = -1/4$. The third coordinate would then be 1, whereas v has third coordinate -1. Neither vector is zero, so the failure of scalar proportionality proves independence.

Conclusion. Independent.

Exercise 1.7.20: Another zero-vector relation

The third vector is zero. The coefficient list $(0, 0, 1)$ therefore solves the column equation nontrivially.

Conclusion. Dependent.

Exercise 1.7.21: Four statements about dependence

(a) **False.** Every homogeneous equation has the trivial solution. Independence requires it to be the *only* solution. The zero matrix has the trivial solution and many others.

(b) **False.** At least one vector is a combination of the others, not necessarily every vector. The list $e_1, 2e_1, e_2$ is dependent, but e_2 is not in the span of the first two vectors.

(c) **True.** Five columns in $\mathbb{R}^4$ give more variables than possible pivots, so the homogeneous system has a free variable.

(d) **True.** A nontrivial relation $ax + by + cz = 0$ must have $c \neq 0$; otherwise independence of x, y would give $a = b = 0$. Solving gives $z = -(a/c)x - (b/c)y$.

Exercise 1.7.22: Avoid converses that have not been proved

(a) **True.** If $w = au + bv$, then $au + bv - w = 0$ is a relation with coefficient -1 on w, hence a nontrivial relation.

(b) **False as written.** A plane need not pass through zero. The independent vectors e_1, e_2, e_3 all lie in the plane $x_1 + x_2 + x_3 = 1$. Vectors lying in a plane *through zero* are dependent when there are three of them, but the question does not impose that condition.

(c) **False.** In $\mathbb{R}^3$, the two vectors $e_1, 2e_1$ are dependent. Having few vectors makes independence possible, not automatic.

(d) **False.** The two vectors $e_1, 2e_1$ in $\mathbb{R}^3$ give a counterexample: dependence with only two vectors, fewer than three.

Exercise 1.7.23: Every rank-deficient two-by-two echelon pattern

Use p for a nonzero pivot and $*$ for an arbitrary entry. Dependent columns mean fewer than two pivots. The possible echelon patterns are

$$\begin{bmatrix} p & * \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & p \\ 0 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

These exhaust rank one and rank zero. A second pivot would force independence.

Exercise 1.7.24: Three independent columns

Independence requires all three columns to have pivots. Since pivots move right as one moves down, their only possible positions are $(1,1),(2,2),(3,3)$. Thus the echelon pattern is

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \end{bmatrix}, \qquad p,q,r \neq 0.$$

The arbitrary entries are above pivots; entries below them are zero.

Exercise 1.7.25: Do not silently assume the first column is nonzero

If $a_1 \neq 0$ and a_2 is not its multiple, the two columns are independent. But the hypothesis also permits $a_1 = 0$, in which case $a_2 \neq 0$ and there is only one pivot, in column two. Thus both patterns are possible:

$$\begin{bmatrix} p & * \\ 0 & q \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & p \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad p,q \neq 0.$$

For the second pattern, no nonzero vector is a multiple of the zero vector, so it really does satisfy the stated condition.

Exercise 1.7.26: A third column outside a plane

Independence of a_1, a_2 supplies two pivots. If the third column failed to add a pivot, it would be a combination of the first two, contrary to the assumption. Hence all three columns are pivot columns:

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix}, \qquad p,q,r \neq 0.$$

Exercise 1.7.27: Count pivots for independence

A 6×4 matrix with independent columns must have **four** pivot columns: one for each variable in $Ax = 0$. Fewer would leave a free variable and hence a nonzero solution. Extra rows do not require extra pivots.

Exercise 1.7.28: Count pivots for spanning

For the columns of a 4×6 matrix to span $\mathbb{R}^4$, $Ax = b$ must be consistent for every four-coordinate b. This requires a pivot in every row, so there must be **four** pivot columns. Six are impossible because there are only four rows.

Exercise 1.7.29: Build dependent and independent examples

Take

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then $A(-2,1)^T = 0$ is a nontrivial solution. On the other hand, $Bx = 0$ has first two equations $x_1 = 0, x_2 = 0$, so its only solution is zero. Both matrices have the required size 3×2.

Exercise 1.7.30: State the exact pivot condition

(a) Fill the blank.

The number is $\boxed{n}$.

(b) Prove both directions.

There are n columns. If all n are pivot columns, $Ax = 0$ has no free variable, so its only solution is zero and the columns are independent. Conversely, if there are fewer than n pivot columns, some variable is free. Set one free variable to 1 and the others to 0, and solve for the pivot variables. This gives a nonzero homogeneous solution, contradicting independence. Thus independence forces n pivot columns.

Exercise 1.7.31: Read a relation from the columns

The given relation is $a_3 = a_1 + a_2$. Move all terms to one side: $a_1 + a_2 - a_3 = 0$. Therefore

$$x = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

is a nonzero solution of $Ax = 0$. No row reduction is needed, because matrix multiplication is the weighted sum of the columns.

Exercise 1.7.32: Read another relation without elimination

Here $a_1 - 3a_2 = a_3$, so $a_1 - 3a_2 - a_3 = 0$. Consequently $x = (1, -3, -1)^T$ is a nontrivial solution. The signs come from moving a_3 to the left, not from changing the columns themselves.

Exercise 1.7.33: A relation already gives the answer

True. The assumption $v_3 = 2v_1 + v_2$ gives $2v_1 + v_2 - v_3 + 0v_4 = 0$. The coefficients are not all zero, so the four-vector list is dependent regardless of v_4.

Exercise 1.7.34: The overlooked zero-vector exception

False. In $\mathbb{R}^4$, take $v_1 = 0$ and $v_2 = e_1$. The nonzero v_2 is not a scalar multiple of v_1, but $1v_1 + 0v_2 = 0$ is a nontrivial dependence relation. For two nonzero vectors the proposed test would be valid; the statement did not require both vectors to be nonzero.

Exercise 1.7.35: A zero column yields a relation

True. If $v_3 = 0$, the coefficient list $(0, 0, 1, 0, 0)$ gives a zero weighted sum and is not itself zero. Thus the list is dependent.

Exercise 1.7.36: The first two vectors might already be dependent

False. Let $v_1 = e_1$, $v_2 = 2e_1$, and $v_3 = e_2$ in $\mathbb{R}^3$. The third vector is outside the span of the first two, but $2v_1 - v_2 + 0v_3 = 0$ proves dependence. Adding a new direction cannot remove a relation that was already present.

Exercise 1.7.37: Dependence survives enlarging the list

True. A nontrivial relation among v_1, v_2, v_3 remains a nontrivial relation among v_1, v_2, v_3, v_4 after assigning coefficient zero to v_4. Therefore the larger list is dependent.

Exercise 1.7.38: Independence survives taking a sublist

True. Suppose $c_1v_1 + c_2v_2 + c_3v_3 = 0$. Then $c_1v_1 + c_2v_2 + c_3v_3 + 0v_4 = 0$. Independence of the four-vector list forces $c_1 = c_2 = c_3 = 0$. This is precisely the independence test for the three-vector sublist.

Exercise 1.7.39: Use the special target zero

What we must show. Prove column independence from the assumption of at most one solution for every b. Apply the assumption to $b = 0$. The zero vector is already one solution of $Ax = 0$, so "at most one" says there is no other solution. Thus every relation among the columns has all coefficients zero, which is the definition of independence. Existence for other b is not needed.

Exercise 1.7.40: No free variable means no second solution

There are n pivot columns and only n columns altogether, so there are no free variables. For a given b, reduction of $[A \mid b]$ may produce a contradiction, giving no solution; if it does not, every variable is a pivot variable with one determined value. Thus there can never be infinitely many solutions or two distinct solutions. Equivalently, if $Ax = Ay = b$, then $A(x - y) = 0$; the n pivots force $x - y = 0$, and hence $x = y$.

Exercise 1.7.41: Choose a maximal independent collection of original columns

What we must show. Retain as many original columns as possible while making $Bx = 0$ have only the zero solution. The source matrix is

$$A = \begin{bmatrix} 3 & -4 & 10 & 7 & -4 \\ -5 & -3 & -7 & -11 & 15 \\ 4 & 3 & 5 & 2 & 1 \\ 8 & -7 & 23 & 4 & 15 \end{bmatrix}.$$

Exact row reduction gives

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its pivot columns are 1, 2, 4. Take these columns from the *original* matrix, not from its reduced matrix:

$$B = \begin{bmatrix} 3 & -4 & 7 \\ -5 & -3 & -11 \\ 4 & 3 & 2 \\ 8 & -7 & 4 \end{bmatrix}.$$

Reducing B gives

$$\operatorname{rref}(B) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $Bx = 0$ forces every one of its 3 variables to be zero. No larger independent subcollection is possible, because the reduced A has only 3 pivot positions; every nonpivot column is a combination of the pivot columns. The next related exercise exhibits such a combination explicitly.

Exercise 1.7.42: Choose a maximal independent collection of original columns

What we must show. Retain as many original columns as possible while making $Bx = 0$ have only the zero solution. The source matrix is

$$A = \begin{bmatrix} 12 & 10 & -6 & 8 & 4 & -14 \\ -7 & -6 & 4 & -5 & -7 & 9 \\ 9 & 9 & -9 & 9 & 9 & -18 \\ -4 & -3 & -1 & 0 & -8 & 1 \\ 8 & 7 & -5 & 6 & 1 & -11 \end{bmatrix}.$$

Exact row reduction gives

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 0 & -2 \\ 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its pivot columns are 1, 2, 3, 5. Take these columns from the *original* matrix, not from its reduced matrix:

$$B = \begin{bmatrix} 12 & 10 & -6 & 4 \\ -7 & -6 & 4 & -7 \\ 9 & 9 & -9 & 9 \\ -4 & -3 & -1 & -8 \\ 8 & 7 & -5 & 1 \end{bmatrix}.$$

Reducing B gives

$$\operatorname{rref}(B) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus $Bx = 0$ forces every one of its 4 variables to be zero. No larger independent subcollection is possible, because the reduced A has only 4 pivot positions; every nonpivot column is a combination of the pivot columns. The next related exercise exhibits such a combination explicitly.

Exercise 1.7.43: Recover an omitted column

For the choice $B = [a_1\ a_2\ a_4]$ in Exercise 41, choose $v = a_3 = (10, -7, 5, 23)^T$. The reduced column relation from that exercise is $a_3 = 2a_1 - a_2$. Directly,

$$2\begin{bmatrix} 3 \\ -5 \\ 4 \\ 8 \end{bmatrix} - \begin{bmatrix} -4 \\ -3 \\ 3 \\ -7 \end{bmatrix} = \begin{bmatrix} 10 \\ -7 \\ 5 \\ 23 \end{bmatrix}.$$

So $Bc = v$ is solved by $c = (2, -1, 0)^T$.

Conclusion. v is in the span of the columns of B.

Exercise 1.7.44: Why maximality forces every omitted column into the span

For Exercise 42, $B = [a_1\ a_2\ a_3\ a_5]$. Choose the omitted column $v = a_4$. Its coordinates in the retained columns are $(1, -1, -1, 0)^T$, because

$$a_1 - a_2 - a_3 = \begin{bmatrix} 12 \\ -7 \\ 9 \\ -4 \\ 8 \end{bmatrix} - \begin{bmatrix} 10 \\ -6 \\ 9 \\ -3 \\ 7 \end{bmatrix} - \begin{bmatrix} -6 \\ 4 \\ -9 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 9 \\ 0 \\ 6 \end{bmatrix} = a_4.$$

Thus $v \in \operatorname{Col} B$. In general, if an omitted column were outside this span, adjoining it to the independent columns of B would produce a larger independent list: a relation with nonzero coefficient on the new column would express it in the old span, and a relation with zero coefficient would contradict independence of the old list. That contradicts the requirement that B use as many columns as possible.

Conclusion. Every omitted column lies in the span of a maximal independent subcollection, not only the particular column checked here.

1.8 Introduction to linear transformations

A matrix as a rule. An $m \times n$ matrix acts on an n-coordinate input and produces an m-coordinate output. The *domain* is $\mathbb{R}^n$, the *codomain* is $\mathbb{R}^m$, and the *range* is the subset of outputs actually attained. A linear transformation satisfies

$$T(cu + dv) = cT(u) + dT(v)$$

for all inputs u, v and scalars c, d. In particular $T(0) = 0$. To find inputs producing b, solve $Ax = b$. To find inputs sent to zero, solve $Ax = 0$. These are different questions: existence concerns whether b is attained; uniqueness concerns whether it is attained by only one input.

Practice problem 1.8.P1: Read matrix dimensions from input and output

Five input coordinates require five columns, so that each column receives one coefficient. Two output coordinates require two rows, one row computation per output entry. Thus A is 2×5.

Practice problem 1.8.P2: Interpret the sign change

Multiplication by $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ takes (x_1, x_2) to $(x_1, -x_2)$. Horizontal position is unchanged and vertical position is reversed, so this is reflection across the x_1-axis. Points on that axis are fixed.

Practice problem 1.8.P3: Map a segment starting at zero

A point of the segment has the form $x = tu$ with $0 \le t \le 1$. Linearity gives $T(x) = tT(u)$, which has exactly the form of a point on the segment from 0 to $T(u)$ with the same parameter. Conversely each such output comes from tu. Thus the image is the whole segment, possibly collapsed to one point if $T(u) = 0$.

Exercise 1.8.1: Multiply by a scalar matrix

Here $A = 2I_2$. Hence

$$T(1, -3)^T = (2, -6)^T, \qquad T(a, b)^T = (2a, 2b)^T.$$

Each output entry is twice the corresponding input entry. Geometrically this doubles lengths from the origin without changing direction.

Exercise 1.8.2: A uniform contraction

Here $A = \frac{1}{3} I_3$, so multiplying each input coordinate by $1/3$ gives

$$T(3, 6, -9)^T = (1, 2, -3)^T, \qquad T(a, b, c)^T = (a/3, b/3, c/3)^T.$$

The factor is $1/3$, not 3: it is the value on each diagonal entry.

Exercise 1.8.3: Solve for an input with the specified image

What we must show. Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & 0 & -3 & -2 \\ -3 & 1 & 6 & 3 \\ 2 & -2 & -1 & -1 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-3)R_1, \quad R_3 \leftarrow R_3 - (2)R_1$. Then

$$\begin{bmatrix} 1 & 0 & -3 & -2 \\ 0 & 1 & -3 & -3 \\ 0 & -2 & 5 & 3 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -3 & -2 \\ 0 & 1 & -3 & -3 \\ 0 & 0 & -1 & -3 \end{bmatrix}.$$

Use $R_3 \leftarrow -1R_3, \quad R_1 \leftarrow R_1 - (-3)R_3, \quad R_2 \leftarrow R_2 - (-3)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

The reduced equations give $x = (7, 6, 3)^T$. Every one of the three variable columns has a pivot, so the solution is unique.

As an independent check, multiplying the displayed particular input by the original A gives $\begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$, exactly the required target.

Exercise 1.8.4: Solve for an input with the specified image

What we must show. Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 2 & -5 & 6 & -5 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (2)R_1$. Then

$$\begin{bmatrix} 1 & -2 & 3 & -6 \\ 0 & 1 & -3 & -4 \\ 0 & -1 & 0 & 7 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-2)R_2, \quad R_3 \leftarrow R_3 - (-1)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -3 & -14 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & -3 & 3 \end{bmatrix}.$$

Use $R_3 \leftarrow -\frac{1}{3}R_3, \quad R_1 \leftarrow R_1 - (-3)R_3, \quad R_2 \leftarrow R_2 - (-3)R_3$. Then

$$\begin{bmatrix} 1 & 0 & 0 & -17 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

The reduced equations give $x = (-17, -7, -1)^T$. Every variable is a pivot variable, so this input is unique.

As an independent check, multiplying the displayed particular input by the original A gives $\begin{bmatrix} -6 \\ -4 \\ -5 \end{bmatrix}$, exactly the required target.

Exercise 1.8.5: Solve for an input with the specified image

What we must show. Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & -5 & -7 & -2 \\ -3 & 7 & 5 & -2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -5 & -7 & -2 \\ 0 & -8 & -16 & -8 \end{bmatrix}.$$

Use $R_2 \leftarrow -\frac{1}{8}R_2, \quad R_1 \leftarrow R_1 - (-5)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \end{bmatrix}.$$

The reduced equations give $x = (3 - 3t, 1 - 2t, t)^T$, $t \in \mathbb{R}$. Here $x_3 = t$ is free. For example $t = 0$ gives $(3, 1, 0)^T$. Varying t gives different inputs with the same output, so uniqueness fails.

As an independent check, multiplying the displayed particular input by the original A gives $\begin{bmatrix} -2 \\ -2 \end{bmatrix}$, exactly the required target.

Exercise 1.8.6: Solve for an input with the specified image

What we must show. Solve $Ax = b$ and decide whether the input is unique. Put the target vector in the last column and apply the same row operations to the whole augmented matrix.

$$[A \mid b] = \begin{bmatrix} 1 & -3 & 2 & 1 \\ 3 & -8 & 8 & 6 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 8 & 10 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (3)R_1, \quad R_4 \leftarrow R_4 - (1)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 2 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 3 & 6 & 9 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-3)R_2, \quad R_3 \leftarrow R_3 - (1)R_2, \quad R_4 \leftarrow R_4 - (3)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 8 & 10 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The reduced equations give $x = (10 - 8t, 3 - 2t, t)^T$, $t \in \mathbb{R}$. Here $x_3 = t$ is free. Taking $t = 0$ gives $(10, 3, 0)^T$, but other values give additional inputs. The zero rows impose no extra restrictions.

As an independent check, multiplying the displayed particular input by the original A gives $\begin{bmatrix} 1 \\ 6 \\ 3 \\ 10 \end{bmatrix}$, exactly the required target.

Exercise 1.8.7: The domain and codomain of a six-by-five matrix

There are five columns, so Ax requires $x \in \mathbb{R}^5$; there are six rows, so $Ax \in \mathbb{R}^6$. Therefore $\boxed{a = 5, b = 6}$. The range can be smaller than $\mathbb{R}^6$; the question asks for the codomain.

Exercise 1.8.8: Construct the required shape

A map $\mathbb{R}^5 \to \mathbb{R}^7$ implemented by Ax needs $\boxed{\text{7 rows and 5 columns}}$. Five entries are consumed and seven entries are produced. Reversing these numbers would define a different map.

Exercise 1.8.9: Find every input sent to zero

What we must show. Solve $Ax = 0$. The right-hand side stays zero under row operations, so we can reduce the coefficient matrix alone.

$$A = \begin{bmatrix} 1 & -3 & 5 & -5 \\ 0 & 1 & -3 & 5 \\ 2 & -4 & 4 & -4 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (2)R_1$. Then

$$\begin{bmatrix} 1 & -3 & 5 & -5 \\ 0 & 1 & -3 & 5 \\ 0 & 2 & -6 & 6 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (-3)R_2, \quad R_3 \leftarrow R_3 - (2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & -4 & 10 \\ 0 & 1 & -3 & 5 \\ 0 & 0 & 0 & -4 \end{bmatrix}.$$

Use $R_3 \leftarrow -\frac{1}{4}R_3, \quad R_1 \leftarrow R_1 - (10)R_3, \quad R_2 \leftarrow R_2 - (5)R_3$. Then

$$\begin{bmatrix} 1 & 0 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The equations are $x_1 - 4x_3 = 0$, $x_2 - 3x_3 = 0$, $x_4 = 0$. Let $x_3 = t$. Then

$$x = t \begin{bmatrix} 4 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Each displayed generator is sent to zero by the original matrix. Conversely, every solution must obey the reduced equations, so no additional inputs are missing.

Exercise 1.8.10: Find every input sent to zero

What we must show. Solve $Ax = 0$. The right-hand side stays zero under row operations, so we can reduce the coefficient matrix alone.

$$A = \begin{bmatrix} 3 & 2 & 10 & -6 \\ 1 & 0 & 2 & -4 \\ 0 & 1 & 2 & 3 \\ 1 & 4 & 10 & 8 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2, \quad R_2 \leftarrow R_2 - (3)R_1, \quad R_4 \leftarrow R_4 - (1)R_1$. Then

$$\begin{bmatrix} 1 & 0 & 2 & -4 \\ 0 & 2 & 4 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 4 & 8 & 12 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3, \quad R_3 \leftarrow R_3 - (2)R_2, \quad R_4 \leftarrow R_4 - (4)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 2 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The equations are $x_1 + 2x_3 - 4x_4 = 0$ and $x_2 + 2x_3 + 3x_4 = 0$. Set $x_3 = s, x_4 = t$. Then

$$x = s \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 4 \\ -3 \\ 0 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

Each displayed generator is sent to zero by the original matrix. Conversely, every solution must obey the reduced equations, so no additional inputs are missing.

Exercise 1.8.11: Test membership in the range

The target belongs to the range exactly when $Ax = b$ is consistent. Reducing the augmented matrix gives

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & -4 & 0 & 2 \\ 0 & 1 & -3 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

The equations give $x_1 = 2 + 4t$, $x_2 = 1 + 3t$, $x_3 = t$, $x_4 = 0$. In particular $x = (2, 1, 0, 0)^T$ produces $b = (-1, 1, 0)^T$. Thus the target is in the range. A free variable does not prevent existence; it only prevents uniqueness.

Exercise 1.8.12: Test membership in the range

The target belongs to the range exactly when $Ax = b$ is consistent. Reducing the augmented matrix gives

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 2 & -4 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

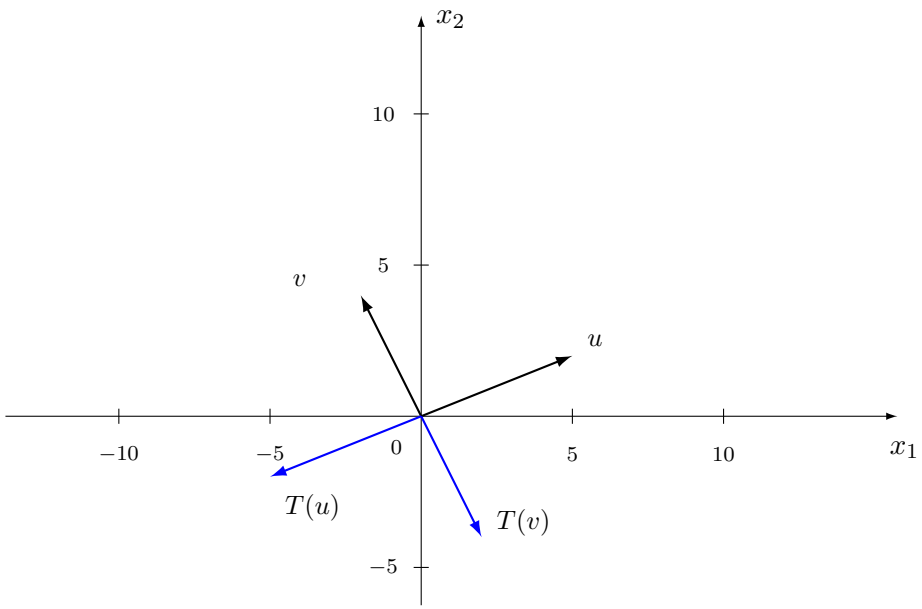
The third row says $0 = 1$, which is impossible. There is no input with image $(-1, 3, -1, 4)^T$, so the target is not in the range. The fact that it lies in the codomain $\mathbb{R}^4$ is not enough.

Exercise 1.8.13: Calculate and sketch the transformed vectors

Multiply each row of A by the input coordinates:

$$T(u) = \begin{bmatrix} -5 \\ -2 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (-x_1, -x_2)$. This is reflection through the origin (equivalently a rotation through 180°). The black arrows show the original vectors; the blue arrows show their images.

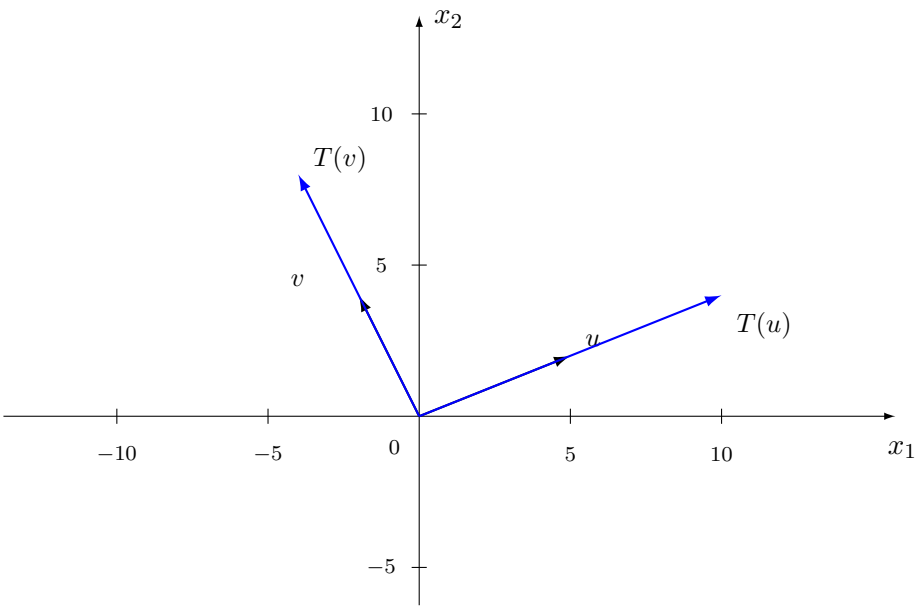


Exercise 1.8.14: Calculate and sketch the transformed vectors

Multiply each row of A by the input coordinates:

$$T(u) = \begin{bmatrix} 10 \\ 4 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} -4 \\ 8 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (2x_1, 2x_2)$. This is expansion by a factor of 2 about the origin. The black arrows show the original vectors; the blue arrows show their images.

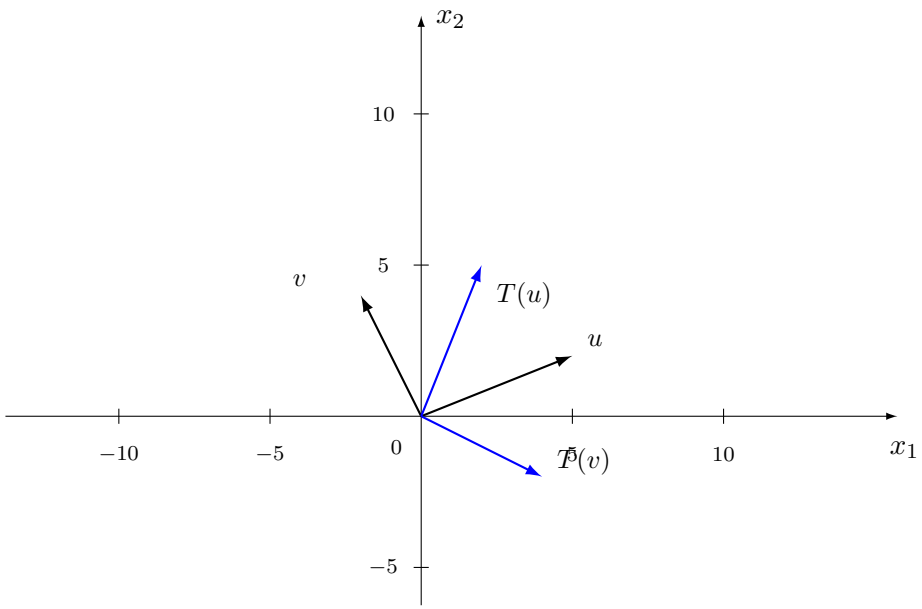


Exercise 1.8.15: Calculate and sketch the transformed vectors

Multiply each row of A by the input coordinates:

$$T(u) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (x_2, x_1)$. This is reflection through the line $x_2 = x_1$. The black arrows show the original vectors; the blue arrows show their images.

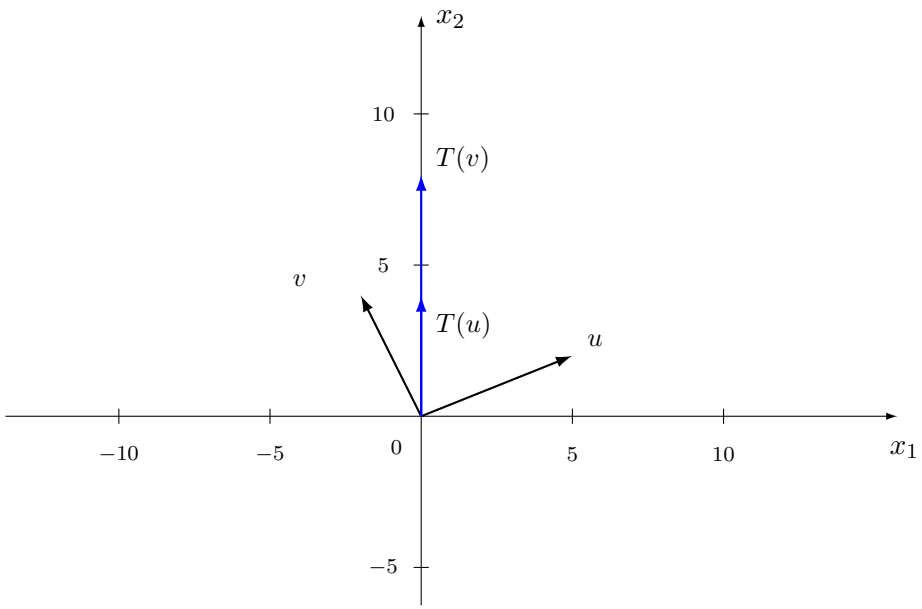


Exercise 1.8.16: Calculate and sketch the transformed vectors

Multiply each row of A by the input coordinates:

$$T(u) = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \qquad T(v) = \begin{bmatrix} 0 \\ 8 \end{bmatrix}.$$

For a general point, $T(x_1, x_2) = (0, 2x_2)$. This is projection onto the x_2-axis followed by multiplication of the remaining coordinate by 2. The black arrows show the original vectors; the blue arrows show their images.



Exercise 1.8.17: Apply linearity before looking for a matrix

The given images are $T(u) = (4, 1)^T$ and $T(v) = (-1, 3)^T$. Homogeneity and then additivity give

$$T(2u) = 2T(u) = (8, 2)^T, \qquad T(3v) = 3T(v) = (-3, 9)^T,$$

$$T(2u + 3v) = 2T(u) + 3T(v) = (8, 2)^T + (-3, 9)^T = (5, 11)^T.$$

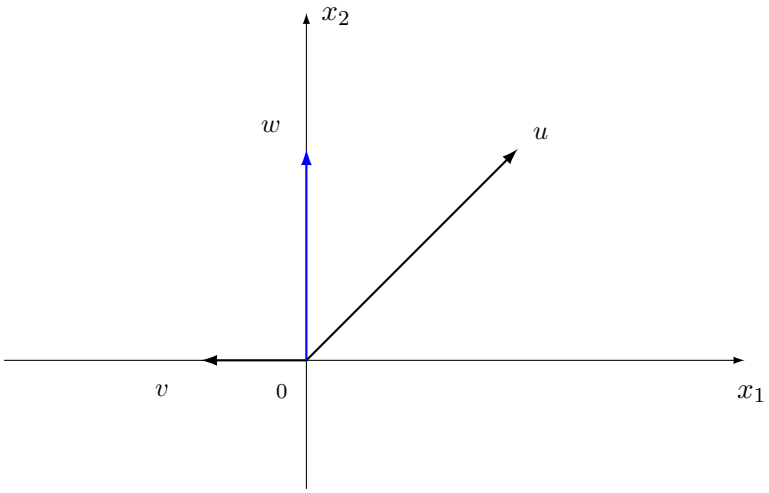
We do not need to reconstruct T on every input to answer these questions.

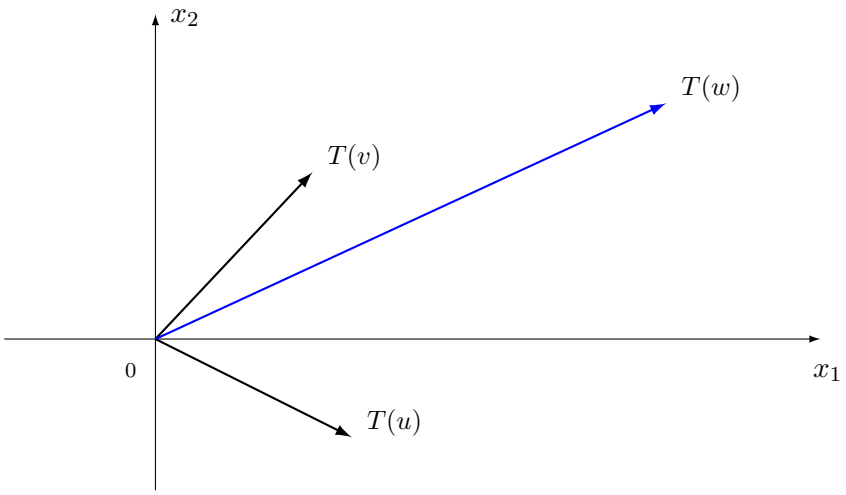
Exercise 1.8.18: Transfer a graphical linear combination

In the input diagram, $w = u+2v$: adding two copies of the left-pointing vector v cancels the horizontal component of u without changing its vertical component. By linearity,

$$T(w) = T(u) + 2T(v).$$

In the output diagram, begin at the tip of $T(u)$, append two copies of $T(v)$ head to tail, and draw the arrow from the origin to the final point. The blue arrow below gives the result. These are schematic coordinate drawings of the relation, not new numerical measurements assigned to the textbook's unlabeled vectors.





Exercise 1.8.19: Use the standard coordinate vectors

Write $(5,-3)^T = 5e_1 - 3e_2$. Then

$$T(5,-3)^T = 5(2,5)^T - 3(-1,6)^T = (13,7)^T.$$

More generally, $x = x_1e_1 + x_2e_2$ gives

$$T(x) = x_1(2,5)^T + x_2(-1,6)^T = (2x_1 - x_2, 5x_1 + 6x_2)^T.$$

The images of the two coordinate vectors determine every image.

Exercise 1.8.20: Make the two image vectors into columns

The rule is $T(x) = x_1(-3,5)^T + x_2(7,-2)^T$. Matrix multiplication forms exactly this column combination, so

$$A = \begin{bmatrix} -3 & 7 \\ 5 & -2 \end{bmatrix}, \qquad T(x) = Ax.$$

Putting the image vectors into rows instead would give the wrong formula.

Exercise 1.8.21: Functions, codomains, ranges, and linearity

(a) **True.** A transformation is a function; a linear transformation is one satisfying additional addition and scaling identities.

(b) **False.** Five columns require inputs in $\mathbb{R}^5$. Three rows determine the codomain $\mathbb{R}^3$, not the domain.

(c) **False.** The range is $\operatorname{Col} A$, which need not equal $\mathbb{R}^m$. For the zero matrix the only output is zero.

(d) **True in the setting of this chapter, $\mathbb{R}^n \to \mathbb{R}^m$.** For any such linear T, write $x = \sum_j x_j e_j$; then $T(x) = \sum_j x_j T(e_j) = [T(e_1) \ \cdots \ T(e_n)]x$. Thus it has a matrix implementation. A transformation on other vector spaces requires a choice of coordinates before this literal formula applies.

(e) **True.** Additivity and homogeneity imply the displayed combination rule. Conversely, setting both scalars to 1 gives additivity, and setting the second scalar to 0 gives homogeneity. Those are exactly the two defining properties.

Exercise 1.8.22: Separate an existence question from a uniqueness question

(a) **True.** The products Ax are precisely all weighted combinations of the columns, by the definition of matrix multiplication.

(b) **True.** $A(cu + dv) = cAu + dAv$, by distributivity and scaling of matrix products.

(c) **False.** Membership of c in the range asks whether *some* input gives c: it is an existence question. A uniqueness question asks whether two different inputs can give that same output.

(d) **True.** These are the defining properties of a linear map.

(e) **True.** $T(0) = T(0+0) = T(0) + T(0)$; subtracting $T(0)$ from both sides gives $T(0) = 0$.

Exercise 1.8.23: A straight-line graph need not define a linear map

(a) Put $b = 0$.

For $f(x) = mx$ and any real c, d, u, v, $f(cu + dv) = m(cu + dv) = c(mu) + d(mv) = cf(u) + df(v)$. Hence it is linear.

(b) Inspect zero when $b \neq 0$.

Then $f(0) = b \neq 0$, violating the necessary condition that a linear transformation sends zero to zero.

(c) Explain the terminology.

In elementary algebra and calculus, "linear function" often means a function whose graph is a straight line, allowing an intercept. Linear algebra uses the stricter operation-preserving meaning. A nonzero intercept makes the function affine rather than linear in this stricter sense.

Exercise 1.8.24: An affine displacement

For $T(x) = Ax + b$, the zero input gives $T(0) = b$. If $b \neq 0$, this cannot equal zero, whereas every linear transformation sends zero to zero. Thus the affine map is not linear. The matrix part is linear; it is the added fixed displacement that violates the condition.

Exercise 1.8.25: The image of an entire line

Write the input line as $x = p + tv$ for all $t \in \mathbb{R}$, with $v \neq 0$. Then $T(x) = T(p) + tT(v)$. If $T(v) \neq 0$, these outputs form the line through $T(p)$ in direction $T(v)$. Every point of that line comes from the same parameter t, so the map is onto that line. If $T(v) = 0$, every input on the line maps to the single point $T(p)$.

Conclusion. The image is a line or a single point.

Exercise 1.8.26: Affine combinations describe lines and segments

(a) Rewrite a line's usual parametrization.

For distinct p, q, the line is $p + t(q - p)$. Distributing gives $(1 - t)p + tq$, for all real t. Conversely these formulas are equal for every t, so they describe the same set. If $p = q$, the formula instead describes the degenerate line consisting of that one point.

(b) Restrict the parameter to the segment.

For $0 \leq t \leq 1$, linearity gives $T((1 - t)p + tq) = (1 - t)T(p) + tT(q)$. As t runs over the whole interval $[0, 1]$, the outputs cover the segment between $T(p)$ and $T(q)$. If these endpoints coincide, that segment is a single point.

Exercise 1.8.27: A plane can collapse in two ways

Every input in P is $su+tv$ with $s, t \in \mathbb{R}$. Its image is $sT(u)+tT(v)$, and conversely every combination of those two image vectors comes from such an input. Thus

$$T(P) = \text{Span}\{T(u), T(v)\}.$$

If the image vectors are independent, their span is a plane through zero. If they are dependent but at least one is nonzero, their span is a line through zero. If both are zero, the span is just $\{0\}$.

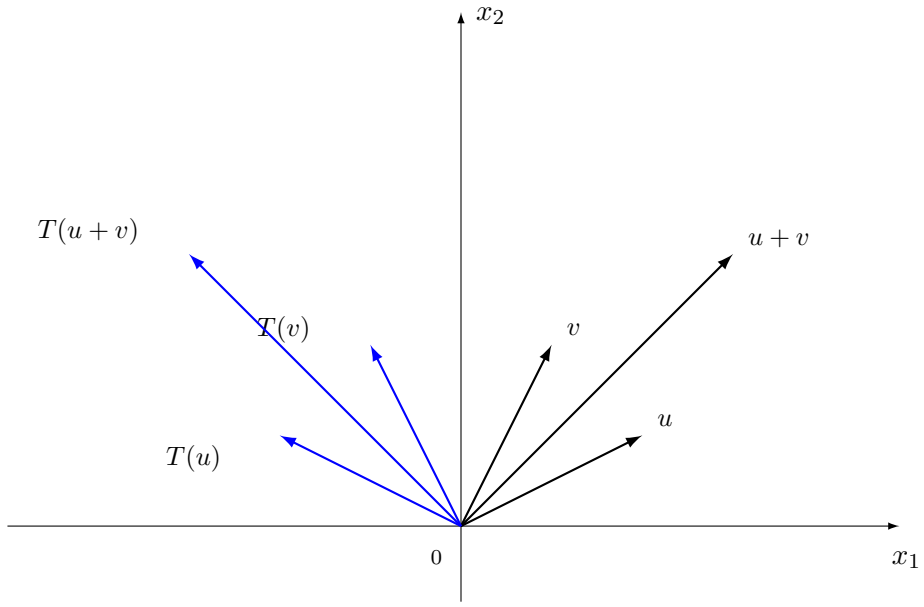
Conclusion. The image is a plane precisely when $T(u), T(v)$ are independent.

Exercise 1.8.28: A parallelogram maps into the corresponding parallelogram

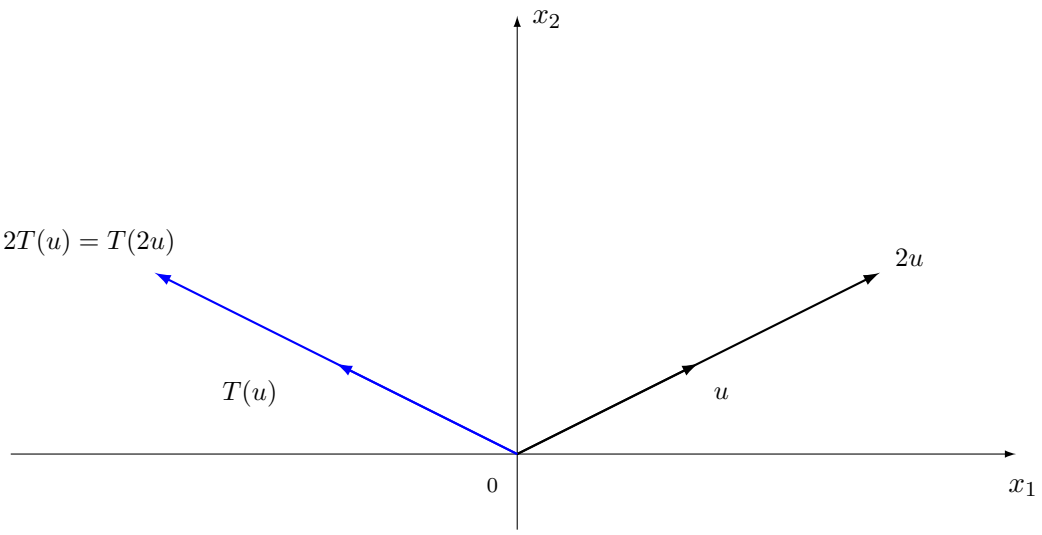
An input in the given parallelogram has the form $au+bv$ with $0 \le a, b \le 1$. Its image is $aT(u)+bT(v)$ with the same two coefficients in the same ranges. This is precisely a point of the parallelogram generated by $T(u), T(v)$. In fact the image is the whole of that set, since every allowed pair a, b also gives an input. The output parallelogram may degenerate to a segment or a point.

Exercise 1.8.29: Two sketches for reflection across the vertical axis

Here $T(x_1, x_2) = (-x_1, x_2)$. For addition, choose $u = (2, 1)$, $v = (1, 2)$. Then $u + v = (3, 3)$, while $T(u) = (-2, 1)$, $T(v) = (-1, 2)$, and $T(u) + T(v) = (-3, 3) = T(u + v)$. The paired arrows below have matching vertical coordinates and opposite horizontal coordinates.



For scaling, take $u = (2, 1)$ and $c = 2$. Then $T(2u) = (-4, 2) = 2T(u)$. Both the original ray and its reflected ray double their lengths; their directions are reflected across the vertical axis.



Exercise 1.8.30: A linear map vanishing on a spanning list is zero

Take any $x \in \mathbb{R}^n$. Since $v_1, \ldots, v_p$ span the domain, there are scalars c_j with $x = \sum_{j=1}^{p} c_j v_j$. Repeated use of linearity gives

$$T(x) = \sum_{j=1}^{p} c_j T(v_j) = \sum_{j=1}^{p} c_j 0 = 0.$$

The choice of x was arbitrary, so T sends every input to zero. The spanning assumption is essential: vanishing on just one line would not generally determine a map on the whole domain.

Exercise 1.8.31: A dependence relation survives applying a linear map

Dependence gives scalars c_1, c_2, c_3, not all zero, such that $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$. Apply T and use $T(0) = 0$:

$$c_1 T(v_1) + c_2 T(v_2) + c_3 T(v_3) = T(c_1 v_1 + c_2 v_2 + c_3 v_3) = 0.$$

The same coefficients are still not all zero. Thus they form a nontrivial relation among the three image vectors, proving dependence.

Exercise 1.8.32: An absolute value violates homogeneity

For $T(x_1, x_2) = (x_1 - 2|x_2|, x_1 - 4x_2)$, choose $u = (0, 1)$. Then $T(u) = (-2, -4)$, so $-T(u) = (2, 4)$. But $T(-u) = T(0, -1) = (-2, 4)$, which differs in its first coordinate. Hence $T(-u) \neq -T(u)$, violating homogeneity for scalar -1.

Conclusion. The transformation is not linear.

Exercise 1.8.33: A constant term violates preservation of zero

The displayed rule gives $T(0,0) = (0,-3,0) \neq (0,0,0)$. Since every linear transformation must map the zero input to the zero output, this single counterexample disproves linearity. No complicated comparison of general inputs is needed.

Exercise 1.8.34: Reflection through a coordinate plane is linear

Let $u, v \in \mathbb{R}^3$ and $c, d \in \mathbb{R}$. For the stated rule,

$$\begin{aligned} T(cu+dv) &= (cu_1+dv_1, cu_2+dv_2, -cu_3-dv_3) \\ &= c(u_1,u_2,-u_3) + d(v_1,v_2,-v_3) \\ &= cT(u) + dT(v). \end{aligned}$$

This is the combined linearity condition for arbitrary inputs and scalars. Equivalently, T is multiplication by $\operatorname{diag}(1,1,-1)$.

Exercise 1.8.35: A coordinate projection is linear

For $T(x_1,x_2,x_3) = (x_1,0,x_3)$,

$$\begin{aligned} T(cu+dv) &= (cu_1+dv_1, 0, cu_3+dv_3) \\ &= c(u_1,0,u_3) + d(v_1,0,v_3) = cT(u) + dT(v). \end{aligned}$$

The middle zero is preserved under addition and scaling. Thus the map is linear; its matrix is $\operatorname{diag}(1,0,1)$.

Exercise 1.8.36: Dependence created in the output forces a nonzero zero-input

Because $T(u), T(v)$ are dependent, some a, b, not both zero, satisfy $aT(u)+bT(v) = 0$. Set $x = au+bv$. Linearity gives $T(x) = 0$. Moreover $x \neq 0$: if $au+bv = 0$, independence of u, v would force $a = b = 0$, contrary to their choice.

Conclusion. x is a nontrivial solution of $T(x) = 0$.

Exercise 1.8.37: Compute all inputs in the kernel

The supplied matrix is

$$A = \begin{bmatrix} 2 & 3 & 5 & -5 \\ -7 & 7 & 0 & 0 \\ -3 & 4 & 1 & 3 \\ -9 & 3 & -6 & -4 \end{bmatrix}.$$

Exact row reduction yields

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Choose the sole free variable to be t. Reading the pivot equations gives

$$x = t \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Direct multiplication of the original A by this generator gives zero in every row. Conversely the pivot equations determine all remaining coordinates from t, so the displayed family is the complete solution set.

Exercise 1.8.38: Compute all inputs in the kernel

The supplied matrix is

$$A = \begin{bmatrix} 3 & 4 & -7 & 0 \\ 5 & -8 & 7 & 4 \\ 6 & -8 & 6 & 4 \\ 9 & -7 & -2 & 0 \end{bmatrix}.$$

Exact row reduction yields

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Choose the sole free variable to be t. Reading the pivot equations gives

$$x = t \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Direct multiplication of the original A by this generator gives zero in every row. Conversely the pivot equations determine all remaining coordinates from t, so the displayed family is the complete solution set.

Exercise 1.8.39: Find an input for a specified output

Use the matrix in Exercise 37 and append the stated target. The exact reduced matrix is

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

No row says $0 = 1$, so the equations are consistent. A particular input is $x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}$. Substitution gives

$$Ax = \begin{bmatrix} 8 \\ 7 \\ 5 \\ -3 \end{bmatrix} = b.$$

All inputs with that same image are

$$x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The homogeneous part changes the input without changing its image, because its product with A is zero.

Exercise 1.8.40: Find an input for a specified output

Use the matrix in Exercise 38 and append the stated target. The exact reduced matrix is

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

No row says $0 = 1$, so the equations are consistent. A particular input is $x = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$. Substitution gives

$$Ax = \begin{bmatrix} 4 \\ -4 \\ -4 \\ -7 \end{bmatrix} = b.$$

All inputs with that same image are

$$x = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

The homogeneous part changes the input without changing its image, because its product with A is zero.

1.9 The matrix of a linear transformation

For a linear map $T : \mathbb{R}^n \to \mathbb{R}^m$, its standard matrix is

$$A = [T(e_1) \ \cdots \ T(e_n)].$$

The image of e_j is column j, not row j. This works because $x = \sum_j x_j e_j$ and linearity gives $T(x) = \sum_j x_j T(e_j) = Ax$. The map is one-to-one exactly when every column has a pivot, so $Ax = 0$ has only the trivial solution. It is onto $\mathbb{R}^m$ exactly when every row has a pivot. One-to-one means at most one input for any output; onto means at least one input for every output in the codomain. For a composition, the transformation applied first corresponds to the matrix farther to the right in the product.

Practice problem 1.9.P1: A shear followed by reflection

The shear fixes e_1 and takes e_2 to $e_2 - \frac{1}{2}e_1$. Reflection across the x_2-axis negates the first coordinate and leaves the second unchanged. Thus the final images are $T(e_1) = (-1, 0)^T$ and $T(e_2) = (1/2, 1)^T$. Therefore

$$A = \begin{bmatrix} -1 & 1/2 \\ 0 & 1 \end{bmatrix}.$$

We follow the image through both operations before making it a column.

Exercise 1.9.1: Put two four-coordinate images into columns

The domain has dimension two and the output has four coordinates. Use the two specified images in their stated order:

$$A = [T(e_1) \ T(e_2)] = \begin{bmatrix} 3 & -5 \\ 1 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix}.$$

Indeed its first column is $T(e_1)$ and its second is $T(e_2)$, which is enough to determine the linear map on every input.

Exercise 1.9.2: Three input coordinates and two output coordinates

There must be three columns and two rows. The standard matrix is

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}.$$

For example, the second column is the given image $(-2, 9)^T$. Multiplying A by any $(x_1, x_2, x_3)^T$ forms the required weighted combination of the three images.

Exercise 1.9.3: A vertical shear

The given images are $T(e_1) = e_1 - 3e_2 = (1, -3)^T$ and $T(e_2) = e_2 = (0, 1)^T$. Hence

$$A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \qquad T(x_1, x_2) = (x_1, x_2 - 3x_1).$$

The horizontal coordinate stays fixed while the vertical coordinate changes in proportion to it.

Exercise 1.9.4: A horizontal shear

Here $T(e_1) = e_1$ and $T(e_2) = e_2 + 2e_1 = (2, 1)^T$. Thus

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad T(x_1, x_2) = (x_1 + 2x_2, x_2).$$

This time the vertical coordinate stays fixed.

Exercise 1.9.5: A counterclockwise quarter turn

Rotation through $\pi/2$ takes e_1 to e_2 and e_2 to $-e_1$. Therefore

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The resulting rule $T(x_1, x_2) = (-x_2, x_1)$ agrees with both coordinate-vector images.

Exercise 1.9.6: A clockwise three-quarter turn

An angle of $-3\pi/2$ differs from $\pi/2$ by a full turn. Consequently the final coordinate-vector images are again $e_1 \mapsto e_2$ and $e_2 \mapsto -e_1$, giving

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The direction of travel is different, but the final transformation is the same.

Exercise 1.9.7: Rotate, then reflect

Rotation through $-3\pi/4$ has matrix

$$R = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}.$$

Reflection across the horizontal axis negates the second output coordinate, so it changes the signs in the second row of R. Thus

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}.$$

In particular the first column is $(-1/\sqrt{2}, 1/\sqrt{2})^T$, agreeing with the hint. The order is important: reflect the rotated image.

Exercise 1.9.8: Shear, then reflect across a sloping line

The shear is $S(x_1, x_2) = (x_1+2x_2, x_2)$. Reflection across $x_2 = -x_1$ takes (a, b) to $(-b, -a)$. Composing these two explicit rules gives

$$T(x_1, x_2) = (-x_2, -x_1 - 2x_2), \qquad A = \begin{bmatrix} 0 & -1 \\ -1 & -2 \end{bmatrix}.$$

The columns can also be checked by applying the two rules to e_1, e_2.

Exercise 1.9.9: Reflect, then rotate clockwise

The first step takes (x_1, x_2) to $(x_1, -x_2)$. A clockwise quarter turn takes (a, b) to $(b, -a)$. Hence the final image is $(-x_2, -x_1)$, and

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

Applying the two steps in the opposite order would produce a different map.

Exercise 1.9.10: Two reflections make a rotation

The horizontal reflection gives $(x_1, -x_2)$; reflection across $x_2 = x_1$ exchanges its coordinates. Thus

$$T(x_1, x_2) = (-x_2, x_1), \qquad A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The result is a counterclockwise quarter turn.

Exercise 1.9.11: Reflect across the two coordinate axes

After horizontal reflection, (x_1, x_2) becomes $(x_1, -x_2)$. Vertical reflection then gives $(-x_1, -x_2)$. The standard matrix is $-I_2$, whose columns are $-e_1, -e_2$. A rotation by π radians has the same two images, so it is the same linear transformation.

Conclusion. A half turn, through π radians (equivalently $-\pi$).

Exercise 1.9.12: Identify the angle from the coordinate-vector images

Exercise 10 gives $e_1 \mapsto e_2$, $e_2 \mapsto -e_1$. These are precisely the images under counterclockwise rotation by $\pi/2$. Since a linear map is determined by its values on the standard coordinate vectors, the two transformations agree on every vector.

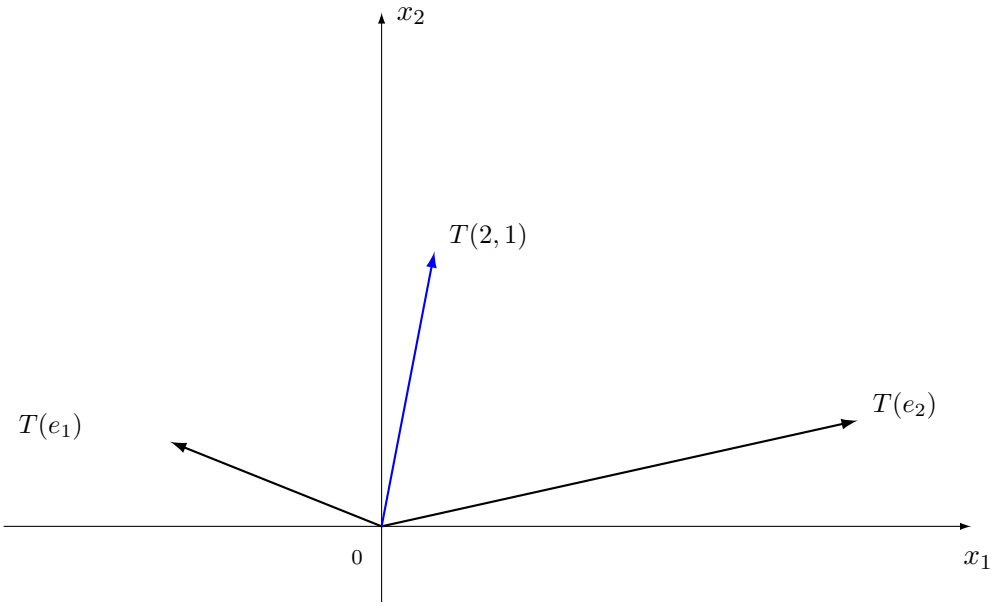
Conclusion. The rotation angle is $\pi/2$ modulo 2π.

Exercise 1.9.13: Draw a weighted sum of the two column images

The vector $(2, 1)^T$ equals $2e_1 + e_2$, so linearity gives

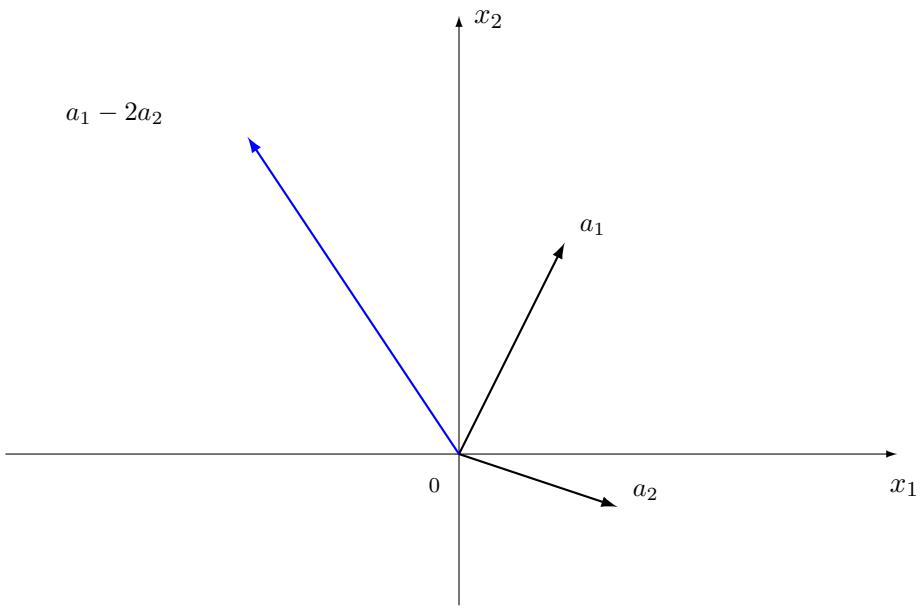
$$T(2, 1)^T = 2T(e_1) + T(e_2).$$

Take two copies of the up-left arrow $T(e_1)$, followed head to tail by one copy of the up-right arrow $T(e_2)$. The final arrow from zero ends high above the origin and slightly to its right. The following schematic preserves the source directions; the exact statement is the linear-combination formula, since the source gives no numerical coordinates for these image vectors.



Exercise 1.9.14: Subtract twice the second column

The product $A(1,-2)^T$ is $a_1 - 2a_2$. Reverse the up-right/down-right source arrow a_2 and double its length, then attach it at the tip of a_1. The resultant points up and to the left. This construction uses the original columns as vectors, not their lengths alone. A schematic is shown below.



Exercise 1.9.15: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 2 & -4 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 2x_1 - 4x_2 \\ x_1 - x_3 \\ -x_2 + 3x_3 \end{bmatrix}.$$

The identity is required for *all* inputs, so these coefficients are forced; one numerical input would not be enough to determine all the entries.

Exercise 1.9.16: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 4 \\ 0 & 1 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 3x_1 - 2x_2 \\ x_1 + 4x_2 \\ x_2 \end{bmatrix}.$$

The identity is required for *all* inputs, so these coefficients are forced; one numerical input would not be enough to determine all the entries.

Exercise 1.9.17: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 + 2x_2 \\ 0 \\ 2x_2 + x_4 \\ x_2 - x_4 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.18: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & 4 \\ 0 & 0 \\ 1 & -3 \\ 1 & 0 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 + 4x_2 \\ 0 \\ x_1 - 3x_2 \\ x_1 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.19: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & -6 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} x_1 - 5x_2 + 4x_3 \\ x_2 - 6x_3 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.20: Read each coefficient into the correct row and column

Each output coordinate becomes one row. In that row, use the coefficients of $x_1, x_2, \ldots$ in order, inserting 0 for a missing variable. This gives

$$A = \begin{bmatrix} 3 & 0 & 4 & -2 \end{bmatrix}.$$

Indeed direct row-by-column multiplication yields

$$Ax = \begin{bmatrix} 3x_1 + 4x_3 - 2x_4 \end{bmatrix}.$$

This is exactly the stated rule $T(x)$, so T is a matrix transformation and therefore linear. For any u, v, c, d, $A(cu + dv) = cAu + dAv$ verifies the definition.

Exercise 1.9.21: Find the input producing two prescribed coordinates

The equations are $x_1 + x_2 = 3$ and $4x_1 + 5x_2 = 8$. Subtract four times the first from the second: $x_2 = -4$. Substitute into the first: $x_1 = 3 - (-4) = 7$. A check gives $T(7, -4) = (7 - 4, 28 - 20) = (3, 8)$.

Conclusion. $x = (7, -4)^T$. The elimination determines both variables uniquely.

Exercise 1.9.22: Check all three output coordinates

From $2x_1 - x_2 = 0$, obtain $x_2 = 2x_1$. The second equation $-3x_1 + x_2 = -1$ then becomes $-x_1 = -1$, so $x_1 = 1$ and $x_2 = 2$. The third equation must still be checked: $2(1) - 3(2) = -4$, as required.

Conclusion. $x = (1, 2)^T$. The third equation is consistent with the values forced by the first two.

Exercise 1.9.23: What coordinate-vector images determine

(a) **True.** Every $x \in \mathbb{R}^n$ is $\sum_j x_j e_j$, so linearity determines $T(x)$ from all the images $T(e_j)$.

(b) **True.** The rotation matrix $\begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$ implements the stated rotation, and multiplication by a matrix is linear.

(c) **False.** A composition of linear maps is always linear: $T(S(cu+dv)) = T(cS(u)+dS(v)) = cT(S(u)) + dT(S(v))$.

(d) **False.** Every map with that codomain sends each input somewhere in $\mathbb{R}^m$, whether or not it is onto. Onto means every *output* in $\mathbb{R}^m$ is reached by some input. The zero map to $\mathbb{R}$ is a counterexample to the claimed implication.

(e) **False.** The 3×2 matrix with columns e_1, e_2 in $\mathbb{R}^3$ has independent columns, so its map is one-to-one. Having more output coordinates than input coordinates does not obstruct injectivity.

Exercise 1.9.24: Distinguish being a function from being one-to-one

(a) **False.** A 4×3 matrix has at most three pivots, so it cannot have a pivot in all four rows. Its columns cannot span $\mathbb{R}^4$.

(b) **True.** The matrix with columns $T(e_j)$ implements any linear map between the stated coordinate spaces.

(c) **True.** This is exactly the column construction of the standard matrix.

(d) **False.** Having one output for each input merely says T is a function. One-to-one requires different inputs to give different outputs. The zero map has a unique output at every input but is not one-to-one on a nontrivial domain.

(e) **False.** A horizontal shear has matrix $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$, with nonzero shear amount k. The displayed diagonal matrices instead change signs of coordinate axes.

Exercise 1.9.25: An unused input coordinate and a zero output coordinate

For the map in Exercise 17, the third column is zero, so $T(e_3) = 0 = T(0)$ with $e_3 \neq 0$. Thus it is **not one-to-one**. The second output coordinate is always zero, so the vector $(0,1,0,0)^T$ is never attained. Therefore it is also **not onto** $\mathbb{R}^4$. These two failures concern different sides of the same matrix: a zero column creates nonuniqueness, and a zero row here prevents some outputs.

Exercise 1.9.26: A wide matrix can be onto but not one-to-one

The matrix in Exercise 2 is $\begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}$. Replacing R_2 by $R_2 - 4R_1$ gives $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 17 & -20 \end{bmatrix}$. There is a pivot in both rows, so the map is **onto** $\mathbb{R}^2$. But three columns and only two pivots leave a free variable, so it is **not one-to-one**. For example $(-11, 20, 17)^T$ is a nonzero input sent to zero, as substitution verifies.

Exercise 1.9.27: Two pivot rows with one free variable

For Exercise 19, the matrix $\begin{bmatrix} 1 & -5 & 4 \\ 0 & 1 & -6 \end{bmatrix}$ is already in echelon form. Its two nonzero rows each have a pivot, making the map **onto** $\mathbb{R}^2$. The homogeneous equations yield $x_2 = 6x_3$ and $x_1 = 26x_3$, so $(26, 6, 1)^T$ is a nonzero input sent to zero. Hence the map is **not one-to-one**.

Exercise 1.9.28: Use the geometry of two column directions

In Exercise 14 the vectors a_1 and a_2 are nonzero and not parallel: the first points up-right and the second down-right. They are therefore independent, giving two pivot columns. In a 2×2 matrix those are also two pivot rows. Thus the map is **one-to-one and onto** $\mathbb{R}^2$. Equivalently, the two independent column directions span the whole plane, and each combination has unique coefficients.

Exercise 1.9.29: A tall one-to-one matrix

For $T : \mathbb{R}^3 \to \mathbb{R}^4$ to be one-to-one, its 4×3 matrix must have a pivot in every column. The sole echelon pattern is

$$\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix}, \qquad p, q, r \neq 0.$$

The last row must be zero because the three available pivots occupy the first three nonzero rows.

Exercise 1.9.30: Every full-row-rank three-by-four echelon pattern

Onto $\mathbb{R}^3$ requires three pivots, one in each row. They can occupy any three of the four columns, subject to moving right as one moves down. Thus the possible patterns are

$$\begin{bmatrix} p & * & * & * \\ 0 & q & * & * \\ 0 & 0 & r & * \end{bmatrix}, \qquad \begin{bmatrix} p & * & * & * \\ 0 & q & * & * \\ 0 & 0 & 0 & r \end{bmatrix},$$

$$\begin{bmatrix} p & * & * & * \\ 0 & 0 & q & * \\ 0 & 0 & 0 & r \end{bmatrix}, \qquad \begin{bmatrix} 0 & p & * & * \\ 0 & 0 & q & * \\ 0 & 0 & 0 & r \end{bmatrix}, \qquad p, q, r \neq 0.$$

Here $*$ denotes any entry. There are no additional patterns, because there are exactly four ways to choose three pivot columns out of four.

Exercise 1.9.31: Count columns for injectivity

The blank is $\boxed{n}$. One-to-one is equivalent to $Ax = 0$ having only zero as a solution. With n variables, this happens exactly when none is free, that is, when all n columns are pivot columns. Conversely any missing column pivot leaves a free variable and thus a nonzero input with the same image as zero.

Exercise 1.9.32: Count rows for surjectivity

The blank is $\boxed{m}$. Theorem 12 says onto is equivalent to columns spanning $\mathbb{R}^m$. Theorem 4 in Section 1.4 says this is equivalent to a pivot in every row. There are m rows, so there must be m pivots and hence m pivot columns. A row without a pivot would allow an inconsistent target after row reduction.

Exercise 1.9.33: The standard matrix is unique

Suppose $T(x) = Ax = Bx$ for every $x \in \mathbb{R}^n$. Substitute $x = e_j$. The vector Ae_j is column j of A and Be_j is column j of B. Both equal $T(e_j)$, so these columns agree for every $j = 1, \ldots, n$. Matrices with identical corresponding columns are identical, proving $A = B$. Equality on just a single nonbasis input would not imply equality of all columns.

Exercise 1.9.34: A composition preserves both operations

Take $u, v \in \mathbb{R}^p$ and scalars c, d. Linearity of S first gives $S(cu + dv) = cS(u) + dS(v)$, where the two image vectors lie in $\mathbb{R}^n$, the domain of T. Applying linearity of T then gives

$$T(S(cu + dv)) = T(cS(u) + dS(v)) = cT(S(u)) + dT(S(v)).$$

This is the combined linearity criterion for $T \circ S$ on $\mathbb{R}^p$, with outputs in $\mathbb{R}^m$. The domain matching is what makes the composition defined.

Exercise 1.9.35: Compare input and output dimensions

If T is onto, its $m \times n$ matrix has m pivot rows, so it needs at least m columns: $\boxed{m \leq n}$. If it is one-to-one, the matrix has n pivot columns and therefore needs at least n rows: $\boxed{n \leq m}$. If both properties hold, both inequalities force $m = n$. These are necessary dimension conditions, not sufficient ones; even a square zero matrix has neither property in positive dimension.

Exercise 1.9.36: Onto asks for the existence of a preimage

Saying T is onto means: for *every* b in the codomain, there *exists* an input x with $T(x) = b$. Thus it asks whether each target can be reached at all. It does not require that such an input be unique. A wide full-row-rank matrix can be onto while each target has infinitely many preimages.

Exercise 1.9.37: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} -5 & 6 & -5 & -6 \\ 8 & 3 & -3 & 8 \\ 2 & 9 & 5 & -12 \\ -3 & 2 & 7 & -12 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are only three pivots in four columns. Taking the free variable $x_4 = 1$ gives the nonzero solution $x = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ of $Ax = 0$. Direct multiplication verifies this relation. Hence T is not one-to-one.

Exercise 1.9.38: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 7 & 5 & 9 & -9 \\ 5 & 6 & 4 & -4 \\ 4 & 8 & 0 & 7 \\ -6 & -6 & 6 & 5 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every one of the four columns has a pivot, so $Ax = 0$ forces $x = 0$. Thus T is one-to-one. Since there is also a pivot in every row, this particular map is onto as well, although injectivity is the requested property.

Exercise 1.9.39: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 4 & -7 & 3 & 7 & 5 \\ 6 & -8 & 5 & 12 & -8 \\ -7 & 10 & -8 & -9 & 14 \\ 3 & -5 & 4 & 2 & -6 \\ -5 & 6 & -6 & -7 & 3 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 5 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are four pivots and five rows. The zero row in the reduced coefficient matrix means there is not a pivot in every row, so the columns cannot span $\mathbb{R}^5$. Therefore T does not map $\mathbb{R}^5$ onto $\mathbb{R}^5$.

Exercise 1.9.40: Use exact reduction to test the requested property

The source matrix and its exact reduced echelon form are

$$A = \begin{bmatrix} 9 & 43 & 5 & 6 & -1 \\ 14 & 15 & -7 & -5 & 4 \\ -8 & -6 & 12 & -5 & -9 \\ -5 & -6 & -4 & 9 & 8 \\ 13 & 14 & 15 & 3 & 11 \end{bmatrix},$$

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There is a pivot in every one of the five rows. Appending any right-hand side cannot create an inconsistent zero-coefficient row, so $Ax = b$ is solvable for every $b \in \mathbb{R}^5$. Thus T maps $\mathbb{R}^5$ onto $\mathbb{R}^5$; it is also one-to-one because every column has a pivot.

1.10 Linear models in business, science, and engineering

A model begins by defining its variables and their units. Matrix columns then describe what one unit of each input contributes, and rows record one output requirement at a time. A mathematically consistent nutritional system is not physically feasible if it requires negative amounts of food. In a circuit, signs depend on the chosen loop directions and battery polarities. In a migration model, column j records where people starting in region j go, so its entries sum to 1 when nobody leaves the system. The labels [M] in the text permit matrix software; exact reduced matrices and substitution checks are supplied below rather than treating a software answer as a substitute for interpreting the model.

Practice problem 1.10.P1: Express the diet example as a matrix equation

Let x_1, x_2, x_3 be the amounts of nonfat milk, soy flour, and whey, respectively, measured in 100-gram units. Put the nutrient contributions per unit into columns and the targets into the right-hand side:

$$A = \begin{bmatrix} 36 & 51 & 13 \\ 52 & 34 & 74 \\ 0 & 7 & 1.1 \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad b = \begin{bmatrix} 33 \\ 45 \\ 3 \end{bmatrix}, \qquad Ax = b.$$

The rows specify protein, carbohydrate, and fat in grams. For example, the first row says $36x_1 + 51x_2 + 13x_3 = 33$, so it adds the protein from all three ingredients. The actual food masses in grams are $100x_j$.

Exercise 1.10.1: A two-cereal mixture with four requirements

(a) Define the serving counts and the vector equation.

Let x be servings of Cheerios and y servings of Natural Cereal. With rows ordered as calories, protein, carbohydrate, and fat,

$$x \begin{bmatrix} 110 \\ 4 \\ 20 \\ 2 \end{bmatrix} + y \begin{bmatrix} 130 \\ 3 \\ 18 \\ 5 \end{bmatrix} = \begin{bmatrix} 295 \\ 9 \\ 48 \\ 8 \end{bmatrix}.$$

(b) Solve and check all the requirements.

The equivalent matrix equation is

$$\begin{bmatrix} 110 & 130 \\ 4 & 3 \\ 20 & 18 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 295 \\ 9 \\ 48 \\ 8 \end{bmatrix}.$$

Use the protein equation $4x + 3y = 9$ and fat equation $2x + 5y = 8$. Subtract twice the latter from the former: $-7y = -7$, hence $y = 1$. Then $2x + 5 = 8$ gives $x = 3/2$. Check the other rows: $110(3/2) + 130 = 295$ and $20(3/2) + 18 = 48$. Both serving counts are nonnegative, so the solution is physically meaningful.

Conclusion. Mix 1.5 servings of Cheerios with 1 serving of Natural Cereal.

Exercise 1.10.2: A specified mixture and a new nutritional target

(a) Build the contribution matrix.

Order the rows as calories, protein, fiber, and fat, and the columns as Shredded Wheat and Crispix. Then

$$B = \begin{bmatrix} 160 & 110 \\ 5 & 2 \\ 6 & 0.1 \\ 1 & 0.4 \end{bmatrix}, \quad u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad Bu = \begin{bmatrix} 700 \\ 19 \\ 18.2 \\ 3.8 \end{bmatrix}.$$

For instance the fiber amount is $6(3) + 0.1(2) = 18.2$ grams.

(b) Solve for the requested different mixture.

The desired equation is $B(x, y)^T = (130, 3.20, 2.46, 0.64)^T$. Its reduced augmented matrix is

$$\begin{bmatrix} 1 & 0 & 0.4 \\ 0 & 1 & 0.6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $x = 0.4$ and $y = 0.6$. To check all rows, the totals are $64 + 66 = 130$ calories, $2 + 1.2 = 3.20$ protein, $2.4 + 0.06 = 2.46$ fiber, and $0.4 + 0.24 = 0.64$ fat. These nonnegative proportions also add to one serving.

Conclusion. The target is attainable with 0.4 serving of Shredded Wheat and 0.6 serving of Crispix.

Exercise 1.10.3: Compare two versions of a lunch

Let x, y, z be servings of the pasta dish, broccoli, and chicken. Use calories, protein, and fiber as the three rows. Solving each system requires all three targets, not only the calories.

(a) Use the original Mac and Cheese.

$$\begin{bmatrix} 270 & 51 & 70 \\ 10 & 5.4 & 15 \\ 2 & 5.2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 400 \\ 30 \\ 10 \end{bmatrix}.$$

Exact row reduction, treating the decimals as the stated rational numbers, gives

$$[A \mid b] \sim \begin{bmatrix} 1 & 0 & 0 & 8205/8323 \\ 0 & 1 & 0 & 12850/8323 \\ 0 & 0 & 1 & 6550/8323 \end{bmatrix}.$$

Hence $(x, y, z) \approx (0.98582, 1.54391, 0.78698)$. Every variable is determined by a pivot, so the solution is unique. Substituting the exact fractions into the original three rows gives $(400, 30, 10)^T$. All three fractions are positive, so the mixture is feasible.

(b) Replace the pasta column by the whole-wheat-shells data.

Only that ingredient's column changes:

$$\begin{bmatrix} 260 & 51 & 70 \\ 9 & 5.4 & 15 \\ 5 & 5.2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 400 \\ 30 \\ 10 \end{bmatrix}.$$

Reduction gives $(x, y, z) = (5470, 4400, 5180)/5023$, approximately $(1.08899, 0.87597, 1.03126)$. Substitution again gives exactly 400 calories, 30 grams protein, and 10 grams fiber before rounding. The broccoli portion drops from about 1.54 to 0.88 serving, in line with the stated preference.

Conclusion. To two decimal places the mixtures are $(0.99, 1.54, 0.79)$ and $(1.09, 0.88, 1.03)$ servings in the stated ingredient order. Use the exact fractions for exact nutritional totals.

Exercise 1.10.4: A consistent diet equation can be physically impossible

(a) Specify the four variables and four nutrient rows.

Let x_1, x_2, x_3, x_4 be 100-gram units of nonfat milk, soy flour, whey, and isolated soy protein. In the row order protein, carbohydrate, fat, calcium, the equation is

$$\begin{bmatrix} 36 & 51 & 13 & 80 \\ 52 & 34 & 74 & 0 \\ 0 & 7 & 1.1 & 3.4 \\ 1.26 & 0.19 & 0.8 & 0.18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 33 \\ 45 \\ 3 \\ 0.8 \end{bmatrix}.$$

Each coefficient is grams of that nutrient per 100 grams of ingredient.

(b) Solve and interpret the signs.

Exact reduction yields the unique solution

$$x = \begin{bmatrix} 2264687/3531104 \\ 1921341/3531104 \\ -326887/3531104 \\ -1468529/7062208 \end{bmatrix} \approx \begin{bmatrix} 0.64135 \\ 0.54412 \\ -0.09257 \\ -0.20794 \end{bmatrix}.$$

Multiplication of the exact vector by the original matrix reproduces all four targets, so the algebraic equations are consistent. But the whey and isolated-protein amounts are negative. One cannot subtract food already absent from the mixture, so this unique algebraic solution is not a permissible recipe. Because it is unique, there is no different nonnegative solution to these same equations.

Conclusion. These four ingredients cannot meet all four targets with nonnegative amounts under the supplied nutrient data.

Circuit convention for Exercises 5–8. Use the counterclockwise loop directions drawn in the textbook. For each loop, add all resistances on its boundary to get the diagonal coefficient. A shared resistor carries the difference of the adjacent loop currents, so its resistance enters the neighboring column with a minus sign. On the right, crossing a battery from its short plate to its long plate contributes a positive voltage; the reverse crossing contributes a negative voltage. These rules explain every sign in $Ri = v$. A negative current means the actual direction is opposite to the assumed counterclockwise direction.

Exercise 1.10.5: Derive the loop equations and solve them

The four diagonal resistance sums are $1+1+4+5=11$, $3+5+1+1=10$, $4+1+2+2=9$, and $1+2+3+4=10$. The successive shared resistances are $5, 1, 2$. Battery crossings give $20+30=50$, $-30-10=-40$, $10+20=30$, and $-20-10=-30$ volts.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 11 & -5 & 0 & 0 \\ -5 & 10 & -1 & 0 \\ 0 & -1 & 9 & -2 \\ 0 & 0 & -2 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 50 \\ -40 \\ 30 \\ -30 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{265}{72} \\ -\frac{137}{72} \\ \frac{185}{72} \\ -\frac{179}{72} \end{bmatrix} \approx \begin{bmatrix} 3.6806 \\ -1.9028 \\ 2.5694 \\ -2.4861 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 50, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Exercise 1.10.6: Derive the loop equations and solve them

The diagonal sums are $3+2+1=6$, $1+4+4=9$, $4+1+2=7$, and $2+2+3=7$. Adjacent loops share resistances $1, 4, 2$. Every battery on the left is crossed from short plate to long plate in the chosen direction, giving voltages $30, 20, 40, 10$.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 6 & -1 & 0 & 0 \\ -1 & 9 & -4 & 0 \\ 0 & -4 & 7 & -2 \\ 0 & 0 & -2 & 7 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 30 \\ 20 \\ 40 \\ 10 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{3630}{571} \\ \frac{4650}{571} \\ \frac{6700}{571} \\ \frac{2730}{571} \end{bmatrix} \approx \begin{bmatrix} 6.3573 \\ 8.1436 \\ 11.734 \\ 4.7811 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 30, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Exercise 1.10.7: Derive the loop equations and solve them

Keep the order I_1 upper left, I_2 lower left, I_3 lower right, I_4 upper right. The diagonal sums are $1+7+4=12$, $7+6+2=15$, $6+5+3=14$, and $4+5+4=13$. Shared resistances are 7 between loops 1 and 2, 6 between 2 and 3, 5 between 3 and 4, and 4 between 4 and 1. The source voltages are $40, 30, 20, -10$; the last is negative because loop 4 crosses that battery against its positive direction.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 12 & -7 & 0 & -4 \\ -7 & 15 & -6 & 0 \\ 0 & -6 & 14 & -5 \\ -4 & 0 & -5 & 13 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 40 \\ 30 \\ 20 \\ -10 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{119910}{10487} \\ \frac{110640}{10487} \\ \frac{84270}{10487} \\ \frac{61240}{10487} \end{bmatrix} \approx \begin{bmatrix} 11.434 \\ 10.55 \\ 8.0357 \\ 5.8396 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 40, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Exercise 1.10.8: Derive the loop equations and solve them

The loop order is the four outer regions $1, 2, 3, 4$, then the central diamond 5. Their diagonal sums are $3+1+4+1=9$, $1+1+3+2=7$, $2+2+3+3=10$, $1+1+2+3=7$, and $4+3+3+2=12$. The central loop shares resistances $4, 3, 3, 2$ with the four outer loops. Additional shared stems are 1 between 1 and 4, 1 between 1 and 2, 2 between 2 and 3, and 3 between 3 and 4. Its own battery contribution is zero; the outer contributions are $50, -30, 20, -40$.

Thus, in amperes, the matrix equation is

$$\begin{bmatrix} 9 & -1 & 0 & -1 & -4 \\ -1 & 7 & -2 & 0 & -3 \\ 0 & -2 & 10 & -3 & -3 \\ -1 & 0 & -3 & 7 & -2 \\ -4 & -3 & -3 & -2 & 12 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 50 \\ -30 \\ 20 \\ -40 \\ 0 \end{bmatrix}.$$

Exact elimination of this augmented system gives a pivot in every variable column and hence the unique solution

$$i = \begin{bmatrix} \frac{3019}{755} \\ -\frac{3309}{755} \\ -\frac{683}{755} \\ -\frac{4382}{755} \\ -\frac{722}{755} \end{bmatrix} \approx \begin{bmatrix} 3.9987 \\ -4.3828 \\ -0.90464 \\ -5.804 \\ -0.95629 \end{bmatrix}.$$

The displayed exact fractions satisfy $Ri = v$ in every row. For example, the first row computes 50, the first specified net voltage. All remaining rows were checked by the same substitution. Interpret a minus sign as a direction reversal rather than a negative physical magnitude.

Migration convention for Exercises 9–13. Put the current populations into a column vector. In each migration matrix, column j tells what fractions of starting population j arrive in the various destinations. Thus $x_{k+1} = Mx_k$ and two years later means $x_2 = M(Mx_0)$, not Mx_0.

Exercise 1.10.9: A seven-percent outward migration rate

Let $x_k = (c_k, s_k)^T$ denote city and suburban populations k years after 2010. The matrix and initial population are

$$M = \begin{bmatrix} 0.93 & 0.05 \\ 0.07 & 0.95 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 800000 \\ 500000 \end{bmatrix}, \qquad x_{k+1} = Mx_k.$$

For example $c_{k+1} = 0.93c_k + 0.05s_k$: city residents who stay plus suburban residents who move into the city. First multiplication gives $x_1 = (769000, 531000)^T$. Multiplying once more gives

$$x_2 = \begin{bmatrix} 741720 \\ 558280 \end{bmatrix}.$$

The total remains 1300000, as a check on the column sums and arithmetic.

Conclusion. In 2012 the model predicts 741720 city residents and 558280 suburban residents.

Exercise 1.10.10: A six-percent outward migration rate

With city first and suburbs second,

$$M = \begin{bmatrix} 0.94 & 0.04 \\ 0.06 & 0.96 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 10000000 \\ 800000 \end{bmatrix}, \quad x_{k+1} = Mx_k.$$

Here the first column says 94% of city residents stay and 6% move to the suburbs. The second says 4% move in and 96% stay in the suburbs. Successive multiplications give

$$x_1 = \begin{bmatrix} 9432000 \\ 1368000 \end{bmatrix}, \qquad x_2 = \begin{bmatrix} 8920800 \\ 1879200 \end{bmatrix}.$$

Both totals equal the initial 10800000.

Conclusion. The 2012 prediction is 8920800 in the city and 1879200 in the suburbs, subject to the model's exclusions.

Exercise 1.10.11: Estimate migration rates before iterating

(a) Divide each flow by its population of origin.

The fraction leaving California is $516100/31524000 \approx 0.0163716533$, hence 0.01637 to five decimal places. The fraction moving into California from elsewhere is $381262/228680000 \approx 0.0016672293$, hence 0.00167. Put California first. The rounded migration model is

$$M = \begin{bmatrix} 0.98363 & 0.00167 \\ 0.01637 & 0.99833 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 31524000 \\ 228680000 \end{bmatrix}.$$

The diagonal entries are the complements of the corresponding outward rates, not independently rounded unrelated quantities.

(b) Apply six annual steps to reach 2000.

Compute $x_{k+1} = Mx_k$ six times, retaining full arithmetic with the specified five-place coefficients. The result is

$$x_6 \approx \begin{bmatrix} 30754526.4814 \\ 229449473.5186 \end{bmatrix}.$$

To the nearest person, these are 30754526 in California and 229449474 elsewhere. Their sum is the unchanged total 260204000. The fractional entries before rounding are population estimates, not literal fractional people. Using unrounded migration rates would yield slightly different estimates, so it is important to state which rates were used.

Conclusion. The model predicts about 30.755 million in California and 229.449 million elsewhere, with births, deaths, and international migration excluded as required.

Exercise 1.10.12: Follow a fleet through two daily transitions

Use the order airport, east, west. The columns correspond to the rental origin and rows to the return location, so

$$M = \begin{bmatrix} 0.97 & 0.05 & 0.10 \\ 0 & 0.90 & 0.05 \\ 0.03 & 0.05 & 0.85 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 295 \\ 55 \\ 150 \end{bmatrix}.$$

Monday to Tuesday gives $x_1 = Mx_0 = (303.9, 57, 139.1)^T$. Tuesday to Wednesday gives

$$x_2 = Mx_1 = \begin{bmatrix} 311.543 \\ 58.255 \\ 130.202 \end{bmatrix}.$$

For example the first entry is $0.97(303.9) + 0.05(57) + 0.10(139.1)$. The three entries sum to 500 before rounding, matching the fleet size.

Conclusion. Approximately 312 cars at the airport, 58 at the east location, and 130 at the west location on Wednesday.

Exercise 1.10.13: Twenty iterations and their limiting pattern

The source model uses $M = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$. The total population is 1000000 in both parts. Therefore, if c_k is the city population, the suburb population is $1000000 - c_k$, and the first row of the recurrence becomes

$$c_{k+1} = 0.95c_k + 0.03(1000000 - c_k) = 0.92c_k + 30000.$$

Subtract 375000 from both sides to obtain $c_{k+1} - 375000 = 0.92(c_k - 375000)$. Repeated substitution gives

$$c_k = 375000 + (c_0 - 375000)(0.92)^k, \qquad s_k = 1000000 - c_k.$$

This formula both checks the matrix iterations and explains their pattern.

(a) Start from $(600000, 400000)^T$.

The city component is $375000 + 225000(0.92)^k$.

(b) Start from $(350000, 650000)^T$.

The city component is $375000 - 25000(0.92)^k$. The following table lists all twenty requested vectors, with entries rounded to the nearest person only for display.

	Part (a)		Part (b)	
k	City	Suburbs	City	Suburbs
1	582000	418000	352000	648000
2	565440	434560	353840	646160
3	550205	449795	355533	644467
4	536188	463812	357090	642910
5	523293	476707	358523	641477
6	511430	488570	359841	640159
7	500515	499485	361054	638946
8	490474	509526	362170	637830
9	481236	518764	363196	636804
10	472737	527263	364140	635860
11	464918	535082	365009	634991
12	457725	542275	365808	634192
13	451107	548893	366544	633456
14	445018	554982	367220	632780
15	439417	560583	367843	632157
16	434264	565736	368415	631585
17	429522	570478	368942	631058
18	425161	574839	369427	630573
19	421148	578852	369872	630128
20	417456	582544	370283	629717

Since $0 < 0.92 < 1$, its powers tend to zero. In (a) the city population decreases toward 375000, passing below the suburban population between years 7 and 8 (first below at the year-8 update). In (b) it increases toward the same 375000. The suburb populations approach 625000 from the opposite sides. The common limiting proportions come from the migration rule, not from the initial distribution.

Conclusion. Both initial distributions tend to $(375000, 625000)^T$, but they approach it from different sides.

Exercise 1.10.14: Temperature models obey superposition

(a) Write one averaging equation for each interior point.

Order the points as 1 upper left, 2 upper right, 3 lower right, 4 lower left, as in the diagrams. Moving

the two interior-neighbor temperatures to the left gives the common coefficient matrix

$$K = \begin{bmatrix} 4 & -1 & 0 & -1 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 4 & -1 \\ -1 & 0 & -1 & 4 \end{bmatrix}.$$

For plate A, each point touches boundary temperatures 20 and 0, so $KT_A = (20, 20, 20, 20)^T$. Direct substitution gives $T_A = (10, 10, 10, 10)^T$. For plate B, the two relevant boundary values sum to $10, 40, 50, 20$ in point order. Thus $KT_B = (10, 40, 50, 20)^T$, and solving gives

$$T_B = (10, 17.5, 20, 12.5)^T.$$

For example its first equation is $4(10) - 17.5 - 12.5 = 10$, and its third is $4(20) - 17.5 - 12.5 = 50$. Their sum is

$$T_A + T_B = (20, 27.5, 30, 22.5)^T,$$

exactly the temperature list cited from Section 1.1. The boundary data there are the sums of these two plates' boundary data, so the same additivity appears at the interior points.

(b) Triple all of plate A's boundary temperatures.

The new right-hand side is $(60, 60, 60, 60)^T$, three times the old one. Hence $T = (30, 30, 30, 30)^T = 3T_A$. This is also checked directly: each row is $4(30) - 30 - 30 = 60$.

(c) Formulate and justify the general rule.

The map from the eight boundary values to the four interior values is linear: adding boundary data adds interior solutions, and scaling boundary data scales the interior solution. To justify uniqueness, if two solutions existed, their difference z would satisfy $Kz = 0$. Choose a component whose absolute value is largest, say $|z_i| = L$. Its equation gives $4L \le 2L$, since there are two neighboring components, each of absolute value at most L. Thus $L = 0$ and $z = 0$. There cannot be two distinct solutions. Existence for every right-hand side b can be verified directly by substitution in the explicit solution formula

$$T = \frac{1}{24} \begin{bmatrix} 7 & 2 & 1 & 2 \\ 2 & 7 & 2 & 1 \\ 1 & 2 & 7 & 2 \\ 2 & 1 & 2 & 7 \end{bmatrix} b.$$

Each b_i is a sum of two boundary values, so both stages of the construction use only fixed linear combinations.

Conclusion. The interior-temperature map is linear. Its additivity explains part (a), and its homogeneity explains part (b).

1.S Chapter 1 supplementary exercises

These exercises combine consistency, parametrization, independence, and transformations. Keep existence and uniqueness separate: a missing variable pivot creates freedom only in a *consistent* system. When a problem describes physical quantities such as floors, solve the real linear equations first, then impose the required integer and nonnegative restrictions.

Exercise 1.S.1: Twenty-six statements and their logical pitfalls

(a) **False.** Reduced echelon form is unique; ordinary echelon form is not. The one-by-one matrices [1] and [2] are row equivalent and both in echelon form, but are different.

(b) **False.** For $n = 1$, the equation $0x = 0$ has infinitely many solutions, more than one. A square system can have zero, one, or infinitely many solutions.

(c) **True.** If $Ax = Ay = b$ and $x \neq y$, then $A(x + t(y - x)) = b$ for every real t. Different t give distinct solutions because $y - x \neq 0$.

(d) **False.** No free variables guarantees at most one solution, not consistency. The equations $x = 0$ and $x = 1$ have no free variable and no solution.

(e) **True.** Elementary row operations are reversible, so each operation preserves the solution set in both directions.

(f) **True.** If $Ax = Ay = b$ with $x \neq y$, then $A(x - y) = 0$ with $x - y \neq 0$. Thus the homogeneous system has both zero and a distinct nonzero solution.

(g) **False.** Every A has a consistent system for $b = 0$. For the zero 2×2 matrix the columns span only $\{0\}$, not $\mathbb{R}^2$.

(h) **False.** Every matrix has a reduced echelon form, including inconsistent augmented matrices. The single equation $0x = 1$ is already a counterexample.

(i) **True.** Each row-equivalence class has one reduced echelon form. Reducing either member therefore gives the same matrix.

(j) **False.** $Ax = 0$ always has the trivial solution, including when free variables exist. Take $A = [0]$.

(k) **True.** Consistency for every $b \in \mathbb{R}^m$ means a pivot in every one of the m rows. There are exactly m pivots and thus m pivot columns.

(l) **False.** Row pivots imply existence for every target, not uniqueness. The matrix [1 0] has a pivot in every row, but $x_1 = b$, with x_2 arbitrary, gives infinitely many solutions.

(m) **True.** In a square $n \times n$ matrix, n pivots occupy every column and every row. Reduced pivots are ones with zeros elsewhere in their columns, so the reduced matrix is I_n.

(n) **True.** Both matrices reduce to I_3. Reduce A to I_3, then reverse the operations reducing B to I_3 to reach B.

(o) **True.** The first system supplies a nonzero z with $Az = 0$. If $Ay = c$, then $A(y + tz) = c$ for every real t, giving infinitely many distinct solutions of the second system.

(p) **True.** Row-equivalent matrices have the same number of pivots. If A spans $\mathbb{R}^m$, its m pivots fill all rows; the same is true of B, so its columns also span $\mathbb{R}^m$. This does not assert equality of their actual column vectors.

(q) **False.** The list $e_1, e_2, e_1 + e_2$ in $\mathbb{R}^3$ has no proportional pair but has the nontrivial relation $e_1 + e_2 - (e_1 + e_2) = 0$.

(r) **True.** Three vectors in $\mathbb{R}^2$ cannot be independent: a 2×3 matrix has at most two pivots. Thus the assumed independence excludes that ambient space.

(s) **False.** Four columns supply at most four pivots, not the five row pivots required to span $\mathbb{R}^5$.

(t) **True.** $-u = (-1)u + 0v$, a permitted linear combination.

(u) **False.** Let $u = e_1$, $v = 2e_1$, $w = e_2$ in $\mathbb{R}^2$. All are nonzero, but the first two span only the horizontal axis, which does not contain w.

(v) **False.** Take $u = e_1$, $v = e_2$, and $w = e_2$. Then $w = 0u + v$, but u is outside Span$\{v, w\}$.

(w) **False.** The hypothesis does not exclude $v_1 = 0$. Take $v_1 = 0$, $v_2 = e_1$, $v_3 = e_2$ in $\mathbb{R}^5$. The second is not a multiple of the first, and the third is outside their span, but the zero vector makes the list dependent.

(x) **True.** A linear transformation is a function with additional operation-preserving properties.

(y) **True.** A 6×5 matrix cannot have six pivots, so its columns cannot span $\mathbb{R}^6$.

(z) **False.** m pivot columns mean onto $\mathbb{R}^m$. For one-to-one we need n pivots. Again [1 0] has $m = 1$ pivot but sends every $(0, t)^T$ to zero.

Exercise 1.S.2: Every possible solution set of one equation

If $a \neq 0$, division gives the unique solution $x = b/a$. If $a = 0$ but $b \neq 0$, the equation says $0 = b$ and has no solution. If $a = b = 0$, the equation is true for every real x, so its solution set is all of $\mathbb{R}$. These three cases exhaust the values of a, b.

Exercise 1.S.3: Three configurations of planes

(a) Use $x = 0$, $y = 0$, $x + y = 0$. The first two force $x = y = 0$, while z is arbitrary; the third adds no restriction. The intersection is the z-axis, a single line.

(b) Use $x = 0$, $y = 0$, $z = 0$. Their only common point is the origin.

(c) Use $x = 0$, $x = 1$, $y = 0$. No point can satisfy the first two simultaneously, so there is no common intersection.

Every listed equation has at least one nonzero coefficient and therefore really does describe a plane, rather than the whole space or an empty set.

Exercise 1.S.4: Why the square size matters

Three pivot columns in a 3×3 coefficient matrix also mean three pivot rows. Thus appending any right-hand side gives a consistent system, because no zero-coefficient row remains to create a contradiction. There are no free variables, so the solution is unique. Both properties are needed: column pivots alone in a taller matrix would not ensure existence.

Exercise 1.S.5: Classify two parameter-dependent systems

(a) Eliminate the first variable.

Subtract four times $x_1 + 3x_2 = k$ from the second equation. The result is $(h - 12)x_2 = 8 - 4k$. If $h \neq 12$, this determines x_2 uniquely, then the first equation determines x_1. If $h = 12$, the result is $0 = 8 - 4k$, consistent exactly when $k = 2$.

Conclusion. (i) No solutions: $h = 12, k \neq 2$. (ii) One solution: $h \neq 12$ with any k. (iii) Infinitely many: $h = 12, k = 2$.

(b) Add three times the first equation to the second.

The eliminated equation is $(k + 3h)x_2 = 1$. If $k + 3h \neq 0$, then $x_2 = 1/(k + 3h)$ and $x_1 = (hx_2 - 1)/2$ are uniquely determined. If $k + 3h = 0$, we obtain the contradiction $0 = 1$.

Conclusion. (i) No solutions: $k = -3h$. (ii) One solution: $k \neq -3h$. (iii) No parameter values give infinitely many solutions.

Exercise 1.S.6: Restate one consistency question three ways

Let $a_1 = (4, 8)^T$, $a_2 = (-2, -3)^T$, $a_3 = (7, 10)^T$, and $b = (-5, -3)^T$.

(a) Use a linear combination.

Ask whether $b = x_1a_1 + x_2a_2 + x_3a_3$ for some real weights. Eliminating twice the first scalar equation from the second gives $x_2 - 4x_3 = 7$. Let $x_3 = t$; then $x_2 = 7 + 4t$ and $4x_1 = 9 + t$. Thus the full solution family is

$$x = (9/4 + t/4,\ 7 + 4t,\ t)^T, \qquad t \in \mathbb{R}.$$

In particular $(9/4, 7, 0)^T$ works, proving consistency.

(b) Use matrix columns.

For $A = \begin{bmatrix} 4 & -2 & 7 \\ 8 & -3 & 10 \end{bmatrix}$, ask whether b lies in the span of the columns of A. Part (a) answers yes.

(c) Use a transformation.

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T(x) = Ax$. Ask whether b is in the range of T, or equivalently whether any input has image b. The displayed family supplies all such inputs.

Exercise 1.S.7: Restate consistency for every right-hand side

Set $v_1 = (2, -5, 7)^T$, $v_2 = (-4, 1, -5)^T$, and $v_3 = (-2, 1, -3)^T$, and put $A = [v_1\ v_2\ v_3]$.

(a) Ask whether the span is all of $\mathbb{R}^3$.

Row three of A is row one minus row two. Hence every vector b in the column span satisfies $b_3 = b_1 - b_2$. The first two rows have two independent leading columns: elimination of x_1 from them leaves a nonzero coefficient -9 on x_2 after multiplying the first row by $5/2$ and adding it to row two. Thus the first two equations can be solved for any b_1, b_2, but the third requires exactly $b_3 = b_1 - b_2$. Consequently $\mathrm{Span}\{v_1, v_2, v_3\}$ is this plane, not $\mathbb{R}^3$.

(b) State the column version.

The question asks whether the columns of A span $\mathbb{R}^3$. They do not; for example $(0, 0, 1)^T$ is outside their span.

(c) State the transformation version.

For $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x) = Ax$, the question is whether T is onto. It is not, because its range is the proper plane above.

Conclusion. The system is consistent precisely for $b_3 = b_1 - b_2$, so it is not consistent for all triples of right-hand sides.

Exercise 1.S.8: Echelon patterns for spanning columns

Use p, q, r for nonzero pivots and $*$ for arbitrary entries.

(a) Two rows and three columns.

Spanning $\mathbb{R}^2$ requires two row pivots, which can occur in columns $(1, 2)$, $(1, 3)$, or $(2, 3)$:

$$\begin{bmatrix} p & * & * \\ 0 & q & * \end{bmatrix}, \quad \begin{bmatrix} p & * & * \\ 0 & 0 & q \end{bmatrix}, \quad \begin{bmatrix} 0 & p & * \\ 0 & 0 & q \end{bmatrix}.$$

(b) Three rows and three columns.

All three columns must also be pivot columns, leaving the sole pattern $\begin{bmatrix} p & * & * \\ 0 & q & * \\ 0 & 0 & r \end{bmatrix}$.

Exercise 1.S.9: Decompose a vector along two different lines

A vector on $y = 2x$ is $(s, 2s)$; one on $y = x/2$ is $(t, t/2)$. Their sum equals $(5, 6)$ when $s + t = 5$ and $2s + t/2 = 6$. Substituting $t = 5 - s$ into the latter and multiplying by two gives $4s + 5 - s = 12$, so $s = 7/3$ and $t = 8/3$. Therefore

$$(5, 6) = (7/3, 14/3) + (8/3, 4/3).$$

Each summand satisfies its respective line equation, and their coordinate sums check the requested result.

Exercise 1.S.10: Infer existence and uniqueness from the diagram

The two source arrows a_1, a_2 are nonzero and not parallel, so they are independent. Their two column directions span $\mathbb{R}^2$. Thus any b shown in that plane is attained, and its two coefficients are unique. Negative coefficients are allowed; being outside the wedge of positive combinations does not prevent a solution of $Ax = b$.

Conclusion. Yes, there is exactly one solution.

Exercise 1.S.11: A nonechelon matrix with a line of zero-images

Choose

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Both rows lead in column one, so it is not in echelon form. The second equation in $Ax = 0$ gives $x_1 = 0$, then the first gives $x_2 = 0$, while x_3 is free. Thus the solution set is $\{(0, 0, t)^T : t \in \mathbb{R}\}$, the required line in $\mathbb{R}^3$.

Exercise 1.S.12: A nonechelon matrix with a plane of zero-images

Choose $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$. The second row again leads under the first, so the matrix is not echelon. Its second equation is twice the first, leaving only $x_1 + 2x_2 + 3x_3 = 0$. Its solutions are

$$x = s(-2, 1, 0)^T + t(-3, 0, 1)^T, \qquad s, t \in \mathbb{R}.$$

The two generators are independent, so this is a plane, not merely a line.

Exercise 1.S.13: A prescribed null vector determines the missing entries

The pivot information forces reduced echelon form $\begin{bmatrix} 1 & 0 & u \\ 0 & 1 & v \\ 0 & 0 & 0 \end{bmatrix}$. Applying it to $(3, -2, 1)^T$ gives $(3 + u, -2 + v, 0)^T$. For this to be zero, $u = -3$ and $v = 2$. Hence the required form is

$$\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Row operations preserve the null space, so the prescribed vector must satisfy these reduced equations even if the original A was different.

Exercise 1.S.14: Parameter values for two-vector independence

Put the two vectors into columns and replace R_2 by $R_2 - aR_1$:

$$\begin{bmatrix} 1 & a \\ a & a + 2 \end{bmatrix} \sim \begin{bmatrix} 1 & a \\ 0 & a + 2 - a^2 \end{bmatrix}.$$

There are two pivots precisely when $a+2-a^2 \neq 0$. Since $a+2-a^2 = -(a-2)(a+1)$, independence holds for $\boxed{a \neq 2, -1}$. At either excluded value there is one pivot, so a free coefficient supplies dependence.

Exercise 1.S.15: Parameters in triangular-looking vector lists

(a) Three vectors in $\mathbb{R}^3$.

The column matrix is $\begin{bmatrix} a & b & d \\ 0 & c & e \\ 0 & 0 & f \end{bmatrix}$. Independence requires and is guaranteed by $a, c, f \neq 0$; the other entries b, d, e may be any real numbers. For sufficiency, read a homogeneous relation from bottom to top: $fz = 0$ gives $z = 0$, then $cy = 0$ gives $y = 0$, then $ax = 0$ gives $x = 0$. For necessity, if $a = 0$ the first vector is zero; if $c = 0$, the first two vectors are dependent; if $f = 0$, all three vectors lie in the two-dimensional coordinate plane with third coordinate zero.

(b) Three vectors in $\mathbb{R}^4$.

There are *no restrictions* on a, b, c, d, e, f. A relation among the columns, read from the last coordinate upward, first forces the third coefficient to zero because that coordinate is 1 only in column three. The third coordinate then forces the second coefficient to zero, and the second coordinate forces the first to zero. Hence the list is independent for every choice of the six numbers.

Exercise 1.S.16: Use the preceding-column characterization

The supplied matrix has nonzero diagonal entries $1, 5, 8, 10$ and zeros above the diagonal. Column one is nonzero. For column $j > 1$, suppose it were a combination of the preceding columns. Its first $j-1$ coordinates are zero. Reading those coordinates from top to bottom forces every coefficient of the preceding columns to be zero, because each relevant diagonal entry is nonzero. The proposed combination would then be zero, contradicting the nonzero jth coordinate of column j. Thus no later column is a combination of preceding ones. The characterization in Theorem 7 of Section 1.7 now proves that all four columns are independent.

Exercise 1.S.17: Adjoin a vector outside the old span

Suppose $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$. If $c_4 \neq 0$, solving for v_4 would put it in $\text{Span}\{v_1, v_2, v_3\}$, contrary to the hypothesis. So $c_4 = 0$. Independence of the first three then forces $c_1 = c_2 = c_3 = 0$. All coefficients vanish, proving independence of the four-vector list.

Exercise 1.S.18: Add one basis vector to another

A relation $av_1 + b(v_1 + v_2) = 0$ can be collected as $(a+b)v_1 + bv_2 = 0$. Independence of v_1, v_2 forces $b = 0$ and $a + b = 0$, hence $a = 0$. Thus the new pair $v_1, v_1 + v_2$ is also independent. The argument proves the result for all coefficients, not just a selected numerical example.

Exercise 1.S.19: Three collinear points give dependent position vectors

Because the points are distinct, $v_2 - v_1 \neq 0$ gives a direction for their common line. Some real t satisfies $v_3 = v_1 + t(v_2 - v_1) = (1-t)v_1 + tv_2$. Therefore

$$(1-t)v_1 + tv_2 - v_3 = 0.$$

The coefficient of v_3 is -1, so this relation is nontrivial. It proves dependence even if the line does not pass through the origin: the line's position vectors still lie in the span of two of its points.

Exercise 1.S.20: Reverse an input using homogeneity

Since $-u = (-1)u$, linearity gives $T(-u) = (-1)T(u) = -v$. This uses the scaling rule with the scalar -1 and does not require T to be invertible.

Exercise 1.S.21: A coordinate-plane reflection

The images of e_1, e_2, e_3 are $e_1, -e_2, e_3$. Put them into columns to get

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

It leaves the first and third coordinates fixed and reverses the second, exactly the specified reflection through $x_2 = 0$.

Exercise 1.S.22: Surjectivity forces injectivity for this square size

Onto $\mathbb{R}^3$ means a pivot in every row of the 3×3 matrix. These three pivots also occupy every column, so there is no free variable in $Ax = 0$. Its only solution is zero, giving one-to-one. The equality of the input and output dimensions is crucial here.

Exercise 1.S.23: A Givens rotation in two coordinates

Multiplying the displayed matrix by $(4, 3)^T$ gives $(4a - 3b, 4b + 3a)^T$. The target requires $4a - 3b = 5$ and $3a + 4b = 0$. From the second equation, $b = -3a/4$; substituting in the first gives $25a/4 = 5$. Thus $\boxed{a = 4/5, b = -3/5}$. The condition $a^2 + b^2 = 16/25 + 9/25 = 1$ is satisfied, and the image is $(5, 0)^T$, so both direction and length check.

Exercise 1.S.24: Zero one coordinate without changing the middle one

The first and third coordinates give $2a - 4b = 2\sqrt{5}$ and $2b + 4a = 0$. The latter says $b = -2a$, so $10a = 2\sqrt{5}$. Therefore

$$a = 1/\sqrt{5}, \qquad b = -2/\sqrt{5}.$$

The middle coordinate remains 3 automatically, and $a^2 + b^2 = 1/5 + 4/5 = 1$. Substituting in the first and third coordinates gives $2\sqrt{5}$ and 0, as required.

Exercise 1.S.25: A building design needs whole, nonnegative floor counts

(a) Interpret a scalar multiple of the recipe column.

The vector $x_1(3,7,8)^T$ gives the numbers of three-, two-, and one-bedroom apartments on the x_1 floors using plan A.

(b) Add the contributions from all three plans.

The total apartment-count vector is

$$x_1\begin{bmatrix}3\\7\\8\end{bmatrix}+x_2\begin{bmatrix}4\\4\\8\end{bmatrix}+x_3\begin{bmatrix}5\\3\\9\end{bmatrix}.$$

(c) Solve the target equations, then impose the physical restrictions.

Appending $(66,74,136)^T$ and reducing gives

$$\left[\begin{array}{ccc|c}3&4&5&66\\7&4&3&74\\8&8&9&136\end{array}\right]\sim\left[\begin{array}{ccc|c}1&0&-1/2&2\\0&1&13/8&15\\0&0&0&0\end{array}\right].$$

Thus the real solutions are $x_1=2+t/2$, $x_2=15-13t/8$, $x_3=t$. Floors must be nonnegative integers. Integer t makes x_2 an integer only when t is a multiple of 8, since 13 and 8 have no common factor. Nonnegativity requires $0\le t\le 120/13<10$. Hence the only allowed values are $t=0,8$, giving $(x_1,x_2,x_3)=(2,15,0)$ or $(6,2,8)$. Substitution yields the requested apartment totals in both cases.

Conclusion. Yes. There are exactly two designs with whole nonnegative floor counts, although the unrestricted real system has infinitely many solutions.

2.1 Matrix operations

A matrix has a *shape*: an $m \times n$ matrix has m rows and n columns. Addition requires identical shapes. In AB, the number of columns of A must equal the number of rows of B; the answer then has the outside shape. If A is $m \times n$ and B is $n \times p$, then

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad AB = [Ab_1 \ \cdots \ Ab_p].$$

The first formula takes a row against a column; the second says that one column of the answer can be computed without the others. The transpose interchanges rows and columns. In particular $(AB)^T = B^T A^T$: its reversal of order is essential. Matrix multiplication generally does not commute.

Practice problem 2.1.P1: A vector and its transpose

With $A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}$ and $x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$, first compute $Ax = (5-9, -10+12)^T = (-4, 2)^T$. Thus

$$(Ax)^T = [-4 \ \ 2], \qquad x^T A^T = [5 \ \ 3] \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} = [-4 \ \ 2].$$

The outer product multiplies each entry in the column by the whole row:

$$xx^T = \begin{bmatrix} 25 & 15 \\ 15 & 9 \end{bmatrix}, \qquad x^T x = 5^2 + 3^2 = 34.$$

Finally A^T is 2×2 and x^T is 1×2. The inner sizes $2, 1$ do not agree, so $A^T x^T$ is undefined. An outer product and an inner product are different operations, even when they use the same entries.

Practice problem 2.1.P2: Choose the order of a computation

What we must show. Compute $A^2 x$ efficiently for A of shape 4×4. Associativity permits $A(Ax)$. A matrix–vector product uses four multiplications for each of four entries, or 16 total. Computing Ax then $A(Ax)$ uses $16 + 16 = 32$. In contrast, A^2 has 16 entries, each requiring four multiplications: 64; the final multiplication by x uses 16 more. The second route uses 80. Both routes give the same vector, but the first avoids computing matrix entries that are not needed.

Exercise 2.1.1: Check shapes before calculating

For the matrices supplied in the exercise, A, B are 2×3 and C, D are 2×2. Scalar multiplication and subtraction act entrywise:

$$-2A = \begin{bmatrix} -4 & 0 & 2 \\ -8 & 10 & -4 \end{bmatrix}, \qquad B - 2A = \begin{bmatrix} 7-4 & -5-0 & 1+2 \\ 1-8 & -4+10 & -3-4 \end{bmatrix} = \begin{bmatrix} 3 & -5 & 3 \\ -7 & 6 & -7 \end{bmatrix}.$$

The product AC is undefined: the inner sizes are 3 and 2. For CD, the inner sizes match, and taking rows against columns gives

$$CD = \begin{bmatrix} 1 \cdot 3 + 2(-1) & 1 \cdot 5 + 2 \cdot 4 \\ -2 \cdot 3 + 1(-1) & -2 \cdot 5 + 1 \cdot 4 \end{bmatrix} = \begin{bmatrix} 1 & 13 \\ -7 & -6 \end{bmatrix}.$$

Exercise 2.1.2: Two defined expressions and two shape failures

Entrywise addition gives

$$A+3B=\begin{bmatrix}2+21 & 0-15 & -1+3\\ 4+3 & -5-12 & 2-9\end{bmatrix}=\begin{bmatrix}23 & -15 & 2\\ 7 & -17 & -7\end{bmatrix}.$$

The vector E is 2×1, so $2C-3E$ is undefined: subtraction cannot combine a 2×2 matrix with a 2×1 matrix. For the defined product,

$$DB=\begin{bmatrix}3\cdot 7+5\cdot 1 & 3(-5)+5(-4) & 3\cdot 1+5(-3)\\ -7+4 & 5-16 & -1-12\end{bmatrix}=\begin{bmatrix}26 & -35 & -12\\ -3 & -11 & -13\end{bmatrix}.$$

Finally EC would have inner sizes 1 and 2, so it is undefined.

Exercise 2.1.3: A scalar identity matrix in a sum and a product

For $A=\begin{bmatrix}2 & -5\\ 3 & -2\end{bmatrix}$,

$$3I_2-A=\begin{bmatrix}3-2 & 0-(-5)\\ 0-3 & 3-(-2)\end{bmatrix}=\begin{bmatrix}1 & 5\\ -3 & 5\end{bmatrix},\qquad (3I_2)A=3A=\begin{bmatrix}6 & -15\\ 9 & -6\end{bmatrix}.$$

Subtracting a scalar multiple of I changes diagonal entries only; multiplying by that matrix scales *every* entry.

Exercise 2.1.4: The same distinction in three dimensions

Subtract 5 from each diagonal entry of the supplied A:

$$A-5I_3=\begin{bmatrix}0 & -1 & 3\\ -4 & -2 & -6\\ -3 & 1 & -3\end{bmatrix}.$$

Multiplication instead scales each row, giving

$$(5I_3)A=5A=\begin{bmatrix}25 & -5 & 15\\ -20 & 15 & -30\\ -15 & 5 & 10\end{bmatrix}.$$

Exercise 2.1.5: Compute a product by columns and by entries

(a) Use the two columns of B.

For $A=\begin{bmatrix}-1 & 3\\ 2 & 4\\ 5 & -3\end{bmatrix}$,

$$Ab_1=4\begin{bmatrix}-1\\ 2\\ 5\end{bmatrix}-2\begin{bmatrix}3\\ 4\\ -3\end{bmatrix}=\begin{bmatrix}-10\\ 0\\ 26\end{bmatrix},\qquad Ab_2=-2\begin{bmatrix}-1\\ 2\\ 5\end{bmatrix}+3\begin{bmatrix}3\\ 4\\ -3\end{bmatrix}=\begin{bmatrix}11\\ 8\\ -19\end{bmatrix}.$$

Place them in order: $AB = \begin{bmatrix} -10 & 11 \\ 0 & 8 \\ 26 & -19 \end{bmatrix}$.

(b) Check by the row–column rule.

The six dot products are $-1(4)+3(-2) = -10$, $-1(-2)+3(3) = 11$, $2(4)+4(-2) = 0$, $2(-2)+4(3) = 8$, $5(4)-3(-2) = 26$, and $5(-2)-3(3) = -19$. They give exactly the same matrix, as the two definitions require.

Exercise 2.1.6: A second product, with zeros included

(a) Compute the columns separately.

The columns of A are $(4,-3,0)^T$ and $(-3,5,1)^T$. Hence

$$Ab_1 = \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} + 3\begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 12 \\ 3 \end{bmatrix}, \qquad Ab_2 = 4\begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} - 2\begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 22 \\ -22 \\ -2 \end{bmatrix}.$$

(b) Verify the entries.

The row–column products are $4-9 = -5$, $16+6 = 22$, $-3+15 = 12$, $-12-10 = -22$, $0+3 = 3$, and $0-2 = -2$. Thus $AB = \begin{bmatrix} -5 & 22 \\ 12 & -22 \\ 3 & -2 \end{bmatrix}$.

Exercise 2.1.7: Recover an unknown shape

A 5×3 matrix can multiply B only if B has 3 rows. The product has as many columns as B; since AB has 7 columns, B has shape $\boxed{3 \times 7}$.

Exercise 2.1.8: Rows of a product

The number of rows of BC is the number of rows of B. Therefore B has $\boxed{5}$ rows. The given product does not determine how many columns B has; that number only has to match the rows of C.

Exercise 2.1.9: A parameter making two products equal

Multiplying in the two orders gives

$$AB = \begin{bmatrix} -7 & 3k+18 \\ -4 & k-9 \end{bmatrix}, \qquad BA = \begin{bmatrix} -7 & 12 \\ -k-6 & k-9 \end{bmatrix}.$$

Equality requires equality in *every* position. The upper-right position gives $3k+18 = 12$, or $k = -2$; the lower-left gives $-4 = -k-6$, also $k = -2$. The diagonal entries already match. Thus $\boxed{k = -2}$ is both necessary and sufficient.

Exercise 2.1.10: Why cancellation can fail

Direct row–column multiplication gives

$$AB = \begin{bmatrix} -21 & -21 \\ 7 & 7 \end{bmatrix} = AC,$$

although, for example, $B_{11} = -1 \neq -3 = C_{11}$. Indeed $B - C = \begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$ and each column is a multiple of $(2, 1)^T$, which A sends to zero. Thus $A(B - C) = 0$ need not imply $B - C = 0$. Cancelling a matrix requires additional hypotheses.

Exercise 2.1.11: Diagonal multiplication and a commuting example

$$AD = \begin{bmatrix} 5 & 6 & 6 \\ 10 & 12 & 10 \\ 15 & 15 & 12 \end{bmatrix}, \qquad DA = \begin{bmatrix} 5 & 10 & 15 \\ 6 & 12 & 15 \\ 6 & 10 & 12 \end{bmatrix}.$$

The jth column of AD is the jth column of A multiplied by d_{jj}; only the jth entry of the jth column of D is nonzero. Similarly, left multiplication scales row i by d_{ii}. For the last request, choose $B = 2I_3$. This is neither zero nor the identity, and $AB = 2A = BA$ by the identity and scalar-multiplication rules.

Exercise 2.1.12: Build a nonzero matrix annihilated by another

For $A = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}$, the equation $Ax = 0$ reduces to $x_1 = 2x_2$. Choose two different nonzero solutions, $(2, 1)^T$ and $(4, 2)^T$, and make them columns:

$$B = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}, \qquad AB = [A(2, 1)^T \;\; A(4, 2)^T] = [0 \;\; 0].$$

Having nonzero columns does not prevent their images from being zero.

Exercise 2.1.13: Pack several matrix–vector products together

Set $R = [r_1 \;\cdots\; r_p]$, an $n \times p$ matrix. The column definition of multiplication gives

$$[Qr_1 \;\cdots\; Qr_p] = QR.$$

The factors are the supplied matrix Q and the matrix of input columns, not auxiliary identity matrices. The sizes are $(m \times n)(n \times p)$. If one of these supplied factors happens itself to be an identity matrix, the literal requirement that neither factor be an identity can still be met: write the product as $(\lambda Q)(\lambda^{-1}R)$. Choose a nonzero scalar λ avoiding any value for which either factor is an identity. Each factor excludes at most one scalar, so at least one of $2, 3, 4$ works.

Exercise 2.1.14: Read the units in a cost product

Column j of Q gives dollars of output of the two products in quarter j. Multiplication by U converts that output vector into three dollar costs: materials, labor, and overhead. Thus UQ is 3×4; its entry (i, j) is the total dollar cost in category i during quarter j. The sum over products is exactly the row–column rule: cost per output dollar times output dollars, added over the two products.

Exercise 2.1.15: Five true–false statements about products and transposes

(a) **False.** Two column vectors a_1, b_1 of shape 2×1 cannot be multiplied as $a_1 b_1$. The correct column formula is $AB = [Ab_1 \;\; Ab_2]$.

(b) **False.** Columns of AB are combinations of columns of A, with weights from columns of B. The roles in the statement are reversed. For example, take $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. The actual first column of AB is $(1, 2)^T$, whereas the proposed combination using the first column of A as weights gives $b_1 = (1, 1)^T$.

(c) **True.** The left distributive law says $A(B + C) = AB + AC$.

(d) **True.** Entry (i, j) of both $A^T + B^T$ and $(A + B)^T$ is $a_{ji} + b_{ji}$.

(e) **False.** Transposing a product reverses order: $(AB)^T = B^T A^T$. For example, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ give different $B^T A^T$ and $A^T B^T$.

Exercise 2.1.16: Five more structural statements

(a) **True.** Entry j of the first row of AB is $\sum_k a_{1k} b_{kj}$, exactly entry j of $(\text{row}_1 A)B$.

(b) **False.** The proposed expression has one column, whereas AB has three. Its correct form is $[Ab_1 \;\; Ab_2 \;\; Ab_3]$.

(c) **True.** $(A^2)^T = (AA)^T = A^T A^T = (A^T)^2$.

(d) **False.** Two reversals give $(ABC)^T = C^T B^T A^T$. The stated middle order is incorrect; taking $C = I$ reduces the claim to the false same-order product rule in the preceding exercise.

(e) **True.** Transpose distributes over sums, entry by entry.

Exercise 2.1.17: Find two unknown input columns

Write $B = [b_1 \; b_2]$. Since $AB = [Ab_1 \; Ab_2]$, solve the two systems with right sides $(-3, 1)^T$ and $(-11, 17)^T$. For a right side $(c, d)^T$, the equations are $x_1 - 3x_2 = c$ and $-3x_1 + 5x_2 = d$. Replacing the second equation by itself plus three times the first gives $-4x_2 = d + 3c$. For $(c, d) = (-3, 1)$, this gives $x_2 = 2$, $x_1 = 3$; for $(-11, 17)$, it gives $x_2 = 4$, $x_1 = 1$. Thus

$$b_1 = (3, 2)^T, \qquad b_2 = (1, 4)^T, \qquad B = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}.$$

Substitution into both original systems checks the two columns.

Exercise 2.1.18: A zero input column

The third column is Ab_3. If $b_3 = 0$, then $Ab_3 = A0 = 0$. Thus the third column of AB is zero, regardless of the other columns.

Exercise 2.1.19: A sum relation survives multiplication

If $b_3 = b_1 + b_2$, then $Ab_3 = A(b_1 + b_2) = Ab_1 + Ab_2$ by distributivity. Therefore column 3 of AB is the sum of columns 1 and 2.

Exercise 2.1.20: Equal columns have equal images

If $b_1 = b_2$, multiplying the equality by A gives $Ab_1 = Ab_2$. These are precisely the first two columns of AB; they are equal. No conclusion is forced about any later column.

Exercise 2.1.21: A nonzero vector sent to zero

Let b be the last column of B. By hypothesis $b \neq 0$, but the last column of AB is $Ab = 0$. Writing this as a linear combination of the columns of A, the coefficients are the entries of b, not all zero. That is a nontrivial dependence relation. Thus the columns of A are linearly dependent.

Exercise 2.1.22: A dependence relation cannot be destroyed by a linear map

If $b_1, \ldots, b_p$ are dependent, there are scalars c_j, not all zero, with $\sum_j c_j b_j = 0$. Multiply by A and distribute:

$$\sum_j c_j(Ab_j) = A\left(\sum_j c_j b_j\right) = A0 = 0.$$

The *same* nonzero coefficient vector witnesses dependence of the columns Ab_j of AB. Some images may even be zero; that only makes dependence easier, not harder.

Exercise 2.1.23: A left inverse forbids free variables

Suppose $Ax = 0$ and $CA = I_n$. Multiplying by C yields $x = I_n x = (CA)x = C(Ax) = C0 = 0$. Thus the homogeneous system has no free variables: all n columns of A must be pivot columns. Since there is at most one pivot in each row, A must have at least n rows. Therefore it cannot have more columns than rows.

Exercise 2.1.24: Construct a right inverse one column at a time

Since the columns of A span $\mathbb{R}^3$, each equation $Ad_j = e_j$ is consistent for $j = 1, 2, 3$. Choose one solution of each, for example by row reduction, and set $D = [d_1\ d_2\ d_3]$. Each d_j has n entries, so D is

$n \times 3$. Then $AD = [Ad_1\ Ad_2\ Ad_3] = [e_1\ e_2\ e_3] = I_3$. If solutions are nonunique, any choices work; uniqueness was not assumed.

Exercise 2.1.25: A left and right inverse force a square matrix

From $CA = I_n$, the argument of Exercise 23 gives $n \le m$. From $AD = I_m$, every $b \in \mathbb{R}^m$ has solution $x = Db$, so A has a pivot in every row and $m \le n$. Thus $m = n$. Associativity then gives

$$C = CI_m = C(AD) = (CA)D = I_nD = D.$$

One must first respect the sizes; the two inequalities explain why both sorts of inverse can occur together only in the square case.

Exercise 2.1.26: A right inverse guarantees all outputs

For arbitrary $b \in \mathbb{R}^m$, choose $x = Db$. Then $Ax = A(Db) = (AD)b = I_mb = b$. This proves consistency for every right side. Hence A has a pivot in each of its m rows. There cannot be more pivots than columns, so $m \le n$: no more rows than columns.

Exercise 2.1.27: Inner and outer products in coordinates

For $u = (-3, 2, -5)^T$ and $v = (a, b, c)^T$, multiplication gives

$$u^Tv = v^Tu = -3a + 2b - 5c,$$

$$uv^T = \begin{bmatrix} -3a & -3b & -3c \\ 2a & 2b & 2c \\ -5a & -5b & -5c \end{bmatrix}, \qquad vu^T = \begin{bmatrix} -3a & 2a & -5a \\ -3b & 2b & -5b \\ -3c & 2c & -5c \end{bmatrix}.$$

The scalar product pairs corresponding coordinates and adds; the outer product keeps all pairwise products, which explains its different shape.

Exercise 2.1.28: Relationships between the two orders

For real vectors, $u^Tv = \sum_i u_iv_i = \sum_i v_iu_i = v^Tu$. The outer products need not be equal, but $(uv^T)^T = (v^T)^Tu^T = vu^T$. Their (i, j) and (j, i) entries are both u_iv_j. The real-scalar assumption matters: complex inner products usually use a conjugate transpose instead.

Exercise 2.1.29: Prove both distributive laws entrywise

What we must show. Establish Theorem 2(b) and 2(c), with all products of matching sizes. For left distributivity, take an arbitrary entry and expand a finite sum:

$$[A(B + C)]_{ij} = \sum_k a_{ik}(b_{kj} + c_{kj}) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj} = [AB + AC]_{ij}.$$

For right distributivity the same scalar rule gives

$$[(B + C)A]_{ij} = \sum_k (b_{ik} + c_{ik})a_{kj} = [BA + CA]_{ij}.$$

The index k runs over the common inner dimension in each formula. Because every entry agrees, the matrices agree.

Exercise 2.1.30: Move a scalar through a product

For an arbitrary position (i, j),

$$[(rA)B]_{ij} = \sum_k (ra_{ik})b_{kj} = r\sum_k a_{ik}b_{kj} = \sum_k a_{ik}(rb_{kj}) = [A(rB)]_{ij}.$$

The middle quantity is also $[r(AB)]_{ij}$. We used only distributivity of finite sums and associativity of real multiplication. Thus $r(AB) = (rA)B = A(rB)$.

Exercise 2.1.31: The identity on the left

Write $A = [a_1 \ \cdots \ a_n]$, where every $a_j \in \mathbb{R}^m$. By the column definition, $I_mA = [I_ma_1 \ \cdots \ I_ma_n] = [a_1 \ \cdots \ a_n] = A$, using the supplied fact that $I_mx = x$ for each vector in $\mathbb{R}^m$.

Exercise 2.1.32: The identity on the right

The jth column of I_n is e_j, whose only nonzero entry is a 1 in position j. Thus $Ae_j = 0a_1 + \cdots + 1a_j + \cdots + 0a_n = a_j$. Consequently $AI_n = [Ae_1 \ \cdots \ Ae_n] = A$.

Exercise 2.1.33: Prove the reversal rule for transposes

Let A be $m \times n$ and B be $n \times p$. Both $(AB)^T$ and B^TA^T have size $p \times m$. At position (i, j),

$$[(AB)^T]_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk}b_{ki} = \sum_{k=1}^{n} (B^T)_{ik}(A^T)_{kj} = [B^TA^T]_{ij}.$$

The middle reordering uses commutativity of scalar entries, not commutativity of matrix multiplication. Equality of all entries proves the rule.

Exercise 2.1.34: Transpose a product with a vector

Treat x as a one-column matrix and reverse the order of all three factors:

$$(ABx)^T = x^T(AB)^T = \boxed{x^TB^TA^T}.$$

If A is $m \times n$, B is $n \times p$, and x is $p \times 1$, the final shape is $1 \times m$, as a transposed output vector should have.

Exercise 2.1.35: Create standard matrices in a matrix program

One concrete choice of software is Python with NumPy. These commands create the four requested arrays without entering every entry:

```
import numpy as np
np.zeros((4,5))
np.ones((5,3))
np.eye(5)
np.diag([3,4,2,5])
```

A tuple such as $(4, 5)$ specifies rows first, columns second. The diagonal constructor places the listed numbers on the diagonal and zeros elsewhere. These commands were executed to verify their shapes.

Exercise 2.1.36: Generate reproducible random arrays

The following uses a fixed seed so another reader can repeat the experiment:

```
rng = np.random.default_rng(21036)
R = rng.random((5,6))
Z = rng.integers(-9,10,size=(4,4))
```

Entries of R lie in $[0, 1)$; the endpoint convention differs slightly from the idealized $0 < x < 1$ in the hint and does not affect the task. Entries of Z are integers in $\{-9, -8, \ldots, 9\}$: the upper limit 10 is excluded. The dedicated integer generator avoids having to round scaled real numbers. A seed controls reproducibility, not a mathematical claim that all resulting matrices have a special property.

Exercise 2.1.37: Test commutativity with four random pairs

```
rng = np.random.default_rng(21037)
for j in range(4):
    A = rng.integers(-9,10,(4,4))
    B = rng.integers(-9,10,(4,4))
    print(np.max(np.abs(A@B-B@A)))
```

The executed outputs are $163, 136, 158, 156$. Here $\max|M|$ means the largest absolute entry, not a determinant. Each positive result proves that that particular commutator $AB - BA$ is nonzero, so $AB \neq BA$ in all four trials. One counterexample already disproves a universal commutativity claim. The trials do not prove that matrices *never* commute: I commutes with every square matrix.

Exercise 2.1.38: A valid and an invalid difference-of-squares identity

Use integer arrays to avoid roundoff in testing these polynomial expressions:

```
rng = np.random.default_rng(21038)
for j in range(3):
    A = rng.integers(-9,10,(5,5)); I = np.eye(5,dtype=int)
    print(np.max(np.abs((A+I)@(A-I)-(A@A-I))))
for j in range(3):
    A = rng.integers(-9,10,(4,4))
    B = rng.integers(-9,10,(4,4))
    print(np.max(np.abs((A+B)@(A-B)-(A@A-B@B))))
```

The first three outputs are $0, 0, 0$; the next three are $182, 188, 189$. Algebra explains the difference:

$$(A + I)(A - I) = A^2 - A + A - I = A^2 - I,$$

whereas

$$(A + B)(A - B) = A^2 - B^2 + BA - AB.$$

Thus the latter proposed identity holds precisely when $AB = BA$; the experiments found three pairs for which this extra condition fails.

Exercise 2.1.39: Test three transpose rules

For three random pairs, execute

```
rng = np.random.default_rng(21039)
for j in range(3):
    A = rng.integers(-9,10,(4,4)); B = rng.integers(-9,10,(4,4))
    errors = [(A+B).T-A.T-B.T, (A@B).T-B.T@A.T,
              (A@B).T-A.T@B.T]
    print([np.max(np.abs(E)) for E in errors])
```

The outputs are $(0, 0, 173)$, $(0, 0, 141)$, and $(0, 0, 223)$. Thus the sum rule and reversed-product rule pass all three exact-integer checks, while the same-order product rule fails all three. The proofs in the preceding exercises establish the first two rules for every matching pair; random tests by themselves would not do that.

Exercise 2.1.40: Powers of a shift matrix

The given S satisfies $S(x_1, x_2, x_3, x_4, x_5)^T = (x_2, x_3, x_4, x_5, 0)^T$. Each multiplication shifts entries one more place and inserts another zero. Consequently

$$S^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad S^3 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$S^4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad S^5 = S^6 = 0_{5\times 5}.$$

Equivalently $(S^k)_{ij} = 1$ exactly when $j = i + k$ is a valid column index. Once $k \geq 5$ there is no such position; the matrix is nilpotent.

Exercise 2.1.41: Powers that settle toward an averaging matrix

For the supplied symmetric matrix A, exact rational powers followed by decimal conversion give, for example,

$$A^5 \approx \begin{bmatrix} .333912 & .334877 & .331211 \\ .334877 & .335134 & .329990 \\ .331211 & .329990 & .338799 \end{bmatrix}, \quad A^{10} \approx \begin{bmatrix} .333341 & .333344 & .333315 \\ .333344 & .333350 & .333306 \\ .333315 & .333306 & .333379 \end{bmatrix}.$$

By A^{20} and A^{30} every entry rounds to 0.333333 at six decimal places. The observed behavior is convergence toward the matrix with every entry $1/3$: repeated multiplication increasingly averages the coordinates. For a reproducible numerical calculation, enter the fractions rather than rounded initial entries and use `np.linalg.matrix_power(A,k)`. As an optional explanation using later eigenvalue theory, A has eigenvalues $1, (1+\sqrt{13})/12, (1-\sqrt{13})/12$. The last two have magnitude less than 1 and their powers tend to zero; the surviving eigenspace is the line spanned by $(1,1,1)^T$. Its orthogonal projection is exactly the averaging matrix. This explains the trend rather than extrapolating from a few powers.

2.2 The inverse of a matrix

An inverse is a *two-sided* undoing operation: for a square matrix, $AB = BA = I$ means $B = A^{-1}$. Multiplication order matters. In particular, $(AB)^{-1} = B^{-1}A^{-1}$ when both inverses exist. For a 2×2 matrix,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{provided } ad - bc \neq 0.$$

For larger matrices, apply the *same row operations* to both halves of $[A \mid I]$. If the left half becomes I, the right half is A^{-1}. If a pivot is missing on the left, the inverse does not exist. To solve $Ax = b$, one usually row reduces $[A \mid b]$ rather than calculating an entire inverse; both routes are explained where requested below.

Practice problem 2.2.P1: Three determinant tests

Compute the scalar $ad - bc$ in each case:

(a) For $\begin{bmatrix} 3 & -9 \\ 2 & 6 \end{bmatrix}$ it is $3 \cdot 6 - (-9)2 = 36 \neq 0$: invertible.

(b) For $\begin{bmatrix} 4 & -9 \\ 0 & 5 \end{bmatrix}$ it is $4 \cdot 5 - (-9)0 = 20 \neq 0$: invertible.

(c) For $\begin{bmatrix} 6 & -9 \\ -4 & 6 \end{bmatrix}$ it is $36 - 36 = 0$: not invertible. Indeed row 2 is $-2/3$ times row 1.

Practice problem 2.2.P2: Detect singularity while seeking an inverse

Begin with the augmented matrix and eliminate its first column:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 5 & 6 & 0 & 1 & 0 \\ 5 & -4 & 5 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 5R_1]{R_2 \leftarrow R_2 + R_1} \left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 6 & 10 & -5 & 0 & 1 \end{array}\right].$$

Now $R_3 \leftarrow R_3 - 2R_2$ makes row 3 equal to $[0\ 0\ 0 \mid -7\ -2\ 1]$. The coefficient half has only two pivots, so it cannot reduce to I_3. Therefore the given matrix has no inverse. A nonzero row in the *right* half does not repair a missing pivot in A.

Exercise 2.2.1: An inverse with determinant two

Here $ad - bc = 8 \cdot 4 - 6 \cdot 5 = 2$. Swap the diagonal entries, negate the other two, and divide by 2:

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}.$$

For example, the first row of AA^{-1} is $(16-15, -24+24) = (1, 0)$, and the second is $(10-10, -15+16) = (0, 1)$. The reverse product is also I_2, either by direct multiplication or the verified 2×2 inverse formula.

Exercise 2.2.2: An inverse with determinant negative one

For $A = \begin{bmatrix} 3 & 2 \\ 8 & 5 \end{bmatrix}$, $ad - bc = 15 - 16 = -1$. Hence

$$A^{-1} = -\begin{bmatrix} 5 & -2 \\ -8 & 3 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 8 & -3 \end{bmatrix}.$$

The negative denominator changes *every* sign, not just the diagonal. Multiplication gives diagonal entries $-15 + 16 = 1$ and $16 - 15 = 1$, and off-diagonal entries $6 - 6 = 0$ and $-40 + 40 = 0$.

Exercise 2.2.3: Keep track of a negative determinant

For $A = \begin{bmatrix} 7 & 3 \\ -6 & -3 \end{bmatrix}$, $ad - bc = -21 + 18 = -3$. Therefore

$$A^{-1} = -\frac{1}{3}\begin{bmatrix} -3 & -3 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & -7/3 \end{bmatrix}.$$

As a check, AA^{-1} has rows $(7 - 6, 7 - 7) = (1, 0)$ and $(-6 + 6, -6 + 7) = (0, 1)$.

Exercise 2.2.4: An inverse with fractional entries

The determinant is $2(-6) - (-4)4 = 4 \neq 0$, so

$$A^{-1} = \frac{1}{4}\begin{bmatrix} -6 & 4 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} -3/2 & 1 \\ -1 & 1/2 \end{bmatrix}.$$

Fractions cause no problem: inverse entries need not be integers even when all entries of the original matrix are integers.

Exercise 2.2.5: Use the first inverse to solve a system

Multiply the right side by the inverse from Exercise 1:

$$x = A^{-1}b = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 + 3 \\ -5 - 4 \end{bmatrix} = \begin{bmatrix} 7 \\ -9 \end{bmatrix}.$$

Check in the original equations: $8(7) + 6(-9) = 2$ and $5(7) + 4(-9) = -1$. Since A is invertible, this solution is unique.

Exercise 2.2.6: Solve using the inverse from Exercise 3

$$x = \begin{bmatrix} 1 & 1 \\ -2 & -7/3 \end{bmatrix}\begin{bmatrix} -9 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ 18 - 28/3 \end{bmatrix} = \begin{bmatrix} -5 \\ 26/3 \end{bmatrix}.$$

Substitution gives $7(-5) + 3(26/3) = -9$ and $-6(-5) - 3(26/3) = 4$. Both equations, including their signs, are satisfied.

Exercise 2.2.7: Four right sides, one set of row operations

(a) Compute the inverse and multiply each right side.

The determinant of $A = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix}$ is $12 - 10 = 2$, so $A^{-1} = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix}$. For any $b = (c, d)^T$, its solution is $(6c - d, (-5c + d)/2)^T$. Applying this formula to the four supplied columns gives, in order,

$$(-9, 4)^T, \quad (11, -5)^T, \quad (6, -2)^T, \quad (13, -5)^T.$$

(b) Solve the four systems together.

Use $R_2 \leftarrow R_2 - 5R_1$, then $R_2 \leftarrow R_2/2$, then $R_1 \leftarrow R_1 - 2R_2$:

$$\left[\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 5 & 12 & 3 & -5 & 6 & 5 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 2 & 8 & -10 & -4 & -10 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{cc|cccc} 1 & 0 & -9 & 11 & 6 & 13 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{array}\right].$$

Every column on the right has undergone the same operations as its own separate augmented system, so the final block lists exactly the four solutions.

Exercise 2.2.8: Undo a similarity expression

From $A = PBP^{-1}$, multiply on the left by P^{-1} and on the right by P:

$$P^{-1}AP = P^{-1}(PBP^{-1})P = (P^{-1}P)B(P^{-1}P) = B.$$

Thus $\boxed{B = P^{-1}AP}$. Putting both new factors on the same side would not cancel the original factors.

Exercise 2.2.9: True–false tests for inverses

(a) **True.** The definition of inverse requires $AB = BA = I$. For square matrices one equality eventually implies the other, but both equalities are still true when B is the inverse.

(b) **False.** The correct order is $(AB)^{-1} = B^{-1}A^{-1}$. For instance take $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \operatorname{diag}(1, 2)$. Then $A^{-1}B^{-1}$ is $\begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}$, whereas the actual inverse is $\begin{bmatrix} 1 & -1 \\ 0 & 1/2 \end{bmatrix}$.

(c) **False.** The determinant is $ad - bc$, not $ab - cd$. The singular matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has $ab - cd = 2 - 8 = -6 \neq 0$, disproving the assertion.

(d) **True.** For each b, $x = A^{-1}b$ satisfies $Ax = AA^{-1}b = b$.

(e) **True.** Reverse the underlying row operation: a swap undoes itself, nonzero scaling is undone by its reciprocal, and adding r times another row is undone by adding $-r$ times that row. The corresponding elementary matrix is the inverse.

Exercise 2.2.10: Five further inverse statements

(a) **False.** Operations taking A to I collectively multiply by A^{-1}. Applied to A^{-1} they give A^{-2}, usually not I. For $A = 2I$, dividing rows by 2 takes $A^{-1} = I/2$ to $I/4$.

(b) **True.** $A^{-1}A = AA^{-1} = I$ says precisely that A is the inverse of A^{-1}.

(c) **False.** The product is invertible, but inverses occur in reverse order; the counterexample in Exercise 9(b) applies.

(d) **True.** Solvability for each e_j lets us place solutions in columns of D with $AD = I$. Alternatively, every $b = \sum_j b_j e_j$ is an output of A by linearity, so A has a pivot in every row. Since A is square, its reduction is I, proving invertibility.

(e) **True.** This is the row-reduction characterization: applying the same operations to I produces the inverse.

Exercise 2.2.11: Solve a matrix equation and prove uniqueness

If $AX = B$, multiply on the left by A^{-1} to obtain $X = A^{-1}B$. This proves every solution must have that form. Conversely, $A(A^{-1}B) = (AA^{-1})B = B$, so that form really is a solution. The two arguments together prove existence and uniqueness. Shapes agree: $A^{-1}B$ is $n \times p$, just as X must be.

Exercise 2.2.12: A right inverse of an invertible matrix is its inverse

Multiply $AD = I$ on the left by A^{-1}:

$$D = (A^{-1}A)D = A^{-1}(AD) = A^{-1}I = A^{-1}.$$

The existing inverse of A justifies the cancellation; it is not a general rule for arbitrary rectangular or singular matrices.

Exercise 2.2.13: Cancellation and the necessity of invertibility

If $AB = AC$ and A is invertible, multiply both sides on the left by A^{-1} to get $B = C$. Without invertibility this need not hold. For example, $A = \operatorname{diag}(1, 0)$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, and $C = 0$ satisfy $AB = 0 = AC$ but $B \neq C$. Thus the hypothesis performs real work.

Exercise 2.2.14: Cancel on the correct side

Starting from $(B - C)D = 0$, multiply on the *right* by D^{-1}: $B - C = (B - C)(DD^{-1}) = 0D^{-1} = 0$. Therefore $B = C$. Left multiplication would not put D next to its inverse.

Exercise 2.2.15: What the augmented reduction computes

Let the row operations be represented by elementary matrices $E_1, \ldots, E_k$, applied in that order. Their combined multiplier is $E = E_k \cdots E_1$. Reducing $[A \mid B]$ to $[I \mid X]$ means $EA = I$ and $EB = X$.

Since A is invertible, $E = EAA^{-1} = A^{-1}$, whence $X = A^{-1}B$. The right block has been transformed along with A, avoiding a separate full inverse calculation.

Exercise 2.2.16: Recover an invertible factor from its product

Put $C = AB$, which is invertible by hypothesis. Right multiplication by B^{-1} gives $A = CB^{-1}$. This is a product of two matrices whose inverses are known: C^{-1} and B. Indeed BC^{-1} is an inverse of A, because

$$(CB^{-1})(BC^{-1}) = I, \qquad (BC^{-1})(CB^{-1}) = I.$$

Thus A is invertible without assuming the desired result in advance.

Exercise 2.2.17: The inverse of three factors

Choose $D = C^{-1}B^{-1}A^{-1}$. Associativity allows the adjacent cancellations

$$(ABC)D = AB(CC^{-1})B^{-1}A^{-1} = AA^{-1} = I,$$

$$D(ABC) = C^{-1}B^{-1}(A^{-1}A)BC = C^{-1}C = I.$$

Both sides have been checked, so D is an inverse of ABC.

Exercise 2.2.18: Solve without pretending that factors commute

From $AB = BC$, right-multiply by B^{-1}: $\boxed{A = BCB^{-1}}$. Substitution checks $(BCB^{-1})B = BC$. In general BCB^{-1} is not C; cancelling the two B factors across C would be invalid.

Exercise 2.2.19: An equation with two outside inverses

Multiply $C^{-1}(A+X)B^{-1} = I$ first on the left by C, then on the right by B. This gives $A+X = CB$, hence $\boxed{X = CB - A}$. Substitution gives $C^{-1}(CB)B^{-1} = I$, proving that a solution always exists under the stated hypotheses. The rearrangement also proves uniqueness.

Exercise 2.2.20: An unknown matrix appearing inside an inverse

(a) First prove that the required inverse exists.

From $(A - AX)^{-1} = X^{-1}B$, left multiplication by X gives $B = X(A - AX)^{-1}$. Both factors on the right are invertible by hypothesis, so B is invertible.

(b) Remove the inverse, then collect the unknown on one side.

The same equality gives $B(A - AX) = X$, or $BA - BAX = X$. Therefore

$$(I + BA)X = BA.$$

This also shows $I + BA = BAX^{-1}$, a product of invertible matrices, so the next inverse is justified. Left multiplication yields

$$\boxed{X = (I + BA)^{-1}BA}.$$

The original assumptions already require an admissible solution X; we used them to prove that every inverse in this expression exists.

Exercise 2.2.21: Invertibility implies independent columns

Suppose a combination of the columns is zero, equivalently $Ax = 0$. Multiplying by A^{-1} gives $x = 0$. Thus all coefficients in the combination are zero, which is exactly linear independence.

Exercise 2.2.22: Invertibility implies spanning

For every $b \in \mathbb{R}^n$, the vector $x = A^{-1}b$ satisfies $Ax = b$. But Ax is the linear combination of A's columns with coefficients from x. Thus every b is such a combination: the columns span $\mathbb{R}^n$.

Exercise 2.2.23: No homogeneous freedom means every column is a pivot

A nonpivot column corresponds to a free variable and would produce a nontrivial solution of $Ax = 0$. The hypothesis rules out that possibility, so all n columns are pivot columns. In an $n \times n$ matrix this means all rows have pivots too. Its reduced echelon form has a leading 1 in each column and zeros elsewhere in those columns, hence is I_n. The inverse algorithm then proves that A is invertible.

Exercise 2.2.24: Every right side means every row has a pivot

If a coefficient row lacked a pivot, some right side would produce a contradictory augmented row $[0 \ \cdots \ 0 \mid 1]$. Consistency for every right side therefore forces n pivot rows. With only n columns there are also n pivot columns, so the reduced echelon form is I_n. Again the inverse algorithm proves invertibility.

Exercise 2.2.25: A zero determinant produces a nonzero null vector

What we must show. Prove singularity directly when $ad - bc = 0$. If $(a, b) \neq (0, 0)$, use $x = (-b, a)^T \neq 0$. Multiplication gives $Ax = (-ab + ba, -bc + da)^T = 0$. If $a = b = 0$ but $(c, d) \neq (0, 0)$, use $x = (-d, c)^T \neq 0$; both rows again give zero. If all four entries vanish, any nonzero x, such as $(1, 0)^T$, works. Thus both $x = 0$ and a nonzero x solve $Ax = 0$. An inverse would imply $x = A^{-1}0 = 0$, contradicting the nonzero solution. Hence A is not invertible.

Exercise 2.2.26: Verify the two-by-two formula

Let $\Delta = ad - bc \neq 0$ and $B = \Delta^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Then

$$AB = \frac{1}{\Delta} \begin{bmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{bmatrix} = I_2,$$

$$BA = \frac{1}{\Delta} \begin{bmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{bmatrix} = I_2.$$

Division by Δ is legal by hypothesis. Both products equal the identity, so $B = A^{-1}$ by definition.

Exercise 2.2.27: Elementary matrices select and change rows

(a) Row i of I_3 has a 1 in position i and zeros elsewhere. Consequently $(\text{row}_i\, I_3)A$ is 1 times row i of A plus zero times the other rows: it equals $\text{row}_i\, A$.

(b) Set $E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The three rows of EA are, by part (a), rows $2, 1, 3$ of A. Thus it swaps rows 1 and 2, exactly as the same operation on I_3 does.

(c) Set $E = \text{diag}(1, 1, 5)$. Its rows select row 1 of A, row 2 of A, and five times row 3 of A. Thus EA performs the desired scaling, and E is obtained by scaling row 3 of I_3 by 5.

Exercise 2.2.28: An elementary row replacement and a printed typo

The intended elementary matrix is

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Its rows show that EA has rows $\text{row}_1\, A$, $-3\,\text{row}_1\, A + \text{row}_2\, A$, and $\text{row}_3\, A$, as required. To construct E from I, the replacement must be $\text{row}_2(I) \leftarrow \text{row}_2(I) - 3\,\text{row}_1(I)$. The supplied edition prints $\text{row}_1(A)$ in that last construction. Read literally that is incorrect: it would make E depend on A and need not be the elementary matrix for this operation. The computation above states and verifies the corrected version explicitly.

Exercise 2.2.29: Find an inverse by the augmented algorithm

Start with $[A \mid I]$ and perform the indicated operations:

$$\left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ 4 & -9 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ 0 & 3 & -4 & 1 \end{array}\right].$$

Divide row 2 by 3, then add three times the new row 2 to row 1. The result is $\left[\begin{array}{cc|cc} 1 & 0 & -3 & 1 \\ 0 & 1 & -4/3 & 1/3 \end{array}\right]$.

Thus $A^{-1} = \begin{bmatrix} -3 & 1 \\ -4/3 & 1/3 \end{bmatrix}$.

Exercise 2.2.30: A second augmented inverse calculation

For $[A \mid I]$, first use $R_1 \leftarrow R_1/3$, then $R_2 \leftarrow R_2 - 4R_1$:

$$\left[\begin{array}{cc|cc} 3 & 6 & 1 & 0 \\ 4 & 7 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 2 & 1/3 & 0 \\ 0 & -1 & -4/3 & 1 \end{array}\right].$$

Negate row 2 and replace row 1 by $R_1 - 2R_2$. This gives

$$[I \mid A^{-1}] = \left[\begin{array}{cc|cc} 1 & 0 & -7/3 & 2 \\ 0 & 1 & 4/3 & -1 \end{array}\right].$$

Both left-hand pivots exist, so the inverse is the displayed right block.

Exercise 2.2.31: A full three-by-three inverse

Eliminate below the first pivot of $[A \mid I]$ with $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - 2R_1$, then $R_3 \leftarrow R_3 + 3R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right].$$

Next divide row 3 by 2 and add twice this new row to each of rows 1 and 2. The coefficient block is now I_3, and the right block is

$$\boxed{A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ 7/2 & 3/2 & 1/2 \end{bmatrix}}.$$

For instance the first row $[1, 0, -2]$ of A times this block is $[8, 3, 1] - 2[7/2, 3/2, 1/2] = [1, 0, 0]$; the other two rows similarly check the identity. Exact multiplication verifies all nine entries.

Exercise 2.2.32: A failed inverse algorithm is useful information

Apply $R_2 \leftarrow R_2 + 4R_1$ and $R_3 \leftarrow R_3 + 2R_1$ to $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ -4 & -7 & 3 & 0 & 1 & 0 \\ -2 & -6 & 4 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 4 & 1 & 0 \\ 0 & -2 & 2 & 2 & 0 & 1 \end{array}\right].$$

Now $R_3 \leftarrow R_3 + 2R_2$ gives a zero coefficient row, with right block row $[10, 2, 1]$. The coefficient block has only two pivots and cannot become I_3. Therefore the inverse does not exist. Stopping here is a conclusion about singularity, not an unfinished calculation.

Exercise 2.2.33: Invert a cumulative-sum matrix

For the 3×3 lower-triangular matrix of ones, apply $R_3 \leftarrow R_3 - R_2$ first and then $R_2 \leftarrow R_2 - R_1$ to $[A \mid I]$. The order ensures that the subtracted rows are still the original rows. The left block becomes I and the right becomes

$$B_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

For the 4×4 case, work upward with $R_4 \leftarrow R_4 - R_3$, $R_3 \leftarrow R_3 - R_2$, $R_2 \leftarrow R_2 - R_1$:

$$B_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

In general B has 1 on the diagonal, -1 immediately below it, and zero elsewhere. For $j < n$, column j of AB is $Ae_j - Ae_{j+1}$. The two cumulative columns agree below row j, and differ by 1 exactly in row j, so this is e_j. The last column is $Ae_n = e_n$. Thus $AB = I_n$; the same upward row-reduction construction shows B is indeed the inverse for every n.

Exercise 2.2.34: A scaled cumulative-sum matrix

Let L be the lower-triangular ones matrix and $D = \operatorname{diag}(1, 2, \ldots, n)$. The given A is DL, because row i consists of i's up through its diagonal entry. Thus its inverse is $B = L^{-1}D^{-1}$. Explicitly,

$$b_{ii} = 1/i, \qquad b_{i,i-1} = -1/(i-1) \ (i \geq 2),$$

with every other entry zero. For example, $B_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/2 & 0 \\ 0 & -1/2 & 1/3 \end{bmatrix}$; B_4 appends a last row $(0, 0, -1/3, 1/4)$ to the corresponding four-column pattern. For $j < n$, column j of AB is $(Ae_j - Ae_{j+1})/j = e_j$, since the two columns differ only in row j, where their difference is j. The last column is $Ae_n/n = e_n$. Hence $AB = I_n$, verifying the formula.

Exercise 2.2.35: Find one inverse column without the others

The third column z of A^{-1} satisfies $Az = e_3$, since $AA^{-1} = I$. Its equations are

$$-z_1 - 7z_2 - 3z_3 = 0, \quad 2z_1 + 15z_2 + 6z_3 = 0, \quad z_1 + 3z_2 + 2z_3 = 1.$$

Add twice the first equation to the second: $z_2 = 0$. Add the first to the third: $-4z_2 - z_3 = 1$, so $z_3 = -1$. The first then gives $z_1 = 3$. Thus the requested column is $\boxed{(3, 0, -1)^T}$. Substitution verifies all three equations. The coefficient matrix has pivots in all three columns by this elimination, so the use of its inverse is justified.

Exercise 2.2.36: Two inverse columns in one augmented system

Solve $Az = e_2$ and $Aw = e_3$ together. No first inverse column is needed. Start with $[A \mid e_2 \ e_3]$. To avoid early fractions, multiply row 2 by 25 and add 536 times row 1; multiply row 3 by 25 and add 154 times row 1. These are compositions of legal elementary operations:

$$\left[\begin{array}{ccc|cc} -25 & -9 & -27 & 0 & 0 \\ 536 & 185 & 537 & 1 & 0 \\ 154 & 52 & 143 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|cc} -25 & -9 & -27 & 0 & 0 \\ 0 & -199 & -1047 & 25 & 0 \\ 0 & -86 & -583 & 0 & 25 \end{array}\right].$$

Replacing row 3 by $199R_3 - 86R_2$ gives its coefficient row $[0, 0, -25975]$ and right side $[-2150, 4975]$. Back-substitution gives

$$z = \frac{1}{1039}\begin{bmatrix} 117 \\ -583 \\ 86 \end{bmatrix}, \qquad w = \frac{1}{1039}\begin{bmatrix} -162 \\ 1047 \\ -199 \end{bmatrix}.$$

For example $z_3 = 2150/25975 = 86/1039$; the preceding row determines z_2, and the first row determines z_1. The same steps give w. All three triangular pivots are nonzero; exact substitution gives $Az = e_2$ and $Aw = e_3$ and verifies both requested columns.

Exercise 2.2.37: A rectangular left inverse need not be a right inverse

For the supplied 3×2 matrix, choose

$$C = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \end{bmatrix}.$$

All entries are allowed. In CA, its first row gives $[1,2] + [1,3] - [1,5] = [1,0]$; the second gives $-[1,2] + [1,3] = [0,1]$. Thus $CA = I_2$. In the other order,

$$AC = \begin{bmatrix} -1 & 3 & -1 \\ -2 & 4 & -1 \\ -4 & 6 & -1 \end{bmatrix} \neq I_3.$$

The unequal dimensions prevent a general reversal of the identity.

Exercise 2.2.38: A rectangular right inverse

The first and fourth columns of the given A are e_1 and e_2. Select them by taking

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Its entries are only $0, 1$, and $AD = [e_1\ e_2] = I_2$. There cannot be a 4×2 matrix C with $CA = I_4$. Indeed the nonzero vector $(0,1,1,0)^T$ satisfies $Ax = 0$. If $CA = I_4$, then $x = CAx = C0 = 0$, a contradiction.

Exercise 2.2.39: Convert forces into deflections

The model is $y = Df$: flexibility times force gives deflection. With $f = (40, 50, 30)^T$ pounds, the three entries are

$$y_1 = .011(40) + .003(50) + .001(30) = .62,$$

$$y_2 = .003(40) + .009(50) + .003(30) = .66, \qquad y_3 = .001(40) + .003(50) + .011(30) = .52.$$

Thus the deflections are $\boxed{(.62, .66, .52)^T}$ inches. Multiplying units, inches per pound times pounds gives inches.

Exercise 2.2.40: Stiffness and a prescribed third deflection

Row reduction, or exact rational arithmetic on the decimal entries, gives

$$D^{-1} = \begin{bmatrix} 100 & -100/3 & 0 \\ -100/3 & 400/3 & -100/3 \\ 0 & -100/3 & 100 \end{bmatrix}.$$

To check rather than trust the calculation, write $D = 10^{-3}\begin{bmatrix}11 & 3 & 1\\ 3 & 9 & 3\\ 1 & 3 & 11\end{bmatrix}$. Its first row times the displayed stiffness matrix gives $[1100-100, -1100/3+400-100/3, -100+100] = [1000, 0, 0]$; the other rows give $[0, 1000, 0]$ and $[0, 0, 1000]$. The factor 10^{-3} therefore makes the product I. For $y = .04e_3$, use only column 3 of D^{-1}:

$$f = D^{-1}y = .04(0, -100/3, 100)^T = (0, -4/3, 4)^T \text{ pounds.}$$

The negative second force points opposite to the chosen positive direction; it is needed to keep the second deflection at zero.

Exercise 2.2.41: Four measured deflections determine four forces

Let $S = 1000D$, an integer matrix, and use the measured y in centimeters. Then $Df = y$ is equivalent to

$$\begin{bmatrix}13 & 5 & 2 & 1\\ 5 & 10 & 4 & 2\\ 2 & 4 & 10 & 5\\ 1 & 2 & 5 & 13\end{bmatrix} f = \begin{bmatrix}70\\ 120\\ 160\\ 120\end{bmatrix}.$$

Exact row reduction produces

$$[I \mid f], \qquad \boxed{f = (20/21, 130/21, 80/7, 80/21)^T} \text{ newtons.}$$

For a transparent check with integers, write $f = (20, 130, 240, 80)^T/21$. Multiplying by S gives $(1470, 2520, 3360, 2520)^T/21 = (70, 120, 160, 120)^T$, which verifies all four equations. The inverse used in this computation is

$$D^{-1} = \frac{1000}{21}\begin{bmatrix}2 & -1 & 0 & 0\\ -1 & 3 & -1 & 0\\ 0 & -1 & 3 & -1\\ 0 & 0 & -1 & 2\end{bmatrix}.$$

Multiplication by the given D gives I_4, so the forces are unique. The form of this inverse makes the next requested calculation especially simple.

Exercise 2.2.42: Prescribe only the second deflection

A unit deflection e_2 requires $D^{-1}e_2$, precisely column 2 of the stiffness matrix. By linearity, a deflection $.22e_2$ requires .22 times that column. Using the inverse from Exercise 41,

$$f = .22\frac{1000}{21}(-1, 3, -1, 0)^T = \boxed{(-220/21, 220/7, -220/21, 0)^T} \text{ newtons.}$$

Multiplication by D gives $(0, .22, 0, 0)^T$ centimeters. The last force is zero, while two forces are negative: forcing a single deflection does not mean that only one applied force is needed.

2.3 Characterizations of invertible matrices

The Invertible Matrix Theorem (IMT) is a collection of equivalent tests for an $n \times n$ matrix. Invertibility, n pivots, independent columns, columns spanning $\mathbb{R}^n$, a trivial null space, and a one-to-one and onto map $x \mapsto Ax$ all describe the same situation. The word *square* is essential: a rectangular matrix can have independent columns without spanning its whole codomain.

For implications, distinguish a proof from a plausible example. A true implication needs an argument for *every* allowed matrix; a false implication needs just one counterexample with true premise and false conclusion. The zero matrix and identity matrix are useful tests, but their special properties should not be mistaken for universal properties.

Practice problem 2.3.P1: Repeated columns prevent invertibility

The three columns of the supplied matrix are $(2, 2, 2)^T$, $(3, 3, 3)^T$, and $(4, 4, 4)^T$. In particular column 2 is $3/2$ times column 1, so the columns are dependent. The IMT says an invertible square matrix has independent columns. Therefore this matrix is not invertible.

Practice problem 2.3.P2: Negate an existence statement correctly

Statement (g) says: for *every* $b \in \mathbb{R}^n$ there exists a solution to $Ax = b$. Its negation is: there exists *at least one* b for which $Ax = b$ is inconsistent. It does not say every system is inconsistent; indeed $Ax = 0$ always has $x = 0$ as a solution. The IMT also implies that this A is not invertible.

Practice problem 2.3.P3: Recognize a noninvertible product

The matrix AB is square. A nontrivial solution of $(AB)x = 0$ violates the trivial-null-space condition in the IMT, so AB is not invertible. We cannot conclude that *both* A and B are singular; one singular factor suffices to make a product singular. For instance $A = 0$, $B = I$ has exactly one singular factor.

Exercise 2.3.1: A two-by-two determinant test

For $A = \begin{bmatrix} 5 & 7 \\ -3 & -6 \end{bmatrix}$, $ad - bc = -30 + 21 = -9 \neq 0$. The two-by-two inverse criterion from Section 2.2 therefore gives **invertible**. No complete inverse calculation is needed.

Exercise 2.3.2: A multiple row

The second row $(6, -3)$ is $-3/2$ times the first row $(-4, 2)$. Eliminating it leaves only one pivot, fewer than two. Thus the matrix is **not invertible**. Equivalently the determinant is $12 - 12 = 0$.

Exercise 2.3.3: A triangular matrix with nonzero diagonal

The matrix is lower triangular with diagonal $3, -4, -3$. The first pivot 3 removes the entries below it; the second pivot -4 removes the next entry below it; the last pivot remains -3. None vanishes.

There is a pivot in every row and column, so the IMT gives **invertible**. The triangular structure saves a full reduction.

Exercise 2.3.4: A zero row

The second row is all zeros. Thus there can be at most two pivot rows in this 3×3 matrix. Since invertibility would require three pivots, the matrix is **not invertible**.

Exercise 2.3.5: A zero column

Column 2 is zero, so $Ae_2 = 0$ with $e_2 \neq 0$. This is a nontrivial null vector. The IMT therefore gives **not invertible**; equivalently the columns are dependent because one of them is the zero vector.

Exercise 2.3.6: A short elimination gives the last pivot

Use $R_3 \leftarrow R_3 + 3R_1$ to obtain the last row $(0, -3, -18)$, then $R_3 \leftarrow R_3 + (3/4)R_2$. The resulting echelon matrix is

$$\begin{bmatrix} 1 & -3 & -6 \\ 0 & 4 & 3 \\ 0 & 0 & -63/4 \end{bmatrix}.$$

Its three diagonal entries are nonzero pivots, so the original matrix is **invertible**. The operations preserve row equivalence.

Exercise 2.3.7: Four pivots from a few row replacements

Apply $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - 2R_1$, and then $R_4 \leftarrow R_4 - (1/4)R_2$. This gives

$$\begin{bmatrix} -1 & -3 & 0 & 1 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every diagonal position is a pivot. Hence the original 4×4 matrix is **invertible**. There is no need to clear the entries above the pivots.

Exercise 2.3.8: Read pivots from an upper triangular matrix

The given matrix is already in echelon form with nonzero diagonal entries $3, 1, 2, 1$. It has four pivots, so it is **invertible** by the IMT. The entries above the diagonal do not change this conclusion.

Exercise 2.3.9: A computer-assisted exact pivot test

Using rational row reduction on the supplied integer matrix gives an echelon matrix

$$\begin{bmatrix} 4 & 0 & -3 & -7 \\ 0 & 9 & 9/2 & -3/2 \\ 0 & 0 & 71/4 & 365/12 \\ 0 & 0 & 0 & -1336/213 \end{bmatrix}.$$

For reproducibility, elimination here uses $R_i \leftarrow R_i - (a_{ik}/a_{kk})R_k$ below each successive pivot. All four pivots are nonzero; therefore the matrix is **invertible**. An exact program check is `Matrix(A).rref()`, using SymPy, which returns I_4 and pivot columns $(0, 1, 2, 3)$ (zero-based indices). These exact calculations avoid confusing a very small pivot with zero.

Exercise 2.3.10: Five pivots despite a small determinant

For the supplied 5×5 integer matrix, exact elimination, including one row interchange when a candidate pivot is zero, gives

$$\begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 0 & 2/5 & 4/5 & -2/5 & -94/5 \\ 0 & 0 & 1 & -21 & 7 \\ 0 & 0 & 0 & 1 & 34 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

There are five nonzero pivots, hence the matrix is **invertible**. Its inverse can nevertheless be numerically sensitive; invertibility is an exact yes/no statement, while conditioning measures sensitivity. Exercise 43 quantifies that distinction. Exact SymPy row reduction returns I_5 with all five pivot columns.

Exercise 2.3.11: Five implications, with the square hypothesis retained

(a) **True.** A trivial null space means no free variables, so all n columns have pivots. For a square matrix that forces the reduced form to be I_n; this is an IMT equivalence.

(b) **True.** Spanning $\mathbb{R}^n$ forces n pivot rows. In an $n \times n$ matrix these are also n pivot columns, giving independence.

(c) **False.** Merely being $n \times n$ does not guarantee consistency for every right side. Take $A = 0$ and $b = e_1$; the equation $0x = e_1$ has no solution.

(d) **True.** A nontrivial homogeneous solution requires a free variable, so some column is not a pivot column. Thus there are fewer than n pivots.

(e) **True.** If A were invertible, then $(A^{-1})^T$ would be an inverse of A^T. The given noninvertibility of A^T excludes that possibility, so A is not invertible.

Exercise 2.3.12: Pay attention to onto versus into

(a) **True.** $AD = I$ is the right-inverse condition in the IMT, so A is invertible. Left multiplication by A^{-1} gives $D = A^{-1}$, and therefore $DA = I$.

(b) **False.** Every $n \times n$ matrix maps $\mathbb{R}^n$ *into* $\mathbb{R}^n$, including $A = 0$, whose reduced form is not I. Mapping *onto* would be a stronger premise and would imply the conclusion.

(c) **True.** For a square matrix, independent columns and columns spanning $\mathbb{R}^n$ are equivalent by the IMT.

(d) **False.** Consistency for every b implies invertibility, hence the transformation *is* one-to-one. The identity matrix is a concrete counterexample to the claimed negative conclusion.

(e) **False.** Consistency for just one right side says nothing about uniqueness. For $A = 0$, $b = 0$, every vector is a solution.

Exercise 2.3.13: When an upper triangular matrix is invertible

It is invertible *if and only if every diagonal entry is nonzero.* If all are nonzero, eliminate above the diagonal from the last row upward, then scale the rows; this reduces the matrix to I_n. For the converse, suppose diagonal entry k is zero. In an upper triangular matrix, each of the first $k-1$ columns has zero entries from row k downward. Column k also has zero entries from row k downward: its diagonal is zero, and its lower entries vanish by triangularity. Thus the first k columns live entirely in their first $k-1$ coordinates. A homogeneous system with k variables but only $k-1$ potentially nonzero equations has a free variable, so these k columns are dependent. Their dependence extends to all columns by assigning zero coefficients to the remaining columns. The IMT therefore excludes invertibility.

Exercise 2.3.14: The lower triangular case

Again the criterion is that every diagonal entry be nonzero. One way to avoid repeating the argument is to transpose: A is lower triangular exactly when A^T is upper triangular, with the same diagonal. The IMT says A is invertible exactly when A^T is invertible. Applying Exercise 13 to A^T proves both directions of the claimed criterion.

Exercise 2.3.15: Nonspanning columns exclude an inverse

No. The IMT says an invertible 4×4 matrix has columns spanning $\mathbb{R}^4$. The proposed failure of spanning is the negation of one of its equivalent conditions, so it rules out invertibility.

Exercise 2.3.16: Independence of the transposed columns

If A is invertible, then A^T is invertible, with inverse $(A^{-1})^T$. Applying the independent-column condition of the IMT to A^T shows that its columns are independent. Equivalently the rows of A are independent.

Exercise 2.3.17: Identical columns produce an explicit dependence

No. If columns i, j are equal with $i \neq j$, then $A(e_i - e_j) = a_i - a_j = 0$. The vector $e_i - e_j$ is nonzero, so the null space is nontrivial and A cannot be invertible.

Exercise 2.3.18: Identical rows also rule out invertibility

No. Equal rows of A become equal columns of A^T. Exercise 17 makes A^T singular, and the transpose equivalence in the IMT then makes A singular too.

Exercise 2.3.19: Seven independent columns give every right side exactly once

Since D is 7×7 and has independent columns, the IMT makes it invertible. For *every* $b \in \mathbb{R}^7$, the system $Dx = b$ therefore has the unique solution $x = D^{-1}b$. Both existence and uniqueness are part of the conclusion; neither needs a special choice of b.

Exercise 2.3.20: Universal consistency for a square system implies uniqueness

It is not possible to have more than one solution for any b. Universal consistency gives a pivot in every one of the five rows. Because there are five columns too, all five variables are pivot variables, leaving no free variable. The matrix is invertible, and every system has exactly one solution.

Exercise 2.3.21: Multiple solutions exclude spanning

If $Cu = v$ has two different solutions u_1, u_2, then $C(u_1 - u_2) = v - v = 0$ with $u_1 - u_2 \neq 0$. Thus C is singular by the IMT. Its columns therefore cannot span $\mathbb{R}^n$. Notice that the original right side v is still attained; the failure concerns some other right sides.

Exercise 2.3.22: A one-sided square inverse forces commutation

From $EF = I$, the IMT makes E invertible. Multiplying on the left by E^{-1} gives $F = E^{-1}$, so $FE = I$ as well. Hence $EF = FE$, which is exactly the definition of commuting matrices. This conclusion uses squareness; rectangular one-sided inverses need not reverse.

Exercise 2.3.23: A missed output implies a nontrivial null vector

If $Fx = y$ is inconsistent for some y, then F is not onto. By the IMT this square matrix is singular, so $Fx = 0$ has a nontrivial solution. In fact if $z \neq 0$ is such a solution, every tz, $t \in \mathbb{R}$, is a solution; thus the homogeneous system has infinitely many solutions.

Exercise 2.3.24: A failure to reduce to the identity

If G cannot be reduced to I_n, the IMT makes it singular. Consequently its columns are dependent and do not span $\mathbb{R}^n$. There are fewer than n pivots. These are equivalent descriptions of the same failure, not alternative cases to choose between.

Exercise 2.3.25: Verify the boxed fact about a one-sided inverse

Assume square matrices A, B satisfy $AB = I$. Then A has a right inverse, so the IMT implies that A is invertible. The equation gives $B = A^{-1}AB = A^{-1}$. Therefore B is invertible too, with inverse A,

and $BA = I$. This proves every assertion in the boxed statement preceding Example 1, rather than merely assuming a right inverse is automatically two-sided.

Exercise 2.3.26: Independent columns lead to a spanning square

Independence of the columns of square A makes A invertible. The product $A^2 = AA$ is then invertible, with inverse $A^{-1}A^{-1}$. Applying the spanning condition of the IMT to A^2 proves that its columns span $\mathbb{R}^n$. Equivalently, A maps onto $\mathbb{R}^n$ and doing so twice still reaches every vector.

Exercise 2.3.27: An invertible product forces the first factor invertible

Let $W = (AB)^{-1}$. Then $A(BW) = (AB)W = I$. Thus A has a square right inverse BW, and the IMT makes A invertible. This proof does *not* apply the inverse-of-a-product formula to unverified inverses of A, B; their existence is what must be proved.

Exercise 2.3.28: An invertible product forces the second factor invertible

Suppose $Bx = 0$. Multiplying by A gives $(AB)x = 0$. Invertibility of AB implies $x = 0$. Hence B has trivial null space, and since B is square the IMT proves it is invertible. Alternatively, $(WA)B = I$ with $W = (AB)^{-1}$ gives a left-inverse test.

Exercise 2.3.29: One-to-one endomorphisms are also onto

The one-to-one condition says $Ax = 0$ has only $x = 0$, or equivalently the columns are independent. For $n \times n$ matrices the IMT makes A invertible and hence onto $\mathbb{R}^n$. Thus the transformation is a bijection with inverse $x \mapsto A^{-1}x$. The equal dimensions are crucial.

Exercise 2.3.30: Failure of injectivity also forces failure of surjectivity

Two solutions of $Ax = b$ give a nonzero homogeneous solution by subtraction. Therefore A is singular. By the IMT the associated transformation is not onto $\mathbb{R}^n$ either: at least one right side has no preimage. It can still attain the particular b mentioned in the premise.

Exercise 2.3.31: Prove uniqueness directly from pivot counting

What we must show. Use row reduction, without Theorems 5 or 8. Consistency for all b forces a pivot in every row; otherwise an appropriate right side creates an inconsistent zero-coefficient row. The square matrix has n rows and n columns, so these n pivots use all n columns. There are no free variables. Thus, for any fixed b, back-substitution determines each variable uniquely. The hypothesis gives existence, and this pivot argument gives uniqueness, as requested.

Exercise 2.3.32: Prove existence directly from a trivial null space

What we must show. Avoid invoking the IMT. If a column were not a pivot column, its variable could be chosen freely in the homogeneous system, producing a nonzero solution. Thus all n columns

have pivots. The n pivots occupy different rows, and there are only n rows, so every row has a pivot. Hence for every augmented right side the system is consistent: no zero coefficient row can create a contradiction. This gives a solution of $Ax = b$ for every b directly.

Exercise 2.3.33: Invert a two-dimensional linear transformation

The standard matrix is $A = \begin{bmatrix} -5 & 9 \\ 4 & -7 \end{bmatrix}$, whose determinant is $35 - 36 = -1$. Thus $A^{-1} = \begin{bmatrix} 7 & 9 \\ 4 & 5 \end{bmatrix}$. For an output (y_1, y_2), its unique input is

$$\boxed{T^{-1}(y_1, y_2) = (7y_1 + 9y_2,\ 4y_1 + 5y_2)}.$$

Multiplying either pair of matrices gives I_2, which verifies both compositions $T^{-1} \circ T$ and $T \circ T^{-1}$ are the identity.

Exercise 2.3.34: A second inverse transformation

Now $A = \begin{bmatrix} 2 & -8 \\ -2 & 7 \end{bmatrix}$ has determinant $14 - 16 = -2$, so $A^{-1} = \begin{bmatrix} -7/2 & -4 \\ -1 & -1 \end{bmatrix}$. Consequently

$$\boxed{T^{-1}(y_1, y_2) = (-\tfrac{7}{2}y_1 - 4y_2,\ -y_1 - y_2)}.$$

For example applying T to this pair gives first coordinate $-7y_1 - 8y_2 + 8y_1 + 8y_2 = y_1$ and second coordinate $7y_1 + 8y_2 - 7y_1 - 7y_2 = y_2$. The reverse composition follows from the same inverse matrix calculation.

Exercise 2.3.35: Use both inverse identities directly

Let $S = T^{-1}$, so $S(T(x)) = x$ and $T(S(y)) = y$. If $T(u) = T(v)$, apply S to get $u = v$; thus T is one-to-one. For every desired output y, the input $S(y)$ satisfies $T(S(y)) = y$; thus T is onto. As a second proof, Theorem 9 identifies invertibility of T with invertibility of its square standard matrix, and the IMT then gives both injectivity and surjectivity.

Exercise 2.3.36: Two inputs sharing an output rule out onto

The equality $T(u) = T(v)$ for $u \neq v$ means T is not one-to-one. Its square standard matrix therefore fails one IMT condition, so it also fails the onto condition. Hence T cannot map $\mathbb{R}^n$ onto $\mathbb{R}^n$. One can also subtract to see $T(u - v) = 0$ with $u - v \neq 0$.

Exercise 2.3.37: A one-sided linear inverse in equal dimensions

Let A, B be the standard matrices of T, U. The hypothesis $T(U(x)) = x$ for all x implies $ABx = x$ for all x. Testing the standard basis vectors gives $AB = I$. The boxed result proved in Exercise 25 yields $BA = I$. Thus $U(T(x)) = BAx = x$ for every x. The conclusion depends on both linearity and the common finite dimension.

Exercise 2.3.38: Why the inverse function is unique

Since T is invertible, it is onto. Given arbitrary $v \in \mathbb{R}^n$, choose x with $T(x) = v$. Then

$$S(v) = S(T(x)) = x = U(T(x)) = U(v).$$

Every v was allowed, so $S = U$ as functions. We did not need to assume that S, U were linear; the surjectivity of T ensured their arguments covered the entire domain.

Exercise 2.3.39: An onto linear map in equal dimensions has an onto inverse

By the IMT applied to its standard $n \times n$ matrix, an onto T is invertible. For any x, taking $y = T(x)$ gives $T^{-1}(y) = x$; therefore T^{-1} is onto. If $T^{-1}(y_1) = T^{-1}(y_2)$, apply T to obtain $y_1 = y_2$; therefore it is one-to-one as well. So both T and its inverse are bijections of $\mathbb{R}^n$.

Exercise 2.3.40: Prove linearity of the inverse without assuming it

Let $x = S(u)$ and $y = S(v)$. The identity $T \circ S = \mathrm{id}$ gives $T(x) = u$, $T(y) = v$. By linearity of T, $T(x + y) = u + v$. Applying S and using $S \circ T = \mathrm{id}$ yields

$$S(u + v) = S(T(x + y)) = x + y = S(u) + S(v).$$

Likewise $T(cx) = cT(x) = cu$ gives $S(cu) = S(T(cx)) = cx = cS(u)$. These are additivity and homogeneity, so S is linear. Both inverse identities were used explicitly.

Exercise 2.3.41: Tiny data perturbations, large solution changes

(a) Solve both systems exactly.

For either right side (b_1, b_2), the inverse of the coefficient matrix is

$$A^{-1} = \frac{1}{4.5(1.1) - 3.1(1.6)} \begin{bmatrix} 1.1 & -3.1 \\ -1.6 & 4.5 \end{bmatrix} = \begin{bmatrix} -110 & 310 \\ 160 & -450 \end{bmatrix},$$

because $4.95 - 4.96 = -.01$. For the unrounded right side,

$$x_1 = -110(19.249) + 310(6.843) = 3.94, \qquad x_2 = 160(19.249) - 450(6.843) = .49.$$

For the rounded right side, the same products give $(2.90, 2.00)^T$. All these decimals are exact terminating decimals, not numerical approximations.

(b) Specify the meaning of percentage error.

Coordinatewise relative percentage errors are

$$100\frac{|2.90 - 3.94|}{3.94} = \frac{5200}{197}\% \approx 26.40\%, \qquad 100\frac{|2.00 - .49|}{.49} = \frac{15100}{49}\% \approx 308.16\%.$$

If a single vector error is wanted instead, the Euclidean relative error is $100\|\widetilde{x} - x\|_2/\|x\|_2 \approx 46.18\%$. Stating the norm matters: these are different legitimate summaries.

(c) Compute and interpret conditioning.

Executing `np.linalg.cond(A)` in NumPy gives the spectral (2-norm) condition number $\kappa_2(A) \approx 3362.9997$. The 1-norm and infinity-norm condition numbers both equal 4636; software conventions therefore need to be identified. The large value explains why very small changes of the measured right side can cause large changes in the recovered solution.

Exercise 2.3.42: Measure roundoff for the matrix in Exercise 9

What we must show. Compare a known vector with a solution recovered from its computed output. The following is an executed NumPy experiment using binary64 arithmetic:

```
import numpy as np
A = np.array([[4,0,-3,-7],[-6,9,9,9],
              [7,-5,10,19],[-1,2,4,-1]],dtype=float)
x = np.random.default_rng(23042).random(4)
b = A@x
x1 = np.linalg.solve(A,b)
print(np.linalg.cond(A))
print(np.max(np.abs(x1-x))/np.max(np.abs(x)))
print(np.abs((x1-x)/x))
```

The condition number is 11.3525089, and the normwise relative error $\|x_1 - x\|_\infty / \|x\|_\infty$ is approximately $4.52 \cdot 10^{-16}$. The chosen vector, rounded here for readability, is

$$x \approx (.98344656, .12402037, .75512517, .48561218)^T.$$

The worst coordinatewise relative error is about $2.13 \cdot 10^{-15}$; one coordinate was recovered exactly in this run.

Binary64 has 53 binary significant bits, about 15–16 decimal significant digits. Thus the observed normwise agreement is roughly 15 digits, with roughly 14–15 in the most sensitive coordinate. The expected worst-case loss suggested by $\log_{10} \kappa_2(A)$ is about 1.06 digits. The measured error is an observation for this vector, not a promise that every possible right side has identical accuracy. Libraries and hardware may alter the last few floating-point bits.

Exercise 2.3.43: Repeat with a more sensitive matrix

Use the same procedure but with the matrix in Exercise 10 and the seed below:

```
A = np.array([[5,3,1,7,9],[6,4,2,8,-8],
              [7,5,3,10,9],[9,6,4,-9,-5],
              [8,5,2,11,4]],dtype=float)
x = np.random.default_rng(23043).random(5)
b = A@x
x1 = np.linalg.solve(A,b)
print(np.linalg.cond(A))
print(np.max(np.abs(x1-x))/np.max(np.abs(x)))
```

The executed condition number is $\kappa_2(A) \approx 68622.4548$; the normwise relative error is approximately $6.01 \cdot 10^{-12}$. Thus this run retains about 11 decimal digits normwise, losing about 4–5 of the approximately 15–16 available digits. The theoretical sensitivity indicator $\log_{10} \kappa_2(A) \approx 4.84$ is consistent with that behavior. The exact pivot calculation in Exercise 10 proved invertibility, but it did not guarantee numerical robustness.

Exercise 2.3.44: The last inverse column of a fifth-order Hilbert matrix

To obtain column 5 of H^{-1} without the other columns, solve $Hx = e_5$, where $H_{ij} = 1/(i + j - 1)$ for $1 \le i, j \le 5$. An exact rational solve gives

$$\boxed{x = (630, -12600, 56700, -88200, 44100)^T}.$$

It can be verified by multiplying the five rows of H by this column; the results are $0, 0, 0, 0, 1$ respectively. For example the first is $630 - 6300 + 18900 - 22050 + 8820 = 0$. A numerical version is

```
H = np.array([[1/(i+j+1) for j in range(5)]
              for i in range(5)])
x = np.linalg.solve(H,np.eye(5)[:,4])
print(np.linalg.cond(H))
```

Here $\kappa_2(H) \approx 476607.25 \approx 10^{5.68}$. Starting with about 15–16 decimal digits, the condition estimate allows a loss of about six digits, so about 9–10 digits is a conservative expectation. Our executed solve has largest relative coordinate error about $1.93 \cdot 10^{-12}$ against the exact column, roughly 11–12 digits of agreement. A bound on possible loss need not be reached for this particular right side.

Exercise 2.3.45: A twelfth-order Hilbert inverse can be numerically unreliable

Use a numerical inverse, rather than an exact rational one, for this experiment:

```
H = np.array([[1/(i+j+1) for j in range(12)]
              for i in range(12)])
B = np.linalg.inv(H)
R = H@B-np.eye(12)
print(np.linalg.cond(H))
print(np.max(np.abs(R)))
print(np.linalg.norm(R,np.inf))
```

In the executed NumPy binary64 run, the outputs are approximately $1.64 \cdot 10^{16}$, 0.0892, and 0.320, respectively. Thus the largest entry of the computed product's error is nearly 0.09: HB is visibly different from the identity. The computed B_{11} is approximately 142.655, whereas the exact Hilbert inverse has $(H^{-1})_{11} = 12^2 = 144$.

The exact rational Hilbert matrix is invertible, but its condition number is comparable to the reciprocal of binary64 rounding precision. Ordinary floating-point arithmetic therefore has essentially no reliable significant digits guaranteed in an inverse computation of this size. Different numerical libraries may produce somewhat different inaccurate answers; none of that contradicts exact invertibility. Using higher precision and checking against exact rational arithmetic are ways to investigate further, not reasons to silently report the floating-point product as I.

2.4 Partitioned matrices

A block is itself a matrix. Block addition and multiplication follow the ordinary row–column rules, but every block product must have matching inner dimensions. Preserve the order of factors: AB cannot in general be changed to BA. An identity block has the size needed in its own row and column, so two symbols I in one display may have different sizes.

Practice problem 2.4.P1: Invert a block row-replacement matrix

Let $T = \begin{bmatrix} I & 0 \\ A & I \end{bmatrix}$. Try reversing the addition of A times the first block row to the second:

$$S = \begin{bmatrix} I & 0 \\ -A & I \end{bmatrix}.$$

Block multiplication gives $TS = \begin{bmatrix} I & 0 \\ A - A & I \end{bmatrix} = I$. In the reverse order ST has lower-left block $-A + A = 0$ and all other identity blocks. Thus $S = T^{-1}$.

Practice problem 2.4.P2: Transpose a horizontal partition before multiplying

For $X = [X_1 \; X_2]$, its transpose is vertical:

$$X^T X = \begin{bmatrix} X_1^T \\ X_2^T \end{bmatrix} [X_1 \; X_2] = \begin{bmatrix} X_1^T X_1 & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{bmatrix}.$$

The off-diagonal blocks do not disappear; they measure cross-products between the two groups of columns.

Exercise 2.4.1: Add a multiple of one block row

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} I & 0 \\ E & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ EA + C & EB + D \end{bmatrix}$$

The lower-left block is $EA + IC = EA + C$; the upper row is unchanged.

Exercise 2.4.2: Multiply by a block diagonal matrix

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} E & 0 \\ 0 & F \end{bmatrix} \begin{bmatrix} P & Q \\ R & S \end{bmatrix} = \begin{bmatrix} EP & EQ \\ FR & FS \end{bmatrix}$$

The first block row is multiplied by E and the second by F; the zero blocks remove the other terms.

Exercise 2.4.3: Interchange two block rows

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$$

For example the upper-left product is $0A + IC = C$. This is a block row interchange.

Exercise 2.4.4: Subtract a block multiple

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} I & 0 \\ -E & I \end{bmatrix} \begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} W & X \\ Y - EW & Z - EX \end{bmatrix}$$

The minus sign acts on both products in the lower block row. The order EW, rather than WE, comes from the row–column rule.

Exercise 2.4.5: Equate the four blocks before choosing inverses

Multiplication of the two given matrices gives

$$\begin{bmatrix} A + BX & BY \\ C & 0 \end{bmatrix} = \begin{bmatrix} 0 & I \\ Z & 0 \end{bmatrix}.$$

Thus the exact requirements are $BX = -A$, $BY = I$, and $Z = C$. If B is square, these requirements force B invertible, and

$$X = -B^{-1}A, \qquad Y = B^{-1}, \qquad Z = C.$$

Indeed $B(-B^{-1}A) = -A$ and $BB^{-1} = I$, checking both upper blocks. More generally, if B is rectangular and has a right inverse Y, the choice $X = -YA$ still works. Without such a right inverse, $BY = I$ is impossible, so no solution exists.

Exercise 2.4.6: Invert a block lower triangular matrix

The product is

$$\begin{bmatrix} XA & 0 \\ YA + ZB & ZC \end{bmatrix} = I.$$

Assume A and C are square and invertible. Then the diagonal blocks give $X = A^{-1}$ and $Z = C^{-1}$. The lower-left equation is $YA = -C^{-1}B$, so right multiplication by A^{-1} gives

$$Y = -C^{-1}BA^{-1}.$$

Inserting these formulas makes $YA + ZB = -C^{-1}B + C^{-1}B = 0$. For square diagonal blocks, their invertibility is also necessary because $XA = I$ and $ZC = I$ imply it.

Exercise 2.4.7: A zero middle block does not contribute

The block product is

$$\begin{bmatrix} XA & XZ \\ YA+B & YZ+I \end{bmatrix} = I.$$

With A square and invertible, $XA = I$ gives $X = A^{-1}$. Then $XZ = 0$ forces $Z = 0$, and $YA + B = 0$ gives $Y = -BA^{-1}$. The remaining equation is $YZ + I = I$, which is now satisfied.

Conclusion. $X = A^{-1}$, $Y = -BA^{-1}$, and $Z = 0$ under the stated square-invertible assumption.

Exercise 2.4.8: Match three block columns

Multiplication gives

$$\begin{bmatrix} AX & AY & AZ+B \\ 0 & 0 & I \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & 0 & I \end{bmatrix}.$$

If A is square and invertible, the first row requires

$$X = A^{-1}, \qquad Y = 0, \qquad Z = -A^{-1}B.$$

Each formula follows by multiplying its block equation on the left by A^{-1}. If A is rectangular with a right inverse X, one possible solution is instead $Y = 0, Z = -XB$; $AX = I$ checks it. An A without a right inverse cannot satisfy the first equation.

Exercise 2.4.9: Eliminate two blocks below a leading block

The second block row starts with $A_{21}B_{11} + B_{21}$. Setting it to zero and multiplying on the right by B_{11}^{-1} gives $A_{21} = -B_{21}B_{11}^{-1}$. Likewise $A_{31} = -B_{31}B_{11}^{-1}$. The second block in the new second row is consequently

$$C_{22} = A_{21}B_{12} + B_{22} = B_{22} - B_{21}B_{11}^{-1}B_{12}.$$

For completeness $C_{11} = B_{11}$, $C_{12} = B_{12}$, and $C_{32} = B_{32} - B_{31}B_{11}^{-1}B_{12}$. The middle inverse cannot be moved past its neighboring factors.

Exercise 2.4.10: Invert three block levels in order

Multiply the original matrix by the proposed inverse. Its lower blocks must satisfy

$$A + P = 0, \qquad D + R = 0, \qquad B + DP + Q = 0.$$

Thus $P = -A$, $R = -D$, and $Q = DA - B$. In the reverse product the lower-left block is $Q + RA + B = (DA - B) - DA + B = 0$, while the other off-diagonal blocks also vanish. Hence both products are the identity.

Conclusion. $P = -A$, $Q = DA - B$, and $R = -D$.

Exercise 2.4.11: Check conformability rather than only the number of blocks

(a) **True.** Corresponding blocks have equal sizes, so adding entries gives $[A_1 + B_1 \;\; A_2 + B_2]$.

(b) **False.** The drawn partitions do not specify the actual sizes. The number of columns of each A_{ij} must equal the number of rows of the corresponding B_j. For example, take all A_{ij} scalar but each B_j a 2×1 column. The displayed block pattern exists but multiplication is not conformable.

Exercise 2.4.12: Dimensions and transposes of partitions

(a) **False.** A has size $2n \times n$ and B has size $n \times 2n$. Both BA $(n \times n)$ and AB $(2n \times 2n)$ are defined. The blocks of AB are A_iB_j.

(b) **False.** Transposition interchanges block positions as well as transposing each block: $A^T = \begin{bmatrix} P^T & R^T \\ Q^T & S^T \end{bmatrix}$. The stated version puts the off-diagonal blocks in the wrong places.

Exercise 2.4.13: Invertibility of a block diagonal matrix

If B and C are invertible, block multiplication shows that $\operatorname{diag}(B^{-1}, C^{-1})$ is a two-sided inverse. Conversely suppose $\operatorname{diag}(B, C)$ is invertible. If $Bu = 0$, then the block vector $(u, 0)^T$ is in the kernel of the whole matrix; injectivity forces $u = 0$. Since B is square, the Invertible Matrix Theorem makes B invertible. The same argument with $(0, v)^T$ proves C invertible.

Conclusion. The equivalence holds in both directions.

Exercise 2.4.14: Invertibility of a block upper triangular matrix

Write $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ with square diagonal blocks. If they are invertible, the two-sided inverse is

$$\begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}.$$

The upper-right block of the product in the original order is $-A_{12}A_{22}^{-1} + A_{12}A_{22}^{-1} = 0$; in the other order it is $A_{11}^{-1}A_{12} - A_{11}^{-1}A_{12} = 0$. For necessity, invertibility of A first forces A_{11} invertible: a nonzero u with $A_{11}u = 0$ would give kernel vector $(u, 0)^T$. If $A_{22}v = 0$ with $v \neq 0$, put $u = -A_{11}^{-1}A_{12}v$. Then $A(u, v)^T = 0$, another contradiction. Thus A_{22} also has trivial kernel and is invertible.

Exercise 2.4.15: Update an accumulated outer-product sum

The column–row rule gives

$$G_k = X_kX_k^T = \sum_{j=1}^{k} x_jx_j^T.$$

Appending the new column yields

$$G_{k+1} = [X_k \;\; x_{k+1}] \begin{bmatrix} X_k^T \\ x_{k+1}^T \end{bmatrix} = G_k + x_{k+1}x_{k+1}^T.$$

Only one new outer product is needed, followed by entrywise addition into the stored G_k. If each x_j has m entries, the new product uses m^2 scalar products of entries, rather than recomputing $k+1$ outer products. The already processed vectors need not be revisited.

Exercise 2.4.16: Derive the Schur-complement factorization

Multiply the last two factors first:

$$\begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix} = \begin{bmatrix} A_{11} & A_{11}Y \\ 0 & S \end{bmatrix}.$$

Multiplying the left factor then gives $\begin{bmatrix} A_{11} & A_{11}Y \\ XA_{11} & XA_{11}Y + S \end{bmatrix}$. Choose $Y = A_{11}^{-1}A_{12}$ and $X = A_{21}A_{11}^{-1}$. The lower-right block becomes $A_{21}A_{11}^{-1}A_{12} + S = A_{22}$ by the definition of S. All four blocks now agree with the required matrix.

Exercise 2.4.17: Invertibility passes to the Schur complement

The outside factors from Exercise 2.4.16 are invertible: reverse X or Y in their off-diagonal block, just as in Practice Problem 1. If $A = L\,\mathrm{diag}(A_{11}, S)U$ is invertible, then $\mathrm{diag}(A_{11}, S) = L^{-1}AU^{-1}$ is invertible. Exercise 2.4.13 implies S is invertible. Inverting the factors in reverse order also gives the explicit formula

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{bmatrix}.$$

This expression follows by multiplying U^{-1}, the block diagonal inverse, and L^{-1}, in that order.

Exercise 2.4.18: Append one data column to a cross-product matrix

For $W = [X\ x_0]$, transposition and block multiplication give

$$W^TW = \begin{bmatrix} X^TX & X^Tx_0 \\ x_0^TX & x_0^Tx_0 \end{bmatrix}.$$

Its Schur complement of the invertible upper-left block is the scalar

$$x_0^Tx_0 - x_0^TX(X^TX)^{-1}X^Tx_0 = x_0^T[I - X(X^TX)^{-1}X^T]x_0 = x_0^TMx_0.$$

The asserted reciprocal entry requires this scalar to be nonzero; otherwise W^TW is singular, and an inverse entry is not defined. The question's requested identity itself holds regardless of whether the scalar vanishes.

Exercise 2.4.19: Eliminate the state vector to obtain the transfer function

Put $H = A - sI_n$, assumed invertible. The top block equation is $Hx + Bu = 0$, so $x = -H^{-1}Bu$. Insert this into the bottom equation $Cx + u = y$ to obtain

$$[I_m - CH^{-1}B]u = y.$$

Thus $W(s) = I_m - C(A - sI_n)^{-1}B$. It is exactly the Schur complement of $A - sI_n$ in the original block system matrix. Its size is $m \times m$, so it acts on the input vector $u \in \mathbb{R}^m$.

Exercise 2.4.20: The inverse transfer function is another Schur complement

The Schur complement of $K = A - BC - sI_n$ in the displayed matrix is

$$I_m - (-C)K^{-1}B = I_m + C(A - BC - sI_n)^{-1}B.$$

This is the requested $W(s)^{-1}$ when the stipulated inverses exist. To check the inverse claim, write $K = H - BC$, $H = A - sI_n$. From $K = H - BC$, multiply $KK^{-1}B = B$ on the left by H^{-1} and rearrange to get the identity

$$K^{-1}B = H^{-1}B + H^{-1}BCK^{-1}B.$$

Multiplying it by C shows $CK^{-1}B = CH^{-1}B + CH^{-1}BCK^{-1}B$. Expanding $(I - CH^{-1}B)(I + CK^{-1}B)$ now cancels these three terms and gives I, confirming the formula. The invertibility of H and W also forces K invertible: if $Kv = 0$, then $v = H^{-1}BCv$ and $W(Cv) = 0$; hence $Cv = 0$ and then $v = 0$. The square-matrix theorem completes this check.

Exercise 2.4.21: Verify an involution using blocks

(a) Multiply the 2×2 matrix.

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 2-2 & 1 \end{bmatrix} = I_2.$$

(b) Recognize the four-dimensional block structure.

For this same A, the given M is $\begin{bmatrix} A & 0 \\ I_2 & -A \end{bmatrix}$. Consequently

$$M^2 = \begin{bmatrix} A^2 & 0 \\ A - A & A^2 \end{bmatrix} = I_4.$$

The lower-left cancellation is essential; squaring only the diagonal blocks would not verify the entire product.

Exercise 2.4.22: Construct a six-dimensional involution

Choose $A = B = I_2$, $D = -I_2$, and $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq 0$. Then

$$M = \begin{bmatrix} I_2 & 0 & 0 \\ 0 & I_2 & 0 \\ C & 0 & -I_2 \end{bmatrix}.$$

It has size 6×6 as required. Its square has identity diagonal blocks and the only possibly nonzero off-diagonal block is $CA + DC = C - C = 0$. Thus $M^2 = I_6$. This construction works with any nonzero 2×2 choice for C.

Exercise 2.4.23: Induct on the dimension of lower triangular matrices

For dimension one the product of two scalars is lower triangular. Assume products of $k \times k$ lower triangular matrices are lower triangular. Partition two matrices of dimension $k+1$ as

$$A_1 = \begin{bmatrix} a & 0^T \\ v & A \end{bmatrix}, \qquad B_1 = \begin{bmatrix} b & 0^T \\ w & B \end{bmatrix}.$$

Their product is

$$A_1 B_1 = \begin{bmatrix} ab & 0^T \\ vb + Aw & AB \end{bmatrix}.$$

The top-right block is zero and AB is lower triangular by the induction hypothesis. These facts account for every entry above the main diagonal. Thus the product in dimension $k+1$ is lower triangular, completing the induction.

Exercise 2.4.24: Invert cumulative sums by successive differences

Let A_n be the lower triangular matrix of ones and B_n the matrix with ones on the diagonal and -1 immediately below it. For $n = 2$, direct multiplication of $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ in either order gives I_2. Suppose $A_k B_k = B_k A_k = I_k$. With $\mathbf{1}$ the k-entry ones vector and $e_1 = (1, 0, \ldots, 0)^T$, partition

$$A_{k+1} = \begin{bmatrix} 1 & 0^T \\ \mathbf{1} & A_k \end{bmatrix}, \quad B_{k+1} = \begin{bmatrix} 1 & 0^T \\ -e_1 & B_k \end{bmatrix}.$$

The lower-left block of $A_{k+1}B_{k+1}$ is $\mathbf{1} - A_k e_1 = 0$, since $A_k e_1$ is the first column, all ones. The lower-left block of the reverse product is $-e_1 + B_k \mathbf{1} = 0$: successive differences of ones are $(1, 0, \ldots, 0)^T$. The diagonal blocks are identities by induction and the upper-right blocks are zero. Both products are I_{k+1}, which proves the assertion for every $n \geq 2$.

Exercise 2.4.25: Invert the three diagonal blocks separately

The given matrix is block diagonal with blocks $B = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, $[2]$, and $C = \begin{bmatrix} 7 & 8 \\ 5 & 6 \end{bmatrix}$. The 2×2 formula gives

$$B^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}, \qquad C^{-1} = \begin{bmatrix} 3 & -4 \\ -5/2 & 7/2 \end{bmatrix}.$$

Thus, without row reduction,

$$A^{-1} = \begin{bmatrix} -5 & 2 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & -5/2 & 7/2 \end{bmatrix}.$$

Each product block is an identity, so the whole product is I_5.

Exercise 2.4.26: Concrete commands for extracting and inserting blocks

Using Python/NumPy's zero-based, end-exclusive slices:

(a) `A[4:10, 14:20]` displays rows 5–10 and columns 15–20. The result has size 6×6.

(b) `A[4:9, 9:19] = B` inserts the 5×10 block in rows 5–9 and columns 10–19. The first inserted entry is at row 5, column 10, as required.

(c) `scipy.linalg.block_diag(A, A.T)` constructs the desired 50×50 block diagonal matrix. Alternatively initialize a 50×50 zero array, assign A to `C[:20, :30]` and A^T to `C[20:, 30:]`. The off-diagonal blocks then already have the necessary zero entries.

These are the requested operations, not a request for unspecified data. Their output sizes follow from the slice endpoints and the given sizes.

Exercise 2.4.27: Work entirely with blocks below the size limit

Keep the large matrices on disk or as separately stored blocks; assembling a full 50×50 array would itself violate the limit.

(a) Partition both matrices into four 25×25 blocks. Compute each sum block $C_{ij} = A_{ij} + B_{ij}$ separately and store the four results as the sum's block representation.

(b) With that same partition compute $C_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j}$ for $i, j \in \{1, 2\}$. Every individual multiplication and addition is 25×25, well within the 32×32 bound.

(c) Use the stipulated $20 + 30$ partition, with $b = (b_1, b_2)^T$ and $x = (x_1, x_2)^T$. Since $A_{12} = 0$, first solve the 20×20 system $A_{11}x_1 = b_1$. Compute $r = b_2 - A_{21}x_1$ (A_{21} is 30×20), then solve $A_{22}x_2 = r$, a 30×30 system. Both diagonal matrices are invertible, so both solves have unique answers. Store x_1, x_2 separately as the solution; no explicit inverse or overlarge matrix is needed.

2.5 Matrix factorizations

An LU factorization writes $A = LU$, where L is unit lower triangular and U is in echelon form. "Unit" means the diagonal entries of L are 1. In elimination, if we replace row i by $R_i - mR_j$ to clear an entry below the current pivot, record m in the corresponding lower column of L. The same row replacements change A to U and L to I; reversing them proves $A = LU$. No row interchanges are needed for the matrices factored below.

To solve $Ax = b$, first solve $Ly = b$ from top to bottom, and then $Ux = y$ from bottom to top. Reuse L, U when the right side changes; do not compute an inverse merely to solve one system.

Practice problem 2.5.P1: A rectangular matrix with only three pivot columns

The given matrix has five rows and four columns. Its first-column multipliers are $3, 1, 2, -3$. After that elimination, the second-column multipliers are $-1, 2, -3$. Column 3 then has no new pivot; the next pivot is the entry 5 in column 4 of row 3. Its multipliers are $-1, 2$. The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 \\ 2 & 2 & -1 & 1 & 0 \\ -3 & -3 & 2 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The unused fourth and fifth columns of L are those of I_5. Multiplying these displayed factors gives the original 5×4 matrix exactly. In particular, L is 5×5, not 5×4; U carries the rectangular dimensions and its last two rows are zero.

Exercise 2.5.1: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= -7 \\ y_2 &= 5 - \big[(-1)(-7)\big] = -2 \\ y_3 &= 2 - \big[(2)(-7) + (-5)(-2)\big] = 6 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$\begin{aligned} x_3 &= \frac{6}{-1} = -6 \\ x_2 &= \frac{-2 - \big[(-1)(-6)\big]}{-2} = 4 \\ x_1 &= \frac{-7 - \big[(-7)(4) + (-2)(-6)\big]}{3} = 3 \end{aligned}$$

Also carry out ordinary elimination.

Apply these row replacements to the augmented matrix $[A \mid b]$:

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1 \\ R_3 &\leftarrow R_3 - (2)R_1 \\ R_3 &\leftarrow R_3 + (5)R_2 \end{aligned}$$

They produce $[U \mid y]$, with the same U, y displayed above. The backward substitution therefore gives the same vector as the factorization method. This explicitly performs the requested second method.

Conclusion. $x = \begin{bmatrix} 3 \\ 4 \\ -6 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.2: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 2 & -6 & 4 \\ -4 & 8 & 0 \\ 0 & -4 & 6 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -6 & 4 \\ 0 & -4 & 8 \\ 0 & 0 & -2 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= 2 \\ y_2 &= -4 - [(-2)(2)] = 0 \\ y_3 &= 6 - [(1)(0)] = 6 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$\begin{aligned} x_3 &= \frac{6}{-2} = -3 \\ x_2 &= \frac{0 - [(8)(-3)]}{-4} = -6 \\ x_1 &= \frac{2 - [(-6)(-6) + (4)(-3)]}{2} = -11 \end{aligned}$$

Also carry out ordinary elimination.

Apply these row replacements to the augmented matrix $[A \mid b]$:

$$\begin{aligned} R_2 &\leftarrow R_2 + (2)R_1 \\ R_3 &\leftarrow R_3 - R_2 \end{aligned}$$

They produce $[U \mid y]$, with the same U, y displayed above. The backward substitution therefore gives the same vector as the factorization method. This explicitly performs the requested second method.

Conclusion. $x = \begin{bmatrix} -11 \\ -6 \\ -3 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.3: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 2 & -4 & 2 \\ -4 & 5 & 2 \\ 6 & -9 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -4 & 2 \\ 0 & -3 & 6 \\ 0 & 0 & 1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= 6 \\ y_2 &= 0 - \big[(-2)(6)\big] = 12 \\ y_3 &= 6 - \big[(3)(6) + (-1)(12)\big] = 0 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$\begin{aligned} x_3 &= 0 = 0 \\ x_2 &= \frac{12 - \big[(6)(0)\big]}{-3} = -4 \\ x_1 &= \frac{6 - \big[(-4)(-4) + (2)(0)\big]}{2} = -5 \end{aligned}$$

Conclusion. $x = \begin{bmatrix} -5 \\ -4 \\ 0 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.4: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}, \qquad b = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & -5 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & -1 & 2 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= 0 \\ y_2 &= -5 - \big[(1)(0)\big] = -5 \\ y_3 &= 7 - \big[(3)(0) + (-5)(-5)\big] = -18 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$x_3 = \frac{-18}{-6} = 3$$
$$x_2 = \frac{-5 - [(-1)(3)]}{-2} = 1$$
$$x_1 = 0 - [(-1)(1) + (2)(3)] = -5$$

Conclusion. $x = \begin{bmatrix} -5 \\ 1 \\ 3 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.5: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 1 & -2 & -2 & -3 \\ 3 & -9 & 0 & -9 \\ -1 & 2 & 4 & 7 \\ -3 & -6 & 26 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 6 \\ 0 \\ 3 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -3 & 4 & -2 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & -2 & -2 & -3 \\ 0 & -3 & 6 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$y_1 = 1$$
$$y_2 = 6 - [(3)(1)] = 3$$
$$y_3 = 0 - [(-1)(1)] = 1$$
$$y_4 = 3 - [(-3)(1) + (4)(3) + (-2)(1)] = -4$$

Then solve $Ux = y$ by backward substitution.

$$x_4 = -4 = -4$$
$$x_3 = \frac{1 - [(4)(-4)]}{2} = \frac{17}{2}$$

$$x_2 = \frac{3 - [(6)(\frac{17}{2})]}{-3} = 16$$
$$x_1 = 1 - [(-2)(16) + (-2)(\frac{17}{2}) + (-3)(-4)] = 38$$

Conclusion. $x = \begin{bmatrix} 38 \\ 16 \\ \frac{17}{2} \\ -4 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.6: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 1 & 3 & 2 & 0 \\ -2 & -3 & -4 & 12 \\ 3 & 0 & 4 & -36 \\ -5 & -3 & -8 & 49 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 2 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 3 & -3 & 1 & 0 \\ -5 & 4 & -1 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 3 & 0 & 12 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$y_1 = 1$$
$$y_2 = -2 - [(-2)(1)] = 0$$
$$y_3 = -1 - [(3)(1) + (-3)(0)] = -4$$
$$y_4 = 2 - [(-5)(1) + (4)(0) + (-1)(-4)] = 3$$

Then solve $Ux = y$ by backward substitution.

$$x_4 = 3 = 3$$
$$x_3 = \frac{-4}{-2} = 2$$
$$x_2 = \frac{0 - [(12)(3)]}{3} = -12$$

$$x_1 = 1 - [(3)(-12) + (2)(2)] = 33$$

Conclusion. $x = \begin{bmatrix} 33 \\ -12 \\ 2 \\ 3 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.7: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$R_2 \leftarrow R_2 + (\frac{3}{2})R_1$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 \\ -\frac{3}{2} & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & 5 \\ 0 & \frac{7}{2} \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.8: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 6 & 4 \\ 12 & 5 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$R_2 \leftarrow R_2 - (2)R_1$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 6 & 4 \\ 0 & -3 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.9: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 3 & 1 & 2 \\ -9 & 0 & -4 \\ 9 & 9 & 14 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + (3)R_1 \\ R_3 &\leftarrow R_3 - (3)R_1 \\ R_3 &\leftarrow R_3 - (2)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.10: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} -5 & 0 & 4 \\ 10 & 2 & -5 \\ 10 & 10 & 16 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + (2)R_1 \\ R_3 &\leftarrow R_3 + (2)R_1 \\ R_3 &\leftarrow R_3 - (5)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 5 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} -5 & 0 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 9 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.11: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 3 & 7 & 2 \\ 6 & 19 & 4 \\ -3 & -2 & 3 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 - (2)R_1 \\ R_3 &\leftarrow R_3 + R_1 \\ R_3 &\leftarrow R_3 - R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & 7 & 2 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.12: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 13 & 9 \\ -6 & 5 & 4 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 - (2)R_1 \\ R_3 &\leftarrow R_3 + (3)R_1 \\ R_3 &\leftarrow R_3 - (2)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 2 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & 3 & 2 \\ 0 & 7 & 5 \\ 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible.

Exercise 2.5.13: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 1 & 3 & -5 & -3 \\ -1 & -5 & 8 & 4 \\ 4 & 2 & -5 & -7 \\ -2 & -4 & 7 & 5 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1 \\ R_3 &\leftarrow R_3 - (4)R_1 \\ R_4 &\leftarrow R_4 + (2)R_1 \\ R_3 &\leftarrow R_3 - (5)R_2 \\ R_4 &\leftarrow R_4 + R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 4 & 5 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & 3 & -5 & -3 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible.

Exercise 2.5.14: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 1 & 3 & 1 & 5 \\ 5 & 20 & 6 & 31 \\ -2 & -1 & -1 & -4 \\ -1 & 7 & 1 & 7 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 - (5)R_1 \\ R_3 &\leftarrow R_3 + (2)R_1 \\ R_4 &\leftarrow R_4 + R_1 \\ R_3 &\leftarrow R_3 - R_2 \\ R_4 &\leftarrow R_4 - (2)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 5 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & 3 & 1 & 5 \\ 0 & 5 & 1 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible.

Exercise 2.5.15: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & 0 & 5 & 2 \\ -6 & 3 & -13 & -3 \\ 4 & 9 & 16 & 17 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$R_2 \leftarrow R_2 + (3)R_1$$
$$R_3 \leftarrow R_3 - (2)R_1$$
$$R_3 \leftarrow R_3 - (3)R_2$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & 0 & 5 & 2 \\ 0 & 3 & 2 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. Because A is rectangular, L is 3×3 and U has the same 3×4 size as A.

Exercise 2.5.16: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & -3 & 4 \\ -4 & 8 & -7 \\ 6 & -5 & 14 \\ -6 & 9 & -12 \\ 8 & -6 & 19 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + (2)R_1 \\ R_3 &\leftarrow R_3 - (3)R_1 \\ R_4 &\leftarrow R_4 + (3)R_1 \\ R_5 &\leftarrow R_5 - (4)R_1 \\ R_3 &\leftarrow R_3 - (2)R_2 \\ R_5 &\leftarrow R_5 - (3)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ -3 & 0 & 0 & 1 & 0 \\ 4 & 3 & 0 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -3 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible. Because A is rectangular, L is 5×5 and U has the same 5×3 size as A.

Exercise 2.5.17: Invert the two triangular factors in reverse order

Use the factors from Exercise 2.5.2. Solving $Lz = e_j$ column by column (forward substitution), and $Uw = e_j$ column by column (backward substitution), gives

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & -1 & 1 \end{bmatrix}, \qquad U^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{4} & -2 \\ 0 & -\frac{1}{4} & -1 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix}.$$

This is the augmented-inverse algorithm: reduce $[L \mid I]$ and $[U \mid I]$ to $[I \mid L^{-1}]$ and $[I \mid U^{-1}]$ using their triangular equations. Each column is checked by multiplying it by its original factor and obtaining the corresponding coordinate vector. Since $(LU)^{-1} = U^{-1}L^{-1}$, the required inverse is

$$A^{-1} = U^{-1}L^{-1} = \begin{bmatrix} 3 & \frac{5}{4} & -2 \\ \frac{3}{2} & \frac{3}{4} & -1 \\ 1 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

Both AA^{-1} and $A^{-1}A$ equal the identity. The order of the two inverse factors matters.

Exercise 2.5.18: Invert the two triangular factors in reverse order

Use the factors from Exercise 2.5.3. Solving $Lz = e_j$ column by column (forward substitution), and $Uw = e_j$ column by column (backward substitution), gives

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, \qquad U^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{2}{3} & 3 \\ 0 & -\frac{1}{3} & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

This is the augmented-inverse algorithm: reduce $[L \mid I]$ and $[U \mid I]$ to $[I \mid L^{-1}]$ and $[I \mid U^{-1}]$ using their triangular equations. Each column is checked by multiplying it by its original factor and obtaining the corresponding coordinate vector. Since $(LU)^{-1} = U^{-1}L^{-1}$, the required inverse is

$$A^{-1} = U^{-1}L^{-1} = \begin{bmatrix} -\frac{23}{6} & \frac{7}{3} & 3 \\ -\frac{8}{3} & \frac{5}{3} & 2 \\ -1 & 1 & 1 \end{bmatrix}.$$

Both AA^{-1} and $A^{-1}A$ equal the identity. The order of the two inverse factors matters.

Exercise 2.5.19: Invert a triangular matrix without losing triangular form

Every diagonal entry is nonzero, so forward substitution solves $Ax = b$ uniquely for every b. Hence A is invertible by the Invertible Matrix Theorem. To follow the suggested row-operation argument, clear entries below the successive diagonal pivots using replacements by earlier rows, then scale the diagonal pivots to 1. No swaps or additions of later rows into earlier rows are needed. Applying these same operations to I keeps it lower triangular: an earlier lower triangular row has zeros in every column beyond its own index, hence also beyond the index of the later row it is added to. Scaling preserves these zeros. Thus reduction of $[A \mid I]$ to $[I \mid A^{-1}]$ leaves the right block lower triangular. This proves the requested structure as well as invertibility.

Exercise 2.5.20: Reduce the unit lower factor to the identity

Because L is unit lower triangular, its diagonal entries are already 1. Clear below each diagonal using only replacements of a later row by itself minus a multiple of an earlier row. Let E be the product of these elementary row-operation matrices. Then $EL = I$. Applying the same replacements to $A = LU$ gives

$$EA = ELU = IU = U.$$

Thus only row replacements are needed, even if A or U is rectangular or singular.

Exercise 2.5.21: One sequence of row operations acts on every factorization

Let E be the product of all elementary matrices for the stated operations. Since they reduce B to I, $EB = I$; therefore

$$EA = E(BC) = (EB)C = IC = C.$$

This proves the claim for any such sequence, not merely a particular algorithm. The converse can fail: take $A = 0$, any invertible $B \neq I$, and $C = 0$. The empty sequence already leaves $A = C$, but does not change B into I.

Exercise 2.5.22: Discard the zero rows and their unused multiplier columns

For the Practice Problem factors, take the first three columns of L and the first three rows of U:

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 2 & -1 \\ -3 & -3 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

They have the required sizes 5×3 and 3×4, and their product is A. To see why, write $L = [L_1 \; L_2]$ and $U = \begin{bmatrix} U_1 \\ 0 \end{bmatrix}$. Block multiplication gives $LU = L_1U_1 + L_2(0) = L_1U_1$. The same argument works for any $m \times n$ matrix whose U has exactly three nonzero rows: use an $m \times 3$ factor and a $3 \times n$ factor. Those rows are first because U is echelon.

Exercise 2.5.23: A compact representation by four outer products

(a) Write $C = [c_1 \; c_2 \; c_3 \; c_4]$ and let the rows of D be $d_1^T, \ldots, d_4^T$. The column–row rule gives $A = c_1d_1^T + c_2d_2^T + c_3d_3^T + c_4d_4^T$. Each term is an $m \times n$ outer product.

(b) A 400×100 array stores 40,000 numbers. The factors require $400(4) + 4(100) = 2{,}000$ numbers, a factor of 20 less storage, ignoring small bookkeeping costs. For a product Ax, first compute Dx (four entries) and then $C(Dx)$; storing the full matrix is unnecessary for that use.

Exercise 2.5.24: Solve a QR-factored system by an orthogonal multiplication

From $Q^TQ = I$ and the fact that Q is square, Q is invertible with inverse Q^T. Thus $QRx = b$ is equivalent to $Rx = Q^Tb$. Compute $c = Q^Tb$, then solve the upper triangular system $Rx = c$ by backward substitution. Invertibility of R guarantees a nonzero diagonal and a unique solution.

Conclusion. For every b, this process yields the unique x. It requires no explicit inverse of R.

Exercise 2.5.25: Invert a singular-value factorization with positive diagonal

The square matrices U, V are invertible because $U^TU = V^TV = I$; their inverses are U^T, V^T. Each $\sigma_j > 0$, so $D^{-1} = \operatorname{diag}(1/\sigma_1, \ldots, 1/\sigma_n)$ exists. Reverse the factor order:

$$A^{-1} = (V^T)^{-1}D^{-1}U^{-1} = VD^{-1}U^T.$$

The product check is $AA^{-1} = UDV^TVD^{-1}U^T = UU^T = I$, and the reverse order similarly gives $VV^T = I$. Strict positivity is stronger than the nonzero condition needed for the inverse, but supplies it here.

Exercise 2.5.26: Cancel neighboring change-of-basis factors in powers

Associativity and $P^{-1}P = I$ give

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1},$$

$$A^3 = PD^3P^{-1}, \qquad A^k = PD^kP^{-1}.$$

The general formula follows inductively by multiplying the formula for A^k by PDP^{-1}. Since $D = \operatorname{diag}(2, 3, 1)$,

$$A^2 = P\operatorname{diag}(4, 9, 1)P^{-1}, \quad A^3 = P\operatorname{diag}(8, 27, 1)P^{-1},$$
$$A^k = P\operatorname{diag}(2^k, 3^k, 1)P^{-1}.$$

Only scalar powers of three diagonal entries need repeated powering.

Circuit convention for the next four exercises. A series resistor of resistance R has transfer matrix $S(R) = \begin{bmatrix} 1 & -R \\ 0 & 1 \end{bmatrix}$; a shunt resistor has $T(R) = \begin{bmatrix} 1 & 0 \\ -1/R & 1 \end{bmatrix}$. The state is $(v, i)^T$ and is mapped from input to output. If the signal passes through circuit 1 and then circuit 2, the total matrix is A_2A_1, so physical order is reversed in the matrix product.

Exercise 2.5.27: Two different networks with the specified input and output

First design: a series resistor followed by a shunt.

A series resistance $R = 1/2$ changes $(12, 6)^T$ to $(12 - (1/2)6, 6)^T = (9, 6)^T$. A shunt resistance $9/2$ then changes this to $(9, 6 - 9/(9/2))^T = (9, 4)^T$. Its matrix is $T(9/2)S(1/2)$.

Second design: a shunt followed by a series resistor.

A shunt resistance 6 changes the input to $(12, 6 - 12/6)^T = (12, 4)^T$. A series resistance $3/4$ then gives $(12 - (3/4)4, 4)^T = (9, 4)^T$. Its matrix is $S(3/4)T(6)$. All four resistances are positive and both calculations verify the required output, despite the different physical orders and resistance values.

Exercise 2.5.28: Add conductances for three successive shunt circuits

Multiplying two shunt matrices adds their lower-left entries. Thus three of them give

$$T(R_3)T(R_2)T(R_1) = \begin{bmatrix} 1 & 0 \\ -(1/R_1 + 1/R_2 + 1/R_3) & 1 \end{bmatrix}.$$

This is $T(R)$ with

$$R = \frac{1}{1/R_1 + 1/R_2 + 1/R_3}.$$

The positive-resistance assumption makes the denominator positive. One may also write $R = R_1R_2R_3/(R_1R_2 + R_1R_3 + R_2R_3)$.

Exercise 2.5.29: Factor the given series–shunt–series network

(a) Multiply in the physical order's reverse.

The signal passes through R_1 in series, R_2 as a shunt, and R_3 in series. Hence

$$S(R_3)T(R_2)S(R_1) = \begin{bmatrix} 1 + R_3/R_2 & -R_1 - R_3 - R_1R_3/R_2 \\ -1/R_2 & 1 + R_1/R_2 \end{bmatrix}.$$

For example, multiplying the right two factors first gives $\begin{bmatrix} 1 & -R_1 \\ -1/R_2 & 1 + R_1/R_2 \end{bmatrix}$; the final left multiplication gives the displayed first row.

(b) Match entries of the target matrix.

The lower-left entry $-1/3$ forces $R_2 = 3$. The lower-right entry $5/3$ then gives $R_1 = 2$. The upper-left entry 3 gives $R_3 = 6$. The remaining entry checks: $-2 - 6 - 2(6)/3 = -12$.

Conclusion. Use a 2-ohm series resistor, a 3-ohm shunt, then a 6-ohm series resistor. Their product is exactly the specified matrix.

Exercise 2.5.30: A shunt–series–shunt realization of the same matrix

Try a shunt of resistance a, then a series resistance b, then a shunt of resistance c. Multiplying gives

$$T(c)S(b)T(a) = \begin{bmatrix} 1 + b/a & -b \\ -1/a - (1 + b/a)/c & 1 + b/c \end{bmatrix}.$$

Match the target's upper-right entry to obtain $b = 12$. Then $1 + 12/a = 3$ gives $a = 6$, and $1 + 12/c = 5/3$ gives $c = 18$. The lower-left entry checks as $-1/6 - 3/18 = -1/3$.

Conclusion. The distinct physical design is a 6-ohm shunt, a 12-ohm series resistor, and an 18-ohm shunt, with exactly the same overall transfer matrix as in Exercise 2.5.29.

Exercise 2.5.31: Factor and solve the supplied eight-node heat plate

All omitted entries in the printed matrix are zero; the actual coefficient matrix used here is

$$A = \begin{bmatrix} 4 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 4 & -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 4 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 4 & -1 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 4 \end{bmatrix}.$$

(a) Perform elimination without row swaps.

The successive pivots are the diagonal entries of the following U. At pivot row j, subtract l_{ij} times that row from row i; the multipliers l_{ij} are precisely the displayed entries of L.

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & -\frac{1}{15} & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4}{15} & -\frac{2}{7} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{15}{56} & -\frac{1}{12} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{7}{24} & -\frac{26}{89} & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{24}{89} & -\frac{208}{2415} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{712}{2415} & -\frac{2623}{8948} & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 4 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{15}{4} & -\frac{1}{4} & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{56}{15} & -\frac{16}{15} & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{24}{7} & -\frac{2}{7} & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{89}{24} & -\frac{13}{12} & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2415}{712} & -\frac{26}{89} & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{8948}{2415} & -\frac{2623}{2415} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{30305}{8948} \end{bmatrix}.$$

Both factors have zeros more than two positions below or above their main diagonals. Exact multiplication gives $LU - A = 0_{8\times 8}$, checking every entry rather than just the diagonals.

(b) Solve using the factors.

For $b = (5, 15, 0, 10, 0, 10, 20, 30)^T$, forward substitution gives

$$y = \begin{bmatrix} 5 \\ \frac{65}{4} \\ \frac{7}{3} \\ 15 \\ \frac{15}{8} \\ \frac{10625}{712} \\ \frac{10525}{483} \\ \frac{364965}{8948} \end{bmatrix}.$$

The rule is $y_i = b_i - \sum_{j<i} l_{ij} y_j$, starting at row 1. Backward substitution uses $x_i = (y_i - \sum_{j>i} u_{ij} x_j)/u_{ii}$, starting at row 8. It gives

$$x = \frac{1}{209} \begin{bmatrix} 827 \\ 1377 \\ 886 \\ 1546 \\ 1171 \\ 1831 \\ 1967 \\ 2517 \end{bmatrix}.$$

Substitution verifies $Ax = b$ exactly. For instance row 1 gives $(4(827) - 1377 - 886)/209 = 1045/209 = 5$.

(c) Compare the inverse with the banded factors.

The exact inverse is

$$A^{-1} = \frac{1}{30305} \begin{bmatrix} 8948 & 2623 & 2864 & 1544 & 964 & 689 & 303 & 248 \\ 2623 & 8948 & 1544 & 2864 & 689 & 964 & 248 & 303 \\ 2864 & 1544 & 9912 & 3312 & 3167 & 1792 & 964 & 689 \\ 1544 & 2864 & 3312 & 9912 & 1792 & 3167 & 689 & 964 \\ 964 & 689 & 3167 & 1792 & 9912 & 3312 & 2864 & 1544 \\ 689 & 964 & 1792 & 3167 & 3312 & 9912 & 1544 & 2864 \\ 303 & 248 & 964 & 689 & 2864 & 1544 & 8948 & 2623 \\ 248 & 303 & 689 & 964 & 1544 & 2864 & 2623 & 8948 \end{bmatrix}.$$

Every entry is nonzero. Unlike L, U, it has no empty outer bands. For an $n \times n$ matrix with fixed band width, storing its factors requires a number of entries proportional to n, whereas a dense inverse

requires n^2. This example verifies the stated storage advantage with actual factors and an actual inverse.

Exercise 2.5.32: Reuse one bidiagonal factorization for four time steps

(a) Set $C = 1$ before factoring.

Then A has diagonal 3 and adjacent off-diagonal entries -1. Its factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 & 0 \\ 0 & -\frac{3}{8} & 1 & 0 \\ 0 & 0 & -\frac{8}{21} & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & -1 & 0 & 0 \\ 0 & \frac{8}{3} & -1 & 0 \\ 0 & 0 & \frac{21}{8} & -1 \\ 0 & 0 & 0 & \frac{55}{21} \end{bmatrix}.$$

Their off-diagonal nonzero entries lie on just one neighboring diagonal, so both factors are bidiagonal. Multiplication checks $LU = A$.

(b) Solve the same two triangular systems at each step.

Starting with $t_0 = (10, 15, 15, 10)^T$, repeatedly solve $Ly = t_k$ and $Ut_{k+1} = y$. The results are

k	$(t_k)_1$	$(t_k)_2$	$(t_k)_3$	$(t_k)_4$
1	7	11	11	7
2	5	8	8	5
3	18/5	29/5	29/5	18/5
4	13/5	21/5	21/5	13/5

.

A short independent check uses symmetry. For a vector $(a, b, b, a)^T$, $A(a, b, b, a)^T = (3a-b, 2b-a, 2b-a, 3a-b)^T$. Thus if (a_0, b_0) is the preceding pair, the next pair is $a = (2a_0 + b_0)/5$, $b = (a_0 + 3b_0)/5$. These formulas reproduce all four rows and verify $At_{k+1} = t_k$ at each step. The factorization is computed once, not recomputed for each new temperature distribution.

2.6 The Leontief input–output model

The production vector x must supply both intermediate consumption Cx and final demand d. Thus

$$x = Cx + d, \qquad (I - C)x = d.$$

Column j of C lists the amounts each sector supplies as inputs per unit produced in sector j. These columns need not sum to one; the remaining part includes value added. This is a different model from the closed exchange economy of Section 1.6.

The vector d is the external demand, not the total required production. When $I - C$ is invertible, $x = (I - C)^{-1}d$. For a single right side, solving $(I - C)x = d$ by elimination is often more efficient than computing an inverse. Respect the instructions against forming inverses in Exercises 2 and 3 below.

Practice problem 2.6.P1: Write each unit-consumption vector as a column

The Goods column is $(.2, .5)^T$ and the Services column is $(.4, .3)^T$. Hence

$$x = \begin{bmatrix} .2 & .4 \\ .5 & .3 \end{bmatrix} x + \begin{bmatrix} 20 \\ 30 \end{bmatrix}, \quad \text{or} \quad \begin{bmatrix} .8 & -.4 \\ -.5 & .7 \end{bmatrix} x = \begin{bmatrix} 20 \\ 30 \end{bmatrix}.$$

The first row accounts for all Goods required, and the second accounts for all Services required. This sets up the requested model.

Exercise 2.6.1: Construct the three-sector consumption matrix

Use Manufacturing, Agriculture, Services as the order of rows and columns. The input columns described in the exercise give

$$C = \begin{bmatrix} .10 & .60 & .60 \\ .30 & .20 & 0 \\ .30 & .10 & .10 \end{bmatrix}.$$

For example producing one agricultural unit requires .60 unit from Manufacturing, .20 from Agriculture, and .10 from Services. If Agriculture produces 100 units, its intermediate-demand vector is 100 times that column:

$$(60, 20, 10)^T.$$

These are the inputs needed for that production plan, not a final-demand vector or a complete equilibrium production plan for all sectors.

Exercise 2.6.2: Meet agricultural final demand without computing an inverse

The final-demand vector is $(0, 20, 0)^T$. Multiplying the equations $(I - C)x = d$ by 10 gives

$$\left[\begin{array}{rrr|r} 9 & -6 & -6 & 0 \\ -3 & 8 & 0 & 200 \\ -3 & -1 & 9 & 0 \end{array}\right].$$

Swap rows 1 and 2. Then use $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - R_1$, and $R_3 \leftarrow 2R_3 + R_2$. The result is

$$\left[\begin{array}{ccc|c} -3 & 8 & 0 & 200 \\ 0 & 18 & -6 & 600 \\ 0 & 0 & 12 & 200 \end{array}\right].$$

Back substitution gives $x_S = 50/3$, $x_A = (600+6(50/3))/18 = 350/9$, and $x_M = (8(350/9)-200)/3 = 1000/27$.

Conclusion. $x = (1000/27, 350/9, 50/3)^T$. Substitution into the original model gives final demand $(0, 20, 0)^T$ exactly.

Exercise 2.6.3: Meet manufacturing final demand by the same elimination

Now $d = (20, 0, 0)^T$, so the initial augmented right side after multiplying by 10 is $(200, 0, 0)^T$. Use the same row swap and replacements listed in Exercise 2.6.2. They produce

$$\left[\begin{array}{ccc|c} -3 & 8 & 0 & 0 \\ 0 & 18 & -6 & 200 \\ 0 & 0 & 12 & 200 \end{array}\right].$$

Thus $x_S = 50/3$, $x_A = (200 + 100)/18 = 50/3$, and $x_M = 8(50/3)/3 = 400/9$.

Conclusion. $x = (400/9, 50/3, 50/3)^T$. No inverse matrix was used.

Exercise 2.6.4: Add the production responses to two demand vectors

The new demand $(20, 20, 0)^T$ is the sum of the demands in the two previous exercises. If $(I-C)x^{(a)} = d^{(a)}$ and $(I-C)x^{(b)} = d^{(b)}$, additivity gives $(I-C)(x^{(a)}+x^{(b)}) = d^{(a)}+d^{(b)}$. Add the actual production vectors:

$$x = \begin{bmatrix} 1000/27 \\ 350/9 \\ 50/3 \end{bmatrix} + \begin{bmatrix} 400/9 \\ 50/3 \\ 50/3 \end{bmatrix} = \begin{bmatrix} 2200/27 \\ 500/9 \\ 100/3 \end{bmatrix}.$$

The elimination pivots in Exercises 2 and 3 are nonzero, so the coefficient matrix is invertible and this solution is unique.

Exercise 2.6.5: Use the inverse requested in the problem

Here

$$I - C = \begin{bmatrix} 1 & -1/2 \\ -3/5 & 4/5 \end{bmatrix}, \qquad \det(I - C) = \frac{4}{5} - \frac{3}{10} = \frac{1}{2}.$$

The 2×2 inverse formula therefore gives

$$(I - C)^{-1} = \begin{bmatrix} 8/5 & 1 \\ 6/5 & 2 \end{bmatrix}.$$

Multiplication by $(50, 30)^T$ yields

$$x = (80 + 30, 60 + 60)^T = (110, 120)^T.$$

Check intermediate demand: $Cx = (60, 90)^T$, so $Cx + d = (60 + 50, 90 + 30)^T = x$.

Exercise 2.6.6: An inverse with a different determinant

Now

$$I - C = \begin{bmatrix} 4/5 & -1/2 \\ -3/5 & 9/10 \end{bmatrix}, \qquad \det(I - C) = \frac{36}{50} - \frac{3}{10} = \frac{21}{50}.$$

Consequently

$$(I - C)^{-1} = \frac{50}{21} \begin{bmatrix} 9/10 & 1/2 \\ 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 15/7 & 25/21 \\ 10/7 & 40/21 \end{bmatrix}.$$

With $d = (16, 12)^T$, the two coordinates are $240/7 + 100/7 = 340/7$ and $160/7 + 160/7 = 320/7$.

Conclusion. $x = (340/7, 320/7)^T$. This vector satisfies $(I - C)x = (16, 12)^T$.

Exercise 2.6.7: The effect of increasing final demand by one unit

Use the inverse from Exercise 2.6.5.

(a) For demand $(1, 0)^T$, the production vector is the first column of the inverse: $(8/5, 6/5)^T$.

(b) For demand $(51, 30)^T$, multiplication gives $x = ((8/5)51 + 30, (6/5)51 + 60)^T = (558/5, 606/5)^T$.

(c) The new demand is $(50, 30)^T + (1, 0)^T$. By distributivity of matrix multiplication, the new production is $(110, 120)^T + (8/5, 6/5)^T = (558/5, 606/5)^T$. Thus the first inverse column records the whole economy's extra production needed for the one-unit increase in sector 1's final demand.

Exercise 2.6.8: Prove superposition and interpret an inverse column

The column-sum hypothesis guarantees $I - C$ invertible for a nonnegative consumption matrix. For instance if $q < 1$ is the largest column sum, $z = Cz$ would give $\|z\|_1 \le q\|z\|_1$, forcing $z = 0$; the square-matrix theorem then gives invertibility.

(a) The given equations are $(I-C)x = d$ and $(I-C)\Delta x = \Delta d$. Adding them gives $(I-C)(x+\Delta x) = d + \Delta d$. Uniqueness makes $x + \Delta x$ the new production vector.

(b) If $\Delta d = e_1 = (1, 0, \ldots, 0)^T$, then $\Delta x = (I - C)^{-1}e_1$. Multiplication by e_1 selects column 1 of any matrix, so it selects exactly the claimed inverse column. The same interpretation applies to column j for an increase in sector j's final demand.

Exercise 2.6.9: Solve a three-sector numerical production model

Multiply $(I - C)x = d$ by 10:

$$8x_1 - 2x_2 = 400, \qquad -3x_1 + 9x_2 - 3x_3 = 600, \qquad -x_1 + 8x_3 = 800.$$

The third equation gives $x_1 = 8x_3 - 800$. Substituting into the first gives $x_2 = 32x_3 - 3400$. Insert both expressions in the middle equation:

$$-3(8x_3 - 800) + 9(32x_3 - 3400) - 3x_3 = 600.$$

Thus $261x_3 = 28800$, giving $x_3 = 3200/29$, $x_1 = 2400/29$, and $x_2 = 3800/29$.

Conclusion. $x = (2400/29, 3800/29, 3200/29)^T$; direct substitution checks all three equations.

Exercise 2.6.10: A demand increase affects every sector when every inverse entry is positive

An increase $h > 0$ in final demand for sector j is $\Delta d = he_j$. The production change is $\Delta x = h(I - C)^{-1}e_j$, namely h times inverse column j. Every entry of that column is strictly positive by hypothesis.

Conclusion. Production must rise in every sector, including sectors whose final demand did not change directly. Intermediate purchases transmit the increase throughout this model.

Exercise 2.6.11: Verify the two GDP expressions algebraically

The production equation gives $d = (I-C)x$. The dual price equation gives $v = p-C^Tp$, so transposing yields $v^T = p^T - p^TC = p^T(I - C)$. Therefore

$$p^Td = p^T(I - C)x = v^Tx.$$

This uses associativity and the transpose identity $(C^Tp)^T = p^TC$; all three expressions are scalar values. It proves the equality without assuming an inverse for either equation.

Exercise 2.6.12: An iterative inverse from finite geometric sums

By definition $D_m = I + C + \cdots + C^m$. Multiplying by C shifts every power up by one, so

$$D_{m+1} = I + CD_m, \qquad D_0 = I.$$

Equivalently $D_{m+1} - D_m = C^{m+1}$. Thus repeatedly multiply the current matrix by C and add I to obtain the next approximation. To verify the limit, the finite telescoping identity is $(I - C)D_m = I - C^{m+1}$. The hypothesis $C^m \to 0$ implies $I - C$ invertible: if $z = Cz$, then $z = C^mz \to 0$, so $z = 0$. Consequently

$$D_m = (I - C)^{-1}(I - C^{m+1}) \longrightarrow (I - C)^{-1}.$$

The recurrence therefore approximates the requested inverse, not just an unexplained matrix sequence.

Exercise 2.6.13: Compute production for the supplied seven-sector economy

Enter the supplied consumption data exactly.

To specify the data without ambiguous rounding, write

$$C = \frac{1}{10000}\begin{bmatrix} 1588 & 64 & 25 & 304 & 14 & 83 & 1594 \\ 57 & 2645 & 436 & 99 & 83 & 201 & 3413 \\ 264 & 1506 & 3557 & 139 & 142 & 70 & 236 \\ 3299 & 565 & 495 & 3636 & 204 & 483 & 649 \\ 89 & 81 & 333 & 295 & 3412 & 237 & 20 \\ 1190 & 901 & 996 & 1260 & 1722 & 2368 & 3369 \\ 63 & 126 & 196 & 98 & 64 & 132 & 12 \end{bmatrix}.$$

Use the demand vector $d = (74000, 56000, 10500, 25000, 17500, 196000, 5000)^T$ in the same sector order as the problem. Solve $Mx = d$ with $M = I - C$; this is a seven-equation linear system, not a demand for external data.

Solve and check the residual before rounding.

Exact rational elimination gives the following values, rounded to six decimal places for display (units: millions of dollars):

$$x \approx \begin{bmatrix} 99575.653398 \\ 97703.022863 \\ 51230.523166 \\ 131569.921929 \\ 49488.491372 \\ 329554.452570 \\ 13835.335715 \end{bmatrix}.$$

The exact rational solution satisfies $Mx - d = 0$ in every coordinate. The six-decimal display is only an approximation and need not have an identically zero numerical residual. To reproduce that exact calculation, define C using the integer matrix above divided by 10000, and run:

```
from sympy import Matrix, eye
# C is the displayed integer matrix divided by 10000.
d = Matrix([74000,56000,10500,25000,17500,196000,5000])
M = eye(7) - C
x = M.LUsolve(d)      # exact rational elimination
assert M*x == d
print(x.evalf(12))
```

This solves the system without constructing an inverse. The displayed vector is the actual production plan, not merely an algorithm for finding it.

Exercise 2.6.14: Use the same coefficient matrix for a new demand

Keep the consumption matrix from Exercise 2.6.13 and change only the demand to

$$d = (99640, 75548, 14444, 33501, 23527, 263985, 6526)^T.$$

Solving $(I - C)x = d$ by exact arithmetic, then displaying six decimals, gives (again in millions of dollars)

$$x \approx \begin{bmatrix} 134033.615328 \\ 131686.642968 \\ 69471.699079 \\ 176912.002023 \\ 66595.556706 \\ 443772.902077 \\ 18431.177683 \end{bmatrix}.$$

The same reproduction code works with this new d. Checking the exact solution gives $(I-C)x-d = 0$ coordinate by coordinate. A previously computed LU factorization of $I - C$ can be reused; only its two triangular solves need be repeated.

Exercise 2.6.15: Count the actual number of iteration steps

Start with the specified $x^{(0)} = d$ from Exercise 2.6.13 and iterate $x^{(k+1)} = d + Cx^{(k)}$ without rounding intermediate values. We interpret "to four significant figures" as agreement, in all seven coordinates,

with rounding the exact solution to four significant figures. The resulting target vector is

$$\begin{bmatrix} 9.958 \times 10^4 \\ 9.770 \times 10^4 \\ 5.123 \times 10^4 \\ 1.316 \times 10^5 \\ 4.949 \times 10^4 \\ 3.296 \times 10^5 \\ 1.384 \times 10^4 \end{bmatrix}.$$

Actual iteration gives these checks:

k	$\max_i \lvert x_i^{(k)} - x_i \rvert / \lvert x_i \rvert$	all rounded entries agree?
0	8.09987×10^{-1}	no
5	4.04043×10^{-2}	no
10	1.62676×10^{-3}	no
15	6.49063×10^{-5}	no
16	3.40686×10^{-5}	yes

.

At step 16, before rounding,

$$x^{(16)} \approx \begin{bmatrix} 99575.105594 \\ 97701.855867 \\ 51228.777814 \\ 131566.362702 \\ 49487.009311 \\ 329550.480997 \\ 13835.050858 \end{bmatrix}.$$

No preceding iterate has all seven required rounded values. At step 15, for instance, the first coordinate is 99574.609463, which rounds to 9.957×10^4, not 9.958×10^4. Here is a reproducible counting procedure using the exact C, d:

```
x_exact = (eye(7)-C).LUsolve(d)
four = lambda v: [format(float(z), '.4g') for z in v]
target = four(x_exact)
xk = d
k = 0
while four(xk) != target:
    xk = d + C*xk
    k += 1
assert k == 16
```

The rational arithmetic prevents iteration-roundoff accumulation; the final formatting is used only for the specified comparison.

Conclusion. With $x^{(0)} = d$ counted as step zero, 16 updates are needed. The usual relative-error check $\max_i \lvert x_i^{(k)} - x_i \rvert / \lvert x_i \rvert < 5 \times 10^{-5}$ also first holds at step 16 for these data.

2.7 Applications to computer graphics

For two-dimensional graphics, write a point as $(x, y, 1)^T$. An affine transformation is then represented by

$$H = \begin{bmatrix} a & b & h \\ c & d & k \\ 0 & 0 & 1 \end{bmatrix}, \qquad H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ax + by + h \\ cx + dy + k \\ 1 \end{bmatrix}.$$

The upper-left block controls the linear part and the last column controls translation. A positive angle denotes counterclockwise rotation in the usual oriented plane. To perform transformation F first and G second, multiply their matrices as GF: the first action is the rightmost factor. We use $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and write $T_p = \begin{bmatrix} I_2 & p \\ 0^T & 1 \end{bmatrix}$ for translation.

Practice problem 2.7.P1: Rotate about a point other than the origin

Let $p = (-2, 6)^T$ and $\theta = -30°$. Subtract p to move the center to zero, rotate, then add p back. Algebraically this is $x \mapsto R_\theta(x - p) + p = R_\theta x + (p - R_\theta p)$. With $c = \sqrt{3}/2$, $s = -1/2$, the matrix is

$$T_p \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} T_{-p} = \boxed{\begin{bmatrix} \sqrt{3}/2 & 1/2 & \sqrt{3} - 5 \\ -1/2 & \sqrt{3}/2 & 5 - 3\sqrt{3} \\ 0 & 0 & 1 \end{bmatrix}}.$$

For example $R_\theta p = (-\sqrt{3} + 3, 1 + 3\sqrt{3})^T$, which explains the translation column. Substituting $x = p$ gives p again, as a rotation center must be fixed.

Exercise 2.7.1: Embed the shear into homogeneous coordinates

Example 2 uses $(x, y) \mapsto (x + y/4, y)$. Preserve the final coordinate 1:

$$\boxed{\begin{bmatrix} 1 & 1/4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

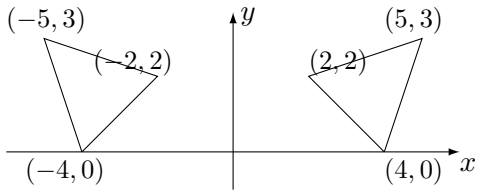
Multiplication by $(x, y, 1)^T$ gives $(x + y/4, y, 1)^T$, so the extra coordinate introduces no translation and does not change the original shear.

Exercise 2.7.2: Reflect and sketch a triangle

Reflection through the y-axis negates x and preserves y. Thus

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 5 \\ 0 & 2 & 3 \end{bmatrix} = \begin{bmatrix} -4 & -2 & -5 \\ 0 & 2 & 3 \end{bmatrix}.$$

The original vertices $(4, 0), (2, 2), (5, 3)$ therefore become $(-4, 0), (-2, 2), (-5, 3)$. Connect corresponding triples of vertices; reflection preserves straight segments and lengths.



Exercise 2.7.3: Translate first, rotate second

Translation by $(2,1)$ takes (x,y) to $(x+2,y+1)$. A 90° rotation then gives $(-(y+1),x+2)$. Reading off coefficients yields

$$R_{90^\circ}T_{(2,1)} = \boxed{\begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}},$$

where the rotation matrix is understood in its homogeneous 3×3 form. The translated origin goes to $(-1,2)$, checking the last column.

Exercise 2.7.4: Translate before unequal scaling

The first action gives $(x-1,y+4)$, and the second gives $((x-1)/2, 3(y+4)/2)$. Therefore the matrix is

$$\boxed{\begin{bmatrix} 1/2 & 0 & -1/2 \\ 0 & 3/2 & 6 \\ 0 & 0 & 1 \end{bmatrix}}.$$

The translation column is scaled too, because the scaling is performed *after* translation.

Exercise 2.7.5: Reflect, then rotate

Reflection in the x-axis sends (x,y) to $(x,-y)$. Put $a=\sqrt{2}/2$. Rotation by 45° gives $(ax+ay, ax-ay)$, so

$$\boxed{\begin{bmatrix} a & a & 0 \\ a & -a & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

Equivalently its upper-left block is $R_{45^\circ}\,\mathrm{diag}(1,-1)$.

Exercise 2.7.6: Reverse the preceding order

First rotation gives $(ax-ay, ax+ay)$. Reflecting through the x-axis now gives $(ax-ay, -ax-ay)$. Thus the matrix is

$$\boxed{\begin{bmatrix} a & -a & 0 \\ -a & -a & 0 \\ 0 & 0 & 1 \end{bmatrix}}, \qquad a=\sqrt{2}/2.$$

It differs from Exercise 5, illustrating why the order cannot be ignored.

Exercise 2.7.7: Rotate sixty degrees around a fixed point

Use the general formula $x \mapsto R_\theta x + p - R_\theta p$ with $p = (6,8)^T$, $\theta = 60°$. Since $R_\theta p = (3-4\sqrt{3}, 3\sqrt{3}+4)^T$, the result is

$$\boxed{\begin{bmatrix} 1/2 & -\sqrt{3}/2 & 3+4\sqrt{3} \\ \sqrt{3}/2 & 1/2 & 4-3\sqrt{3} \\ 0 & 0 & 1 \end{bmatrix}}.$$

Its action fixes $(6,8)$, which independently checks the translation terms.

Exercise 2.7.8: Rotate forty-five degrees around another point

Now $p = (3,7)^T$, and $R_{45°}p = (-2\sqrt{2}, 5\sqrt{2})^T$. Therefore

$$\boxed{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 3+2\sqrt{2} \\ \sqrt{2}/2 & \sqrt{2}/2 & 7-5\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}}.$$

This is $T_p R_{45°} T_{-p}$, not $R_{45°} T_p$.

Exercise 2.7.9: Precompute a composite transformation

For $A(BD)$, the 2×100 product BD has 200 entries, each requiring two multiplications: 400 multiplications. Multiplication by A costs another 400, for 800 total. For $(AB)D$, forming the four entries of AB costs $4 \cdot 2 = 8$, and multiplying its result by D costs 400, for 408 total. Thus precomputing the 2×2 composite saves 392 multiplications. The advantage grows when the same transformations are applied to more vertices: combine the small transformation matrices once, then transform all data points with their product.

Exercise 2.7.10: Which geometric transformations commute?

Let $D(x) = \lambda x$, $R(x) = Rx$, and $T(x) = x + p$. Then $D(Rx) = \lambda Rx = R(\lambda x)$, so a uniform dilation and an origin-centered rotation always commute. By contrast,

$$D(Tx) = \lambda x + \lambda p, \qquad T(Dx) = \lambda x + p.$$

They commute exactly when $(\lambda - 1)p = 0$, and generally do not. Likewise $R(Tx) = Rx + Rp$ and $T(Rx) = Rx + p$, so rotation and translation commute exactly when $Rp = p$. A nontrivial planar rotation fixes only the zero vector, making a nonzero translation fail this test. For concrete failures take $p = (1,0)$, $\lambda = 2$, and a $90°$ rotation, and evaluate the two orders at $x = 0$.

Exercise 2.7.11: Two shear-and-scale factors produce a rotation

The first action is A_1, the second is A_2, so calculate $A_2 A_1$. Its upper-left block is

$$\begin{bmatrix} \sec\varphi & -\tan\varphi \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \sec\varphi - \tan\varphi \sin\varphi & -\tan\varphi\cos\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}.$$

The first simplification uses $(1-\sin^2\varphi)/\cos\varphi = \cos\varphi$. The last homogeneous row and column remain $(0,0,1)$, so this is rotation by φ. The stated factors require $\cos\varphi \neq 0$; the rotation itself still exists at other angles, but these particular factors do not.

Exercise 2.7.12: Three shears can also implement rotation

Write $t = \tan(\varphi/2)$, $s = \sin\varphi$. Ignoring the unchanged homogeneous coordinate for a moment, multiply the three 2×2 blocks:

$$\begin{bmatrix}1 & -t\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ s & 1\end{bmatrix}\begin{bmatrix}1 & -t\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1-ts & -2t+t^2s\\ s & 1-st\end{bmatrix}.$$

Half-angle identities give $st = 1-\cos\varphi$ and $-2t+t^2s = -\sin\varphi$, so the result is R_φ. For instance $\sin\varphi = 2t/(1+t^2)$ makes the second identity $-2t+2t^3/(1+t^2) = -2t/(1+t^2)$. Appending the homogeneous row and column yields precisely the claimed 3×3 rotation matrix. Each shear needs one nontrivial multiplication per point, whereas the usual rotation needs four. The factorization requires $\tan(\varphi/2)$ to be defined, excluding odd multiples of $180°$.

Exercise 2.7.13: Factor a general affine transformation

Block multiplication gives

$$\begin{bmatrix}A & p\\ 0^T & 1\end{bmatrix} = \begin{bmatrix}I_2 & p\\ 0^T & 1\end{bmatrix}\begin{bmatrix}A & 0\\ 0^T & 1\end{bmatrix}.$$

The right factor acts first and sends $(x,1)$ to $(Ax,1)$. The left then adds p, giving $(Ax+p,1)$. Thus the transformation is a linear transformation of the plane followed by a translation, as requested. No inverse or invertibility of A is needed.

Exercise 2.7.14: Separate the linear rotation from its final translation

The transformation from Exercise 7 is $x \mapsto R_{60°}x + (3+4\sqrt{3}, 4-3\sqrt{3})^T$. Hence it is rotation about zero followed by translation by

$$\boxed{p = (3+4\sqrt{3},\ 4-3\sqrt{3})^T}.$$

The translation is the original center minus its rotated image; it is not simply the center $(6,8)$ itself.

Exercise 2.7.15: Normalize a homogeneous four-vector

A homogeneous point (X,Y,Z,w) with $w \neq 0$ represents $(X/w, Y/w, Z/w)$. Dividing by $1/24$ multiplies the first three entries by 24, so the represented vector is $\boxed{(12,-6,-3)^T}$ in $\mathbb{R}^3$.

Exercise 2.7.16: Different representatives of one point

Yes: $(10,-20,-30,40) = 10(1,-2,-3,4)$. Dividing either representative by its final coordinate gives $(1/4,-1/2,-3/4)$. A common nonzero scale changes the homogeneous representative but not the point it represents.

Exercise 2.7.17: A three-dimensional rotation about the x-axis

The x-coordinate is unchanged, while the (y, z) coordinates undergo the usual 60° planar rotation in the orientation shown. The homogeneous matrix is

$$\boxed{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 & 0 \\ 0 & \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

It sends e_2 toward e_3, confirming the positive orientation about e_1 by the right-hand convention.

Exercise 2.7.18: Rotate about z, then translate

The rotation leaves z fixed and has planar angle -30°. Translation after rotation appends $(5, -2, 1)^T$ as the last column:

$$\boxed{\begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 & 5 \\ -1/2 & \sqrt{3}/2 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

For example the origin is sent to $(5, -2, 1)$, checking the order of actions.

Exercise 2.7.19: Project a triangle from a finite viewing point

A line from $c = (0, 0, 10)$ through $v = (x, y, z)$ is $c + t(v - c) = (tx, ty, 10 + t(z - 10))$. Its intersection with the image plane $z = 0$ has $t = 10/(10 - z)$. Consequently the projection formula is

$$(x, y, z) \longmapsto \left(\frac{10x}{10 - z}, \frac{10y}{10 - z}, 0 \right).$$

For the three given vertices the scale factors are $5/3, 5/4, 5/2$, so their images are

$$\boxed{(7, 2, 0), \quad (15/2, 5, 0), \quad (5, 5, 0)}.$$

All original points have $z < 10$, so each line meets the image plane, and the triangular region projects to the triangle with these vertices.

Exercise 2.7.20: A second perspective triangle

Apply the same formula. At $(7, 3, -5)$ the scale factor is $2/3$; at $(12, 8, 2)$ it is $5/4$; at $(1, 2, 1)$ it is $10/9$. Thus the image triangle has vertices

$$\boxed{(14/3, 2, 0), \quad (15, 10, 0), \quad (10/9, 20/9, 0)}.$$

Perspective projection is not a single ordinary 3×3 linear map on the original three coordinates: the required division depends on z. Homogeneous coordinates encode that division by a final normalization.

Exercise 2.7.21: Convert standard color data into screen data

Let M denote the stated CIE conversion matrix, so $M(R,G,B)^T = (X,Y,Z)^T$. To recover the gun data solve this system:

$$\boxed{\begin{bmatrix} R \\ G \\ B \end{bmatrix} = M^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}}, \qquad M^{-1} = \frac{1}{20224} \begin{bmatrix} 45677 & -21023 & -7023 \\ -27293 & 47407 & 1407 \\ 1840 & -6160 & 25840 \end{bmatrix}.$$

This exact rational inverse treats the printed decimals as exact input. Its decimal approximation is

$$\begin{bmatrix} 2.258554 & -1.039508 & -.347261 \\ -1.349535 & 2.344096 & .069571 \\ .090981 & -.304589 & 1.277690 \end{bmatrix}.$$

Exact matrix multiplication checks $MM^{-1} = I$; in particular the conversion is uniquely reversible. A matrix program can reproduce it with SymPy rational decimal entries and `M.inv()`; do not reverse the conversion by multiplying by M a second time.

Exercise 2.7.22: Recover RGB from a television signal

Write the supplied matrix as N, so the transmitted vector is $(Y,I,Q)^T = N(R,G,B)^T$. Solving for RGB gives

$$\boxed{\begin{bmatrix} R \\ G \\ B \end{bmatrix} = N^{-1} \begin{bmatrix} Y \\ I \\ Q \end{bmatrix}},$$

$$N^{-1} = \frac{1}{50845523} \begin{bmatrix} 51002600 & 48549800 & 31415400 \\ 50681600 & -13764200 & -32784600 \\ 51277600 & -56463200 & 86415400 \end{bmatrix} \approx \begin{bmatrix} 1.003089 & .954849 & .617860 \\ .996776 & -.270706 & -.644788 \\ 1.008498 & -1.110485 & 1.699568 \end{bmatrix}.$$

The exact form was calculated and verified by rational matrix multiplication. It uses the coefficients as printed, including $.212, -.528, .311$ in the last row. Different engineering conventions or more accurate calibration coefficients would produce a different numerical inverse; no such change has been silently substituted here.

2.8 Subspaces of Rn

A subspace contains zero and is closed under *every* vector sum and *every* scalar multiple of its members. A single failure disproves subspace status. A span is always a subspace. For an $m \times n$ matrix,

$$\operatorname{Col} A = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \operatorname{Nul} A = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n.$$

Column space concerns attainable *outputs*; null space concerns *inputs* with zero output. A basis is an independent spanning set. Row reduction preserves null space and dependence relations between columns, but generally changes the column space itself. To construct a column-space basis, take the *original* columns in the pivot positions. To construct a null-space basis, solve $Ax = 0$, choose the free variables as independent parameters, and read off their coefficient vectors.

Practice problem 2.8.P1: Membership in two very different spaces

For the supplied A and $u = (-7, 3, 2)^T$,

$$Au = \begin{bmatrix} -7 - 3 + 10 \\ -14 + 14 \\ 21 - 15 - 6 \end{bmatrix} = 0.$$

Thus $u \in \operatorname{Nul} A$. It is not in $\operatorname{Col} A$. Indeed row 3 of A equals five times row 1 minus four times row 2. Every possible output $b = Ax$ must consequently obey $b_3 = 5b_1 - 4b_2$. For u, that would require $2 = 5(-7) - 4(3) = -47$, which is false. Thus the same vector can belong to the null space but not the column space, even when these spaces happen to share the same ambient dimension.

Practice problem 2.8.P2: Read a shift matrix

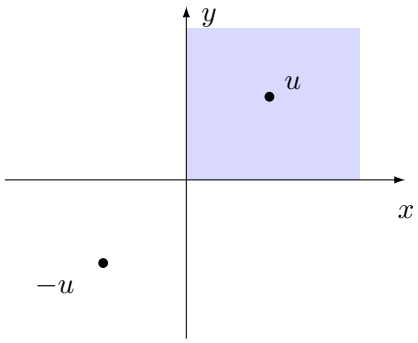
Here $A(x_1, x_2, x_3)^T = (x_2, x_3, 0)^T$. Thus $e_1 = (1, 0, 0)^T$ is a nonzero null vector. The second column of A is e_1, so e_1 is also a nonzero vector in $\operatorname{Col} A$. Alternatively e_2 belongs to the column space as column 3. It is legitimate for a vector to lie in both spaces in this example.

Practice problem 2.8.P3: The spaces of an invertible matrix

If A is invertible, every $b \in \mathbb{R}^n$ equals $A(A^{-1}b)$, so $\operatorname{Col} A = \mathbb{R}^n$. Also $Ax = 0$ implies $x = A^{-1}0 = 0$, so $\operatorname{Nul} A = \{0\}$. The zero subspace contains one vector; it is not an empty set.

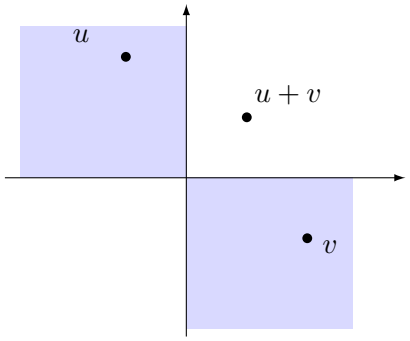
Exercise 2.8.1: A quadrant fails negative scaling

The shaded set is the first quadrant, including its axes. The vector $u = (1, 1)$ belongs to it, but $-u = (-1, -1)$ does not. Since -1 is an allowed scalar, closure under scalar multiplication fails. The following is a finite-window sketch; the quadrant continues outward.



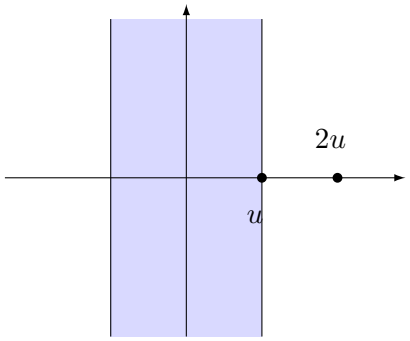
Exercise 2.8.2: Two allowed quadrants fail addition

The set consists of quadrants II and IV, with boundary axes. Take $u = (-1, 2)$ in quadrant II and $v = (2, -1)$ in quadrant IV. Their sum is $(1, 1)$ in quadrant I, outside the set. Thus addition is not closed.



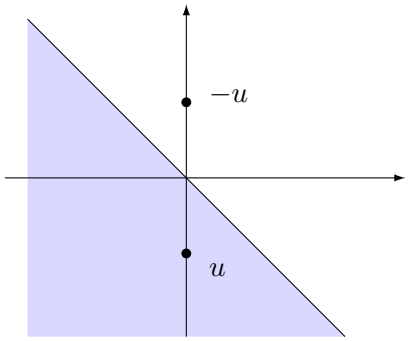
Exercise 2.8.3: A strip fails large scaling

Let the positive vertical boundary have coordinate $x = a > 0$. The point $u = (a, 0)$ lies on that included boundary, but $2u = (2a, 0)$ lies to its right, outside the strip. Hence closure under scalar multiplication fails. The finite upper and lower ends in the sketch are viewing limits, not extra boundaries.



Exercise 2.8.4: One side of a line is not a vector space

The shaded half-plane includes the region below a line through the origin. Take a point on the negative vertical axis, such as $u = (0, -1)$. It belongs to the shaded region, but $-u = (0, 1)$ lies on the unshaded side. Thus multiplying by -1 leaves the set, so it is not a subspace.



Exercise 2.8.5: Exhibit the required linear combination

Compute

$$v_1 + 2v_2 = (1, 3, -4)^T + 2(-2, -3, 7)^T = (-3, -3, 10)^T = w.$$

Thus $w \in \text{Span}\{v_1, v_2\}$. Giving coefficients that reproduce every coordinate proves membership directly; merely noticing that all vectors have three entries would not prove it.

Exercise 2.8.6: A consistency obstruction in four dimensions

Form $A = [v_1\ v_2\ v_3]$. Its four rows obey the relation $-11\,\text{row}_1 A + 5\,\text{row}_2 A + 10\,\text{row}_3 A + 2\,\text{row}_4 A = 0$. This identity is checked by its three entries: $-11 - 15 + 20 + 6 = 0$, $-44 - 20 + 50 + 14 = 0$, and $-55 - 15 + 60 + 10 = 0$. If $Ac = u$ had a solution, the same combination of its right-hand sides would be zero. Instead it equals $-11(-1) + 5(-7) + 10(-1) + 2(2) = -30$. This contradiction proves $u \notin \text{Span}\{v_1, v_2, v_3\}$. Equivalently, elimination of $[A \mid u]$ creates an inconsistent row.

Exercise 2.8.7: A finite generating set versus its infinite span

(a) The set $\{v_1, v_2, v_3\}$ contains exactly three distinct vectors.

(b) Its column space contains infinitely many vectors: all multiples tv_1 are in it, and $v_1 \neq 0$ makes these distinct for distinct real t.

(c) Yes, $p \in \text{Col}\, A$. Solving $Ac = p$ gives the reduced equations $c_1 + (7/4)c_3 = -9/4$, $c_2 + (5/2)c_3 = -7/2$. Choosing the free variable $c_3 = 0$ yields $p = (-9/4)v_1 - (7/2)v_2$. Its coordinates are $(6, -10, 11)$, as required.

Exercise 2.8.8: Another column-space membership test

A convenient combination with integer coefficients is

$$v_1 + 2v_2 + v_3 = (-2, 0, 6)^T + 2(-2, 3, 3)^T + (0, -5, 5)^T = (-6, 1, 17)^T = p.$$

Thus $p \in \operatorname{Col} A$. One could instead reduce $[A \mid p]$; the displayed coefficients give a direct, independently checkable witness.

Exercise 2.8.9: Column membership is not null membership

With A, p from Exercise 7, compute

$$Ap = \begin{bmatrix} 2(6) - 3(-10) - 4(11) \\ -8(6) + 8(-10) + 6(11) \\ 6(6) - 7(-10) - 7(11) \end{bmatrix} = \begin{bmatrix} -2 \\ -62 \\ 29 \end{bmatrix} \neq 0.$$

Therefore $p \notin \operatorname{Nul} A$, even though Exercise 7 placed p in $\operatorname{Col} A$.

Exercise 2.8.10: A nonzero null vector

For $u = (-5, 5, 3)^T$ and the matrix in Exercise 8,

$$Au = \begin{bmatrix} 10 - 10 + 0 \\ 0 + 15 - 15 \\ -30 + 15 + 15 \end{bmatrix} = 0.$$

Hence $u \in \operatorname{Nul} A$. Every coordinate has been checked; the nonzero input being sent to zero is exactly the definition of null-space membership.

Exercise 2.8.11: Ambient dimensions from the matrix shape

The given matrix has three rows and four columns. An input to A has four entries and an output has three, so $\operatorname{Nul} A \subseteq \mathbb{R}^4$ and $\operatorname{Col} A \subseteq \mathbb{R}^3$. Thus $\boxed{p = 4, q = 3}$. These are ambient dimensions, not necessarily the dimensions of the two subspaces themselves.

Exercise 2.8.12: Five outputs and three inputs

Here A is 5×3. Consequently $\operatorname{Nul} A \subseteq \mathbb{R}^3$ and $\operatorname{Col} A \subseteq \mathbb{R}^5$, giving $\boxed{p = 3, q = 5}$. Count columns for inputs and rows for outputs; these roles should not be reversed.

Exercise 2.8.13: One null vector and one column vector

For the matrix in Exercise 11, use

$$z = (1, -2, 1, 0)^T, \qquad a_1 = (3, -9, 9)^T.$$

Multiplication gives $Az = (3 - 4 + 1, -9 + 8 + 1, 9 - 4 - 5)^T = 0$, so the nonzero z is in $\operatorname{Nul} A$. The nonzero a_1 belongs to $\operatorname{Col} A$ because it is an actual column, namely Ae_1. The two vectors have different numbers of entries, as their respective ambient spaces require.

Exercise 2.8.14: Choose a null vector with integer entries

For the 5×3 matrix in Exercise 12, take $z = (1, -5, 3)^T$. Its row products are $1 - 10 + 9$, $4 - 25 + 21$, $-5 + 5 + 0$, $2 - 35 + 33$, and $3 - 15 + 12$, all zero. Thus $z \neq 0$ belongs to $\operatorname{Nul} A$. A nonzero column-space vector is $a_1 = (1, 4, -5, 2, 3)^T$. It belongs to $\operatorname{Col} A$ by its definition as column 1.

Exercise 2.8.15: A basis of the plane

The two vectors $(4, -2)^T$ and $(16, -3)^T$ have determinant $4(-3) - 16(-2) = 20 \neq 0$. Their 2×2 column matrix is invertible, so they are independent and span $\mathbb{R}^2$. Therefore they form a basis for $\mathbb{R}^2$.

Exercise 2.8.16: A dependent pair cannot be a basis

The second vector $(4, -10)^T$ is -2 times the first $(-2, 5)^T$. Thus the set is dependent and spans only one line. It is not a basis for $\mathbb{R}^2$, which requires two independent directions.

Exercise 2.8.17: Three pivots after a cyclic row reordering

Place the given vectors into columns: $A = \begin{bmatrix} 0 & 5 & 6 \\ 0 & 0 & 3 \\ -2 & 4 & 2 \end{bmatrix}$. Moving row 3 to the top gives the echelon matrix $\begin{bmatrix} -2 & 4 & 2 \\ 0 & 5 & 6 \\ 0 & 0 & 3 \end{bmatrix}$. Its three pivots show independence and spanning, so the given vectors form a basis for $\mathbb{R}^3$.

Exercise 2.8.18: An explicit dependence in three dimensions

The third vector equals twice the first plus the second: $2(1, 1, -3)^T + (3, -1, 2)^T = (5, 1, -4)^T$. Thus $2v_1 + v_2 - v_3 = 0$ is a nontrivial dependence relation. The set is not a basis for $\mathbb{R}^3$.

Exercise 2.8.19: Too few vectors to span three-dimensional space

Only two vectors are supplied. Their 3×2 column matrix has at most two pivots, so it cannot have a pivot in every one of its three rows. They do not span $\mathbb{R}^3$ and therefore do not form a basis for it. Their own span can have a basis of two vectors; that is a different question.

Exercise 2.8.20: Too many vectors for independence in three dimensions

Four vectors in $\mathbb{R}^3$ are necessarily dependent: the associated 3×4 homogeneous system has at least one free variable. Hence these four vectors do not form a basis for $\mathbb{R}^3$, regardless of whether they happen to span the whole space.

Exercise 2.8.21: Subspace definitions and preserved dependence

(a) **False as written.** The closure requirements must say *for every* $u, v \in H$, $u + v \in H$ and *for every* $u \in H$ *and every scalar* c, $cu \in H$. Simply listing that some $u, v, u + v, cu$ are in H is insufficient. For example $H = \{0, e_1, -e_1\}$ has zero and satisfies those listed facts with $u = v = 0$, but $2e_1 \notin H$. With the universal conditional quantifiers restored, the statement is exactly the correct subspace definition.

(b) **True.** $[v_1 \ \cdots \ v_p]x$ is precisely a linear combination of the listed vectors, so its set of possible values is their span.

(c) **False.** There are n unknowns, so the solution vectors have n entries; the subspace is in $\mathbb{R}^n$, not generally $\mathbb{R}^m$.

(d) **True.** The IMT supplies both independence and spanning for the columns of an invertible $n \times n$ matrix.

(e) **True.** If row operations give $B = EA$ with invertible E, then $Ac = 0$ if and only if $Bc = 0$. Thus precisely the same coefficient vectors express dependencies among the corresponding columns.

Exercise 2.8.22: Distinguish spaces from their descriptions

(a) **False.** $H = \{0, e_1\}$ contains zero but not $2e_1$, so it fails scalar closure.

(b) **False.** Pivot columns must be selected from A itself. For example $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ reduces to $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. Its pivot column e_1 is not in $\operatorname{Col} A = \operatorname{Span}\{e_2\}$.

(c) **True.** The set contains zero (all coefficients zero), and sums or scalar multiples of linear combinations are linear combinations. These are exactly the three subspace tests.

(d) **False.** Closure requires both added vectors in H. Take $H = \operatorname{Span}\{e_1\} \subset \mathbb{R}^2$, $x = 0$, and $y = e_2$; then $x + y = e_2 \notin H$.

(e) **False.** $\operatorname{Col} A$ is the set of right sides b for which $Ax = b$ has a solution. It is not the set of solution inputs x to one fixed equation; input and output spaces can even have different dimensions.

Exercise 2.8.23: Two pivot columns and two free variables

The given echelon form has pivots in columns 1 and 2. Therefore a column-space basis is

$$\mathcal{C} = \{(4, 6, 3)^T, (5, 5, 4)^T\},$$

taken from the *original* matrix. For $Ax = 0$, the echelon equations are $x_1 + 2x_2 + 6x_3 - 5x_4 = 0$ and $x_2 + 5x_3 - 6x_4 = 0$. Set $x_3 = s$, $x_4 = t$; then $x_2 = -5s + 6t$ and $x_1 = 4s - 7t$. Thus

$$x = s(4, -5, 1, 0)^T + t(-7, 6, 0, 1)^T,$$

and the two displayed coefficient vectors form a basis for $\operatorname{Nul} A$. They span because this is the general solution, and they are independent because their free coordinates x_3, x_4 force $s = t = 0$ in a zero combination.

Exercise 2.8.24: A nonadjacent pair of pivot columns

The pivot columns are 1 and 3, so a basis for $\operatorname{Col} A$ is $\{(3,2,3)^T, (9,7,6)^T\}$. The echelon equations reduce to $x_3 = -2x_4$ and $x_1 - 2x_2 - 6x_4 = 0$. Let $x_2 = s$, $x_4 = t$. Then

$$x = s(2,1,0,0)^T + t(6,0,-2,1)^T.$$

The coefficient vectors form a null-space basis: they describe all solutions and their free-coordinate entries make them independent.

Exercise 2.8.25: Three pivot columns among five inputs

The echelon form has pivot columns 1,2,4. A column-space basis is

$$\{(1,-1,-2,3)^T, (4,2,2,6)^T, (-3,3,5,-5)^T\}.$$

The three equations are $x_1 + 4x_2 + 8x_3 + 5x_5 = 0$, $2x_2 + 5x_3 - x_5 = 0$, and $x_4 + 4x_5 = 0$. Taking $x_3 = s$, $x_5 = t$ gives $x_2 = -5s/2 + t/2$, $x_4 = -4t$, $x_1 = 2s - 7t$. Hence a basis for $\operatorname{Nul} A$ is

$$\{(2,-5/2,1,0,0)^T, (-7,1/2,0,-4,1)^T\}.$$

Substituting either vector into the three echelon equations yields zero; independence follows from their third and fifth coordinates.

Exercise 2.8.26: Another three-pivot example

The pivot positions are in columns 1,2,4 of the original matrix. A column-space basis is

$$\{(3,3,0,6)^T, (-1,1,3,3)^T, (-1,0,-1,-2)^T\}.$$

The echelon equations give $x_4 = 2x_5$, $x_2 = -3x_3 + 2x_5$, and $3x_1 - x_2 - 3x_3 + 6x_5 = 0$, so $x_1 = -4x_5/3$. Thus a null-space basis is

$$\{(0,-3,1,0,0)^T, (-4/3,2,0,2,1)^T\}.$$

The free variables are x_3, x_5; this explains both why there are two basis vectors and why they are independent.

Exercise 2.8.27: A column combination different from every column

Choose $A = I_3$ and $b = (1,1,0)^T$. Then $b = e_1 + e_2 = A(1,1,0)^T \in \operatorname{Col} A$, and $b \neq 0$. It differs from each of e_1, e_2, e_3, the columns of A. Column space contains all combinations, not just the listed columns.

Exercise 2.8.28: Construct an unattainable output

Take $A = \operatorname{diag}(1,1,0)$ and $b = e_3$. Every output Ax has third coordinate zero, whereas b has third coordinate one. Hence $b \notin \operatorname{Col} A$.

Exercise 2.8.29: A nonzero input annihilated by a nonzero matrix

Take $A = \operatorname{diag}(1, 0, 0)$ and $b = e_2$. Both A and b are nonzero, but $Ab = 0$, so $b \in \operatorname{Nul} A$. This does not violate any rule about scalars: nonzero matrices can annihilate nonzero vectors.

Exercise 2.8.30: Independent columns are already a basis for their own span

By definition $\operatorname{Col} A = \operatorname{Span}\{a_1, \ldots, a_p\}$, so the columns span this space. The given assumption supplies independence. These are exactly the two conditions for a basis, hence the columns are a basis for $\operatorname{Col} A$. No claim that the column space equals the whole codomain is needed.

Exercise 2.8.31: A proper column space of a square matrix

Since $\operatorname{Col} F \neq \mathbb{R}^5$, the 5×5 matrix is not onto. The IMT makes it singular, so $\operatorname{Nul} F$ contains a nonzero vector. It is a subspace of $\mathbb{R}^5$ and contains infinitely many vectors, since all real multiples of any nonzero null vector are distinct members. The exact size of a basis is not determined by the premise.

Exercise 2.8.32: Full column space in seven dimensions

$\operatorname{Col} B = \mathbb{R}^7$ says every b is attained. Because B is square, the IMT makes it invertible. Thus for every $b \in \mathbb{R}^7$, $Bx = b$ has exactly one solution, $B^{-1}b$. In particular $\operatorname{Nul} B = \{0\}$; there are no free variables.

Exercise 2.8.33: A trivial null space in six dimensions

The only homogeneous solution is zero, so the square matrix C is invertible by the IMT. Therefore for every $b \in \mathbb{R}^6$ the equation $Cx = b$ has a unique solution $C^{-1}b$, and $\operatorname{Col} C = \mathbb{R}^6$. The conclusion would require qualification for a nonsquare matrix.

Exercise 2.8.34: A column basis of the entire codomain fixes the shape

Spanning $\mathbb{R}^m$ requires a pivot in every row, so there are m pivots. Independence requires every column to be a pivot column, so there are n pivots. Hence $m = n$: A is square. It is then invertible by the IMT. Both basis requirements, rather than spanning alone, forced equality.

Exercise 2.8.35: A nontrivial null space implies a proper column space

The given nonzero null vector makes the 5×5 matrix singular. Therefore its column space is a proper subspace of $\mathbb{R}^5$ by the IMT. Some right sides are unattainable. Every attainable right side has infinitely many preimages: if $Bx = b$ and $Bz = 0$ with $z \neq 0$, then $B(x + tz) = b$ for every real t.

Exercise 2.8.36: Independent columns of a tall matrix

The four columns are independent, so $Cx = 0$ has only $x = 0$. Hence $\boxed{\operatorname{Nul} C = \{0\} \subseteq \mathbb{R}^4}$. Although there are six output coordinates, inputs and null vectors have four coordinates. The independent columns do not span $\mathbb{R}^6$, so not every right side is attained, despite uniqueness whenever one is attained.

Exercise 2.8.37: An exact computer-assisted pair of bases

Exact row reduction of the supplied matrix gives

$$R = \begin{bmatrix} 1 & 0 & 5/2 & -9/2 & 7/2 \\ 0 & 1 & 3/2 & -5/2 & 3/2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1 and 2, so $\{(3,-7,-5,3)^T, (-5,9,7,-7)^T\}$ is a column-space basis. For the null space, solve $x_1 = -5x_3/2 + 9x_4/2 - 7x_5/2$ and $x_2 = -3x_3/2 + 5x_4/2 - 3x_5/2$. Taking each free variable in turn equal to 2 gives the integer basis

$$\{(-5,-3,2,0,0)^T, (9,5,0,2,0)^T, (-7,-3,0,0,2)^T\}.$$

Scaling basis vectors by 2 does not change their span or independence. These were checked by exact multiplication by the original A. A reproducible computation is `R,pivots=Matrix(A).rref()` in SymPy, followed by `Matrix(A).nullspace()`; the latter may return scalar multiples of the chosen vectors.

Exercise 2.8.38: Four pivot columns and one free coordinate

Exact row reduction gives

$$R = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Thus original columns 1,2,4,5 form a column-space basis. Explicitly they are the columns of

$$C = \begin{bmatrix} 5 & 3 & -6 & -8 \\ 4 & 1 & -8 & -7 \\ 5 & 1 & 5 & 19 \\ -7 & -5 & 8 & 5 \end{bmatrix}.$$

For $Rx = 0$, $x_1 = -x_3$, $x_2 = x_3$, and $x_4 = x_5 = 0$. Hence $\{(-1,1,1,0,0)^T\}$ is a basis for $\operatorname{Nul} A$. Multiplication by the original matrix checks the null vector, and the four pivot positions establish independence of the selected original columns.

2.9 Dimension and rank

For an ordered basis $\mathcal{B} = (b_1, \ldots, b_p)$, the coordinate vector $[x]_{\mathcal{B}} = (c_1, \ldots, c_p)^T$ means $x = c_1 b_1 + \cdots + c_p b_p$. Coordinates depend on the basis and its order. The dimension of a subspace is the number of vectors in any basis for it. For an $m \times n$ matrix, rank counts pivots, while nullity counts free variables; therefore

$$\dim \operatorname{Col} A = \operatorname{rank} A, \qquad \dim \operatorname{Nul} A = n - \operatorname{rank} A.$$

A subspace's dimension is different from the number of coordinates in its vectors. A two-dimensional subspace of $\mathbb{R}^6$ still consists of six-entry vectors, not two-entry vectors.

Practice problem 2.9.P1: Count independent directions rather than generators

Put the three vectors into a matrix. Eliminating below column 1 gives

$$\begin{bmatrix} 2 & 3 & -1 \\ -8 & -7 & 6 \\ 6 & -1 & -7 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & -10 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 2 are pivot columns. Thus a basis for H is $\{(2, -8, 6)^T, (3, -7, -1)^T\}$, and $\dim H = 2$. The third vector is redundant: $v_3 = -11v_1/10 + 2v_2/5$.

Practice problem 2.9.P2: Turn coordinates into a vector

The coordinate entries are weights, so

$$x = 3(1, .2)^T + 2(.2, 1)^T = (3 + .4, .6 + 2)^T = (3.4, 2.6)^T.$$

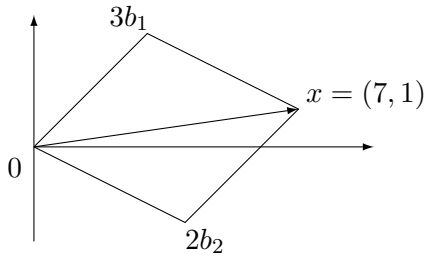
This ordinary coordinate vector differs from $(3, 2)^T$, because the basis is not the standard basis.

Practice problem 2.9.P3: A subspace cannot have more independent directions than its ambient space

No subspace of $\mathbb{R}^3$ can have dimension four. A four-dimensional subspace would have a basis of four independent vectors in $\mathbb{R}^3$, but any 3×4 homogeneous system has a free variable, making four such columns dependent. In fact possible dimensions are only $0, 1, 2, 3$; a three-dimensional subspace of $\mathbb{R}^3$ must be $\mathbb{R}^3$ itself.

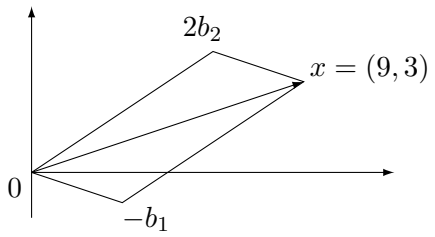
Exercise 2.9.1: Combine two basis vectors and sketch the result

The supplied coordinate vector gives $x = 3(1, 1)^T + 2(2, -1)^T = (7, 1)^T$. The sketch shows the parallelogram with sides $3b_1 = (3, 3)$ and $2b_2 = (4, -2)$; its diagonal is x.



Exercise 2.9.2: A negative basis coordinate reverses a direction

Now $x = -1(-3, 1)^T + 2(3, 2)^T = (3, -1)^T + (6, 4)^T = (9, 3)^T$. A negative coefficient means travel opposite to b_1, not negate the whole answer. The parallelogram below has sides $-b_1$ and $2b_2$.



Exercise 2.9.3: Recover coordinates from two equations

Write $x = c_1b_1 + c_2b_2$. The two coordinates require $2c_1 - c_2 = 0$ and $-3c_1 + 5c_2 = 7$. The first gives $c_2 = 2c_1$; substituting into the second gives $7c_1 = 7$, hence $c_1 = 1, c_2 = 2$. Thus $\boxed{[x]_{\mathcal{B}} = (1, 2)^T}$, and $(2, -3)^T + 2(-1, 5)^T = (0, 7)^T$ verifies the answer.

Exercise 2.9.4: Coordinates may both be negative

The equations are $c_1 - 2c_2 = 1$ and $-5c_1 + 3c_2 = 9$. Substitute $c_1 = 1 + 2c_2$ into the second: $-5 - 10c_2 + 3c_2 = 9$, so $c_2 = -2$ and $c_1 = -3$. Therefore $\boxed{[x]_{\mathcal{B}} = (-3, -2)^T}$. Indeed $-3(1, -5)^T - 2(-2, 3)^T = (1, 9)^T$.

Exercise 2.9.5: Two coordinates describe a three-entry vector

From the first two entries of $c_1b_1 + c_2b_2 = x$, $c_1 - 2c_2 = 2$ and $4c_1 - 7c_2 = 9$. Subtract four times the first equation from the second to get $c_2 = 1$; then $c_1 = 4$. The third entry also matches: $-3(4) + 5(1) = -7$. Thus $\boxed{[x]_{\mathcal{B}} = (4, 1)^T}$. Checking the third coordinate is important: a solution to only two equations would not by itself prove the vector equality in $\mathbb{R}^3$.

Exercise 2.9.6: Verify the remaining coordinate after solving

The first two equations are $-3c_1 + 7c_2 = 5$ and $2c_1 - 3c_2 = 0$. The second gives $c_1 = 3c_2/2$; the first becomes $(5/2)c_2 = 5$. Thus $c_2 = 2, c_1 = 3$. The last coordinate is $-4(3) + 5(2) = -2$, as required.

Therefore $\boxed{[x]_{\mathcal{B}} = (3,2)^T}$.

Exercise 2.9.7: Read the skew grid and confirm algebraically

The diagram suggests two units in the b_1 direction and minus one unit in the b_2 direction for w, and one and a half b_1 units plus half a b_2 unit for x. Algebra confirms

$$[w]_{\mathcal{B}} = (2,-1)^T, \qquad [x]_{\mathcal{B}} = (3/2,1/2)^T.$$

Indeed $2(3,0)^T - (-1,2)^T = (7,-2)^T = w$, and $(3/2)(3,0)^T + (1/2)(-1,2)^T = (4,1)^T = x$. These computations turn approximate visual readings into exact answers.

Exercise 2.9.8: A second skew coordinate grid

Here $c_1 b_1 + c_2 b_2 = (2c_2, 2c_1 + c_2)^T$. Thus for a vector $(u,v)^T$, $c_2 = u/2$ and $c_1 = (v - u/2)/2$. Applying this formula gives

$$[x]_{\mathcal{B}} = (2,-1)^T, \quad [y]_{\mathcal{B}} = (3/2,1)^T, \quad [z]_{\mathcal{B}} = (-1,-1/2)^T.$$

The first matches the picture's movement two b_1 steps and minus one b_2 step. The requested confirmations are $(3/2)(0,2)^T + (2,1)^T = (2,4)^T = y$ and $-(0,2)^T - (1/2)(2,1)^T = (-1,-5/2)^T = z$.

Exercise 2.9.9: Rank three, nullity one

The supplied echelon form has pivots in columns 1,3,4, so an original-column basis is

$$\mathcal{C} = \{(1,3,2,5)^T, (2,1,-1,0)^T, (-6,5,9,14)^T\}.$$

The bottom nonzero equation gives $x_4 = 0$; the next gives $x_3 = 0$; the first then gives $x_1 = -3x_2$. Thus $\mathcal{N} = \{(-3,1,0,0)^T\}$ is a null-space basis. Consequently $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 1$. Their sum is four, the number of columns, not necessarily the number of rows in other examples.

Exercise 2.9.10: Four pivots among five columns

The pivot positions are columns 1,2,4,5. A column-space basis consists of the columns of

$$C = \begin{bmatrix} 1 & -2 & 5 & 4 \\ 2 & -1 & 5 & 6 \\ -2 & 0 & 1 & -6 \\ 3 & 1 & 1 & 5 \end{bmatrix}.$$

Reading the supplied echelon equations from the bottom upward gives $x_5 = 0$, $x_4 = 0$, $x_2 = -x_3$, and $x_1 = -x_3$. Thus $\{(-1,-1,1,0,0)^T\}$ is a null-space basis. The dimensions are $\boxed{4 \text{ and } 1}$ for column and null spaces, respectively. The basis vectors for the column space are from the original A, while the equations for the null space come from its echelon form.

Exercise 2.9.11: Two pivot directions and three degrees of freedom

The pivots are in columns 1 and 3. Thus a column-space basis is

$$\{(2,3,0,-3)^T,(-5,-8,9,-7)^T\}.$$

The second echelon equation gives $x_3=-x_5$. Substitution in the first gives $x_1=-2x_2-x_4-x_5$. Taking x_2,x_4,x_5 as independent parameters gives a null-space basis

$$\{(-2,1,0,0,0)^T,(-1,0,0,1,0)^T,(-1,0,-1,0,1)^T\}.$$

These vectors have independent free-coordinate entries, so they are independent and span all solutions. Therefore $\dim\operatorname{Col}A=2$, $\dim\operatorname{Nul}A=3$.

Exercise 2.9.12: A third free-variable calculation

The echelon form has pivots in columns 1,2,3. A column-space basis is

$$\{(1,5,4,3)^T,(2,1,6,4)^T,(-4,-9,-9,-5)^T\}.$$

The bottom nonzero equation gives $x_3=x_5$. The preceding equation then gives $x_2=-2x_4-x_5$, and the first gives $x_1=0$. Hence a null-space basis is

$$\{(0,-2,0,1,0)^T,(0,-1,1,0,1)^T\}.$$

The dimensions are $\boxed{3 \text{ and } 2}$. Direct substitution into the original A checks both null vectors, and the free coordinates prove their independence.

Exercise 2.9.13: Discard one redundant spanning vector

Place the four supplied vectors into $A=[v_1\ v_2\ v_3\ v_4]$. Exact row reduction gives

$$R=\begin{bmatrix}1&-3&0&0\\0&0&1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}.$$

The pivot columns are 1,3,4, so

$$\{(1,-3,2,-4)^T,(2,-1,4,2)^T,(-4,5,-3,7)^T\}$$

is a basis for the supplied span, whose dimension is 3. The omitted vector is $v_2=-3v_1$. The other three are independent by the pivot test; discarding a multiple does not by itself establish that last fact without the reduction.

Exercise 2.9.14: Select three of five generators

The column matrix of the five vectors reduces exactly to

$$R=\begin{bmatrix}1&0&-2&0&15/2\\0&1&1&0&-7/2\\0&0&0&1&-5/2\\0&0&0&0&0\end{bmatrix}.$$

Thus v_1, v_2, v_4 are pivot columns, giving the basis

$$\{(1,-1,-2,3)^T, (2,-3,-1,4)^T, (-1,4,-7,7)^T\}$$

and dimension 3. For a check on spanning, $v_3 = -2v_1 + v_2$ and $v_5 = (15/2)v_1 - (7/2)v_2 - (5/2)v_4$. Thus the two omitted vectors are indeed combinations of the selected ones.

Exercise 2.9.15: Equal dimensions do not mean equal sets

Four pivot columns in a 4×6 matrix give four pivot rows, so $\operatorname{Col} A = \mathbb{R}^4$. There are $6 - 4 = 2$ free variables, so $\dim \operatorname{Nul} A = 2$. However $\operatorname{Nul} A \neq \mathbb{R}^2$: its vectors have six entries and it is a subspace of $\mathbb{R}^6$. A basis identifies it with $\mathbb{R}^2$ through coordinates; an identification by coordinates is different from literal equality of the sets.

Exercise 2.9.16: A three-dimensional column space inside four dimensions

The column space has dimension 3 but consists of four-entry vectors. It is a proper subspace of $\mathbb{R}^4$, not literally $\mathbb{R}^3$. Rank–nullity gives $\dim \operatorname{Nul} A = 7 - 3 = \boxed{4}$. The five words "three-dimensional subspace of $\mathbb{R}^4$" specify both its internal dimension and its ambient space.

Exercise 2.9.17: True–false questions about dimension

(a) **True.** For an ordered basis, the unique weights in $x = \sum c_i v_i$ are precisely its basis coordinates.

(b) **False.** A line must pass through zero to be a subspace. The line $\{(1, t) : t \in \mathbb{R}\}$ does not contain zero, for instance. A line through zero does form a one-dimensional subspace.

(c) **True.** Pivot columns form a basis for $\operatorname{Col} A$, so their number equals its dimension.

(d) **True.** Pivots plus free variables count all n columns: $\dim \operatorname{Col} A + \dim \operatorname{Nul} A = n$.

(e) **True.** A spanning set of p vectors in a p-dimensional space is a basis. If it were dependent, one vector could be deleted without changing its span, leaving at most $p - 1$ generators for a space of dimension p, which is impossible.

Exercise 2.9.18: Uniqueness, nullity, and coordinate maps

(a) **True.** Two coefficient representations would subtract to a dependence relation among basis vectors; independence forces the corresponding coefficients to be identical.

(b) **False.** The null-space dimension is the number of *free* variables, not all variables. For $A = I_n$ it is zero despite there being n variables.

(c) **True.** Rank is the number of pivots, which is the column-space dimension.

(d) **True.** The coordinate map is a linear bijection to $\mathbb{R}^p$, with inverse $(c_1, \ldots, c_p)^T \mapsto \sum c_i v_i$. Thus it preserves addition and scalar multiplication and assigns one coordinate vector to every vector of H.

(e) **True.** If p independent vectors in a p-dimensional space did not span it, a vector outside their span could be appended to produce $p + 1$ independent vectors, contradicting the dimension.

Exercise 2.9.19: Recover rank from nullity

The null space has a basis of three vectors, so its dimension is 3. There are seven columns. Thus $\operatorname{rank} A = 7 - 3 = \boxed{4}$ by rank–nullity. The five rows impose the separate upper bound $\operatorname{rank} A \leq 5$, which this answer respects.

Exercise 2.9.20: A six-by-eight matrix

There are eight columns and nullity three, so $\operatorname{rank} A = 8 - 3 = \boxed{5}$. This is possible because five pivots fit into six rows.

Exercise 2.9.21: The dimension of a homogeneous solution space

The solution space is $\operatorname{Nul} A$. Rank–nullity with eight columns and rank seven gives $\dim \operatorname{Nul} A = 8 - 7 = \boxed{1}$. Its vectors have eight entries even though only one parameter is needed to describe them.

Exercise 2.9.22: Five generators of a four-dimensional span are dependent

If the five vectors were independent, they would be a basis for their own span, making its dimension five by the definition of dimension. The given dimension is four, a contradiction. Hence the vectors are dependent. The ambient value of n does not change this argument.

Exercise 2.9.23: Construct prescribed rank and nullity

Take

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its column-space basis is $\{e_1, e_2\} \subset \mathbb{R}^3$, so its rank is 2. Its homogeneous solutions are $(0, 0, s, t, u)^T$, so nullity is 3. Thus the requested 3×5 shape and both dimensions are achieved.

Exercise 2.9.24: Construct rank one

Choose

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Every column is a multiple of the nonzero first column, so the column space is exactly one line. Hence A has rank 1; it has the required 3×4 shape.

Exercise 2.9.25: Full column-space dimension forces independent columns

The p columns span $\operatorname{Col} A$ by definition, and this space has dimension p by hypothesis. A spanning set with exactly its dimension many vectors is a basis. Therefore the columns are independent. Equivalently rank p in a matrix with p columns leaves no free variables.

Exercise 2.9.26: A basis need not be the leftmost pivot selection

The five named columns lie in $\operatorname{Col} A$, which has dimension five because $\operatorname{rank} A = 5$. They are independent by hypothesis. An independent set of five vectors in a five-dimensional space is a basis, so these columns span $\operatorname{Col} A$. The pivot-column algorithm is one way to find a basis, not a claim that all bases must use the same indices.

Exercise 2.9.27: More vectors than a spanning set must be dependent

(a) Since $a_j \in W = \operatorname{Span}\{b_1, \ldots, b_p\}$, there are coefficients $c_{1j}, \ldots, c_{pj}$ with $a_j = \sum_i c_{ij} b_i$. Collect them in $c_j \in \mathbb{R}^p$ to obtain $a_j = Bc_j$.

(b) The matrix $C = [c_1 \ \cdots \ c_q]$ has p rows and $q > p$ columns. Its homogeneous system has more variables than possible pivots, so at least one variable is free. Thus some $u \neq 0$ satisfies $Cu = 0$.

(c) The column identities in part (a) give $A = BC$. Therefore $Au = B(Cu) = B0 = 0$. Since $u \neq 0$, its entries furnish a nontrivial linear dependence among $a_1, \ldots, a_q$.

This proves the claim without assuming the two supplied lists were bases.

Exercise 2.9.28: Every basis has the same number of vectors

Let $\mathcal{A}, \mathcal{B}$ be bases of W, with q, p vectors, respectively. If $q > p$, then Exercise 27 applied to the spanning set $\mathcal{B}$ would make $\mathcal{A}$ dependent, contrary to its being a basis. Hence $q \leq p$. Interchanging the roles gives $p \leq q$. Thus $p = q$. This justifies defining dimension using *any* basis.

Exercise 2.9.29: Compute coordinates in a two-dimensional subspace of R4

The vectors v_1, v_2 are independent: their first two coordinates have 2×2 determinant $15(-10) - 14(-5) = -80 \neq 0$. Thus they are a basis for their span H. A direct calculation gives

$$2v_1 - v_2 = (30 - 14, -10 + 10, 24 - 13, 14 - 17)^T = (16, 0, 11, -3)^T = x.$$

Therefore $x \in H$ and $\boxed{[x]_{\mathcal{B}} = (2, -1)^T}$. These exact integer products also verify the computer-assisted answer.

Exercise 2.9.30: Three coordinates in a four-dimensional ambient space

Put $V = [v_1 \ v_2 \ v_3]$ and row reduce the augmented matrix. Exact arithmetic gives

$$[V \mid x] \ \sim \ \left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The three coefficient pivots show that v_1, v_2, v_3 are independent, so $\mathcal{B}$ is a basis for H. The consistent right block gives $[x]_{\mathcal{B}} = \boxed{(-2, 1, 1)^T}$. For an independent check,

$$-2v_1 + v_2 + v_3 = (12 + 8 - 9, -6 + 0 + 4, 18 + 7 - 8, -8 - 3 + 3)^T = (11, -2, 17, -8)^T = x.$$

Thus the coordinate vector has three entries while x itself has four.

2.S Chapter 2 supplementary exercises

These exercises combine matrix shapes, multiplication order, inverse identities, and the geometry of column and null spaces. Unless the statement explicitly says a matrix is square, do not add that hypothesis silently. A rectangular matrix can have a left inverse or a right inverse without having both. For a true–false question, a false statement is settled by one valid counterexample, while a true statement needs a general argument.

Exercise 2.S.1: Sixteen structural true–false statements

(a) **True.** If A, B are $m \times n$, then AB^T has shape $(m \times n)(n \times m)$ and $A^T B$ has shape $(n \times m)(m \times n)$. Both inner dimensions agree.

(b) **False.** The product has as many columns as B, not A. For example a 1×3 matrix times a 3×2 matrix has two columns, even though the first factor has three columns.

(c) **True.** For $A = \operatorname{diag}(a_1, \ldots, a_m)$, entry (i, j) of AB is $a_i b_{ij}$. Thus row i is scaled by a_i.

(d) **False.** Let $B = 0$, $C = I$, $D = 0$ of the same square size. Then $BC = BD = 0$ but $C \neq D$. Cancellation needs an injective left factor.

(e) **False.** $A = \operatorname{diag}(1, 0)$ and $C = \operatorname{diag}(0, 1)$ are nonzero but $AC = 0$.

(f) **False.** Expanding gives $A^2 - B^2 + BA - AB$. For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \operatorname{diag}(1, 2)$, one has $AB \neq BA$, so the extra term does not vanish.

(g) **True.** A row swap or nonzero row scaling of I_n leaves exactly n nonzero entries. Adding a nonzero multiple of a different row introduces one off-diagonal nonzero entry, giving $n + 1$. A zero-multiple replacement leaves the identity with n entries.

(h) **True.** A swap matrix and a scaling matrix are symmetric. The transpose of $I + re_i e_j^T$ is $I + re_j e_i^T$, which is the matrix for the corresponding row replacement with the indices exchanged.

(i) **True.** An elementary matrix is obtained by one row operation on a square identity matrix; its shape remains square.

(j) **False.** Every product of elementary matrices is invertible, whereas a square zero matrix is not. Only invertible square matrices admit such a factorization.

(k) **True.** Three pivots in a 3×3 matrix imply reduced form I_3. List the elementary row operations that carry out that reduction; their matrices $E_1, \ldots, E_p$ give $E_p \cdots E_1 A = I$.

(l) **False without a square hypothesis.** Take $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $AB = I_2$ but A is rectangular and has a nonzero null vector e_3, so it has no two-sided inverse. If A, B were assumed square, the assertion would instead be true by the IMT.

(m) **False.** The inverse is $B^{-1}A^{-1}$, with reversed order. For the noncommuting invertible pair in part (f), the same-order product is not an inverse.

(n) **True.** From $AB = BA$, left-multiply by A^{-1} and then right-multiply by A^{-1}: $BA^{-1} = A^{-1}B$.

(o) **False.** The correct scalar factor is $1/r$: $(rA)((1/r)A^{-1}) = I$. Taking $A = I$, $r = 2$ disproves the claimed factor r.

(p) **True.** Let x_0 be the unique solution to the stated right side. If $Az = 0$, then $A(x_0 + z) = Ax_0$, so uniqueness forces $z = 0$. Thus the square matrix has trivial null space and is invertible.

Exercise 2.S.2: Invert a supplied inverse

Since $C^{-1} = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$ has determinant $28 - 30 = -2$, invert this 2×2 matrix:

$$\boxed{C = \frac{1}{-2}\begin{bmatrix} 7 & -5 \\ -6 & 4 \end{bmatrix} = \begin{bmatrix} -7/2 & 5/2 \\ 3 & -2 \end{bmatrix}}.$$

The inverse of an inverse is the original matrix, which explains why this operation recovers C rather than another new matrix.

Exercise 2.S.3: A nilpotent shift and a finite geometric sum

The given A sends $(x_1, x_2, x_3)^T$ to $(0, x_1, x_2)^T$. Therefore $A^2x = (0, 0, x_1)^T$ and $A^3x = 0$ for every x, proving $A^3 = 0$. Distribute the requested product in its existing order:

$$(I - A)(I + A + A^2) = I + A + A^2 - A - A^2 - A^3 = I.$$

The cancellation works because all terms are powers of the same matrix.

Exercise 2.S.4: Invert identity minus a nilpotent matrix

Let $S = I + A + \cdots + A^{n-1}$. Then telescoping gives

$$(I - A)S = I - A^n = I.$$

Multiplying in the other order gives the same telescoping sum, $S(I - A) = I - A^n = I$, since a matrix commutes with its own powers. Thus $\boxed{(I - A)^{-1} = I + A + \cdots + A^{n-1}}$. No inverse of A was used; in fact a nilpotent matrix of positive size cannot itself be invertible.

Exercise 2.S.5: Reduce higher powers using a quadratic identity

The hypothesis says $A^2 = 2A - I$. Multiply by A and substitute again:

$$A^3 = 2A^2 - A = 2(2A - I) - A = 3A - 2I.$$

Next multiply this result by A:

$$A^4 = 3A^2 - 2A = 3(2A - I) - 2A = 4A - 3I.$$

Every step uses multiplication and addition of already equal matrices; we did not divide by A or assume it was diagonalizable.

Exercise 2.S.6: A pair of anticommuting matrices

Direct multiplication gives

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -AB.$$

Thus both are their own inverses, but they anticommute rather than commute. This is a concrete illustration of behavior unavailable for nonzero real scalars, since scalar multiplication commutes.

Exercise 2.S.7: Solve a two-column matrix equation efficiently

Compute $A^{-1}B$ by solving $AX = B$. Begin with $[A \mid B]$. Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$, then $R_3 \leftarrow R_3 - (1/2)R_2$:

$$\left[\begin{array}{ccc|cc} 1 & 3 & 8 & -3 & 5 \\ 2 & 4 & 11 & 1 & 5 \\ 1 & 2 & 5 & 3 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|cc} 1 & 3 & 8 & -3 & 5 \\ 0 & -2 & -5 & 7 & -5 \\ 0 & 0 & -1/2 & 5/2 & 3/2 \end{array}\right].$$

Back-substitution gives the last row of X as $(-5, -3)$, then the second as $(9, 10)$, and the first as $(10, -1)$. Therefore

$$\boxed{A^{-1}B = \begin{bmatrix} 10 & -1 \\ 9 & 10 \\ -5 & -3 \end{bmatrix}}.$$

The three nonzero pivots verify invertibility of A, and exact multiplication checks $AX = B$. No entire inverse was computed.

Exercise 2.S.8: Determine a map from two prescribed images

Place the input vectors into $X = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and outputs into $Y = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$. The two prescribed equations are exactly $AX = Y$. Since $\det X = 7 - 6 = 1$, $X^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$. Thus

$$\boxed{A = YX^{-1} = \begin{bmatrix} -2 & 1 \\ 4 & -1 \end{bmatrix}}.$$

Indeed $A(1, 3)^T = (1, 1)^T$ and $A(2, 7)^T = (3, 1)^T$. The independent inputs ensure that this matrix is uniquely determined.

Exercise 2.S.9: Recover the left factor of a known product

The given B has determinant $7-6=1$, so $B^{-1}=\begin{bmatrix}1 & -3\\ -2 & 7\end{bmatrix}$. Right multiplication by this inverse yields

$$A=(AB)B^{-1}=\begin{bmatrix}5 & 4\\ -2 & 3\end{bmatrix}\begin{bmatrix}1 & -3\\ -2 & 7\end{bmatrix}=\boxed{\begin{bmatrix}-3 & 13\\ -8 & 27\end{bmatrix}}.$$

For example the first output row checks as $[-3,13]B=[-21+26,-9+13]=[5,4]$.

Exercise 2.S.10: Another formula for an inverse

If A is invertible, so is A^T. Therefore A^TA is a product of invertible square matrices and is invertible. Set $X=(A^TA)^{-1}A^T$. Then $XA=(A^TA)^{-1}(A^TA)=I$. Since A^{-1} already exists, right multiplication by it gives $X=XAA^{-1}=A^{-1}$, establishing

$$\boxed{A^{-1}=(A^TA)^{-1}A^T}.$$

This is an algebraic identity, not a recommendation to compute the inverse by forming A^TA in floating-point arithmetic.

Exercise 2.S.11: Vandermonde matrices and interpolation

(a) Row i of Vc is $c_0+c_1x_i+\cdots+c_{n-1}x_i^{n-1}=p(x_i)$. Thus $Vc=y$ states precisely $p(x_i)=y_i$ for every i.

(b) Suppose $Vc=0$. Then p vanishes at the n distinct numbers $x_1,\ldots,x_n$. A nonzero polynomial of degree at most $n-1$ can have at most $n-1$ distinct roots, so p must be the zero polynomial. All its coefficients c_j are therefore zero. Thus $Vc=0$ has only the trivial solution, proving that the columns of V are independent.

(c) The square matrix V is invertible by the IMT, using part (b). For every arbitrary data vector y, choose $c=V^{-1}y$ and form the polynomial with those coefficients. Part (a) shows it interpolates every data point, and its degree is at most $n-1$. The polynomial is also unique: any second coefficient vector would be another solution of $Vc=y$.

Exercise 2.S.12: Read the first two columns of an LU product

The first column of upper triangular U is $(u_{11},0,\ldots,0)^T$, so $a_1=Lu_1=u_{11}l_1$. The second column of U is $(u_{12},u_{22},0,\ldots,0)^T$, so $a_2=u_{12}l_1+u_{22}l_2$. Thus the first column of A is a multiple of column 1 of L, and the second is a combination of columns 1 and 2 of L. Triangularity, not a mysterious cancellation, makes the later coefficients zero.

Exercise 2.S.13: Projection and reflection identities

Since $u^Tu=1$ is a scalar, associativity gives

$$\text{(a)}\quad P^2=(uu^T)(uu^T)=u(u^Tu)u^T=uu^T=P.$$

For the transpose,

$$\text{(b)} \quad P^T = (uu^T)^T = (u^T)^T u^T = uu^T = P.$$

Finally distribute the product using $P^2 = P$:

$$\text{(c)} \quad Q^2 = (I - 2P)^2 = I - 4P + 4P^2 = I.$$

Thus Q undoes itself: applying the same reflection twice returns every vector to its original position.

Exercise 2.S.14: A coordinate-plane reflection

With $u = e_3$,

$$P = uu^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad Q = I - 2P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

For $x = (1, 5, 3)^T$ this gives $Px = (0, 0, 3)^T$ and $Qx = (1, 5, -3)^T$. The vector $x - Px = (1, 5, 0)^T$ is the component in the coordinate plane; Q preserves it and reverses the perpendicular component Px.

Exercise 2.S.15: A product of elementary operations gives row equivalence

Multiplication by E_1 performs one elementary row operation on B, then E_2 performs another and E_3 a third. Consequently $C = E_3E_2E_1B$ is obtained from B by a finite sequence of row operations, which is the definition of row equivalence. The reverse sequence uses $E_3^{-1}, E_2^{-1}, E_1^{-1}$ in that temporal order, so the relationship works both ways.

Exercise 2.S.16: Construct a nonzero matrix with zero product

Because square A is singular, $Av = 0$ has a nonzero solution v. Choose one by row reduction, then set $B = [v \ 0 \ \cdots \ 0]$. The first column makes $B \neq 0$, whereas $AB = [Av \ A0 \ \cdots \ A0] = 0$. This construction works even if the null space is only one-dimensional.

Exercise 2.S.17: A product that factors through four dimensions

Every column of AB is a combination of the four columns of the 6×4 matrix A. Therefore $\text{Col}(AB) \subseteq \text{Col}\, A$, whose dimension is at most four. It cannot equal $\mathbb{R}^6$, whose dimension is six. The 6×6 matrix AB is thus not onto and cannot be invertible by the IMT.

Exercise 2.S.18: A left inverse gives uniqueness for attainable outputs

If $Ax = b$, then $x = CAx = Cb$. Thus every solution must equal Cb. The premise already says a solution exists, so it is unique. Equivalently, if $Ax_1 = Ax_2$, multiplication by C yields $x_1 = x_2$. The 5×3 matrix need not attain every $b \in \mathbb{R}^5$; the proof establishes uniqueness when existence has separately been supplied.

Exercise 2.S.19: Observe and explain limits of stochastic powers

Both matrices have nonnegative entries and each column sums to one. Such a matrix is called column-stochastic: it preserves the sum of a vector's coordinates. Exact rational powers give

$$A^2 = \begin{bmatrix} .31 & .26 & .30 \\ .39 & .48 & .39 \\ .30 & .26 & .31 \end{bmatrix}, \qquad B^2 = \begin{bmatrix} .29 & .18 & .18 \\ .33 & .44 & .33 \\ .38 & .38 & .49 \end{bmatrix}.$$

For $k = 8$, the powers are approximately

$$A^8 \approx \begin{bmatrix} .285728 & .285696 & .285728 \\ .428543 & .428609 & .428543 \\ .285728 & .285696 & .285728 \end{bmatrix}, \quad B^8 \approx \begin{bmatrix} .202364 & .202218 & .202218 \\ .370732 & .370879 & .370732 \\ .426904 & .426904 & .427050 \end{bmatrix}.$$

By $k = 16$ their columns are close to the respective vectors

$$p_A = (2/7, 3/7, 2/7)^T, \qquad p_B = (18/89, 33/89, 38/89)^T.$$

These are verified exactly by $Ap_A = p_A$, $Bp_B = p_B$, and coordinate sums one. The powers tend toward matrices having every column equal to the corresponding p. An optional explanation from later eigenvalue theory is that A's other eigenvalues are $.3, .1$, and B's are $\pm\sqrt{11}/10$; all have magnitude less than one, so their powers die out.

Here is a further reproducible investigation with positive column-stochastic matrices:

```
import numpy as np
rng = np.random.default_rng(22019)
C = rng.integers(1,10,(3,3)).astype(float)
C = C/C.sum(axis=0)
print(np.linalg.matrix_power(C,16))
```

The executed result has nearly identical columns $(.20882917, .35930902, .43186180)^T$. A precise conjecture supported by these examples is that positive stochastic matrices, and more generally stochastic matrices with some positive power, have powers converging to a matrix with identical columns. It would be false to remove that extra condition: the permutation matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is stochastic but its powers alternate instead of converging. Thus the experiment suggests a qualified theorem, not a claim about every stochastic matrix.

Exercise 2.S.20: A closed formula for inverses of zero-diagonal ones matrices

Let J_n have every entry equal to one. Then $A_n = J_n - I_n$ and $J_n^2 = nJ_n$, since each entry of the square is a sum of n ones. For $n \geq 2$ propose

$$B_n = -I_n + \frac{1}{n-1} J_n.$$

Its diagonal entries are $-(n-2)/(n-1)$ and off-diagonal entries are $1/(n-1)$. In particular the computed inverses for the three requested sizes are

$$B_4 = \frac{1}{3} \begin{bmatrix} -2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 1 \\ 1 & 1 & -2 & 1 \\ 1 & 1 & 1 & -2 \end{bmatrix},$$

$$B_5 = \frac{1}{4}\begin{bmatrix} -3 & 1 & 1 & 1 & 1 \\ 1 & -3 & 1 & 1 & 1 \\ 1 & 1 & -3 & 1 & 1 \\ 1 & 1 & 1 & -3 & 1 \\ 1 & 1 & 1 & 1 & -3 \end{bmatrix},$$

$$B_6 = \frac{1}{5}\begin{bmatrix} -4 & 1 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & 1 & -4 \end{bmatrix}.$$

To turn the conjecture into a proof, multiply symbolically:

$$(J_n - I_n)B_n = I_n + \left(-1 + \frac{n}{n-1} - \frac{1}{n-1}\right) J_n = I_n.$$

The reverse product is identical because both factors are combinations of I_n and J_n. Hence $B_n = A_n^{-1}$ for all $n \geq 2$. For $n = 1$, $A_1 = [0]$ is singular and the denominator $n-1$ correctly signals the excluded case.

3.1 Introduction to determinants

A determinant is a number attached to a square matrix. For a 2×2 matrix it is $ad - bc$. In a larger matrix, delete row i and column j to obtain the minor matrix A_{ij}; its cofactor is the *number* $C_{ij} = (-1)^{i+j} \det A_{ij}$. The signs form a checkerboard, beginning with plus in position $(1, 1)$.

A cofactor expansion multiplies every entry of one chosen row or column by its cofactor, then adds. A zero entry contributes zero. Row or column positions in each new minor must be counted anew: they are not the original positions. For numerical checks below all arithmetic was also performed exactly with rational-number matrix computations.

Practice problem 3.1.P1: A sparse fourth-order determinant

What we must show. Compute the determinant without expanding sixteen terms.

Plan. Column 3 has just one nonzero entry, 2 in row 1. Its sign is $(-1)^{1+3} = 1$.

$$D = 2 \begin{vmatrix} 0 & 3 & -4 \\ -5 & -8 & 3 \\ 0 & 5 & -6 \end{vmatrix}.$$

Expand that minor down its first column. The only nonzero entry is now -5 in position $(2, 1)$, whose cofactor sign is negative. Hence

$$D = 2[-(-5)] \begin{vmatrix} 3 & -4 \\ 5 & -6 \end{vmatrix} = 10[3(-6) - (-4)5] = 10(2) = 20.$$

Conclusion. The determinant is 20. Keeping track of the sign and the negative entry separately prevents a common sign error.

Exercise 3.1.1: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (3)(-1)^{1+1} \begin{vmatrix} 3 & 2 \\ 5 & -1 \end{vmatrix} + (4)(-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 0 & 5 \end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\det A_{11} = (3)(-1) - (2)(5) = -13$$
$$\det A_{13} = (2)(5) - (3)(0) = 10.$$

Check by expanding down column 2, as requested.

$$\det A = (3)(-1)^{2+2}\begin{vmatrix}3 & 4\\ 0 & -1\end{vmatrix} + (5)(-1)^{3+2}\begin{vmatrix}3 & 4\\ 2 & 2\end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\det A_{22} = (3)(-1) - (4)(0) = -3$$
$$\det A_{32} = (3)(2) - (4)(2) = -2.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 1$.

Exercise 3.1.2: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix}0 & 5 & 1\\ 4 & -3 & 0\\ 2 & 4 & 1\end{bmatrix}.$$

Expand across row 1.

$$\det A = (5)(-1)^{1+2}\begin{vmatrix}4 & 0\\ 2 & 1\end{vmatrix} + (1)(-1)^{1+3}\begin{vmatrix}4 & -3\\ 2 & 4\end{vmatrix} = 2.$$

The 2×2 determinants used above are

$$\det A_{12} = (4)(1) - (0)(2) = 4$$
$$\det A_{13} = (4)(4) - (-3)(2) = 22.$$

Check by expanding down column 2, as requested.

$$\det A = (5)(-1)^{1+2}\begin{vmatrix}4 & 0\\ 2 & 1\end{vmatrix} + (-3)(-1)^{2+2}\begin{vmatrix}0 & 1\\ 2 & 1\end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix}0 & 1\\ 4 & 0\end{vmatrix} = 2.$$

The 2×2 determinants used above are

$$\det A_{12} = (4)(1) - (0)(2) = 4$$
$$\det A_{22} = (0)(1) - (1)(2) = -2$$
$$\det A_{32} = (0)(0) - (1)(4) = -4.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 2$.

Exercise 3.1.3: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (2)(-1)^{1+1}\begin{vmatrix} 1 & 2 \\ 4 & -1 \end{vmatrix} + (-4)(-1)^{1+2}\begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + (3)(-1)^{1+3}\begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} = -5.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (1)(-1) - (2)(4) = -9 \\ \det A_{12} &= (3)(-1) - (2)(1) = -5 \\ \det A_{13} &= (3)(4) - (1)(1) = 11. \end{aligned}$$

Check by expanding down column 2, as requested.

$$\det A = (-4)(-1)^{1+2}\begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + (1)(-1)^{2+2}\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{12} &= (3)(-1) - (2)(1) = -5 \\ \det A_{22} &= (2)(-1) - (3)(1) = -5 \\ \det A_{32} &= (2)(2) - (3)(3) = -5. \end{aligned}$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = -5$.

Exercise 3.1.4: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 1 \\ 3 & 4 & 2 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (1)(-1)^{1+1}\begin{vmatrix} 1 & 1 \\ 4 & 2 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (5)(-1)^{1+3}\begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} = 20.$$

The 2×2 determinants used above are

$$\det A_{11} = (1)(2) - (1)(4) = -2$$
$$\det A_{12} = (2)(2) - (1)(3) = 1$$
$$\det A_{13} = (2)(4) - (1)(3) = 5.$$

Check by expanding down column 2, as requested.

$$\det A = (3)(-1)^{1+2}\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (1)(-1)^{2+2}\begin{vmatrix} 1 & 5 \\ 3 & 2 \end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix} = 20.$$

The 2×2 determinants used above are

$$\det A_{12} = (2)(2) - (1)(3) = 1$$
$$\det A_{22} = (1)(2) - (5)(3) = -13$$
$$\det A_{32} = (1)(1) - (5)(2) = -9.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 20$.

Exercise 3.1.5: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 2 & 3 & -4 \\ 4 & 0 & 5 \\ 5 & 1 & 6 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (2)(-1)^{1+1}\begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} + (-4)(-1)^{1+3}\begin{vmatrix} 4 & 0 \\ 5 & 1 \end{vmatrix} = -23.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(6) - (5)(1) = -5$$
$$\det A_{12} = (4)(6) - (5)(5) = -1$$
$$\det A_{13} = (4)(1) - (0)(5) = 4.$$

Conclusion. $\det A = -23$.

Exercise 3.1.6: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 5 & -2 & 4 \\ 0 & 3 & -5 \\ 2 & -4 & 7 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (5)(-1)^{1+1}\begin{vmatrix} 3 & -5 \\ -4 & 7 \end{vmatrix} + (-2)(-1)^{1+2}\begin{vmatrix} 0 & -5 \\ 2 & 7 \end{vmatrix} + (4)(-1)^{1+3}\begin{vmatrix} 0 & 3 \\ 2 & -4 \end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (3)(7) - (-5)(-4) = 1 \\ \det A_{12} &= (0)(7) - (-5)(2) = 10 \\ \det A_{13} &= (0)(-4) - (3)(2) = -6. \end{aligned}$$

Conclusion. $\det A = 1$.

Exercise 3.1.7: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 3 & 0 \\ 6 & 5 & 2 \\ 9 & 7 & 3 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (4)(-1)^{1+1}\begin{vmatrix} 5 & 2 \\ 7 & 3 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 6 & 2 \\ 9 & 3 \end{vmatrix} = 4.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (5)(3) - (2)(7) = 1 \\ \det A_{12} &= (6)(3) - (2)(9) = 0. \end{aligned}$$

Conclusion. $\det A = 4$.

Exercise 3.1.8: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 8 & 1 & 6 \\ 4 & 0 & 3 \\ 3 & -2 & 5 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (8)(-1)^{1+1}\begin{vmatrix} 0 & 3 \\ -2 & 5 \end{vmatrix} + (1)(-1)^{1+2}\begin{vmatrix} 4 & 3 \\ 3 & 5 \end{vmatrix} + (6)(-1)^{1+3}\begin{vmatrix} 4 & 0 \\ 3 & -2 \end{vmatrix} = -11.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(5) - (3)(-2) = 6$$
$$\det A_{12} = (4)(5) - (3)(3) = 11$$
$$\det A_{13} = (4)(-2) - (0)(3) = -8.$$

Conclusion. $\det A = -11$.

Exercise 3.1.9: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix} = (2)(-1)^{3+1} D_2 = 10.$$

$$D_2 = \begin{vmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 8 \end{vmatrix} = (5)(-1)^{1+3} D_3 = 5.$$

$$D_3 = \begin{vmatrix} 7 & 2 \\ 3 & 1 \end{vmatrix} = (7)(1) - (2)(3) = 1.$$

Thus the answer is 10.

Exercise 3.1.10: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix} = (3)(-1)^{2+3} D_2 = -6.$$

$$D_2 = \begin{vmatrix} 1 & -2 & 2 \\ 2 & -6 & 5 \\ 5 & 0 & 4 \end{vmatrix} = (5)(-1)^{3+1} D_3 + (4)(-1)^{3+3} D_4 = 2.$$

$$D_4 = \begin{vmatrix} 1 & -2 \\ 2 & -6 \end{vmatrix} = (1)(-6) - (-2)(2) = -2.$$

$$D_3 = \begin{vmatrix} -2 & 2 \\ -6 & 5 \end{vmatrix} = (-2)(5) - (2)(-6) = 2.$$

Thus the answer is -6.

Exercise 3.1.11: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{vmatrix} = (2)(-1)^{4+4} D_2 = -12.$$

$$D_2 = \begin{vmatrix} 3 & 5 & -8 \\ 0 & -2 & 3 \\ 0 & 0 & 1 \end{vmatrix} = (1)(-1)^{3+3} D_3 = -6.$$

$$D_3 = \begin{vmatrix} 3 & 5 \\ 0 & -2 \end{vmatrix} = (3)(-2) - (5)(0) = -6.$$

Thus the answer is -12.

Exercise 3.1.12: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the

deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{vmatrix} = (4)(-1)^{1+1} D_2 = 36.$$

$$D_2 = \begin{vmatrix} -1 & 0 & 0 \\ 6 & 3 & 0 \\ -8 & 4 & -3 \end{vmatrix} = (-1)(-1)^{1+1} D_3 = 9.$$

$$D_3 = \begin{vmatrix} 3 & 0 \\ 4 & -3 \end{vmatrix} = (3)(-3) - (0)(4) = -9.$$

Thus the answer is 36.

Exercise 3.1.13: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix} = (2)(-1)^{2+3} D_2 = 6.$$

$$D_2 = \begin{vmatrix} 4 & 0 & 3 & -5 \\ 7 & 3 & 4 & -8 \\ 5 & 0 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{vmatrix} = (3)(-1)^{2+2} D_3 = -3.$$

$$D_3 = \begin{vmatrix} 4 & 3 & -5 \\ 5 & 2 & -3 \\ 0 & -1 & 2 \end{vmatrix} = (-1)(-1)^{3+2} D_4 + (2)(-1)^{3+3} D_5 = -1.$$

$$D_5 = \begin{vmatrix} 4 & 3 \\ 5 & 2 \end{vmatrix} = (4)(2) - (3)(5) = -7.$$

$$D_4 = \begin{vmatrix} 4 & -5 \\ 5 & -3 \end{vmatrix} = (4)(-3) - (-5)(5) = 13.$$

Thus the answer is 6.

Exercise 3.1.14: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix} = (3)(-1)^{4+1} D_2 = 9.$$

$$D_2 = \begin{vmatrix} 3 & 2 & 4 & 0 \\ 0 & -4 & 1 & 0 \\ -5 & 6 & 7 & 1 \\ 2 & 3 & 2 & 0 \end{vmatrix} = (1)(-1)^{3+4} D_3 = -3.$$

$$D_3 = \begin{vmatrix} 3 & 2 & 4 \\ 0 & -4 & 1 \\ 2 & 3 & 2 \end{vmatrix} = (-4)(-1)^{2+2} D_4 + (1)(-1)^{2+3} D_5 = 3.$$

$$D_5 = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = (3)(3) - (2)(2) = 5.$$

$$D_4 = \begin{vmatrix} 3 & 4 \\ 2 & 2 \end{vmatrix} = (3)(2) - (4)(2) = -2.$$

Thus the answer is 9.

Exercise 3.1.15: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}$. The three downward products are -9, 0, 40, and the three upward products are 0, 30, 0. Add the downward products and subtract the entire upward sum:

$$\det A = (-9 + 0 + 40) - (0 + 30 + 0) = 1.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.16: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 0 & 5 & 1 \\ 4 & -3 & 0 \\ 2 & 4 & 1 \end{bmatrix}$. The three downward products are 0, 0, 16, and the three upward products are -6, 0, 20. Add the downward products and subtract the entire upward sum:

$$\det A = (0 + 0 + 16) - (-6 + 0 + 20) = 2.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.17: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{bmatrix}$. The three downward products are -2, -8, 36, and the three upward products are 3, 16, 12. Add the downward products and subtract the entire upward sum:

$$\det A = (-2 + -8 + 36) - (3 + 16 + 12) = -5.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.18: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 1 \\ 3 & 4 & 2 \end{bmatrix}$. The three downward products are 2, 9, 40, and the three upward products are 15, 4, 12. Add the downward products and subtract the entire upward sum:

$$\det A = (2 + 9 + 40) - (15 + 4 + 12) = 20.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.19: A row operation and its determinant

The second matrix is obtained by $R_1 \leftrightarrow R_2$. Its determinant is $cb - da = -(ad - bc)$. Thus a row interchange reverses the determinant's sign.

Exercise 3.1.20: A row operation and its determinant

The operation is $R_2 \leftarrow kR_2$. Direct calculation gives $a(kd) - b(kc) = k(ad - bc)$. Multiplying one row by k multiplies the determinant by k.

Exercise 3.1.21: A row operation and its determinant

The operation is $R_2 \leftarrow R_2 + kR_1$. The old determinant is $3 \cdot 6 - 4 \cdot 5 = -2$; the new one is $3(6 + 4k) - 4(5 + 3k) = 18 + 12k - 20 - 12k = -2$. The added terms cancel, so the determinant is unchanged.

Exercise 3.1.22: A row operation and its determinant

The operation is $R_1 \leftarrow R_1 + kR_2$. The new determinant is $(a + kc)d - (b + kd)c = ad + kcd - bc - kdc = ad - bc$. Thus row replacement preserves the determinant.

Exercise 3.1.23: A row operation and its determinant

The first row $(1, 1, 1)$ is replaced by $(k, k, k) = k(1, 1, 1)$. To see the factor directly, expand along row 1: every term in the expansion gains exactly one factor k. Thus the new determinant is k times the old one (also when $k = 0$).

Exercise 3.1.24: A row operation and its determinant

The operation interchanges rows 1 and 2. Interchanging these two rows reverses the sign of every signed product in the determinant, so the new determinant is the negative of the old determinant.

Exercise 3.1.25: An elementary-matrix determinant

This lower triangular matrix has diagonal $1, 1, 1$. Its determinant is their product, 1. It performs $R_3 \leftarrow R_3 + kR_2$ on another matrix.

Exercise 3.1.26: An elementary-matrix determinant

The matrix is lower triangular with diagonal $1, 1, 1$, so its determinant is 1. Its row operation is $R_3 \leftarrow R_3 + kR_1$.

Exercise 3.1.27: An elementary-matrix determinant

The diagonal matrix has diagonal $k, 1, 1$, so its determinant is k. It scales row 1 by k. For an elementary matrix this operation requires $k \neq 0$; the displayed determinant formula is still valid at $k = 0$.

Exercise 3.1.28: An elementary-matrix determinant

The diagonal entries are $1, k, 1$, giving determinant k. The operation scales row 2 by k; an elementary scaling has $k \neq 0$.

Exercise 3.1.29: An elementary-matrix determinant

The matrix is obtained from I_3 by interchanging rows 1 and 2, so its determinant is -1. Direct expansion along row 3 gives $1 \det \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1$.

Exercise 3.1.30: An elementary-matrix determinant

The matrix interchanges rows 1 and 3 of I_3. Its determinant is -1. Expansion across row 2 gives $1(-1)^{2+2} \det \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1$.

Exercise 3.1.31: An elementary-matrix determinant

A row-replacement elementary matrix is obtained from I_n by inserting a scalar k into an off-diagonal position (i, j). Expand by rows different from i, j to reduce to the 2×2 block $\left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$ or its transpose. The determinant is 1. Equivalently, adding k times one identity row to another creates no new nonzero signed product. The examples in Exercises 25 and 26 illustrate this general result.

Exercise 3.1.32: An elementary-matrix determinant

A scaling elementary matrix is diagonal with one diagonal entry $k \neq 0$ and all others 1. The triangular-matrix theorem therefore gives determinant k. It is one row that is scaled, not all n rows.

Exercise 3.1.33: Verify multiplication for one elementary matrix

Here $E = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, so $EA = \left[\begin{smallmatrix} c & d \\ a & b \end{smallmatrix}\right]$. Consequently $\det(EA) = cb - da = -(ad - bc)$, while $\det E = -1$. Thus $\det(EA) = (\det E)(\det A)$.

Exercise 3.1.34: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & 0 \\ 0 & k \end{smallmatrix}\right]$, we have $EA = \left[\begin{smallmatrix} a & b \\ kc & kd \end{smallmatrix}\right]$. Its determinant is $akd - bkc = k(ad - bc)$. Since $\det E = k$, the required equality follows.

Exercise 3.1.35: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$, multiplication gives $EA = \left[\begin{smallmatrix} a+kc & b+kd \\ c & d \end{smallmatrix}\right]$. Its determinant is $ad+kcd-bc-kdc = ad-bc$. Also $\det E = 1$, so the two sides agree.

Exercise 3.1.36: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & 0 \\ k & 1 \end{smallmatrix}\right]$, multiplication gives $EA = \left[\begin{smallmatrix} a & b \\ ka+c & kb+d \end{smallmatrix}\right]$. Its determinant is $a(kb+d)-b(ka+c) = ad-bc$. This equals $(\det E)(\det A)$ because $\det E = 1$.

Exercise 3.1.37: Determinants and their interpretation

Scalar multiplication scales every entry, so $5A = \left[\begin{smallmatrix} 15 & 5 \\ 20 & 10 \end{smallmatrix}\right]$. We find $\det A = 3 \cdot 2 - 1 \cdot 4 = 2$, but $\det(5A) = 15 \cdot 10 - 5 \cdot 20 = 50$, whereas $5 \det A = 10$. Thus the proposed equality is false. Both rows were multiplied by 5, giving a factor 5^2.

Exercise 3.1.38: Determinants and their interpretation

Write $kA = \left[\begin{smallmatrix} ka & kb \\ kc & kd \end{smallmatrix}\right]$. Then $\det(kA) = (ka)(kd) - (kb)(kc) = k^2(ad - bc) = k^2 \det A$. This calculation also covers $k = 0$.

Exercise 3.1.39: Determinants and their interpretation

(a) True, with the usual initial case.

For $n \geq 2$, the recursive definition expands into determinants of $(n-1) \times (n-1)$ minor matrices. A 1×1 determinant is its sole entry, the base case.

(b) False: a cofactor is a scalar.

Deleting a row and column produces the matrix A_{ij}. The cofactor is $(-1)^{i+j} \det A_{ij}$, a signed number, not that submatrix.

Exercise 3.1.40: Determinants and their interpretation

(a) False.

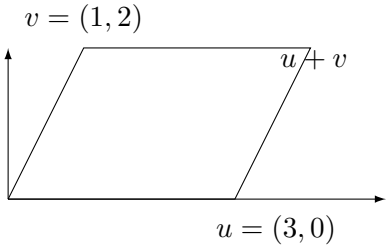
Row and column expansions compute the same number $\det A$, not negatives in general. For example, both expansions of I_2 give 1.

(b) False.

For a triangular matrix the determinant is the *product* of diagonal entries. For I_2 the determinant is 1, whereas the sum of the diagonal is 2.

Exercise 3.1.41: Determinants and their interpretation

What we must show. Compare unsigned area with a determinant, and then vary the shear. The two adjacent edges are $u = (3, 0)$ and $v = (1, 2)$. The horizontal base has length 3 and perpendicular height 2, so the area is $3 \cdot 2 = 6$. Also $\det[u\ v] = 3 \cdot 2 - 0 \cdot 1 = 6$. Replacing v by $(x, 2)$ leaves its height above the horizontal base equal to 2, hence leaves the area 6; the determinant is still $3 \cdot 2 - 0x = 6$.

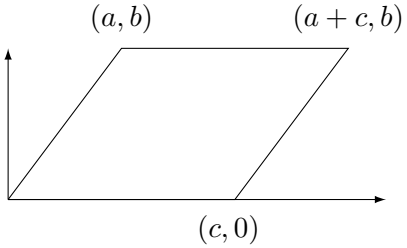


The horizontal displacement changes the slant, not the height. This operation is a shear; the sketch uses $x = 1$, and the argument works for every real x.

Exercise 3.1.42: Determinants and their interpretation

Take the horizontal base $v = (c, 0)$ of length $c > 0$. The other edge $u = (a, b)$ has height $b > 0$, so the area is bc. On the other hand,

$$\det[u\ v] = \begin{vmatrix} a & c \\ b & 0 \end{vmatrix} = -bc, \qquad \det[v\ u] = \begin{vmatrix} c & a \\ 0 & b \end{vmatrix} = bc.$$



The sketch is schematic. Interchanging the order of the columns changes orientation and the determinant's sign, but the ordinary area is nonnegative and equals either determinant's absolute value.

Exercise 3.1.43: A reproducible test of determinant additivity

What we must show. Test additivity with four reproducible randomly generated pairs. A computer experiment supplies evidence, not a proof of a universal identity. Here is an exact integer experiment; a fixed seed lets another reader repeat the same draws.

```
from random import Random
from sympy import Matrix
r = Random(31043)
for n in (5,2,3,4):
    A = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    B = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    print(n, A.det(), B.det(), (A+B).det(),
          (A+B).det()-A.det()-B.det())
```

n	$\det A$	$\det B$	$\det(A+B)$	difference
5	10070	136248	121574	−24744
2	−34	1	−78	−45
3	888	−55	1464	631
4	176	748	−6819	−7743

The nonzero differences disprove additivity. A particularly short exact counterexample is $A = B = I_2$: $\det(A + B) = 4$, but $\det A + \det B = 2$. The determinant is linear in *one row at a time with the other rows fixed*, not linear in the entire matrix.

Exercise 3.1.44: A reproducible test of multiplication

Use the same four random pairs and seed from Exercise 43. Replace the last line of that code by

```
print(n, (A*B).det(), A.det()*B.det(),
      (A*B).det()-A.det()*B.det())
```

The exact outputs are

n	$\det(AB)$	$(\det A)(\det B)$	difference
5	1372017360	1372017360	0
2	−34	−34	0
3	−48840	−48840	0
4	131648	131648	0

All four differences are zero. This suggests $\det(AB) = (\det A)(\det B)$. The next section proves it for all square matrices, so the experiment is not being substituted for a proof.

Exercise 3.1.45: Scaling and transposition in several dimensions

Generate three 4×4 samples and two each of sizes 5 and 6. The following code carries out all comparisons, including transpose, with exact arithmetic.

```
from random import Random
from sympy import Matrix
r = Random(31045)
for n in (4,4,4,5,5,6,6):
    A = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    d = A.det()
    assert A.T.det() == d
    print(n,d,(-A).det(),(2*A).det(),(10*A).det())
```

The transpose check passed for every sample; the remaining outputs are

n	$\det A$	$\det(-A)$	$\det(2A)$	$\det(10A)$
4	2446	2446	39136	24460000
4	3572	3572	57152	35720000
4	2144	2144	34304	21440000
5	150900	−150900	4828800	15090000000
5	29441	−29441	942112	2944100000
6	−122640	−122640	−7848960	−122640000000
6	−191568	−191568	−12260352	−191568000000

The conjectures are $\det A^T = \det A$ and $\det(kA) = k^n \det A$. In particular, negation preserves the determinant in even dimensions and reverses its sign in odd dimensions. Scaling the matrix scales *every* row, so the scale factor occurs n times. The integer calculations above check that description at every sampled size.

Exercise 3.1.46: A determinant of an inverse

Use seed 31046 and the same integer generator as Exercise 45, now at sizes $4, 5, 6$. For each generated matrix, compute

```
d = A.det()
if d != 0:
    print(n,d,A.inv().det())
else:
    print(A.echelon_form())
```

The three sampled matrices were nonsingular and gave

n	$\det A$	$\det(A^{-1})$
4	-1586	$-\frac{1}{1586}$
5	34641	$\frac{1}{34641}$
6	-48084	$-\frac{1}{48084}$

Each last entry is the reciprocal of the preceding one. Thus the conjecture is $\det(A^{-1}) = 1/\det A$ when A is invertible. A zero determinant would instead lead to a missing pivot in echelon form and no inverse; the reciprocal expression would then be undefined. Multiplicativity in the next section proves the conjecture by applying it to $AA^{-1} = I$.

3.2 Properties of determinants

A row replacement preserves a determinant, a row interchange reverses its sign, and multiplying a single row by k multiplies it by k. In the calculations below, keeping the operations visible is how we avoid losing a sign or a scale factor. Once a matrix is triangular its determinant is the product of its diagonal entries.

A square matrix is invertible exactly when its determinant is nonzero. Thus n vectors in $\mathbb{R}^n$ are linearly independent exactly when the determinant of the matrix having those vectors as columns is nonzero. The multiplication theorem says $\det(AB) = \det A \det B$, and transposition preserves a determinant.

Practice problem 3.2.P1: Create a zero row

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_4 \leftarrow R_4 + 3R_1$. Both resulting rows 2 and 4 are $(0, 1, -3, 2)$. Then $R_4 \leftarrow R_4 - R_2$ creates a zero row. All three operations preserve the determinant, and expansion along the zero row gives $\det A = 0$.

Practice problem 3.2.P2: A determinant tests independence

Place the three vectors in columns, not rows:

$$A = \begin{bmatrix} 5 & -3 & 2 \\ -7 & 3 & -7 \\ 9 & -5 & 5 \end{bmatrix}.$$

Replace R_2 by $R_2 + R_1 = (-2, 0, -5)$. Expanding down column 2 now gives

$$\det A = 3\begin{vmatrix} -2 & -5 \\ 9 & 5 \end{vmatrix} + 5\begin{vmatrix} 5 & 2 \\ -2 & -5 \end{vmatrix} = 3(35) + 5(-21) = 0.$$

Thus A is not invertible, and the three columns are linearly dependent by the Invertible Matrix Theorem.

Exercise 3.2.1: Identify the determinant rule

The displayed matrices differ by $R_1 \leftrightarrow R_2$. A single interchange reverses the determinant's sign, which explains the minus sign in the equation.

Exercise 3.2.2: Identify the determinant rule

The first row $(2, -6, 4)$ is $2(1, -3, 2)$, while the other rows are fixed. Linearity in one row gives exactly the displayed factor 2.

Exercise 3.2.3: Identify the determinant rule

The right-hand matrix results from $R_2 \leftarrow R_2 - 2R_1$, since $(2, 0, -3) - 2(1, 3, -4) = (0, -6, 5)$. A row replacement preserves the determinant.

Exercise 3.2.4: Identify the determinant rule

Apply $R_3 \leftarrow R_3 - 3R_1$: $(3,7,4) - 3(1,2,3) = (0,1,-5)$. This is a row replacement and does not change the determinant.

Exercise 3.2.5: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & -6 \\ -1 & -4 & 4 \\ -2 & -7 & 9 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-1)R_1, \quad R_3 \leftarrow R_3 - (-2)R_1$, giving

$$\begin{bmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 3 & -3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (3)R_2$, giving

$$\begin{bmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(3) = 3.$$

Exercise 3.2.6: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & -3 \\ 3 & -3 & 3 \\ 2 & 13 & -7 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (3)R_1, \quad R_3 \leftarrow R_3 - (2)R_1$, giving

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -18 & 12 \\ 0 & 3 & -1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{1}{6})R_2$, giving

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -18 & 12 \\ 0 & 0 & 1 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(-18)(1) = -18.$$

Exercise 3.2.7: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-2)R_1, \quad R_3 \leftarrow R_3 - (3)R_1, \quad R_4 \leftarrow R_4 - (1)R_1$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & -4 & 2 & -5 \\ 0 & -4 & 2 & -5 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-4)R_2, \quad R_4 \leftarrow R_4 - (-4)R_2$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 30 & 27 \\ 0 & 0 & 30 & 27 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (1)R_3$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 30 & 27 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(30)(0) = 0.$$

Exercise 3.2.8: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 2 & 5 & 4 & -3 \\ -3 & -7 & -5 & 2 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3 - (2)R_1, \quad R_4 \leftarrow R_4 - (-3)R_1$, giving

$$\begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 0 & -1 & -2 & 5 \\ 0 & 2 & 4 & -10 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-1)R_2, \quad R_4 \leftarrow R_4 - (2)R_2$, giving

$$\begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(0)(0) = 0.$$

Exercise 3.2.9: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 2 & 8 & 5 \\ 3 & -1 & -2 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3 - (-1)R_1, \quad R_4 \leftarrow R_4 - (3)R_1$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 1 & 5 & 5 \\ 0 & 2 & 7 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (1)R_2, \quad R_4 \leftarrow R_4 - (2)R_2$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -3 & -5 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & -3 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(1)(1)(-3)(1) = 3.$$

Exercise 3.2.10: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ -2 & -6 & 2 & 3 & 9 \\ 3 & 7 & -3 & 8 & -7 \\ 3 & 5 & 5 & 2 & 7 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3-(-2)R_1, \quad R_4 \leftarrow R_4-(3)R_1, \quad R_5 \leftarrow R_5-(3)R_1$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & -2 & 0 & 8 & -1 \\ 0 & -4 & 8 & 2 & 13 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4-(-1)R_2, \quad R_5 \leftarrow R_5-(-2)R_2$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & 0 & -4 & 7 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & -4 & 7 & -7 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(1)(2)(-4)(3)(1) = 24.$$

Exercise 3.2.11: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2-(\frac{3}{2})R_1, \quad R_3 \leftarrow R_3-(-3)R_1, \quad R_4 \leftarrow R_4-(2)R_1$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 15 & -13 & 6 \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3-(-2)R_2$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-1)R_3$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-\frac{15}{2})(-2)(4) = 120.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.12: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 5 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-3)R_1, \quad R_3 \leftarrow R_3 - (-5)R_1, \quad R_4 \leftarrow R_4 - (-4)R_1$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 14 & 21 & 6 \\ 0 & 10 & 16 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{7}{5})R_2, \quad R_4 \leftarrow R_4 - (1)R_2$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 0 & \frac{21}{5} & 6 \\ 0 & 0 & 4 & 3 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (\frac{20}{21})R_3$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 0 & \frac{21}{5} & 6 \\ 0 & 0 & 0 & -\frac{19}{7} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(-1)(10)(\frac{21}{5})(-\frac{19}{7}) = 114.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.13: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 2 & 5 & 4 & 1 \\ 4 & 7 & 6 & 2 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (2)R_1, \quad R_3 \leftarrow R_3 - (3)R_1, \quad R_4 \leftarrow R_4 - (-3)R_1$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & -17 & -16 & -3 \\ 0 & 22 & 19 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{17}{3})R_2, \quad R_4 \leftarrow R_4 - (-\frac{22}{3})R_2$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & 0 & -\frac{14}{3} & -3 \\ 0 & 0 & \frac{13}{3} & 3 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-\frac{13}{14})R_3$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & 0 & -\frac{14}{3} & -3 \\ 0 & 0 & 0 & \frac{3}{14} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-3)(-\frac{14}{3})(\frac{3}{14}) = 6.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.14: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -3 & 4 & -2 & 8 \\ 3 & -4 & 0 & 4 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-\frac{1}{3})R_1, \quad R_3 \leftarrow R_3 - (1)R_1, \quad R_4 \leftarrow R_4 - (-1)R_1$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 6 & -3 & 12 \\ 0 & -6 & 1 & 0 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{18}{7})R_2, \quad R_4 \leftarrow R_4 - (-\frac{18}{7})R_2$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 0 & -\frac{27}{7} & \frac{162}{7} \\ 0 & 0 & \frac{13}{7} & -\frac{78}{7} \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-\frac{13}{27})R_3$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 0 & -\frac{27}{7} & \frac{162}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(-3)(\frac{7}{3})(-\frac{27}{7})(0) = 0.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.15: Transform a known determinant

The third row is scaled by 5 and the other rows are fixed. Thus the determinant is $5 \cdot 7 = 35$.

Exercise 3.2.16: Transform a known determinant

The second row is scaled by 3, so the determinant is $3 \cdot 7 = 21$.

Exercise 3.2.17: Transform a known determinant

Interchanging rows 2 and 3 once gives the displayed matrix. Its determinant is -7.

Exercise 3.2.18: Transform a known determinant

Move the original third row to the top by two interchanges: first interchange rows 2 and 3, then rows 1 and 2. The two sign changes cancel, leaving determinant 7.

Exercise 3.2.19: Transform a known determinant

First scale the second row by 2, multiplying the determinant by 2. Then add row 1 to it, which preserves that new determinant. The answer is $2 \cdot 7 = 14$.

Exercise 3.2.20: Transform a known determinant

The operation $R_1 \leftarrow R_1 + R_2$ preserves the determinant, so the answer is 7.

Exercise 3.2.21: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{2})R_1, \quad R_3 \leftarrow R_3 - (\frac{1}{2})R_1$, giving

$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & \frac{3}{2} & 4 \\ 0 & \frac{1}{2} & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{1}{3})R_2$, giving

$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & \frac{3}{2} & 4 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(\frac{3}{2})(-\frac{1}{3}) = -1.$$

Since the determinant is -1, the matrix is invertible by the determinant criterion.

Exercise 3.2.22: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 5 & 0 & -1 \\ 1 & -3 & -2 \\ 0 & 5 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{5})R_1$, giving

$$\begin{bmatrix} 5 & 0 & -1 \\ 0 & -3 & -\frac{9}{5} \\ 0 & 5 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{5}{3})R_2$, giving

$$\begin{bmatrix} 5 & 0 & -1 \\ 0 & -3 & -\frac{9}{5} \\ 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(5)(-3)(0) = 0.$$

Since the determinant is 0, the matrix is not invertible by the determinant criterion.

Exercise 3.2.23: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 2 & 0 & 0 & 8 \\ 1 & -7 & -5 & 0 \\ 3 & 8 & 6 & 0 \\ 0 & 7 & 5 & 4 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{2})R_1, \quad R_3 \leftarrow R_3 - (\frac{3}{2})R_1$, giving

$$\begin{bmatrix} 2 & 0 & 0 & 8 \\ 0 & -7 & -5 & -4 \\ 0 & 8 & 6 & -12 \\ 0 & 7 & 5 & 4 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{8}{7})R_2, \quad R_4 \leftarrow R_4 - (-1)R_2$, giving

$$\begin{bmatrix} 2 & 0 & 0 & 8 \\ 0 & -7 & -5 & -4 \\ 0 & 0 & \frac{2}{7} & -\frac{116}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-7)(\frac{2}{7})(0) = 0.$$

Since the determinant is 0, the matrix is not invertible by the determinant criterion.

Exercise 3.2.24: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 4 & -7 & -3 \\ 6 & 0 & -5 \\ -7 & 2 & 6 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{3}{2})R_1, \quad R_3 \leftarrow R_3 - (-\frac{7}{4})R_1$, giving

$$\begin{bmatrix} 4 & -7 & -3 \\ 0 & \frac{21}{2} & -\frac{1}{2} \\ 0 & -\frac{41}{4} & \frac{3}{4} \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{41}{42})R_2$, giving

$$\begin{bmatrix} 4 & -7 & -3 \\ 0 & \frac{21}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{11}{42} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(4)(\frac{21}{2})(\frac{11}{42}) = 11.$$

There are 3 vectors in $\mathbb{R}^3$, so the determinant test applies. The determinant is 11; hence the set is linearly independent.

Exercise 3.2.25: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 7 & -8 & 7 \\ -4 & 5 & 0 \\ -6 & 7 & -5 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-\frac{4}{7})R_1, \quad R_3 \leftarrow R_3 - (-\frac{6}{7})R_1$, giving

$$\begin{bmatrix} 7 & -8 & 7 \\ 0 & \frac{3}{7} & 4 \\ 0 & \frac{1}{7} & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{1}{3})R_2$, giving

$$\begin{bmatrix} 7 & -8 & 7 \\ 0 & \frac{3}{7} & 4 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(7)(\frac{3}{7})(-\frac{1}{3}) = -1.$$

There are 3 vectors in $\mathbb{R}^3$, so the determinant test applies. The determinant is -1; hence the set is linearly independent.

Exercise 3.2.26: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 3 & 2 & -2 & 0 \\ 5 & -6 & -1 & 0 \\ -6 & 0 & 3 & 0 \\ 4 & 7 & 0 & -3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{5}{3})R_1, \quad R_3 \leftarrow R_3 - (-2)R_1, \quad R_4 \leftarrow R_4 - (\frac{4}{3})R_1$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 4 & -1 & 0 \\ 0 & \frac{13}{3} & \frac{8}{3} & -3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{3}{7})R_2, \quad R_4 \leftarrow R_4 - (-\frac{13}{28})R_2$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{15}{4} & -3 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 0 & \frac{15}{4} & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(3)(-\frac{28}{3})(\frac{15}{4})(0) = 0.$$

There are 4 vectors in $\mathbb{R}^4$, so the determinant test applies. The determinant is 0; hence the set is linearly dependent.

Exercise 3.2.27: Properties, proofs, and exact calculations

(a) True.

This is the row-replacement part of Theorem 3.

(b) False as stated for an arbitrary echelon form.

The proposed sign correction accounts for row interchanges, but not row scalings. Starting with $A = I_2$, scale row 1 by 2. The resulting echelon form has pivot product 2 and no interchanges, although $\det A = 1$. If reduction uses *only* replacements and interchanges, the stated product with its sign correction is valid.

(c) True.

Dependent columns mean that A is not invertible, by the Invertible Matrix Theorem; Theorem 4 then gives $\det A = 0$.

(d) False.

Take $A = B = I_2$. The left side is $\det(2I_2) = 4$, while the right side is $1 + 1 = 2$.

Exercise 3.2.28: Properties, proofs, and exact calculations

(a) True.

Each interchange multiplies the determinant by -1, so two multiply it by $(-1)^2 = 1$.

(b) False.

The diagonal-product rule needs a triangular matrix. For example, $\left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ has determinant 0, but diagonal product 1.

(c) False.

The matrix $\left[\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{smallmatrix}\right]$ has row 3 equal to row 1 plus row 2 and therefore determinant zero. None of its rows or columns is zero or identical to another.

(d) False.

Theorem 5 gives $\det A^T = \det A$. The identity matrix, whose determinant is 1, disproves the proposed minus sign.

Exercise 3.2.29: Properties, proofs, and exact calculations

For $B = \left[\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{smallmatrix}\right]$, subtract row 1 from rows 2 and 3, giving lower right block $\left[\begin{smallmatrix} 1 & 1 \\ 2 & 0 \end{smallmatrix}\right]$. Thus $\det B = 1(0 - 2) = -2$. Applying multiplicativity repeatedly gives $\det(B^5) = (\det B)^5 = (-2)^5 = -32$.

Exercise 3.2.30: Properties, proofs, and exact calculations

Interchange the two equal rows. The numerical matrix does not change, but Theorem 3 says its determinant changes sign. Thus $d = -d$, where $d = \det A$. Adding d gives $2d = 0$, hence $d = 0$. If two columns are equal, transpose the matrix; its transpose has two equal rows and the same determinant by Theorem 5. This proof does not use the invertibility criterion in Theorem 4.

Exercise 3.2.31: Properties, proofs, and exact calculations

Since A is invertible, Theorem 4 gives $\det A \neq 0$. The multiplication theorem applied to $AA^{-1} = I$ gives $(\det A)(\det A^{-1}) = \det I = 1$. Dividing by the nonzero scalar $\det A$ proves $\det A^{-1} = 1/\det A$.

Exercise 3.2.32: Properties, proofs, and exact calculations

The matrix rA is obtained by multiplying each of the n rows by r. Repeated use of Theorem 3(c), one row at a time, gives $\det(rA) = r^n \det A$. If $r = 0$ and $n \geq 1$, the matrix is zero and both sides are zero, so that case is also covered.

Exercise 3.2.33: Properties, proofs, and exact calculations

Theorem 6 and commutativity of *real-number* multiplication give $\det(AB) = (\det A)(\det B) = (\det B)(\det A) = \det(BA)$. This does not assert that the matrix products themselves commute.

Exercise 3.2.34: Properties, proofs, and exact calculations

By multiplicativity, $\det(PAP^{-1}) = (\det P)(\det A)(\det P^{-1})$. Because P is invertible, Exercise 31 gives $(\det P)(\det P^{-1}) = 1$. The product therefore reduces to $\det A$. Similar matrices have the same determinant.

Exercise 3.2.35: Properties, proofs, and exact calculations

Take determinants of $U^TU = I$. Theorems 5 and 6 give $(\det U)^2 = (\det U^T)(\det U) = 1$. Since $\det U$ is real, solving $d^2 = 1$ gives $d = 1$ or $d = -1$. Both signs occur: I gives 1, and a single coordinate reflection gives -1.

Exercise 3.2.36: Properties, proofs, and exact calculations

Repeated multiplication gives $0 = \det(A^4) = (\det A)^4$. A real number has fourth power zero only when it is zero. Thus $\det A = 0$, so Theorem 4 rules out invertibility.

Exercise 3.2.37: Properties, proofs, and exact calculations

Multiply the matrices first:

$$AB = \begin{bmatrix} 6 & 0 \\ 17 & 4 \end{bmatrix}.$$

Direct 2×2 calculations give $\det A = 3$, $\det B = 8$, and

$$\det(AB) = (6)(4) - (0)(17) = 24 = (\det A)(\det B).$$

This is a direct verification for these matrices, rather than an application of Theorem 6.

Exercise 3.2.38: Properties, proofs, and exact calculations

Multiply the matrices first:

$$AB = \begin{bmatrix} 6 & 0 \\ -2 & 0 \end{bmatrix}.$$

Direct 2×2 calculations give $\det A = 0$, $\det B = -2$, and

$$\det(AB) = (6)(0) - (0)(-2) = 0 = (\det A)(\det B).$$

This is a direct verification for these matrices, rather than an application of Theorem 6.

Exercise 3.2.39: Properties, proofs, and exact calculations

Use the given 3×3 dimension when scaling every row:

(a) $\det(AB) = 4(-3) = -12$, by multiplicativity.

(b) $\det(5A) = 5^3 \cdot 4 = 500$, since all three rows are scaled.

(c) $\det(B^T) = \det B = -3$, by transposition invariance.

(d) $\det(A^{-1}) = 1/4$, since $\det A \neq 0$.

(e) $\det(A^3) = 4^3 = 64$, by applying multiplicativity twice.

Exercise 3.2.40: Properties, proofs, and exact calculations

The matrices are now 4×4:

(a) $\det(AB) = (-1)(2) = -2$.

(b) $\det(B^5) = 2^5 = 32$.

(c) $\det(2A) = 2^4(-1) = -16$.

(d) $\det(A^T A) = \det(A^T) \det A = (-1)^2 = 1$.

(e) $\det(B^{-1}AB) = (1/2)(-1)(2) = -1$; the inverse exists because $\det B = 2 \neq 0$.

Exercise 3.2.41: Properties, proofs, and exact calculations

Expand each 2×2 determinant directly:

$$\det A = (a + e)d - (b + f)c = (ad - bc) + (ed - fc) = \det B + \det C.$$

Only the *first row* was split, while the second row stayed fixed. This is compatible with failure of additivity in the whole matrix.

Exercise 3.2.42: Properties, proofs, and exact calculations

Here $A = I_2$. Direct calculation yields

$$\det(I_2 + B) = (1 + a)(1 + d) - bc = 1 + ad - bc + (a + d).$$

But $\det I_2 + \det B = 1 + ad - bc$. The two expressions are equal if and only if their difference $a + d$ is zero. This proves both implications of the claimed equivalence.

Exercise 3.2.43: Properties, proofs, and exact calculations

Expand each determinant down its third column. The cofactors C_{13}, C_{23}, C_{33} depend only on the first two columns, so they are the same for all three matrices. Therefore

$$\begin{aligned}\det A &= \sum_{i=1}^{3}(u_i + v_i)C_{i3} \\ &= \sum_{i=1}^{3} u_i C_{i3} + \sum_{i=1}^{3} v_i C_{i3} = \det B + \det C.\end{aligned}$$

The matrices do not satisfy $A = B + C$, because adding B and C would double their first two columns. The equality is linearity in one column with the other columns held fixed.

Exercise 3.2.44: Properties, proofs, and exact calculations

Transpose first: $(AE)^T = E^T A^T$. The transpose E^T is an elementary matrix of the corresponding type. Theorem 3, applied to left multiplication by E^T, gives

$$\det(E^T A^T) = (\det E^T)(\det A^T).$$

By Theorem 5, the left side equals $\det(AE)$, and the right side equals $(\det E)(\det A)$. This argument uses row-operation rules and transpose invariance, not the general product theorem.

Exercise 3.2.45: Rectangular matrices and two Gram products

Run the following exact experiment, including two samples at each requested size.

```
from random import Random
from sympy import Matrix
r = Random(32045)
for m,n in ((4,5),(4,5),(5,6),(5,6)):
    A = Matrix(m,n,[r.randint(-5,5) for _ in range(m*n)])
    print(m,n,(A.T*A).det(),(A*A.T).det())
```

The computed outputs are

shape of A	$\det(A^T A)$	$\det(AA^T)$
4×5	0	22645
4×5	0	128772
5×6	0	33391308
5×6	0	1276024

Explain what must hold for every such matrix.

If A has $m < n$, its n columns in $\mathbb{R}^m$ are dependent, so some nonzero x satisfies $Ax = 0$. Then $A^T Ax = 0$, proving that $A^T A$ is singular and its determinant is zero. In contrast, AA^T is invertible exactly when the rows of A are independent: for any y,

$$y^T AA^T y = \|A^T y\|^2.$$

This is zero exactly when $A^T y = 0$. Thus independent rows imply a zero null space for AA^T. A dependent-row sample would also give $\det(AA^T) = 0$; its nonzero values in these four experiments are not a universal guarantee.

Exercise 3.2.46: Small determinant versus near singularity

What we must show. Compare scaling-sensitive determinants with a scaling-invariant condition number. The specified matrix from Section 2.3, Exercise 9 is

$$A = \begin{bmatrix} 4 & 0 & -3 & -7 \\ -6 & 9 & 9 & 9 \\ 7 & -5 & 10 & 19 \\ -1 & 2 & 4 & -1 \end{bmatrix}.$$

Using the spectral (2-norm) condition number, the following double-precision calculation gives the table below.

```
import numpy as np
A = np.array([[4,0,-3,-7],[-6,9,9,9],
              [7,-5,10,19],[-1,2,4,-1]],float)
for B in (A,np.eye(4)):
    for c in (0.1,1.,10.):
        print(c,np.linalg.det(c*B),np.linalg.cond(c*B,2))
```

The determinants in the table were additionally checked in exact rational arithmetic, avoiding cancellation errors in their reported values.

matrix	det	κ_2
$\frac{1}{10}A$	$-\frac{501}{1250}$	11.3525
$1A$	-4008	11.3525
$10A$	-40080000	11.3525
$\frac{1}{10}I_4$	$\frac{1}{10000}$	1
$1I_4$	1	1
$10I_4$	10000	1

Interpret the comparison.

The supplied A is less well-conditioned than the identity, with a moderate condition number about 11.35. The exercise calls it nearly singular, but the actual supplied entries do not exhibit severe ill-conditioning. Its determinant changes by a factor 10^4 when A is multiplied by 10, while its condition number does not change. In general, $\det(cA) = c^4 \det A$ in this dimension. But for nonzero c,

$$\kappa_2(cA) = \|cA\|_2 \|(cA)^{-1}\|_2 = |c| \|A\|_2 \, |c|^{-1} \|A^{-1}\|_2 = \kappa_2(A).$$

Every nonzero scalar multiple of the identity has condition number 1, even if its determinant is made arbitrarily small. Thus determinant size alone, without controlling matrix scale, does not measure relative nearness to singularity. Exact zero still characterizes singularity; "small" is the scale-dependent part.

3.3 Cramer's rule, volume, and linear transformations

Cramer's rule solves $Ax = b$ when the square coefficient matrix satisfies $\det A \neq 0$: replace column i of A by b to form $A_i(b)$, then $x_i = \det A_i(b)/\det A$. Replacing a *row* is not Cramer's rule. The adjugate is the transpose of the cofactor matrix, and $A^{-1} = \operatorname{adj}(A)/\det A$ for an invertible A.

In geometry, determinants carry orientation signs. Ordinary areas and volumes do not: a parallelogram's area is $|\det[u\ v]|$, a parallelepiped's volume is $|\det[u\ v\ w]|$, and a linear transformation multiplies area or volume by the absolute value of its determinant. Translations do not change these measures.

Practice problem 3.3.P1: Transform a parallelogram

The edge matrix is $B = \left[\begin{smallmatrix} 1 & 5 \\ 3 & 1 \end{smallmatrix}\right]$, whose determinant is $1 - 15 = -14$. Thus the original area is 14. The transformation matrix $A = \left[\begin{smallmatrix} 1 & -0.1 \\ 0 & 2 \end{smallmatrix}\right]$ has determinant 2. Consequently the image area is $|\det A|\,|\det B| = 2 \cdot 14 = 28$. As a check, $AB = \left[\begin{smallmatrix} 0.7 & 4.9 \\ 6 & 2 \end{smallmatrix}\right]$ has determinant $1.4 - 29.4 = -28$, whose absolute value is 28.

Exercise 3.3.1: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 5 & 7 \\ 2 & 4 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (5)(4) - (7)(2) = 6.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 3 & 7 \\ 1 & 4 \end{bmatrix}, \qquad \det A_1(b) = 5.$$

The latter determinant is $(3)(4) - (7)(1) = 5$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}, \qquad \det A_2(b) = -1.$$

The latter determinant is $(5)(1) - (3)(2) = -1$.

Divide and verify.

$$x = \begin{bmatrix} \frac{5}{6} \\ -\frac{1}{6} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 6. Substitution gives $Ax = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.2: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 4 & 1 \\ 5 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 6 \\ 7 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (4)(2) - (1)(5) = 3.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 6 & 1 \\ 7 & 2 \end{bmatrix}, \qquad \det A_1(b) = 5.$$

The latter determinant is $(6)(2) - (1)(7) = 5$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 4 & 6 \\ 5 & 7 \end{bmatrix}, \qquad \det A_2(b) = -2.$$

The latter determinant is $(4)(7) - (6)(5) = -2$.

Divide and verify.

$$x = \begin{bmatrix} \frac{5}{3} \\ -\frac{2}{3} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 3. Substitution gives $Ax = \begin{bmatrix} 6 \\ 7 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.3: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 6 \end{bmatrix}, \qquad b = \begin{bmatrix} 7 \\ -5 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (3)(6) - (-2)(-5) = 8.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 7 & -2 \\ -5 & 6 \end{bmatrix}, \qquad \det A_1(b) = 32.$$

The latter determinant is $(7)(6) - (-2)(-5) = 32$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 3 & 7 \\ -5 & -5 \end{bmatrix}, \qquad \det A_2(b) = 20.$$

The latter determinant is $(3)(-5) - (7)(-5) = 20$.

Divide and verify.

$$x = \begin{bmatrix} 4 \\ \frac{5}{2} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 8. Substitution gives $Ax = \begin{bmatrix} 7 \\ -5 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.4: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} -5 & 3 \\ 3 & -1 \end{bmatrix}, \qquad b = \begin{bmatrix} 9 \\ -5 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (-5)(-1) - (3)(3) = -4.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 9 & 3 \\ -5 & -1 \end{bmatrix}, \qquad \det A_1(b) = 6.$$

The latter determinant is $(9)(-1) - (3)(-5) = 6$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} -5 & 9 \\ 3 & -5 \end{bmatrix}, \qquad \det A_2(b) = -2.$$

The latter determinant is $(-5)(-5) - (9)(3) = -2$.

Divide and verify.

$$x = \begin{bmatrix} -\frac{3}{2} \\ \frac{1}{2} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by -4. Substitution gives $Ax = \begin{bmatrix} 9 \\ -5 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.5: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 7 \\ -8 \\ -3 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (2)(-1)^{1+1} \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} + (1)(-1)^{1+2} \begin{vmatrix} -3 & 1 \\ 0 & 2 \end{vmatrix} = 4.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(2) - (1)(1) = -1$$
$$\det A_{12} = (-3)(2) - (1)(0) = -6.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 7 & 1 & 0 \\ -8 & 0 & 1 \\ -3 & 1 & 2 \end{bmatrix}, \qquad \det A_1(b) = 6.$$

Its first-row cofactor expansion evaluates to $(7)(-1) + (1)(13) + (0)(-8) = 6$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 2 & 7 & 0 \\ -3 & -8 & 1 \\ 0 & -3 & 2 \end{bmatrix}, \qquad \det A_2(b) = 16.$$

Its first-row cofactor expansion evaluates to $(2)(-13) + (7)(6) + (0)(9) = 16$. Column 3 is replaced by b:

$$A_3(b) = \begin{bmatrix} 2 & 1 & 7 \\ -3 & 0 & -8 \\ 0 & 1 & -3 \end{bmatrix}, \qquad \det A_3(b) = -14.$$

Its first-row cofactor expansion evaluates to $(2)(8)+(1)(-9)+(7)(-3)=-14$.

Divide and verify.

$$x=\begin{bmatrix}\frac{3}{2}\\4\\-\frac{7}{2}\end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 4. Substitution gives $Ax=\begin{bmatrix}7\\-8\\-3\end{bmatrix}=b$, checking every original equation.

Exercise 3.3.6: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A=\begin{bmatrix}2&1&1\\-1&0&2\\3&1&3\end{bmatrix},\qquad b=\begin{bmatrix}4\\2\\-2\end{bmatrix}.$$

Check the denominator first.

$$\det A=(2)(-1)^{1+1}\begin{vmatrix}0&2\\1&3\end{vmatrix}+(1)(-1)^{1+2}\begin{vmatrix}-1&2\\3&3\end{vmatrix}+(1)(-1)^{1+3}\begin{vmatrix}-1&0\\3&1\end{vmatrix}=4.$$

The 2×2 determinants used above are

$$\begin{aligned}\det A_{11}&=(0)(3)-(2)(1)=-2\\ \det A_{12}&=(-1)(3)-(2)(3)=-9\\ \det A_{13}&=(-1)(1)-(0)(3)=-1.\end{aligned}$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b)=\begin{bmatrix}4&1&1\\2&0&2\\-2&1&3\end{bmatrix},\qquad \det A_1(b)=-16.$$

Its first-row cofactor expansion evaluates to $(4)(-2)+(1)(-10)+(1)(2)=-16$. Column 2 is replaced by b:

$$A_2(b)=\begin{bmatrix}2&4&1\\-1&2&2\\3&-2&3\end{bmatrix},\qquad \det A_2(b)=52.$$

Its first-row cofactor expansion evaluates to $(2)(10)+(4)(9)+(1)(-4)=52$. Column 3 is replaced by b:

$$A_3(b)=\begin{bmatrix}2&1&4\\-1&0&2\\3&1&-2\end{bmatrix},\qquad \det A_3(b)=-4.$$

Its first-row cofactor expansion evaluates to $(2)(-2)+(1)(4)+(4)(-1)=-4$.

Divide and verify.

$$x=\begin{bmatrix}-4\\13\\-1\end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 4. Substitution gives $Ax=\begin{bmatrix}4\\2\\-2\end{bmatrix}=b$, checking every original equation.

Exercise 3.3.7: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A=\begin{bmatrix}6s&4\\9&2s\end{bmatrix},\qquad b=\begin{bmatrix}5\\-2\end{bmatrix}.$$

Its determinant is

$$\det A=12\left(s^2-3\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is $s\neq\sqrt{3},-\sqrt{3}$. Under that condition the two replacement determinants are

$$\det A_1(b)=10s+8,\qquad \det A_2(b)=-12s-45.$$

Cramer's rule therefore gives

$$x_1=\frac{5s+4}{6\left(s^2-3\right)},\qquad x_2=-\frac{4s+15}{4\left(s^2-3\right)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.8: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A=\begin{bmatrix}3s&-5\\9&5s\end{bmatrix},\qquad b=\begin{bmatrix}3\\2\end{bmatrix}.$$

Its determinant is

$$\det A=15\left(s^2+3\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is every real s. Under that condition the two replacement determinants are

$$\det A_1(b) = 15s + 10, \qquad \det A_2(b) = 6s - 27.$$

Cramer's rule therefore gives

$$x_1 = \frac{3s + 2}{3\left(s^2 + 3\right)}, \qquad x_2 = \frac{2s - 9}{5\left(s^2 + 3\right)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.9: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A = \begin{bmatrix} s & -2s \\ 3 & 6s \end{bmatrix}, \qquad b = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

Its determinant is

$$\det A = 6s\left(s + 1\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is $s \neq 0, -1$. Under that condition the two replacement determinants are

$$\det A_1(b) = 2s, \qquad \det A_2(b) = 4s + 3.$$

Cramer's rule therefore gives

$$x_1 = \frac{1}{3\left(s + 1\right)}, \qquad x_2 = \frac{4s + 3}{6s\left(s + 1\right)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.10: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A = \begin{bmatrix} 2s & 1 \\ 3s & 6s \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Its determinant is

$$\det A = 3s\left(4s - 1\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is $s \neq 0, 1/4$. Under that condition the two replacement determinants are

$$\det A_1(b) = 6s - 2, \qquad \det A_2(b) = s.$$

Cramer's rule therefore gives

$$x_1 = \frac{2\,(3s-1)}{3s\,(4s-1)}, \qquad x_2 = \frac{1}{3\,(4s-1)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.11: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 0 & -2 & -1 \\ 3 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned} C_{11} &= (-1)^{1+1}[(0)(1) - (0)(1)] = 0 \\ C_{12} &= (-1)^{1+2}[(3)(1) - (0)(-1)] = -3 \\ C_{13} &= (-1)^{1+3}[(3)(1) - (0)(-1)] = 3 \\ C_{21} &= (-1)^{2+1}[(-2)(1) - (-1)(1)] = 1 \\ C_{22} &= (-1)^{2+2}[(0)(1) - (-1)(-1)] = -1 \\ C_{23} &= (-1)^{2+3}[(0)(1) - (-2)(-1)] = 2 \\ C_{31} &= (-1)^{3+1}[(-2)(0) - (-1)(0)] = 0 \\ C_{32} &= (-1)^{3+2}[(0)(0) - (-1)(3)] = -3 \\ C_{33} &= (-1)^{3+3}[(0)(0) - (-2)(3)] = 6 \end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} 0 & -3 & 3 \\ 1 & -1 & 2 \\ 0 & -3 & 6 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} 0 & 1 & 0 \\ -3 & -1 & -3 \\ 3 & 2 & 6 \end{bmatrix}.$$

The first-row expansion gives $\det A = (0)(0) + (-2)(-3) + (-1)(3) = 3 \neq 0$. Thus

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 0 & 1 & 0 \\ -3 & -1 & -3 \\ 3 & 2 & 6 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 3I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.12: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned} C_{11} &= (-1)^{1+1}[(-2)(0) - (1)(1)] = -1 \\ C_{12} &= (-1)^{1+2}[(2)(0) - (1)(0)] = 0 \\ C_{13} &= (-1)^{1+3}[(2)(1) - (-2)(0)] = 2 \\ C_{21} &= (-1)^{2+1}[(1)(0) - (3)(1)] = 3 \\ C_{22} &= (-1)^{2+2}[(1)(0) - (3)(0)] = 0 \\ C_{23} &= (-1)^{2+3}[(1)(1) - (1)(0)] = -1 \\ C_{31} &= (-1)^{3+1}[(1)(1) - (3)(-2)] = 7 \\ C_{32} &= (-1)^{3+2}[(1)(1) - (3)(2)] = 5 \\ C_{33} &= (-1)^{3+3}[(1)(-2) - (1)(2)] = -4 \end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 0 & -1 \\ 7 & 5 & -4 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} -1 & 3 & 7 \\ 0 & 0 & 5 \\ 2 & -1 & -4 \end{bmatrix}.$$

The first-row expansion gives $\det A = (1)(-1) + (1)(0) + (3)(2) = 5 \neq 0$. Thus

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -1 & 3 & 7 \\ 0 & 0 & 5 \\ 2 & -1 & -4 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 5I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.13: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 3 & 5 & 4 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned}
C_{11} &= (-1)^{1+1}[(0)(1) - (1)(1)] = -1 \\
C_{12} &= (-1)^{1+2}[(1)(1) - (1)(2)] = 1 \\
C_{13} &= (-1)^{1+3}[(1)(1) - (0)(2)] = 1 \\
C_{21} &= (-1)^{2+1}[(5)(1) - (4)(1)] = -1 \\
C_{22} &= (-1)^{2+2}[(3)(1) - (4)(2)] = -5 \\
C_{23} &= (-1)^{2+3}[(3)(1) - (5)(2)] = 7 \\
C_{31} &= (-1)^{3+1}[(5)(1) - (4)(0)] = 5 \\
C_{32} &= (-1)^{3+2}[(3)(1) - (4)(1)] = 1 \\
C_{33} &= (-1)^{3+3}[(3)(0) - (5)(1)] = -5
\end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} -1 & 1 & 1 \\ -1 & -5 & 7 \\ 5 & 1 & -5 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} -1 & -1 & 5 \\ 1 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}.$$

The first-row expansion gives $\det A = (3)(-1) + (5)(1) + (4)(1) = 6 \neq 0$. Thus

$$A^{-1} = \frac{1}{6} \begin{bmatrix} -1 & -1 & 5 \\ 1 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 6I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.14: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 3 & 6 & 7 \\ 0 & 2 & 1 \\ 2 & 3 & 4 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned}
C_{11} &= (-1)^{1+1}[(2)(4) - (1)(3)] = 5 \\
C_{12} &= (-1)^{1+2}[(0)(4) - (1)(2)] = 2 \\
C_{13} &= (-1)^{1+3}[(0)(3) - (2)(2)] = -4 \\
C_{21} &= (-1)^{2+1}[(6)(4) - (7)(3)] = -3 \\
C_{22} &= (-1)^{2+2}[(3)(4) - (7)(2)] = -2 \\
C_{23} &= (-1)^{2+3}[(3)(3) - (6)(2)] = 3 \\
C_{31} &= (-1)^{3+1}[(6)(1) - (7)(2)] = -8 \\
C_{32} &= (-1)^{3+2}[(3)(1) - (7)(0)] = -3 \\
C_{33} &= (-1)^{3+3}[(3)(2) - (6)(0)] = 6
\end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} 5 & 2 & -4 \\ -3 & -2 & 3 \\ -8 & -3 & 6 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}.$$

The first-row expansion gives $\det A = (3)(5) + (6)(2) + (7)(-4) = -1 \neq 0$. Thus

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = -1 I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.15: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 3 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 3 & 2 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$C_{11} = (-1)^{1+1}[(1)(2) - (0)(3)] = 2$$
$$C_{12} = (-1)^{1+2}[(-1)(2) - (0)(-2)] = 2$$
$$C_{13} = (-1)^{1+3}[(-1)(3) - (1)(-2)] = -1$$
$$C_{21} = (-1)^{2+1}[(0)(2) - (0)(3)] = 0$$
$$C_{22} = (-1)^{2+2}[(3)(2) - (0)(-2)] = 6$$
$$C_{23} = (-1)^{2+3}[(3)(3) - (0)(-2)] = -9$$
$$C_{31} = (-1)^{3+1}[(0)(0) - (0)(1)] = 0$$
$$C_{32} = (-1)^{3+2}[(3)(0) - (0)(-1)] = 0$$
$$C_{33} = (-1)^{3+3}[(3)(1) - (0)(-1)] = 3$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} 2 & 2 & -1 \\ 0 & 6 & -9 \\ 0 & 0 & 3 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 6 & 0 \\ -1 & -9 & 3 \end{bmatrix}.$$

The first-row expansion gives $\det A = (3)(2) + (0)(2) + (0)(-1) = 6 \neq 0$. Thus

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 2 & 0 & 0 \\ 2 & 6 & 0 \\ -1 & -9 & 3 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 6I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.16: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$C_{11} = (-1)^{1+1}[(-3)(3) - (1)(0)] = -9$$
$$C_{12} = (-1)^{1+2}[(0)(3) - (1)(0)] = 0$$
$$C_{13} = (-1)^{1+3}[(0)(0) - (-3)(0)] = 0$$
$$C_{21} = (-1)^{2+1}[(2)(3) - (4)(0)] = -6$$
$$C_{22} = (-1)^{2+2}[(1)(3) - (4)(0)] = 3$$
$$C_{23} = (-1)^{2+3}[(1)(0) - (2)(0)] = 0$$
$$C_{31} = (-1)^{3+1}[(2)(1) - (4)(-3)] = 14$$
$$C_{32} = (-1)^{3+2}[(1)(1) - (4)(0)] = -1$$
$$C_{33} = (-1)^{3+3}[(1)(-3) - (2)(0)] = -3$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} -9 & 0 & 0 \\ -6 & 3 & 0 \\ 14 & -1 & -3 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} -9 & -6 & 14 \\ 0 & 3 & -1 \\ 0 & 0 & -3 \end{bmatrix}.$$

The first-row expansion gives $\det A = (1)(-9) + (2)(0) + (4)(0) = -9 \neq 0$. Thus

$$A^{-1} = \frac{1}{-9} \begin{bmatrix} -9 & -6 & 14 \\ 0 & 3 & -1 \\ 0 & 0 & -3 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = -9I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.17: The adjugate formula and its consequences

For $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, the four cofactors are $d, -c, -b, a$, in row order. Transposing gives

$$\operatorname{adj} A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Theorem 8 therefore gives $A^{-1} = (ad - bc)^{-1} \left[\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right]$ when $ad - bc \neq 0$. This is exactly the earlier 2×2 inverse formula. The nonzero-denominator condition is part of both formulas.

Exercise 3.3.18: The adjugate formula and its consequences

Each minor matrix has integer entries. Its determinant is an integer because a determinant is a finite sum of signed products of its entries. Cofactors, and hence every entry of the adjugate, are therefore integers. Since $\det A = 1$, Theorem 8 gives $A^{-1} = \operatorname{adj} A/1$, whose entries are all integers. The hypothesis $\det A = 1$ is essential to this simple argument; division by a general nonzero integer need not leave integers.

Exercise 3.3.19: Area from translated adjacent edges

Translate the vertex $p = (0,\ 0)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \qquad v = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

The remaining listed vertex is $p+u+v = (11,\ 6)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} 5 & 6 \\ 2 & 4 \end{vmatrix} \right\| = |8| = 8.$$

Exercise 3.3.20: Area from translated adjacent edges

Translate the vertex $p = (0,\ 0)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \qquad v = \begin{bmatrix} 4 \\ -5 \end{bmatrix}.$$

The remaining listed vertex is $p + u + v = (3,\ -2)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} -1 & 4 \\ 3 & -5 \end{vmatrix} \right\| = |-7| = 7.$$

Exercise 3.3.21: Area from translated adjacent edges

Translate the vertex $p = (-1,\ 0)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \qquad v = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

The remaining listed vertex is $p+u+v = (2,\ 1)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} 1 & 2 \\ 5 & -4 \end{vmatrix} \right\| = |-14| = 14.$$

Exercise 3.3.22: Area from translated adjacent edges

Translate the vertex $p = (0,\ -2)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} 6 \\ 1 \end{bmatrix}, \qquad v = \begin{bmatrix} -3 \\ 3 \end{bmatrix}.$$

The remaining listed vertex is $p+u+v = (3,\ 2)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} 6 & -3 \\ 1 & 3 \end{vmatrix} \right\| = |21| = 21.$$

Exercise 3.3.23: Volume of a parallelepiped

Place the three adjacent edge vectors in columns:

$$A = \begin{bmatrix} 1 & 1 & 7 \\ 0 & 2 & 1 \\ -2 & 4 & 0 \end{bmatrix}.$$

$$\det A = (1)(-1)^{1+1}\begin{vmatrix} 2 & 1 \\ 4 & 0 \end{vmatrix} + (1)(-1)^{1+2}\begin{vmatrix} 0 & 1 \\ -2 & 0 \end{vmatrix} + (7)(-1)^{1+3}\begin{vmatrix} 0 & 2 \\ -2 & 4 \end{vmatrix} = 22.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (2)(0) - (1)(4) = -4 \\ \det A_{12} &= (0)(0) - (1)(-2) = 2 \\ \det A_{13} &= (0)(4) - (2)(-2) = 4. \end{aligned}$$

The determinant is signed, so the ordinary volume is $|22| = 22$. If the columns are reordered, the signed determinant may change but this volume does not.

Exercise 3.3.24: Volume of a parallelepiped

Place the three adjacent edge vectors in columns:

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 4 & -5 & 2 \\ 0 & 2 & -1 \end{bmatrix}.$$

$$\det A = (1)(-1)^{1+1}\begin{vmatrix} -5 & 2 \\ 2 & -1 \end{vmatrix} + (-2)(-1)^{1+2}\begin{vmatrix} 4 & 2 \\ 0 & -1 \end{vmatrix} + (-1)(-1)^{1+3}\begin{vmatrix} 4 & -5 \\ 0 & 2 \end{vmatrix} = -15.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (-5)(-1) - (2)(2) = 1 \\ \det A_{12} &= (4)(-1) - (2)(0) = -4 \\ \det A_{13} &= (4)(2) - (-5)(0) = 8. \end{aligned}$$

The determinant is signed, so the ordinary volume is $|-15| = 15$. If the columns are reordered, the signed determinant may change but this volume does not.

Exercise 3.3.25: Geometry and linearity

Let the columns be a_1, a_2, a_3. Their parallelepiped has volume $|\det A|$. If they are linearly dependent, all three vectors lie in a plane through the origin (or a smaller subspace), so the parallelepiped is flat and has zero volume. Conversely, three independent vectors form a genuinely three-dimensional parallelepiped: the first two span a base plane of positive area, and the third has positive distance from that plane, giving positive base-area times height. Thus volume is zero exactly when the columns are dependent. By the Invertible Matrix Theorem, dependence is exactly noninvertibility. This geometric proof does not invoke the determinant criterion in Section 3.2, Theorem 4.

Exercise 3.3.26: Geometry and linearity

What we must show. Prove equality of the two sets, which requires both inclusions. A point y belongs to $T(p+S)$ exactly when $y = T(p+s)$ for some $s \in S$. Linearity changes this to $y = T(p)+T(s)$ for some $s \in S$, which is exactly membership in $T(p) + T(S)$. Every step can be reversed, so

$$T(p + S) = T(p) + T(S).$$

The argument also handles $S = \varnothing$, when both sets are empty. A linear map sends a translated set to the corresponding translated image.

Exercise 3.3.27: Area before and after a linear map

The original edge matrix and its image are

$$B = \begin{bmatrix} -2 & -2 \\ 3 & 5 \end{bmatrix}, \qquad AB = \begin{bmatrix} -18 & -22 \\ 12 & 16 \end{bmatrix}.$$

The original area is $|\det B| = |-4| = 4$. Since $\det A = 6$, the image area is

$$|\det A|\,|\det B| = 6 \cdot 4 = 24.$$

A direct determinant of the displayed image edge matrix gives -24, confirming the same absolute area.

Exercise 3.3.28: Area before and after a linear map

The original edge matrix and its image are

$$B = \begin{bmatrix} 4 & 0 \\ -7 & 1 \end{bmatrix}, \qquad AB = \begin{bmatrix} 14 & 2 \\ -3 & 1 \end{bmatrix}.$$

The original area is $|\det B| = |4| = 4$. Since $\det A = 5$, the image area is

$$|\det A|\,|\det B| = 5 \cdot 4 = 20.$$

A direct determinant of the displayed image edge matrix gives 20, confirming the same absolute area.

Exercise 3.3.29: Area and volume formulas

The diagonal from 0 to $v_1 + v_2$ divides the parallelogram into two congruent triangles. Each has half the parallelogram's area, so the triangle with vertices $0, v_1, v_2$ has area

$$\frac{1}{2}\left|\det[v_1 \;\; v_2]\right|.$$

If the two edges are dependent, the parallelogram and triangle are both flat and the same formula gives zero.

Exercise 3.3.30: Area and volume formulas

What we must show. Derive an orientation-independent triangle-area formula.

Translate one vertex to the origin.

Subtract (x_1, y_1) from all vertices. The new edge vectors are $u = (x_2 - x_1, y_2 - y_1)$ and $v = (x_3 - x_1, y_3 - y_1)$. Translation does not change area, so Exercise 29 gives $\frac{1}{2}|\det[u\ v]|$.

Connect this to the displayed determinant.

In the 3×3 matrix in the exercise, subtract row 1 from rows 2 and 3. Expanding down column 3 then gives

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix}.$$

This last matrix is $[u\ v]^T$, with the same determinant as $[u\ v]$.

Conclusion. The ordinary area is one half the *absolute value* of the displayed 3×3 determinant. The supplied print formula omits the absolute-value bars; without them it is a signed area and can be negative. For vertices $(0,0), (0,1), (1,0)$ the displayed determinant is -1, so the omission cannot be valid for ordinary area with arbitrary vertex order.

Exercise 3.3.31: Area and volume formulas

(a) Describe the image in both directions.

The map is $T(u_1, u_2, u_3) = (au_1, bu_2, cu_3)$. Because $a, b, c > 0$, it has inverse $u = (x_1/a, x_2/b, x_3/c)$. Thus

$$u_1^2 + u_2^2 + u_3^2 \le 1 \quad \Longleftrightarrow \quad \frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} \le 1.$$

So the ball maps exactly onto the solid ellipsoid, and equality describes its boundary.

(b) Multiply the original volume by the determinant factor.

The diagonal matrix has determinant $abc > 0$. Hence the image volume is $abc(4\pi/3) = 4\pi abc/3$.

Exercise 3.3.32: Area and volume formulas

(a) Specify the map by its columns.

Take $A = [v_1\ v_2\ v_3]$ and $T(x) = Ax$. Then $T(e_i) = v_i$ and $T(0) = 0$. More explicitly, the standard tetrahedron consists of the points $t_1e_1 + t_2e_2 + t_3e_3$ with $t_i \ge 0$ and $t_1 + t_2 + t_3 \le 1$. Their images are exactly the same combinations of v_1, v_2, v_3, namely S'.

(b) Compute the scale factor from a unit reference volume.

The standard base triangle in the coordinate plane has area $1/2$, and the perpendicular height to e_3 is 1. Thus $\text{vol}(S) = (1/3)(1/2)(1) = 1/6$. The linear map multiplies volume by $|\det A|$, giving

$$\text{vol}(S') = \frac{1}{6}|\det[v_1\ v_2\ v_3]|.$$

The same formula gives zero when the three vectors are dependent and the image tetrahedron is flat.

Exercise 3.3.33: A floating-point adjugate experiment

Use the reproducible random integer matrix

$$A = \begin{bmatrix} -3 & 8 & -2 & 4 \\ 7 & 8 & 9 & -1 \\ 3 & 8 & -9 & -3 \\ 0 & 4 & -5 & -1 \end{bmatrix}.$$

The following computation forms every cofactor independently in double precision, transposes, divides by the determinant, and compares with a numerical inverse.

```
import numpy as np
A = np.array([[-3, 8, -2, 4],
              [7, 8, 9, -1],
              [3, 8, -9, -3],
              [0, 4, -5, -1]],float)
C = np.zeros((4,4))
for i in range(4):
    for j in range(4):
        minor = np.delete(np.delete(A,i,0),j,1)
        C[i,j] = (-1)**(i+j)*np.linalg.det(minor)
B = C.T/np.linalg.det(A)
print(np.max(np.abs(B-np.linalg.inv(A))))
```

In this run the maximum absolute entry of $B - \mathrm{inv}(A)$ was approximately 1.55×10^{-15}. The exact determinant is $-640 \neq 0$, and exact rational arithmetic also gives $A \operatorname{adj} A = (\det A)I$. The floating discrepancy is roundoff, not a failure of the inverse identity; a different numerical library may change the last digits. Double precision stores about 16 decimal significant digits, not infinitely many, so we report a maximum error rather than an unjustified assertion of bit-for-bit equality.

Exercise 3.3.34: Cramer's rule and an inverse on the same random system

For one reproducible random system, take

$$A = \begin{bmatrix} 4 & 6 & 0 & -7 \\ 3 & -6 & 9 & -3 \\ -1 & 6 & -5 & -5 \\ -2 & 0 & 9 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} -7 \\ 9 \\ -2 \\ -5 \end{bmatrix}.$$

The denominator and four column-replacement determinants are

$$\det A = -2448, \qquad (\det A_1, \det A_2, \det A_3, \det A_4) = (2616,\ 5067,\ 1188,\ 3390)\,.$$

Hence Cramer's rule gives

$$x = \begin{bmatrix} -\frac{109}{102} \\ -\frac{563}{272} \\ -\frac{33}{68} \\ -\frac{565}{408} \end{bmatrix}.$$

Direct exact multiplication gives $Ax = b$, and exact inversion gives the identical vector $A^{-1}b$. Here is a matrix-program command for the second component, with Python's zero-based column indexing:

```
from sympy import Matrix
A = Matrix([[4, 6, 0, -7], [3, -6, 9, -3], [-1, 6, -5, -5], [-2, 0, 9, 2]])
b = Matrix([-7, 9, -2, -5])
A2 = A.copy()
A2[:,1] = b
x2 = A2.det()/A.det()
assert x2 == (A.inv()*b)[1]
```

In MATLAB the corresponding column-replacement command is

```
A2 = A; A2(:,2) = b; x2 = det(A2)/det(A);
```

The determinant must be checked nonzero before division; a singular random draw would require a new nonsingular test matrix.

Exercise 3.3.35: Count arithmetic instead of confusing timing with flops

What we must show. Compare operation counts for a 30×30 inverse and an adjugate-based inverse. The question is conditional on having the historical MATLAB `flops` command. Modern MATLAB and NumPy do not provide that same counter, so a timing measurement must not be called a flop count. Instead we give an explicit arithmetic-count model that is reproducible and makes the comparison precise.

Count each scalar addition, subtraction, multiplication, or division as one flop; exclude comparisons, indexing, and memory movement. Assume a dense nonsingular matrix with no zero shortcuts. For elimination at step k, with $r = n - k - 1$ rows remaining, computing multipliers costs r divisions and updating the trailing square costs $2r^2$ flops. Thus

$$L(n) = \sum_{r=1}^{n-1} (r + 2r^2).$$

After one LU factorization, solve n right-hand sides to find the inverse. Unit-lower-triangular forward substitution costs $n(n-1)$ per right-hand side; back substitution costs $n(n-1) + n$. Hence the total is

$$L(n) + n(2n^2 - n).$$

For the cofactor method, form all n^2 determinants of size $n - 1$, each with elimination cost $L(n-1)$ plus $n - 2$ diagonal-product multiplications. Form the original determinant with cost $L(n) + n - 1$, then divide n^2 adjugate entries by it. The cost is

$$n^2[L(n-1) + n - 2] + L(n) + n - 1 + n^2.$$

At $n = 30$, these formulas give 70645 flops for LU plus solves, versus 14294274 for the independently evaluated adjugate formula, about 202.3 times as many. They can be checked by

```
def L(n):
```

```
    return sum(r+2*r*r for r in range(1,n))
n = 30
print(L(n)+n*(2*n*n-n))
print(n*n*(L(n-1)+n-2)+L(n)+n-1+n*n)
```

These are model counts, not measurements from an unavailable old MATLAB counter. Implementations that reuse subdeterminants or use different algorithms have different counts. The main conclusion is that forming all cofactors independently is an $O(n^5)$ method here, while factorization and solves are $O(n^3)$. Cofactor *expansion* without elimination would be still more expensive.

3.S Chapter 3 supplementary exercises

These exercises combine determinant calculations with proofs. A counterexample disproves a universal claim; several successful numerical experiments do not prove one. Throughout this supplement matrices and scalars are real unless the problem says otherwise. It is helpful to keep three facts distinct: row replacement preserves a determinant, scaling *one* row scales it once, and scaling an entire $n \times n$ matrix scales it n times.

Exercise 3.S.1: Sixteen statements with explanations

What we must show. Judge all sixteen statements, giving a reason rather than just a letter.

(a) **True.** Zero determinant means dependent columns. For two columns u, v, dependence makes one a multiple of the other: if a dependence $cu + dv = 0$ has $c \neq 0$, solve for u; otherwise $d \neq 0$ gives $v = 0$, a multiple of u. This includes zero columns.

(b) **True.** Interchanging the equal rows leaves the matrix unchanged but reverses the determinant, so $d = -d$ and $d = 0$.

(c) **False.** Scaling every row of a 3×3 matrix gives $\det(5A) = 5^3 \det A$, not $5 \det A$. For $A = I_3$, the two proposed sides are 125 and 5.

(d) **False.** For $n = 2$, take $A = \operatorname{diag}(2, 1)$ and $B = \operatorname{diag}(3, 1)$. Their determinants are 2 and 3, but $\det(A + B) = \det \operatorname{diag}(5, 2) = 10$, not 5.

(e) **False.** Multiplicativity gives $\det(A^3) = (\det A)^3 = 2^3 = 8$, not 6.

(f) **False.** A single interchange gives $\det B = -\det A$. Interchanging two rows of I_2 gives determinant -1, not 1.

(g) **True.** Scaling row 3 by 5 multiplies the determinant by 5, with all other rows fixed.

(h) **True.** Add the stated multiples one at a time. Each is a row replacement using an unchanged other row, so each preserves the determinant and their finite succession does too.

(i) **False.** Transposition preserves the determinant. The identity matrix disproves a general minus sign.

(j) **False in general.** The correct rule is $\det(-A) = (-1)^n \det A$. For I_2 this is 1, not -1; the proposed rule holds in odd dimensions.

(k) **True.** Multiplicativity and transpose invariance give $\det(A^T A) = (\det A)^2 \geq 0$.

(l) **False.** Cramer's quotient formulas require a nonzero coefficient determinant. The system $0x = 1$, for example, is inconsistent and cannot be solved by dividing replacement determinants by 0.

(m) **False.** The parallelogram has area 10; the stated triangle has half that area, namely 5.

(n) **True.** Taking determinants in $A^3 = 0$ yields $(\det A)^3 = 0$, hence $\det A = 0$.

(o) **False.** The correct formula is $\det(A^{-1}) = 1/\det A$. Taking $A = [2]$ gives $1/2$, not 2.

(p) **True.** Apply multiplicativity to $AA^{-1} = I$: the two determinant factors have product 1.

Exercise 3.S.2: Equally spaced rows

For the matrix with rows $(12, 13, 14)$, $(15, 16, 17)$, and $(18, 19, 20)$, perform $R_3 \leftarrow R_3 - R_2$ first, giving $(3, 3, 3)$ in row 3. Next do $R_2 \leftarrow R_2 - R_1$, also giving $(3, 3, 3)$. Finally $R_3 \leftarrow R_3 - R_2$ creates a zero row. Only row replacements were used, so the determinant is unchanged and equals zero.

Exercise 3.S.3: A symbolic determinant that vanishes

Begin with rows $(1, a, b+c)$, $(1, b, a+c)$, and $(1, c, a+b)$. Subtract row 1 from rows 2 and 3 to obtain

$$\begin{bmatrix} 1 & a & b+c \\ 0 & b-a & a-b \\ 0 & c-a & a-c \end{bmatrix}.$$

If $b = a$, row 2 is already zero. Otherwise replace row 3 by row 3 minus $\frac{c-a}{b-a}$ times row 2. Its second entry becomes zero, and its third becomes

$$(a-c) - \frac{c-a}{b-a}(a-b) = (a-c) + (c-a) = 0.$$

Thus a determinant-preserving operation makes a zero row in every case, proving the determinant is zero. The case split avoids dividing by a possibly zero parameter.

Exercise 3.S.4: Rows translated in a common direction

Subtract row 1 from rows 2 and 3. The matrix becomes

$$\begin{bmatrix} a & b & c \\ x & x & x \\ y & y & y \end{bmatrix}.$$

If $x = 0$, it has a zero row. If $x \neq 0$, replace row 3 by row 3 minus (y/x) times row 2; that gives a zero row. All operations were row replacements, so the original determinant is zero, including the special parameter cases.

Exercise 3.S.5: Choose sparse expansions

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 9 & 1 & 9 & 9 & 9 \\ 9 & 0 & 9 & 9 & 2 \\ 4 & 0 & 0 & 5 & 0 \\ 9 & 0 & 3 & 9 & 0 \\ 6 & 0 & 0 & 7 & 0 \end{vmatrix} = (1)(-1)^{1+2} D_2 = -12.$$

$$D_2 = \begin{vmatrix} 9 & 9 & 9 & 2 \\ 4 & 0 & 5 & 0 \\ 9 & 3 & 9 & 0 \\ 6 & 0 & 7 & 0 \end{vmatrix} = (2)(-1)^{1+4} D_3 = 12.$$

$$D_3 = \begin{vmatrix} 4 & 0 & 5 \\ 9 & 3 & 9 \\ 6 & 0 & 7 \end{vmatrix} = (3)(-1)^{2+2} D_4 = -6.$$

$$D_4 = \begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix} = (4)(7) - (5)(6) = -2.$$

Thus the answer is -12.

Exercise 3.S.6: Choose sparse expansions

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 8 & 8 & 8 & 5 \\ 0 & 1 & 0 & 0 & 0 \\ 6 & 8 & 8 & 8 & 7 \\ 0 & 8 & 8 & 3 & 0 \\ 0 & 8 & 2 & 0 & 0 \end{vmatrix} = (1)(-1)^{2+2} D_2 = 12.$$

$$D_2 = \begin{vmatrix} 4 & 8 & 8 & 5 \\ 6 & 8 & 8 & 7 \\ 0 & 8 & 3 & 0 \\ 0 & 2 & 0 & 0 \end{vmatrix} = (2)(-1)^{4+2} D_3 = 12.$$

$$D_3 = \begin{vmatrix} 4 & 8 & 5 \\ 6 & 8 & 7 \\ 0 & 3 & 0 \end{vmatrix} = (3)(-1)^{3+2} D_4 = 6.$$

$$D_4 = \begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix} = (4)(7) - (5)(6) = -2.$$

Thus the answer is 12.

Exercise 3.S.7: Geometric and structural determinant proofs

Subtract row 2 from rows 1 and 3 in the displayed determinant. Expanding down its first column shows that its zero equation is equivalent to

$$(x - x_1)(y_2 - y_1) - (y - y_1)(x_2 - x_1) = 0.$$

This says the displacement $(x - x_1, y - y_1)$ is parallel to the nonzero direction $(x_2 - x_1, y_2 - y_1)$. To check without a geometric shortcut, if $x_2 \neq x_1$ solve for $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$; if $x_2 = x_1$, distinctness gives $y_2 \neq y_1$, and the equation reduces to $x = x_1$. These are exactly the nonvertical and vertical lines through the two distinct points.

Exercise 3.S.8: Geometric and structural determinant proofs

A direction vector for a line of finite slope m is $(1, m)$. Thus one valid determinant equation is

$$\begin{vmatrix} 1 & x & y \\ 1 & x_1 & y_1 \\ 0 & 1 & m \end{vmatrix} = 0.$$

Subtract row 2 from row 1, then expand down column 1. The left side is $-[(x - x_1)m - (y - y_1)]$, so setting it to zero is precisely $y - y_1 = m(x - x_1)$. Alternatively the third row can be $(1, x_1 + 1, y_1 + m)$, corresponding to a second point on the same line; subtracting row 2 converts it to the displayed version.

Exercise 3.S.9: Geometric and structural determinant proofs

Subtract row 1 from rows 2 and 3 in T. The two new rows are $(0, b - a, b^2 - a^2)$ and $(0, c - a, c^2 - a^2)$. Factor $b^2 - a^2 = (b - a)(b + a)$ and $c^2 - a^2 = (c - a)(c + a)$ to obtain

$$\begin{aligned} \det T &= (b - a)(c - a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b + a \\ 0 & 1 & c + a \end{vmatrix} \\ &= (b - a)(c - a)[(c + a) - (b + a)] \\ &= (b - a)(c - a)(c - b). \end{aligned}$$

Row factoring is a polynomial identity, so no division by $b - a$ or $c - a$ was used; the formula remains valid when two parameters coincide.

Exercise 3.S.10: Geometric and structural determinant proofs

Expand along the first row of $V(t)$. Its cofactors depend only on the fixed numbers x_1, x_2, x_3, so the expansion is a polynomial in t of degree at most 3. The coefficient of t^3 is the cofactor in position $(1, 4)$:

$$-\begin{vmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{vmatrix} = -(x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

It is nonzero because the three numbers are distinct. Hence the degree is exactly 3. At $t = x_i$, the first row equals row $i + 1$, so the determinant is zero. Thus three graph points are $(x_1, 0)$, $(x_2, 0)$, $(x_3, 0)$. In fact these roots and the leading coefficient determine

$$f(t) = -(x_2 - x_1)(x_3 - x_1)(x_3 - x_2) \prod_{i=1}^{3} (t - x_i).$$

Exercise 3.S.11: Geometric and structural determinant proofs

Use $p = (1, 4)$ as a base vertex, and take adjacent displacements $u = (-1, 5) - p = (-2, 1)$ and $v = (5, 8) - p = (4, 4)$. Their sum gives the remaining vertex: $p + u + v = (3, 9)$. This verifies the

parallelogram structure. Equivalently, the diagonals have the same midpoint because $(1,4)+(3,9) = (-1,5)+(5,8)$. Its area is

$$\left\|\begin{vmatrix} -2 & 4 \\ 1 & 4 \end{vmatrix}\right\| = |-8-4| = 12.$$

Exercise 3.S.12: Geometric and structural determinant proofs

A 2×2 matrix is invertible if and only if the parallelogram spanned by its two columns has positive area. Indeed, the columns are independent exactly when they are not parallel, which is exactly when the parallelogram has a nonzero base and positive perpendicular height. The Invertible Matrix Theorem equates that independence with invertibility. In determinant language the same area is $|\det A|$, but the geometric statement itself describes the nonflat image of the unit square.

Exercise 3.S.13: Geometric and structural determinant proofs

Put $d = \det A \neq 0$. The adjugate identity gives $A \operatorname{adj} A = \operatorname{adj} A\, A = dI$. Divide both equations by d:

$$(A/d) \operatorname{adj} A = I, \qquad \operatorname{adj} A\, (A/d) = I.$$

Thus A/d is a two-sided inverse of $\operatorname{adj} A$. This proves both its invertibility and $(\operatorname{adj} A)^{-1} = A/\det A$.

Exercise 3.S.14: Geometric and structural determinant proofs

A block-triangular determinant rule.

For $M = \left[\begin{smallmatrix} A & 0 \\ C & D \end{smallmatrix}\right]$, expand the determinant by its signed products, one entry from each row and column. Any nonzero product must use the first n columns for the top n rows, since the upper-right block is zero. Those top rows use *all* first n columns. The bottom n rows must therefore use the last n columns. Such a product splits into one signed product of A and one of D; the signs also multiply because no columns cross between blocks. Summing gives $\det M = (\det A)(\det D)$. This argument does not require either block to be invertible.

(a) Put $C = 0$ and $D = I_n$.

The rule gives $\det \left[\begin{smallmatrix} A & 0 \\ 0 & I \end{smallmatrix}\right] = (\det A)(\det I) = \det A$.

(b) Put the upper-left block equal to I_n.

Then $\det \left[\begin{smallmatrix} I & 0 \\ C & D \end{smallmatrix}\right] = (\det I)(\det D) = \det D$.

(c) Both block-triangular orientations.

The proved rule gives the lower-block-triangular formula. For the upper-block-triangular matrix, transpose:

$$\det \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det \begin{bmatrix} A^T & 0 \\ B^T & D^T \end{bmatrix} = (\det A^T)(\det D^T) = (\det A)(\det D).$$

The off-diagonal blocks affect entries but not the determinant in these zero-block arrangements.

Exercise 3.S.15: Geometric and structural determinant proofs

(a) Determine the factors by multiplying their blocks.

Block multiplication gives

$$\begin{bmatrix} I & 0 \\ X & I \end{bmatrix} \begin{bmatrix} A & B \\ 0 & Y \end{bmatrix} = \begin{bmatrix} A & B \\ XA & XB+Y \end{bmatrix}.$$

To match the lower-left block C, use $X = CA^{-1}$. To match the lower-right block D, use $Y = D - CA^{-1}B$. Multiplicativity and Exercise 14 now give

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = 1 \cdot (\det A) \det(D - CA^{-1}B).$$

The first factor has determinant 1 because its two diagonal blocks are identities.

(b) Use precisely the commuting hypothesis stated.

If $AC = CA$, then $ACA^{-1} = C$, and hence

$$A(D - CA^{-1}B) = AD - CB.$$

Taking determinants gives $(\det A)\det(D-CA^{-1}B) = \det(AD-CB)$. Substitution in part (a) proves the asserted formula. We have not commuted A with B or D; neither of those extra hypotheses is supplied or needed.

Exercise 3.S.16: Geometric and structural determinant proofs

Let $d = a - b$. The claimed determinant is $d^{n-1}[a + (n-1)b]$.

(a) Subtract successive rows in the correct order.

Perform $R_1 \leftarrow R_1 - R_2$, then $R_2 \leftarrow R_2 - R_3$, continuing downward through $R_{n-1} \leftarrow R_{n-1} - R_n$. At each step the row being subtracted is still its original row. The first $n-1$ rows now have d in position (i,i), $-d$ in position $(i,i+1)$, and zeros elsewhere. The last row remains $(b, b, \ldots, b, a)$. Every operation is a row replacement, so the determinant is unchanged.

(b) Add successive columns, now from left to right.

Apply $C_2 \leftarrow C_2 + C_1$, then $C_3 \leftarrow C_3 + C_2$ using the *new* column 2, and continue. The first $n-1$ rows become zero off their diagonal d entries. In the last row the entries accumulate to $b, 2b, \ldots, (n-1)b, a + (n-1)b$. Column replacements preserve the determinant, by the transpose version of the row rule.

(c) Read a triangular determinant.

The matrix is now lower triangular with diagonal $d, \ldots, d, a+(n-1)b$. Their product is $(a-b)^{n-1}[a + (n-1)b]$. No division by $a-b$ was used, so $a = b$ causes no gap in the proof. The case $n = 1$ simply says the determinant is a.

Exercise 3.S.17: Geometric and structural determinant proofs

What we must show. Reprove the formula by induction without assuming the preceding row-operation argument. For $n = 1$ the determinant is a. Suppose the formula holds at size $n - 1$. By linearity in column 1, with all other columns fixed, $\det A = \det B + \det C$ for the displayed matrices.

In B, expand down column 1: only the first entry $a - b$ is nonzero. The remaining $(n - 1) \times (n - 1)$ matrix has diagonal a and off-diagonal entries b. The induction hypothesis therefore gives

$$\det B = (a - b)^{n-1}[a + (n - 2)b].$$

In C, subtract row 1 from every other row. Column 1 then has b followed by zeros, and the remaining lower-right block is $(a - b)I_{n-1}$. Thus $\det C = b(a - b)^{n-1}$. Adding these two expressions yields

$$\det A = (a - b)^{n-1}[a + (n - 2)b + b] = (a - b)^{n-1}[a + (n - 1)b].$$

This proves the induction step, including zero parameter values because no division occurs.

Exercise 3.S.18: Two constant-off-diagonal examples

For the first matrix, $n = 4$, $a = 3$, $b = 8$. The formula gives

$$(3 - 8)^3[3 + 3 \cdot 8] = (-5)^3(27) = -3375.$$

For the second, $n = 5$, $a = 8$, $b = 3$, so

$$(8 - 3)^4[8 + 4 \cdot 3] = 5^4(20) = 12500.$$

An independent exact computation is

```
from sympy import Matrix
for n,a,b in ((4,3,8),(5,8,3)):
    A = Matrix(n,n,lambda i,j: a if i == j else b)
    print(A.det())
```

It prints -3375 and 12500, respectively. A negative determinant in the first example describes orientation, not a negative ordinary volume.

Exercise 3.S.19: A matrix of successive minima

Let H_n have entry $\min(i, j)$ in position (i, j). The three displayed examples are H_3, H_4, H_5. Exact computation gives determinant 1 for each:

```
from sympy import Matrix
for n in (3,4,5):
    H = Matrix(n,n,lambda i,j: min(i+1,j+1))
    print(H.det())
```

Prove the conjecture for every size.

Starting at the *bottom*, do $R_n \leftarrow R_n - R_{n-1}$, then $R_{n-1} \leftarrow R_{n-1} - R_{n-2}$, and continue to $R_2 \leftarrow R_2 - R_1$. This order ensures that every subtracted row is still its original row. Row 1 is all 1's; new row i has $i - 1$ zeros followed by 1's. Thus the resulting matrix is upper triangular with every diagonal entry 1. No operation changed the determinant, so $\det H_n = 1$ for every $n \geq 1$.

Exercise 3.S.20: A related matrix with a different first increment

What we must show. Determine the determinant of the matrix exactly as printed. Its successive distinct entries are $d_1 = 1$ and $d_i = 3(i-1)$ for $i \geq 2$: in particular $1, 3, 6, 9, \ldots, 3(n-1)$. These are *not* the triangular numbers after the first three entries. The (i, j) entry is $d_{\min(i,j)}$.

Run the same bottom-up differences.

For $i = n, n-1, \ldots, 2$, replace R_i by $R_i - R_{i-1}$. The resulting matrix is upper triangular with diagonal

$$d_1, \quad d_2 - d_1, \quad d_3 - d_2, \ldots, d_n - d_{n-1} = 1, 2, 3, \ldots, 3.$$

Therefore

$$\boxed{\det A = 2 \cdot 3^{n-2} \quad (n \geq 2).}$$

At $n = 1$, the matrix is [1] and has determinant 1. A direct check for $n = 2, 3, 4, 5$ gives $2, 6, 18, 54$:

```
from sympy import Matrix
for n in (2,3,4,5):
    d = [1] + [3*i for i in range(1,n)]
    A = Matrix(n,n,lambda i,j: d[min(i,j)])
    print(A.det())
```

The special first difference $3 - 1 = 2$ is why a guess of 3^{n-1} would be wrong. Equivalently, the triangular matrix after row differences has a leading 2×2 block of determinant 2 and a remaining block equal to 3 times the upper-triangular matrix from Exercise 19; block-triangular determinants give the same answer.

4.1 Vector spaces and subspaces

A vector need not be an arrow: a polynomial, a matrix, or a function can be a vector when addition and scalar multiplication satisfy the vector-space axioms. A subset H of a known vector space is a subspace when (i) it contains the zero vector, (ii) $u + v \in H$ for *every* $u, v \in H$, and (iii) $cu \in H$ for *every* $u \in H$ and real scalar c. One counterexample disproves closure; one successful example does not prove it. A span is automatically a subspace.

Practice problem 4.1.P1: An affine line fails scalar closure

What we must show. For $H = \{(3s, 2 + 5s) : s \in \mathbb{R}\}$, find a vector whose scalar multiple leaves H. Take $s = 0$. Then $u = (0, 2) \in H$. Its multiple $2u = (0, 4)$ is not in H: the equation $3s = 0$ would require $s = 0$, but then the second coordinate would be 2, not 4. Thus scalar multiplication is not closed on H.

Conclusion. H is not a vector space under the usual operations.

Practice problem 4.1.P2: Every generator belongs to its span

For $W = \text{Span}\{v_1, \ldots, v_p\}$ and a fixed k between 1 and p, write

$$v_k = 0v_1 + \cdots + 0v_{k-1} + 1v_k + 0v_{k+1} + \cdots + 0v_p.$$

This is a permitted linear combination of the generators, hence $v_k \in W$. In particular, choose $k = 1$ for the first part of the question. A span contains all its generators, even if some are zero or repeated.

Exercise 4.1.1: The first quadrant

(a) Addition preserves both inequalities.

If $u = (u_1, u_2)$ and $v = (v_1, v_2)$ have nonnegative coordinates, then $u_1 + v_1 \geq 0$ and $u_2 + v_2 \geq 0$. Therefore $u + v$ is still in the first quadrant.

(b) Negative scalars reveal the obstruction.

The vector $u = (1, 0)$ belongs to the set, but $(-1)u = (-1, 0)$ does not.

Conclusion. Addition is closed, but scalar multiplication is not; the first quadrant is not a vector space.

Exercise 4.1.2: The first and third quadrants together

(a) Any scalar multiple stays in the set.

Here $W = \{(x, y) : xy \geq 0\}$. If $u = (x, y) \in W$, then the product of the coordinates of cu is $(cx)(cy) = c^2xy \geq 0$, since $c^2 \geq 0$. This proves closure under multiplication by every real scalar, including zero.

(b) Addition can leave the set.

Choose $u = (2, 1)$ and $v = (-1, -2)$. Both coordinate products equal $2 > 0$, so both vectors are in W. Their sum is $(1, -1)$, whose coordinate product is $-1 < 0$. Thus $u + v \notin W$.

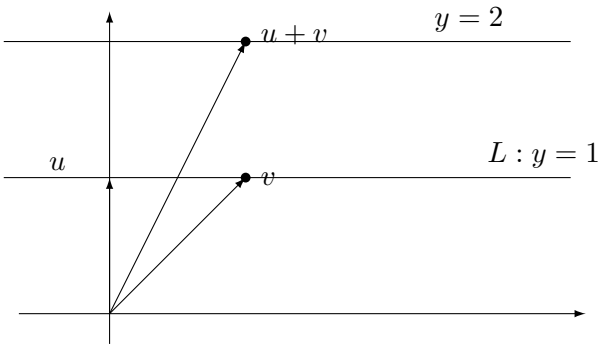
Conclusion. W passes the scalar test but fails the addition test.

Exercise 4.1.3: The closed unit disk

The vector $u = (1, 0)$ is in the disk because $1^2 + 0^2 = 1$. But $2u = (2, 0)$ is outside it, since $2^2 + 0^2 = 4 > 1$. Thus the disk is not closed under scalar multiplication and is not a subspace. It is not enough that the disk contains zero: closure must hold for arbitrarily large as well as small scalars.

Exercise 4.1.4: A translated line is not closed under addition

What we must show. Supply a geometric counterexample, rather than just assert that a line misses zero. Consider the line $L = \{(x, 1) : x \in \mathbb{R}\}$. Take $u = (0, 1)$ and $v = (1, 1)$. Both endpoints lie on L, but $u + v = (1, 2)$ lies on the parallel line $y = 2$. The parallelogram construction for adding u and v therefore ends off L.



More generally write any line not through zero as $p + \text{Span}\{d\}$, where $p \notin \text{Span}\{d\}$. Then $p \in L$ but $2p \notin L$, for otherwise $2p = p + td$ would imply $p = td$, a contradiction. Adding p to itself already fails.

Exercise 4.1.5: Multiples of a single polynomial

The set $\{at^2 : a \in \mathbb{R}\}$ equals $\text{Span}\{t^2\}$, a subspace of P_2 (and of P_n for every $n \geq 2$). Explicitly its zero is obtained with $a = 0$, $at^2 + bt^2 = (a + b)t^2$ stays in the set, and $c(at^2) = (ca)t^2$ stays in the set.

Conclusion. It is a subspace.

Exercise 4.1.6: A fixed quadratic coefficient

The set consists of polynomials $a + t^2$, whose coefficient of t^2 is always 1. It does not contain the zero polynomial: no choice of a makes $a + t^2$ zero for every t. Equivalently multiplying t^2 by zero produces a polynomial outside the set. Thus this subset of P_2 is not a subspace.

Exercise 4.1.7: Integer coefficients and real scalars

The constant polynomial $p(t) = 1$ has integer coefficients and degree at most 3. The real scalar $1/2$ is permitted in the ambient vector space, but $(1/2)p(t) = 1/2$ does not have integer coefficients. Scalar closure fails.

Conclusion. This set is not a real subspace of P_3. The choice of scalar field matters.

Exercise 4.1.8: Polynomials vanishing at zero

Let $H = \{p \in P_n : p(0) = 0\}$. The zero polynomial is in H. If $p, q \in H$, then $(p+q)(0) = p(0)+q(0) = 0$. For real c, $(cp)(0) = cp(0) = 0$. These equations prove both closure properties; sums and scalar multiples still have degree at most n.

Conclusion. H is a subspace. For $n \geq 1$ it is $\text{Span}\{t, t^2, \ldots, t^n\}$; for $n = 0$ it is just the zero polynomial.

Exercise 4.1.9: A one-parameter line in three dimensions

Factor out the arbitrary real parameter:

$$(-2t, 5t, 3t) = t(-2, 5, 3).$$

Thus choose $v = (-2, 5, 3)$, and $H = \text{Span}\{v\}$. This is equality in both directions: every displayed vector is a multiple of v, and every real multiple of v is obtained by choosing that number as t. A span is a subspace, so H is a subspace of $\mathbb{R}^3$.

Exercise 4.1.10: A line with a zero middle coordinate

Here $(3t, 0, -7t) = t(3, 0, -7)$. As t runs through all real numbers, these are exactly all linear combinations of the single vector $(3, 0, -7)$. Consequently $H = \text{Span}\{(3, 0, -7)\}$ is a subspace of $\mathbb{R}^3$. The zero middle coordinate does not make it a different kind of vector space.

Exercise 4.1.11: Separate two free parameters

For arbitrary $b, c \in \mathbb{R}$,

$$\begin{pmatrix} 2b + 3c \\ -b \\ 2c \end{pmatrix} = b \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}.$$

Take $u = (2, -1, 0)$ and $v = (3, 0, 2)$. Every original vector is in their span, and an arbitrary linear combination $bu + cv$ has exactly the required form. Therefore $W = \text{Span}\{u, v\}$, and spans are subspaces of the ambient vector space.

Exercise 4.1.12: Separate two parameters in four dimensions

Collect the coefficients of s and t in each coordinate:

$$\begin{pmatrix} 2s+4t \\ 2s \\ 2s-3t \\ 5t \end{pmatrix} = s\begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \end{pmatrix} + t\begin{pmatrix} 4 \\ 0 \\ -3 \\ 5 \end{pmatrix}.$$

Since s, t are unrestricted, W is precisely the span of the two displayed constant vectors. The span theorem proves that W is a subspace of $\mathbb{R}^4$.

Exercise 4.1.13: A set of generators versus its span

What we must show. Distinguish membership in a finite set from membership in all its linear combinations. The given vectors are $v_1 = (1, 0, -1)$, $v_2 = (2, 1, 3)$, $v_3 = (4, 2, 6)$, and $w = (3, 1, 2)$.

(a) Inspect the three individual vectors.

The vector w equals none of v_1, v_2, v_3, so $w \notin \{v_1, v_2, v_3\}$. Those three vectors are distinct, so that set has exactly three elements.

(b) Count the span.

Since $v_1 \neq 0$, the multiples tv_1 give distinct vectors for distinct real t. Thus the span contains infinitely many vectors.

(c) Find an actual combination.

Adding coordinates gives $v_1 + v_2 = (1+2, 0+1, -1+3) = (3, 1, 2) = w$. Hence w belongs to the span even though it is not one of its three generators.

Exercise 4.1.14: Another vector in the same plane

Use the generators from the preceding exercise and $w = (1, 3, 14)$. Because $v_3 = 2v_2$, combinations of the first two vectors suffice. Set $av_1 + bv_2 = w$. Coordinate comparison gives

$$a + 2b = 1, \qquad b = 3, \qquad -a + 3b = 14.$$

The middle equation gives $b = 3$, the first gives $a = -5$, and the last checks: $-(-5) + 3(3) = 14$. Therefore $w = -5v_1 + 3v_2$ lies in the required span.

Exercise 4.1.15: A coordinate forced to be negative one

Every vector in the set has second coordinate -1. The zero vector has second coordinate zero and is therefore absent. A vector space must contain zero, so no spanning set for this W exists: W is not a vector space.

Exercise 4.1.16: A coordinate forced to be one

Every vector has first coordinate 1, regardless of the values of a, b. Thus zero is absent. This violates the first subspace condition and shows that W is not a vector space under the inherited operations.

Exercise 4.1.17: Three parameters, each with its own coefficient vector

Expanding coordinate by coordinate gives

$$\begin{pmatrix} 2a-b \\ 3b-c \\ 3c-a \\ 3b \end{pmatrix} = a\begin{pmatrix} 2 \\ 0 \\ -1 \\ 0 \end{pmatrix} + b\begin{pmatrix} -1 \\ 3 \\ 0 \\ 3 \end{pmatrix} + c\begin{pmatrix} 0 \\ -1 \\ 3 \\ 0 \end{pmatrix}.$$

With all three parameters arbitrary, W equals the span of these three vectors, so it is a subspace of $\mathbb{R}^4$. These vectors form a valid spanning set; one need not prove independence merely to supply a spanning set.

Exercise 4.1.18: A permanently zero coordinate is harmless

The required decomposition is

$$\begin{pmatrix} 4a+3b \\ 0 \\ a+3b+c \\ 3b-2c \end{pmatrix} = a\begin{pmatrix} 4 \\ 0 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 3 \\ 0 \\ 3 \\ 3 \end{pmatrix} + c\begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}.$$

Every such combination has the specified form and vice versa. Thus the three coefficient vectors span W, which is a subspace. A coordinate fixed at zero is compatible with subspaces; a coordinate fixed at nonzero 1 was not.

Exercise 4.1.19: The space of oscillating functions

Fix ω and let $f(t) = \cos(\omega t)$ and $g(t) = \sin(\omega t)$. The set of functions $c_1 f + c_2 g$, with arbitrary real c_1, c_2, is exactly $\text{Span}\{f, g\}$ inside the vector space of all real-valued functions on $\mathbb{R}$. Hence it is a subspace. Explicitly, adding two functions adds their two coefficients, and scalar multiplication multiplies both coefficients; neither operation changes the fixed frequency ω. Choosing both coefficients zero gives the zero function. The span argument remains valid if $\omega = 0$.

Exercise 4.1.20: Continuous functions and an endpoint condition

(a) State the three required facts.

The function identically zero on $[a, b]$ is continuous. The sum of any two continuous functions on $[a, b]$ is continuous, and every real scalar multiple of a continuous function there is continuous. These are the subspace tests, so $C[a, b]$ is a subspace of all real-valued functions on that interval.

(b) Preserve equality of the endpoint values.

Let $H = \{f \in C[a,b] : f(a) = f(b)\}$. The zero function is in H. If $f, g \in H$, then

$$(f+g)(a) = f(a) + g(a) = f(b) + g(b) = (f+g)(b).$$

For every real c, $(cf)(a) = cf(a) = cf(b) = (cf)(b)$. Both new functions are continuous by part (a), so both belong to H.

Conclusion. H is a subspace of $C[a,b]$.

Exercise 4.1.21: Upper triangular matrices form a subspace

Take $H = \{\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) : a, b, d \in \mathbb{R}\}$. Zero is obtained with $a = b = d = 0$. Adding two such matrices still gives a zero entry in position $(2,1)$, and multiplying by a scalar preserves that zero entry. Therefore H is a subspace of $M_{2\times 2}$. One explicit spanning set is

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Exercise 4.1.22: A homogeneous matrix equation

Let F be the fixed 3×2 matrix, and let $H = \{A \in M_{2\times 4} : FA = 0\}$. The zero 2×4 matrix is in H, because multiplying it by F gives the zero 3×4 matrix. For $A, B \in H$, distributivity gives $F(A+B) = FA + FB = 0$. For real c, $F(cA) = c(FA) = 0$. Thus both $A + B$ and cA belong to H.

Conclusion. H is a subspace of $M_{2\times 4}$. The matrix sizes ensure each product is defined.

Exercise 4.1.23: Five statements about vectors and subspaces

(a) **False.** A zero function must vanish at *every* input. For example $f(t) = t$ has $f(0) = 0$ but is not the zero function.

(b) **False.** Vectors can be functions, matrices, or polynomials, not only arrows in three-dimensional space. An element of P_2, such as t^2, is a vector in that vector space.

(c) **False.** Containing zero is necessary but insufficient. The unit disk contains zero but fails scalar closure, as Exercise 4.1.3 shows.

(d) **True.** A subspace inherits the vector-space operations and all axioms; the subspace conditions ensure the operations stay inside it.

(e) **False.** The chapter introduction describes the major shuttle control systems in terms of *digital*, not analog, signals.

Exercise 4.1.24: Five more statements, including the closure quantifiers

(a) **True.** A vector means an element of a vector space; its physical appearance is irrelevant.

(b) **True.** The identity $(-1)u = -u$ follows from distributivity and uniqueness of the additive inverse; Exercise 4.1.29 supplies the proof.

(c) **True.** Every vector space is a subspace of itself, using its own zero vector and the two operations.

(d) **False.** As defined here, elements of $\mathbb{R}^2$ are ordered pairs, not ordered triples in $\mathbb{R}^3$. The plane $\{(x, y, 0) : x, y \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$ isomorphic to $\mathbb{R}^2$, but it is a different set.

(e) **False as written.** The stated conditions merely say that $u, v, u + v$ are in H and that cu is in H for particular choices. Closure requires these conclusions for *all* $u, v \in H$ and *all* real c. For instance $H = \{(0, 0), (1, 0)\}$ satisfies the listed conditions with $u = v = 0$ and $c = 0$ but is not closed under multiplication by 2. If the omitted universal quantifiers are supplied, the resulting statement is exactly the correct subspace test.

Exercise 4.1.25: Uniqueness of the zero vector

The missing axiom is **4**, the additive-identity axiom (with the commutative form justified by Axiom 2, as the exercise specifies). If w is another additive identity, then using it as an identity gives $0 + w = 0$, whereas using the original zero as an identity gives $0 + w = w$. The same expression equals both 0 and w, hence $w = 0$.

Exercise 4.1.26: Uniqueness of the additive inverse

The blanks are **(a) 3, (b) 5, (c) 4**. To see each step, suppose $u + w = 0$ and add $-u$ to both sides:

$$(-u) + (u + w) = (-u) + 0.$$

Associativity (Axiom 3) changes the left side to $((-u) + u) + w$. The inverse axiom (Axiom 5, with commutativity) changes $(-u)+u$ to zero. The identity axiom (Axiom 4) then changes $0+w = (-u)+0$ to $w = -u$.

Conclusion. There can be only one additive inverse of a given vector.

Exercise 4.1.27: Multiplication by the scalar zero

The blanks are **(a) 8, (b) 3, (c) 5, (d) 4**. Distributivity over scalar addition (Axiom 8) gives $0u = (0 + 0)u = 0u + 0u$. Add the additive inverse $-(0u)$ to both sides. Associativity (3) groups the last two terms on the right as $0u + (-(0u))$; the inverse axiom (5) changes each such pair to zero. The result is $0 = 0u + 0$, which the identity axiom (4) reduces to $0 = 0u$. Here the first zero in $0u$ is a scalar; the zero on the right is a vector.

Exercise 4.1.28: A scalar times the zero vector

The blanks are **(a) 4, (b) 7, (c) 3, (d) 5, (e) 4**. First $0 = 0 + 0$ by the identity axiom (4). Thus $c0 = c(0 + 0) = c0 + c0$ by distributivity over vector addition (7). Add the inverse of $c0$ to both sides. Associativity (3) allows the right-hand inverse pair to be grouped; Axiom 5 reduces the inverse pairs to zero; and Axiom 4 removes the remaining added zero. Hence $0 = c0$. This holds for every scalar c, including negative and zero scalars.

Exercise 4.1.29: The scalar negative one gives the additive inverse

Axiom 10 says $u = 1u$. Axiom 8 then gives

$$u + (-1)u = 1u + (-1)u = (1 + (-1))u = 0u = 0,$$

where the last equality is Exercise 4.1.27. Therefore $(-1)u$ is an additive inverse of u. By uniqueness of that inverse (Exercise 4.1.26), $(-1)u = -u$.

Exercise 4.1.30: A nonzero scalar cannot annihilate a nonzero vector

Suppose $cu = 0$ and $c \neq 0$. Then $1/c$ is an allowed scalar. Multiply both sides by it and use Axioms 9 and 10 and Exercise 4.1.28:

$$u = 1u = ((1/c)c)u = (1/c)(cu) = (1/c)0 = 0.$$

The nonzero hypothesis is needed to form $1/c$; when $c = 0$, the equation $cu = 0$ holds for every u and does not force $u = 0$.

Exercise 4.1.31: The span is the smallest containing subspace

Suppose the subspace H contains u and v. An arbitrary member of $\text{Span}\{u, v\}$ has form $au + bv$ for real a, b. Scalar closure puts au, bv in H, and additive closure puts $au+bv$ in H. Thus $\text{Span}\{u, v\} \subseteq H$. Since the span itself is a subspace that contains u, v, it is the smallest such subspace, in the sense of inclusion.

Exercise 4.1.32: Intersection works; union can fail

Check the intersection's three properties.

Both H and K contain zero, so their intersection does too. If $u, v \in H \cap K$, then they belong to both subspaces; hence $u + v$ belongs to both, so $u + v \in H \cap K$. Likewise any scalar multiple cu belongs to both, hence to their intersection. Thus $H \cap K$ is a subspace.

Give a concrete counterexample for a union.

Let $H = \{(x, 0) : x \in \mathbb{R}\}$ and $K = \{(0, y) : y \in \mathbb{R}\}$, the two coordinate axes. Their union contains $(1, 0)$ and $(0, 1)$, but not their sum $(1, 1)$. Thus a union of subspaces need not be a subspace.

Exercise 4.1.33: The sum of two subspaces

(a) Verify closure, keeping each summand in its own subspace.

Zero is $0 + 0$ with the first zero in H and the second in K. For $x = h_1 + k_1$ and $y = h_2 + k_2$ in $H + K$, associativity and commutativity give

$$x + y = (h_1 + h_2) + (k_1 + k_2) \in H + K,$$

since $h_1 + h_2 \in H$ and $k_1 + k_2 \in K$. For a real scalar c, $cx = ch_1 + ck_1 \in H + K$ by both scalar closures. Therefore $H + K$ is a subspace of V.

(b) Put the missing summand equal to zero.

If $h \in H$, then $h = h + 0$ with $0 \in K$, so $H \subseteq H + K$. If $k \in K$, write $k = 0 + k$ to obtain $K \subseteq H + K$. With their inherited operations, both are subspaces of $H + K$.

Exercise 4.1.34: Spanning the sum by combining the generators

Let $H = \text{Span}\{u_1, \ldots, u_p\}$ and $K = \text{Span}\{v_1, \ldots, v_q\}$.

Every element of the sum belongs to the combined span.

If $x = h + k \in H + K$, write $h = \sum_i a_i u_i$ and $k = \sum_j b_j v_j$. Then $x = \sum_i a_i u_i + \sum_j b_j v_j$, a combination of the combined list.

Every element of the combined span belongs to the sum.

Conversely split any combination of that combined list into its u-part $h \in H$ and its v-part $k \in K$. It then equals $h + k \in H + K$. These two inclusions prove equality of the sets.

Exercise 4.1.35: A checked representation in four dimensions

What we must show. Express $w = (9, -4, -4, 7)$ as a combination of the given v_1, v_2, v_3. Set $av_1 + bv_2 + cv_3 = w$. The augmented system and its reduced form are

$$\left[\begin{array}{rrr|r} 8 & -4 & -7 & 9 \\ -4 & 3 & 6 & -4 \\ -3 & -2 & -5 & -4 \\ 9 & -8 & -18 & 7 \end{array}\right] \sim \left[\begin{array}{ccc|r} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The three coefficient values are $1, -2, 1$. Rather than trust the reduction alone, check them in the original vectors:

$$\begin{pmatrix} 8 \\ -4 \\ -3 \\ 9 \end{pmatrix} - 2\begin{pmatrix} -4 \\ 3 \\ -2 \\ -8 \end{pmatrix} + \begin{pmatrix} -7 \\ 6 \\ -5 \\ -18 \end{pmatrix} = \begin{pmatrix} 8 + 8 - 7 \\ -4 - 6 + 6 \\ -3 + 4 - 5 \\ 9 + 16 - 18 \end{pmatrix} = \begin{pmatrix} 9 \\ -4 \\ -4 \\ 7 \end{pmatrix}.$$

Conclusion. $w = v_1 - 2v_2 + v_3$ belongs to the stated span.

Exercise 4.1.36: Test membership in a column span

The columns of A span precisely the vectors Ax. Therefore solve $Ax = y$:

$$\left[\begin{array}{rrr|r} 3 & -5 & -9 & -4 \\ 8 & 7 & -6 & -8 \\ -5 & -8 & 3 & 6 \\ 2 & -2 & -9 & -5 \end{array}\right] \sim \left[\begin{array}{ccc|r} 1 & 0 & 0 & -1/5 \\ 0 & 1 & 0 & -2/5 \\ 0 & 0 & 1 & 3/5 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

There is no inconsistent row. The candidate is $x = (-1/5, -2/5, 3/5)$. Checking the original multiplication gives

$$Ax = \tfrac{1}{5}(-3 + 10 - 27,\ -8 - 14 - 18,\ 5 + 16 + 9,\ -2 + 4 - 27) = (-4, -8, 6, -5) = y.$$

Conclusion. y is in the subspace spanned by the columns of A.

Exercise 4.1.37: Recognizing two multiple-angle functions

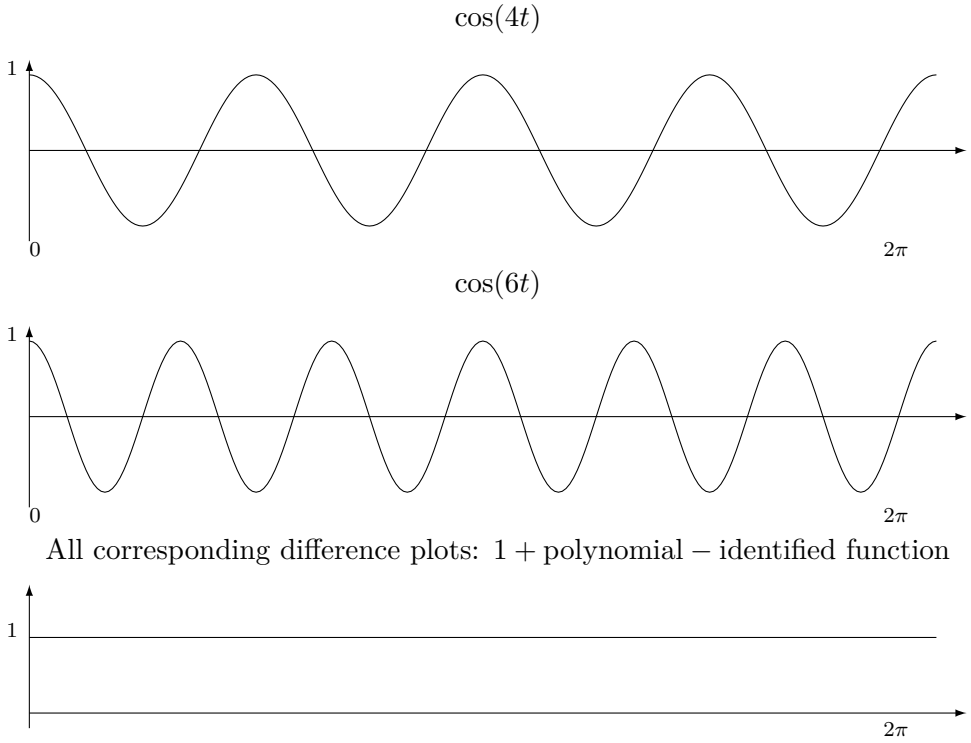
What we must show. Identify the two given polynomials in $\cos t$, then verify them algebraically and graphically. Write $c = \cos t$. The double-angle formula gives $\cos 2t = 2c^2 - 1$. Apply it a second time:

$$\cos 4t = 2(2c^2 - 1)^2 - 1 = 8c^4 - 8c^2 + 1 = f(t).$$

For the other polynomial use $\cos 3u = 4\cos^3 u - 3\cos u$ with $u = 2t$:

$$\cos 6t = 4(2c^2 - 1)^3 - 3(2c^2 - 1) = 32c^6 - 48c^4 + 18c^2 - 1 = g(t).$$

Thus the first graph has four full cosine oscillations on $[0, 2\pi]$ and the second has six. The requested difference graphs are exactly $1 + f(t) - \cos 4t = 1$ and $1 + g(t) - \cos 6t = 1$. A numerical graph supports the recognition, while the displayed identities prove it for every t.



Exercise 4.1.38: Recognizing three polynomials in sine

Put $s = \sin t$. The imaginary part of $(\cos t + i \sin t)^3 = \cos 3t + i \sin 3t$ gives

$$\sin 3t = 3s\cos^2 t - s^3 = 3s(1 - s^2) - s^3 = 3s - 4s^3 = f(t).$$

The double-angle identity $\cos 2t = 1 - 2s^2$ gives

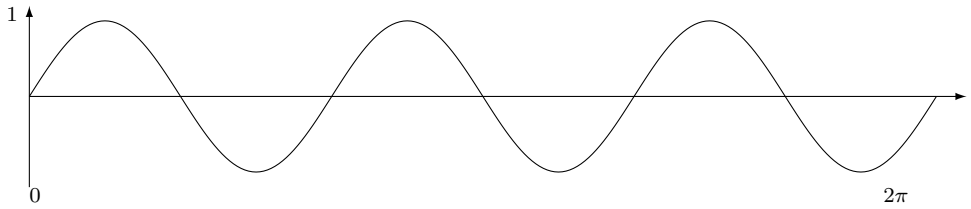
$$\cos 4t = 2(1 - 2s^2)^2 - 1 = 1 - 8s^2 + 8s^4 = g(t).$$

Likewise the imaginary part of the fifth power gives

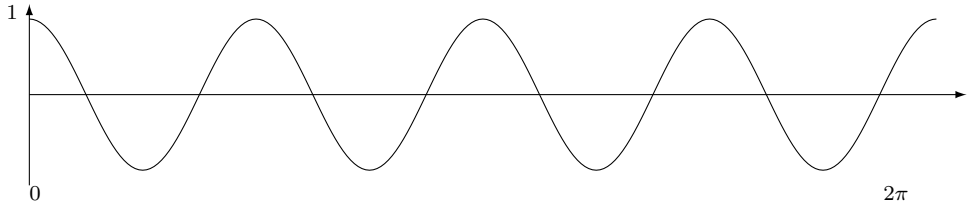
$$\begin{aligned}\sin 5t &= 5\cos^4 t\, s - 10\cos^2 t\, s^3 + s^5 \\ &= 5(1 - s^2)^2 s - 10(1 - s^2)s^3 + s^5 = 5s - 20s^3 + 16s^5 = h(t).\end{aligned}$$

(The power identities used here also follow by repeated angle-addition formulas.) The plots have three sine cycles, four cosine cycles, and five sine cycles, respectively, on $[0, 2\pi]$. Subtracting each identified function from one plus its original polynomial gives the constant 1, as requested.

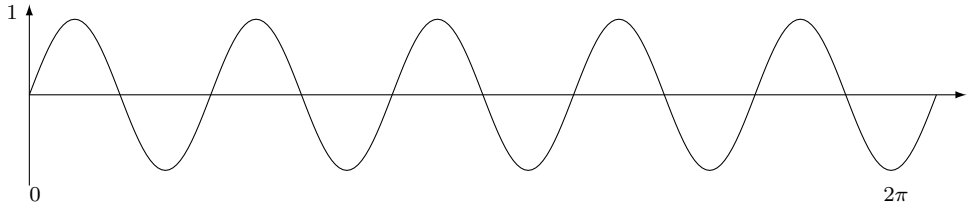
$\sin(3t)$



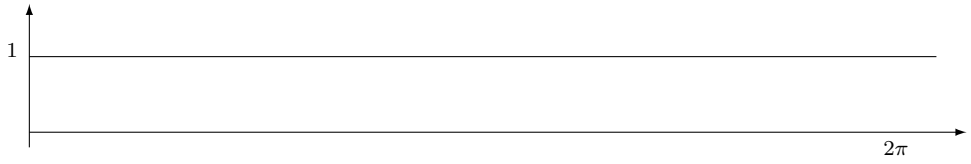
$\cos(4t)$



$\sin(5t)$



All corresponding difference plots: $1 +$ polynomial $-$ identified function



4.2 Null spaces, column spaces, and linear transformations

For an $m \times n$ matrix A, the null space $\operatorname{Nul} A$ contains the *inputs* $x \in \mathbb{R}^n$ satisfying $Ax = 0$. The column space $\operatorname{Col} A$ contains the *outputs* $Ax \in \mathbb{R}^m$. Do not confuse these two roles. To test $w \in \operatorname{Nul} A$, multiply Aw and compare with zero. To test $w \in \operatorname{Col} A$, solve $Ax = w$ and ask whether a solution exists. A homogeneous linear equation defines a subspace; a nonhomogeneous equation usually describes a translated subspace and does not even contain zero. For a linear map $T : V \to W$, the corresponding terms are kernel $\{v : T(v) = 0\}$ and range $\{T(v) : v \in V\}$.

Practice problem 4.2.P1: Two descriptions of one plane

For $W = \{(a, b, c) : a - 3b - c = 0\}$, first observe that $W = \operatorname{Nul}[\,1 \;\; -3 \;\; -1\,]$. The null space theorem therefore makes it a subspace of $\mathbb{R}^3$. Alternatively solve $a = 3b + c$. Then

$$(a, b, c) = b(3, 1, 0) + c(1, 0, 1).$$

As b, c vary freely, this proves $W = \operatorname{Span}\{(3, 1, 0), (1, 0, 1)\}$, a subspace by the span theorem. The first description is implicit (an equation); the second is explicit (generators and arbitrary coefficients).

Practice problem 4.2.P2: Add two attainable outputs

Suppose $Ax = v$ and $Ay = w$ have solutions x, y. By distributivity, $A(x + y) = Ax + Ay = v + w$, so the equation with right-hand side $v + w$ is also consistent. Equivalently $v, w \in \operatorname{Col} A$, and the column space is closed under addition. We need not find either of the original solutions.

Exercise 4.2.1: Test a proposed null-space vector directly

What we must show. Decide whether $w = (1, 3, -4)$ is in $\operatorname{Nul} A$ for the displayed matrix. Multiply using the original rows, retaining the signs:

$$\begin{pmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 3 - 15 + 12 \\ 6 - 6 + 0 \\ -8 + 12 - 4 \end{pmatrix} = 0.$$

Every coordinate is zero, so $w \in \operatorname{Nul} A$. A vector belongs to the null space precisely when this multiplication gives zero, not merely when its coordinates resemble a row or column of A.

Exercise 4.2.2: Another direct multiplication

With $w = (1, -1, 1)$, compute

$$\begin{pmatrix} 2 & 6 & 4 \\ -3 & 2 & 5 \\ -5 & -4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 - 6 + 4 \\ -3 - 2 + 5 \\ -5 + 4 + 1 \end{pmatrix} = 0.$$

Thus $w \in \operatorname{Nul} A$. All three equations must hold; checking only the first row would not have been enough.

Exercise 4.2.3: Describe a null space using two free variables

The homogeneous system for $A = \left(\begin{smallmatrix} 1 & 2 & 4 & 0 \\ 0 & 1 & 3 & -2 \end{smallmatrix}\right)$ is

$$x_1 + 2x_2 + 4x_3 = 0, \qquad x_2 + 3x_3 - 2x_4 = 0.$$

The pivot variables are x_1, x_2; choose $x_3 = s, x_4 = t$ freely. The second equation gives $x_2 = -3s + 2t$, and substitution in the first gives $x_1 = -2(-3s + 2t) - 4s = 2s - 4t$. Therefore

$$x = s \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -4 \\ 2 \\ 0 \\ 1 \end{pmatrix}.$$

Both parameters may be any real number, so the two displayed vectors span $\operatorname{Nul} A$; this formula describes every solution and only solutions.

Exercise 4.2.4: A forced zero coordinate and two free coordinates

For $A = \left(\begin{smallmatrix} 1 & -3 & 2 & 0 \\ 0 & 0 & 3 & 0 \end{smallmatrix}\right)$, the second equation is $3x_3 = 0$, so $x_3 = 0$. The first becomes $x_1 = 3x_2$. Letting $x_2 = s$ and $x_4 = t$ gives

$$\operatorname{Nul} A = \left\{ s \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\}.$$

In particular the fourth column being zero leaves the fourth input free; it does not require the fourth input to be zero.

Exercise 4.2.5: Back substitution in a five-column matrix

The nonzero rows give

$$x_1 - 4x_2 + 2x_4 = 0, \qquad x_3 - 5x_4 = 0, \qquad 2x_5 = 0.$$

The last equation gives $x_5 = 0$. Set the free variables $x_2 = s, x_4 = t$. Then $x_3 = 5t$ and $x_1 = 4s - 2t$. Thus

$$\operatorname{Nul} A = \operatorname{Span} \left\{ \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 5 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Each generating vector is obtained by giving one free variable value 1 and the other value 0; their arbitrary combinations preserve both free variables.

Exercise 4.2.6: Three free variables

The matrix gives two effective equations:

$$x_1 + 3x_2 - 4x_3 - 3x_4 + x_5 = 0, \qquad x_2 - 3x_3 + x_4 = 0.$$

The zero third row adds no condition. Set $x_3 = r, x_4 = s, x_5 = t$. Then $x_2 = 3r - s$ and $x_1 = -3(3r - s) + 4r + 3s - t = -5r + 6s - t$. Consequently

$$\operatorname{Nul} A = \operatorname{Span}\left\{\begin{pmatrix}-5\\3\\1\\0\\0\end{pmatrix}, \begin{pmatrix}6\\-1\\0\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\0\\0\\1\end{pmatrix}\right\}.$$

Exercise 4.2.7: An affine plane misses zero

The equation is $a+b+c=2$. At $(a,b,c)=(0,0,0)$ it would read $0=2$, which is false. Hence the zero vector is not in W; W is not a vector space under the usual operations. The fact that the equation is linear in the variables does not make its *nonhomogeneous* solution set a subspace.

Exercise 4.2.8: Move all variable terms to one side

The relation $3r - 2 = 3s + t$ becomes $3r - 3s - t = 2$. Zero does not satisfy this relation, because its left side is zero. Therefore the given set is not a vector space.

Exercise 4.2.9: Two homogeneous constraints define a null space

Rewrite $p - 3q = 4s$ and $2p = s + 5r$ as

$$p - 3q + 0r - 4s = 0, \qquad 2p + 0q - 5r - s = 0.$$

Thus W is the null space of $\left(\begin{smallmatrix}1 & -3 & 0 & -4\\ 2 & 0 & -5 & -1\end{smallmatrix}\right)$, and the null space theorem makes it a subspace of $\mathbb{R}^4$. Both right-hand sides after rearrangement are zero; this is the essential property.

Exercise 4.2.10: Another homogeneous description

Move all terms to the left: $3a + b - c + 0d = 0$ and $a + b + 2c - 2d = 0$. Then

$$W = \operatorname{Nul}\begin{pmatrix}3 & 1 & -1 & 0\\ 1 & 1 & 2 & -2\end{pmatrix}.$$

The null space of a matrix is a subspace of its input space, here $\mathbb{R}^4$. This proves the requested vector-space property without finding parameters.

Exercise 4.2.11: A hidden constant obstructs the zero vector

For $(s - 2t, 3 + 3s, 3s + t, 2s)$ to be zero, its fourth coordinate forces $s = 0$. But then its second coordinate is 3, not zero. No choice of t can fix that contradiction. Thus zero is absent, and W is not a vector space. Merely setting all parameters zero would give only one candidate; the argument above checks that *no* parameters produce zero.

Exercise 4.2.12: Another parametrization that never produces zero

If $(3p-5q, 4q, p, q+1) = 0$, the third coordinate forces $p = 0$ and the second forces $q = 0$. The fourth would then be 1. This contradiction shows that zero cannot occur, so W is not a vector space.

Exercise 4.2.13: A span in disguise

For arbitrary real c, d,

$$(c-6d, d, c) = c(1,0,1) + d(-6,1,0).$$

Both directions of this parametrization hold, so $W = \text{Span}\{(1,0,1), (-6,1,0)\}$. The span theorem makes it a subspace of $\mathbb{R}^3$.

Exercise 4.2.14: A two-generator subspace

Collecting the coefficients of s, t gives

$$(-s+3t, s-2t, 5s-t) = s(-1,1,5) + t(3,-2,-1).$$

Since s, t are unrestricted real numbers, W is exactly the span of these two vectors, hence is a subspace of $\mathbb{R}^3$.

Exercise 4.2.15: Make the coefficient vectors into columns

What we must show. Find a matrix whose column space is the given parametrized set. Separate the coefficients of r, s, t:

$$\begin{pmatrix} 2s+t \\ r-s+2t \\ 3r+s \\ 2r-s-t \end{pmatrix} = r\begin{pmatrix} 0 \\ 1 \\ 3 \\ 2 \end{pmatrix} + s\begin{pmatrix} 2 \\ -1 \\ 1 \\ -1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ 0 \\ -1 \end{pmatrix}.$$

Therefore one suitable matrix is

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 0 \\ 2 & -1 & -1 \end{pmatrix}.$$

By definition its column space contains exactly the linear combinations just listed. Other matrices can have the same column space; uniqueness is not requested.

Exercise 4.2.16: Construct a matrix from three parameters

Here the required equality is

$$\begin{pmatrix} b-c \\ 2b+3d \\ b+3c-3d \\ c+d \end{pmatrix} = b\begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + c\begin{pmatrix} -1 \\ 0 \\ 3 \\ 1 \end{pmatrix} + d\begin{pmatrix} 0 \\ 3 \\ -3 \\ 1 \end{pmatrix}.$$

Thus take $A = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 3 \\ 1 & 3 & -3 \\ 0 & 1 & 1 \end{pmatrix}$. For $x = (b, c, d)$, the product Ax gives the original form, and every such form is obtained from some x. This proves equality with $\operatorname{Col} A$.

Exercise 4.2.17: Count inputs and outputs

The displayed matrix has four rows and two columns, so its mapping is $A : \mathbb{R}^2 \to \mathbb{R}^4$.

Conclusion. (a) $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^2$, so $k = 2$. (b) $\operatorname{Col} A$ is a subspace of $\mathbb{R}^4$, so $k = 4$.

These are ambient-space dimensions, not necessarily the dimensions of the two subspaces.

Exercise 4.2.18: A rectangular matrix with three inputs

The matrix has four rows and three columns. A vector x in Ax must therefore have three coordinates, whereas the output has four.

Conclusion. (a) $k = 3$ for the ambient space of $\operatorname{Nul} A$. (b) $k = 4$ for the ambient space of $\operatorname{Col} A$.

Exercise 4.2.19: Five inputs and two outputs

The matrix has size 2×5, so Ax is defined for $x \in \mathbb{R}^5$ and belongs to $\mathbb{R}^2$.

Conclusion. (a) $\operatorname{Nul} A \subseteq \mathbb{R}^5$, so $k = 5$. (b) $\operatorname{Col} A \subseteq \mathbb{R}^2$, so $k = 2$.

Exercise 4.2.20: A row matrix

Here $A = [\,1 \;\; -3 \;\; 2 \;\; 0 \;\; -5\,]$ has one row and five columns. It maps a five-coordinate input to one real number.

Conclusion. (a) $k = 5$ for $\operatorname{Nul} A$. (b) $k = 1$ for $\operatorname{Col} A$.

Exercise 4.2.21: Exhibit one vector in each space

Using the matrix in Exercise 4.2.17, its second column equals $-2/3$ times its first. Therefore $2a_1 + 3a_2 = 0$, giving the nonzero null vector $(2, 3)$. A direct check is

$$\begin{pmatrix} 6 & -4 \\ -3 & 2 \\ -9 & 6 \\ 9 & -6 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 12 - 12 \\ -6 + 6 \\ -18 + 18 \\ 18 - 18 \end{pmatrix} = 0.$$

The first column $(6, -3, -9, 9)$ is a nonzero member of $\operatorname{Col} A$. It equals $A(1, 0)$, so its membership follows directly from the definition.

Exercise 4.2.22: A null relation among three columns

For the matrix in Exercise 4.2.18, choose $x = (-1, -1, 1)$. Multiplication verifies

$$Ax = \begin{pmatrix} -5+2+3 \\ 1+0-1 \\ 0+2-2 \\ 5-7+2 \end{pmatrix} = 0.$$

Thus this is a nonzero vector of $\operatorname{Nul} A$. The first column $(5, -1, 0, -5)$ is a nonzero vector in $\operatorname{Col} A$, because it is the output $A(1, 0, 0)$.

Exercise 4.2.23: A vector can belong to both spaces

For $A = \left(\begin{smallmatrix} -2 & 4 \\ -1 & 2 \end{smallmatrix}\right)$ and $w = (2, 1)$, the negative of the first column is w. Hence $A(-1, 0) = w$, proving $w \in \operatorname{Col} A$. For null-space membership test the different multiplication:

$$Aw = (-4+4, -2+2) = (0, 0).$$

Thus $w \in \operatorname{Nul} A$ as well. These two statements are compatible; a square matrix's input and output spaces have the same ambient space, though they serve different roles.

Exercise 4.2.24: Two independent membership checks in four dimensions

The matrix and vector are

$$A = \begin{pmatrix} 10 & -8 & -2 & -2 \\ 0 & 2 & 2 & -2 \\ 1 & -1 & 6 & 0 \\ 1 & 1 & 0 & -2 \end{pmatrix}, \qquad w = \begin{pmatrix} 2 \\ 2 \\ 0 \\ 2 \end{pmatrix}.$$

Column-space membership.

The first two columns add to w, so $A(1, 1, 0, 0) = w$. This is a concrete solution of $Ax = w$, proving $w \in \operatorname{Col} A$.

Null-space membership.

Use w itself as the input:

$$Aw = (20 - 16 - 4,\ 4 - 4,\ 2 - 2,\ 2 + 2 - 4) = (0, 0, 0, 0).$$

Conclusion. w belongs to both $\operatorname{Col} A$ and $\operatorname{Nul} A$.

Exercise 4.2.25: Six statements about null spaces and column spaces

(a) **True.** $\operatorname{Nul} A = \{x : Ax = 0\}$ is the definition.

(b) **False.** An $m \times n$ matrix has n input coordinates, so its null space is in $\mathbb{R}^n$, not generally $\mathbb{R}^m$.

(c) **True.** The possible outputs of $x \mapsto Ax$ are precisely all combinations of the columns, hence form $\operatorname{Col} A$.

(d) **False.** One consistent equation only says that this particular b is in the column space. For instance $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $b = (1, 0)$ give a consistent equation but $\operatorname{Col} A$ is just the horizontal axis.

(e) **True.** The kernel contains zero and is closed under addition and scalar multiplication because the map is linear.

(f) **True.** $\operatorname{Col} A = \{Ax : x \in \mathbb{R}^n\}$; this is the output description of the span of the columns.

Exercise 4.2.26: Six related statements about kernels and ranges

(a) **True.** A null space is a subspace of the input space, hence a vector space.

(b) **True.** Columns of an $m \times n$ matrix have m coordinates; their combinations belong to $\mathbb{R}^m$.

(c) **False.** Solutions of $Ax = b$ are inputs x, not outputs. The column space instead contains those right-hand sides b for which a solution exists.

(d) **True.** The kernel of $x \mapsto Ax$ is, by definition, the set of inputs taken to zero, exactly $\operatorname{Nul} A$.

(e) **True.** The range is a subspace of the target vector space: add two outputs or multiply an output by a scalar and linearity produces another output.

(f) **True.** A homogeneous linear differential equation is $L(f) = 0$ for a linear differential operator L. Its solutions are the kernel of L on the vector space of functions with the required derivatives.

Exercise 4.2.27: Scale a known homogeneous solution

The given solution vector is $u = (3, 2, -1)$. The proposed vector is $10u$. The solution set of the homogeneous system is a null space, which is a vector space and is therefore closed under scalar multiplication. Consequently $10u = (30, 20, -10)$ is another solution. This uses no substitution or row operations, as the exercise requests.

Exercise 4.2.28: Scale an attainable right-hand side

Both systems have the same coefficient matrix A. Their right-hand sides are $b = (0, 1, 9)$ and $5b = (0, 5, 45)$, respectively. Since the first is consistent, there is an input u with $Au = b$. Linearity gives $A(5u) = 5Au = 5b$, so the second system is consistent too. Equivalently $b \in \operatorname{Col} A$ and column spaces are closed under scalar multiplication.

Exercise 4.2.29: Prove the column-space subspace theorem

A general member of $\operatorname{Col} A$ has form Ax, where $x \in \mathbb{R}^n$.

(a) Zero is an output.

Take $x = 0$. Then $Ax = A0 = 0$, so the zero vector of $\mathbb{R}^m$ lies in $\operatorname{Col} A$.

(b) Two outputs can be added.

For inputs $x, w \in \mathbb{R}^n$, distributivity gives $Ax + Aw = A(x + w)$. Since $x + w$ is another allowed input, this is an element of $\operatorname{Col} A$.

(c) An output can be multiplied by any scalar.

For real c, the identity $c(Ax) = A(cx)$ expresses it as an output as well.

Conclusion. The three conditions prove that $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

Exercise 4.2.30: The range of a linear transformation is a subspace

Let $R = \{T(x) : x \in V\} \subseteq W$. First $T(0) = 0$, so $0 \in R$. For arbitrary $y = T(x)$ and $z = T(v)$ in R, linearity gives

$$y + z = T(x) + T(v) = T(x + v) \in R.$$

For any real scalar c, $cy = cT(x) = T(cx) \in R$. The inputs $x + v, cx$ are in V because V is a vector space. Thus R passes all subspace tests inside W. One need not assume that T is onto W.

Exercise 4.2.31: Evaluate a quadratic at two inputs

The map is $T(p) = (p(0), p(1))$ from P_2 into $\mathbb{R}^2$.

(a) Check the two linearity identities at both coordinates.

For polynomials p, q and scalar c,

$$T(p + q) = (p(0) + q(0), p(1) + q(1)) = T(p) + T(q), \qquad T(cp) = cT(p).$$

Thus T is linear.

(b) Solve the kernel equations and then describe the range.

Write $p(t) = a + bt + ct^2$. The equations $p(0) = p(1) = 0$ give $a = 0$ and $b + c = 0$, hence $p(t) = c(t^2 - t)$. The kernel is therefore $\operatorname{Span}\{t^2 - t\}$. For any desired output (r, s), the polynomial $p(t) = r + (s - r)t$ satisfies $p(0) = r$ and $p(1) = s$. Therefore the range is all of $\mathbb{R}^2$.

Conclusion. One kernel generator is $t^2 - t$, and T is onto $\mathbb{R}^2$.

Exercise 4.2.32: Repeat the same evaluation twice

Here $T(p) = (p(0), p(0))$. With $p(t) = a + bt + ct^2$, the output is (a, a). It is zero exactly when $a = 0$, leaving b, c arbitrary. Thus

$$\ker T = \operatorname{Span}\{t, t^2\}, \qquad \operatorname{range} T = \{(a, a) : a \in \mathbb{R}\} = \operatorname{Span}\{(1, 1)\}.$$

The output has two coordinates, but those coordinates must be equal; therefore the map is not onto the whole target $\mathbb{R}^2$.

Exercise 4.2.33: Add a matrix to its transpose

Define $T(A) = A + A^T$ on $M_{2\times 2}$.

(a) Verify linearity using transpose rules.

For matrices A, C and real scalar r,

$$T(A+C) = A + C + (A+C)^T = (A + A^T) + (C + C^T), \qquad T(rA) = r(A + A^T).$$

These are exactly the two linearity identities.

(b) Produce a preimage of any symmetric matrix.

If $B^T = B$, choose $A = B/2$. Then $T(A) = B/2 + B^T/2 = B$.

(c) Establish equality of the range with the symmetric matrices.

Every output satisfies $(A + A^T)^T = A^T + A = A + A^T$, so every output is symmetric. Conversely part (b) realizes every symmetric matrix as an output. Both inclusions prove the claimed equality of sets.

(d) Solve the zero-output condition entry by entry.

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$T(A) = \begin{pmatrix} 2a & b+c \\ b+c & 2d \end{pmatrix}.$$

This is zero exactly when $a = d = 0$ and $c = -b$. Therefore

$$\ker T = \left\{ b \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : b \in \mathbb{R} \right\}.$$

Exercise 4.2.34: The antiderivative with a fixed initial value

For continuous f on $[0, 1]$, the specified antiderivative is uniquely

$$T(f)(x) = \int_0^x f(t)\, dt.$$

Existence follows from the Fundamental Theorem of Calculus. For uniqueness, two such antiderivatives differ by a constant, and their value zero at the left endpoint forces that constant to be zero. Linearity of the integral gives $T(f+g)(x) = T(f)(x) + T(g)(x)$ and $T(cf)(x) = cT(f)(x)$ for every x. Thus these are equalities of functions, and T is a linear transformation. If $T(f)$ is the zero function, differentiating it gives $f(x) = 0$ on $(0, 1)$, and continuity gives zero at the endpoints too. Conversely the zero function has zero antiderivative with the prescribed initial value.

Conclusion. The kernel consists only of the zero function.

Exercise 4.2.35: The image of a subspace

Let U be a subspace of V. Since $0 \in U$ and $T(0) = 0$, zero is in $T(U)$. For $y = T(u)$ and $z = T(v)$ with $u, v \in U$, linearity gives $y + z = T(u+v) \in T(U)$ because $u + v \in U$. Likewise $cy = T(cu) \in T(U)$ since $cu \in U$ for every real c. These three tests prove that $T(U)$ is a subspace of W. The conclusion concerns the image of the whole subspace, not just a list of selected outputs.

Exercise 4.2.36: The inverse image of a subspace

Let $U = \{x \in V : T(x) \in Z\}$, where Z is a subspace of W. Since $T(0) = 0 \in Z$, zero belongs to U. If $x, y \in U$, then $T(x), T(y) \in Z$, and $T(x + y) = T(x) + T(y) \in Z$ by additive closure of Z. Thus $x + y \in U$. For real c, $T(cx) = cT(x) \in Z$, so $cx \in U$. Therefore U is a subspace of V. This argument uses the preimage as a set; it does not require T to have an inverse function.

Exercise 4.2.37: An output that is not a zero-output input

The displayed data are

$$A = \begin{pmatrix} 7 & 6 & -4 & 1 \\ -5 & -1 & 0 & -2 \\ 9 & -11 & 7 & -3 \\ 19 & -9 & 7 & 1 \end{pmatrix}, \qquad w = (1, 1, -1, -3).$$

To test null-space membership, multiply:

$$Aw = (7 + 6 + 4 - 3,\ -5 - 1 + 6,\ 9 - 11 - 7 + 9,\ 19 - 9 - 7 - 3) = (14, 0, 0, 0) \neq 0.$$

Therefore $w \notin \operatorname{Nul} A$. For the column space, the second column plus the third minus the fourth is

$$(6, -1, -11, -9) + (-4, 0, 7, 7) - (1, -2, -3, 1) = (1, 1, -1, -3) = w.$$

Thus $A(0, 1, 1, -1) = w$, which proves $w \in \operatorname{Col} A$.

Conclusion. w is in the column space only.

Exercise 4.2.38: Another vector lying in both spaces

Here

$$A = \begin{pmatrix} -8 & 5 & -2 & 0 \\ -5 & 2 & 1 & -2 \\ 10 & -8 & 6 & -3 \\ 3 & -2 & 1 & 0 \end{pmatrix}, \qquad w = (1, 2, 1, 0).$$

First $Aw = (-8 + 10 - 2,\ -5 + 4 + 1,\ 10 - 16 + 6,\ 3 - 4 + 1) = 0$, so $w \in \operatorname{Nul} A$. For the distinct column-space test use $x = (-2, -3, 0, 1)$:

$$Ax = (16 - 15,\ 10 - 6 - 2,\ -20 + 24 - 3,\ -6 + 6) = (1, 2, 1, 0) = w.$$

Conclusion. w belongs to both spaces, verified by two different products.

Exercise 4.2.39: Recover column relations and the entire null space

The given matrix is

$$A = \begin{pmatrix} 5 & 1 & 2 & 2 & 0 \\ 3 & 3 & 2 & -1 & -12 \\ 8 & 4 & 4 & -5 & 12 \\ 2 & 1 & 1 & 0 & -2 \end{pmatrix}, \qquad B = [\, a_1\ a_2\ a_4 \,].$$

(a) Exhibit the two column combinations.

Coordinate-wise verification gives

$$a_3 = \tfrac{1}{3}a_1 + \tfrac{1}{3}a_2, \qquad a_5 = \tfrac{10}{3}a_1 - \tfrac{26}{3}a_2 - 4a_4.$$

Both are combinations of columns of B, so $a_3, a_5 \in \operatorname{Col} B$.

(b) Solve all homogeneous equations.

Exact row reduction gives

$$A \sim \begin{pmatrix} 1 & 0 & 1/3 & 0 & 10/3 \\ 0 & 1 & 1/3 & 0 & -26/3 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Take the free variables $x_3 = s, x_5 = t$. Then $x_1 = -s/3 - 10t/3$, $x_2 = -s/3 + 26t/3$, and $x_4 = 4t$. Hence

$$\operatorname{Nul} A = \operatorname{Span}\left\{ \begin{pmatrix} -1/3 \\ -1/3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -10/3 \\ 26/3 \\ 0 \\ 4 \\ 1 \end{pmatrix} \right\}.$$

These formulas also check the two column relations in part (a): moving all terms to one side gives the corresponding zero combinations.

(c) Explain both failures of invertibility separately.

The first null-space generator is nonzero, so T is not one-to-one: it maps that generator and zero to the same output. There are only three pivots among four rows, so the columns do not span $\mathbb{R}^4$; hence T is not onto $\mathbb{R}^4$. Its having five inputs does not by itself make it onto.

Exercise 4.2.40: Find the line where two planes meet

What we must show. Construct a nonzero vector in both spans and show that its span is the entire intersection. Let $w = c_1v_1 + c_2v_2 = c_3v_3 + c_4v_4$. Move the last two terms left and solve the homogeneous system with columns $v_1, v_2, -v_3, -v_4$:

$$\begin{pmatrix} 5 & 1 & -2 & 0 \\ 3 & 3 & 1 & 12 \\ 8 & 4 & -5 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -10/3 \\ 0 & 1 & 0 & 26/3 \\ 0 & 0 & 1 & -4 \end{pmatrix}.$$

There is one free variable, $c_4 = t$, and $c_1 = 10t/3$, $c_2 = -26t/3$, $c_3 = 4t$. Taking $t = 1$ gives

$$w = \tfrac{10}{3}(5, 3, 8) - \tfrac{26}{3}(1, 3, 4) = (8, -16, -8).$$

Independently the other plane gives $4(2, -1, 5) + (0, -12, -28) = (8, -16, -8)$, confirming both memberships. Every possible set of coefficients in the homogeneous system is a multiple of this one, so every common vector is a multiple of w.

Conclusion. $H \cap K = \operatorname{Span}\{(8, -16, -8)\} = \operatorname{Span}\{(1, -2, -1)\}$.

4.3 Linearly independent sets; bases

A basis for H must consist of vectors *in* H, span exactly H, and be linearly independent. Spanning a larger space is not enough. Independence means that the only zero combination has every coefficient zero. Removing a generator already in the span of the others does not change the span.

For a matrix, row operations preserve the solutions of its homogeneous system and all linear relations between columns. Consequently the pivot indices in RREF identify which *original* columns form a basis for the column space. Reduced columns themselves need not belong to the original column space. In the numerical work below, columns are denoted v_j in their printed order; a displayed RREF and the explicit column relations provide checkable elimination and spanning certificates.

Practice problem 4.3.P1: A basis must have the right ambient space

The two vectors are not scalar multiples: the first would require multiplier -2, but then its second entry would be 4, not 7. They are independent, but two vectors cannot span $\mathbb{R}^3$ because a 3×2 coefficient matrix cannot have a pivot in every row. Nor can they be a basis of $\mathbb{R}^2$: they have three coordinates and are not elements of that space.

Conclusion. They are a basis for their two-dimensional span in $\mathbb{R}^3$, not for $\mathbb{R}^3$ or $\mathbb{R}^2$.

Practice problem 4.3.P2: Remove redundant spanning vectors

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 6 & 2 & -4 \\ -3 & 2 & -2 & -8 \\ 4 & -1 & 3 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{4}{5} & 2 \\ 0 & 1 & \frac{1}{5} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (\frac{4}{5})v_1 + (\frac{1}{5})v_2$$
$$v_4 = (2)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Practice problem 4.3.P3: Spanning every vector of a line is not enough

Neither $e_1 = (1,0,0)^T$ nor $e_2 = (0,1,0)^T$ lies in $H = \{(s,s,0)^T : s \in \mathbb{R}\}$, since their first two coordinates differ. Their span is the entire xy-plane, which is larger than H. Thus they cannot be a basis of H, despite the valid expression $(s,s,0)^T = se_1 + se_2$. A basis of H is instead $\{(1,1,0)^T\}$: it is nonzero and its multiples are exactly H.

Conclusion. No. Membership in H and equality of the span with H are missing.

Exercise 4.3.1: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. The set is a basis for $\mathbb{R}^3$: it is independent and spans.

Exercise 4.3.2: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_2 = 0$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and does not span $\mathbb{R}^3$.

Exercise 4.3.3: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & -1 \\ -3 & -4 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (1)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and does not span $\mathbb{R}^3$.

Exercise 4.3.4: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 2 & -8 \\ -1 & -3 & 5 \\ 1 & 2 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} -8 \\ 5 \\ 4 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. The set is a basis for $\mathbb{R}^3$: it is independent and spans.

Exercise 4.3.5: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 3 & -3 & 0 & 0 \\ -3 & 7 & 0 & -3 \\ 0 & 0 & 0 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = 0$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and spans $\mathbb{R}^3$.

Exercise 4.3.6: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 3 \\ -4 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 6 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is independent and does not span $\mathbb{R}^3$.

Exercise 4.3.7: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -2 & 6 \\ 3 & -1 \\ 0 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -1 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is independent and does not span $\mathbb{R}^3$.

Exercise 4.3.8: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ -2 & 3 & -1 & 0 \\ 3 & -1 & 5 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (2)v_1 + (1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and spans $\mathbb{R}^3$.

Exercise 4.3.9: A null-space basis from one free variable

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & 1 & 4 \\ 3 & -1 & -7 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The free variable columns are 3. The pivot equations are

$$\begin{aligned} x_1 &= 2x_3 \\ x_2 &= -x_3 \\ x_4 &= 0 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Exercise 4.3.10: A null-space basis in five coordinates

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 1 & -2 & 1 & 5 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -8 & 0 & 16 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & -2 \end{bmatrix}.$$

The free variable columns are 4, 5. The pivot equations are

$$\begin{aligned} x_1 &= -2x_4 - 3x_5 \\ x_2 &= x_4 + 2x_5 \\ x_3 &= 2x_5 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Exercise 4.3.11: A basis for a plane

Solve the single homogeneous equation for x: $x = 3y - 2z$. Then

$$(x, y, z)^T = y(3, 1, 0)^T + z(-2, 0, 1)^T.$$

Every vector in the plane has this form. The displayed vectors lie in the plane, and a zero combination has second coordinate $y = 0$ and third coordinate $z = 0$. Thus they are independent and span precisely the plane.

Conclusion. $\{(3, 1, 0)^T, (-2, 0, 1)^T\}$ is a basis.

Exercise 4.3.12: A basis for a line

The line equation $y = -3x$ means that every vector is $(x, -3x)^T = x(1, -3)^T$. The vector $(1, -3)^T$ is nonzero, so the one-element set containing it is independent. Its multiples are exactly the line, so it spans the required space.

Conclusion. $\{(1, -3)^T\}$ is a basis.

Exercise 4.3.13: Null space and original column space

Null space: use the row-equivalent matrix.

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & \frac{5}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The free variable columns are 3, 4. The pivot equations are

$$x_1 = -6x_3 - 5x_4$$
$$x_2 = -\frac{5x_3}{2} - \frac{3x_4}{2}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -6 \\ -\frac{5}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -\frac{3}{2} \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Column space: return to the original matrix.

The pivot indices are 1 and 2, so a column-space basis is $\{(-2, 2, -3)^T, (4, -6, 8)^T\}$. Do not take columns of the echelon matrix instead: row operations need not preserve the actual column space.

Exercise 4.3.14: Two different spaces from one reduction

Null space: solve the homogeneous equations.

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 2 & 3 & -4 & 8 \\ 1 & 2 & 0 & 2 & 8 \\ 2 & 4 & -3 & 10 & 9 \\ 3 & 6 & 0 & 6 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The free variable columns are 2, 4. The pivot equations are

$$\begin{aligned} x_1 &= -2x_2 - 2x_4 \\ x_3 &= 2x_4 \\ x_5 &= 0 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Column space: retain the original pivot columns.

The pivot indices are 1,3,5. Thus a basis for $\operatorname{Col} A$ is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 8 \\ 8 \\ 9 \\ 9 \end{bmatrix} \right\}.$$

Exercise 4.3.15: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 2 & 2 & 3 \\ 0 & 1 & -2 & -1 & -1 \\ -2 & 2 & -8 & 10 & -6 \\ 3 & 3 & 0 & 3 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 10 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -6 \\ 9 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (2)v_1 + (-2)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.16: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -2 & 3 & 5 & 2 \\ 0 & 0 & -1 & -3 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 1 & 2 & -1 & -4 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -\frac{5}{2} & 0 \\ 0 & 1 & 0 & \frac{3}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-\frac{5}{2})v_1 + (\frac{3}{4})v_2 + (3)v_3$$
$$v_5 = (\frac{1}{2})v_2 + (1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Exercise 4.3.17: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 4 & -2 & 8 & -8 \\ 0 & 0 & -4 & 4 & 4 \\ -4 & 2 & 0 & 8 & 0 \\ -6 & -4 & 1 & -3 & 0 \\ 0 & 4 & -7 & 15 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 0 \\ -4 \\ -6 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 2 \\ -4 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ 0 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -8 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-1)v_1 + (2)v_2 + (-1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.18: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -3 & 3 & 0 & 6 & -6 \\ 2 & 0 & 2 & -2 & 3 \\ 6 & -9 & -4 & -14 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ -7 & 6 & -1 & 13 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -3 \\ 2 \\ 6 \\ 0 \\ -7 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -9 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -4 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -6 \\ 3 \\ 0 \\ -1 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (2)v_2 + (-1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.19: Delete a generator using a known relation

The supplied relation gives $v_3 = (4/3)v_1 + (5/3)v_2$. Therefore every combination of all three vectors is a combination of v_1, v_2, so those two still span H. They are not scalar multiples: if $v_2 = cv_1$, the first coordinate gives $c = 1/4$, but the second would be $-3/4$ rather than 9. Thus they are independent.

Conclusion. $\{(4, -3, 7)^T, (1, 9, -2)^T\}$ is a basis for H.

Exercise 4.3.20: Another known dependence relation

Rearrange $2v_1 - v_2 - v_3 = 0$ as $v_3 = 2v_1 - v_2$. The third generator is redundant, so $H = \text{Span}\{v_1, v_2\}$. If $v_2 = cv_1$, the first coordinate gives $c = 4/3$, while the second coordinate would require $3 = (4/3)4 = 16/3$, a contradiction. Hence the two retained vectors are independent.

Conclusion. $\{(3, 4, -2, -5)^T, (4, 3, 2, 4)^T\}$ is a basis for H.

Exercise 4.3.21: Basis definitions and row operations

(a) False. A singleton $\{v\}$ is independent whenever $v \neq 0$, because $cv = 0$ then forces $c = 0$; only the zero singleton is dependent.

(b) False. Spanning is only one requirement. The set $\{e_1, 2e_1\}$ spans a line but is dependent and hence is not a basis.

(c) True. An invertible square matrix has a pivot in every row and column, so its columns both span $\mathbb{R}^n$ and are independent.

(d) False. One can append redundant vectors to a spanning set. A basis is a *minimal* spanning set, equivalently a maximal independent set, not a largest spanning set.

(e) False. If A is replaced by an elementary row-equivalent matrix EA, the relations $Ac = 0$ and $EAc = 0$ are equivalent because E is invertible. Thus precisely the same coefficient relations remain.

Exercise 4.3.22: Minimal spanning and maximal independence

(a) False. $\{e_1\}$ is independent in $\mathbb{R}^2$ but does not span it.

(b) True. Repeatedly delete a vector expressible in terms of the others. Each deletion preserves the span, and finiteness ensures the process stops at an independent spanning subset, hence a basis.

(c) True, with "as large as possible" meaning that no further vector of the space can be added while preserving independence. If an independent set did not span, a vector outside its span could be adjoined without creating dependence. Thus a maximal independent set spans and is a basis.

(d) False. The usual free-variable construction always gives independent vectors: in their free-coordinate positions they have distinct unit entries.

(e) False. Use pivot columns of A, not of its row-reduced matrix. For example $A = (1, 1)^T$ reduces to $B = (1, 0)^T$; their column spaces are different lines.

Exercise 4.3.23: Four spanning vectors in four dimensions

Put the four spanning vectors into a 4×4 matrix A. Because they span $\mathbb{R}^4$, its echelon form has a pivot in every row. There are four rows, hence four pivots; all four columns are pivot columns. Therefore $Ac = 0$ has no free variable and only the zero solution. The columns are independent as well as spanning.

Conclusion. The four vectors form a basis for $\mathbb{R}^4$.

Exercise 4.3.24: An independent square family

The n independent columns of their $n \times n$ matrix give a pivot in every column. Hence there are n pivots, one in each row too. A pivot in every row means the columns span $\mathbb{R}^n$. They were already independent, so they satisfy both basis requirements.

Conclusion. $\{v_1, \ldots, v_n\}$ is a basis for $\mathbb{R}^n$.

Exercise 4.3.25: A proposed basis outside the subspace

A basis of H must be a subset of H. Here $v_1 = (1, 0, 1)^T$ and $v_3 = (0, 1, 0)^T$ have unequal second and third coordinates, so neither belongs to H. The supplied expansion does indeed express every $(s, t, t)^T$ using the three vectors, but it does not show that their span is contained in H. Their span is larger: the three vectors are actually independent and span $\mathbb{R}^3$. A correct basis of H is $\{(1, 0, 0)^T, (0, 1, 1)^T\}$, since $(s, t, t)^T = s(1, 0, 0)^T + t(0, 1, 1)^T$ uniquely.

Conclusion. No. Unique representation of the vectors of H does not repair the failure of the proposed basis vectors to lie in H.

Exercise 4.3.26: A trigonometric spanning set

Use the identity $\sin 2t = 2\sin t\cos t$. It shows the third function is half the second, so $\sin t$ and $\sin 2t$ span the same subspace as all three. If $a\sin t + b\sin 2t = 0$ for every t, evaluate at $t = \pi/2$ to obtain $a = 0$, then at $t = \pi/4$ to obtain $b = 0$. Thus the two retained functions are independent.

Conclusion. $\{\sin t, \sin 2t\}$ is a basis of the specified function space.

Exercise 4.3.27: The mass–spring vibration space

Exercise 4.1.19 gives all displacements as $y(t) = c_1\cos(\omega t) + c_2\sin(\omega t)$, with fixed physical frequency $\omega > 0$. This already proves spanning by the two displayed functions. If $a\cos(\omega t) + b\sin(\omega t) = 0$ identically, $t = 0$ gives $a = 0$, and $t = \pi/(2\omega)$ gives $b = 0$.

Conclusion. $\{\cos(\omega t), \sin(\omega t)\}$ is a basis for the vibration space. If a nonphysical zero frequency were allowed, the sine function would vanish and the space would instead have basis $\{1\}$.

Exercise 4.3.28: A critically damped circuit space

The given description $v(t) = e^{-bt}(c_1 + c_2 t)$ means $H = \text{Span}\{e^{-bt}, te^{-bt}\}$. To check independence, suppose $ae^{-bt} + cte^{-bt} = 0$ for every t. Since $e^{-bt} > 0$, division gives $a + ct = 0$ identically. At $t = 0$ we get $a = 0$; at $t = 1$ we then get $c = 0$.

Conclusion. $\{e^{-bt}, te^{-bt}\}$ is a basis for the critically damped circuit's solution space H.

Exercise 4.3.29: Too few vectors to span

Place the $k < n$ vectors into an $n \times k$ matrix. At most k pivots are possible, so at least one of the n rows has no pivot. By the spanning criterion of Section 1.4, the columns cannot span $\mathbb{R}^n$. A basis must span the whole space.

Conclusion. A set with fewer than n vectors cannot be a basis of $\mathbb{R}^n$.

Exercise 4.3.30: Too many vectors to be independent

There are $k > n$ columns but only n rows. The homogeneous system with these columns therefore has at least one free variable, so it has a nonzero coefficient solution. This is a nontrivial dependence relation among the k vectors. A basis must be independent.

Conclusion. A set with more than n vectors cannot be a basis of $\mathbb{R}^n$.

Exercise 4.3.31: Dependence survives a linear map

Dependence provides scalars $c_1, \ldots, c_p$, not all zero, such that $\sum_j c_j v_j = 0$. Apply the linear map T:

$$\sum_j c_j T(v_j) = T\left(\sum_j c_j v_j\right) = T(0) = 0.$$

The coefficients have not changed, so they are still not all zero. This is a nontrivial dependence relation for the image vectors. Taking the contrapositive, independent images force the original vectors to be independent.

Conclusion. Linear maps preserve dependence, although they can turn an independent family into a dependent one.

Exercise 4.3.32: Injectivity reflects dependence

A dependence relation among the images has the form $\sum_j c_j T(v_j) = 0$ with not all c_j zero. By linearity,

$$T\left(\sum_j c_j v_j\right) = 0 = T(0).$$

Injectivity says equal images have equal inputs, so $\sum_j c_j v_j = 0$. The same nonzero coefficient list proves original dependence. The contrapositive states that an injective linear map carries an independent family to an independent image family.

Conclusion. Injectivity prevents a nonzero input combination from being collapsed to zero.

Exercise 4.3.33: Independence of two quadratic polynomials

Suppose $a(1+t^2) + b(1-t^2) = 0$ as a polynomial identity. Its constant coefficient is $a+b$, and its t^2 coefficient is $a-b$. Both must be zero, so addition and subtraction give $a = b = 0$. The polynomials belong to $\mathcal{P}_3$ because their degrees are at most three, even though their actual degree is only two.

Conclusion. Yes, the two polynomials are linearly independent in $\mathcal{P}_3$; they do not form a basis of that larger space.

Exercise 4.3.34: A polynomial dependence by inspection

The identity $(1+t) + (1-t) - 2 = 0$ gives $p_1 + p_2 - p_3 = 0$. Thus p_3 is redundant. To check the remaining pair, a zero combination $a(1+t) + b(1-t) = 0$ has constant coefficient $a+b = 0$ and linear coefficient $a-b = 0$, so $a = b = 0$.

Conclusion. $\{1+t, 1-t\}$ is a basis of the specified span, and the displayed relation proves the original three-function set dependent.

Exercise 4.3.35: Construct a four-vector family with a two-vector basis

Set $v_1 = u_1$, $v_2 = u_1 + u_2$, $v_3 = u_2$, and $v_4 = u_1 - u_2$. Every v_j lies in $\text{Span}\{u_1, u_2\}$, and $v_1 = u_1, v_3 = u_2$ already generate that space. Since a subset of an independent set is independent, $\{u_1, u_2\}$ is independent.

Conclusion. These four choices give $\{v_1, v_3\} = \{u_1, u_2\}$ as a basis of their common span. The additional u_3, u_4 need not be used.

Exercise 4.3.36: Bases of two spaces and their sum

Find a basis for H.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 4 \\ 0 & -1 & 1 \\ -1 & 1 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (3)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Find a basis for K.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -2 & 2 & -1 \\ -2 & 3 & 4 \\ -1 & 2 & 6 \\ 3 & -6 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ -2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 6 \\ -2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Find a basis for their sum.

The sum $H + K$ consists exactly of combinations of all six generators. Use the concatenated matrix, numbering its columns $1, \ldots, 6$ in the order $u_1, u_2, u_3, v_1, v_2, v_3$.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 3 & -2 & 2 & -1 \\ 2 & 2 & 4 & -2 & 3 & 4 \\ 0 & -1 & 1 & -1 & 2 & 6 \\ -1 & 1 & -4 & 3 & -6 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 3 & -2 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 5, 6. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 6 \\ -2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (3)v_1 + (-1)v_2$$
$$v_4 = (-2)v_1 + (1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.37: Four independent functions by exact evaluations

Assume $c_1 t + c_2 \sin t + c_3 \cos 2t + c_4 \sin t \cos t = 0$ for every real t. Evaluating at the exact inputs $0, \pi, \pi/2, \pi/4$ gives, successively,

$$\begin{aligned} c_3 &= 0, \\ \pi c_1 + c_3 &= 0, \\ (\pi/2)c_1 + c_2 - c_3 &= 0, \\ (\pi/4)c_1 + (\sqrt{2}/2)c_2 + (1/2)c_4 &= 0. \end{aligned}$$

The first equation gives $c_3 = 0$, the second $c_1 = 0$, the third $c_2 = 0$, and the last $c_4 = 0$. All divisions are by nonzero constants. Thus the only identical zero combination is trivial.

Conclusion. The four functions are linearly independent. Exact evaluation points avoid confusing small numerical roundoff with an actual zero.

Exercise 4.3.38: Seven cosine powers are independent

Suppose $c_0 + c_1 \cos t + \cdots + c_6 \cos^6 t = 0$ for every real t. Choose seven exact inputs $t_j = \arccos r_j$, where

$$(r_1, \ldots, r_7) = (-1, -2/3, -1/3, 0, 1/3, 2/3, 1).$$

Each is allowed since $r_j \in [-1, 1]$. Their evaluation equations are $\sum_{k=0}^{6} c_k r_j^k = 0$ for $j = 1, \ldots, 7$. This is the 7×7 system whose jth row is $[1, r_j, r_j^2, \ldots, r_j^6]$. Its RREF is I_7; equivalently, the polynomial $q(r) = \sum_{k=0}^{6} c_k r^k$ has seven distinct roots but degree at most six, so it must be identically zero. All c_k therefore vanish.

Conclusion. $\{1, \cos t, \ldots, \cos^6 t\}$ is an independent family. The seven evaluations are a finite exact certificate, as requested.

4.4 Coordinate systems

An ordered basis $\mathcal{B} = (b_1, \ldots, b_n)$ assigns to each vector x its unique list of weights $[x]_{\mathcal{B}} = (c_1, \ldots, c_n)^T$ in the expression $x = \sum_j c_j b_j$. Changing the basis order changes the meaning of this list.

For a basis in $\mathbb{R}^n$, put $P_{\mathcal{B}} = [b_1 \ \cdots \ b_n]$. The two directions are

$$x = P_{\mathcal{B}}[x]_{\mathcal{B}}, \qquad [x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x.$$

The first assembles a vector from weights; the second finds its weights. For polynomials use the standard coefficient order $1, t, t^2, \ldots$. A missing power has coefficient zero and must still occupy its position.

Practice problem 4.4.P1: Both directions of a basis-coordinate matrix

(a) Check independence and spanning.

The matrix

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -3 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 3 \end{bmatrix}$$

is triangular with three nonzero pivots. It is invertible, so its columns are independent and span $\mathbb{R}^3$; they are a basis.

(b)–(c) State the matrix and direction correctly.

The displayed matrix is the change from $\mathcal{B}$ coordinates to standard coordinates, and the equation is $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$.

(d) Solve for the weights.

For $x = (-8, 2, 3)^T$, the last equation is $3c_3 = 3$, so $c_3 = 1$. The middle equation $4c_2 - 6c_3 = 2$ gives $c_2 = 2$. The first $c_1 - 3c_2 + 3c_3 = -8$ gives $c_1 = -5$.

Conclusion. $[x]_{\mathcal{B}} = (-5, 2, 1)^T$, and multiplication by the matrix gives back $(-8, 2, 3)^T$.

Practice problem 4.4.P2: Coordinates of a quadratic polynomial

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & -1 & 1 & -3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 5 \\ c_2 = 1 \\ c_3 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 1 \\ -2 \end{bmatrix}$.

Exercise 4.4.1: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 3 & -4 \\ -5 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (3)(5) + (-4)(3) \\ (-5)(5) + (6)(3) \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.2: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 3 & -4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -26 \\ 1 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (3)(-2) + (-4)(5) \\ (2)(-2) + (1)(5) \end{bmatrix} = \begin{bmatrix} -26 \\ 1 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.3: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 1 & 5 & 4 \\ -2 & 0 & -3 \\ 3 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ 3 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (1)(1) + (5)(0) + (4)(-2) \\ (-2)(1) + (0)(0) + (-3)(-2) \\ (3)(1) + (-2)(0) + (0)(-2) \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ 3 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.4: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} -2 & 3 & 4 \\ 2 & 0 & -1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 1 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (-2)(-3) + (3)(2) + (4)(-1) \\ (2)(-3) + (0)(2) + (-1)(-1) \\ (0)(-3) + (2)(2) + (3)(-1) \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 1 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.5: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rr|r} 1 & 3 & -1 \\ -2 & -5 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{rr|r} 1 & 3 & -1 \\ 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_2$ to obtain

$$\left[\begin{array}{rr|r} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 2 \\ c_2 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Exercise 4.4.6: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rr|r} 1 & 2 & -1 \\ -4 & -3 & -6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-4)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 5 & -10 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{5})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

Exercise 4.4.7: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ -1 & 4 & -2 & -9 \\ -3 & 9 & 4 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ -3 & 9 & 4 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 10 & 30 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{10})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = -1 \\ c_2 = -1 \\ c_3 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}$.

Exercise 4.4.8: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 3 & 8 & 3 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -2 & -2 & 0 \\ 3 & 8 & 3 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 2 & 0 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -2 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 1 \\ c_2 = -1 \\ c_3 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Exercise 4.4.9: Coordinates and basis representations

The map from basis coordinates to standard coordinates places each ordered basis vector in the corresponding matrix column:

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ -3 & -5 \end{bmatrix}.$$

Indeed, multiplying by $(c_1, \ldots, c_n)^T$ gives $c_1 b_1 + \cdots + c_n b_n = x$, so this matrix has exactly the required direction.

Exercise 4.4.10: Coordinates and basis representations

The map from basis coordinates to standard coordinates places each ordered basis vector in the corresponding matrix column:

$$P_{\mathcal{B}} = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & -2 \\ 6 & -4 & 3 \end{bmatrix}.$$

Indeed, multiplying by $(c_1, \ldots, c_n)^T$ gives $c_1 b_1 + \cdots + c_n b_n = x$, so this matrix has exactly the required direction.

Exercise 4.4.11: Coordinates and basis representations

Invert the ordered basis matrix.

For a 2×2 matrix, the inverse is the adjugate divided by the nonzero determinant. Here

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -3 \\ -2 & 5 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = -1, \qquad P_{\mathcal{B}}^{-1} = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix}.$$

Apply the inverse to the actual standard vector.

$$[x]_{\mathcal{B}} = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Multiplying the answer by $P_{\mathcal{B}}$ returns x, so the direction of the change has been checked.

Exercise 4.4.12: Coordinates and basis representations

Invert the ordered basis matrix.

For a 2×2 matrix, the inverse is the adjugate divided by the nonzero determinant. Here

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = 1, \qquad P_{\mathcal{B}}^{-1} = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}.$$

Apply the inverse to the actual standard vector.

$$[x]_{\mathcal{B}} = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -8 \\ 5 \end{bmatrix}.$$

Multiplying the answer by $P_{\mathcal{B}}$ returns x, so the direction of the change has been checked.

Exercise 4.4.13: Coordinates and basis representations

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 1 & 1 & 1 & 7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 0 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -2 & 2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 2 \\ c_2 = 6 \\ c_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}$.

Exercise 4.4.14: Coordinates and basis representations

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ -1 & -1 & 1 & -6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & -1 & 3 & -5 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 2 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = 2 \\ c_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$.

Exercise 4.4.15: Coordinates and basis representations

(a) True. A basis with n ordered members produces exactly n real weights, hence a coordinate vector in $\mathbb{R}^n$, whatever type of object x itself may be.

(b) False. For a basis of $\mathbb{R}^n$, $P_{\mathcal{B}}$ maps weights to standard vectors: $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$. The coordinate map uses $P_{\mathcal{B}}^{-1}$ instead.

(c) False. The four independent polynomials $1, t, t^2, t^3$ would have independent images under an isomorphism, but four vectors cannot be independent in $\mathbb{R}^3$. In fact $\mathcal{P}_3$ corresponds to $\mathbb{R}^4$.

Exercise 4.4.16: Coordinates and basis representations

(a) True. For the standard basis, $x = x_1e_1 + \cdots + x_ne_n$, so its basis weights are exactly its usual coordinates.

(b) False. The coordinate mapping goes from x to $[x]_{\mathcal{B}}$. The displayed reverse direction is its inverse, assembling a vector from its weights.

(c) True. Every two-dimensional plane through the origin has a two-vector basis. Its coordinate map is a linear bijection onto $\mathbb{R}^2$. An affine plane not through zero would not be a vector space under the usual operations.

Exercise 4.4.17: Coordinates and basis representations

One representation is $5v_1 - 2v_2 + 0v_3 = (1, 1)^T$. There is also the nontrivial relation $5v_1 - v_2 + v_3 = 0$, since

$$5(1, -3)^T - (2, -8)^T + (-3, 7)^T = (0, 0)^T.$$

Add that zero relation to the first representation. It gives $10v_1 - 3v_2 + v_3 = (1, 1)^T$. The two weight triples $(5, -2, 0)$ and $(10, -3, 1)$ differ, and direct substitution verifies both.

Conclusion. The failure of independence permits distinct representations of the same vector.

Exercise 4.4.18: Coordinates and basis representations

The jth basis vector has the representation

$$b_j = 0b_1 + \cdots + 0b_{j-1} + 1b_j + 0b_{j+1} + \cdots + 0b_n.$$

A basis makes representations unique, so these are exactly its coordinates. The coefficient column is e_j, with a 1 in place j and zero elsewhere.

Conclusion. $[b_j]_{\mathcal{B}} = e_j$ for each j, the columns of I_n.

Exercise 4.4.19: Coordinates and basis representations

The assumed existence of a representation for every vector says Span $S = V$. To establish independence, compare two representations of zero. One uses all coefficients zero. If a nontrivial zero combination existed, it would give a different representation of zero, contradicting uniqueness. Thus the only zero combination is trivial.

Conclusion. S is independent and spans V, hence is a basis.

Exercise 4.4.20: Coordinates and basis representations

Because the four generators span, write $w = \sum_{j=1}^{4} k_j v_j$. Their dependence supplies $a_1, \ldots, a_4$, not all zero, with $\sum_j a_j v_j = 0$. For every real t,

$$w = \sum_{j=1}^{4} (k_j + t a_j) v_j.$$

At least one a_j is nonzero, so changing t changes that coefficient. The resulting coefficient lists are distinct, although the resulting vector remains w.

Conclusion. Every $w \in V$ has infinitely many representations in the dependent spanning family, in particular more than one.

Exercise 4.4.21: Coordinates and basis representations

The basis matrix and its determinant are

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -2 \\ -4 & 9 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = 9 - 8 = 1.$$

Thus the matrix implementing the coordinate map is its inverse:

$$A = P_{\mathcal{B}}^{-1} = \begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}.$$

Indeed $AP_{\mathcal{B}} = I_2$, so $Ax = AP_{\mathcal{B}}[x]_{\mathcal{B}} = [x]_{\mathcal{B}}$ for every x.

Conclusion. $A = \begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}$.

Exercise 4.4.22: Coordinates and basis representations

Form $P_{\mathcal{B}} = [b_1 \ \cdots \ b_n]$. Its columns are a basis, so it is invertible. Since $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$, multiply by the inverse to obtain

$$[x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x.$$

Conclusion. The implementing matrix is $A = [b_1 \ \cdots \ b_n]^{-1}$, not the basis matrix itself.

Exercise 4.4.23: Coordinates and basis representations

Suppose $[u]_{\mathcal{B}} = [w]_{\mathcal{B}} = (c_1, \ldots, c_n)^T$. By the definition of coordinates, $u = \sum_j c_j b_j$ and $w = \sum_j c_j b_j$. The right sides are identical, so $u = w$.

Conclusion. Equal coordinate outputs imply equal vector inputs; the coordinate map is one-to-one.

Exercise 4.4.24: Coordinates and basis representations

Take any $y = (y_1, \ldots, y_n)^T \in \mathbb{R}^n$ and define $u = y_1 b_1 + \cdots + y_n b_n \in V$. This is a legitimate vector because V is closed under sums and scalar multiplication. The defining representation has weight list y, and its uniqueness gives $[u]_{\mathcal{B}} = y$.

Conclusion. Every vector of $\mathbb{R}^n$ is reached, so the coordinate map is onto.

Exercise 4.4.25: Coordinates and basis representations

Let $C(x) = [x]_{\mathcal{B}}$. It is linear and one-to-one. For any scalars c_j,

$$\sum_j c_j u_j = 0 \quad \Longleftrightarrow \quad C\left(\sum_j c_j u_j\right) = 0 \quad \Longleftrightarrow \quad \sum_j c_j C(u_j) = 0.$$

The forward implication in the first equivalence uses $C(0) = 0$; the reverse uses injectivity. The second equivalence uses linearity. Thus the two homogeneous equations have exactly the same coefficient solutions. One family has only the trivial zero combination if and only if the other does.

Conclusion. The original vectors are independent exactly when their coordinate vectors are independent.

Exercise 4.4.26: Coordinates and basis representations

If $w = \sum_j c_j u_j$, linearity gives $[w]_{\mathcal{B}} = \sum_j c_j [u_j]_{\mathcal{B}}$. Conversely suppose the latter relation holds. Linearity rewrites it as $C(w) = C(\sum_j c_j u_j)$. Injectivity of C then implies $w = \sum_j c_j u_j$. The coefficients themselves are the same in both representations.

Conclusion. Span membership is preserved in both directions by coordinates.

Exercise 4.4.27: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 2 \\ 2 & 0 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a pivot in every column, so the coefficient columns are independent. The coordinate map is injective and linear, so the polynomials are independent as well.

Conclusion. The set is linearly independent.

Exercise 4.4.28: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -2 & 0 & -2 \\ -1 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a pivot in every column, so the coefficient columns are independent. The coordinate map is injective and linear, so the polynomials are independent as well.

Conclusion. The set is linearly independent.

Exercise 4.4.29: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -3 \\ 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(-1)p_1 + (1)p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The set is linearly dependent.

Exercise 4.4.30: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 8 & 9 & 1 \\ -12 & -6 & 6 \\ 6 & 1 & -5 \\ -1 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(1)p_1 + (-1)p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The set is linearly dependent.

Exercise 4.4.31: Coordinates and basis representations

Use the coefficient order $(1, t, t^2)$. Spanning $\mathcal{P}_2$ is equivalent to having a pivot in every row of the coefficient matrix.

(a) Reduce the coefficient matrix.

$$\begin{bmatrix} 1 & -3 & -4 & 1 \\ -3 & 5 & 5 & 0 \\ 5 & -7 & -6 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{5}{4} & -\frac{5}{4} \\ 0 & 1 & \frac{7}{4} & -\frac{3}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Only 2 rows contain pivots, so the set does not span $\mathcal{P}_2$.

(b) Reduce the coefficient matrix.

$$\begin{bmatrix} 0 & 1 & -3 & 2 \\ 5 & -8 & 4 & -3 \\ 1 & -2 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

All three rows contain pivots, so the set spans $\mathcal{P}_2$.

Exercise 4.4.32: Coordinates and basis representations

(a) Check the three coefficient columns.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -3 \end{bmatrix}, \qquad \det A = -1 \neq 0.$$

Therefore these columns form a basis of $\mathbb{R}^3$, so the polynomials form a basis of $\mathcal{P}_2$.

(b) Assemble the specified polynomial.

$$q = -p_1 + p_2 + 2p_3 = -(1 + t^2) + (t - 3t^2) + 2(1 + t - 3t^2) = 1 + 3t - 10t^2.$$

Conclusion. $q(t) = 1 + 3t - 10t^2$.

Exercise 4.4.33: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 3 & 5 & 0 & 1 \\ 7 & 1 & 1 & 16 \\ 0 & 0 & -2 & -6 \\ 0 & -2 & 0 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -2 \\ 1 \\ -3 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(-2)p_1 + (1)p_2 + (-3)p_3 + (1)p_4 = 0.$$

Thus the polynomials are dependent.

Conclusion. The polynomials do not form a basis of $\mathcal{P}_3$, since they are dependent.

Exercise 4.4.34: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 5 & 9 & 6 & 0 \\ -3 & 1 & -2 & 0 \\ 4 & 8 & 5 & 0 \\ 2 & -6 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{3}{4} & 0 \\ 0 & 1 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -\frac{3}{4} \\ -\frac{1}{4} \\ 1 \\ 0 \end{bmatrix}$. It gives the polynomial relation

$$(-\frac{3}{4})p_1 + (-\frac{1}{4})p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The polynomials do not form a basis of $\mathcal{P}_3$, since they are dependent.

Exercise 4.4.35: Coordinates and basis representations

Check that the ordered generators are independent.

The coefficient matrix has a pivot in every column, as the reduction below verifies. Hence the generators form a basis of their stated span H.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 11 & 14 & 19 \\ -5 & -8 & -13 \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{11})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ -5 & -8 & -13 \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-5)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (10)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (7)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{11}{18})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (\frac{3}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (\frac{12}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{14}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -\frac{5}{3} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = -\frac{5}{3} \\ c_2 = \frac{8}{3} \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} -\frac{5}{3} \\ \frac{8}{3} \end{bmatrix}$.

Verify membership directly.

The computed weights satisfy $\begin{bmatrix} 11 & 14 \\ -5 & -8 \\ 10 & 13 \\ 7 & 10 \end{bmatrix} \begin{bmatrix} -\frac{5}{3} \\ \frac{8}{3} \end{bmatrix} = \begin{bmatrix} 19 \\ -13 \\ 18 \\ 15 \end{bmatrix}$, so $x \in H$ and the listed coordinates are the unique basis weights.

Exercise 4.4.36: Coordinates and basis representations

Check that the ordered generators are independent.

The coefficient matrix has a pivot in every column, as the reduction below verifies. Hence the generators form a basis of their stated span H.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} -6 & 8 & -9 & 4 \\ 4 & -3 & 5 & 7 \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_1 \leftarrow (-\frac{1}{6})R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 4 & -3 & 5 & 7 \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (4)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-9)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ 0 & -5 & \frac{11}{2} & -14 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (4)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ 0 & -5 & \frac{11}{2} & -14 \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_2 \leftarrow (\frac{3}{7})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & -5 & \frac{11}{2} & -14 \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-5)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & \frac{47}{14} & \frac{47}{7} \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (\frac{7}{3})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & \frac{47}{14} & \frac{47}{7} \\ 0 & 0 & -2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{14}{47})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -2 & -4 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (-2)R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{3}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{4}{3} & 0 & -\frac{11}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{3}{7})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{4}{3} & 0 & -\frac{11}{3} \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-\frac{4}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = 5 \\ c_3 = 2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$.

Verify membership directly.

The computed weights satisfy $\begin{bmatrix} -6 & 8 & -9 \\ 4 & -3 & 5 \\ -9 & 7 & -8 \\ 4 & -3 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \\ -8 \\ 3 \end{bmatrix}$, so $x \in H$ and the listed coordinates are the unique basis weights.

Exercise 4.4.37: Coordinates and basis representations

What we must show. Convert lattice weights into ordinary spatial coordinates. The basis columns are given in angstroms, so use $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ rather than the inverse direction:

$$x = \begin{bmatrix} \frac{13}{5} & 0 & 0 \\ -\frac{3}{2} & 3 & 0 \\ 0 & 0 & \frac{24}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{4} \\ \frac{1}{6} \end{bmatrix} = \begin{bmatrix} \frac{13}{10} \\ 0 \\ \frac{4}{5} \end{bmatrix}.$$

Coordinatewise, the first value is $2.6(1/2) = 1.3$, the second is the sum of the contributions from the first two basis columns, and the third is 4.8 times the third lattice coordinate.

Conclusion. The standard-coordinate location is $(\frac{13}{10}, 0, \frac{4}{5})^T$ angstroms.

Exercise 4.4.38: Coordinates and basis representations

What we must show. Convert lattice weights into ordinary spatial coordinates. The basis columns are given in angstroms, so use $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ rather than the inverse direction:

$$x = \begin{bmatrix} \frac{13}{5} & 0 & 0 \\ -\frac{3}{2} & 3 & 0 \\ 0 & 0 & \frac{24}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{13}{10} \\ \frac{3}{4} \\ \frac{8}{5} \end{bmatrix}.$$

Coordinatewise, the first value is $2.6(1/2) = 1.3$, the second is the sum of the contributions from the first two basis columns, and the third is 4.8 times the third lattice coordinate.

Conclusion. The standard-coordinate location is $(\frac{13}{10}, \frac{3}{4}, \frac{8}{5})^T$ angstroms.

4.5 The dimension of a vector space

The dimension of a finite-dimensional space is the number of vectors in any basis. A spanning list need not have that size: it can contain redundant vectors. An independent list can be too short to span.

To analyze a parametrized subspace, split its vector expression into one coefficient vector per parameter, then remove any dependent generators. For A with n columns and r pivots, $\dim \operatorname{Col} A = r$ and $\dim \operatorname{Nul} A = n - r$: the null-space basis has one vector per free variable. The zero space has the empty basis and dimension zero.

Practice problem 4.5.P1: Dependence alone does not bound the list size

False. The singleton $S = \{0\}$ is dependent because $1 \cdot 0 = 0$ is a nontrivial relation, yet it has only one element. In the nonzero space V with dimension $p \geq 1$, one is not greater than p.

Conclusion. A short set can be dependent; being too large is sufficient for dependence but is not necessary.

Practice problem 4.5.P2: A larger set than a spanning set is dependent

The finite spanning set S contains a basis, obtained by deleting redundant vectors. Thus $\dim V \leq |S|$. If T contains more vectors than S, then $|T| > |S| \geq \dim V$. Any family with more vectors than the dimension is dependent. If T is infinite, take already a finite subset of more than $\dim V$ elements to obtain a dependence relation.

Conclusion. True. The conclusion uses that S spans, not that S itself must be a basis.

Exercise 4.5.1: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 1 \\ 0 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 2.

Exercise 4.5.2: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -4 \\ -2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ -4 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 2.

Exercise 4.5.3: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 0 & 0 & 2 \\ 1 & -1 & 0 \\ 0 & 1 & -3 \\ 1 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.4: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & -1 \\ 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 2.

Exercise 4.5.5: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 0 & 5 \\ 0 & -2 & 2 \\ -3 & 0 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 2 \\ 6 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.6: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 0 & -1 & -3 \\ -7 & 6 & 5 \\ -3 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 0 \\ -7 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 5 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.7: A homogeneous system defining the zero space

The second equation gives $b = 2c$. Substitution into the third, $2b - c = 0$, gives $4c - c = 3c = 0$, so $c = 0$ and then $b = 0$. The first equation now yields $a = 0$. Thus the subspace contains only $(0, 0, 0)$.

Conclusion. (a) The empty set is a basis of the zero subspace. (b) Its dimension is 0. The singleton containing zero is not a basis, because it is dependent.

Exercise 4.5.8: One constraint in four coordinates

Solve for a: $a = 3b - c$, with b, c, d unrestricted. Then

$$(a, b, c, d)^T = b(3, 1, 0, 0)^T + c(-1, 0, 1, 0)^T + d(0, 0, 0, 1)^T.$$

This proves spanning. For a zero combination, coordinates 2,3,4 force the three coefficients to be zero, proving independence.

Conclusion. (a) The three displayed coefficient vectors form a basis. (b) The dimension is 3.

Exercise 4.5.9: An equal-coordinate plane

A vector whose first and third coordinates agree has form $(s, t, s)^T = s(1, 0, 1)^T + t(0, 1, 0)^T$. These two coefficient vectors are independent: a zero combination has first coordinate equal to the first weight and second coordinate equal to the second weight. Both therefore vanish.

Conclusion. The dimension is 2, with basis $\{(1, 0, 1)^T, (0, 1, 0)^T\}$.

Exercise 4.5.10: Three generators of one line

The second and third generators are -2 and -3 times $(1, -5)^T$. Thus all three generate the same line as this one nonzero vector. A nonzero singleton is independent, so $\{(1, -5)^T\}$ is a basis.

Conclusion. $\dim H = 1$, not three: redundant generators do not add dimensions.

Exercise 4.5.11: Find a dimension from a checked basis

Find a basis rather than counting the generators.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 3 & -2 & 5 \\ 0 & 1 & -1 & 2 \\ 2 & 1 & 1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (1)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.12: Find a dimension from a checked basis

Find a basis rather than counting the generators.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -3 & -2 & -3 \\ -2 & -6 & 3 & 5 \\ 0 & 0 & 5 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ -6 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-1)v_1 + (1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.13: Nullity and column dimension from pivots

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 21 & 0 & \frac{164}{5} \\ 0 & 1 & 2 & 0 & \frac{29}{5} \\ 0 & 0 & 0 & 1 & \frac{1}{5} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are 5 variable columns and 3 pivots, in columns 1, 2, 4. The free-variable columns are 3, 5.

Conclusion. $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 5 - 3 = 2$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.14: Seven columns with four pivots

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & 2 & -4 & 3 & -2 & 6 & 0 \\ 0 & 0 & 0 & 1 & 0 & -3 & 7 \\ 0 & 0 & 0 & 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & -4 & 0 & 0 & 23 & 0 \\ 0 & 0 & 0 & 1 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There are 7 variable columns and 4 pivots, in columns 1, 4, 5, 7. The free-variable columns are 2, 3, 6.

Conclusion. $\dim \operatorname{Col} A = 4$ and $\dim \operatorname{Nul} A = 7 - 4 = 3$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.15: A nonconsecutive pivot pattern

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

There are 5 variable columns and 3 pivots, in columns 1, 3, 4. The free-variable columns are 2, 5.

Conclusion. $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 5 - 3 = 2$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.16: An invertible two-by-two matrix

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 3 & 2 \\ -6 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

There are 2 variable columns and 2 pivots, in columns 1, 2. There are no free-variable columns.

Conclusion. $\dim \operatorname{Col} A = 2$ and $\dim \operatorname{Nul} A = 2 - 2 = 0$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.17: Three nonzero triangular pivots

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

There are 3 variable columns and 3 pivots, in columns 1, 2, 3. There are no free-variable columns.

Conclusion. $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 3 - 3 = 0$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.18: Two pivots and a zero row

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

There are 3 variable columns and 2 pivots, in columns 1, 2. The free-variable columns are 3.

Conclusion. $\dim \operatorname{Col} A = 2$ and $\dim \operatorname{Nul} A = 3 - 2 = 1$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.19: Dimension definitions and missing hypotheses

(a) True. The original pivot columns form a basis of the column space, so their number equals its dimension.

(b) False as stated: a plane not through zero is not a subspace at all. For example $z = 1$ is a plane in $\mathbb{R}^3$ but excludes the zero vector. A plane *through the origin* is a two-dimensional subspace.

(c) False. $\mathcal{P}_4$ has basis $1, t, t^2, t^3, t^4$, of size five.

(d) False. An independent set can be shorter than n; for example $\{e_1\}$ is independent in $\mathbb{R}^2$ but does not span it.

(e) True. A spanning family of p vectors gives $\dim V \leq p$. More than p vectors means more than the dimension, forcing dependence.

Exercise 4.5.20: Subspace identity and spanning-set cautions

(a) False. Elements of $\mathbb{R}^2$ have two coordinates and are not literally elements of $\mathbb{R}^3$. The embedded set $\{(x, y, 0)\}$ is a subspace isomorphic to $\mathbb{R}^2$, which is a different assertion.

(b) False. Nullity counts *free* variables. For $A = I_n$ there are n variables but no free variables, so $\dim \operatorname{Nul} A = 0$.

(c) False. A one-dimensional line is spanned by its infinitely many vectors; an infinite spanning set need not be independent or minimal.

(d) False. A spanning set can be redundant; $\{e_1, e_2, e_1 + e_2\}$ spans $\mathbb{R}^2$ but is dependent.

(e) True. A three-vector basis of such a subspace is independent in $\mathbb{R}^3$, so it also spans $\mathbb{R}^3$. Hence the subspace is the whole space.

Exercise 4.5.21: Hermite polynomial coefficients

Record every coefficient, including zeros.

Relative to the standard basis $(1, t, t^2, t^3)$, the polynomial coefficient matrix is

$$A = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}.$$

It is triangular, with nonzero diagonal entries 1, 2, 4, 8. Thus its determinant is $64 \neq 0$, and its four columns are independent and span $\mathbb{R}^4$. The standard coefficient map is a linear bijection between $\mathcal{P}_3$ and $\mathbb{R}^4$, so the same two properties hold for the four polynomials.

Conclusion. The four Hermite polynomials form a basis of $\mathcal{P}_3$.

Exercise 4.5.22: Laguerre polynomial coefficients

Record every coefficient, including zeros.

Relative to the standard basis $(1, t, t^2, t^3)$, the polynomial coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 & 2 & 6 \\ 0 & -1 & -4 & -18 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

It is triangular, with nonzero diagonal entries 1, -1, 1, -1. Thus its determinant is $1 \neq 0$, and its four columns are independent and span $\mathbb{R}^4$. The standard coefficient map is a linear bijection between $\mathcal{P}_3$ and $\mathbb{R}^4$, so the same two properties hold for the four polynomials.

Conclusion. The four Laguerre polynomials form a basis of $\mathcal{P}_3$.

Exercise 4.5.23: Hermite coordinates of a cubic

What we must show. Express the given polynomial in the specified ordered basis. The coefficient equation uses the same ordered coefficient columns as in the preceding basis calculation:

$$\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix} c = \begin{bmatrix} -1 \\ 0 \\ 8 \\ 8 \end{bmatrix}.$$

Solve from the last coefficient upward. This gives $c_4 = 1$, $c_3 = 2$, $c_2 = 6$, $c_1 = 3$. For a direct check, multiply the coefficient matrix by these weights; the result is exactly the target coefficient column displayed on the right.

Conclusion. $[p]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 6 \\ 2 \\ 1 \end{bmatrix}$.

Exercise 4.5.24: Laguerre coordinates of a quadratic

What we must show. Express the given polynomial in the specified ordered basis. The coefficient equation uses the same ordered coefficient columns as in the preceding basis calculation:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & -4 \\ 0 & 0 & 1 \end{bmatrix} c = \begin{bmatrix} 5 \\ 5 \\ -2 \end{bmatrix}.$$

Solve from the last coefficient upward. This gives $c_3 = -2$, $c_2 = 3$, $c_1 = 6$. For a direct check, multiply the coefficient matrix by these weights; the result is exactly the target coefficient column displayed on the right.

Conclusion. $[p]_{\mathcal{B}} = \begin{bmatrix} 6 \\ 3 \\ -2 \end{bmatrix}$.

Exercise 4.5.25: A short set cannot span

Suppose S did span V. A finite spanning set contains a basis, obtained by deleting redundant vectors. That basis would have at most $|S| < n$ members, contradicting the fact that every basis of V has n members. Thus the assumed spanning property is impossible.

Conclusion. Fewer than n vectors cannot span an n-dimensional space.

Exercise 4.5.26: Equal finite dimensions force equality

Choose a basis $h_1, \ldots, h_n$ for H. Because $H \subseteq V$, these are also n independent vectors in the n-dimensional space V. The basis theorem says they span V. But their span is already H, so $V = H$.

Conclusion. A subspace with the full finite ambient dimension is the entire space.

Exercise 4.5.27: Why all polynomials have no finite basis

Take any finite set of polynomials. Choose an integer $d \geq 0$ at least as large as all their degrees. Every linear combination of them still has degree at most d, since addition and scalar multiplication cannot create a nonzero coefficient of t^{d+1}. Thus this set cannot span the polynomial t^{d+1}. No finite spanning set, and hence no finite basis, exists for all polynomials.

Conclusion. The polynomial space $\mathcal{P}$ is infinite-dimensional.

Exercise 4.5.28: Continuous functions contain arbitrarily large independent families

For every integer $m \geq 0$, the $m+1$ functions $1, t, t^2, \ldots, t^m$ are continuous and linearly independent on $\mathbb{R}$: an identical zero polynomial has every coefficient zero. If $C(\mathbb{R})$ had finite dimension d, take $m = d$. It would then contain $d+1$ independent vectors, contradicting the dimension bound.

Conclusion. $C(\mathbb{R})$ is infinite-dimensional because it contains independent families of arbitrarily large finite size.

Exercise 4.5.29: Compare spanning and independent family sizes

(a) True. Deleting redundancies from the p spanning vectors gives a basis with at most p members, so $\dim V \leq p$.

(b) True. An independent family extends to a basis; the basis therefore contains at least its original p vectors, so $\dim V \geq p$.

(c) True. Take any p-vector basis and adjoin the zero vector. Since $V \neq \{0\}$, we have $p \geq 1$, and zero is not a basis member. The resulting $p+1$ distinct vectors still span V.

Exercise 4.5.30: Counterexamples to incorrect dimension converses

(a) False. In $V = \mathbb{R}^{p+1}$, the p distinct vectors $0, e_1, \ldots, e_{p-1}$ are dependent, but $\dim V = p+1 > p$. For $p = 1$ this example is simply the singleton $\{0\}$ in $\mathbb{R}^2$.

(b) True. If instead $\dim V = d \leq p$, take a d-vector basis and, if necessary, add distinct redundant multiples of one nonzero basis vector until there are p distinct members. This p-element set spans, contradicting the assumption. Thus $d > p$.

(c) False in general. For $p = 3$, the two nonzero vectors $e_1, 2e_1$ in $\mathbb{R}^3$ are dependent, despite there being $p - 1 = 2$ of them. The special boundary case $p = 2$ would be true, since a single nonzero vector is independent; it does not make the general statement true.

Exercise 4.5.31: A linear image cannot gain dimension

Let $h_1, \ldots, h_d$ be a basis of H, where $d = \dim H$. For any $h \in H$, write $h = \sum_j c_j h_j$. Linearity gives $T(h) = \sum_j c_j T(h_j)$, so the d vectors $T(h_j)$ span $T(H)$. Removing redundant images gives a basis with at most d vectors. They may all vanish, in which case the image is the zero space of dimension zero.

Conclusion. $\dim T(H) \le d = \dim H$. Images cannot need more independent directions than the inputs supplying them.

Exercise 4.5.32: An injective image preserves dimension

As in Exercise 31, the images of a basis $h_1, \dots, h_d$ span $T(H)$. If $\sum_j c_j T(h_j) = 0$, linearity gives $T(\sum_j c_j h_j) = T(0)$. Injectivity yields $\sum_j c_j h_j = 0$; independence of the basis forces every $c_j = 0$. Thus the images are also independent and form a d-element basis of $T(H)$. Taking $H = V$ and using surjectivity when it is assumed gives $T(V) = W$, so $\dim W = \dim V$.

Conclusion. $\dim T(H) = \dim H$ for an injective T; in particular finite- dimensional isomorphic vector spaces have equal dimensions.

Exercise 4.5.33: Extend an independent family to a basis

(a) Append the standard coordinate directions.

Form $A = [v_1\ v_2\ v_3\ e_1\ \cdots\ e_5]$. The following reduction and pivot-column certificate give a basis; columns 4 through 8 refer to the standard vectors in order.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -9 & 9 & 6 & 1 & 0 & 0 & 0 & 0 \\ -7 & 4 & 7 & 0 & 1 & 0 & 0 & 0 \\ 8 & 1 & -8 & 0 & 0 & 1 & 0 & 0 \\ -5 & 6 & 5 & 0 & 0 & 0 & 1 & 0 \\ 7 & -7 & -7 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 1 & \frac{3}{7} \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & \frac{5}{7} \\ 0 & 0 & 1 & -\frac{1}{3} & 0 & 0 & 0 & -\frac{3}{7} \\ 0 & 0 & 0 & 0 & 1 & 0 & 3 & \frac{22}{7} \\ 0 & 0 & 0 & 0 & 0 & 1 & -9 & -\frac{53}{7} \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5, 6. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -9 \\ -7 \\ 8 \\ -5 \\ 7 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ 1 \\ 6 \\ -7 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \\ -8 \\ 5 \\ -7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$\begin{aligned} v_4 &= (-\frac{1}{3})v_1 + (-\frac{1}{3})v_3 \\ v_7 &= (1)v_1 + (1)v_2 + (3)v_5 + (-9)v_6 \\ v_8 &= (\frac{3}{7})v_1 + (\frac{5}{7})v_2 + (-\frac{3}{7})v_3 + (\frac{22}{7})v_5 + (-\frac{53}{7})v_6 \end{aligned}$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 5.

(b) Explain why the construction works generally.

The original independent columns $v_1, \ldots, v_k$ each add a new pivot as the columns are processed left to right: none lies in the span of its predecessors. Thus all are retained among the pivot columns. Appending every e_j makes the full column space $\mathbb{R}^n$, because those columns alone already span $\mathbb{R}^n$. The pivot columns therefore form a basis of the whole space that contains all the original independent vectors.

Exercise 4.5.34: Cosine multiples give another basis

(a) Use the given identities to read coefficient columns.

In the ordered basis $\mathcal{B} = (1, \cos t, \ldots, \cos^6 t)$, the coordinate columns of $\mathcal{C} = (1, \cos t, \cos 2t, \ldots, \cos 6t)$ are

$$P = \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 & -8 & 0 & 18 \\ 0 & 0 & 0 & 4 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 & -48 \\ 0 & 0 & 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 32 \end{bmatrix}.$$

The jth column is the coordinate vector of the jth function in $\mathcal{C}$. The matrix is triangular with diagonal $1, 1, 2, 4, 8, 16, 32$, whose product is $32768 \neq 0$. Hence its columns are independent. Because coordinates preserve dependence relations, the functions in $\mathcal{C}$ are independent too.

(b) Compare the number of independent functions with the dimension.

By Exercise 4.3.38, $\mathcal{B}$ is a seven-element basis of H, so $\dim H = 7$. The seven independent members of $\mathcal{C}$ lie in H by the supplied identities. An independent family of seven vectors in a seven-dimensional space is a basis.

Conclusion. $\mathcal{C}$ is a basis for H. The displayed columns give all requested $\mathcal{B}$ coordinates.

4.6 Rank

For an $m \times n$ matrix, the rank r is the number of pivots, the dimension of its column space, and the dimension of its row space. The rank–nullity equation is

$$\dim \operatorname{Nul} A = n - r.$$

Notice the number of *columns* on the right. Row operations preserve the row space and null space, but may change the column space. Consequently use pivot columns of the *original* matrix for a column-space basis; use nonzero rows of an echelon form for a row-space basis. A basis for a null space comes from solving the homogeneous equations with free variables.

Practice problem 4.6.P1: Rank and nullity from the given echelon form

The given echelon matrix B has two nonzero rows and two pivot positions. Thus $\operatorname{rank} A = 2$. Since A has five columns, rank–nullity gives $\dim \operatorname{Nul} A = 5 - 2 = 3$.

Practice problem 4.6.P2: Choose the correct rows and columns

The pivot columns of B are columns $1, 2$. Therefore a column-space basis uses the original columns $(2, 1, -7, 4)$ and $(-1, -2, 8, -5)$ of A. A row-space basis uses the two nonzero rows of B:

$$(1, -2, -4, 3, -2), \qquad (0, 3, 9, -12, 12).$$

The former vectors have four entries and live in $\mathbb{R}^4$; the latter have five entries and live in $\mathbb{R}^5$. This is a useful check on their roles.

Practice problem 4.6.P3: The next step toward a null-space basis

Continue reducing B to reduced echelon form: divide its second row by 3, then add twice that new row to its first. The nonzero rows become $(1, 0, 2, -5, 6)$ and $(0, 1, 3, -4, 4)$. From there take x_3, x_4, x_5 free, express x_1, x_2 in terms of them, and separate their coefficient vectors. It is the free variables, not the pivot columns, that directly produce this basis.

Practice problem 4.6.P4: Transpose does not change rank

The rank theorem gives $\operatorname{rank} A^T = \operatorname{rank} A = 2$. Therefore every row echelon form of A^T has exactly two pivot columns.

Exercise 4.6.1: Three bases from the supplied echelon form

The supplied B has 2 pivots, so $\operatorname{rank} A = 2$ and $\dim \operatorname{Nul} A = 4 - 2 = 2$, without row reducing A. Its pivot columns are 1, 2.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & -4 \\ -1 & 2 \\ 5 & -6 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 0 & -1 & 5 \\ 0 & -2 & 5 & -6 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} 2 & -5 \\ 5 & -3 \\ 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^2\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.2: Three bases from the supplied echelon form

The supplied B has 3 pivots, so $\operatorname{rank} A = 3$ and $\dim \operatorname{Nul} A = 5 - 3 = 2$, without row reducing A. Its pivot columns are 1, 3, 5.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 6 & -3 \\ 3 & 3 & -3 \\ 3 & 0 & 0 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 3 & 4 & -1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & -5 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} -3 & -3 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^2\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.3: Three bases from the supplied echelon form

The supplied B has 3 pivots, so $\operatorname{rank} A = 3$ and $\dim \operatorname{Nul} A = 6 - 3 = 3$, without row reducing A. Its pivot columns are 1, 2, 5.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 2 & 6 & 3 \\ -2 & -3 & 0 \\ 4 & 9 & 3 \\ -2 & 3 & 3 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 2 & 6 & -6 & 6 & 3 & 6 \\ 0 & 3 & 0 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} 3 & 0 & -3 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^3\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.4: Three bases from the supplied echelon form

The supplied B has 5 pivots, so $\operatorname{rank} A = 5$ and $\dim \operatorname{Nul} A = 6 - 5 = 1$, without row reducing A. Its pivot columns are 1, 2, 3, 5, 6.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & 1 & -2 & 1 & -2 \\ 1 & 2 & -3 & -2 & -3 \\ 1 & -1 & 0 & 1 & 6 \\ 1 & -2 & 2 & -3 & 0 \\ 1 & -2 & 1 & 2 & -1 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 1 & -2 & 0 & 1 & -2 \\ 0 & 1 & -1 & 0 & -3 & -1 \\ 0 & 0 & 1 & 1 & -13 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^1\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.5: A rank-three matrix with seven columns

Rank–nullity gives $\dim \operatorname{Nul} A = 7 - 3 = 4$. The row-space dimension equals rank, so $\dim \operatorname{Row} A = 3$. Transpose preserves rank, so $\operatorname{rank} A^T = 3$.

Exercise 4.6.6: A rank-two matrix with five columns

There are $5 - 2 = 3$ nonpivot columns, giving $\dim \operatorname{Nul} A = 3$. The rank theorem gives $\dim \operatorname{Row} A = 2$ and $\operatorname{rank} A^T = 2$.

Exercise 4.6.7: Full row rank does not identify a null space with its coordinate space

Four pivot columns in a 4×7 matrix mean rank 4, so its columns span all of $\mathbb{R}^4$: $\operatorname{Col} A = \mathbb{R}^4$. Rank–nullity gives nullity $7 - 4 = 3$. But $\operatorname{Nul} A$ is a three-dimensional subspace of $\mathbb{R}^7$, not the set $\mathbb{R}^3$ of triples. Equality $\operatorname{Nul} A = \mathbb{R}^3$ is therefore false; a choice of basis gives an isomorphism with $\mathbb{R}^3$, not literal equality of the sets.

Exercise 4.6.8: Rank four with six output coordinates

The nullity is $8-4=4$. The column space is a four-dimensional subspace of $\mathbb{R}^6$, because the columns have six entries. Thus it is not literally $\mathbb{R}^4$; dimension does not change the number of entries in its vectors.

Exercise 4.6.9: Recover rank from a known nullity

The matrix has six columns and nullity three, so its rank is $6-3=3$. Thus $\dim \operatorname{Col} A = 3$. But the columns have four entries, so $\operatorname{Col} A$ is a subspace of $\mathbb{R}^4$, not literally $\mathbb{R}^3$.

Exercise 4.6.10: Seven columns and five free coordinates

By rank–nullity, $\operatorname{rank} A = 7-5=2$. Column-space dimension equals rank, so the requested dimension is 2. The eight rows specify the ambient output space but do not replace the seven in the rank–nullity equation.

Exercise 4.6.11: From nullity to row-space dimension

Here $n=5$ and nullity is 3, so rank is $5-3=2$. The row-space dimension equals rank and is therefore 2.

Exercise 4.6.12: A second row-space dimension

There are four columns and nullity 2. Hence rank, and therefore row-space dimension, equals $4-2=2$.

Exercise 4.6.13: The largest possible rank

There can be no more pivots than rows or columns, so an $m \times n$ matrix has rank at most $\min(m,n)$. Both a 7×5 and a 5×7 matrix therefore have largest possible rank 5. This is attainable by placing a 5×5 identity matrix in the first five rows or columns and filling remaining entries with zeros.

Exercise 4.6.14: The largest row-space dimension

The row-space dimension is rank. For sizes 5×4 and 4×5, the maximum rank is 4 in each case. Identity blocks with an additional zero row or zero column show that this bound is attained.

Exercise 4.6.15: The smallest nullity for a wide matrix

A 3×7 matrix has rank at most 3, so its nullity is $7-\operatorname{rank} A \geq 7-3=4$. The matrix $[\,I_3\ 0\,]$ attains rank 3 and nullity 4. Thus the smallest possible nullity is 4.

Exercise 4.6.16: The smallest nullity for a tall matrix

A 7×5 matrix can have rank 5, so its nullity can be $5 - 5 = 0$. For example stack two zero rows below I_5. Nullity cannot be negative, so zero is the minimum.

Exercise 4.6.17: Five rank statements

(a) **True.** The columns of A^T are exactly the transposed rows of A; their spans are the same coordinate vectors.

(b) **False.** The first original rows need not be independent. For $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ there are three pivots but the first three rows are dependent.

(c) **True.** Both dimensions equal rank, including for rectangular matrices.

(d) **False.** The sum equals the number of *columns*, n, not the number of rows m.

(e) **True.** Finite-precision rounding can turn an exact zero pivot into a small nonzero value or the reverse. Exact elementary operations preserve rank; its apparent computed value may change, as the Numerical Note explains.

Exercise 4.6.18: Five statements about row operations

(a) **False.** Use pivot columns of A for $\operatorname{Col} A$, not generally those of its echelon form. For example swapping the rows of $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ changes its column span.

(b) **False.** The row span is preserved, but the particular relations among the rows can change. Rows $(1, 0), (1, 0)$ satisfy $r_2 - r_1 = 0$; a row replacement gives rows $(1, 0), (0, 0)$, for which that relation no longer holds.

(c) **True.** Nullity equals the number of free variables, namely the number of nonpivot columns.

(d) **True.** Transposing the rows of A^T gives the columns of A.

(e) **True.** Every new row is a combination of old rows, and invertibility of elementary operations gives the reverse inclusion of row spans.

Exercise 4.6.19: A one-dimensional kernel of a five-row system

All homogeneous solutions are multiples of one nonzero solution, so nullity is 1. With six columns, rank is $6 - 1 = 5$, equal to the number of rows. Thus there is a pivot in every row, and $Ax = b$ is consistent for every right-hand side $b \in \mathbb{R}^5$. The answer is yes.

Exercise 4.6.20: Two free variables in an eight-variable system

A consistent system with eight variables and two free variables has six pivots. There are also six equations, so every row has a pivot. Therefore $Ax = b$ is consistent for every $b \in \mathbb{R}^6$; changing constants alone cannot make it inconsistent. The answer is no.

Exercise 4.6.21: Onto with just one free direction

Consistency for every right-hand side in a nine-row system means rank 9. With ten variables the nullity is $10 - 9 = 1$. Thus all homogeneous solutions are multiples of one vector. It is impossible to find two nonzero homogeneous solutions that are not multiples of each other.

Exercise 4.6.22: Too many variables for a one-dimensional kernel

For ten equations in twelve variables, rank is at most ten. Nullity is therefore at least $12 - 10 = 2$. It cannot be 1, so not all homogeneous solutions can be multiples of one fixed nonzero vector.

Exercise 4.6.23: Replace redundant equations

The two given nonproportional solutions are independent and span the kernel, so nullity is 2. There are eight variables, giving rank $8 - 2 = 6$. Row reduction therefore leaves six nonzero rows and six redundant zero rows. Keeping just the six nonzero homogeneous equations describes exactly the same solutions. Fewer than six cannot suffice: a matrix with fewer rows would have rank below six and nullity above two.

Exercise 4.6.24: A unique solution for one right-hand side versus all

A 7×6 matrix may have rank 6, giving zero kernel. Then every consistent system has a unique solution; for example $A = \begin{pmatrix} I_6 \\ 0 \end{pmatrix}$ and $b = 0$ do. But rank is at most $6 < 7$, so the column space cannot be all of $\mathbb{R}^7$. Thus a unique solution for some right-hand side is possible, while a unique solution for every right-hand side is impossible.

Exercise 4.6.25: One missing pivot row leaves some constants unattainable

Twelve variables and three free variables mean rank 9. The system has ten rows, so it lacks a pivot in one row. Its column space is a proper subspace of $\mathbb{R}^{10}$, and some right-hand sides lie outside it. Thus the scientist cannot guarantee a solution after changing the constants; indeed there necessarily exist choices of constants that make the system inconsistent.

Exercise 4.6.26: Full rank for a tall matrix

When $m > n$, the largest possible rank is n. The columns are independent if and only if all n columns are pivot columns, which is equivalent to rank n. This is precisely full rank for the given shape of matrix.

Exercise 4.6.27: Locate the four fundamental spaces

Rows of A and inputs to A have n entries; columns of A and inputs to A^T have m entries. Consequently

$$\begin{array}{c|cc} \mathbb{R}^n & \operatorname{Row} A = \operatorname{Col} A^T, & \operatorname{Nul} A \\ \mathbb{R}^m & \operatorname{Col} A = \operatorname{Row} A^T, & \operatorname{Nul} A^T \end{array}$$

The six names reduce to four fundamental spaces after the two displayed identifications. For special matrices some of these four may coincide as sets (for example several can be the zero space); "four" refers to the four roles, not a guarantee of four different sets in every example.

Exercise 4.6.28: Two rank–nullity identities

(a) Apply rank–nullity to A.

Since $\dim \operatorname{Row} A = \operatorname{rank} A$ and $\dim \operatorname{Nul} A = n - \operatorname{rank} A$, their sum is n.

(b) Apply rank–nullity to A^T.

This transpose has m columns and rank equal to that of A. Thus $\dim \operatorname{Nul} A^T = m - \operatorname{rank} A$; adding $\dim \operatorname{Col} A = \operatorname{rank} A$ gives m.

Exercise 4.6.29: Onto is equivalent to a trivial transpose kernel

The equation $Ax = b$ is solvable for every $b \in \mathbb{R}^m$ if and only if $\dim \operatorname{Col} A = m$. By Exercise 4.6.28(b), this is equivalent to $\dim \operatorname{Nul} A^T = 0$, which is equivalent to $\operatorname{Nul} A^T = \{0\}$. That last condition says exactly that $A^T x = 0$ has only the trivial solution.

Exercise 4.6.30: The augmented-rank consistency criterion

Adding b as a new column increases rank precisely when b is outside the span of the old columns. Therefore $Ax = b$ is consistent if and only if

$$\operatorname{rank}[\, A \;\; b \,] = \operatorname{rank} A.$$

Equivalently, an inconsistent row in the reduced augmented matrix gives an extra pivot in its last column. If ranks agree, that extra pivot is absent.

Exercise 4.6.31: An outer product has at most one independent column

For $u = (2, -3, 5)$ and $v = (a, b, c)$,

$$uv^T = \begin{pmatrix} 2a & 2b & 2c \\ -3a & -3b & -3c \\ 5a & 5b & 5c \end{pmatrix}.$$

Its columns are au, bu, cu, so its column space is contained in $\operatorname{Span}\{u\}$, a one-dimensional space. Hence rank is at most 1. It is zero when $v = 0$, and otherwise is exactly 1, since $u \neq 0$.

Exercise 4.6.32: Factor a rank-one matrix

With $u = (1, 2)$ the matrix uv^T has first row v^T and second row $2v^T$. Thus choose $v = (1, -3, 4)$, giving

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} [\, 1 \;\; -3 \;\; 4 \,] = \begin{pmatrix} 1 & -3 & 4 \\ 2 & -6 & 8 \end{pmatrix}.$$

Exercise 4.6.33: Every nonzero column generates a rank-one column space

If rank is 1 and the first column u is nonzero, it is a basis for the column space. Thus the three columns have form $u, \alpha u, \beta u$. Taking $v = (1, \alpha, \beta)$ gives $A = uv^T$. If the first column is zero, choose any other nonzero column as u (one exists because rank is 1). Express each column as $v_j u$, including coefficient zero for a zero column, and set $v = (v_1, v_2, v_3)$. The same outer-product formula then holds.

Exercise 4.6.34: A rank-r matrix is a sum of r rank-one matrices

Since U is obtained by elementary row operations, $U = FA$ for an invertible product F of elementary matrices. Set $E = F^{-1}$ to get $A = EU$. Write $E = [\, e_1 \ \cdots \ e_m \,]$ and let u_i^T be row i of U. Matrix multiplication by columns and rows gives

$$A = EU = \sum_{i=1}^{m} e_i u_i^T = \sum_{i=1}^{r} e_i u_i^T.$$

The last equality uses the zero rows below the first r rows of the echelon matrix. Each of those first r rows is nonzero, and each column e_i of an invertible matrix is nonzero. Thus every displayed outer product has rank exactly 1, proving the assertion.

Exercise 4.6.35: Four fundamental bases and two invertible assembled matrices

$$A = \begin{pmatrix} 7 & -9 & -4 & 5 & 3 & -3 & -7 \\ -4 & 6 & 7 & -2 & -6 & -5 & 5 \\ 5 & -7 & -6 & 5 & -6 & 2 & 8 \\ -3 & 5 & 8 & -1 & -7 & -4 & 8 \\ 6 & -8 & -5 & 4 & 4 & 9 & 3 \end{pmatrix}.$$

(a) Supply and verify the three requested bases.

The pivot columns are 1, 2, 4, 6, so rank is 4. The following matrices give respectively a column-space basis in their columns, a null-space basis in their columns, and a row-space basis in their rows:

$$C = \begin{pmatrix} 7 & -9 & 5 & -3 \\ -4 & 6 & -2 & -5 \\ 5 & -7 & 5 & 2 \\ -3 & 5 & -1 & -4 \\ 6 & -8 & 4 & 9 \end{pmatrix}.$$

$$N = \begin{pmatrix} -13 & -10 & 3 \\ -11 & -1 & -2 \\ 2 & 0 & 0 \\ 0 & 11 & -7 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$R = \begin{pmatrix} 1 & 0 & \frac{13}{2} & 0 & 5 & 0 & -3 \\ 0 & 1 & \frac{11}{2} & 0 & \frac{1}{2} & 0 & 2 \\ 0 & 0 & 0 & 1 & -\frac{11}{2} & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

The rows of R are the nonzero rows of the reduced echelon form. The 3 columns of N result from its free-variable equations; exact multiplication gives $AN = 0$, and their free coordinates show independence. Thus these are bases, not only spanning lists.

(b) Supply the transpose-kernel basis and check invertibility.

$$M = \begin{pmatrix} 2 \\ 41 \\ 0 \\ -28 \\ 11 \end{pmatrix}.$$

Here $A^T M = 0$, and its 1 columns are independent. Form $S = [\, R^T \; N \,]$ and $T = [\, C \; M \,]$ using the displayed blocks. There are $4 + 3 = 7$ columns in $S \in M_{7\times 7}$ and $4 + 1 = 5$ columns in $T \in M_{5\times 5}$. Exact determinants are

$$\det S = -\frac{88369}{2}, \qquad \det T = 10360.$$

Both are nonzero, verifying that both square matrices are invertible. Conceptually, a vector in both $\operatorname{Row} A$ and $\operatorname{Nul} A$ has form $x = A^T y$ and satisfies $Ax = 0$, so $x^T x = y^T Ax = 0$. Since $x^T x$ is a sum of squares, $x = 0$. The same argument with A^T treats $\operatorname{Col} A$ and $\operatorname{Nul} A^T$; their dimensions add to the full ambient dimensions. This explains why assembling their bases produces invertible matrices.

Exercise 4.6.36: Four fundamental bases and two invertible assembled matrices

The problem permits a generated example. Here is one reproducible sample: with Python `random.Random(46)`, draw the entries of a 6×2 matrix J and then a 2×7 matrix K successively from `randint(-1,1)`, and set $A = JK$. This guarantees rank at most 2, which in particular is at most 4. The resulting matrix is displayed below; its first two columns are independent, so its actual rank is 2.

$$A = \begin{pmatrix} 1 & 1 & 0 & -1 & 1 & 1 & 1 \\ 2 & 1 & 1 & -1 & 2 & 0 & 1 \\ -2 & -1 & -1 & 1 & -2 & 0 & -1 \\ 0 & -1 & 1 & 1 & 0 & -2 & -1 \\ 2 & 1 & 1 & -1 & 2 & 0 & 1 \\ -2 & -1 & -1 & 1 & -2 & 0 & -1 \end{pmatrix}.$$

(a) Supply and verify the three requested bases.

The pivot columns are 1, 2, so rank is 2. The following matrices give respectively a column-space

basis in their columns, a null-space basis in their columns, and a row-space basis in their rows:

$$C = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ -2 & -1 \\ 0 & -1 \\ 2 & 1 \\ -2 & -1 \end{pmatrix}.$$

$$N = \begin{pmatrix} -1 & 0 & -1 & 1 & 0 \\ 1 & 1 & 0 & -2 & -1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$R = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 2 & 1 \end{pmatrix}.$$

The rows of R are the nonzero rows of the reduced echelon form. The 5 columns of N result from its free-variable equations; exact multiplication gives $AN = 0$, and their free coordinates show independence. Thus these are bases, not only spanning lists.

(b) Supply the transpose-kernel basis and check invertibility.

$$M = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Here $A^T M = 0$, and its 4 columns are independent. Form $S = [\, R^T \; N \,]$ and $T = [\, C \; M \,]$ using the displayed blocks. There are $2 + 5 = 7$ columns in $S \in M_{7\times 7}$ and $2 + 4 = 6$ columns in $T \in M_{6\times 6}$. Exact determinants are

$$\det S = 23, \qquad \det T = -21.$$

Both are nonzero, verifying that both square matrices are invertible. Conceptually, a vector in both $\operatorname{Row} A$ and $\operatorname{Nul} A$ has form $x = A^T y$ and satisfies $Ax = 0$, so $x^T x = y^T Ax = 0$. Since $x^T x$ is a sum of squares, $x = 0$. The same argument with A^T treats $\operatorname{Col} A$ and $\operatorname{Nul} A^T$; their dimensions add to the full ambient dimensions. This explains why assembling their bases produces invertible matrices.

Exercise 4.6.37: Reconstruct the original matrix from pivot columns

Use the C and R displayed in Exercise 4.6.35. Those are precisely the pivot columns of the original A and the nonzero rows of its reduced echelon form. Multiplication gives

$$CR = \begin{pmatrix} 7 & -9 & -4 & 5 & 3 & -3 & -7 \\ -4 & 6 & 7 & -2 & -6 & -5 & 5 \\ 5 & -7 & -6 & 5 & -6 & 2 & 8 \\ -3 & 5 & 8 & -1 & -7 & -4 & 8 \\ 6 & -8 & -5 & 4 & 4 & 9 & 3 \end{pmatrix} = A.$$

For example, the third column of R is $(13/2, 11/2, 0, 0)$, so the third column of CR is $(13/2)a_1 + (11/2)a_2 = a_3$. The reduced matrix records the coefficients of every original column in terms of the pivot columns. Thus this product restores all columns exactly.

Exercise 4.6.38: Three generated rank factorizations and a general proof

For a reproducible family, use `random.Random(4600+r)` for each $r = 5, 4, 3$. Form C by stacking I_r over a $(5 - r) \times r$ array of successive `randint(-2,2)` draws, and form R by placing an $r \times (7 - r)$ array of subsequent draws after I_r. Set $A = CR$. The first r columns of A are C; its reduced echelon form has nonzero rows R. Thus these are actual examples of the exercise's construction, with exact rank r, rather than uncontrolled random ranks.

The rank-5 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \end{pmatrix}.$$

The rank-4 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 0 & 2 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 2 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 2 \\ -2 & 0 & 2 & 1 & -3 & 2 & 2 \end{pmatrix}.$$

The rank-3 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 2 & -2 \\ 1 & 2 & 2 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & -2 & 1 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & -2 & 1 \\ 2 & 2 & -2 & 2 & -2 & 4 & -4 \\ 1 & 2 & 2 & 9 & -6 & -2 & 0 \end{pmatrix}.$$

Prove the observed equality for every matrix.

Let C consist of the original pivot columns of a rank-r matrix A, and R of the nonzero rows of its reduced echelon form U. In U, the pivot columns are the first r coordinate vectors above the zero rows. Therefore the entries of column j of R express column j of U as a combination of its pivot columns. Elementary row operations preserve all column dependence relations: if $U = EA$ with E invertible, then $U\alpha = 0$ if and only if $A\alpha = 0$. The same coefficients thus express column j of A using columns of C. This means column j of CR is column j of A, for every j; hence $CR = A$.

4.7 Change of basis

A vector stays the same when a basis changes; its list of coordinates changes. We use $P_{C\leftarrow B}$ to mean *from B-coordinates to C-coordinates*:

$$[x]_C = P_{C\leftarrow B}[x]_B, \qquad P_{C\leftarrow B} = [\,[b_1]_C \ \cdots \ [b_n]_C\,].$$

In $\mathbb{R}^n$, if B_0, C_0 are the matrices of the basis vectors in standard coordinates, then $P_{C\leftarrow B} = C_0^{-1}B_0$. In practice reduce $[\,C_0 \ B_0\,]$ to $[\,I \ P_{C\leftarrow B}\,]$. Reversing the change uses the inverse matrix; swapping the labels without inverting is incorrect.

Practice problem 4.7.P1: Read the coordinate labels on the columns

The columns of P are $[f_1]_G, [f_2]_G$. If $v = af_1+bf_2$, linearity of coordinates gives $[v]_G = a[f_1]_G+b[f_2]_G = P(a,b)^T = P[v]_F$. Thus the correct choice is **(ii)**.

Practice problem 4.7.P2: Reverse the change from the example

Example 1 gives $P_{C\leftarrow B} = \left(\begin{smallmatrix} 4 & -6 \\ 1 & 1 \end{smallmatrix}\right)$. Its determinant is $4 + 6 = 10$, so the inverse is

$$P_{B\leftarrow C} = \frac{1}{10}\begin{pmatrix} 1 & 6 \\ -1 & 4 \end{pmatrix}.$$

Multiplying it by the original matrix gives I_2, confirming that changing there and back restores every coordinate vector.

Exercise 4.7.1: Construct columns directly from the given equations

(a) Record the C-coordinates of each b-vector.

From $b_1 = 6c_1 - 2c_2$ and $b_2 = 9c_1 - 4c_2$,

$$P_{C\leftarrow B} = \begin{pmatrix} 6 & 9 \\ -2 & -4 \end{pmatrix}.$$

(b) Apply that change to the specified vector.

For $x = -3b_1 + 2b_2$, its B-coordinate vector is $(-3, 2)$, so

$$[x]_C = \begin{pmatrix} 6 & 9 \\ -2 & -4 \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -18+18 \\ 6-8 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}.$$

Exercise 4.7.2: A singular matrix reveals a printed inconsistency

The supplied relations are $b_1 = -2c_1 + 4c_2$ and $b_2 = 3c_1 - 6c_2$. But then $b_2 = -\frac{3}{2}b_1$, so $\{b_1, b_2\}$ is dependent and cannot be a basis. These relations are incompatible with the printed assumption that B and C are both bases. This is visible in the formal column matrix

$$Q = \begin{pmatrix} -2 & 3 \\ 4 & -6 \end{pmatrix}, \qquad \det Q = 12 - 12 = 0.$$

(a) State what the formal calculation does and does not mean.

The columns are the given C-coordinates, but Q is not a change-of-basis matrix between two bases and has no inverse. There is no valid instance satisfying all the printed assumptions.

(b) Evaluate the specified linear combination nevertheless.

The vector $x = 2b_1 + 3b_2$ is meaningful even for dependent generators:

$$x = (-4 + 9)c_1 + (8 - 18)c_2 = 5c_1 - 10c_2.$$

Thus $[x]_C = (5, -10)$. The coefficient list $(2, 3)$ is one representation using b_1, b_2, but cannot be called a unique B-coordinate vector.

Exercise 4.7.3: The output coordinate system matches the columns

The columns are $[u_1]_W, [u_2]_W$, so multiplying by the coefficients of x in U forms the W-coordinates of x. Hence $[x]_W = P[x]_U$, choice **(ii)**. The reverse equation would need P^{-1}.

Exercise 4.7.4: A three-vector version of the same rule

Here P has columns $[d_1]_A, [d_2]_A, [d_3]_A$. Therefore the input coefficients are D-coordinates, and the output is an A-coordinate vector: $[x]_A = P[x]_D$. The correct choice is **(i)**.

Exercise 4.7.5: Translate three linear-combination equations

(a) Put the B-coordinates of a_1, a_2, a_3 into columns.

The equations give

$$P_{B \leftarrow A} = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

(b) Multiply by the A-coordinate vector.

For $x = 3a_1 + 4a_2 + a_3$,

$$[x]_B = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \\ 2 \end{pmatrix}.$$

For example, the first output entry is $4(3) - 1(4) + 0(1) = 8$.

Exercise 4.7.6: Change from F to D

The columns must be $[f_1]_D, [f_2]_D, [f_3]_D$, giving

$$P_{D \leftarrow F} = \begin{pmatrix} 2 & 0 & -3 \\ -1 & 3 & 0 \\ 1 & 1 & 2 \end{pmatrix}.$$

For $x = f_1 - 2f_2 + 2f_3$, multiply by $(1, -2, 2)$ to obtain

$$[x]_D = (2 - 6,\ -1 - 6,\ 1 - 2 + 4) = (-4, -7, 3).$$

This answers (a) and (b), keeping the direction of the coordinate change explicit.

Exercise 4.7.7: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 7 & -3 \\ 5 & -1 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 1 & -2 \\ -5 & 2 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} -3 & 1 \\ -5 & 2 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} -2 & 1 \\ -5 & 3 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.8: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} -1 & 1 \\ 8 & -7 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} 9 & -8 \\ -10 & 9 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} 9 & 8 \\ 10 & 9 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.9: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 4 & 8 \\ 4 & 4 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} 2 & 3 \\ 0 & -1 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \\ 0 & -1 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.10: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 6 & 4 \\ -12 & 2 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 4 & 3 \\ 2 & 9 \end{pmatrix}$. Solve $C_0P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C\leftarrow B} = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}, \qquad P_{B\leftarrow C} = \begin{pmatrix} 0 & -\frac{1}{2} \\ 1 & \frac{3}{2} \end{pmatrix}.$$

The first identity is checked by $C_0P_{C\leftarrow B} = B_0$; the second is the inverse, checked by $P_{B\leftarrow C}P_{C\leftarrow B} = I_2$. For example, column one of C_0P reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.11: Two statements about the direction of a change

(a) False.

The columns of $P_{C\leftarrow B}$ are the C-coordinates of the B vectors, $[b_j]_C$, not the B-coordinates of the C vectors. The latter columns belong to its inverse $P_{B\leftarrow C}$.

(b) True.

If C is the standard basis of $\mathbb{R}^n$, $[x]_C = x$ and $[b_j]_C = b_j$. Thus $P_{C\leftarrow B} = [\,b_1\ \cdots\ b_n\,] = P_B$.

Exercise 4.7.12: Independence and an algorithm's direction

(a) True.

A change-of-coordinates matrix between bases is invertible: composing it with the reverse change gives identity. By the Invertible Matrix Theorem, its columns are therefore independent.

(b) False.

Reducing $[\,c_1\ c_2\ b_1\ b_2\,]$ to $[\,I\ P\,]$ computes $P = C_0^{-1}B_0$. It therefore satisfies $[x]_C = P[x]_B$, the reverse of the printed assertion. The printed equation would use P^{-1} instead.

Exercise 4.7.13: Change from a polynomial basis to coefficients

The standard coordinate vector of $a + bt + ct^2$ is (a, b, c). Consequently, reading coefficients of the three given polynomials gives

$$P_{C\leftarrow B} = \begin{pmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{pmatrix}.$$

To find $[-1 + 2t]_B$, solve $Pz = (-1, 2, 0)$. Exact elimination gives $z = (5, -2, 1)$. Check directly:

$$5(1 - 2t + t^2) - 2(3 - 5t + 4t^2) + (2t + 3t^2) = -1 + 2t.$$

Thus the coordinate vector is $(5, -2, 1)$, and the displayed matrix is the requested change to the standard basis.

Exercise 4.7.14: Represent t^2 using three different polynomials

Reading constant, linear, and quadratic coefficients gives

$$P = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ -3 & -5 & 0 \end{pmatrix}.$$

Solve $Pz = (0, 0, 1)$. Its first two equations are $z_1 + 2z_2 + z_3 = 0$ and $z_2 + 2z_3 = 0$, so $z_2 = -2z_3$ and $z_1 = 3z_3$. The last equation gives $-9z_3 + 10z_3 = 1$, hence $z_3 = 1$. Therefore

$$t^2 = 3(1 - 3t^2) - 2(2 + t - 5t^2) + (1 + 2t).$$

The coefficients $(3, -2, 1)$ can be checked by canceling constant and linear terms.

Exercise 4.7.15: Justify the construction of the change matrix

(a) B spans V, because B is a basis. Therefore v has an expansion $v = \sum_j x_j b_j$.

(b) The coordinate mapping determined by a basis is linear. It takes that expansion to $[v]_C = \sum_j x_j [b_j]_C$.

(c) This is the definition of a matrix-vector product as a linear combination of the matrix's columns.

(d) The column $(x_1, \ldots, x_n)$ is $[v]_B$, by the definition and uniqueness of coordinates relative to B.

These observations prove $[v]_C = P_{C \leftarrow B}[v]_B$ for every vector v.

Exercise 4.7.16: Justify uniqueness of the change matrix

(a) $[b_1]_B = e_1$, so $Q[b_1]_B = Qe_1$ is the first column of Q.

(b) For a general k, column k must be $[b_k]_C$.

(c) Indeed $[b_k]_B = e_k$, and multiplying Q by e_k selects its kth column.

Thus every column of Q is forced. It must equal $[\,[b_1]_C \ \cdots \ [b_n]_C\,] = P_{C \leftarrow B}$.

Exercise 4.7.17: Change between powers of cosine and multiple angles

Write $x_j(t) = \cos^j t$, $y_j(t) = \cos(jt)$ for $j = 0, \ldots, 6$.

(a) Form P from the multiple-angle identities, then invert it.

The angle-addition formulas recursively give the polynomials $T_0(u) = 1$, $T_1(u) = u$, $T_{j+1}(u) =$

$2uT_j(u) - T_{j-1}(u)$ with $y_j(t) = T_j(\cos t)$. Taking their coefficients as columns gives

$$P = \begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 & -8 & 0 & 18 \\ 0 & 0 & 0 & 4 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 & -48 \\ 0 & 0 & 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 32 \end{pmatrix}.$$

$$P^{-1} = \begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 & \frac{3}{8} & 0 & \frac{5}{16} \\ 0 & 1 & 0 & \frac{3}{4} & 0 & \frac{5}{8} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & \frac{15}{32} \\ 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{5}{16} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & \frac{3}{16} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{32} \end{pmatrix}.$$

Exact multiplication gives $PP^{-1} = P^{-1}P = I_7$.

(b) Interpret every column, rather than just listing numbers.

Since $P = P_{B \leftarrow C}$, its inverse converts B-coordinates to C-coordinates. The jth x-vector has B-coordinates equal to the corresponding standard coordinate vector, so its C-coordinates are a column of P^{-1}. Reading those seven columns gives

$$\begin{aligned} 1 &= \cos 0t, \\ \cos t &= \cos t, \\ \cos^2 t &= \tfrac{1}{2} + \tfrac{1}{2}\cos 2t, \\ \cos^3 t &= \tfrac{3}{4}\cos t + \tfrac{1}{4}\cos 3t, \\ \cos^4 t &= \tfrac{3}{8} + \tfrac{1}{2}\cos 2t + \tfrac{1}{8}\cos 4t, \\ \cos^5 t &= \tfrac{5}{8}\cos t + \tfrac{5}{16}\cos 3t + \tfrac{1}{16}\cos 5t, \\ \cos^6 t &= \tfrac{5}{16} + \tfrac{15}{32}\cos 2t + \tfrac{3}{16}\cos 4t + \tfrac{1}{32}\cos 6t. \end{aligned}$$

Exercise 4.7.18: Integrate after changing the function basis

The integrand has B-coordinates $(0, 0, 0, 5, -6, 5, -12)$ in the notation of Exercise 4.7.17. Multiplication by P^{-1} gives its C-coordinates

$$(-6,\ 55/8,\ -69/8,\ 45/16,\ -3,\ 5/16,\ -3/8).$$

Thus it equals

$$-6 + \tfrac{55}{8}\cos t - \tfrac{69}{8}\cos 2t + \tfrac{45}{16}\cos 3t - 3\cos 4t + \tfrac{5}{16}\cos 5t - \tfrac{3}{8}\cos 6t.$$

Integrate these simple cosine terms using $\int \cos(jt)\,dt = \sin(jt)/j$ for $j > 0$:

$$\int (5\cos^3 t - 6\cos^4 t + 5\cos^5 t - 12\cos^6 t)\,dt$$

$$= -6t + \tfrac{55}{8}\sin t - \tfrac{69}{16}\sin 2t + \tfrac{15}{16}\sin 3t - \tfrac{3}{4}\sin 4t + \tfrac{1}{16}\sin 5t - \tfrac{1}{16}\sin 6t + C.$$

Differentiating the result restores the multiple-angle expression, which Exercise 4.7.17 identifies with the original polynomial in cosine.

Exercise 4.7.19: Find bases when a change matrix is prescribed

Let $V_0 = [\, v_1 \; v_2 \; v_3 \,]$ and let P be the supplied matrix.

(a) Its columns give the new u-vectors in v-coordinates.

For $P = P_{V \leftarrow U}$ we need $U_0 = V_0 P$. Multiplication gives

$$U_0 = \begin{pmatrix} -6 & -6 & -5 \\ -5 & -9 & 0 \\ 21 & 32 & 3 \end{pmatrix}.$$

Take its columns as u_1, u_2, u_3. Since V_0 and P are invertible (the latter has determinant -1), their product is invertible, so these columns are a basis.

(b) Solve the reversed requirement.

Now $P = P_{W \leftarrow V}$ means $V_0 = W_0 P$, so $W_0 = V_0 P^{-1}$. This gives

$$W_0 = \begin{pmatrix} 28 & 38 & 21 \\ -9 & -13 & -7 \\ -3 & 2 & 3 \end{pmatrix}.$$

Take these three columns as w_1, w_2, w_3. Their matrix is invertible for the same reason, and $W_0 P = V_0$ checks the required direction.

Exercise 4.7.20: Compose changes in the same order as their actions

(a) Apply the first change, then the second.

For every $x \in V$,

$$[x]_D = P_{D \leftarrow C}[x]_C = P_{D \leftarrow C} P_{C \leftarrow B}[x]_B.$$

By uniqueness of the change matrix, $P_{D \leftarrow B} = P_{D \leftarrow C} P_{C \leftarrow B}$. The rightmost factor acts first, exactly as in composition of functions.

(b) A concrete exact check.

Use the bases with standard-coordinate matrices

$$B_0 = \begin{pmatrix} 7 & -3 \\ 5 & -1 \end{pmatrix}, \quad C_0 = \begin{pmatrix} 1 & -2 \\ -5 & 2 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -1 & 1 \\ 8 & -7 \end{pmatrix}.$$

The program calculation $C_0^{-1}B_0$, $D_0^{-1}C_0$, $D_0^{-1}B_0$ (using exact matrix inversion) gives

$$\begin{pmatrix} 2 & -12 \\ 3 & -14 \end{pmatrix} \begin{pmatrix} -3 & 1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 54 & -22 \\ 61 & -25 \end{pmatrix}.$$

The right side is $D_0^{-1}B_0$, independently verifying the composition rule.

4.8 Applications to difference equations

A signal in this section is a sequence $\{y_k\}$ defined for every integer k, including negative integers. For an order-n constant-coefficient homogeneous equation with nonzero first and last coefficients, specifying n consecutive values determines exactly one signal (Theorem 16). Its solution space therefore has dimension n (Theorem 17). To prove that n proposed solutions form a basis, verify the equations when necessary and prove independence. One convenient independence test is the Casorati matrix whose columns are the signals evaluated at n consecutive indices. If that matrix is invertible at one index, the signals are independent. Fewer than n solutions cannot span an n-dimensional space.

Practice problem 4.8.P1: A three-signal basis

The problem already supplies that 2^k, $3^k \sin(k\pi/2)$, and $3^k \cos(k\pi/2)$ solve the third-order equation. Their values at $0, 1, 2$ form the matrix

$$C(0) = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 4 & 0 & -9 \end{pmatrix}, \qquad \det C(0) = -39 \neq 0.$$

A relation among the three signals would give $C(0)c = 0$, forcing all coefficients to vanish. The solution space has dimension three; these three independent solutions are therefore a basis.

Exercise 4.8.1: Substitute two exponential signals

For $y_k = r^k$, the left side of $y_{k+2} + 2y_{k+1} - 8y_k = 0$ is $r^k(r^2 + 2r - 8) = r^k(r-2)(r+4)$. It vanishes for $r = 2$ and $r = -4$. Both are nonzero, so these signals are defined at negative indices as well. Thus 2^k and $(-4)^k$ are solutions for every integer k.

Exercise 4.8.2: Two roots with opposite signs

For $y_k = r^k$, the equation $y_{k+2} - 25y_k = 0$ becomes $r^k(r^2 - 25) = 0$. The roots $r = 5$ and $r = -5$ make it hold for every integer k, verifying the two proposed solutions.

Exercise 4.8.3: Show the first two verified solutions span

By Exercise 4.8.1 the signals solve the equation. Their values at $k = 0, 1$ form $\left(\begin{smallmatrix} 1 & 1 \\ 2 & -4 \end{smallmatrix}\right)$, with determinant $-6 \neq 0$. They are independent. The equation is of order 2 and has nonzero last coefficient -8, so its solution space has dimension two. Thus the two signals form a basis, and all solutions are $y_k = c_1 2^k + c_2(-4)^k$.

Exercise 4.8.4: A basis for the second equation

The solutions $5^k, (-5)^k$ from Exercise 4.8.2 give initial-value matrix $\left(\begin{smallmatrix} 1 & 1 \\ 5 & -5 \end{smallmatrix}\right)$, whose determinant is -10. Thus they are independent and, since the solution space has dimension two, form a basis. Every solution is $y_k = c_1 5^k + c_2(-5)^k$.

Exercise 4.8.5: A repeated root produces a factor of k

For $u_k = (-2)^k$, the substitution factor is $r^2 + 4r + 4 = (r+2)^2$, so u solves the equation. For $v_k = k(-2)^k$, substitution gives

$$v_{k+2} + 4v_{k+1} + 4v_k = (-2)^k[4(k+2) - 8(k+1) + 4k] = 0.$$

The initial-value matrix $\left(\begin{smallmatrix} 1 & 0 \\ -2 & -2 \end{smallmatrix}\right)$ has determinant $-2 \neq 0$, proving independence. There are two independent solutions in a two-dimensional solution space; hence all solutions are $y_k = (c_1 + c_2 k)(-2)^k$.

Exercise 4.8.6: A cosine-sine pair

For either sine or cosine, adding π to the angle changes its sign. Thus the signals $u_k = 4^k \cos(k\pi/2)$ and $v_k = 4^k \sin(k\pi/2)$ satisfy $u_{k+2} = -16u_k$ and $v_{k+2} = -16v_k$. The initial-value matrix is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$ with determinant 4. They are independent and, by the order-two dimension theorem, a basis. Every solution is $4^k[c_1 \cos(k\pi/2) + c_2 \sin(k\pi/2)]$.

Exercise 4.8.7: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -2 \\ 1 & 4 & 4 \end{pmatrix}, \qquad \det C(0) = 12 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.8: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}, \qquad \det C(0) = 12 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.9: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 5 \\ 4 & -25 & 0 \end{pmatrix}, \qquad \det C(0) = 145 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.10: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 0 & 1 \\ -2 & -2 & 3 \\ 4 & 8 & 9 \end{pmatrix}, \qquad \det C(0) = -50 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.11: Too few solutions for a third-order equation

The equation has order 3 and nonzero last coefficient 4, so its solution space has dimension 3. Only two signals are listed. Even if independent, their span has dimension at most 2, so they cannot form a basis for the whole solution space. The answer is no.

Exercise 4.8.12: Too few solutions for a fourth-order equation

Here the highest and lowest shifts differ by four and the last coefficient is $36 \neq 0$. The solution space therefore has dimension four. Two listed signals cannot span it. Thus they are not a basis, despite being solutions.

Exercise 4.8.13: Two distinct fractional roots

The auxiliary polynomial factors as $r^2 - r + 2/9 = (r - 1/3)(r - 2/3)$. Hence $(1/3)^k$ and $(2/3)^k$ solve the equation. Their initial-value matrix has determinant $2/3 - 1/3 = 1/3 \neq 0$, so they are independent. The dimension-two theorem proves that they span:

$$y_k = c_1(1/3)^k + c_2(2/3)^k.$$

Exercise 4.8.14: Two positive integer roots

The auxiliary polynomial is $r^2 - 5r + 6 = (r - 2)(r - 3)$. The solutions $2^k, 3^k$ have initial-value determinant $3 - 2 = 1 \neq 0$. They are therefore independent; since the solution space has dimension two, all solutions are $y_k = c_1 2^k + c_2 3^k$.

Exercise 4.8.15: A leading coefficient different from one

The auxiliary polynomial is $6r^2 + r - 2 = (2r - 1)(3r + 2)$, with roots $1/2, -2/3$. Dividing the original equation by 6 gives an equivalent order-two equation with nonzero last coefficient, so its solution

space has dimension two. The initial-value determinant is $-2/3 - 1/2 = -7/6 \neq 0$. Thus a basis is $\{(1/2)^k, (-2/3)^k\}$ and every solution is their linear combination.

Exercise 4.8.16: A familiar difference equation

The polynomial $r^2 - 25$ has roots $5, -5$. Their exponential signals solve the equation and have initial-value determinant $-10 \neq 0$. Consequently they form a basis by the dimension-two theorem: $y_k = c_1 5^k + c_2(-5)^k$.

Exercise 4.8.17: Income approaches its equilibrium

For $a = 9/10, b = 4/9$ the equation becomes

$$Y_{k+2} - \tfrac{13}{10} Y_{k+1} + \tfrac{2}{5} Y_k = 1.$$

A constant solution $Y_k = T$ satisfies $(1 - 13/10 + 2/5)T = 1$, giving $T = 10$. The homogeneous polynomial factors as $(r - 4/5)(r - 1/2)$. Its two distinct roots give independent exponential solutions, hence span the homogeneous solution space. Subtracting the particular solution 10 from any nonhomogeneous solution gives a homogeneous solution. Therefore

$$Y_k = 10 + c_1(4/5)^k + c_2(1/2)^k.$$

Both geometric terms tend to zero as $k \to +\infty$, so $Y_k \to 10$, regardless of the fixed initial values.

Exercise 4.8.18: A second stable income model

For $a = 9/10, b = 1/2$, the equation is $Y_{k+2} - (27/20)Y_{k+1} + (9/20)Y_k = 1$. A constant particular solution is again 10, since $1 - 27/20 + 9/20 = 1/10$. The characteristic roots are $3/4$ and $3/5$, as $(r - 3/4)(r - 3/5) = r^2 - (27/20)r + 9/20$. Thus the full solution is

$$Y_k = 10 + c_1(3/4)^k + c_2(3/5)^k.$$

The distinct-root initial-value matrix is invertible, so these terms give every homogeneous solution, not just examples. In particular this model also tends to 10 as k increases.

Exercise 4.8.19: Two real roots for the bending-moment recurrence

For $y_{k+2} + 4y_{k+1} + y_k = 0$, the roots of $r^2 + 4r + 1$ are $r = -2 + \sqrt{3}$ and $s = -2 - \sqrt{3}$. They are distinct and nonzero. The initial-value matrix for r^k, s^k has determinant $s - r = -2\sqrt{3} \neq 0$. The order-two theorem therefore gives the general solution

$$y_k = c_1 r^k + c_2 s^k.$$

These two terms can equivalently be written with exponent $k - 1$ by absorbing one nonzero factor into each arbitrary coefficient.

Exercise 4.8.20: Impose the two boundary values

Assume $N \geq 2$; for $N = 1$ the two stated boundary values would conflict. Write $y_k = Ar^{k-1} + Bs^{k-1}$ using the roots in Exercise 4.8.19. Then $y_1 = 5000$ and $y_N = 0$ give $A + B = 5000$ and $Ar^{N-1} + Bs^{N-1} = 0$. Eliminating one coefficient gives

$$y_k = 5000\,\frac{r^{k-1}s^{N-1} - s^{k-1}r^{N-1}}{s^{N-1} - r^{N-1}}.$$

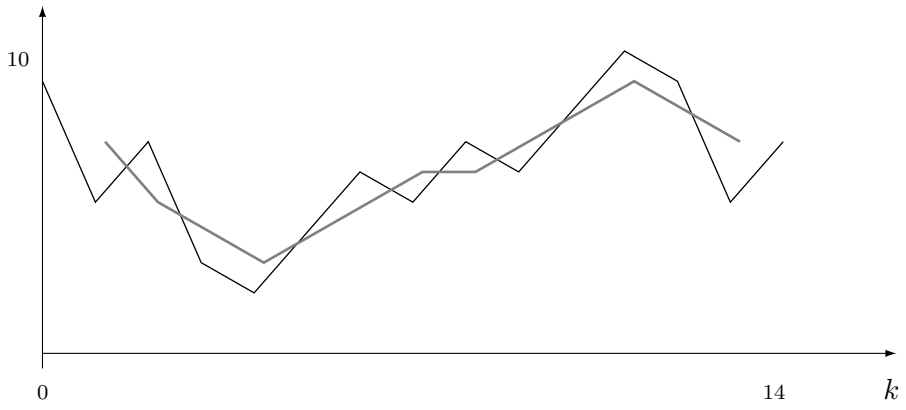
The denominator is nonzero because $|r| \neq |s|$ and $N - 1 > 0$. At $k = 1$ the fraction is 1; at $k = N$ its numerator is zero. Every term solves the recurrence, so their combination does too. Thus this is the required, uniquely determined bending-moment sequence on the support indices.

Exercise 4.8.21: Compute and plot a moving average

Each output is the average of the entry and its two neighbors: $z_k = (y_{k-1} + y_k + y_{k+1})/3$. For instance $z_1 = (9 + 5 + 7)/3 = 7$ and $z_2 = (5 + 7 + 3)/3 = 5$. Continuing through the supplied fifteen entries gives

$$(z_1, \ldots, z_{13}) = (7, 5, 4, 3, 4, 5, 6, 6, 7, 8, 9, 8, 7).$$

The original signal has indices 0 through 14, but the smoothed signal here only has indices 1 through 13; there is no invented endpoint average. In the plot below, the solid broken line is the original signal and the thicker gray broken line is the moving average, using the same axes.



Exercise 4.8.22: A filter separates two frequencies

The filter of Example 3 is $z_k = (\sqrt{2}/4)y_{k+2} + (1/2)y_{k+1} + (\sqrt{2}/4)y_k$. The decimals .35 and .7 in the example abbreviate exact radicals; treating them as rounded decimal inputs would introduce artificial error.

(a) Compute the exact output.

The input is $y_k = 2\cos(k\pi/4) + \cos(3k\pi/4)$. For $u_k = \cos(k\theta)$, angle addition gives

$$u_{k+2} + u_k = 2\cos\theta\cos((k+1)\theta).$$

Thus its filtered output is $[1/2+(\sqrt{2}/2)\cos\theta]\cos((k+1)\theta)$. For $\theta=\pi/4$ the bracket is 1; for $\theta=3\pi/4$ it is 0. Therefore

$$z_k = 2\cos((k+1)\pi/4),$$

with successive values, starting at $k=0$, $\sqrt{2}, 0, -\sqrt{2}, -2, -\sqrt{2}, 0, \sqrt{2}, 2, \ldots$.

(b) Explain the relation to the example.

The filter is linear, so it processes the two input components separately. Example 3 passes the lower-frequency component with a one-index advance and removes the higher-frequency component. The coefficient 2 simply doubles the former output; the latter still contributes zero.

Exercise 4.8.23: A loan paid off by a final smaller payment

(a) Model interest before each payment.

The balance obeys $y_{k+1} = 1.01y_k - 450$, with $y_0 = 10000$, until the last payment. Solving gives $y_k = 45000 - 35000(1.01)^k$ while full payments are made.

(b) A complete balance table and a reproducible program.

The table uses full-precision internal balances and displays dollars rounded to cents; the recurrence itself does not round the balance after each month.

Month	Balance	Month	Balance	Month	Balance
0	10,000.00	9	6,721.02	18	3,134.84
1	9,650.00	10	6,338.23	19	2,716.19
2	9,296.50	11	5,951.61	20	2,293.35
3	8,939.46	12	5,561.12	21	1,866.28
4	8,578.86	13	5,166.74	22	1,434.94
5	8,214.65	14	4,768.40	23	999.29
6	7,846.79	15	4,366.09	24	559.29
7	7,475.26	16	3,959.75	25	114.88
8	7,100.02	17	3,549.34	26	0.00

The following Python code generates the table and adjusts the last payment:

```
from decimal import Decimal, getcontext
getcontext().prec = 50
y = Decimal('10000'); payment = Decimal('450'); k = 0
print(k, f'{y:.2f}')
while y * Decimal('1.01') > payment:
    y = y * Decimal('1.01') - payment
    k += 1
    print(k, f'{y:.2f}')
last = y * Decimal('1.01')
print(k + 1, '0.00', 'last payment', f'{last:.2f}')
```

(c) Find the last payment and the total amount paid.

After payment 25, the unrounded balance is about 114.8801743968. One more month's interest makes the amount due 116.0289761408. Thus the final payment is at month 26, for approximately \$116.03; there have been 25 full \$450 payments. The total paid is $25(450) + 116.0289761408 \approx \$11{,}366.03$. Continuing to subtract a full \$450 after payoff would produce a negative balance and would not model the actual final payment.

Exercise 4.8.24: Monthly deposits and compound interest

(a) Apply interest, then make the new deposit.

The recurrence is $y_{k+1} = 1.005y_k + 200$ with $y_0 = 1000$. Its closed form is $y_k = 41000(1.005)^k - 40000$: substituting it checks both the initial value and the recurrence.

(b) The complete table for months zero through sixty.

Month	Balance	Month	Balance	Month	Balance
0	1,000.00	21	5,527.22	42	10,554.34
1	1,205.00	22	5,754.86	43	10,807.11
2	1,411.02	23	5,983.63	44	11,061.15
3	1,618.08	24	6,213.55	45	11,316.45
4	1,826.17	25	6,444.62	46	11,573.04
5	2,035.30	26	6,676.84	47	11,830.90
6	2,245.48	27	6,910.23	48	12,090.06
7	2,456.71	28	7,144.78	49	12,350.51
8	2,668.99	29	7,380.50	50	12,612.26
9	2,882.33	30	7,617.40	51	12,875.32
10	3,096.75	31	7,855.49	52	13,139.70
11	3,312.23	32	8,094.77	53	13,405.39
12	3,528.79	33	8,335.24	54	13,672.42
13	3,746.43	34	8,576.92	55	13,940.78
14	3,965.17	35	8,819.80	56	14,210.49
15	4,184.99	36	9,063.90	57	14,481.54
16	4,405.92	37	9,309.22	58	14,753.95
17	4,627.95	38	9,555.77	59	15,027.72
18	4,851.09	39	9,803.55	60	15,302.86
19	5,075.34	40	10,052.56		
20	5,300.72	41	10,302.83		

Here is the program used, retaining internal precision until display:

```
from decimal import Decimal, getcontext
getcontext().prec = 50
y = Decimal('1000')
for k in range(61):
    print(k, f'{y:.2f}')
    y = y * Decimal('1.005') + Decimal('200')
```

(c) Separate deposits from interest.

At months $24, 48, 60$, the balances are respectively \$6,213.55, \$12,090.06, and \$15,302.86, rounded to cents. By month 60 the deposits total $1000 + 60(200) = 13000$. Thus interest accounts for $15302.8562545 - 13000 \approx \$2{,}302.86$.

Exercise 4.8.25: A quadratic particular solution

For $y_k = k^2$, substitution gives

$$(k+2)^2 + 3(k+1)^2 - 4k^2 = 10k + 7,$$

which is the required right-hand side. The homogeneous polynomial is $r^2 + 3r - 4 = (r-1)(r+4)$, with independent exponential solutions $1, (-4)^k$. Any two particular solutions differ by a homogeneous solution, so the general solution is $y_k = k^2 + c_1 + c_2(-4)^k$.

Exercise 4.8.26: A linear particular solution

For $y_k = 1 + k$, the left side is $(k+3) - 6(k+2) + 5(k+1) = -4$. The homogeneous roots of $r^2 - 6r + 5$ are $1, 5$, so its solutions are $c_1 + c_2 5^k$. Hence all solutions are $y_k = 1 + k + c_1 + c_2 5^k$. The constant 1 could be absorbed into c_1, but retaining it makes the supplied particular solution visible.

Exercise 4.8.27: Another linear particular solution

For $y_k = k - 2$, compute $y_{k+2} - 4y_k = k - 4(k-2) = 8 - 3k$. The homogeneous roots are $2, -2$. Thus the full solution is $y_k = k - 2 + c_1 2^k + c_2(-2)^k$. The two exponential solutions are independent because their values at indices zero and one have determinant $-4 \neq 0$.

Exercise 4.8.28: An affine particular solution with roots of magnitude five

For $y_k = 1 + 2k$, the left side is $(1 + 2(k+2)) - 25(1 + 2k) = -48k - 20$, exactly as required. The associated homogeneous equation has roots $5, -5$. Therefore the general solution is $y_k = 1 + 2k + c_1 5^k + c_2(-5)^k$.

Exercise 4.8.29: A fourth-order equation as a first-order vector recurrence

Choose the state $x_k = (y_k, y_{k+1}, y_{k+2}, y_{k+3})^T$. The first three entries of x_{k+1} just shift those coordinates left. Solving the recurrence for y_{k+4} gives $y_{k+4} = 2y_k - 6y_{k+1} + 8y_{k+2} - 3y_{k+3}$. Thus

$$x_{k+1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & -6 & 8 & -3 \end{pmatrix} x_k.$$

The state definition is part of the answer: reversing the order of its coordinates would change the companion matrix.

Exercise 4.8.30: A third-order companion system

Set $x_k = (y_k, y_{k+1}, y_{k+2})^T$. Since $y_{k+3} = 5y_{k+2} - 8y_k$, the desired system is

$$x_{k+1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & 0 & 5 \end{pmatrix} x_k.$$

The middle coefficient is zero because there is no y_{k+1} term in the original equation.

Exercise 4.8.31: Remove an irrelevant common shift

The equation $y_{k+3} + 5y_{k+2} + 6y_{k+1} = 0$ has no y_k term. Put $j = k + 1$. Because the equation holds for *all* integers k, it is equivalent to $y_{j+2} + 5y_{j+1} + 6y_j = 0$ for all integers j. Its order is therefore 2, not 3: the largest and smallest shifts differ by two and both endpoint coefficients are nonzero.

Exercise 4.8.32: The order depends on the last nonzero coefficient

For $y_{k+3} + a_1y_{k+2} + a_2y_{k+1} + a_3y_k = 0$: if $a_3 \neq 0$ the order is 3; if $a_3 = 0, a_2 \neq 0$ a shift makes the order 2; if $a_3 = a_2 = 0, a_1 \neq 0$ it is 1. If all three are zero, it says $y_{k+3} = 0$ at every integer, which is just the zero-signal condition, of order 0 in this shift-width convention. Without a nonzero hypothesis on the final coefficient, one cannot simply call every equation in the displayed form third order.

Exercise 4.8.33: Independence does not require an invertible Casorati sample for arbitrary signals

Let $y_k = k^2$ and $z_k = 2k|k|$. If $ay_k + bz_k = 0$ for every integer k, using $k = 1$ gives $a + 2b = 0$ and using $k = -1$ gives $a - 2b = 0$. Subtracting gives $b = 0$, then $a = 0$. Thus the signals are independent on all integers (they would be dependent if restricted to nonnegative indices). The requested matrices are

$$C(0) = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}, \quad C(-1) = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}, \quad C(-2) = \begin{pmatrix} 4 & -8 \\ 1 & -2 \end{pmatrix}.$$

All are singular. Invertibility of one Casorati matrix is a sufficient independence test, but its failure alone does not prove dependence for arbitrary signals. No common second-order recurrence hypothesis was given here.

Exercise 4.8.34: Sampling can destroy independence

Take $f(t) = 1$, $g(t) = t$, and $h(t) = \sin(\pi t)$. They are independent functions: a relation $a + bt + c\sin(\pi t) = 0$ for every real t, evaluated at $0, 1, 1/2$, successively gives $a = b = c = 0$. But sampling at integers turns h into the zero signal. The three sampled signals are therefore dependent. Thus the answer is no: distinct continuous behaviors can become indistinguishable at the chosen sample points.

Exercise 4.8.35: A linear difference operator on signals

For two signals u, v and a scalar c, evaluate each index:

$$\begin{aligned}(T(u+v))_k &= (u_{k+2}+v_{k+2}) + a(u_{k+1}+v_{k+1}) + b(u_k+v_k)\\ &= (Tu)_k + (Tv)_k,\\ (T(cu))_k &= cu_{k+2} + acu_{k+1} + bcu_k = c(Tu)_k.\end{aligned}$$

The resulting sequences are defined at every integer, so T maps S into S. The two coordinate-wise identities prove linearity as an equality of signals.

Exercise 4.8.36: Add a homogeneous solution to a particular solution

Suppose $T(x_p) = z$ and $u \in \ker T$, so $T(u) = 0$. Linearity gives $T(u+x_p) = T(u)+T(x_p) = 0+z = z$. Thus $u + x_p$ solves the same nonhomogeneous equation. Conversely, if $T(x) = z$, then $T(x - x_p) = z - z = 0$, explaining why all solutions are precisely a particular solution plus kernel vectors.

Exercise 4.8.37: One-sided shifts have only a one-sided inverse

Apply the maps directly to an arbitrary sequence $y = (y_0, y_1, \ldots)$:

$$TD(y) = T(0, y_0, y_1, \ldots) = (y_0, y_1, \ldots) = y.$$

Hence $TD = I$. But $DT(y) = D(y_1, y_2, \ldots) = (0, y_1, y_2, \ldots)$, which need not equal y. For example $y = (1, 0, 0, \ldots)$ is sent to the zero sequence by DT. Thus $DT \neq I$, illustrating why the finite-dimensional square-matrix conclusion about one-sided inverses does not extend to this infinite-dimensional space.

4.9 Applications to Markov chains

We use column probability vectors. Entry p_{ij} is the probability of moving *from state j to state i*, so each column of the transition matrix sums to 1 and its entries are nonnegative. The distribution evolves by $x_{k+1} = Px_k$. A steady-state vector is a probability vector q with $Pq = q$; solve $(P-I)q = 0$ together with $\sum_i q_i = 1$. A stochastic matrix is regular if some positive integer power has every entry strictly positive. For a regular matrix, Theorem 18 guarantees a unique steady-state vector and convergence $P^k x_0 \to q$ for every initial probability vector.

Practice problem 4.9.P1: A city resident's next two locations

The migration matrix from Example 1 is $M = \left(\begin{smallmatrix} .95 & .03 \\ .05 & .97 \end{smallmatrix}\right)$, with city first and suburbs second. Starting from $x_0 = (1, 0)$, $x_1 = Mx_0 = (.95, .05)$. Thus (a) the probability of living in the suburbs next year is 5%. A second multiplication gives $x_2 = (.95^2 + .03(.05), .05(.95) + .97(.05)) = (.904, .096)$. Thus (b) the probability after two years is 9.6%.

Practice problem 4.9.P2: Test the steady-state equation rather than its appearance

For the given P, q, direct multiplication gives

$$\begin{pmatrix} .6 & .2 \\ .4 & .8 \end{pmatrix} \begin{pmatrix} .3 \\ .7 \end{pmatrix} = \begin{pmatrix} .18 + .14 \\ .12 + .56 \end{pmatrix} = \begin{pmatrix} .32 \\ .68 \end{pmatrix} \neq q.$$

Although q is a probability vector, it is not a steady-state vector.

Practice problem 4.9.P3: A long-run migration percentage

Solve $Mq = q$: the first equation reduces to $.05q_1 = .03q_2$. Together with $q_1 + q_2 = 1$, this gives $q = (3/8, 5/8)$. Every entry of M is positive, so it is regular; Theorem 18 makes this the limiting distribution. Thus 62.5% will live in the suburbs after many years.

Exercise 4.9.1: Radio listeners at two station breaks

(a) Build columns from current stations.

Order the states as news, music for both rows (next station) and columns (current station). The probabilities give

$$P = \begin{pmatrix} .7 & .6 \\ .3 & .4 \end{pmatrix}.$$

Column one describes a news listener; column two describes a music listener. Each column sums to 1.

(b) Use the stated initial audience.

Everyone initially listens to news, so $x_0 = (1, 0)$.

(c) Count the breaks, not the elapsed minutes.

At 9:25 there have been two breaks, at 8:30 and 9:00. Then $x_1 = (.7, .3)$ and $x_2 = (.7(.7) + .6(.3), .3(.7) + .4(.3)) = (.67, .33)$. Thus 33% will be listening to music.

Exercise 4.9.2: Three equally treated food choices

(a) Use foods 1, 2, 3 in both state lists.

The probability of retaining the current food is .6; the other two possibilities each have probability .2. Thus

$$P = \begin{pmatrix} .6 & .2 & .2 \\ .2 & .6 & .2 \\ .2 & .2 & .6 \end{pmatrix}.$$

(b) Apply two transitions after the initial trial.

Starting at food 1 gives $x_0 = (1, 0, 0)$ and $x_1 = (.6, .2, .2)$. The second component of x_2 is $.2(.6) + .6(.2) + .2(.2) = .12 + .12 + .04 = .28$. The requested probability is 28%; a single multiplication would answer the first trial after the initial trial, not the second.

Exercise 4.9.3: Healthy and ill students

(a) Define the state order and fill the columns.

Use healthy, ill. A healthy student becomes ill with probability .05; an ill student becomes healthy with probability $1 - .55 = .45$. Therefore

$$P = \begin{pmatrix} .95 & .45 \\ .05 & .55 \end{pmatrix}.$$

(b) Advance Monday's probability vector twice.

Monday's $x_0 = (.8, .2)$ gives Tuesday $x_1 = (.95(.8) + .45(.2), .05(.8) + .55(.2)) = (.85, .15)$. Wednesday gives $x_2 = (.875, .125)$. The ill percentages are 15% and 12.5%.

(c) Use a different initial condition for a currently healthy student.

From $(1, 0)$, after one day the vector is $(.95, .05)$, and after two days the healthy probability is $.95^2 + .45(.05) = .925$. Thus the requested probability is 92.5%.

Exercise 4.9.4: Good, indifferent, and bad weather

(a) Supply probabilities missing from the wording.

Use good, indifferent, bad. Each column sums to one, so the remaining indifferent-to-bad probability is .3 and bad-to-bad is .3. The matrix is

$$P = \begin{pmatrix} .4 & .5 & .3 \\ .3 & .2 & .4 \\ .3 & .3 & .3 \end{pmatrix}.$$

(b) Tomorrow's bad-weather probability.

Starting from $(.5, .5, 0)$, its bad component after one transition is $.3(.5) + .3(.5) + .3(0) = .3$, or 30%.

(c) Wednesday is two transitions from Monday.

From Monday $(0, .6, .4)$, Tuesday is $(.42, .28, .30)$. Wednesday's good component is $.4(.42) + .5(.28) + .3(.30) = .398$, so the answer is 39.8%.

Exercise 4.9.5: Normalize a two-state balance equation

For $P = \left(\begin{smallmatrix} .1 & .5 \\ .9 & .5 \end{smallmatrix}\right)$, $Pq = q$ reduces to $.9q_1 = .5q_2$, or $q_2 = (9/5)q_1$. Impose $q_1 + q_2 = 1$ to get $(14/5)q_1 = 1$.

Conclusion. $q = (5/14, 9/14)$. Both entries are nonnegative and sum to one.

Exercise 4.9.6: A second two-state equilibrium

Here the balance equation is $.6q_1 = .8q_2$, so $3q_1 = 4q_2$. Together with normalization this gives

$$q = (4/7, 3/7).$$

Substitution gives $Pq = q$, confirming the calculation.

Exercise 4.9.7: Three-state balance with two useful equations

From the first row, $.3q_1 = .1q_2 + .1q_3 = .1(1 - q_1)$, so $.4q_1 = .1$ and $q_1 = 1/4$. From the second, $.2q_2 = .2q_1 + .2q_3 = .2(1 - q_2)$, so $q_2 = 1/2$. Normalization then gives $q_3 = 1/4$.

Conclusion. $q = (1/4, 1/2, 1/4)$. The remaining row equation is satisfied as well.

Exercise 4.9.8: A steady state despite zero transition entries

The second and third row equations give $.5q_2 = .1q_3$ and $.9q_3 = .6q_1$. Thus $q_2 = q_3/5$ and $q_3 = 2q_1/3$. Normalizing gives

$$q_1(1 + 2/15 + 2/3) = 1, \qquad q = (5/9, 2/27, 10/27).$$

Substituting these fractions into the first row gives $5/9$; their sum is one, so they form the required probability vector.

Exercise 4.9.9: One zero can disappear under squaring

The given matrix is stochastic, since it has nonnegative entries and unit column sums. Its square is

$$P^2 = \begin{pmatrix} .2 & 1 \\ .8 & 0 \end{pmatrix}^2 = \begin{pmatrix} .84 & .2 \\ .16 & .8 \end{pmatrix}.$$

Every entry of this power is strictly positive. Therefore P is regular, even though P itself has a zero entry.

Exercise 4.9.10: An absorbing state prevents regularity

For $P = \left(\begin{smallmatrix} 1 & .3 \\ 0 & .7 \end{smallmatrix}\right)$, the first column is $(1, 0)$, and $P(1, 0) = (1, 0)$. Inductively the first column of every positive power remains $(1, 0)$. Thus each power has a zero entry, so P is not regular.

Exercise 4.9.11: Long-run radio preferences

For the matrix in Exercise 4.9.1, the balance equation is $.3q_N = .6q_M$. Thus $q_N = 2q_M$, and normalization gives (a) $q = (2/3, 1/3)$. All entries of P are positive, so it is regular. Theorem 18 implies (b) that the long-run fraction listening to the news is $2/3$, regardless of the initial audience. The finite two-break answer need not equal this limit.

Exercise 4.9.12: No long-run preference between symmetric foods

The matrix of Exercise 4.9.2 sends $(1/3, 1/3, 1/3)$ to itself; every row also sums to one. All entries are positive, so Theorem 18 makes this the unique limiting probability vector. Hence none of the three foods is preferred in the long run: each is chosen with probability $1/3$.

Exercise 4.9.13: The long-run illness rate

For the healthy/ill matrix, $.05q_H = .45q_I$ implies $q_H = 9q_I$. Normalization gives (a) $q = (.9, .1)$. Since this matrix has all positive entries, it is regular; thus (b) the long-run probability that a specific student is ill is .1, or 10%. Theorem 18 applies to every initial probability vector, including certainty of being healthy or ill today, so today's state does not change that limit.

Exercise 4.9.14: Long-run good weather

The third row of the weather matrix gives $q_B = .3$ immediately because $q_G + q_I + q_B = 1$. Then $q_G + q_I = .7$ and $q_I = .3q_G + .2q_I + .4(.3)$. Substitute $q_I = .7 - q_G$ to obtain $.8(.7 - q_G) = .3q_G + .12$, whence $q_G = .4$ and $q_I = .3$. The matrix is strictly positive, so the long-run good-weather probability is 40%.

Exercise 4.9.15: A migration percentage in the long run

With states California and rest of the United States, use

$$P = \begin{pmatrix} .9821 & .0029 \\ .0179 & .9971 \end{pmatrix}.$$

The steady-state equation gives $.0179q_{CA} = .0029q_{rest}$. Since $q_{rest} = 1 - q_{CA}$,

$$q_{CA} = \frac{.0029}{.0179 + .0029} = \frac{29}{208} \approx .13942308.$$

All entries are positive, hence the matrix is regular. The eventual share is approximately 13.9423%. The initial 11.7% affects the transient distributions, but not this limit, under the stipulated constant migration model.

Exercise 4.9.16: Allocate a car fleet across three locations

Order states as City Airport, Downtown, Metro Airport, using the printed *columns* as rental locations and rows as return locations:

$$P = \begin{pmatrix} .90 & .01 & .09 \\ .01 & .90 & .01 \\ .09 & .09 & .90 \end{pmatrix}.$$

The downtown balance gives $.10q_D = .01q_C + .01q_M = .01(1 - q_D)$, so $.11q_D = .01$ and $q_D = 1/11$. The full steady state is $(91/209, 1/11, 9/19)$; its entries sum to one and satisfy $Pq = q$. Because the matrix is positive, it is regular and this steady distribution describes a typical long-run day under the model. For about 2000 cars, downtown gets $2000/11 \approx 181.82$ cars, or approximately **182 cars** rented or ready to rent there.

Exercise 4.9.17: A stochastic matrix has eigenvalue one

(a) Every column of P sums to one, while every column of I sums to one. Thus every column of $P - I$ sums to zero. Adding all other rows to the bottom row produces that column-sum row, which is zero.

(b) The sum of all original rows of $P - I$ is zero. The coefficients of this relation are all 1, not all zero, so the rows are dependent.

(c) A dependent list of n rows spans a space of dimension less than n. Equivalently $\operatorname{rank}(P - I) < n$.

(d) By rank–nullity, $\dim \operatorname{Nul}(P - I) = n - \operatorname{rank}(P - I) > 0$. Thus there is a nonzero x satisfying $(P - I)x = 0$, or $Px = x$.

This proves existence of a nonzero fixed vector; by itself it does not prove its entries can be chosen nonnegative and normalized, which is the additional point mentioned in the exercise.

Exercise 4.9.18: An explicit steady state for every two-state matrix

For $P = \left(\begin{smallmatrix} 1-\alpha & \beta \\ \alpha & 1-\beta \end{smallmatrix}\right)$, the equation $Pq = q$ is $\alpha q_1 = \beta q_2$. If $\alpha + \beta > 0$, choose

$$q = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta}\right).$$

Both entries are nonnegative and sum to one, and the balance equation holds, so this is a steady-state vector. It is the unique normalized solution because the balance and sum equations determine both entries. If $\alpha = \beta = 0$, then $P = I_2$. Every vector $(t, 1 - t)$ with $0 \le t \le 1$ is a steady state; in particular $(1, 0), (0, 1)$ are two independent ones. Thus the parenthetical claim means two independent steady-state directions, not only two probability vectors in this case.

Exercise 4.9.19: A row of ones records total probability

(a) Interpret the scalar Sx.

For $S = [\,1 \ \cdots \ 1\,]$, $Sx = \sum_i x_i$. Therefore x is a probability vector exactly when its entries are nonnegative and $Sx = 1$.

(b) Interpret the entries of SP.

Entry j is the sum of column j of P, which equals 1. Hence $SP = S$.

(c) Preserve nonnegativity and the total separately.

Every entry of Px is a sum of products of nonnegative entries, so it is nonnegative. Also $S(Px) = (SP)x = Sx = 1$. By part (a), Px is a probability vector.

Exercise 4.9.20: The square is stochastic

Every column p_j of P is a probability vector. Column j of P^2 is Pp_j, which is a probability vector by Exercise 4.9.19(c). Thus each column of P^2 has nonnegative entries and sums to one, which is exactly the definition of a stochastic matrix.

Exercise 4.9.21: Observe convergence of all columns of powers

(a) Four powers of the supplied positive 4×4 matrix.

Keep full internal precision in the products and round only the displayed entries to four decimal places. The results are

$$P^2 = \begin{pmatrix} 0.2779 & 0.2780 & 0.2803 & 0.2941 \\ 0.3368 & 0.3355 & 0.3357 & 0.3335 \\ 0.1847 & 0.1861 & 0.1833 & 0.1697 \\ 0.2005 & 0.2004 & 0.2007 & 0.2027 \end{pmatrix}.$$

$$P^3 = \begin{pmatrix} 0.2817 & 0.2817 & 0.2817 & 0.2814 \\ 0.3356 & 0.3356 & 0.3355 & 0.3352 \\ 0.1817 & 0.1817 & 0.1819 & 0.1825 \\ 0.2010 & 0.2010 & 0.2010 & 0.2009 \end{pmatrix}.$$

$$P^4 = \begin{pmatrix} 0.2816 & 0.2816 & 0.2816 & 0.2816 \\ 0.3355 & 0.3355 & 0.3355 & 0.3355 \\ 0.1819 & 0.1819 & 0.1819 & 0.1819 \\ 0.2009 & 0.2009 & 0.2009 & 0.2009 \end{pmatrix}.$$

$$P^5 = \begin{pmatrix} 0.2816 & 0.2816 & 0.2816 & 0.2816 \\ 0.3355 & 0.3355 & 0.3355 & 0.3355 \\ 0.1819 & 0.1819 & 0.1819 & 0.1819 \\ 0.2009 & 0.2009 & 0.2009 & 0.2009 \end{pmatrix}.$$

Solving $(P - I)q = 0$ together with $q_1 + q_2 + q_3 + q_4 = 1$ gives

$$q \approx \begin{pmatrix} 0.2816326912 \\ 0.3355052569 \\ 0.181921411 \\ 0.2009406409 \end{pmatrix}.$$

The columns are rapidly becoming nearly equal to this fixed probability vector.

(b) Powers of the more slowly mixing 3×3 matrix.

$$Q^{10} = \begin{pmatrix} 0.8222 & 0.4044 & 0.5385 \\ 0.0324 & 0.3966 & 0.1666 \\ 0.1453 & 0.1990 & 0.2949 \end{pmatrix}.$$

$$Q^{20} = \begin{pmatrix} 0.7674 & 0.6000 & 0.6690 \\ 0.0637 & 0.2036 & 0.1326 \\ 0.1688 & 0.1964 & 0.1984 \end{pmatrix}.$$

$$Q^{30} = \begin{pmatrix} 0.7477 & 0.6815 & 0.7105 \\ 0.0783 & 0.1329 & 0.1074 \\ 0.1740 & 0.1856 & 0.1821 \end{pmatrix}.$$

$$Q^{40} = \begin{pmatrix} 0.7401 & 0.7140 & 0.7257 \\ 0.0843 & 0.1057 & 0.0960 \\ 0.1756 & 0.1802 & 0.1783 \end{pmatrix}.$$

$$Q^{50} = \begin{pmatrix} 0.7372 & 0.7269 & 0.7315 \\ 0.0867 & 0.0951 & 0.0913 \\ 0.1761 & 0.1780 & 0.1772 \end{pmatrix}.$$

$$Q^{60} = \begin{pmatrix} 0.7360 & 0.7320 & 0.7338 \\ 0.0876 & 0.0909 & 0.0894 \\ 0.1763 & 0.1771 & 0.1767 \end{pmatrix}.$$

$$Q^{70} = \begin{pmatrix} 0.7356 & 0.7340 & 0.7347 \\ 0.0880 & 0.0893 & 0.0887 \\ 0.1764 & 0.1767 & 0.1766 \end{pmatrix}.$$

$$Q^{80} = \begin{pmatrix} 0.7354 & 0.7348 & 0.7351 \\ 0.0881 & 0.0887 & 0.0884 \\ 0.1764 & 0.1766 & 0.1765 \end{pmatrix}.$$

Its second equilibrium equation gives $q_2 = q_3/2$, and its first gives $.03q_1 = .05q_2 + .10q_3$. Normalization then yields

$$q = (25/34, 3/34, 3/17) \approx (.73529412, .08823529, .17647059).$$

The columns approach this vector more slowly, so an arbitrary fixed power such as 100 is not a universal certificate of four-place accuracy.

(c) Explain the observation with Theorem 18.

P is positive, hence regular. Q^2 has all entries positive (the only zero in Q acquires a positive two-step route), so Q is regular too. Column j of P^k is $P^k e_j$, the evolving probability vector from the initial state e_j. Theorem 18 says each such vector tends to the same unique steady state. The same reasoning applies to Q^k and explains the observed column limits for any regular stochastic matrix.

Exercise 4.9.22: A reproducible comparison of two numerical methods

What we must show. Run both methods, report measured times, and distinguish numerical approximation from a mathematical convergence guarantee. A matrix program such as NumPy has no fixed largest permitted order; the practical limit depends on memory and desired runtime. The

following experiment uses positive random matrices through order 1024, the largest order tested here, rather than claiming an intrinsic software size limit. Normalize each random positive column to obtain a regular stochastic matrix. For method 1 solve $(P - I)q = 0$ with its final equation replaced by $\sum_i q_i = 1$. For method 2 use column one of P^{100}.

The following table records actual execution times in seconds on this workspace, with one BLAS thread; input generation is outside the timing. Each method is invoked by one matrix-program call after the shared setup. Human typing time is user dependent and is not fabricated as an execution measurement.

Order	Direct solve	Power 100	Maximum difference
64	0.000149	0.000120	$6.09e-16$
128	0.000173	0.000829	$3.93e-16$
256	0.000788	0.005619	$5.89e-16$
512	0.004078	0.043638	$7.01e-16$
1024	0.028829	0.329893	$7.84e-16$

Here is the complete experimental core; start Python with `OPENBLAS_NUM_THREADS=1` for the stated thread setting.

```
import numpy as np
from time import perf_counter
rng = np.random.default_rng(4922)
for n in [64, 128, 256, 512, 1024]:
    P = rng.uniform(.1, 1, (n, n)); P /= P.sum(axis=0)
    A = P - np.eye(n); A[-1, :] = 1
    b = np.zeros(n); b[-1] = 1
    start = perf_counter()
    q = np.linalg.solve(A, b)
    direct = perf_counter() - start
    start = perf_counter()
    v = np.linalg.matrix_power(P, 100)[:, 0]
    power = perf_counter() - start
    print(n, direct, power, np.max(abs(q-v)))
```

For these well-mixed positive samples, the two vectors agree to less than 8×10^{-16} in maximum coordinate error. This empirical accuracy is not guaranteed for every regular stochastic matrix: a slowly mixing matrix can require far more than 100 steps, as Exercise 4.9.21 illustrates. Solving the normalized linear system avoids choosing a mixing time and was faster for the larger tested orders. Computing a power is easy to request and can reveal transients, but a large power performs more matrix products and still needs a convergence check. For finding only the steady state, the direct normalized solve is preferable in this experiment.

4.S Chapter 4 supplementary exercises

These exercises connect spanning sets, bases, coordinate changes, and rank. When proving two sets equal, establish both inclusions. When proving a universal claim false, give one concrete counterexample. For controllability, form $[\,B\ AB\ \cdots\ A^{n-1}B\,]$ and check whether its columns span all of the state space $\mathbb{R}^n$.

Exercise 4.S.1: Twenty statements tying the chapter together

(a) **True.** The set is $\text{Span}\{v_1, \ldots, v_p\}$, a subspace of V by the span theorem.

(b) **True.** Adding another generator cannot remove combinations already obtainable. A spanning subset remains spanning after a vector is added.

(c) **False.** Start with the independent list $\{e_1\}$ in $\mathbb{R}^2$ and add e_1 again; alternatively add $2e_1$. The enlarged list is dependent.

(d) **False.** In $\mathbb{R}^2$, $\{e_1\}$ is independent but does not span $\mathbb{R}^2$, so it is not a basis.

(e) **True.** The Spanning Set Theorem removes any redundant vectors from a finite spanning list, leaving an independent spanning subset.

(f) **True.** A list of exactly $\dim V = p$ vectors that spans V is a basis by the Basis Theorem, hence is independent.

(g) **False.** A plane not through zero is not a subspace. For example the plane $z = 1$ misses the zero vector.

(h) **False.** In $A = [\,I_2\ I_2\,]$ the nonpivot columns are e_1, e_2, which are independent. Each is a combination of pivot columns, but that does not force a dependence among the nonpivot columns themselves.

(i) **True.** Rows $(1, 0), (1, 0)$ have the relation $r_2 - r_1 = 0$; replacing row two by row two minus row one changes the list to $(1, 0), (0, 0)$, which no longer has that same relation.

(j) **False.** Elementary row operations replace $Ax = 0$ by an equivalent system, so they preserve the null space.

(k) **False.** The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 0 \end{smallmatrix}\right)$ has two nonzero rows but rank one. Count nonzero rows only in an echelon form.

(l) **False.** Nullity is the number of columns minus rank, $n - k$, not generally $m - k$.

(m) **True.** Row-equivalent matrices have the same row space, so their row-space dimensions, and hence ranks, agree.

(n) **False.** Nonzero rows need not be independent, as the matrix in part (k) shows. Nonzero rows of an echelon form do give a row-space basis.

(o) **True.** Row operations preserve row spaces, and both row spaces equal that of the common reduced echelon form.

(p) **True.** Choose a basis $h_1, \dots, h_d$ for H, where $d \le 3$, and use these as columns of a 3×3 matrix, filling any remaining columns with zeros. Its column space is H. If $H = \{0\}$, use the zero matrix.

(q) **False.** Rank m means onto, not necessarily one-to-one. For $A = [\,1\ 0\,]$, rank is $m = 1$ but $(0, 1)$ is a nonzero kernel vector.

(r) **True.** Onto $\mathbb{R}^m$ means the column space has dimension m, which is exactly rank m.

(s) **True.** The reverse change-of-coordinates matrix is the inverse, since changing from one basis to the other and back restores every vector.

(t) **False.** Column j of $P_{C \leftarrow B}$ is $[b_j]_C$, not $[c_j]_B$. The latter columns occur in the inverse matrix.

Exercise 4.S.2: A parametrization has a redundant generator

Collect the coefficients of a, b, c to get generators

$$u = (1, 2, -1, 3), \quad v = (-2, 5, -4, 1), \quad w = (5, -8, 7, 1).$$

But $w = u - 2v$, so the third generator is redundant. The first two are independent: if $\alpha u + \beta v = 0$, the first two coordinates give $\alpha - 2\beta = 0$, $2\alpha + 5\beta = 0$, hence $9\beta = 0$ and both vanish.

Conclusion. A basis is $\{(1, 2, -1, 3), (-2, 5, -4, 1)\}$. There are three parameters in the original expression, but only two independent directions.

Exercise 4.S.3: Describe a plane by a homogeneous equation

Both $u_1 = (-2, 4, -6)$ and $u_2 = (1, 2, -5)$ satisfy $x_1 + 2x_2 + x_3 = 0$. They are not multiples, so their span is a two-dimensional subspace contained in the plane defined by that equation. To verify equality without relying only on dimensions, let b satisfy $b_1 + 2b_2 + b_3 = 0$. Solve the first two equations of $au_1 + cu_2 = b$: they give

$$a = (b_2 - 2b_1)/8, \qquad c = b_1 + 2a.$$

With these choices the third coordinate is $-6a - 5c = -b_1 - 2b_2 = b_3$.

Conclusion. $W = \{b \in \mathbb{R}^3 : b_1 + 2b_2 + b_3 = 0\}$.

Exercise 4.S.4: A scalar multiplier cannot depend on the input variable

The equality $g(t) = tf(t)$ uses the function t, not one fixed real scalar. Linear dependence would require constants a, b, not both zero, with $a(3 + t) + b(3t + t^2) = 0$ for every t. Comparing the t^2 coefficient gives $b = 0$; comparing the constant gives $3a = 0$, so $a = 0$ as well. Thus the pair is independent, and the proposed discussion incorrectly treats a variable multiplier as a scalar.

Exercise 4.S.5: Delete redundant polynomials from the given list

The polynomials are $p_1 = 1 + t$, $p_2 = 1 - t$, $p_3 = 4$, $p_4 = t + t^2$, $p_5 = 1 + 2t + t^2$. First $p_3 = 2p_1 + 2p_2$, so deleting p_3 does not alter the span. Next $p_5 = p_1 + p_4$, so delete p_5. The remaining p_1, p_2, p_4 are independent: in $ap_1 + bp_2 + cp_4 = 0$, the quadratic coefficient gives $c = 0$; then the constant and linear coefficients give $a + b = 0$ and $a - b = 0$, forcing $a = b = 0$.

Conclusion. $\{1+t, 1-t, t+t^2\}$ is a basis for H.

Exercise 4.S.6: A basis of a known two-dimensional span

At least one of the four spanning polynomials is nonzero; choose it as p. Since their span has dimension two, at least one other polynomial q is not a scalar multiple of p. Find such a q by comparing coefficients. Two nonproportional nonzero vectors are independent, so $\{p, q\}$ is an independent pair in the two-dimensional space H. By the Basis Theorem it is already a basis. There is no need to row reduce all four coefficient vectors when the dimension is known in advance.

Exercise 4.S.7: The homogeneous information that certifies every right-hand side

For eighteen equations in twenty variables, every nonhomogeneous right-hand side is attainable precisely when rank is eighteen. By rank–nullity this is equivalent to nullity $20 - 18 = 2$. Thus it suffices, and is necessary, to know that the homogeneous solution set is two-dimensional: exactly two independent homogeneous solutions span it.

Exercise 4.S.8: Equal finite dimensions force equality of nested spaces

Let a basis of H have n vectors. Because $H \subseteq V$, these are n independent vectors in the n-dimensional space V. The Basis Theorem says they span V. But their span is also H, so $H = V$. The containment hypothesis matters: distinct subspaces can have equal dimensions without either containing the other.

Exercise 4.S.9: Rank and nullity of injective and surjective maps

(a) If T is one-to-one.

Its kernel is zero, so rank–nullity gives $\dim \operatorname{range} T = n$. This also requires $n \leq m$, since the range is contained in $\mathbb{R}^m$.

(b) If T maps onto $\mathbb{R}^m$.

Its range has dimension m. Thus its kernel has dimension $n - m$, which in particular requires $n \geq m$.

Exercise 4.S.10: A maximal independent set must span

Suppose the independent set S failed to span V. Choose $v \in V \setminus \operatorname{Span} S$. Then adjoining v still leaves an independent set: if $av + \sum_i c_i s_i = 0$ is a finite relation and $a \neq 0$, it would express v as a combination of elements of S, a contradiction. Thus $a = 0$, and independence of S forces all $c_i = 0$. This contradicts maximality. Hence S spans, and being independent as well, it is a basis. The argument covers possibly infinite S because linear combinations always involve only finitely many terms.

Exercise 4.S.11: A minimal finite spanning set is independent

Suppose a finite spanning set were dependent. Then a nontrivial relation would allow one vector with a nonzero coefficient to be solved for as a combination of the others. Deleting that vector would preserve every original linear combination, and thus preserve the span V. That contradicts minimality. Therefore the set is independent; since it already spans V, it is a basis.

Exercise 4.S.12: Rank cannot increase beyond either factor's rank

(a) Compare column spaces.

Every output $ABx = A(Bx)$ belongs to $\operatorname{Col} A$, so $\operatorname{Col}(AB) \subseteq \operatorname{Col} A$. Taking dimensions gives $\operatorname{rank}(AB) \leq \operatorname{rank} A$.

(b) Apply that result to the transposes in reversed order.

Using $(AB)^T = B^T A^T$ and transpose invariance of rank,

$$\operatorname{rank}(AB) = \operatorname{rank}(B^T A^T) \leq \operatorname{rank} B^T = \operatorname{rank} B.$$

Both inequalities follow. Repeated application proves the corresponding bound for a product with more than two factors.

Exercise 4.S.13: An invertible left factor preserves rank

The product inequality gives $\operatorname{rank}(PA) \leq \operatorname{rank} A$. Since $A = P^{-1}(PA)$, applying it again gives $\operatorname{rank} A \leq \operatorname{rank}(PA)$. The two opposite inequalities imply equality. Invertibility is what supplies the reverse factorization.

Exercise 4.S.14: An invertible right factor preserves rank

The product inequality first gives $\operatorname{rank}(AQ) \leq \operatorname{rank} A$. The identity $A = (AQ)Q^{-1}$ gives the reverse inequality $\operatorname{rank} A \leq \operatorname{rank}(AQ)$, hence equality. Equivalently one can apply Exercise 4.S.13 to $(AQ)^T = Q^T A^T$.

Exercise 4.S.15: A zero product limits two ranks

If $AB = 0$, every column of B is in $\operatorname{Nul} A$. Thus $\operatorname{Col} B \subseteq \operatorname{Nul} A$, and dimensions give

$$\operatorname{rank} B \leq \dim \operatorname{Nul} A = n - \operatorname{rank} A.$$

Rearrangement yields $\operatorname{rank} A + \operatorname{rank} B \leq n$. The shared n is the number of inputs of A and outputs of B, which makes the containment meaningful.

Exercise 4.S.16: Rank is subadditive

Take rank factorizations $A = C_A R_A$, $B = C_B R_B$, with ranks r, s. Then block multiplication gives

$$A + B = \begin{bmatrix} C_A & C_B \end{bmatrix} \begin{pmatrix} R_A \\ R_B \end{pmatrix}.$$

The left block matrix has $r+s$ columns and therefore rank at most $r+s$. By the product inequality, $\operatorname{rank}(A+B) \le r+s = \operatorname{rank} A + \operatorname{rank} B$. If either matrix has rank zero, it is zero and the inequality is immediate, so the proof also covers those degenerate cases.

Exercise 4.S.17: Find an invertible square submatrix of full rank

(a) Select independent columns.

Choose the r pivot columns of A to obtain an $m \times r$ matrix A_1. They are independent, so $\operatorname{rank} A_1 = r$.

(b) Select independent rows of that submatrix.

Its row rank also equals r, so choose r independent rows of A_1. The resulting $r \times r$ matrix A_2 has r independent rows, hence rank r and is invertible by the Invertible Matrix Theorem. This proves the requested existence; the case $r = 0$ has the usual empty square-matrix convention or may be excluded as trivial.

Exercise 4.S.18: Expand four controlled time steps

(a) Compute each state before collecting the blocks.

From $x_0 = 0$ and $x_{k+1} = Ax_k + Bu_k$,

$$\begin{aligned} x_1 &= Bu_0, \\ x_2 &= ABu_0 + Bu_1, \\ x_3 &= A^2Bu_0 + ABu_1 + Bu_2, \\ x_4 &= A^3Bu_0 + A^2Bu_1 + ABu_2 + Bu_3 \\ &= \underbrace{[\,B \;\; AB \;\; A^2B \;\; A^3B\,]}_{M} \begin{pmatrix} u_3 \\ u_2 \\ u_1 \\ u_0 \end{pmatrix}. \end{aligned}$$

The order of the input blocks is reversed to match the stated order in M. Each u_j has two entries, so the stacked input has eight entries, matching the 4×8 matrix M.

(b) Use full row rank to realize an arbitrary target.

Controllability means rank $M = 4$, so M maps $\mathbb{R}^8$ onto $\mathbb{R}^4$. Given $v \in \mathbb{R}^4$, solve $Mz = v$, then split its eight entries into four two-entry blocks (u_3, u_2, u_1, u_0). The computation in part (a) shows that this control sequence gives $x_4 = v$.

Exercise 4.S.19: Check the rank of the controllability matrix

Here the state dimension is 3 and the input has one coordinate, so form $M = [\,B \;\; AB \;\; A^2B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -\frac{9}{10} & \frac{81}{100} \\ 1 & \frac{1}{2} & \frac{1}{4} \end{pmatrix}, \qquad \det M = \frac{14}{25}.$$

Since this determinant is nonzero, M has rank 3 and its columns span the state space. Therefore the pair (A, B) is controllable.

Exercise 4.S.20: Check the rank of the controllability matrix

Here the state dimension is 3 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & \frac{1}{2} & \frac{19}{100} \\ 1 & \frac{7}{10} & \frac{9}{20} \\ 0 & 0 & 0 \end{pmatrix}, \qquad \det M = 0.$$

The determinant is zero, so M does not have full rank and (A, B) is not controllable. Its actual rank is 2. Its last row is zero, while the leading 2×2 minor has determinant $1/5 \neq 0$, verifying rank two.

Exercise 4.S.21: Check the rank of the controllability matrix

Here the state dimension is 4 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\ A^3B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & \frac{8}{5} \\ 0 & -1 & \frac{8}{5} & -\frac{24}{25} \\ -1 & \frac{8}{5} & -\frac{24}{25} & -\frac{3}{125} \end{pmatrix}, \qquad \det M = 0.$$

The determinant is zero, so M does not have full rank and (A, B) is not controllable. Its actual rank is 3. For example its leading 3×3 minor has determinant -1, showing rank at least three; the zero full determinant then gives exactly three.

Exercise 4.S.22: Check the rank of the controllability matrix

Here the state dimension is 4 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\ A^3B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{229}{20} \\ -1 & \frac{1}{2} & \frac{229}{20} & -\frac{411}{40} \end{pmatrix}, \qquad \det M = -\frac{3}{10}.$$

Since this determinant is nonzero, M has rank 4 and its columns span the state space. Therefore the pair (A, B) is controllable.

5.1 Eigenvectors and eigenvalues

An eigenvector of A is a *nonzero* vector v for which $Av = \lambda v$ for some scalar λ. The scalar is the eigenvalue: the direction stays on the same line, while the vector is scaled, possibly reversed or sent to zero. Rearranging gives

$$(A - \lambda I)v = 0.$$

Thus the eigenspace E_λ is $\mathrm{Nul}(A - \lambda I)$, and λ is an eigenvalue exactly when this null space contains a nonzero vector. The eigenspace includes zero; zero itself is never an eigenvector. To find eigenvectors for a specified eigenvalue, reduce $A - \lambda I$, not A. Row reduction of A itself generally changes its eigenvalues.

Practice problem 5.1.P1: Test five without guessing an eigenvector

The supplied matrix gives

$$A - 5I = \begin{bmatrix} 1 & -3 & 1 \\ 3 & -5 & 5 \\ 2 & 2 & 1 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - 3R_1$, $R_3 \leftarrow R_3 - 2R_1$:

$$\begin{bmatrix} 1 & -3 & 1 \\ 0 & 4 & 2 \\ 0 & 8 & -1 \end{bmatrix}.$$

Then $R_3 \leftarrow R_3 - 2R_2$ gives last row $(0, 0, -5)$. There are three pivots and no free variables, so $(A - 5I)x = 0$ has only zero as a solution. Hence 5 is not an eigenvalue.

Practice problem 5.1.P2: A power acting on an eigenvector

If $Ax = \lambda x$, then $A^2x = A(\lambda x) = \lambda Ax = \lambda^2 x$. Applying A once more gives $A^3x = \lambda^2 Ax = \lambda^3 x$. Each step uses linearity to pull out a scalar. Repeating the argument proves $A^k x = \lambda^k x$ for every nonnegative integer k.

Practice problem 5.1.P3: Several independent vectors within one eigenspace

Yes, all four vectors are independent. Suppose $c_1b_1 + c_2b_2 + c_3b_3 + c_4b_4 = 0$. Apply $A - \lambda_3 I$:

$$c_1(\lambda_1 - \lambda_3)b_1 + c_2(\lambda_2 - \lambda_3)b_2 = 0.$$

The two vectors b_1, b_2 correspond to distinct eigenvalues and are independent by Theorem 2. Since the eigenvalue differences are nonzero, $c_1 = c_2 = 0$. The original relation now reduces to $c_3b_3 + c_4b_4 = 0$, and their assumed independence forces $c_3 = c_4 = 0$. This proof also handles the possibility that $c_3b_3 + c_4b_4$ is zero; zero must not be called an eigenvector.

Exercise 5.1.1: Test a proposed eigenvalue

What we must show. Decide whether $\lambda = 2$ has a nonzero eigenvector. Reduce $A - \lambda I$, shown here as B:

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (3)R_1$. Then

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The reduced equation is $x_1 + 2x_2 = 0$, with x_2 free. Taking $x_2 = 1$ gives $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \neq 0$. Directly $Av = \begin{bmatrix} -4 \\ 2 \end{bmatrix} = 2v$.

Conclusion. Yes, 2 is an eigenvalue; all nonzero multiples of v are corresponding eigenvectors.

Exercise 5.1.2: Test a proposed eigenvalue

What we must show. Decide whether $\lambda = -3$ has a nonzero eigenvector. Reduce $A - \lambda I$, shown here as B:

$$B = \begin{bmatrix} 2 & 4 \\ 6 & 12 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{2}R_1, \quad R_2 \leftarrow R_2 - (6)R_1$. Then

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The reduced equation is $x_1 + 2x_2 = 0$, with x_2 free. Taking $x_2 = 1$ gives $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \neq 0$. Directly $Av = \begin{bmatrix} 6 \\ -3 \end{bmatrix} = -3v$.

Conclusion. Yes, -3 is an eigenvalue; all nonzero multiples of v are corresponding eigenvectors.

Exercise 5.1.3: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 1 & -1 \\ 6 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \end{bmatrix}.$$

This equals $-2v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{-2}$.

Exercise 5.1.4: Check a specified vector directly

The candidate is $v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \end{bmatrix}.$$

This equals $3v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{3}$.

Exercise 5.1.5: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} -4 & 3 & 3 \\ 2 & -3 & -2 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -15 \\ 10 \\ -5 \end{bmatrix}.$$

This equals $-5v$ in every coordinate. Therefore v is an eigenvector with eigenvalue $\boxed{-5}$.

Exercise 5.1.6: Check a specified vector directly

The candidate is $v = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$, which is nonzero. Multiplying by the source matrix gives

$$Av = \begin{bmatrix} 3 & 6 & 7 \\ 3 & 2 & 7 \\ 5 & 6 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \\ 1 \end{bmatrix}.$$

The first coordinate would force $\lambda = 5$, but the second would then have to be -10, not 13. A single common multiplier must work in every coordinate, so v is not an eigenvector.

Exercise 5.1.7: Produce a nonzero vector in the shifted null space

Solve $(A - 4I)x = 0$; the coefficient reduction is

$$A - 4I = \begin{bmatrix} -1 & 0 & -1 \\ 2 & -1 & 1 \\ -3 & 4 & 1 \end{bmatrix}.$$

Use $R_1 \leftrightarrow R_2$, $\quad R_1 \leftarrow \frac{1}{2}R_1$, $\quad R_2 \leftarrow R_2 - (-1)R_1$, $\quad R_3 \leftarrow R_3 - (-3)R_1$. Then

$$\begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{5}{2} & \frac{5}{2} \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{2}{5}R_2$, $R_1 \leftarrow R_1 - (-\frac{1}{2})R_2$, $R_3 \leftarrow R_3 - (-\frac{1}{2})R_2$. Then

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The free variable gives, for example, $v = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$. This is nonzero, and substitution gives $Av = \begin{bmatrix} -4 \\ -4 \\ 4 \end{bmatrix} = 4v$.

Conclusion. Yes. The specified scalar is an eigenvalue and v is a corresponding eigenvector.

Exercise 5.1.8: Produce a nonzero vector in the shifted null space

Solve $(A - 1I)x = 0$; the coefficient reduction is

$$A - 1I = \begin{bmatrix} 3 & -2 & 3 \\ 0 & -2 & 3 \\ -1 & 2 & -3 \end{bmatrix}.$$

Use $R_1 \leftarrow \frac{1}{3}R_1$, $R_3 \leftarrow R_3 - (-1)R_1$. Then

$$\begin{bmatrix} 1 & -\frac{2}{3} & 1 \\ 0 & -2 & 3 \\ 0 & \frac{4}{3} & -2 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$, $R_2 \leftarrow \frac{3}{4}R_2$, $R_1 \leftarrow R_1 - (-\frac{2}{3})R_2$, $R_3 \leftarrow R_3 - (-2)R_2$. Then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix}.$$

The free variable gives, for example, $v = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}$. This is nonzero, and substitution gives $Av = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} = 1v$.

Conclusion. Yes. The specified scalar is an eigenvalue and v is a corresponding eigenvector.

Exercise 5.1.9: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 1$.

$$A - 1I = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_1 = \text{Span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix}0 & 0\\2 & -2\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -1\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \text{Span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.10: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}-4 & 2\\3 & 1\end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -5$.

$$A - (-5)I = \begin{bmatrix}1 & 2\\3 & 6\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & 2\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{-5} = \text{Span}\left\{\begin{bmatrix}-2\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = -5v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.11: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}1 & -3\\-4 & 5\end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -1$.

$$A - (-1)I = \begin{bmatrix}2 & -3\\-4 & 6\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -\frac{3}{2}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_{-1} = \text{Span}\left\{\begin{bmatrix}3\\2\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = -1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 7$.

$$A - 7I = \begin{bmatrix}-6 & -3\\-4 & -2\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & \frac{1}{2}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_7 = \text{Span}\left\{\begin{bmatrix}-1\\2\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 7v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.12: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix}4 & 1\\3 & 6\end{bmatrix}$, solve $(A-\lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix}1 & 1\\3 & 3\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \text{Span}\left\{\begin{bmatrix}-1\\1\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 7$.

$$A - 7I = \begin{bmatrix}-3 & 1\\3 & -1\end{bmatrix}, \qquad \text{rref}(A - \lambda I) = \begin{bmatrix}1 & -\frac{1}{3}\\0 & 0\end{bmatrix}.$$

Reading the free variables gives

$$E_7 = \text{Span}\left\{\begin{bmatrix}1\\3\end{bmatrix}\right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 7v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.13: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 1$.

$$A - 1I = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 1v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 2$.

$$A - 2I = \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 2v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.14: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 4 & 0 & -1 \\ 3 & 0 & 3 \\ 2 & -2 & 5 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 3$.

$$A - 3I = \begin{bmatrix} 1 & 0 & -1 \\ 3 & -3 & 3 \\ 2 & -2 & 2 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vector is nonzero; the pivot equations show that they span all solutions. Each satisfies $Av = 3v$ by direct substitution. Thus the displayed list is a basis, of dimension 1.

Exercise 5.1.15: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} -4 & 1 & 1 \\ 2 & -3 & 2 \\ 3 & 3 & -2 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = -5$.

$$A - (-5)I = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_{-5} = \operatorname{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent because they arise from distinct choices of a free coordinate; the pivot equations show that they span all solutions. Each satisfies $Av = -5v$ by direct substitution. Thus the displayed list is a basis, of dimension 2.

Exercise 5.1.16: Find a basis for each listed eigenspace

For the source matrix $A = \begin{bmatrix} 5 & 0 & -1 & 0 \\ 1 & 3 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 4 & -2 & -2 & 4 \end{bmatrix}$, solve $(A - \lambda I)x = 0$ separately for each listed scalar. An eigenspace basis must span every solution, not merely provide one example.

The value $\lambda = 4$.

$$A - 4I = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 2 & -1 & -1 & 0 \\ 4 & -2 & -2 & 0 \end{bmatrix}, \qquad \operatorname{rref}(A - \lambda I) = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Reading the free variables gives

$$E_4 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent because they arise from distinct choices of a free coordinate; the pivot equations show that they span all solutions. Each satisfies $Av = 4v$ by direct substitution. Thus the displayed list is a basis, of dimension 2.

Exercise 5.1.17: Read a triangular matrix's eigenvalues

The source matrix is upper triangular with diagonal entries $0, 3, -2$. For an upper triangular matrix, the eigenvalues are precisely its diagonal entries: $A - \lambda I$ loses a pivot exactly when one of these entries minus λ becomes zero.

Conclusion. The eigenvalues are $0, 3, -2$.

Exercise 5.1.18: The lower-triangular case

The source matrix is lower triangular with diagonal entries $5, 0, 3$. The triangular-matrix theorem therefore gives the eigenvalues directly.

Conclusion. The eigenvalues are $5, 0, 3$. Off-diagonal entries do not change this conclusion for a triangular matrix.

Exercise 5.1.19: A singular matrix has eigenvalue zero

All three columns are multiples of $(1, 1, 1)^T$, so they are dependent. Thus $Ax = 0$ has a nonzero solution and 0 is an eigenvalue. For example $(-2, 1, 0)^T$ gives a zero column combination, without needing elimination. The problem asks for one eigenvalue, not the complete list.

Exercise 5.1.20: Two independent vectors with coordinate sum zero

Each column of A is $(2,2,2)^T$. Therefore any input whose coordinates sum to zero is sent to zero. In particular

$$v_1 = (1,-1,0)^T, \qquad v_2 = (1,0,-1)^T$$

are eigenvectors for $\lambda = 0$. They are independent: in $av_1 + bv_2 = 0$, coordinate two forces $a = 0$ and coordinate three forces $b = 0$.

Conclusion. One eigenvalue is 0, with the two independent eigenvectors above.

Exercise 5.1.21: Do not omit the nonzero-vector requirement

(a) **False.** Taking $x = 0$ makes $Ax = \lambda x$ true for any scalar, even one that is not an eigenvalue. The definition requires $x \neq 0$.

(b) **True.** Noninvertibility is equivalent to $Ax = 0$ having a nonzero solution, precisely an eigenvector for eigenvalue zero.

(c) **True.** Rearranging $Ax = cx$ gives $(A - cI)x = 0$, with the same nonzero x required in both formulations.

(d) **True.** Compute Ax and check whether it equals one scalar multiple of the specified nonzero x. Finding the vector from scratch is a different task.

(e) **False.** Row operations on A need not preserve its eigenvalues. For example $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$ has eigenvalues $2, 0$, but replacing row two by row two minus row one gives triangular $\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}$, with eigenvalues $1, 0$.

Exercise 5.1.22: Eigenspaces and repeated eigenvalues

(a) **False.** Zero solves the equality but is not an eigenvector. The condition $x \neq 0$ is missing.

(b) **False.** The independent vectors e_1, e_2 are both eigenvectors of I_2 with the same eigenvalue 1.

(c) **True.** A steady-state probability vector q is nonzero and satisfies $Mq = q$, so it is an eigenvector for eigenvalue 1.

(d) **False in general.** The matrix $\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$ has eigenvalues $1, -1$, not the diagonal entry 0. The diagonal rule applies to triangular matrices.

(e) **True.** $E_\lambda = \mathrm{Nul}(A - \lambda I)$ by rearranging the eigenvector equation; the null space also includes zero.

Exercise 5.1.23: Bound the number of distinct eigenvalues without a polynomial

If an $n \times n$ matrix had p distinct eigenvalues, choose one nonzero eigenvector for each. Theorem 2 says those p vectors are independent. A list of vectors in $\mathbb{R}^n$ cannot be independent when $p > n$.

Hence $p \leq n$. In particular a 2×2 matrix has at most two distinct eigenvalues. This bounds distinct values, not the number of different eigenvectors, which can be infinite.

Exercise 5.1.24: An example with one distinct eigenvalue

Take $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I_2$. Every nonzero vector satisfies $Ax = 2x$. If $Ax = \lambda x$ with $x \neq 0$, then $(2 - \lambda)x = 0$ forces $\lambda = 2$. Thus the only distinct eigenvalue is 2, although there are infinitely many corresponding eigenvectors.

Exercise 5.1.25: Invert the eigenvalue as well as the matrix

Choose $x \neq 0$ with $Ax = \lambda x$. Since A is invertible, $\lambda \neq 0$: otherwise $Ax = 0$ would force $x = 0$. Multiply the eigenvector equation by A^{-1}: $x = \lambda A^{-1}x$, so $A^{-1}x = \lambda^{-1}x$ after division by the nonzero scalar. The vector x is still nonzero, proving that λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector.

Exercise 5.1.26: A square-zero matrix can have only eigenvalue zero

If $Ax = \lambda x$ with $x \neq 0$, applying A again gives $0 = A^2x = \lambda^2 x$. A nonzero vector times a scalar is zero only when the scalar is zero, so $\lambda = 0$. Zero really is an eigenvalue: if $A = 0$, every nonzero vector works; if $A \neq 0$, choose y with $Ay \neq 0$ and set $x = Ay$. Then $x \neq 0$ but $Ax = A^2y = 0$.

Conclusion. The only eigenvalue is 0.

Exercise 5.1.27: Transpose preserves eigenvalues

The identity $A^T - \lambda I = (A - \lambda I)^T$ is the key. A matrix B is invertible if and only if B^T is invertible: if B^{-1} exists, $(B^{-1})^T$ is an inverse of B^T, and applying the same argument to B^T proves the converse. Therefore $A - \lambda I$ is singular exactly when $A^T - \lambda I$ is singular. Singularity of the shifted matrix is equivalent to the existence of a nonzero eigenvector.

Conclusion. A and A^T have the same eigenvalues, though their eigenvectors need not agree.

Exercise 5.1.28: Deduce the lower-triangular theorem by transposing

If A is lower triangular, A^T is upper triangular and has the same diagonal entries. The established upper-triangular case says that the eigenvalues of A^T are precisely these entries. Exercise 27 says A has exactly the same eigenvalues, completing the lower-triangular case without repeating an elimination argument.

Exercise 5.1.29: Equal row sums produce an obvious eigenvector

Let $\mathbf{1} = (1, \ldots, 1)^T \neq 0$. Entry i of $A\mathbf{1}$ is the sum of row i, which is s by hypothesis. Thus $A\mathbf{1} = (s, \ldots, s)^T = s\mathbf{1}$.

Conclusion. s is an eigenvalue, with eigenvector $\mathbf{1}$.

Exercise 5.1.30: Equal column sums use the transpose

The column sums of A are the row sums of A^T. Exercise 29 therefore makes s an eigenvalue of A^T. By Exercise 27 it is also an eigenvalue of A. The all-ones vector is an eigenvector of A^T here; it need not itself be an eigenvector of A, so the transpose step is substantive.

Exercise 5.1.31: Read eigenvalues from a reflection

Every nonzero vector on the reflecting line L is fixed, so $\lambda = 1$ is an eigenvalue and $E_1 = L$. A vector perpendicular to L is reversed, so -1 is the other eigenvalue, with $E_{-1} = L^{\perp}$. No vector outside L can be fixed: decompose it into its along-line and perpendicular parts; the reflection reverses the latter, so equality requires that part to be zero.

Exercise 5.1.32: Read a fixed eigenspace from a rotation

Every vector on the rotation axis L is fixed, so 1 is an eigenvalue. For a nontrivial rotation, the perpendicular-plane part of a nonzero vector is rotated and cannot remain unchanged, giving $E_1 = L$. If the angle is a whole multiple of 2π, the transformation is instead the identity and $E_1 = \mathbb{R}^3$. This qualification handles the degenerate zero-angle rotation, which the wording does not exclude.

Exercise 5.1.33: Combine two eigenvector solutions of a recurrence

(a) Shift the index in the definition.

The next term is $x_{k+1} = c_1\lambda^{k+1}u + c_2\mu^{k+1}v$.

(b) Apply A to the stated expression.

Using $Au = \lambda u$, $Av = \mu v$, and linearity,

$$Ax_k = c_1\lambda^k Au + c_2\mu^k Av = c_1\lambda^{k+1}u + c_2\mu^{k+1}v = x_{k+1}.$$

Thus the sequence satisfies the difference equation at every nonnegative index. Zeroth powers are taken as 1, including for a zero eigenvalue, as required by the initial term.

Exercise 5.1.34: Try expanding the initial state in eigenvectors

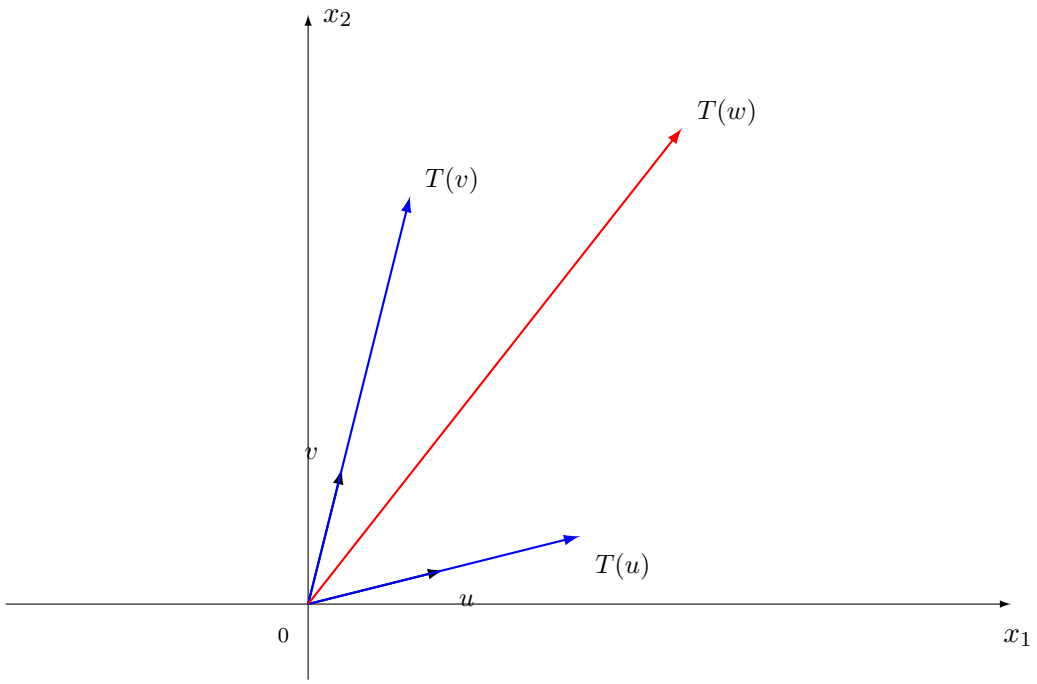
Find eigenvectors v_j of A and try solving $x_0 = \sum_j c_j v_j$ for coefficients c_j. If this is possible, set $x_k = \sum_j c_j \lambda_j^k v_j$, using the corresponding eigenvalues. At $k = 0$ it equals the prescribed state, and the calculation in Exercise 33 proves $Ax_k = x_{k+1}$ term by term. This method works whenever x_0 lies in the span of the available eigenvectors; in particular it works for every x_0 when there is an eigenvector basis. Not every matrix has such a basis, so this is a useful construction rather than an unconditional guarantee.

Exercise 5.1.35: Sketch the action on two eigendirections

The eigenvector equations give $T(u) = 2u$ and $T(v) = 3v$. Since $w = u + v$, linearity gives $T(w) = 2u + 3v$. Thus double the arrow along u, triple the arrow along v, and add those two new

arrows head to tail.

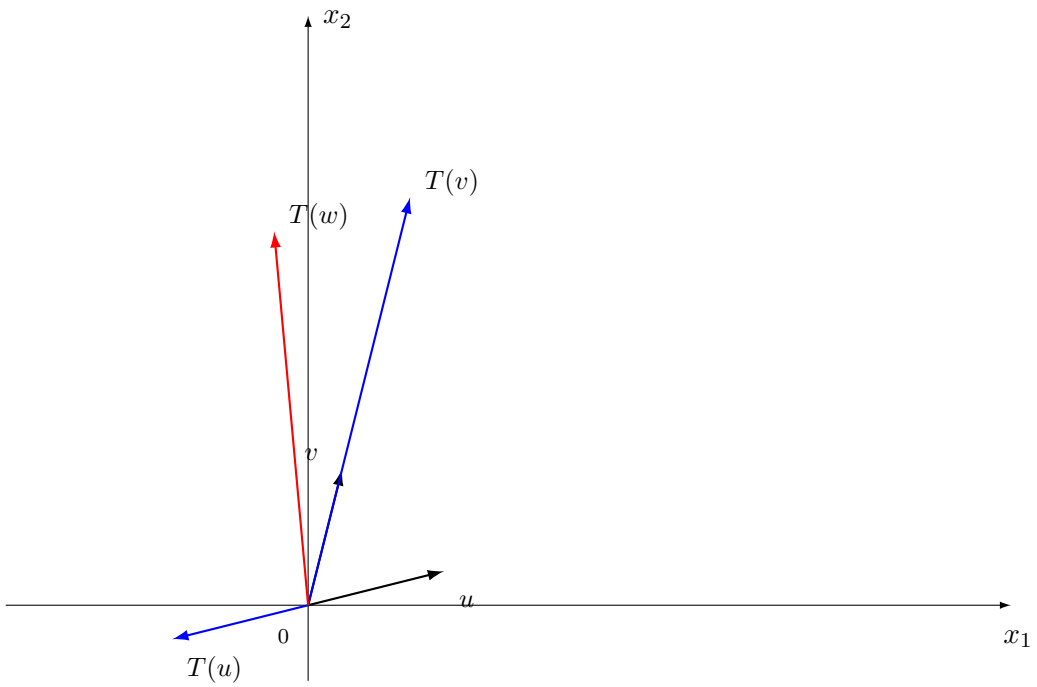
The black arrows reproduce the two source directions schematically; blue shows the separate images and red their sum. No numerical coordinates are assumed for the textbook's unlabeled vectors.



Exercise 5.1.36: Sketch the action on two eigendirections

The eigenvector equations give $T(u) = -1u$ and $T(v) = 3v$. Since $w = u + v$, linearity gives $T(w) = -1u + 3v$. Reverse the arrow u without changing its length, triple the arrow v, and add the resulting arrows head to tail.

The black arrows reproduce the two source directions schematically; blue shows the separate images and red their sum. No numerical coordinates are assumed for the textbook's unlabeled vectors.



Exercise 5.1.37: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A = \begin{bmatrix} 12 & 1 & 4 \\ 2 & 11 & 4 \\ 1 & 3 & 7 \end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A - \lambda I) = -(\lambda - 15)(\lambda - 10)(\lambda - 5).$$

Eigenvalue 5**.**

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_5 = \left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 5v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 10**.**

$$\operatorname{rref}(A-(10)I)=\begin{bmatrix}1&0&3\\0&1&-2\\0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{10}=\left\{\begin{bmatrix}-3\\2\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=10v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 15**.**

$$\operatorname{rref}(A-(15)I)=\begin{bmatrix}1&0&-2\\0&1&-2\\0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{15}=\left\{\begin{bmatrix}2\\2\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=15v$ by exact substitution. The eigenspace dimension is 1.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.38: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A=\begin{bmatrix}5&-2&2&-4\\7&-4&2&-4\\4&-4&2&0\\3&-1&1&-3\end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A-\lambda I)=(\lambda-2)\,(\lambda-1)\,(\lambda+1)\,(\lambda+2)\,.$$

Eigenvalue -2**.**

$$\operatorname{rref}(A-(-2)I)=\begin{bmatrix}1&0&0&0\\0&1&-1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-2}=\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=-2v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue -1**.**

$$\operatorname{rref}(A-(-1)I)=\begin{bmatrix}1&0&0&-1\\0&1&0&-1\\0&0&1&0\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-1}=\left\{\begin{bmatrix}1\\1\\0\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=-1v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 1**.**

$$\operatorname{rref}(A-(1)I)=\begin{bmatrix}1&0&0&-2\\0&1&0&-2\\0&0&1&0\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{1}=\left\{\begin{bmatrix}2\\2\\0\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=1v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 2**.**

$$\operatorname{rref}(A-(2)I)=\begin{bmatrix}1&0&0&-2\\0&1&0&-2\\0&0&1&-1\\0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_2=\left\{\begin{bmatrix}2\\2\\1\\1\end{bmatrix}\right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av=2v$ by exact substitution. The eigenspace dimension is 1.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.39: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A=\begin{bmatrix}12&-90&30&30&30\\8&-49&15&15&15\\16&-52&12&0&20\\0&-30&10&22&10\\8&-41&15&15&7\end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A-\lambda I)=-\left(\lambda-12\right)^2\left(\lambda+4\right)\left(\lambda+8\right)^2.$$

Eigenvalue -8**.**

$$\operatorname{rref}(A-(-8)I)=\begin{bmatrix}1&0&0&-3&0\\0&1&0&-\frac{3}{2}&0\\0&0&1&-\frac{3}{2}&1\\0&0&0&0&0\\0&0&0&0&0\end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-8}=\left\{\begin{bmatrix}6\\3\\3\\2\\0\end{bmatrix},\ \begin{bmatrix}0\\0\\-1\\0\\1\end{bmatrix}\right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -8v$ by exact substitution. The eigenspace dimension is 2.

Eigenvalue -4**.**

$$\operatorname{rref}(A - (-4)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-4} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Its 1 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -4v$ by exact substitution. The eigenspace dimension is 1.

Eigenvalue 12**.**

$$\operatorname{rref}(A - (12)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{12} = \left\{ \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 12v$ by exact substitution. The eigenspace dimension is 2.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

Exercise 5.1.40: Compute eigenvalues, then reduce each shifted matrix

For the supplied matrix

$$A = \begin{bmatrix} -23 & 57 & -9 & -15 & -59 \\ -10 & 12 & -10 & 2 & -22 \\ 11 & 5 & -3 & -19 & -15 \\ -27 & 31 & -27 & 25 & -37 \\ -5 & -15 & -5 & 1 & 31 \end{bmatrix},$$

exact computation gives the factored characteristic polynomial

$$\det(A - \lambda I) = -(\lambda - 42)^2 (\lambda + 14)^3.$$

Eigenvalue -14**.**

$$\operatorname{rref}(A - (-14)I) = \begin{bmatrix} 1 & 0 & 1 & 0 & -\frac{10}{3} \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 & -\frac{5}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{-14} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 10 \\ 6 \\ 0 \\ 5 \\ 3 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = -14v$ by exact substitution. The eigenspace dimension is 2.

Eigenvalue 42**.**

$$\operatorname{rref}(A - (42)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{1}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{3}{5} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Assigning the free coordinates in turn gives the eigenspace basis

$$\mathcal{B}_{42} = \left\{ \begin{bmatrix} 0 \\ 1 \\ -2 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -1 \\ -3 \\ 0 \\ 5 \end{bmatrix} \right\}.$$

Its 2 vectors are independent, span every solution of the reduced system, and satisfy the original $Av = 42v$ by exact substitution. The eigenspace dimension is 2.

The listed roots exhaust the characteristic polynomial, so no eigenvalues are omitted. A repeated root need not supply the same number of independent eigenvectors as its multiplicity; the null-space calculation, not multiplicity alone, determines the basis.

5.2 The characteristic equation

We use the book's convention $p(\lambda) = \det(A - \lambda I)$. Some software instead reports $\det(\lambda I - A)$; they differ by $(-1)^n$ and have the same zeros. An eigenvalue is a zero of p, and its algebraic multiplicity is its multiplicity as a polynomial root. That multiplicity need not equal the dimension of its eigenspace. For a triangular matrix, the characteristic determinant is the product of the shifted diagonal entries. For a general matrix, use the full determinant rather than multiplying the original diagonal entries.

Practice problem 5.2.P1: A characteristic equation with no real roots

For the given matrix,

$$p(\lambda) = \det \begin{bmatrix} 1-\lambda & -4 \\ 4 & 2-\lambda \end{bmatrix} = (1-\lambda)(2-\lambda) + 16 = \lambda^2 - 3\lambda + 18.$$

Its discriminant is $9 - 72 = -63 < 0$, so the characteristic equation has no real root and A has no real eigenvalue. If complex scalars are allowed, its eigenvalues are $(3 \pm i\sqrt{63})/2$.

Exercise 5.2.1: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (2-\lambda)(2-\lambda) - (7)(7) = \lambda^2 - 4\lambda - 45.$$

Factoring gives $p(\lambda) = (\lambda - 9)\,(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 9$. Each listed root has multiplicity one.

Exercise 5.2.2: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & -1 \\ 6 & 1 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(1-\lambda) - (-1)(6) = \lambda^2 + 3\lambda + 2.$$

Factoring gives $p(\lambda) = (\lambda + 1)\,(\lambda + 2)$. Its real roots, hence the eigenvalues, are $-2,\ -1$. Each listed root has multiplicity one.

Exercise 5.2.3: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & 2 \\ 6 & 7 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(7-\lambda) - (2)(6) = \lambda^2 - 3\lambda - 40.$$

Factoring gives $p(\lambda) = (\lambda - 8)\,(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 8$. Each listed root has multiplicity one.

Exercise 5.2.4: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 8 & 2 \\ 3 & 3 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (8 - \lambda)(3 - \lambda) - (2)(3) = \lambda^2 - 11\lambda + 18.$$

Factoring gives $p(\lambda) = (\lambda - 9)(\lambda - 2)$. Its real roots, hence the eigenvalues, are $2,\ 9$. Each listed root has multiplicity one.

Exercise 5.2.5: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (8 - \lambda)(8 - \lambda) - (4)(4) = \lambda^2 - 16\lambda + 48.$$

Factoring gives $p(\lambda) = (\lambda - 12)(\lambda - 4)$. Its real roots, hence the eigenvalues, are $4,\ 12$. Each listed root has multiplicity one.

Exercise 5.2.6: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 9 & -2 \\ 2 & 5 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (9 - \lambda)(5 - \lambda) - (-2)(2) = \lambda^2 - 14\lambda + 49.$$

Factoring gives $p(\lambda) = (\lambda - 7)^2$. Its real roots, hence the eigenvalues, are 7. The root 7 has multiplicity two.

Exercise 5.2.7: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 5 & 3 \\ -4 & 4 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (5 - \lambda)(4 - \lambda) - (3)(-4) = \lambda^2 - 9\lambda + 32.$$

The discriminant is $(-9)^2 - 4(32) = -47 < 0$. Consequently there are no real eigenvalues. The two complex roots, if desired, are $(9 \pm i\sqrt{47})/2$.

Exercise 5.2.8: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & 3 \\ 2 & 1 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(1 - \lambda) - (3)(2) = \lambda^2 + 3\lambda - 10.$$

Factoring gives $p(\lambda) = (\lambda - 2)(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 2$. Each listed root has multiplicity one.

Exercise 5.2.9: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}4-\lambda & 0 & -1\\ 0 & 4-\lambda & -1\\ 1 & 0 & 2-\lambda\end{bmatrix}.$$

Expand down column two; only its middle entry is nonzero. Thus

$$p(\lambda)=(4-\lambda)[(4-\lambda)(2-\lambda)+1].$$

Multiplying the factors gives

$$\boxed{p(\lambda)=-\lambda^3+10\lambda^2-33\lambda+36}.$$

A useful factored check is $p(\lambda)=-(\lambda-4)(\lambda-3)^2$. Expanding this expression reproduces the same cubic and, at $\lambda=0$, the original determinant.

Exercise 5.2.10: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}3-\lambda & 1 & 1\\ 0 & 5-\lambda & 0\\ -2 & 0 & 7-\lambda\end{bmatrix}.$$

Expand across row two; only its middle entry is nonzero. Thus

$$p(\lambda)=(5-\lambda)[(3-\lambda)(7-\lambda)+2].$$

Multiplying the factors gives

$$\boxed{p(\lambda)=-\lambda^3+15\lambda^2-73\lambda+115}.$$

A useful factored check is $p(\lambda)=-(\lambda-5)(\lambda^2-10\lambda+23)$. Expanding this expression reproduces the same cubic and, at $\lambda=0$, the original determinant.

Exercise 5.2.11: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}3-\lambda & 0 & 0\\ 2 & 1-\lambda & 4\\ 1 & 0 & 4-\lambda\end{bmatrix}.$$

Expand down column two; only its middle entry is nonzero, and the remaining minor is triangular. Thus

$$p(\lambda)=(1-\lambda)(3-\lambda)(4-\lambda).$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 8\lambda^2 - 19\lambda + 12}.$$

A useful factored check is $p(\lambda) = -(\lambda - 4)(\lambda - 3)(\lambda - 1)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.12: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} -\lambda - 1 & 0 & 2 \\ 3 & 1 - \lambda & 0 \\ 0 & 1 & 2 - \lambda \end{bmatrix}.$$

Expand across row one. The middle term is zero; the last cofactor contributes $2(3) = 6$. Thus

$$p(\lambda) = (-1 - \lambda)(1 - \lambda)(2 - \lambda) + 6.$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 2\lambda^2 + \lambda + 4}.$$

Factoring is not required here. As checks, the leading term is $-\lambda^3$ under our convention, and substituting $\lambda = 0$ gives $\det A = 4$.

Exercise 5.2.13: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} 6 - \lambda & -2 & 0 \\ -2 & 9 - \lambda & 0 \\ 5 & 8 & 3 - \lambda \end{bmatrix}.$$

Expand down column three; only its last entry is nonzero. Thus

$$p(\lambda) = (3 - \lambda)[(6 - \lambda)(9 - \lambda) - 4].$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 18\lambda^2 - 95\lambda + 150}.$$

A useful factored check is $p(\lambda) = -(\lambda - 10)(\lambda - 5)(\lambda - 3)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.14: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 0 & -1 \\ -1 & -\lambda & 4 \\ 0 & 2 & 3-\lambda \end{bmatrix}.$$

Expand across row one. The first minor gives $(-\lambda)(3-\lambda) - 8$, and the last cofactor contributes $(-1)(-2) = 2$. Thus

$$p(\lambda) = (4-\lambda)[(-\lambda)(3-\lambda) - 8] + 2.$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 7\lambda^2 - 4\lambda - 30}.$$

A useful factored check is $p(\lambda) = -(\lambda - 5)(\lambda^2 - 2\lambda - 6)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.15: Read triangular eigenvalues with repetition

This matrix is upper triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = (\lambda - 5)^2 (\lambda - 3)(\lambda - 2).$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list 2, 3, 5, 5. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.16: Read triangular eigenvalues with repetition

This matrix is lower triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = (\lambda - 6)(\lambda - 3)(\lambda - 2)(\lambda + 5).$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list -5, 2, 3, 6. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.17: Read triangular eigenvalues with repetition

This matrix is lower triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = -\lambda (\lambda - 3)^2 (\lambda - 1)^2.$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list 0, 1, 1, 3, 3. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.18: Force a two-dimensional eigenspace for four

Subtract $4I$ to obtain

$$A - 4I = \begin{bmatrix} 0 & 2 & 3 & 3 \\ 0 & -2 & h & 3 \\ 0 & 0 & 0 & 14 \\ 0 & 0 & 0 & -2 \end{bmatrix}.$$

The last row of $(A - 4I)x = 0$ forces $x_4 = 0$. The first row then gives $2x_2 + 3x_3 = 0$, so $x_2 = -3x_3/2$. Substituting into the second gives $(h+3)x_3 = 0$. The coordinate x_1 is always free. For the eigenspace to have a second free coordinate, we need $h + 3 = 0$. At $h = -3$ its basis can be chosen as

$$(1, 0, 0, 0)^T, \qquad (0, -3, 2, 0)^T.$$

For any other h, $x_3 = x_2 = x_4 = 0$, leaving only one free coordinate.

Conclusion. $\boxed{h = -3}$.

Exercise 5.2.19: The constant coefficient is the determinant

The supplied factorization is $p(\lambda) = \prod_{j=1}^{n}(\lambda_j - \lambda)$, with eigenvalues repeated according to multiplicity. Substitute $\lambda = 0$:

$$\det A = \det(A - 0I) = p(0) = \lambda_1 \cdots \lambda_n.$$

No sign change is needed under this convention; using $\det(\lambda I - A)$ would introduce the corresponding constant-term sign. The same evaluation works over complex numbers when nonreal roots are present.

Exercise 5.2.20: Transpose preserves the whole characteristic polynomial

Since $I^T = I$ and transposition commutes with subtraction, $A^T - \lambda I = (A - \lambda I)^T$. The determinant of a transpose equals the original determinant. Hence

$$\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$$

for every λ. Thus the polynomials are identical, so the roots agree with their algebraic multiplicities, not merely as distinct sets.

Exercise 5.2.21: Determinants and the sign of a root

(a) **False.** For $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\det A = -1$, but the product of the diagonal entries is 0. The diagonal-product rule is valid for triangular matrices.

(b) **False.** A row interchange negates the determinant; scaling a row by c multiplies it by c. Only row replacement preserves it unchanged in general.

(c) **True.** This is the determinant product law for square matrices.

(d) **False.** A factor $\lambda + 5$ vanishes at $\lambda = -5$, not 5. For example $A = [-5]$ has that factor up to the harmless leading sign, but has only eigenvalue -5.

Exercise 5.2.22: Volume, transpose, multiplicity, and row operations

(a) **False as stated.** The geometric volume is $|\det A|$. The determinant can be negative because it also records orientation.

(b) **False.** $\det A^T = \det A$, with no factor -1. The identity matrix gives 1, not -1.

(c) **True.** This is the definition of algebraic multiplicity.

(d) **False.** Start with $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, whose eigenvalues are $2, 0$. Replacing row two by row two minus row one gives $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, whose eigenvalues are $1, 0$. Row replacement preserves the determinant, not every eigenvalue.

Exercise 5.2.23: Interchanging two factors gives a similar matrix

Assume $A = QR$ with Q invertible. Then

$$Q^{-1}AQ = Q^{-1}(QR)Q = (Q^{-1}Q)RQ = RQ = A_1.$$

Thus $A_1 = Q^{-1}AQ$, precisely the definition of similarity. Only Q needs to be invertible; R need not be. This is why interchanging the factors in the QR iteration preserves eigenvalues.

Exercise 5.2.24: Similarity preserves determinants

If $B = P^{-1}AP$ with P invertible, the product law gives

$$\det B = \det(P^{-1}) \det A \det P.$$

Since $\det(P^{-1}) \det P = \det I = 1$, the two outer factors cancel and $\det B = \det A$. This proof also covers singular A: only P, not A, is required to have an inverse.

Exercise 5.2.25: Resolve a probability vector into two eigenvectors

(a) Find a second eigendirection.

The stated $v_1 = (3/7, 4/7)^T$ satisfies $Av_1 = v_1$. For $v_2 = (1, -1)^T$, multiplication gives $Av_2 = (0.3, -0.3)^T = 0.3v_2$. The vectors correspond to distinct eigenvalues 1 and 0.3, so they are independent and form a basis of $\mathbb{R}^2$.

(b) Expand the initial vector.

$$x_0 - v_1 = (1/2 - 3/7, 1/2 - 4/7)^T = (1/14, -1/14)^T = \tfrac{1}{14}v_2.$$

Thus $x_0 = v_1 + (1/14)v_2$.

(c) Use the eigenvalue powers.

Direct multiplication gives $x_1 = (0.45, 0.55)^T$ and $x_2 = (0.435, 0.565)^T$. For every $k \geq 0$,

$$x_k = A^k x_0 = v_1 + \frac{1}{14}(0.3)^k v_2.$$

Because $|0.3| < 1$, the scalar factor tends to zero, so $x_k \to v_1$. This proves convergence, rather than merely inferring it from two similar iterates.

Exercise 5.2.26: Derive the two-by-two determinant from elimination

Case 1: $a \neq 0$**.**

Replace row two by row two minus (c/a) times row one. This operation does not change the determinant, and it produces $\begin{bmatrix} a & b \\ 0 & d - cb/a \end{bmatrix}$. The triangular formula gives $a(d - cb/a) = ad - bc$.

Case 2: $a = 0$**.**

If also $c = 0$, the first column is zero, so the determinant is zero, agreeing with $ad - bc$. If $c \neq 0$, interchange rows to produce $\begin{bmatrix} c & d \\ 0 & b \end{bmatrix}$. Its determinant is cb, and one interchange reverses the sign, so the original determinant is $-cb = ad - bc$. These cases include all possible values of a, c.

Exercise 5.2.27: Three eigendirections determine a steady-state limit

(a) Check the three given images.

With the matrix in the exercise, direct multiplication gives

$$Av_1 = (0.3, 0.6, 0.1)^T = v_1,$$

$$Av_2 = (0.5, -1.5, 1)^T = \tfrac{1}{2}v_2, \qquad Av_3 = (-0.2, 0, 0.2)^T = \tfrac{1}{5}v_3.$$

All three vectors are nonzero, so these really are eigenvector equations.

(b) Use both a basis and the coordinate sum.

The eigenvalues $1, 1/2, 1/5$ are distinct, so the three eigenvectors are independent and form a basis of $\mathbb{R}^3$. Thus any x_0 has unique coefficients c_j in $x_0 = c_1v_1 + c_2v_2 + c_3v_3$. For $w = (1, 1, 1)^T$, the probability hypothesis gives $w^T x_0 = 1$. But $w^T v_1 = 1$, $w^T v_2 = 0$, and $w^T v_3 = 0$. Applying w^T to the expansion therefore yields $1 = c_1$.

(c) Let the contracting components vanish.

$$x_k = A^k x_0 = v_1 + c_2(1/2)^k v_2 + c_3(1/5)^k v_3.$$

The two powers tend to zero, so $x_k \to v_1$. The two coefficients may depend on the initial distribution, but the limit does not, because its coefficient was fixed by total probability one.

Exercise 5.2.28: Reproducible random examples and the transpose distinction

To make the experiment reproducible, draw upper-triangular entries uniformly from the integers $-3, \ldots, 4$ with pseudorandom seed 52028, setting entries below the diagonal to zero. Successive draws of sizes four and five give the following valid random integer examples. The triangular choice makes exact checks of every multiplicity transparent.

$$A_4 = \begin{bmatrix} -1 & 0 & 0 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Exact calculation gives the same polynomial for A_4 and A_4^T:

$$p(\lambda) = (\lambda - 2)(\lambda - 1)^2(\lambda + 1).$$

Its eigenvalues, with multiplicity, are $-1,\ 1,\ 1,\ 2$. However e_1 is an eigenvector of A_4 with eigenvalue -1, whereas $A_4^T e_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 2 \end{bmatrix}$ is not a multiple of e_1. Thus their eigenvectors are not the same.

$$A_5 = \begin{bmatrix} 1 & 3 & 2 & 3 & 0 \\ 0 & -3 & -3 & 4 & -3 \\ 0 & 0 & 4 & -3 & 4 \\ 0 & 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Exact calculation gives the same polynomial for A_5 and A_5^T:

$$p(\lambda) = -\lambda(\lambda - 4)^2(\lambda - 1)(\lambda + 3).$$

Its eigenvalues, with multiplicity, are $-3,\ 0,\ 1,\ 4,\ 4$. However e_1 is an eigenvector of A_5 with eigenvalue 1, whereas $A_5^T e_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 3 \\ 0 \end{bmatrix}$ is not a multiple of e_1. Thus their eigenvectors are not the same.

The equality of characteristic polynomials is guaranteed in general by Exercise 20; equal eigenspaces are not guaranteed. If a particular matrix is symmetric, its transpose is identical and its eigenspaces do agree, so one must not claim they always differ.

Exercise 5.2.29: Compare a determinant with pivot and eigenvalue products

One reproducible choice of a random integer matrix is

$$A = \begin{bmatrix} 2 & 3 & -3 & -2 \\ 5 & -3 & 0 & 3 \\ -3 & -4 & 0 & 2 \\ 1 & 1 & 4 & 1 \end{bmatrix}.$$

It was selected from pseudorandom entries in $-4, \dots, 5$ (seed 52029) with nonzero pivots and real eigenvalues, so every requested quantity can be displayed in real decimal form.

(a) Eliminate without row scaling.

First use $R_2 \leftarrow R_2 - (5/2)R_1$, $R_3 \leftarrow R_3 + (3/2)R_1$, and $R_4 \leftarrow R_4 - (1/2)R_1$. Continue replacing rows to clear entries below the second and third pivots. No swaps or row scalings are needed. The resulting echelon matrix is

$$U = \begin{bmatrix} 2 & 3 & -3 & -2 \\ 0 & -21/2 & 15/2 & 8 \\ 0 & 0 & -29/7 & -13/21 \\ 0 & 0 & 0 & 74/87 \end{bmatrix}.$$

Row replacement preserves the determinant, so

$$\det A = 2(-21/2)(-29/7)(74/87) = 74.$$

(b) Compute and multiply the eigenvalues.

Exact computation gives $p(\lambda) = \lambda^4 - 40\lambda^2 - 22\lambda + 74$. Its four real roots, computed with guard digits, are approximately

$$-5.8358727568, \quad -1.7435586500, \quad 1.1271321671, \quad 6.4522992397.$$

Their product using unrounded roots is 74 to numerical precision, as also follows exactly from the polynomial's constant coefficient.

(c) Give the requested four-decimal report and comparison.

The pivots are $2.0000, -10.5000, -4.1429, 0.8506$; the eigenvalues are $-5.8359, -1.7436, 1.1271, 6.4523$. The direct matrix determinant is exactly 74, agreeing with both products before rounding. Products formed only from the four-place display may differ slightly because the displayed numbers have been rounded.

Exercise 5.2.30: A small entry change can destroy two real eigenvalues

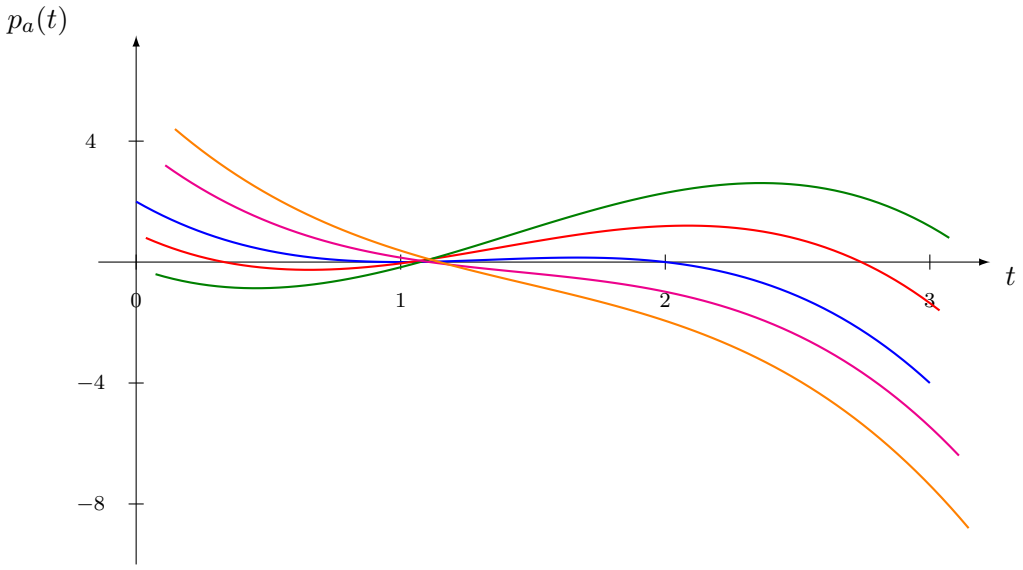
For the stated matrix, expansion of the shifted determinant gives

$$p_a(t) = -t^3 + 4t^2 + (379 - 12a)t + (12a - 382) = -(t-1)(t^2 - 3t + 12a - 382).$$

Thus 1 is always an eigenvalue, and the other two are $(3 \pm \sqrt{1537 - 48a})/2$. Substituting each requested value gives

a	$p_a(t)$	eigenvalues (approximately)
32	$-t^3 + 4t^2 - 5t + 2$	$1, 1, 2$
31.9	$-t^3 + 4t^2 - 3.8t + 0.8$	$0.295841, 1, 2.704159$
31.8	$-t^3 + 4t^2 - 2.6t - 0.4$	$-0.127882, 1, 3.127882$
32.1	$-t^3 + 4t^2 - 6.2t + 3.2$	$1, 1.5 \pm 0.974679i$
32.2	$-t^3 + 4t^2 - 7.4t + 4.4$	$1, 1.5 \pm 1.466288i$

The complex entries in the last two rows are the two nonreal eigenvalues; the only real eigenvalue in those cases is 1. Here are all five graphs on the requested common interval $0 \le t \le 3$.



$a = 32$, $a = 31.9$, $a = 31.8$, $a = 32.1$, $a = 32.2$.

Every graph passes through $(1, 0)$. At $a = 32$, the root 1 is double, so its graph touches the axis there and also crosses at 2. For $a = 31.9$, the three real crossings separate. For $a = 31.8$, two real roots move outside the plotted window; the graph within $[0, 3]$ shows only the crossing at 1, even though all three roots remain real. For $a = 32.1, 32.2$, the quadratic discriminant is negative, so those two roots are genuinely nonreal, not merely off screen. The coalescence threshold for that pair is $a = 1537/48 \approx 32.020833$.

Conclusion. The exact factorization explains the changes. A graph on a limited window alone cannot distinguish a root outside the window from a nonreal root; the discriminant and computed values supply that distinction.

5.3 Diagonalization

In this section diagonalization is over $\mathbb{R}$: only real eigenvectors can form the required basis of $\mathbb{R}^n$. Diagonalization is a change to an eigenvector coordinate system. If the columns of an invertible matrix P are eigenvectors and the corresponding eigenvalues are placed in the same order on the diagonal of D, then $AP = PD$ and $A = PDP^{-1}$. The requirement is *enough independent eigenvectors*, not merely enough vectors or roots of the characteristic polynomial.

The algebraic multiplicity of an eigenvalue is its number of appearances as a characteristic-polynomial root. Its geometric multiplicity is the dimension of its eigenspace. These are different notions. A repeated root must be checked by solving a homogeneous system; it neither guarantees nor forbids diagonalization.

Practice problem 5.3.P1: A high power from two eigenvectors

The characteristic polynomial is $(t-2)(t-1)$. Solving the two homogeneous systems gives eigenvectors $(3,2)$ for 2 and $(1,1)$ for 1. Thus

$$P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}.$$

In $(PDP^{-1})^8$, every adjacent $P^{-1}P$ cancels to I, giving $A^8 = PD^8P^{-1}$. Since $D^8 = \operatorname{diag}(256, 1)$,

$$A^8 = \begin{bmatrix} 766 & -765 \\ 510 & -509 \end{bmatrix}.$$

This result was also checked by eightfold exact matrix multiplication.

Practice problem 5.3.P2: Use the supplied eigenvectors

Multiplying by $A = \left[\begin{smallmatrix} -3 & 12 \\ -2 & 7 \end{smallmatrix}\right]$ gives $A(3,1)^T = (3,1)^T$ and $A(2,1)^T = 3(2,1)^T$. The eigenvalues are therefore 1 and 3, in that order. Take $P = \left[\begin{smallmatrix} 3 & 2 \\ 1 & 1 \end{smallmatrix}\right]$ and $D = \operatorname{diag}(1,3)$. Since $\det P = 1 \neq 0$ and $AP = PD$, we obtain $A = PDP^{-1}$.

Practice problem 5.3.P3: Count eigenspace dimensions

Yes. The eigenspace for 3 supplies two independent vectors. The distinct eigenvalues 5 and -2 each supply at least one nonzero eigenvector. Combining bases from distinct eigenspaces gives independent vectors, hence at least $2 + 1 + 1 = 4$ independent vectors in $\mathbb{R}^4$. There cannot be more than four. Thus they form an eigenvector basis and the matrix is diagonalizable.

Exercise 5.3.1: Cancel the inner change-of-basis factors

What we must show. Compute A^4 from the given $A = PDP^{-1}$. Associativity gives $(PDP^{-1})^4 = PD(P^{-1}P)D(P^{-1}P)D(P^{-1}P)DP^{-1} = PD^4P^{-1}$. Here

$$P = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}, \quad D^4 = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}.$$

Multiplying these three displayed matrices gives

$$A^4 = \begin{bmatrix} 226 & -525 \\ 90 & -209 \end{bmatrix}.$$

The calculation involves taking fourth powers of the diagonal entries of D, not taking fourth powers entry by entry in A.

Exercise 5.3.2: Cancel the inner change-of-basis factors

What we must show. Compute A^4 from the given $A = PDP^{-1}$. Associativity gives $(PDP^{-1})^4 = PD(P^{-1}P)D(P^{-1}P)D(P^{-1}P)DP^{-1} = PD^4P^{-1}$. Here

$$P = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad D^4 = \begin{bmatrix} 1 & 0 \\ 0 & 81 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}.$$

Multiplying these three displayed matrices gives

$$A^4 = \begin{bmatrix} 321 & -160 \\ 480 & -239 \end{bmatrix}.$$

The calculation involves taking fourth powers of the diagonal entries of D, not taking fourth powers entry by entry in A.

Exercise 5.3.3: An arbitrary power of a parameterized triangular matrix

Use the supplied factorization with $P = \left[\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right]$, $D = \operatorname{diag}(a, b)$, and $P^{-1} = \left[\begin{smallmatrix} 1 & 0 \\ -2 & 1 \end{smallmatrix}\right]$. For every positive integer k,

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a^k & 0 \\ 0 & b^k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} a^k & 0 \\ 2(a^k - b^k) & b^k \end{bmatrix}.$$

There is no division by $a - b$, so this remains true when $a = b$ or one parameter is zero.

Exercise 5.3.4: An arbitrary power with negative eigenvalues

Put $\alpha = (-3)^k$ and $\beta = (-2)^k$. The supplied matrices give

$$A^k = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} -3\alpha + 4\beta & 6\alpha - 6\beta \\ -2\alpha + 2\beta & 4\alpha - 3\beta \end{bmatrix}.$$

The signs in $(-3)^k$ and $(-2)^k$ depend on the parity of k and must remain inside parentheses. Setting $k = 1$ recovers the original matrix, a useful quick check.

Exercise 5.3.5: Read eigenspaces from an existing factorization

Read matching columns and diagonal entries in the supplied factorization. If p_j is column j of P, then $Ap_j = d_{jj}p_j$. Here

$$P = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

For $\lambda = 2$, a basis is $\{\begin{bmatrix}-1\\1\\-1\end{bmatrix}\}$. For $\lambda = 3$, a basis is $\{\begin{bmatrix}1\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\-1\\1\end{bmatrix}\}$. Because P is invertible, its columns are independent. All columns at a repeated diagonal value are needed; retaining only one of them would lose an eigenspace direction.

Exercise 5.3.6: Read eigenspaces from an existing factorization

Read matching columns and diagonal entries in the supplied factorization. If p_j is column j of P, then $Ap_j = d_{jj}p_j$. Here

$$P = \begin{bmatrix}3 & 0 & -1\\0 & 1 & -3\\1 & 0 & 0\end{bmatrix}, \qquad D = \begin{bmatrix}3 & 0 & 0\\0 & 4 & 0\\0 & 0 & 3\end{bmatrix}.$$

For $\lambda = 3$, a basis is $\{\begin{bmatrix}3\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\-3\\0\end{bmatrix}\}$. For $\lambda = 4$, a basis is $\{\begin{bmatrix}0\\1\\0\end{bmatrix}\}$. Because P is invertible, its columns are independent. All columns at a repeated diagonal value are needed; retaining only one of them would lose an eigenspace direction.

Exercise 5.3.7: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix}1 & 0\\6 & -1\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-1)(t+1).$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The exact checks are $\det P = -1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.8: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 2 \\ 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 3)^2.$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 1 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 2, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.9: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)^2.$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 2, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.10: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t+2).$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & -\frac{3}{4} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.11: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 3 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t+1)^2.$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -\frac{1}{3} \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & -1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 6 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.12: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)\,(t - 2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \text{Span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 1 & 1\\ -1 & 0 & 1\\ 0 & -1 & 1\end{bmatrix}, \qquad D = \begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 5\end{bmatrix}.$$

The exact checks are $\det P = 3 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.13: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix}2 & 2 & -1\\ 1 & 3 & -1\\ -1 & -2 & 2\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-1)^2.$$

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (1)I) = \begin{bmatrix}1 & 2 & -1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \text{Span}\left\{\begin{bmatrix}2\\-1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = -4 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.14: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 1 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 3)^2 (t - 2).$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 1. Solve $(A-\lambda I)x=0$; row reduction gives

$$\operatorname{rref}(A-(2)I)=\begin{bmatrix}1&1&0\\0&0&1\\0&0&0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2=\operatorname{Span}\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av=(2)v$ for each listed vector.

Find the eigenspace for $\lambda=3$.

The algebraic multiplicity is 2. Solve $(A-\lambda I)x=0$; row reduction gives

$$\operatorname{rref}(A-(3)I)=\begin{bmatrix}1&0&2\\0&0&0\\0&0&0\end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3=\operatorname{Span}\left\{\begin{bmatrix}0\\1\\0\end{bmatrix},\ \begin{bmatrix}2\\0\\-1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av=(3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1+2=3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P=\begin{bmatrix}1&0&2\\-1&1&0\\0&0&-1\end{bmatrix},\qquad D=\begin{bmatrix}2&0&0\\0&3&0\\0&0&3\end{bmatrix}.$$

The exact checks are $\det P=-1\neq 0$ and $AP=PD$. Multiplying the latter by P^{-1} on the right proves $A=PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.15: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A=\begin{bmatrix}0&-1&-1\\1&2&1\\-1&-1&0\end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\,(t-1)^2\,.$$

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (0)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_0 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (1)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 2 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The exact checks are $\det P = 1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.16: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & -2 \\ 1 & 3 & 1 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\left(t^2 - 7t + 16\right).$$

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (0)I) = \begin{bmatrix} 1 & 0 & -11 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_0 = \operatorname{Span}\left\{ \begin{bmatrix} 11 \\ -4 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 3, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction. The remaining quadratic factor is $t^2 - 7t + 16$, whose discriminant is $49 - 64 = -15 < 0$. Thus its other eigenvalues are nonreal. They do not supply real eigenvectors for a basis of $\mathbb{R}^3$; complex diagonalization is a different question introduced in Section 5.5.

Exercise 5.3.17: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-2)^3 .$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} .$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (2)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 = 1$. This is smaller than 3, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.18: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & -2 & -2 \\ 3 & -3 & -2 \\ 2 & -2 & -2 \end{bmatrix} .$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = t\,(t+1)\,(t+2) .$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} .$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \text{Span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (-1)I) = \begin{bmatrix}1 & 0 & -1\\0 & 1 & -\frac{1}{2}\\0 & 0 & 0\end{bmatrix}.$$

Columns 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \text{Span}\left\{\begin{bmatrix}2\\1\\2\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 0$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (0)I) = \begin{bmatrix}1 & -1 & 0\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{0} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (0)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 + 1 = 3$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 2 & 1\\1 & 1 & 1\\1 & 2 & 0\end{bmatrix}, \qquad D = \begin{bmatrix}-2 & 0 & 0\\0 & -1 & 0\\0 & 0 & 0\end{bmatrix}.$$

The exact checks are $\det P = 1 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.19: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 5 & -3 & 0 & 9 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-3)(t-2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -\frac{3}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 1 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 & 3 & 1 \\ 1 & -2 & 2 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = -2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P.

Exercise 5.3.20: Find enough independent eigenvectors

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)^2 (t-2)^2.$$

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 2, 3 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_2 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (2)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 = 3$. This is smaller than 4, so no eigenvector basis exists. Hence A is not diagonalizable. Repeating or rescaling one of these vectors cannot supply a missing independent direction.

Exercise 5.3.21: Diagonalization criteria and constructions

(a) False.

The condition must say that D is *diagonal*. Without that, every matrix has the factorization $A = IAI^{-1}$, including a nondiagonalizable Jordan block.

(b) True.

A basis consists of n independent vectors; the Diagonalization Theorem applies to that eigenvector basis.

(c) False.

Counting roots with multiplicities does not count independent eigenvectors. For instance $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ has characteristic polynomial $(t-1)^2$ but only the one-dimensional eigenspace $\text{Span}\{(1,0)\}$.

(d) False.

A diagonal matrix can have a zero diagonal entry. The zero matrix is diagonalizable but not invertible.

Exercise 5.3.22: Diagonalization criteria and constructions

(a) False.

The vectors must be linearly independent. One eigenvector and its nonzero scalar multiples give arbitrarily many eigenvectors but only one direction.

(b) False.

I_n is diagonalizable and has only the eigenvalue 1 when $n > 1$.

(c) True.

Taking column j of $AP = PD$ gives $Ap_j = d_{jj}p_j$. If $p_j \neq 0$, this is exactly an eigenvector equation; zero columns are excluded by the definition.

(d) False.

The Jordan block $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ has determinant 1 and is invertible, but its eigenspace has dimension 1, so it is not diagonalizable.

Exercise 5.3.23: Diagonalization criteria and constructions

Yes. Choose a basis of three vectors from the first eigenspace and a basis of two from the other. Eigenvectors from different eigenspaces combine independently, so the union has $3+2=5$ independent vectors in $\mathbb{R}^5$. It is an eigenvector basis, giving diagonalizability.

Exercise 5.3.24: Diagonalization criteria and constructions

No. The dimensions of the only two eigenspaces add to $1+1=2$, which is less than 3. Every eigenvector belongs to one of those spaces, so one cannot obtain a third independent eigenvector. Thus the Diagonalization Theorem fails.

Exercise 5.3.25: Diagonalization criteria and constructions

No: under these assumptions the matrix must be diagonalizable. The two known dimensions are 1 and 2. The third eigenspace has at least dimension 1, giving at least $1+2+1=4$ independent

eigenvectors from the three distinct eigenspaces. Since the ambient space has dimension 4, this is exactly a basis.

Exercise 5.3.26: Diagonalization criteria and constructions

Yes, failure is possible. The known dimensions add to $2 + 3 = 5$, and the third can have dimension 1 or 2. If it has dimension 1, the total is $6 < 7$. For a concrete example take

$$A = \operatorname{diag}(0, 0, 1, 1, 1, J), \qquad J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

The eigenspace dimensions for $0, 1, 2$ are $2, 3, 1$, respectively, so this 7×7 matrix is not diagonalizable. If the last block were $2I_2$, the third dimension would instead be 2 and the matrix would be diagonalizable. The supplied data do not decide which occurs.

Exercise 5.3.27: Diagonalization criteria and constructions

Write $A = PDP^{-1}$. Invertibility of A implies no diagonal entry of D is zero. Inverting the product in reverse order gives $A^{-1} = PD^{-1}P^{-1}$, where D^{-1} is again diagonal. Thus A^{-1} is diagonalizable, and it is invertible because its inverse is A. Equivalently, $Av = \lambda v$ implies $A^{-1}v = \lambda^{-1}v$ for the same eigenvector basis.

Exercise 5.3.28: Diagonalization criteria and constructions

The hypothesis means $A = PDP^{-1}$ with D diagonal. Transpose this equality:

$$A^T = (P^{-1})^T D P^T = QDQ^{-1}, \qquad Q = (P^{-1})^T.$$

The matrix Q is invertible, and its columns therefore form n independent eigenvectors of A^T. The transpose generally does not have the *same* individual eigenvectors; the point is that it has its own eigenvector basis.

Exercise 5.3.29: Diagonalization criteria and constructions

In Example 2, $A = \left[\begin{smallmatrix} 7 & 2 \\ -4 & 1 \end{smallmatrix}\right]$, with eigenvectors $(1, -1)$ for 5 and $(1, -2)$ for 3. The new $D_1 = \operatorname{diag}(3, 5)$ reverses the eigenvalue order, so reverse the columns as well:

$$P_1 = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}.$$

Its determinant is $1 \neq 0$, and multiplication verifies $AP_1 = P_1 D_1$. Thus $A = P_1 D_1 P_1^{-1}$. Reordering eigenvalues without reordering eigenvectors would generally be wrong.

Exercise 5.3.30: Diagonalization criteria and constructions

Keep $D = \operatorname{diag}(5, 3)$ from Example 2, but rescale its first eigenvector. For instance,

$$P_2 = \begin{bmatrix} 2 & 1 \\ -2 & -2 \end{bmatrix}$$

has columns $2(1,-1)$ and $(1,-2)$ and determinant $-2 \neq 0$. It differs from the displayed P, but $AP_2 = P_2D$ still holds, so $A = P_2DP_2^{-1}$. Any nonzero rescaling of a basis eigenvector gives another valid factorization with that eigenvalue order.

Exercise 5.3.31: Diagonalization criteria and constructions

Take $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$. Its determinant is 1, so it is invertible and certainly nonzero. Its only eigenvalue is 1, and $(A - I)x = 0$ says $x_2 = 0$, leaving only a one-dimensional eigenspace. Therefore it is not diagonalizable.

Exercise 5.3.32: Diagonalization criteria and constructions

Take $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right]$. It is nondiagonal and has determinant 0, so is not invertible. Its distinct eigenvalues are 1 and 0; corresponding independent eigenvectors are $(1,0)$ and $(-1,1)$. Thus with $P = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right]$ and $D = \operatorname{diag}(1,0)$, we have $AP = PD$ and $A = PDP^{-1}$.

Exercise 5.3.33: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 9 & -4 & -2 & -4 \\ -56 & 32 & -28 & 44 \\ -14 & -14 & 6 & -14 \\ 42 & -33 & 21 & -45 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 13)^2 (t + 12)^2.$$

Find the eigenspace for $\lambda = -12$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-12)I) = \begin{bmatrix} 1 & 0 & -\frac{2}{7} & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-12} = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 7 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-12)v$ for each listed vector.

Find the eigenspace for $\lambda = 13$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (13)I) = \begin{bmatrix} 1 & 0 & \frac{1}{2} & -\frac{1}{3} \\ 0 & 1 & 0 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{13} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \\ 0 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (13)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 2 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 0 & 1 & 1 \\ 7 & 1 & 0 & -4 \\ 7 & 0 & -2 & 0 \\ 0 & -1 & 0 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} -12 & 0 & 0 & 0 \\ 0 & -12 & 0 & 0 \\ 0 & 0 & 13 & 0 \\ 0 & 0 & 0 & 13 \end{bmatrix}.$$

The exact checks are $\det P = 25 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.34: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 4 & -9 & -7 & 8 & 2 \\ -7 & -9 & 0 & 7 & 14 \\ 5 & 10 & 5 & -5 & -10 \\ -2 & 3 & 7 & 0 & 4 \\ -3 & -13 & -7 & 10 & 11 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)^3 (t+2)^2 .$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & 0 & \frac{2}{5} & -\frac{3}{5} \\ 0 & 1 & 0 & -\frac{7}{5} & -\frac{2}{5} \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} .$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span} \left\{ \begin{bmatrix} 2 \\ -7 \\ 5 \\ -5 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ -5 \\ 0 \\ 5 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & -2 & 1 & -2 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} .$$

Columns 3, 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span} \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} .$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 3. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 3 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 3 & 2 & 1 & 2 \\ -7 & 7 & -1 & -1 & 0 \\ 5 & -5 & 1 & 0 & 0 \\ -5 & 0 & 0 & -1 & 0 \\ 0 & 5 & 0 & 0 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 35 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.35: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 13 & -12 & 9 & -15 & 9 \\ 6 & -5 & 9 & -15 & 9 \\ 6 & -12 & -5 & 6 & 9 \\ 6 & -12 & 9 & -8 & 9 \\ -6 & 12 & 12 & -6 & -2 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 7)^3 (t + 14)^2.$$

Find the eigenspace for $\lambda = -14$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-14)I) = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-14} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\\-1\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-14)v$ for each listed vector.

Find the eigenspace for $\lambda = 7$.

The algebraic multiplicity is 3. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (7)I) = \begin{bmatrix}1 & -2 & 0 & -1 & \frac{3}{2}\\0 & 0 & 1 & -1 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix}.$$

Columns 2, 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_7 = \text{Span}\left\{\begin{bmatrix}2\\1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\0\\0\\-2\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 3. Direct multiplication checks $Av = (7)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 3 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}1 & 0 & 2 & 1 & 3\\1 & 0 & 1 & 0 & 0\\0 & 1 & 0 & 1 & 0\\1 & 0 & 0 & 1 & 0\\0 & -1 & 0 & 0 & -2\end{bmatrix}, \qquad D = \begin{bmatrix}-14 & 0 & 0 & 0 & 0\\0 & -14 & 0 & 0 & 0\\0 & 0 & 7 & 0 & 0\\0 & 0 & 0 & 7 & 0\\0 & 0 & 0 & 0 & 7\end{bmatrix}.$$

The exact checks are $\det P = -7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

Exercise 5.3.36: Exact computer-assisted diagonalization

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 24 & -6 & 2 & 6 & 2 \\ 72 & 51 & 9 & -99 & 9 \\ 0 & -63 & 15 & 63 & 63 \\ 72 & 15 & 9 & -63 & 9 \\ 0 & 63 & 21 & -63 & -27 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 36)^2 (t - 24) (t + 48)^2 .$$

Find the eigenspace for $\lambda = -48$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-48)I) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-48} = \operatorname{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-48)v$ for each listed vector.

Find the eigenspace for $\lambda = 24$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (24)I) = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{24} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\\1\\0\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (24)v$ for each listed vector.

Find the eigenspace for $\lambda = 36$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\text{rref}(A - (36)I) = \begin{bmatrix}1 & 0 & 0 & -1 & \frac{1}{3}\\0 & 1 & 0 & 0 & -1\\0 & 0 & 1 & -3 & 0\\0 & 0 & 0 & 0 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix}.$$

Columns 4, 5 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{36} = \text{Span}\left\{\begin{bmatrix}1\\0\\3\\1\\0\end{bmatrix}, \begin{bmatrix}1\\-3\\0\\0\\-3\end{bmatrix}\right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (36)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 2 = 5$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix}0 & 0 & 1 & 1 & 1\\1 & 0 & 1 & 0 & -3\\0 & 1 & 0 & 3 & 0\\1 & 0 & 1 & 1 & 0\\0 & -1 & 0 & 0 & -3\end{bmatrix}, \qquad D = \begin{bmatrix}-48 & 0 & 0 & 0 & 0\\0 & -48 & 0 & 0 & 0\\0 & 0 & 24 & 0 & 0\\0 & 0 & 0 & 36 & 0\\0 & 0 & 0 & 0 & 36\end{bmatrix}.$$

The exact checks are $\det P = -12 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.

5.4 Eigenvectors and linear transformations

The matrix of a linear map depends on its *ordered* domain and codomain bases. Column j is the coordinate vector of the image of domain basis vector j, written in the codomain basis. For $T(x) = Ax$ on $\mathbb{R}^n$, if P has the basis vectors as its columns, then $[T]_{\mathcal{B}} = P^{-1}AP$.

The relation $[T(x)]_{\mathcal{B}} = [T]_{\mathcal{B}}[x]_{\mathcal{B}}$ uses coordinates on both sides. A coordinate column is not automatically the same as the vector it represents; they coincide only when using the standard basis. Similarity changes coordinates without changing the underlying linear transformation.

Practice problem 5.4.P1: Recover a polynomial from its coordinate column

Relative to $(1, t, t^2)$, the polynomial $p = a_0 + a_1t + a_2t^2$ has column $(a_0, a_1, a_2)^T$. Multiply by the supplied matrix:

$$[T(p)]_{\mathcal{B}} = \begin{bmatrix} 3 & 4 & 0 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3a_0 + 4a_1 \\ 5a_1 - a_2 \\ a_0 - 2a_1 + 7a_2 \end{bmatrix}.$$

Translate this column back into the polynomial $T(p) = (3a_0 + 4a_1) + (5a_1 - a_2)t + (a_0 - 2a_1 + 7a_2)t^2$.

Practice problem 5.4.P2: Reflexivity and transitivity of similarity

(a) Every matrix is similar to itself.

Choose $P = I$; then $P^{-1}AP = A$ and P is invertible.

(b) Compose two changes of basis.

If $B = P^{-1}AP$ and $C = Q^{-1}BQ$, substitute the first equation into the second:

$$C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ).$$

The product PQ is invertible, so this is exactly the definition saying C is similar to A. The reversed order in $(PQ)^{-1} = Q^{-1}P^{-1}$ is essential.

Exercise 5.4.1: Construct a matrix from basis-vector images

The codomain coordinate vectors of $T(b_1)$, $T(b_2)$, $T(b_3)$ are $(3, -5)^T$, $(-1, 6)^T$, and $(0, 4)^T$. Put them in columns, in that order:

$$[T]_{\mathcal{D}\leftarrow\mathcal{B}} = \begin{bmatrix} 3 & -1 & 0 \\ -5 & 6 & 4 \end{bmatrix}.$$

It has two rows (the codomain dimension) and three columns (the domain dimension). Multiplication by a column $(x_1, x_2, x_3)^T$ produces the coordinates of $x_1T(b_1) + x_2T(b_2) + x_3T(b_3)$, explaining the column rule.

Exercise 5.4.2: Construct a matrix from basis-vector images

The domain basis is now $\mathcal{D}$ and the codomain basis is $\mathcal{B}$. The supplied images have coordinate columns $(3,-3)^T$ and $(-2,5)^T$ in $\mathcal{B}$. Therefore

$$[T]_{\mathcal{B}\leftarrow\mathcal{D}} = \begin{bmatrix} 3 & -2 \\ -3 & 5 \end{bmatrix}.$$

The arrow in the subscript records the direction of coordinates and helps avoid reversing the bases.

Exercise 5.4.3: Construct a matrix from basis-vector images

(a) Substitute the standard basis vectors one at a time.

The formula gives $T(e_1) = b_3$, $T(e_2) = -b_1 - 2b_2$, and $T(e_3) = 2b_1 + 3b_3$.

(b) Read their coefficients in $\mathcal{B}$.

These columns are $(0,0,1)^T$, $(-1,-2,0)^T$, and $(2,0,3)^T$.

(c) Put them in the original basis order.

Thus the matrix relative to the standard domain basis and codomain basis $\mathcal{B}$ is

$$\begin{bmatrix} 0 & -1 & 2 \\ 0 & -2 & 0 \\ 1 & 0 & 3 \end{bmatrix}.$$

Multiplying by $(x_1,x_2,x_3)^T$ recovers exactly $(2x_3 - x_2, -2x_2, x_1 + 3x_3)^T$, the coefficients in the given transformation.

Exercise 5.4.4: Construct a matrix from basis-vector images

The input column in $\mathcal{B}$ is $(x_1,x_2,x_3)^T$, while the output is already in standard coordinates of $\mathbb{R}^2$. Reading the coefficient of each input scalar gives

$$[T]_{\mathcal{E}\leftarrow\mathcal{B}} = \begin{bmatrix} 2 & -3 & 1 \\ -2 & 0 & 5 \end{bmatrix}.$$

In particular the zero in row 2, column 2 is needed because the second output component does not contain x_2.

Exercise 5.4.5: Multiplication of polynomials as a linear map

(a) Multiply the specified polynomial, combining like powers.

$$T(3 - 2t + t^2) = (t+3)(3 - 2t + t^2) = t^3 + t^2 - 3t + 9.$$

(b) Verify linearity for arbitrary inputs and scalars.

For polynomials p, q and scalars c, d, distributivity gives

$$T(cp + dq) = (t + 3)(cp + dq) = cT(p) + dT(q).$$

The degree is at most 3, so the image does belong to the stated codomain.

(c) Apply the map to each monomial in the domain basis.

$T(1) = t + 3$. $T(t) = t^2 + 3t$. $T(t^2) = t^3 + 3t^2$. Writing those coefficients in increasing powers of t gives the matrix

$$C = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

The column lengths reflect the codomain degree bound, even when one image has lower degree and hence trailing zero coefficients.

Exercise 5.4.6: Multiplication of polynomials as a linear map

(a) Multiply the specified polynomial, combining like powers.

$$T(3 - 2t + t^2) = (2t^2 + 1)(3 - 2t + t^2) = 2t^4 - 4t^3 + 7t^2 - 2t + 3.$$

(b) Verify linearity for arbitrary inputs and scalars.

For polynomials p, q and scalars c, d, distributivity gives

$$T(cp + dq) = (2t^2 + 1)(cp + dq) = cT(p) + dT(q).$$

The degree is at most 4, so the image does belong to the stated codomain.

(c) Apply the map to each monomial in the domain basis.

$T(1) = 2t^2 + 1$. $T(t) = 2t^3 + t$. $T(t^2) = 2t^4 + t^2$. Writing those coefficients in increasing powers of t gives the matrix

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

The column lengths reflect the codomain degree bound, even when one image has lower degree and hence trailing zero coefficients.

Exercise 5.4.7: Coordinates determine a linear transformation

Evaluate the formula at the three basis polynomials: $T(1) = 3 + 5t$, $T(t) = -2t + 4t^2$, $T(t^2) = t^2$. Their coefficient columns give

$$[T]_{\mathcal{B}} = \begin{bmatrix} 3 & 0 & 0 \\ 5 & -2 & 0 \\ 0 & 4 & 1 \end{bmatrix}.$$

As a check, multiplying this by $(a_0, a_1, a_2)^T$ yields $(3a_0, 5a_0 - 2a_1, 4a_1 + a_2)^T$, exactly the supplied coefficient rule.

Exercise 5.4.8: Coordinates determine a linear transformation

The input has coordinate vector $(4, -3, 0)^T$. Thus

$$[T(4b_1 - 3b_2)]_{\mathcal{B}} = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & -2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ -5 \end{bmatrix}.$$

Translate back to the vector space: $T(4b_1 - 3b_2) = 5b_2 - 5b_3$. The result is not the coordinate column itself unless these basis vectors happen to be standard coordinate vectors.

Exercise 5.4.9: Evaluation at three input values

(a) Evaluate the given polynomial.

For $p(t) = 5 + 3t$, the three values are $p(-1) = 2$, $p(0) = 5$, and $p(1) = 8$, so $T(p) = (2, 5, 8)^T$.

(b) Establish linearity.

For any fixed input s, $(cp + dq)(s) = cp(s) + dq(s)$. Applying this at $s = -1, 0, 1$ componentwise proves $T(cp + dq) = cT(p) + dT(q)$.

(c) Evaluate the three basis polynomials.

The image columns for $1, t, t^2$ are $(1, 1, 1)^T$, $(-1, 0, 1)^T$, and $(1, 0, 1)^T$. Therefore

$$[T] = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

For example this matrix times $(5, 3, 0)^T$ returns the value found in part (a).

Exercise 5.4.10: Evaluation of a cubic at four points

(a) Linearity holds componentwise.

Evaluation at any fixed input distributes over polynomial addition and scalar multiplication. Applying this at $-2, 3, 1, 0$ proves the vector-valued map is linear.

(b) Respect the given order of evaluation points.

Evaluate each of $1, t, t^2, t^3$, putting its four values in a column. The matrix is

$$[T] = \begin{bmatrix} 1 & -2 & 4 & -8 \\ 1 & 3 & 9 & 27 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

For a general coefficient column $(a_0, a_1, a_2, a_3)^T$, multiplication gives the polynomial's values at those four points. The row for input 0 is last, not first, because the output order was specified that way.

Exercise 5.4.11: Change both input and output coordinates

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 2 & 3 \\ -4 & -5 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -2 & -2 \\ 0 & -1 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order.

Exercise 5.4.12: Change both input and output coordinates

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} -2 & 2 \\ 0 & -4 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -4 & 0 \\ 2 & -2 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order.

Exercise 5.4.13: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 0 & 1 \\ -3 & 4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)(t-1).$$

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

The exact checks are $\det P = 2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.14: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)(t + 1).$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{5} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.15: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-2)(t+5).$$

Find the eigenspace for $\lambda = -5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-5)I) = \begin{bmatrix} 1 & \frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-5} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-5)v$ for each listed vector.

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{2} = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (2)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 2 \\ -3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -5 & 0 \\ 0 & 2 \end{bmatrix}.$$

The exact checks are $\det P = 7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.16: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 4 & -2 \\ -1 & 5 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-6)(t-3).$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Find the eigenspace for $\lambda = 6$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (6)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_6 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (6)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}.$$

The exact checks are $\det P = -3 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.17: An eigenvector without an eigenvector basis

(a) Verify the stated eigenvector and inspect multiplicity.

Multiplication gives $A(1,-1)^T = (3,-3)^T = 3(1,-1)^T$. But

$$\det(tI - A) = (t-3)^2, \qquad A - 3I = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

The homogeneous equation is $x_1 + x_2 = 0$, so its eigenspace has only dimension 1. It cannot supply a basis for $\mathbb{R}^2$; thus A is not diagonalizable.

(b) Use the specified basis anyway.

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 3 & -2 \\ -3 & 5 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. The resulting matrix is not diagonal. A valid change of basis exists even when no diagonalizing change of basis exists.

Exercise 5.4.18: Similarity and coordinate identities

The eigenvalues alone do not decide the question. A diagonal basis matrix exists precisely when A is diagonalizable, which here means the eigenspace for 5 has dimension 2. The eigenspace for -2, whose algebraic multiplicity is 1, has dimension 1.

For a positive example take $A = \operatorname{diag}(5,5,-2)$; the standard basis works. For a negative example take $A = \left[\begin{smallmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -2 \end{smallmatrix}\right]$. It has the same eigenvalues, but its 5-eigenspace is only one-dimensional. Thus one must compute an eigenspace or equivalent rank information before answering for a particular A.

Exercise 5.4.19: Similarity and coordinate identities

Write $B = P^{-1}AP$ with P invertible. The product is invertible because all three factors are, and reversing the factors gives

$$B^{-1} = P^{-1}A^{-1}P.$$

Thus the same change of basis $Q = P$ exhibits similarity of the inverses. It is legitimate to invert this expression because invertibility was established first.

Exercise 5.4.20: Similarity and coordinate identities

If $B = P^{-1}AP$, then $B^2 = P^{-1}AP\,P^{-1}AP = P^{-1}A^2P$, since the middle $PP^{-1} = I$. This is the required similarity factorization for the squares. The same cancellation would prove the result for any positive integer power.

Exercise 5.4.21: Similarity and coordinate identities

Write $B = P^{-1}AP$ and $C = Q^{-1}AQ$. Solving the second identity for A gives $A = QCQ^{-1}$. Substitution yields

$$B = P^{-1}QCQ^{-1}P = R^{-1}CR, \qquad R = Q^{-1}P.$$

The matrix R is invertible as a product of invertible matrices, so B and C are similar. This argument is the symmetry-plus-transitivity property in explicit form.

Exercise 5.4.22: Similarity and coordinate identities

Suppose $A = PDP^{-1}$ with D diagonal and $B = Q^{-1}AQ$. Then

$$B = (Q^{-1}P)D(P^{-1}Q).$$

The last factor is $(Q^{-1}P)^{-1}$. Thus this is a diagonalization of B, with an invertible change-of-basis matrix $Q^{-1}P$.

Exercise 5.4.23: Similarity and coordinate identities

Set $y = P^{-1}x$. This vector is nonzero: if $y = 0$, multiplication by P would force $x = 0$, contradicting the eigenvector definition. Now

$$By = P^{-1}AP\,P^{-1}x = P^{-1}Ax = P^{-1}(\lambda x) = \lambda y.$$

Therefore y is an eigenvector of B for the same eigenvalue. The nonzero check is an essential part of the proof.

Exercise 5.4.24: Similarity and coordinate identities

For $B = P^{-1}AP$, first note that AP has the same column space as A. Every column of AP is a combination of columns of A, and $A = (AP)P^{-1}$ supplies the reverse inclusion. Thus $\operatorname{rank}(AP) = \operatorname{rank} A$. Next, multiplication on the left by P^{-1} is an invertible linear map on the codomain. It maps the column space of AP bijectively onto that of B, preserving its dimension. Consequently $\operatorname{rank} B = \operatorname{rank} A$.

Exercise 5.4.25: Similarity and coordinate identities

Use the supplied cyclic identity on two square factors, $F = P^{-1}A$ and $G = P$:

$$\operatorname{tr}(B) = \operatorname{tr}((P^{-1}A)P) = \operatorname{tr}(P(P^{-1}A)) = \operatorname{tr}(A).$$

Only associativity and $PP^{-1} = I$ are used; this does not incorrectly commute arbitrary matrix factors.

Exercise 5.4.26: Similarity and coordinate identities

If $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, Exercise 25 gives $\operatorname{tr} A = \operatorname{tr} D$. The latter is the sum of the diagonal entries, hence $\sum_{j=1}^{n} \lambda_j$. These diagonal entries are the eigenvalues counted with their algebraic multiplicities in a diagonalization, proving the assertion in the requested case.

Exercise 5.4.27: Similarity and coordinate identities

The identity map sends the jth domain basis vector b_j to the same vector b_j, now expressed in the standard codomain basis. Thus its columns are $b_1, \dots, b_n$:

$$[I]_{\mathcal{E} \leftarrow \mathcal{B}} = [b_1 \ \cdots \ b_n] = P_{\mathcal{B}}.$$

This is the change-of-coordinates matrix from $\mathcal{B}$ coordinates to standard coordinates, also called the basis matrix in Section 4.4. Its action is $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$.

Exercise 5.4.28: Similarity and coordinate identities

The identity sends b_j to b_j, whose codomain coordinate column is $[b_j]_{\mathcal{C}}$. Therefore

$$[I]_{\mathcal{C} \leftarrow \mathcal{B}} = \big[[b_1]_{\mathcal{C}} \ \cdots \ [b_n]_{\mathcal{C}}\big].$$

This is the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$ in Section 4.7. Its defining equation is $[x]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}}[x]_{\mathcal{B}}$; the vector x itself does not change.

Exercise 5.4.29: Similarity and coordinate identities

With the same basis on both sides, $[I(b_j)]_{\mathcal{B}} = [b_j]_{\mathcal{B}} = e_j$. The matrix with those columns is exactly I_n. Hence $[I]_{\mathcal{B}} = I_n$ for every ordered basis, not merely the standard one.

Exercise 5.4.30: An exact basis-matrix computation

The standard matrix is

$$A = \begin{bmatrix} 6 & -2 & -2 \\ 3 & 1 & -2 \\ 2 & -2 & 2 \end{bmatrix}.$$

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & -1 \\ 1 & 3 & 0 \end{bmatrix}, \qquad \det P = -1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 2 & 4 & -4 \\ 2 & 1 & -4 \\ 2 & 8 & 0 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3 & 0 \\ 0 & 1 & 4 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. This computer-assisted calculation can be reproduced exactly by setting the displayed matrices in SymPy and evaluating

```
C = P.inv()*A*P
assert P*C == A*P
print(C)
```

Solving the augmented system instead of explicitly forming the inverse gives the same result.

Exercise 5.4.31: An exact basis-matrix computation

The standard matrix is

$$A = \begin{bmatrix} -7 & -48 & -16 \\ 1 & 14 & 6 \\ -3 & -45 & -19 \end{bmatrix}.$$

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} -3 & -2 & 3 \\ 1 & 1 & -1 \\ -3 & -3 & 0 \end{bmatrix}, \qquad \det P = 3 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 21 & 14 & 27 \\ -7 & -6 & -11 \\ 21 & 18 & 36 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -7 & -2 & -6 \\ 0 & -4 & -6 \\ 0 & 0 & -1 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. This computer-assisted calculation can be reproduced exactly by setting the displayed matrices in SymPy and evaluating

```
C = P.inv()*A*P
assert P*C == A*P
print(C)
```

Solving the augmented system instead of explicitly forming the inverse gives the same result.

Exercise 5.4.32: Find a diagonal coordinate representation

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} -6 & 4 & 0 & 9 \\ -3 & 0 & 1 & 6 \\ -1 & -2 & 1 & 0 \\ -4 & 4 & 0 & 7 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-1)(t+2)^2.$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & -1 & -\frac{3}{2} \\ 0 & 1 & -1 & \frac{3}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -3 \\ 0 \\ 4 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{7}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ -1 \\ -7 \\ 2 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 1 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 6 & 2 & 2 \\ 1 & -3 & -1 & 1 \\ 1 & 0 & -7 & -1 \\ 0 & 4 & 2 & 2 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 84 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.The columns of the displayed P form the requested basis; the corresponding matrix of the transformation is D.

5.5 Complex eigenvalues

A real matrix can have nonreal eigenvalues when it acts on complex coordinate vectors. Complex arithmetic uses $i^2 = -1$ and conjugation $\overline{a+bi} = a-bi$. For a real matrix, conjugating $Av = \lambda v$ gives $A\overline{v} = \overline{\lambda}\,\overline{v}$, so nonreal eigenvalues occur in conjugate pairs.

The real matrix $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ is a scaling followed by a rotation: $C = r \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$, where $r = \sqrt{a^2+b^2}$, $\cos\varphi = a/r$, and $\sin\varphi = b/r$. The sign of b matters for clockwise versus counterclockwise rotation. We use angles in $(-\pi, \pi]$.

Practice problem 5.5.P1: Verify the two complex eigenpairs directly

Multiplication gives

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} a+bi \\ b-ai \end{bmatrix} = (a+bi) \begin{bmatrix} 1 \\ -i \end{bmatrix}.$$

Likewise $A(1,i)^T = (a-bi)(1,i)^T$. Both vectors are nonzero, so these are genuine eigenpairs. They are conjugate to one another. If $b = 0$, the eigenvalues coincide at a; the verified equations still hold, and in fact every nonzero vector is an eigenvector of aI_2.

Exercise 5.5.1: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 4t + 5 = 0$, whose two roots are $2-i,\ 2+i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 2-i$, the first equation of $(A-\lambda I)x = 0$ is

$$(-1+i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1-i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2-i} = \mathrm{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1-i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 2+i$, the first equation of $(A-\lambda I)x = 0$ is

$$(-1-i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1+i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2+i} = \mathrm{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1+i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.2: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 3 & -3 \\ 3 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 6t + 18 = 0$, whose two roots are $3 - 3i,\ 3 + 3i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 3 - 3i$, the first equation of $(A - \lambda I)x = 0$ is

$$(3i)x_1 + (-3)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -3s$ and $x_2 = (-3i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3-3i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -3 \\ -3i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 3 + 3i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-3i)x_1 + (-3)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -3s$ and $x_2 = (3i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3+3i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -3 \\ 3i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.3: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 5 & 1 \\ -8 & 1 \end{bmatrix}.$$

The characteristic equation is $t^2 - 6t + 13 = 0$, whose two roots are $3 - 2i,\ 3 + 2i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 3 - 2i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 + 2i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (-2 - 2i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3-2i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ -2 - 2i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 3 + 2i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 - 2i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (-2 + 2i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{3+2i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ -2 + 2i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.4: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 4t + 5 = 0$, whose two roots are $2 - i,\ 2 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 2 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 + i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 2 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 - i)x_1 + (-2)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -2s$ and $x_2 = (1 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{2+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -2 \\ 1 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. The printed matrix in Exercise 4 is identical to that in Exercise 1; the repeated answer reflects the actual supplied edition, rather than an altered matrix.

Exercise 5.5.5: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 5 \end{bmatrix}.$$

The characteristic equation is $t^2 - 8t + 17 = 0$, whose two roots are $4 - i,\ 4 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 4 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 + i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (1 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{4-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ 1 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 4 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(-1 - i)x_1 + (1)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = 1s$ and $x_2 = (1 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{4+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} 1 \\ 1 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.6: Complex roots and their one-dimensional eigenspaces

What we must show. Find the eigenspaces over $\mathbb{C}$, where

$$A = \begin{bmatrix} 7 & -5 \\ 1 & 3 \end{bmatrix}.$$

The characteristic equation is $t^2 - 10t + 26 = 0$, whose two roots are $5 - i,\ 5 + i$. These roots are distinct because their imaginary parts are nonzero. For $\lambda = 5 - i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 + i)x_1 + (-5)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -5s$ and $x_2 = (-2 - i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{5-i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -5 \\ -2 - i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace. For $\lambda = 5 + i$, the first equation of $(A - \lambda I)x = 0$ is

$$(2 - i)x_1 + (-5)x_2 = 0.$$

The second equation is dependent on this one because the coefficient determinant is zero. Set $x_1 = -5s$ and $x_2 = (-2 + i)s$, where $s \in \mathbb{C}$. Every solution has that form, since the coefficient of x_2 in the first equation is nonzero. Thus

$$E_{5+i} = \text{Span}_{\mathbb{C}} \left\{ \begin{bmatrix} -5 \\ -2 + i \end{bmatrix} \right\}.$$

Substitution gives $Av = \lambda v$ exactly, and $v \neq 0$, so this is a one-vector basis of the one-dimensional complex eigenspace.

Exercise 5.5.7: Rotation angle and positive scale

The matrix has the canonical entries $a = \sqrt{3}$ and $b = 1$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = \sqrt{3} \pm (1)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = \frac{1}{2}$. In the requested interval these give

$$\varphi = \frac{\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.8: Rotation angle and positive scale

The matrix has the canonical entries $a = 3$ and $b = -3\sqrt{3}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 3 \pm (-3\sqrt{3})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 6.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{1}{2}$ and $\sin\varphi = b/r = -\frac{\sqrt{3}}{2}$. In the requested interval these give

$$\varphi = -\frac{\pi}{3}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.9: Rotation angle and positive scale

The matrix has the canonical entries $a = 0$ and $b = -2$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 0 \pm (-2)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = 0$ and $\sin\varphi = b/r = -1$. In the requested interval these give

$$\varphi = -\frac{\pi}{2}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.10: Rotation angle and positive scale

The matrix has the canonical entries $a = 0$ and $b = -\frac{1}{2}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 0 \pm (-\frac{1}{2})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = \frac{1}{2}.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = 0$ and $\sin\varphi = b/r = -1$. In the requested interval these give

$$\varphi = -\frac{\pi}{2}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.11: Rotation angle and positive scale

The matrix has the canonical entries $a = -\sqrt{3}$ and $b = -1$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = -\sqrt{3} \pm (-1)i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = -\frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = -\frac{1}{2}$. In the requested interval these give

$$\varphi = -\frac{5\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.12: Rotation angle and positive scale

The matrix has the canonical entries $a = 3$ and $b = \sqrt{3}$ in the real rotation-scaling form. Hence the eigenvalues are

$$\lambda = a \pm bi = 3 \pm (\sqrt{3})i.$$

The positive scale factor is

$$r = \sqrt{a^2 + b^2} = 2\sqrt{3}.$$

To determine the angle without a quadrant error, require both $\cos\varphi = a/r = \frac{\sqrt{3}}{2}$ and $\sin\varphi = b/r = \frac{1}{2}$. In the requested interval these give

$$\varphi = \frac{\pi}{6}.$$

The sign of this angle gives the direction of rotation. Substitution in the rotation matrix recovers all four original entries, checking both its quadrant and scale.

Exercise 5.5.13: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

The characteristic roots are $2 \pm (1)i$. Choose the root $\lambda = a - bi = 2 - i$ with $b = 1 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -2 \\ 1 - i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -2 & 0 \\ 1 & -1 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Here $\det P = 2 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.14: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix}.$$

The characteristic roots are $2 \pm (\sqrt{2})i$. Choose the root $\lambda = a - bi = 2 - \sqrt{2}i$ with $b = \sqrt{2} > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -3 \\ -1 - \sqrt{2}i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -\sqrt{2} \end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}-3 & 0\\ -1 & -\sqrt{2}\end{bmatrix},\qquad C=\begin{bmatrix}2 & -\sqrt{2}\\ \sqrt{2} & 2\end{bmatrix}.$$

Here $\det P=3\sqrt{2}\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.15: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A=\begin{bmatrix}0 & 5\\ -2 & 2\end{bmatrix}.$$

The characteristic roots are $1\pm(3)i$. Choose the root $\lambda=a-bi=1-3i$ with $b=3>0$. Solving its first homogeneous equation as above gives the eigenvector

$$v=\begin{bmatrix}5\\ 1-3i\end{bmatrix}=u+iw,\qquad u=\begin{bmatrix}5\\ 1\end{bmatrix},\quad w=\begin{bmatrix}0\\ -3\end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}5 & 0\\ 1 & -3\end{bmatrix},\qquad C=\begin{bmatrix}1 & -3\\ 3 & 1\end{bmatrix}.$$

Here $\det P=-15\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.16: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A=\begin{bmatrix}4 & -2\\ 1 & 6\end{bmatrix}.$$

The characteristic roots are $5\pm(1)i$. Choose the root $\lambda=a-bi=5-i$ with $b=1>0$. Solving its first homogeneous equation as above gives the eigenvector

$$v=\begin{bmatrix}-2\\ 1-i\end{bmatrix}=u+iw,\qquad u=\begin{bmatrix}-2\\ 1\end{bmatrix},\quad w=\begin{bmatrix}0\\ -1\end{bmatrix}.$$

Equate real and imaginary parts in $A(u+iw)=(a-bi)(u+iw)$. This yields $Au=au+bw$ and $Aw=-bu+aw$, so in the ordered real basis (u,w) the matrix is C. Consequently take

$$P=\begin{bmatrix}-2 & 0\\ 1 & -1\end{bmatrix},\qquad C=\begin{bmatrix}5 & -1\\ 1 & 5\end{bmatrix}.$$

Here $\det P=2\neq 0$. Direct exact multiplication verifies $AP=PC$, which, after multiplying by P^{-1}, gives the required $A=PCP^{-1}$.

Exercise 5.5.17: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} -11 & -4 \\ 20 & 5 \end{bmatrix}.$$

The characteristic roots are $-3 \pm (4)i$. Choose the root $\lambda = a - bi = -3 - 4i$ with $b = 4 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -4 \\ 8 - 4i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -4 \\ 8 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -4 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -4 & 0 \\ 8 & -4 \end{bmatrix}, \qquad C = \begin{bmatrix} -3 & -4 \\ 4 & -3 \end{bmatrix}.$$

Here $\det P = 16 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.18: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} 3 & -5 \\ 2 & 5 \end{bmatrix}.$$

The characteristic roots are $4 \pm (3)i$. Choose the root $\lambda = a - bi = 4 - 3i$ with $b = 3 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -5 \\ 1 - 3i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -5 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -3 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -5 & 0 \\ 1 & -3 \end{bmatrix}, \qquad C = \begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}.$$

Here $\det P = 15 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.19: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} \frac{38}{25} & -\frac{7}{10} \\ \frac{14}{25} & \frac{2}{5} \end{bmatrix}.$$

The characteristic roots are $\frac{24}{25} \pm (\frac{7}{25})i$. Choose the root $\lambda = a - bi = \frac{24}{25} - \frac{7i}{25}$ with $b = \frac{7}{25} > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -\frac{7}{10} \\ -\frac{14}{25} - \frac{7i}{25} \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -\frac{7}{10} \\ -\frac{14}{25} \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -\frac{7}{25} \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -\frac{7}{10} & 0 \\ -\frac{14}{25} & -\frac{7}{25} \end{bmatrix}, \qquad C = \begin{bmatrix} \frac{24}{25} & -\frac{7}{25} \\ \frac{7}{25} & \frac{24}{25} \end{bmatrix}.$$

Here $\det P = \frac{49}{250} \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$. All decimals in the question were treated as exact terminating rationals, so these entries are exact, not rounded numerical eigenpairs.

Exercise 5.5.20: A real rotation-scaling block in a new basis

What we must show. Find a real change of basis for

$$A = \begin{bmatrix} -3 & -8 \\ 4 & 5 \end{bmatrix}.$$

The characteristic roots are $1 \pm (4)i$. Choose the root $\lambda = a - bi = 1 - 4i$ with $b = 4 > 0$. Solving its first homogeneous equation as above gives the eigenvector

$$v = \begin{bmatrix} -8 \\ 4 - 4i \end{bmatrix} = u + iw, \qquad u = \begin{bmatrix} -8 \\ 4 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ -4 \end{bmatrix}.$$

Equate real and imaginary parts in $A(u + iw) = (a - bi)(u + iw)$. This yields $Au = au + bw$ and $Aw = -bu + aw$, so in the ordered real basis (u, w) the matrix is C. Consequently take

$$P = \begin{bmatrix} -8 & 0 \\ 4 & -4 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & -4 \\ 4 & 1 \end{bmatrix}.$$

Here $\det P = 32 \neq 0$. Direct exact multiplication verifies $AP = PC$, which, after multiplying by P^{-1}, gives the required $A = PCP^{-1}$.

Exercise 5.5.21: Conjugation and real invariant directions

The first equation in Example 2 is $(-0.3 + 0.6i)x_1 - 0.6x_2 = 0$. Solve it for the second component:

$$x_2 = \frac{-0.3 + 0.6i}{0.6} x_1 = (-1/2 + i)x_1.$$

Taking $x_1 = 2$ gives $y = (2, -1 + 2i)^T$, as claimed. The original vector was $v_1 = (-2 - 4i, 5)^T$. With $c = (-1 + 2i)/5$, direct multiplication gives

$$c(-2 - 4i) = 2, \qquad c \cdot 5 = -1 + 2i.$$

Thus $y = cv_1$, with $c \neq 0$. They span the same complex eigenspace; different choices of a free parameter do not produce different eigenspace directions.

Exercise 5.5.22: Conjugation and real invariant directions

Suppose $Ax = \lambda x$ with $x \neq 0$ and choose a nonzero complex scalar μ. Complex linearity gives $A(\mu x) = \mu Ax = \mu\lambda x = \lambda(\mu x)$. Also $\mu x \neq 0$, because multiplying a zero result by $1/\mu$ would imply $x = 0$. Thus μx is an eigenvector for the same eigenvalue. Both its eigenvector equation and nonzero status are required.

Exercise 5.5.23: Conjugation and real invariant directions

Write $x^* = \overline{x}^T$ and $q = x^*Ax$, a 1×1 scalar. Here is the complete chain with the reasons for all five transitions:

$$\begin{aligned}
\overline{q} &= \overline{\overline{x}^T A x} \\
&= x^T \overline{A}\,\overline{x} && \text{(a): conjugation distributes over products;} \\
&= x^T A \overline{x} && \text{(b): } A \text{ has real entries;} \\
&= (x^T A \overline{x})^T && \text{(c): a scalar equals its transpose;} \\
&= \overline{x}^T A^T x && \text{(d): transpose reverses product order;} \\
&= \overline{x}^T A x = q && \text{(e): } A^T = A.
\end{aligned}$$

A complex number equals its conjugate exactly when its imaginary part is zero, so q is real. Conjugation does not reverse product order in (a); transposition does in (d).

Exercise 5.5.24: Conjugation and real invariant directions

Prove the eigenvalue is real without assuming a spectral theorem.

Let $x \neq 0$ and $Ax = \lambda x$. Multiply on the left by $x^* = \overline{x}^T$:

$$x^*Ax = \lambda x^*x, \qquad x^*x = \sum_{j=1}^{n} |x_j|^2 > 0.$$

The numerator is real by Exercise 23, so $\lambda = (x^*Ax)/(x^*x)$ is a real number.

Separate the real and imaginary components, checking nonzero status.

Write $x = u + iv$ with real u, v. Since A and λ are real, equating parts gives $Au = \lambda u$ and $Av = \lambda v$. At least one of u, v is nonzero because $x \neq 0$; every nonzero one is therefore a real eigenvector.

Conclusion. The eigenvalue is real, and $\operatorname{Re} x$ is an eigenvector *if it is nonzero*. The printed unconditional claim about the real part needs this qualification: for $A = I_2$ and $x = (i, 0)^T$, the real part is zero and is not an eigenvector. If the real part vanishes, the nonzero imaginary part supplies a real eigenvector instead; multiplying x by $-i$ makes that vector its real part.

Exercise 5.5.25: Conjugation and real invariant directions

Write $x = u + iv$ with $u = \operatorname{Re} x$ and $v = \operatorname{Im} x$. A real matrix times a real vector is real, and linearity gives $Ax = A(u + iv) = Au + iAv$. Uniqueness of the real and imaginary parts of each component

therefore yields

$$\operatorname{Re}(Ax) = A \operatorname{Re} x, \qquad \operatorname{Im}(Ax) = A \operatorname{Im} x.$$

The real-entry hypothesis matters: a matrix containing i could mix the real and imaginary parts.

Exercise 5.5.26: Conjugation and real invariant directions

(a) Expand the eigenvector equation before separating parts.

Let $v = u + iw$, with u, w real, and use the given $\lambda = a - bi$:

$$A(u + iw) = (a - bi)(u + iw) = (au + bw) + i(aw - bu).$$

Since A is real, comparison gives $Au = au + bw$ and $Aw = -bu + aw$, the required identities.

(b) Match the two columns of the matrix products.

With $P = [u\ w]$ and $C = \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$,

$$AP = [Au\ Aw] = [au + bw\ \ -bu + aw] = PC.$$

For completeness, u, w are independent when $b \neq 0$: if v were a complex scalar multiple of one real vector, the eigenvector equation would force that nonzero real vector to be mapped by the real matrix to a nonreal scalar multiple of itself, impossible. Thus P is invertible, which is the additional fact needed to pass from $AP = PC$ to $A = PCP^{-1}$.

Exercise 5.5.27: Two real invariant planes from complex eigenpairs

What we must show. Produce an exact real block factorization of

$$A = \begin{bmatrix} 26 & 33 & 23 & 20 \\ -6 & -8 & -1 & -13 \\ -14 & -19 & -16 & 3 \\ -20 & -20 & -20 & -14 \end{bmatrix}.$$

The characteristic polynomial factors as $(t^2 + 4t + 29)\,(t^2 + 8t + 116)$. An exact eigenspace computation gives the following eigenvectors for one root in each conjugate pair:

$$\lambda = -2 - 5i, \qquad v = \begin{bmatrix} -1 - i \\ i \\ 1 \\ 0 \end{bmatrix}.$$

$$\lambda = -4 - 10i, \qquad v = \begin{bmatrix} -1 + i \\ -2i \\ 2i \\ 2 \end{bmatrix}.$$

For each pair choose the root with negative imaginary part. Write its eigenvector as $u + iw$, then put u, w into two adjacent columns of P. The identities in Exercise 26 supply the corresponding 2×2

block of C. This gives

$$P = \begin{bmatrix} -1 & -1 & -1 & 1 \\ 0 & 1 & 0 & -2 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{bmatrix}.$$

$$C = \begin{bmatrix} -2 & -5 & 0 & 0 \\ 5 & -2 & 0 & 0 \\ 0 & 0 & -4 & -10 \\ 0 & 0 & 10 & -4 \end{bmatrix}.$$

The exact determinant is $\det P = 2 \neq 0$, and multiplying the displayed matrices gives $AP = PC$ with zero residual in every entry. Therefore $A = PCP^{-1}$. The calculation can be reproduced with exact SymPy arithmetic: factor `A.charpoly().as_expr()`, then for each selected root `lam` compute `(A-lam*eye(4)).nullspace()`. Extract componentwise real and imaginary parts for the two columns. Multiplying those columns by a common nonzero real scalar does not alter the block identity, which is why fractional vectors may be cleared to integers.

Exercise 5.5.28: Two real invariant planes from complex eigenpairs

What we must show. Produce an exact real block factorization of

$$A = \begin{bmatrix} 7 & 11 & 20 & 17 \\ -20 & -40 & -86 & -74 \\ 0 & -5 & -10 & -10 \\ 10 & 28 & 60 & 53 \end{bmatrix}.$$

The characteristic polynomial factors as $(t^2 - 6t + 10)\,(t^2 - 4t + 29)$. An exact eigenspace computation gives the following eigenvectors for one root in each conjugate pair:

$$\lambda = 3 - i, \qquad v = \begin{bmatrix} -2 \\ 2i \\ -3 - i \\ 4 \end{bmatrix}.$$

$$\lambda = 2 - 5i, \qquad v = \begin{bmatrix} 1 - i \\ -4 \\ 0 \\ 2 \end{bmatrix}.$$

For each pair choose the root with negative imaginary part. Write its eigenvector as $u + iw$, then put u, w into two adjacent columns of P. The identities in Exercise 26 supply the corresponding 2×2 block of C. This gives

$$P = \begin{bmatrix} -2 & 0 & 1 & -1 \\ 0 & 2 & -4 & 0 \\ -3 & -1 & 0 & 0 \\ 4 & 0 & 2 & 0 \end{bmatrix}.$$

$$C = \begin{bmatrix} 3 & -1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 5 & 2 \end{bmatrix}.$$

The exact determinant is $\det P = -4 \neq 0$, and multiplying the displayed matrices gives $AP = PC$ with zero residual in every entry. Therefore $A = PCP^{-1}$. The calculation can be reproduced with exact SymPy arithmetic: factor `A.charpoly().as_expr()`, then for each selected root `lam` compute `(A-lam*eye(4)).nullspace()`. Extract componentwise real and imaginary parts for the two columns. Multiplying those columns by a common nonzero real scalar does not alter the block identity, which is why fractional vectors may be cleared to integers.

5.6 Discrete dynamical systems

For $x_{k+1} = Ax_k$, start by expanding x_0 in an eigenbasis: $x_0 = \sum_j c_j v_j$, $Av_j = \lambda_j v_j$. Then

$$x_k = \sum_j c_j \lambda_j^k v_j.$$

This follows by induction: multiplication by A adds one power of each eigenvalue. A mode with $|\lambda| < 1$ decays; one with $|\lambda| > 1$ grows. A negative eigenvalue also reverses its mode's sign at each step. The direction of fastest decay corresponds to the smallest modulus, whereas fastest growth corresponds to the largest modulus. These discrete-time tests concern modulus, not merely the sign of an eigenvalue.

Practice problem 5.6.P1: Determine the three initial coefficients

The given eigenvectors are $v_1 = (-2, 2, 1)^T$, $v_2 = (2, 1, 2)^T$, $v_3 = (1, 2, -2)^T$. Solve $c_1v_1 + c_2v_2 + c_3v_3 = (1, 11, -2)^T$:

$$-2c_1 + 2c_2 + c_3 = 1, \quad 2c_1 + c_2 + 2c_3 = 11, \quad c_1 + 2c_2 - 2c_3 = -2.$$

Elimination gives $(c_1, c_2, c_3) = (2, 1, 3)$; substitution checks all three coordinates. Therefore $x_k = 2v_1 + (2/3)^k v_2 + 3(1/3)^k v_3$.

Practice problem 5.6.P2: Retain only the nondecaying mode

The powers $(2/3)^k$ and $(1/3)^k$ tend to zero. Thus the preceding solution tends to $2v_1 = (-4, 4, 2)^T$, rather than zero. The eigenvalue 1 preserves its component indefinitely.

Exercise 5.6.1: Use the given eigenvectors without reconstructing the matrix

Solve $c_1(1, 1)^T + c_2(-1, 1)^T = (9, 1)^T$. The coordinate equations $c_1 - c_2 = 9$, $c_1 + c_2 = 1$ give $c_1 = 5, c_2 = -4$.

(a) Apply A once to the decomposition.

$x_1 = 5(3)(1, 1)^T - 4(1/3)(-1, 1)^T = (49/3, 41/3)^T$.

(b) Apply the same rule k times.

$$x_k = 5\,3^k(1, 1)^T - 4\,(1/3)^k(-1, 1)^T.$$

At $k = 0$ this gives the specified initial vector; multiplication by A verifies the recurrence at every subsequent step.

Exercise 5.6.2: Separate the growing mode from two decaying modes

The coefficient system $[v_1\ v_2\ v_3]c = x_0$ is

$$c_1 + 2c_2 - 3c_3 = -2, \quad c_2 - 3c_3 = -5, \quad -3c_1 - 5c_2 + 7c_3 = 3.$$

It gives $c = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. Hence

$$x_k = 2 \cdot 3^k v_1 + (4/5)^k v_2 + 2(3/5)^k v_3.$$

The last two modes decay, whereas the first grows. More precisely, $x_k/3^k \to 2v_1 = (2, 0, -6)^T$. Thus the sequence is unbounded and its direction approaches the line through v_1; its second coordinate tends to zero while its first and third grow in opposite signs.

Exercise 5.6.3: Increase predation in the owl–wood-rat model

For $p = .2$ the matrix is $A = \begin{bmatrix} .5 & .4 \\ -.2 & 1.1 \end{bmatrix}$. The characteristic polynomial is $\lambda^2 - 1.6\lambda + .63$ with roots $.9, .7$. Eigenvectors are $(1, 1)^T$ and $(2, 1)^T$, as direct multiplication verifies. Thus

$$x_k = c_1(.9)^k(1, 1)^T + c_2(.7)^k(2, 1)^T.$$

Both modes decay. Under the exercise's $c_1 > 0$ assumption, the long-term positive populations both decline with factor approaching .9 per month. The first component counts owls; the second is the number of wood rats in thousands, as in Example 1.

Exercise 5.6.4: The borderline between eventual decline and growth

With $p = .125$, the matrix has eigenvalues $1, .6$ and eigenvectors $(4, 5)^T, (4, 1)^T$. For instance $A(4, 5)^T = (4, 5)^T$ and $A(4, 1)^T = .6(4, 1)^T$. Consequently

$$x_k = c_1(4, 5)^T + c_2(.6)^k(4, 1)^T \longrightarrow c_1(4, 5)^T.$$

The populations approach constant levels rather than tending to zero or growing indefinitely. Small parameter changes can move the largest eigenvalue below or above 1, changing the prediction to decline or growth. This explains the sensitivity called an unstable equilibrium in the exercise; the equilibrium line is not a robust attracting equilibrium under model changes. Coordinate ratio $4 : 5$ means 4 owls per 5 thousand rats under the model's units.

Exercise 5.6.5: Long-term growth and the dominant population ratio

For $p = .325$, $A = \begin{bmatrix} .4 & .3 \\ -.325 & 1.2 \end{bmatrix}$. Its eigenvalues solve $\lambda^2 - 1.6\lambda + .5775 = 0$, giving $1.05, .55$. A dominant eigenvector is $(6, 13)^T$, because $A(6, 13)^T = (6.3, 13.65)^T = 1.05(6, 13)^T$. Under the stated positive dominant-coefficient assumption, x_k is asymptotic to $c_1(1.05)^k(6, 13)^T$.

Conclusion. Both populations eventually grow about 5% per step, and their coordinate ratio tends to $6 : 13$. If the prey coordinate uses the thousands convention of Example 1, this is 6 owls per 13 thousand squirrels; coordinate scaling must be retained when interpreting a physical ratio.

Exercise 5.6.6: Choose the predation rate that preserves population levels

At $p = .5$, the characteristic polynomial is $\lambda^2 - 1.6\lambda + .63$, whose roots are $.9, .7$. Both powers tend to zero, so both populations perish in this model. For constant limiting levels, require an eigenvalue 1:

$$\det(A - I) = (-.6)(.2) - (.3)(-p) = -.12 + .3p = 0.$$

Thus $p = .4$. The other eigenvalue is .6, and a 1-eigenvector is $(1, 2)^T$. Therefore the solutions tend to $c_1(1, 2)^T$.

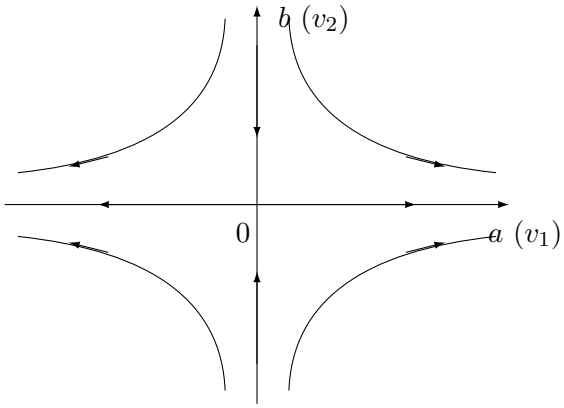
Conclusion. The preserving parameter is $p = .4$, and the limiting coordinate ratio is $1 : 2$, with prey-unit scaling interpreted as in the preceding exercise.

Exercise 5.6.7: A saddle with one attracting and one repelling direction

(a) Since $3 > 1$ but $1/3 < 1$, the origin is a saddle point.

(b) The line spanned by $v_1 = (1, 1)^T$ is repelling, with factor 3 each step. The line spanned by $v_2 = (-1, 1)^T$ is attracting, with factor $1/3$. These are also the greatest repulsion and attraction directions, respectively.

(c) In eigen-coordinates $x = av_1 + bv_2$, the update is $(a, b) \mapsto (3a, b/3)$. The sketch uses those coordinates: horizontal arrows point away from zero and vertical arrows toward zero. Off-axis trajectories approach the repelling axis while moving outward. The curves are schematic connections of discrete points, not continuous-time trajectories; ab remains constant here.



Exercise 5.6.8: Two stable modes and one unstable mode

The eigenvalues from Exercise 5.6.2 are $3, 4/5, 3/5$. A mode expands along $(1, 0, -3)^T$, while the two-dimensional span of the other eigenvectors is attracted to zero. The fastest contraction has the smaller modulus $3/5$, along $(-3, -3, 7)^T$; the greatest repulsion has modulus 3, along $(1, 0, -3)^T$.

Conclusion. The origin is a saddle. The initial vector in Exercise 2 has coefficient $c_1 = 2$, so its trajectory is repelled. Only initial vectors in the stable plane ($c_1 = 0$) tend to zero.

Exercise 5.6.9: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{17}{10} & -\frac{3}{10} \\ -\frac{6}{5} & \frac{4}{5} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(\lambda - 2)(2\lambda - 1)}{2} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{1}{2},\ v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}; \qquad \lambda = 2,\ v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **saddle point**. The attracting direction is $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ (factor $\frac{1}{2}$), and the repelling direction is $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ (factor 2).

Exercise 5.6.10: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{3}{10} & \frac{2}{5} \\ -\frac{3}{10} & \frac{11}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(2\lambda - 1)(10\lambda - 9)}{20} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{1}{2},\ v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{9}{10},\ v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **attractor**. The greatest attraction is along $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, for the smaller eigenvalue $\frac{1}{2}$. There is no repelling eigendirection.

Exercise 5.6.11: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{2}{5} & \frac{1}{2} \\ -\frac{2}{5} & \frac{13}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 4)(10\lambda - 9)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{4}{5},\ v = \begin{bmatrix} 5 \\ 4 \end{bmatrix}; \qquad \lambda = \frac{9}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **attractor**. The greatest attraction is along $\begin{bmatrix} 5 \\ 4 \end{bmatrix}$, for the smaller eigenvalue $\frac{4}{5}$. There is no repelling eigendirection.

Exercise 5.6.12: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{1}{2} & \frac{3}{5} \\ -\frac{3}{10} & \frac{7}{5} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 4)(10\lambda - 11)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{4}{5},\ v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{11}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **saddle point**. The attracting direction is $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (factor $\frac{4}{5}$), and the repelling direction is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (factor $\frac{11}{10}$).

Exercise 5.6.13: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{4}{5} & \frac{3}{10} \\ -\frac{2}{5} & \frac{3}{2} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(5\lambda - 6)(10\lambda - 11)}{50} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{11}{10},\ v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{6}{5},\ v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **repeller**. The greatest repulsion is along $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$, for the larger eigenvalue $\frac{6}{5}$. There is no attracting eigendirection.

Exercise 5.6.14: Classify the origin by the two eigenvalue moduli

For $A = \begin{bmatrix} \frac{17}{10} & \frac{3}{5} \\ -\frac{2}{5} & \frac{7}{10} \end{bmatrix}$, the characteristic polynomial factors as

$$\frac{(10\lambda - 13)(10\lambda - 11)}{100} = 0.$$

The eigenpairs, checked by $Av = \lambda v$, are

$$\lambda = \frac{11}{10},\ v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \qquad \lambda = \frac{13}{10},\ v = \begin{bmatrix} -3 \\ 2 \end{bmatrix}.$$

A decomposition in these two independent directions evolves by multiplying their coefficients by the corresponding eigenvalues at each step. Comparing their moduli with 1 makes the origin an **repeller**. The greatest repulsion is along $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$, for the larger eigenvalue $\frac{13}{10}$. There is no attracting eigendirection.

Exercise 5.6.15: A stochastic model with a nonzero limiting vector

The columns of A sum to 1. The given $v_1 = (.1, .6, .3)^T$ has $Av_1 = v_1$. Eigenvectors for $.5, .2$ are $v_2 = (2, -3, 1)^T$ and $v_3 = (-1, 0, 1)^T$. Solving $c_1v_1 + c_2v_2 + c_3v_3 = (0, .3, .7)^T$ gives $c_1 = 1, c_2 = 1/10, c_3 = 3/10$. Thus

$$x_k = v_1 + \tfrac{1}{10}(.5)^k v_2 + \tfrac{3}{10}(.2)^k v_3.$$

At zero the coordinates are $(.1 + .2 - .3, .6 - .3, .3 + .1 + .3)$, as required. The last two modes tend to zero, so $x_k \to v_1$. The limiting proportions are $10\%, 60\%, 30\%$.

Exercise 5.6.16: The full Hertz three-location solution

In City Airport, Downtown, Metro Airport order, the matrix from Section 4.9 is

$$A = \frac{1}{100}\begin{bmatrix} 90 & 1 & 9 \\ 1 & 90 & 1 \\ 9 & 9 & 90 \end{bmatrix}.$$

Exact calculation gives three eigenpairs:

$$1,\ (91, 19, 99)^T; \qquad .89,\ (-1, 1, 0)^T; \qquad .81,\ (-1, 0, 1)^T.$$

The eigenvalues are distinct, so their eigenvectors form a basis. Therefore the general solution is

$$x_k = c_1(91, 19, 99)^T + c_2(.89)^k(-1, 1, 0)^T + c_3(.81)^k(-1, 0, 1)^T.$$

The constants are determined by the initial fleet distribution. For total fleet size N, adding the coordinates gives $N = 209c_1$, so the limit is $(N/209)(91, 19, 99)^T$. At $N = 2000$ the downtown limit is $2000(19)/209 \approx 181.818$ cars. This recovers the steady-state interpretation while providing the requested general dynamics.

Exercise 5.6.17: A two-stage growing population with all four requested graphs

(a) Translate births and survival into matrix entries.

Juveniles next year are born to this year's adults, while next year's adults are surviving juveniles plus surviving adults. Hence

$$A = \begin{bmatrix} 0 & 1.6 \\ .3 & .8 \end{bmatrix}.$$

(b) Find growth and the eventual ratio.

The characteristic polynomial is $\lambda^2 - .8\lambda - .48$, with roots $1.2, -.4$. A dominant eigenvector is $(4, 3)^T$; a second is $(-4, 1)^T$. The growing mode therefore has factor 1.2, a 20% annual growth rate, and limiting juvenile/adult ratio 4/3. For a nonzero nonnegative initial population the growing coefficient is positive: solving $x_0 = c_1(4, 3)^T + c_2(-4, 1)^T$ gives $c_1 = (j_0 + 4a_0)/16 > 0$.

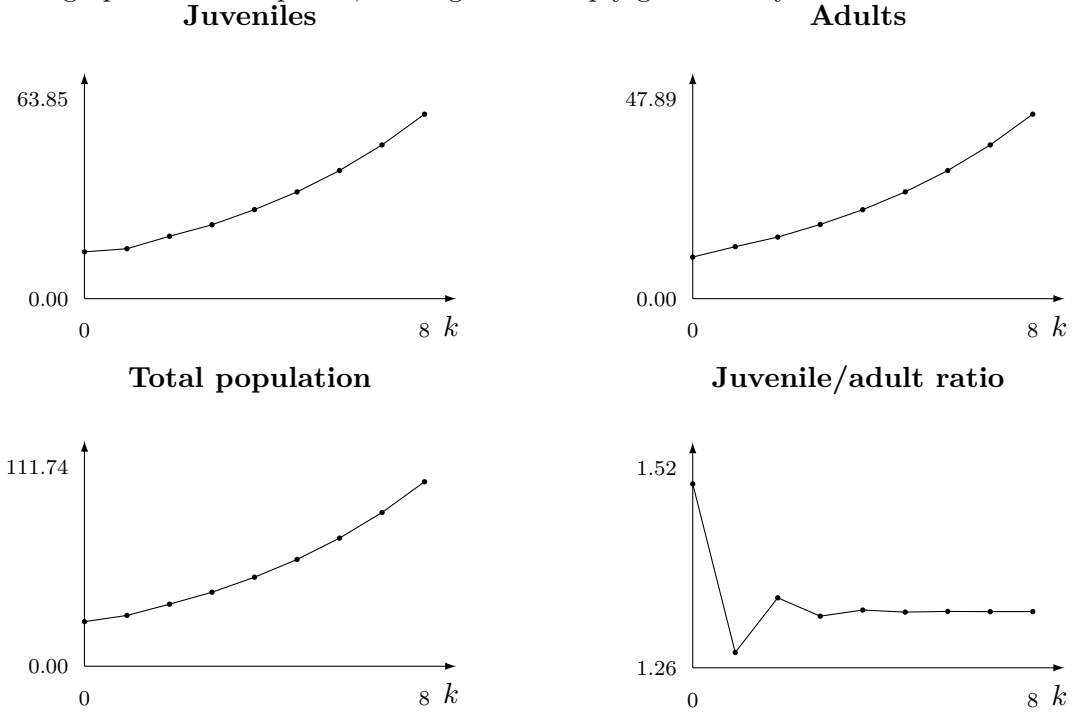
(c) Calculate the eight-year trajectory and plot it.

Beginning at $(15, 10)^T$, the actual iterates are

k	j_k	a_k	j_k/a_k
0	15.000000	10.000000	1.500000
1	16.000000	12.500000	1.280000
2	20.000000	14.800000	1.351351
3	23.680000	17.840000	1.327354
4	28.544000	21.376000	1.335329
5	34.201600	25.664000	1.332668
6	41.062400	30.791680	1.333555
7	49.266688	36.952064	1.333259
8	59.123302	44.341658	1.333358

.

The four graphs use these points; line segments simply guide the eye between successive integer years.



The ratio oscillates toward 4/3 because the secondary eigenvalue is negative. It is already close by years 3–4 and differs by less than .001 from 4/3 by year 5. The underlying relative mode shrinks by $|-.4/1.2| = 1/3$ each year. A complete plotting program, including the requested total and ratio panels, is:

```
import numpy as np
import matplotlib.pyplot as plt
A = np.array([[0.,1.6],[.3,.8]])
x = np.array([15.,10.])
values = [x.copy()]
for k in range(8):
    x = A @ x
    values.append(x.copy())
v = np.array(values)
series = [v[:,0],v[:,1],v.sum(axis=1),v[:,0]/v[:,1]]
```

```
labels = ['Juveniles','Adults','Total','Juveniles/adults']
fig, axes = plt.subplots(2,2)
for ax, y, label in zip(axes.flat, series, labels):
    ax.plot(range(9), y, 'o-')
    ax.set(xlabel='Year', ylabel=label)
fig.tight_layout()
plt.show()
```

Exercise 5.6.18: A three-stage buffalo model

(a) Construct the stage matrix.

Only adults give birth. Calves survive into the yearling stage and yearlings into adulthood. Thus

$$A = \begin{bmatrix} 0 & 0 & .42 \\ .60 & 0 & 0 \\ 0 & .75 & .95 \end{bmatrix}.$$

(b) Compute the dominant mode and its composition.

The characteristic polynomial is $\lambda^3 - .95\lambda^2 - .189$. Its real positive root is $\lambda_1 \approx 1.104834392$; the other two roots are complex conjugates of modulus less than 1. Since $\lambda_1 > 1$, the nonzero nonnegative population grows asymptotically by factor 1.104834392, about 10.483439% per year. Normalize the dominant eigenvector to 100 adults. The first row of $Av = \lambda_1 v$ gives $c = 42/\lambda_1$, and the second gives $y = .6c/\lambda_1$. Therefore the eventual composition per 100 adults is approximately

$$(c, y, a) = (38.014747, 20.644586, 100).$$

Substituting these formulas into the third row reproduces the characteristic equation; this checks the eigenvector normalization.

Show that every nonzero nonnegative initial population has a growing mode.

An initial state could in principle have zero coefficient in a particular eigenvector direction, so we must verify that the dominant coefficient is positive here. A useful row multiplier is the positive left eigenvector

$$w = \begin{bmatrix} .45/\lambda_1^2 \\ .75/\lambda_1 \\ 1 \end{bmatrix}.$$

Indeed direct multiplication gives

$$w^T A = \begin{bmatrix} .45/\lambda_1 & .75 & .189/\lambda_1^2 + .95 \end{bmatrix} = \lambda_1 w^T.$$

The last equality uses the characteristic equation, divided by λ_1^2: $\lambda_1 = .95 + .189/\lambda_1^2$. All three entries of w are strictly positive because $\lambda_1 > 0$.

The three distinct eigenvalues give an eigenvector expansion $x_0 = c_1 v_1 + c_2 v_2 + c_3 v_3$, allowing complex coefficients for the conjugate pair. For $j = 2, 3$, calculate the same scalar in two ways:

$$w^T A v_j = \lambda_1 w^T v_j = \lambda_j w^T v_j.$$

Since $\lambda_j \neq \lambda_1$, this forces $w^T v_j = 0$. Multiplying the expansion of x_0 by w^T therefore isolates

$$c_1 = \frac{w^T x_0}{w^T v_1} > 0.$$

The numerator is positive whenever x_0 has nonnegative entries and at least one positive entry; the denominator is positive because v_1 has the positive components computed above. Consequently

$$\frac{x_k}{\lambda_1^k} = c_1 v_1 + c_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k v_2 + c_3 \left(\frac{\lambda_3}{\lambda_1}\right)^k v_3 \longrightarrow c_1 v_1.$$

Both remaining ratios have modulus less than 1, so their powers tend to zero. Thus the dominant mode really is present for every nonzero nonnegative initial population, justifying both the growth rate and the eventual stage proportions claimed above.

5.7 Applications to differential equations

For continuous time the rule is $x'(t) = Ax(t)$. If $Av = \lambda v$, then $ve^{\lambda t}$ is a solution because its derivative is $\lambda ve^{\lambda t} = Ave^{\lambda t}$. For a real eigenbasis, expand the initial vector in that basis and replace each coefficient's evolution by $e^{\lambda t}$.

Complex eigenvalues. If $\lambda = a + bi$ and $v = p + iq$ with p, q real and $b > 0$, a conjugate pair of complex eigenfunctions is $ve^{(a+bi)t}$ and $\overline{v}e^{(a-bi)t}$. A real basis for the same solution space is

$$r_1(t) = e^{at}(p\cos bt - q\sin bt), \qquad r_2(t) = e^{at}(p\sin bt + q\cos bt).$$

This follows by multiplying out Euler's identity $e^{ibt} = \cos bt + i\sin bt$ and taking real and imaginary parts. For real initial data use arbitrary real coefficients of r_1, r_2. The sign of the real part a, rather than $|\lambda|$, decides continuous-time decay or growth.

Practice problem 5.7.P1: Diagonalize over the complex numbers

The three given eigenvalues $-.5, .2+.3i, .2-.3i$ are distinct. Their eigenvectors are consequently linearly independent over $\mathbb{C}$. The matrix $P = [v_1\ v_2\ v_3]$ is invertible and $A = P\,\mathrm{diag}(-.5, .2+.3i, .2-.3i)P^{-1}$.

Practice problem 5.7.P2: Convert the complex general solution to a real one

The complex general solution is $x = c_1v_1e^{-.5t} + c_2v_2e^{(.2+.3i)t} + c_3v_3e^{(.2-.3i)t}$, with complex constants. For real solutions use $v_1 = (1, -2, 1)^T$, $p = (1, 0, 2)^T$, and $q = (2, 4, 0)^T$. The real general solution is

$$x = d_1e^{-.5t}(1, -2, 1)^T + d_2r_1(t) + d_3r_2(t),$$

where r_1, r_2 are the formulas above with $a = .2, b = .3$ and all d_j real. This gives every real solution, since at zero the vectors v_1, p, q are linearly independent.

Practice problem 5.7.P3: Describe the stable line and unstable spiral plane

On the line spanned by v_1 trajectories approach zero like $e^{-.5t}$. In the plane spanned by p, q, they spiral outward with exponential factor $e^{.2t}$. A typical three-dimensional trajectory has both components: its stable-line component decays while its rotating planar component expands. The origin is a spiral saddle, not an attractor.

Exercise 5.7.1: Determine the constants from the initial position

Write $x(0) = c_1v_1 + c_2v_2$. In coordinates this is

$$\begin{bmatrix} -3 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}.$$

Solving the two scalar equations gives $c_1 = \frac{5}{2}$, $c_2 = -\frac{3}{2}$. Therefore

$$x(t) = \frac{5}{2}e^{4t}\begin{bmatrix} -3 \\ 1 \end{bmatrix} + \left(-\frac{3}{2}\right)e^{2t}\begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

At $t = 0$ the coefficients reproduce the specified vector. Differentiating each term and using $Av_j = \lambda_jv_j$ verifies $x' = Ax$.

Exercise 5.7.2: Determine the constants from the initial position

Write $x(0) = c_1v_1 + c_2v_2$. In coordinates this is

$$\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Solving the two scalar equations gives $c_1 = \frac{1}{2}$, $c_2 = \frac{5}{2}$. Therefore

$$x(t) = \frac{1}{2}e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \left(\frac{5}{2}\right) e^{-1t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

At $t = 0$ the coefficients reproduce the specified vector. Differentiating each term and using $Av_j = \lambda_j v_j$ verifies $x' = Ax$.

Exercise 5.7.3: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 1)(\lambda + 1) = 0.$$

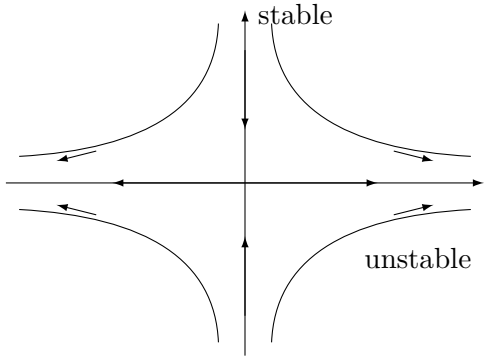
Its two eigenpairs are

$$\lambda_1 = -1,\ v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \quad \lambda_2 = 1,\ v_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

Solving $c_1v_1 + c_2v_2 = (3, 2)^T$ gives $(c_1, c_2) = (\frac{9}{2}, -\frac{5}{2})$. Hence the required initial-value solution is

$$x(t) = \left(\frac{9}{2}\right) e^{-1t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \left(-\frac{5}{2}\right) e^{1t} \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **saddle point**. The negative eigenvalue's eigenvector v_1 gives the attracting direction; the positive eigenvalue's v_2 gives the repelling direction. The schematic sketch is in these eigen-coordinates: vertical is stable, horizontal unstable. Transforming it by $[v_2\ v_1]$ gives the corresponding picture in the original plane.



Exercise 5.7.4: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} -2 & -5 \\ 1 & 4 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 3)(\lambda + 1) = 0.$$

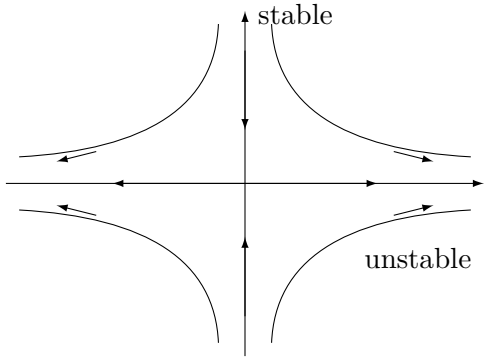
Its two eigenpairs are

$$\lambda_1 = -1,\ v_1 = \begin{bmatrix} -5 \\ 1 \end{bmatrix}; \quad \lambda_2 = 3,\ v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-\frac{5}{4}, \frac{13}{4})$. Hence the required initial-value solution is

$$x(t) = \left(-\frac{5}{4}\right) e^{-1t} \begin{bmatrix} -5 \\ 1 \end{bmatrix} + \left(\frac{13}{4}\right) e^{3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **saddle point**. The negative eigenvalue's eigenvector v_1 gives the attracting direction; the positive eigenvalue's v_2 gives the repelling direction. The schematic sketch is in these eigen-coordinates: vertical is stable, horizontal unstable. Transforming it by $[v_2\ v_1]$ gives the corresponding picture in the original plane.



Exercise 5.7.5: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 7 & -1 \\ 3 & 3 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda - 6)(\lambda - 4) = 0.$$

Its two eigenpairs are

$$\lambda_1 = 4,\ v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}; \quad \lambda_2 = 6,\ v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-\frac{1}{2}, \frac{7}{2})$. Hence the required initial-value solution is

$$x(t) = \left(-\frac{1}{2}\right) e^{4t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \left(\frac{7}{2}\right) e^{6t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **repeller**. Both exponential factors grow; the greater eigenvalue 6 gives the fastest repulsion, along $(1, 1)^T$. There is no attracting eigendirection.

Exercise 5.7.6: Solve, classify, and identify the principal directions

The coefficient matrix is $A = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$. Factoring its characteristic polynomial gives

$$(\lambda + 1)(\lambda + 2) = 0.$$

Its two eigenpairs are

$$\lambda_1 = -2,\ v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}; \quad \lambda_2 = -1,\ v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solving $c_1 v_1 + c_2 v_2 = (3, 2)^T$ gives $(c_1, c_2) = (-1, 5)$. Hence the required initial-value solution is

$$x(t) = (-1)\, e^{-2t} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + (5)\, e^{-1t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Both the initial value and $x' = Ax$ check by substitution. The origin is an **attractor**. Both exponents are negative. The fastest attraction is along $(2, 3)^T$, for eigenvalue -2; the other mode decays more slowly like e^{-t}. There is no repelling eigendirection.

Exercise 5.7.7: A constant eigenbasis decouples the differential equations

Choose the eigenvectors from Exercise 5.7.5 as columns:

$$P = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 \\ 0 & 6 \end{bmatrix}.$$

Direct multiplication gives $AP = PD$, and $\det P = -2 \neq 0$. Put $x(t) = Py(t)$. Since P is constant, $x' = Py'$, so

$$Py' = APy = PDy, \qquad y' = Dy.$$

The last equality uses multiplication by P^{-1}. Thus $y_1' = 4y_1$ and $y_2' = 6y_2$, with no cross-coupling. Their scalar solutions are exponentials, and $x = Py$ converts back.

Exercise 5.7.8: A constant eigenbasis decouples the differential equations

Choose the eigenvectors from Exercise 5.7.6 as columns:

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}.$$

Direct multiplication gives $AP = PD$, and $\det P = -1 \neq 0$. Put $x(t) = Py(t)$. Since P is constant, $x' = Py'$, so

$$Py' = APy = PDy, \qquad y' = Dy.$$

The last equality uses multiplication by P^{-1}. Thus $y_1' = -2y_1$ and $y_2' = -1y_2$, with no cross-coupling. Their scalar solutions are exponentials, and $x = Py$ converts back.

Exercise 5.7.9: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -3 & 2 \\ -1 & -1 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 4\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = -2 + i, \qquad v = \begin{bmatrix} 1 - i \\ 1 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(-2+i)t} + c_2 \overline{v} e^{(-2-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{-2t}\big[p\cos(1t) - q\sin(1t)\big],$$
$$r_2 = e^{-2t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation in the original plane, and the negative real part produces decay. Every nonzero trajectory spirals inward to the origin.

Exercise 5.7.10: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 - 4\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = 2 + i, \qquad v = \begin{bmatrix} -1 - i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(2+i)t} + c_2 \overline{v} e^{(2-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{2t}\big[p\cos(1t) - q\sin(1t)\big],$$
$$r_2 = e^{2t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation, while the positive real part expands its size. Nonzero trajectories spiral outward from the origin.

Exercise 5.7.11: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -3 & -9 \\ 2 & 3 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 9.$$

A positive-imaginary eigenpair is

$$\lambda = 3i, \qquad v = \begin{bmatrix} -3 + 3i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} -3 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(3i)t} + c_2 \overline{v} e^{(-3i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{0t}\big[p\cos(3t) - q\sin(3t)\big],$$
$$r_2 = e^{0t}\big[p\sin(3t) + q\cos(3t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is no exponential growth or decay. The real solutions trace closed ellipses around the origin, with period $2\pi/3$; the origin is a center.

Exercise 5.7.12: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -7 & 10 \\ -4 & 5 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 2\lambda + 5.$$

A positive-imaginary eigenpair is

$$\lambda = -1 + 2i, \qquad v = \begin{bmatrix} 3 - i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(-1+2i)t} + c_2 \overline{v} e^{(-1-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{-1t}\big[p\cos(2t) - q\sin(2t)\big],$$
$$r_2 = e^{-1t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation in the original plane, and the negative real part produces decay. Every nonzero trajectory spirals inward to the origin.

Exercise 5.7.13: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 4 & -3 \\ 6 & -2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 - 2\lambda + 10.$$

A positive-imaginary eigenpair is

$$\lambda = 1 + 3i, \qquad v = \begin{bmatrix} 1+i \\ 2 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(1+3i)t} + c_2 \overline{v} e^{(1-3i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{1t}\big[p\cos(3t) - q\sin(3t)\big],$$
$$r_2 = e^{1t}\big[p\sin(3t) + q\cos(3t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

The sine and cosine produce elliptical rotation, while the positive real part expands its size. Nonzero trajectories spiral outward from the origin.

Exercise 5.7.14: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -2 & 1 \\ -8 & 2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = \lambda^2 + 4.$$

A positive-imaginary eigenpair is

$$\lambda = 2i, \qquad v = \begin{bmatrix} 1-i \\ 4 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \qquad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(2i)t} + c_2 \overline{v} e^{(-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{0t}\big[p\cos(2t) - q\sin(2t)\big],$$
$$r_2 = e^{0t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is no exponential growth or decay. The real solutions trace closed ellipses around the origin, with period $2\pi/2$; the origin is a center.

Exercise 5.7.15: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -8 & -12 & -6 \\ 2 & 1 & 2 \\ 7 & 12 & 5 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 1)(\lambda + 1)(\lambda + 2).$$

All eigenvalues in this particular computer exercise are real, which is a special case of the requested complex-eigenfunction method. A complete eigenbasis is

$$\lambda_1 = -2, \quad v_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\lambda_2 = -1, \quad v_2 = \begin{bmatrix} -6 \\ 1 \\ 5 \end{bmatrix}.$$

$$\lambda_3 = 1, \quad v_3 = \begin{bmatrix} -4 \\ 1 \\ 4 \end{bmatrix}.$$

Thus the general complex solution is

$$x(t) = \sum_{j=1}^{3} c_j e^{\lambda_j t} v_j, \qquad c_j \in \mathbb{C}.$$

The general real solution is the same formula with $c_j \in \mathbb{R}$, because this basis consists of real vectors. Each eigenpair checks by $Av_j = \lambda_j v_j$, and distinct eigenvalues give a full basis. There are two negative eigenvalues and one positive eigenvalue. The stable plane is $\mathrm{Span}\{v_1, v_2\}$ and the repelling line is $\mathrm{Span}\{v_3\}$. Typical trajectories first combine these modes and eventually move outward in the positive-eigenvalue direction: the origin is a saddle, with no oscillatory spiral in this example.

Exercise 5.7.16: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} -6 & -11 & 16 \\ 2 & 5 & -4 \\ -4 & -5 & 10 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 4)(\lambda - 3)(\lambda - 2).$$

All eigenvalues in this particular computer exercise are real, which is a special case of the requested complex-eigenfunction method. A complete eigenbasis is

$$\lambda_1 = 2, \quad v_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

$$\lambda_2 = 3, \quad v_2 = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}.$$

$$\lambda_3 = 4, \quad v_3 = \begin{bmatrix} 7 \\ -2 \\ 3 \end{bmatrix}.$$

Thus the general complex solution is

$$x(t) = \sum_{j=1}^{3} c_j e^{\lambda_j t} v_j, \qquad c_j \in \mathbb{C}.$$

The general real solution is the same formula with $c_j \in \mathbb{R}$, because this basis consists of real vectors. Each eigenpair checks by $Av_j = \lambda_j v_j$, and distinct eigenvalues give a full basis. All three eigenvalues are positive. The origin is a repeller; generic trajectories eventually align with the fastest growing $\lambda = 4$ eigenvector, while exact invariant eigenspaces retain their own exponential behavior. There is no spiral because the spectrum is real.

Exercise 5.7.17: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 30 & 64 & 23 \\ -11 & -23 & -9 \\ 6 & 15 & 4 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda - 1)\left(\lambda^2 - 10\lambda + 29\right).$$

A positive-imaginary eigenpair is

$$\lambda = 5 + 2i, \qquad v = \begin{bmatrix} 23 - 34i \\ -9 + 14i \\ 3 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 23 \\ -9 \\ 3 \end{bmatrix}, \qquad q = \begin{bmatrix} -34 \\ 14 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(5+2i)t} + c_2 \overline{v} e^{(5-2i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{5t}\big[p\cos(2t) - q\sin(2t)\big],$$

$$r_2 = e^{5t}\big[p\sin(2t) + q\cos(2t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is also the real eigenpair $\lambda_0 = 1$, $w = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$. Add $c_0 e^{1t} w$ to the complex solution and $d_0 e^{1t} w$ to the real one. These three independent modes give the entire solution space. The real mode grows like e^t and the spiral-plane modes grow like e^{5t}. Thus the origin is repelling: generic trajectories spiral outward increasingly close to the invariant plane $\mathrm{Span}\{p, q\}$; only the real eigendirection has a nonrotating trajectory.

Exercise 5.7.18: Complex eigenfunctions and a full real solution basis

For the given matrix $A = \begin{bmatrix} 53 & -30 & -2 \\ 90 & -52 & -3 \\ 20 & -10 & 2 \end{bmatrix}$, exact calculation gives

$$\det(\lambda I - A) = (\lambda + 7)\left(\lambda^2 - 10\lambda + 26\right).$$

A positive-imaginary eigenpair is

$$\lambda = 5 + i, \qquad v = \begin{bmatrix} 6 + 2i \\ 9 + 3i \\ 10 \end{bmatrix} = p + iq,$$

where

$$p = \begin{bmatrix} 6 \\ 9 \\ 10 \end{bmatrix}, \qquad q = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}.$$

Checking $Av = \lambda v$ verifies the pair; since A is real, $\overline{v}, \overline{\lambda}$ give the conjugate pair. The corresponding complex solution is

$$c_1 v e^{(5+i)t} + c_2 \overline{v} e^{(5-i)t}.$$

Its real version is $d_1 r_1(t) + d_2 r_2(t)$, where explicitly

$$r_1 = e^{5t}\big[p\cos(1t) - q\sin(1t)\big],$$

$$r_2 = e^{5t}\big[p\sin(1t) + q\cos(1t)\big], \qquad d_1, d_2 \in \mathbb{R}.$$

There is also the real eigenpair $\lambda_0 = -7$, $w = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$. Add $c_0 e^{-7t} w$ to the complex solution and $d_0 e^{-7t} w$ to the real one. These three independent modes give the entire solution space. The line $\mathrm{Span}\{w\}$ is stable, decaying like e^{-7t}. The plane $\mathrm{Span}\{p, q\}$ spirals outward like e^{5t}. Thus this is a spiral saddle: only points on the stable line approach zero, and other trajectories eventually spiral outward.

Exercise 5.7.19: Substitute the circuit constants before solving the voltage system

Example 1 gives

$$A = \begin{bmatrix} -(1/R_1 + 1/R_2)/C_1 & 1/(R_2 C_1) \\ 1/(R_2 C_2) & -1/(R_2 C_2) \end{bmatrix} = \begin{bmatrix} -2 & \frac{3}{4} \\ 1 & -1 \end{bmatrix}.$$

Its eigenvalues are $-\frac{5}{2}, -\frac{1}{2}$, with eigenvector matrix $P = \begin{bmatrix} -3 & 1 \\ 2 & 2 \end{bmatrix}$. Solving $Pc = (4, 4)^T$ gives $c = \begin{bmatrix} -\frac{1}{2} \\ \frac{5}{2} \end{bmatrix}$. Therefore the requested voltage vector is

$$\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \left(-\frac{1}{2}\right) e^{-\frac{5}{2}t} \begin{bmatrix} -3 \\ 2 \end{bmatrix} + \left(\frac{5}{2}\right) e^{-\frac{1}{2}t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Each component at zero is the stipulated initial voltage, and differentiating verifies the given circuit equation. Both exponentials decay, consistent with a passive resistive circuit.

Exercise 5.7.20: Substitute the circuit constants before solving the voltage system

Example 1 gives

$$A=\begin{bmatrix}-(1/R_1+1/R_2)/C_1 & 1/(R_2C_1)\\ 1/(R_2C_2) & -1/(R_2C_2)\end{bmatrix}=\begin{bmatrix}-2 & \frac{1}{3}\\ \frac{3}{2} & -\frac{3}{2}\end{bmatrix}.$$

Its eigenvalues are $-\frac{5}{2},-1$, with eigenvector matrix $P=\begin{bmatrix}-2 & 1\\ 3 & 3\end{bmatrix}$. Solving $Pc=(3,3)^T$ gives $c=\begin{bmatrix}-\frac{2}{3}\\ \frac{5}{3}\end{bmatrix}$. Therefore the requested voltage vector is

$$\begin{bmatrix}v_1(t)\\ v_2(t)\end{bmatrix}=\left(-\frac{2}{3}\right)e^{-\frac{5}{2}t}\begin{bmatrix}-2\\ 3\end{bmatrix}+\left(\frac{5}{3}\right)e^{-1t}\begin{bmatrix}1\\ 3\end{bmatrix}.$$

Each component at zero is the stipulated initial voltage, and differentiating verifies the given circuit equation. Both exponentials decay, consistent with a passive resistive circuit.

Exercise 5.7.21: A damped oscillation in current and capacitor voltage

Using Example 3's circuit matrix with the specified constants gives

$$A=\begin{bmatrix}-R_2/L & -1/L\\ 1/C & -1/(R_1C)\end{bmatrix}=\begin{bmatrix}-1 & -8\\ 5 & -5\end{bmatrix}.$$

Its eigenvalues are $-3\pm 6i$. Put $B=A+3I$; direct multiplication gives $B^2=-36I$. Consequently the solution for $x_0=(0,15)^T$ is

$$x(t)=e^{-3t}\big[x_0\cos 6t+(Bx_0/6)\sin 6t\big].$$

Here $Bx_0=(-120,-30)^T$, so

$$i_L(t)=-20e^{-3t}\sin 6t,\qquad v_C(t)=e^{-3t}(15\cos 6t-5\sin 6t).$$

At zero these give 0 amps and 15 volts. Differentiating gives $x'=Ax$; the identity $B^2=-36I$ verifies the formula directly.

Exercise 5.7.22: Solve the second circuit with its stated sign convention

The problem supplies a different circuit matrix; with its constants,

$$A=\begin{bmatrix}0 & 1/L\\ -1/C & -1/(RC)\end{bmatrix}=\begin{bmatrix}0 & 2\\ -2/5 & -4/5\end{bmatrix}.$$

Its eigenvalues are $-2/5\pm(4/5)i$. Set $a=-2/5, b=4/5$ and $B=A-aI$. Then $B^2=-b^2I$ and $Bx_0=(24,-24/5)^T$ for $x_0=(0,12)^T$. The same cosine–sine construction gives

$$i_L(t)=30e^{-2t/5}\sin(4t/5),$$

$$v_C(t)=e^{-2t/5}\big[12\cos(4t/5)-6\sin(4t/5)\big].$$

The initial data and the differential equation both check directly. The signs differ from Exercise 21 because the supplied circuit and current-direction convention differ.

5.8 Iterative estimates for eigenvalues

Follow the book's indexing carefully. Starting with x_0, compute $y_k = Ax_k$, choose an entry μ_k of y_k with largest absolute value (retain its sign), and set $x_{k+1} = y_k/\mu_k$. Thus μ_1 comes from Ax_1, not Ax_0. A negative dominant eigenvalue need not cause the normalized vectors to alternate, because the signed normalization removes the reversal.

For inverse iteration with a shift α, solve $(A-\alpha I)y_k = x_k$ instead. Normalize by its largest-magnitude entry μ_k, and estimate the original eigenvalue by $\theta_k = \alpha + 1/\mu_k$. Reuse a factorization of $A - \alpha I$; do not explicitly form its inverse at every step. A small residual $\|Av - \theta v\|$ checks a proposed eigenpair. Agreement of two consecutive estimates alone can be misleading.

Practice problem 5.8.P1: Test proportionality and inspect the residual

The given vector $x = (1, -4.3, 8.1)^T$ gives $Ax = (3, -13, 24.5)^T$. The entrywise ratios are 3, $130/43 \approx 3.023256$, and $245/81 \approx 3.024691$. Their close agreement suggests an eigenvalue near 3.024. The Rayleigh estimate is

$$R(x) = \frac{x^T Ax}{x^T x} = \frac{25735}{8510} \approx 3.024089.$$

For $\theta = R(x)$, inspect $Ax - \theta x$ relative to the size of Ax or x. A residual near zero, rather than the raw size of x, is the meaningful approximation test. Exact proportionality would give zero residual and an exact eigenvector.

Exercise 5.8.1: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} 1 \\ \frac{1663}{5000} \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} \frac{24989}{5000} \\ \frac{4163}{2500} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 4.997800$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$, and direct multiplication checks $Av = 5v$ exactly.

Conclusion. The dominant eigenvalue is 5, with eigenvector $\begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$.

Exercise 5.8.2: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} -\frac{63}{250} \\ 1 \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} -\frac{1567}{1250} \\ \frac{3129}{625} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 5.006400$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} -\frac{1}{4} \\ 1 \end{bmatrix}$, and direct multiplication checks $Av = 5v$ exactly.

Conclusion. The dominant eigenvalue is 5, with eigenvector $\begin{bmatrix} -\frac{1}{4} \\ 1 \end{bmatrix}$.

Exercise 5.8.3: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} \frac{1297}{2500} \\ 1 \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} \frac{2297}{5000} \\ \frac{2836}{3125} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 0.907520$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$, and direct multiplication checks $Av = \frac{9}{10}v$ exactly.

Conclusion. The dominant eigenvalue is $\frac{9}{10}$, with eigenvector $\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$.

Exercise 5.8.4: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} 1 \\ \frac{3751}{5000} \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} -\frac{1003}{2500} \\ -\frac{3761}{12500} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx -0.401200$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix}$, and direct multiplication checks $Av = -\frac{2}{5}v$ exactly.

Conclusion. The dominant eigenvalue is $-\frac{2}{5}$, with eigenvector $\begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix}$.

Here "largest" in the exercise means largest in *absolute value*: the other eigenvalue is .1, which is algebraically larger but not dominant.

Exercise 5.8.5: Normalize by the second component of the supplied iterate

Dividing the last vector by its second entry gives

$$x = \begin{bmatrix} -\frac{24991}{31241} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -\frac{7999}{10000} \\ 1 \end{bmatrix}.$$

Use the requested four-decimal vector for a check:

$$A \begin{bmatrix} -\frac{7999}{10000} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{8003}{2000} \\ -\frac{2501}{500} \end{bmatrix}.$$

The second entry estimates the eigenvalue by $-\frac{2501}{500}$, since the corresponding input entry is 1. The nearby exact eigenpair is $\lambda = -5$, $v = \begin{bmatrix} -\frac{4}{5} \\ 1 \end{bmatrix}$, verified by direct multiplication. The rounded estimate is approximate, so exact equality $Ax = \lambda x$ is not expected until using the exact vector.

Exercise 5.8.6: Normalize by the second component of the supplied iterate

Dividing the last vector by its second entry gives

$$x = \begin{bmatrix} -\frac{2045}{4093} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -\frac{1249}{2500} \\ 1 \end{bmatrix}.$$

Use the requested four-decimal vector for a check:

$$A \begin{bmatrix} -\frac{1249}{2500} \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{2501}{1250} \\ \frac{5003}{1250} \end{bmatrix}.$$

The second entry estimates the eigenvalue by $\frac{5003}{1250}$, since the corresponding input entry is 1. The nearby exact eigenpair is $\lambda = 4$, $v = \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix}$, verified by direct multiplication. The rounded estimate is approximate, so exact equality $Ax = \lambda x$ is not expected until using the exact vector.

Exercise 5.8.7: Compute the requested five indexed power-method rows

Use the signed largest-magnitude entry for normalization and do not round intermediate vectors. For $A = \begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix}$ and $x_0 = (1, 0)^T$, the results are

k	$(x_k)_1$	$(x_k)_2$	μ_k
1	0.750000000	1.000000000	11.500000000
2	1.000000000	0.956521739	12.782608696
3	0.993197279	1.000000000	12.959183673
4	1.000000000	0.998950131	12.994750656
5	0.999838416	1.000000000	12.999030499

.

For an indexing check, the first multiplication gives $Ax_0 = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$ and $\mu_0 = 8$. Dividing by this scalar produces x_1, after which μ_1 is computed from Ax_1. The values converge toward the dominant eigenvalue 13, with eigenvector direction $(1, 1)^T$.

Exercise 5.8.8: Compute the requested five indexed power-method rows

Use the signed largest-magnitude entry for normalization and do not round intermediate vectors. For $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$ and $x_0 = (1, 0)^T$, the results are

k	$(x_k)_1$	$(x_k)_2$	μ_k
1	0.500000000	1.000000000	7.000000000
2	0.285714286	1.000000000	6.142857143
3	0.255813953	1.000000000	6.023255814
4	0.250965251	1.000000000	6.003861004
5	0.250160772	1.000000000	6.000643087

.

For an indexing check, the first multiplication gives $Ax_0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ and $\mu_0 = 4$. Dividing by this scalar produces x_1, after which μ_1 is computed from Ax_1. The values converge toward the dominant eigenvalue 6, with eigenvector direction $(1/4, 1)^T$.

Exercise 5.8.9: Continue the same indexed algorithm to the sixth estimate

For the given matrix

$$A = \begin{bmatrix} 8 & 0 & 12 \\ 1 & -2 & 1 \\ 0 & 3 & 0 \end{bmatrix}, \qquad x_0 = (1, 0, 0)^T,$$

repeatedly compute Ax_k, retain its signed largest-magnitude entry, and divide to form x_{k+1}. This gives

$$\mu_5 \approx 8.4233109850, \qquad \mu_6 \approx 8.4246301920.$$

The accompanying current vectors, which make these estimates reproducible, are

$$x_5 \approx \begin{bmatrix} 1.0 \\ 0.0993553378 \\ 0.0352759154 \end{bmatrix}, \qquad x_6 \approx \begin{bmatrix} 1.0 \\ 0.0993154879 \\ 0.0353858493 \end{bmatrix}.$$

Multiplying each vector by A and selecting its largest-magnitude entry checks the listed scalars. The vectors have been rounded only for display, so the check agrees to the displayed precision.

Exercise 5.8.10: Continue the same indexed algorithm to the sixth estimate

For the given matrix

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 9 \\ 0 & 1 & 9 \end{bmatrix}, \qquad x_0 = (1, 0, 0)^T,$$

repeatedly compute Ax_k, retain its signed largest-magnitude entry, and divide to form x_{k+1}. This gives

$$\mu_5 \approx 9.9319371728, \qquad \mu_6 \approx 9.9871727289.$$

The accompanying current vectors, which make these estimates reproducible, are

$$x_5 \approx \begin{bmatrix} 0.0183246073 \\ 1.0 \\ 0.9904013962 \end{bmatrix}, \qquad x_6 \approx \begin{bmatrix} 0.0037778949 \\ 1.0 \\ 0.9981549815 \end{bmatrix}.$$

Multiplying each vector by A and selecting its largest-magnitude entry checks the listed scalars. The vectors have been rounded only for display, so the check agrees to the displayed precision.

Exercise 5.8.11: Compare the scaling estimate with the Rayleigh quotient

The exact dominant eigenvalue, checked from the characteristic polynomial, is 6. Compute each Rayleigh quotient from the current x_k using $R(x_k) = x_k^T A x_k / (x_k^T x_k)$:

k	μ_k	$R(x_k)$	$\lvert\mu_k - \lambda\rvert$	$\lvert R(x_k) - \lambda\rvert$
1	5.800000000000	5.965517241379	$2.000e-01$	$3.448e-02$
2	5.965517241379	5.999035679846	$3.448e-02$	$9.643e-04$
3	5.994219653179	5.999973208305	$5.780e-03$	$2.679e-05$
4	5.999035679846	5.999999255782	$9.643e-04$	$7.442e-07$

.

The quotient's error is substantially smaller at the same iterate. Both matrices are symmetric, so this demonstrates the improved accuracy claimed in the exercise. Signed normalization is especially important for the negative dominant eigenvalue in Exercise 12.

Exercise 5.8.12: Compare the scaling estimate with the Rayleigh quotient

The exact dominant eigenvalue, checked from the characteristic polynomial, is -4. Compute each Rayleigh quotient from the current x_k using $R(x_k) = x_k^T A x_k / (x_k^T x_k)$:

k	μ_k	$R(x_k)$	$\lvert\mu_k - \lambda\rvert$	$\lvert R(x_k) - \lambda\rvert$
1	-4.333333333333	-3.923076923077	$3.333e-01$	$7.692e-02$
2	-3.923076923077	-3.995121951220	$7.692e-02$	$4.878e-03$
3	-4.019607843137	-3.999694842844	$1.961e-02$	$3.052e-04$
4	-3.995121951220	-3.999980926586	$4.878e-03$	$1.907e-05$

.

The quotient's error is substantially smaller at the same iterate. Both matrices are symmetric, so this demonstrates the improved accuracy claimed in the exercise. Signed normalization is especially important for the negative dominant eigenvalue in Exercise 12.

Exercise 5.8.13: A theoretically convergent method can be impractically slow

If the two leading eigenvalues have different moduli, there is a strictly dominant one, so the power method converges for an initial vector with a nonzero component in that eigendirection (under the section's eigenbasis hypotheses). But the ratio of the other large modulus to the dominant modulus is near $4/4 = 1$. The unwanted mode therefore decays slowly relative to the desired one.

Conclusion. It can work, but may require many iterations and be less useful than shifted inverse iteration. An initial vector with no dominant component is an exception to the usual convergence claim.

Exercise 5.8.14: Break the tie by shifting toward the positive eigenvalue

If the leading eigenvalues are exactly λ and $-\lambda$, their powers have equal moduli; ordinary power iteration need not select a single direction. Choose a shift α near the positive eigenvalue (for example 4.1, if suitably close) but not exactly an eigenvalue. Inverse iteration applies to $(A-\alpha I)^{-1}$. Its eigenvalues are $1/(\lambda_j-\alpha)$, so the desired positive eigenvalue is dominant when its distance from the shift is uniquely smallest. Solving shifted systems and normalizing then gives a sequence for that positive eigenpair. Alternatively, adding a suitable positive multiple of I can create a unique positive dominant modulus; subtract that shift from the resulting eigenvalue estimate. The inverse method is usually faster when an accurate initial estimate is available.

Exercise 5.8.15: Transform an eigenvalue under a shifted inverse

From $Ax=\lambda x$ subtract αx to get $(A-\alpha I)x=(\lambda-\alpha)x$. Since α is not an eigenvalue, the matrix is invertible and $\lambda-\alpha\neq 0$. Multiply by $(A-\alpha I)^{-1}$:

$$x=(\lambda-\alpha)(A-\alpha I)^{-1}x, \qquad (A-\alpha I)^{-1}x=\frac{1}{\lambda-\alpha}x.$$

Because $x\neq 0$, this is exactly the claimed eigenpair.

Exercise 5.8.16: Recover the original eigenvalue from the inverse one

Multiply $(A-\alpha I)^{-1}x=\mu x$ by $A-\alpha I$: $x=\mu(A-\alpha I)x$. The inverse is invertible, so its eigenvalue μ is nonzero. Dividing by μ and rearranging gives

$$Ax=\left(\alpha+\frac{1}{\mu}\right)x.$$

Thus $\alpha+1/\mu$ is an eigenvalue of A, with the same nonzero eigenvector. This proves the scalar update in inverse iteration.

Exercise 5.8.17: Run shifted inverse iteration and verify its residual

Use

$$A=\begin{bmatrix}10&-8&-4\\-8&13&4\\-4&5&4\end{bmatrix}, \quad \alpha=3.3, \quad x_0=(1,0,0)^T.$$

At each step solve $(A-\alpha I)y_k=x_k$, choose a signed largest-magnitude component μ_k, and set $x_{k+1}=y_k/\mu_k$, $\theta_k=\alpha+1/\mu_k$. Using high-precision arithmetic, iteration index $k=7$ gives

$$\theta_k\approx 3.3212201246571, \qquad x_{k+1}\approx\begin{bmatrix}1.0\\0.787012129501\\0.0956707098337\end{bmatrix}.$$

Before rounding, $\|Ax_{k+1}-\theta_k x_{k+1}\|_\infty<10^{-10}$ and the last two scalar estimates differ by less than 10^{-12}. The eigenvalue to four decimal places is $\boxed{3.3212}$. Independent roots of the exact characteristic polynomial confirm this choice. The shift makes this eigenvalue uniquely closest, so inverse iteration selects it rather than the dominant eigenvalue of A.

Exercise 5.8.18: Run shifted inverse iteration and verify its residual

Use

$$A = \begin{bmatrix} 8 & 0 & 12 \\ 1 & -2 & 1 \\ 0 & 3 & 0 \end{bmatrix}, \quad \alpha = -1.4, \quad x_0 = (1, 0, 0)^T.$$

At each step solve $(A - \alpha I)y_k = x_k$, choose a signed largest-magnitude component μ_k, and set $x_{k+1} = y_k/\mu_k$, $\theta_k = \alpha + 1/\mu_k$. Using high-precision arithmetic, iteration index $k = 10$ gives

$$\theta_k \approx -1.424428900898, \qquad x_{k+1} \approx \begin{bmatrix} 1.0 \\ 0.372900802803 \\ -0.785369075075 \end{bmatrix}.$$

Before rounding, $\|Ax_{k+1} - \theta_k x_{k+1}\|_\infty < 10^{-10}$ and the last two scalar estimates differ by less than 10^{-12}. The eigenvalue to four decimal places is $\boxed{-1.4244}$. Independent roots of the exact characteristic polynomial confirm this choice. The shift makes this eigenvalue uniquely closest, so inverse iteration selects it rather than the dominant eigenvalue of A.

Exercise 5.8.19: Compute both extreme-modulus eigenpairs from the required start

Use $x_0 = (1, 0, 0, 0)^T$ for both computations.

(a) Ordinary power iteration.

With signed largest-component normalization, the scalar estimate converges to

$$\lambda_{\max} \approx 30.288685345802, \quad v_{\max} \approx \begin{bmatrix} 0.95762878047 \\ 0.68893692056 \\ 1.0 \\ 0.94378150874 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{30.2887}$.

(b) Inverse iteration with shift zero.

Repeatedly solve $Ay_k = x_k$ instead of explicitly multiplying by A^{-1}. The smallest-modulus eigenpair is

$$\lambda_{\text{near }0} \approx 0.010150048397894, \quad v_{\text{near }0} \approx \begin{bmatrix} -0.60397234234 \\ 1.0 \\ -0.25113513051 \\ 0.14895344556 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{0.0102}$. The stopping tests were both a scalar-estimate change below 10^{-12} and an unrounded residual below 10^{-10}. They hold at indices $k = 16$ and $k = 6$, respectively, with $k = 0$ representing the first multiplication or solve. Independent characteristic-polynomial roots verify which eigenvalues are largest and nearest zero. The displayed eigenvectors are scaled to make their largest-magnitude component equal to 1.

Exercise 5.8.20: Compute both extreme-modulus eigenpairs from the required start

Use $x_0 = (1, 0, 0, 0)^T$ for both computations.

(a) Ordinary power iteration.

With signed largest-component normalization, the scalar estimate converges to

$$\lambda_{\max} \approx 19.182036763332, \quad v_{\max} \approx \begin{bmatrix} 0.1844163575 \\ 1.0 \\ 0.17961525893 \\ 0.40710960748 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{19.1820}$.

(b) Inverse iteration with shift zero.

Repeatedly solve $Ay_k = x_k$ instead of explicitly multiplying by A^{-1}. The smallest-modulus eigenpair is

$$\lambda_{\text{near } 0} \approx 0.012205562828844, \quad v_{\text{near } 0} \approx \begin{bmatrix} 1.0 \\ 0.22260978915 \\ -0.91799334087 \\ 0.66048300357 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{0.0122}$. The stopping tests were both a scalar-estimate change below 10^{-12} and an unrounded residual below 10^{-10}. They hold at indices $k = 18$ and $k = 6$, respectively, with $k = 0$ representing the first multiplication or solve. Independent characteristic-polynomial roots verify which eigenvalues are largest and nearest zero. The displayed eigenvectors are scaled to make their largest-magnitude component equal to 1.

Exercise 5.8.21: Separate vector size from convergence of direction

For $x = (.5, .5)^T$, diagonal multiplication is explicit:

(a) $A^k x = (.5(.8)^k, .5(.2)^k)^T \to 0$. The zero vector is not an eigenvector. Yet the second coordinate relative to the first is $(1/4)^k \to 0$, so the direction tends to the first coordinate axis.

(b) $A^k x = (.5, .5(.8)^k)^T \to (.5, 0)^T$, a nonzero eigenvector. Here the dominant eigenvalue is exactly 1, so its scale is preserved.

(c) $A^k x = (.5(8)^k, .5(2)^k)^T$ grows without bound. Its direction again tends to the first coordinate axis, but it does not converge to a finite vector.

The common conclusion is about normalized direction, not convergence of $A^k x$ itself. Moreover, a vector exactly missing the dominant mode cannot reveal that eigendirection. Appropriate normalization is therefore an essential part of the power method, not an optional cosmetic step.

Reproducing the iterative computations. The following complete functions implement the signed indexing used in the tables. Enter the matrix printed in the chosen exercise and its stated initial vector as NumPy arrays. For 19–20 and inverse iteration, continue until both the estimate change and eigenpair residual are small.

```
import numpy as np
def power_rows(A, x, count):
    for k in range(count+1):
        y = A @ x
        mu = y[np.argmax(np.abs(y))]
        rayleigh = (x @ A @ x)/(x @ x)
        yield k, x.copy(), mu, rayleigh
        x = y/mu

def inverse_rows(A, alpha, x, count):
    H = A-alpha*np.eye(len(x))
    for k in range(count+1):
        y = np.linalg.solve(H, x)
        mu = y[np.argmax(np.abs(y))]
        xnew = y/mu
        theta = alpha+1/mu
        residual = np.max(np.abs(A@xnew-theta*xnew))
        yield k, xnew.copy(), theta, residual
        x = xnew
```

All numerical results above were actually computed from the supplied matrices; the code is provided to reproduce them. Extra working precision was used for the final residual checks in Exercises 17–20.

5.S Chapter 5 supplementary exercises

These exercises connect eigenvectors, similarity, polynomial expressions in matrices, and diagonalization. Keep three distinctions in view: row equivalence does not preserve eigenvalues; similarity does. An eigenvalue may be zero, but an eigenvector must be nonzero. A repeated eigenvalue does not by itself decide diagonalizability: the total number of independent eigenvectors decides it. All matrices are real unless complex roots or complex diagonalization are explicitly being considered.

Exercise 5.S.1: Twenty-four statements with proofs or counterexamples

(a) **True.** If $Ax = x$ and $x \neq 0$, multiplication by A^{-1} gives $A^{-1}x = x$.

(b) **False.** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is invertible and row equivalent to I, but has only one independent eigenvector, since its 1-eigenspace is $\text{Span}\{(1,0)^T\}$.

(c) **True.** A zero column makes $Ae_j = 0$ for some nonzero coordinate vector. A zero row makes the matrix singular as well; by the Invertible Matrix Theorem its kernel contains a nonzero vector.

(d) **False.** The scalar matrix [2] has eigenvalue 2, whereas its square [4] has only eigenvalue 4.

(e) **True.** $Ax = \lambda x$ implies $A^2x = \lambda^2 x$, with the same nonzero x.

(f) **True.** An invertible A has $\lambda \neq 0$; $Ax = \lambda x$ gives $A^{-1}x = \lambda^{-1}x$.

(g) **False.** The zero matrix has eigenvalue zero.

(h) **True.** Nonzeroness is part of the eigenvector definition; otherwise $A0 = \lambda 0$ would make every scalar an eigenvalue.

(i) **False.** For I_2, the independent vectors e_1, e_2 have the same eigenvalue 1.

(j) **True.** Similarity gives $\det(P^{-1}(A - \lambda I)P) = \det(A - \lambda I)$, so the characteristic polynomials, including multiplicities, agree.

(k) **False.** For $A = \text{diag}(1, 2)$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ the eigenvalues agree and the matrices are similar (both have independent eigenbases), but e_2 is an eigenvector of A and not of B.

(l) **False.** For $A = \text{diag}(1, 2)$, $e_1 + e_2$ is not an eigenvector. Even within one eigenspace, the sum of v and $-v$ is zero and hence not an eigenvector.

(m) **False.** All diagonal entries of an upper triangular matrix are eigenvalues, including zeros.

(n) **True.** $\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$.

(o) **False.** I_5 has only one distinct eigenvalue but is already diagonal; its five coordinate vectors form an eigenbasis.

(p) **True.** The real rotation matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues $\pm i$ and no real eigenvectors.

(q) **False.** $\text{diag}(0, 1)$ is diagonalizable but its first column is zero, so its columns are dependent.

(r) **True.** If $Av = \lambda v = \mu v$ and $v \neq 0$, then $(\lambda - \mu)v = 0$ forces $\lambda = \mu$.

(s) **False.** Invertibility and diagonalizability are different. The non-diagonalizable invertible example in (b) refutes one direction; $\operatorname{diag}(0, 1)$ refutes the reverse direction.

(t) **True.** If $Ae_j = \lambda_j e_j$, column j has only its diagonal entry possibly nonzero, so A is diagonal.

(u) **True.** If $A = SBS^{-1}$ and $B = PDP^{-1}$, then $A = (SP)D(SP)^{-1}$, a diagonalization.

(v) **True.** $A^{-1}(AB)A = BA$, so the products are similar. In fact invertibility of A alone suffices.

(w) **False.** The zero matrix has n independent eigenvectors but is not invertible.

(x) **True.** Diagonalizability supplies an eigenvector basis, and the definition of a basis says every vector is a linear combination of its vectors (over $\mathbb{R}$ for real diagonalization here).

Exercise 5.S.2: Move an eigenvector through the right-hand factor

Suppose $(AB)x = \lambda x$ and $Bx \neq 0$. Associativity gives

$$(BA)(Bx) = B(ABx) = B(\lambda x) = \lambda(Bx).$$

Because Bx is nonzero by assumption, it is an eigenvector of BA for the same eigenvalue. The condition $Bx \neq 0$ is what prevents the zero vector from being incorrectly called an eigenvector.

Exercise 5.S.3: Apply two polynomials to an eigenvector

(a) $(5I - A)x = 5x - Ax = (5 - \lambda)x$. Thus x remains a nonzero eigenvector, with eigenvalue $5 - \lambda$.

(b) First $A^2x = A(\lambda x) = \lambda^2 x$. Consequently $(5I - 3A + A^2)x = (5 - 3\lambda + \lambda^2)x$. The new eigenvalue is $5 - 3\lambda + \lambda^2$.

Exercise 5.S.4: Induct on the matrix power

For $m = 1$ the assertion is the assumed eigenvector equation. Suppose $A^m x = \lambda^m x$. Multiplying by A gives $A^{m+1}x = A(\lambda^m x) = \lambda^m Ax = \lambda^{m+1}x$. The same x remains nonzero at every stage, so each displayed scalar is genuinely an eigenvalue of the corresponding power. This proves the claim for every positive integer m.

Exercise 5.S.5: Polynomial evaluation preserves eigenvectors

Exercise 5.S.4 gives $A^j x = \lambda^j x$ for every positive power; for $j = 0$ the same equality is $Ix = x$. Therefore

$$p(A)x = \sum_{j=0}^{n} c_j A^j x = \sum_{j=0}^{n} c_j \lambda^j x = p(\lambda)x.$$

Since $x \neq 0$, $p(\lambda)$ is an eigenvalue of $p(A)$. Different eigenvalues of A can map to the same polynomial value; that does not invalidate the conclusion.

Exercise 5.S.6: Diagonalize polynomial expressions using the original basis

(a) With $A = PDP^{-1}$, neighboring factors $P^{-1}P$ cancel in A^2. Since $I = PIP^{-1}$, $5I - 3A + A^2 = P(5I - 3D + D^2)P^{-1}$. At diagonal entries $2, 7$, the polynomial has values $3, 33$. Thus $B = P\,\mathrm{diag}(3, 33)P^{-1}$ is a diagonalization.

(b) The same cancellation gives $A^j = PD^jP^{-1}$ for every power. Adding the finitely many terms gives $p(A) = Pp(D)P^{-1}$, with diagonal entries $p(2), p(7)$. This remains a diagonalization even if those two values coincide.

Exercise 5.S.7: Cayley–Hamilton for a diagonalizable matrix

Write $A = PDP^{-1}$. Every diagonal entry λ_j of D is an eigenvalue of A and hence a root of its characteristic polynomial p. By the polynomial identity in Exercise 5.S.6,

$$p(A) = Pp(D)P^{-1} = P\,\mathrm{diag}(p(\lambda_1), \ldots, p(\lambda_n))P^{-1} = P0P^{-1} = 0.$$

This proves the theorem under the stated diagonalizability assumption; it does not pretend that every square matrix is already known to be diagonalizable.

Exercise 5.S.8: One eigenvalue plus diagonalizability forces a scalar matrix

(a) If the only eigenvalue is λ with multiplicity n, a diagonalization has $D = \lambda I$. Therefore $A = P(\lambda I)P^{-1} = \lambda PP^{-1} = \lambda I$.

(b) The given $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ has only eigenvalue 3, twice, but is not $3I$. By (a), it cannot be diagonalizable. Directly, $(A - 3I)x = 0$ forces $x_2 = 0$, leaving only one independent eigenvector.

Exercise 5.S.9: An eigenvalue inside the unit disk cannot equal one

If $I - A$ were singular, a nonzero v would satisfy $(I - A)v = 0$. Then $Av = v$, so 1 would be an eigenvalue of A. Its modulus is 1, contradicting the strict bound on every eigenvalue. Hence $I - A$ is invertible. This argument requires no diagonalizability assumption.

Exercise 5.S.10: Every eigenmode decays and therefore the whole matrix does

Use $A = PDP^{-1}$. Its powers satisfy $A^k = PD^kP^{-1}$. Every diagonal entry of D^k is $\lambda_j^k \to 0$ because $|\lambda_j| < 1$. Thus each entry of PD^kP^{-1} is a finite sum of fixed constants times quantities tending to zero, so it also tends to zero. Equivalently, apply the mode decomposition to each coordinate vector e_j; then $A^k e_j \to 0$, which says each column of A^k tends to zero.

Conclusion. $A^k \to 0$ entry by entry.

Exercise 5.S.11: Invariant lines and real eigenvectors

(a) Every $x \in H$ has the form $x = cu$. Then $Ax = cAu = c\lambda u \in H$. This is exactly invariance.

(b) Choose a nonzero u spanning the one-dimensional invariant subspace K. Invariance gives $Au \in K$, so $Au = \lambda u$ for some scalar λ. Since $u \neq 0$, it is an eigenvector.

Exercise 5.S.12: Factor a block triangular characteristic polynomial

In the permutation expansion of $G = \begin{bmatrix} A & X \\ 0 & B \end{bmatrix}$, a nonzero term must assign the bottom block rows to bottom block columns; otherwise it uses a zero from the lower-left block. The remaining top rows use top block columns. The products and signs therefore separate into the determinant sums for A and B, giving $\det G = \det A \det B$. The same argument applied to $G - \lambda I = \begin{bmatrix} A - \lambda I & X \\ 0 & B - \lambda I \end{bmatrix}$ gives the product of the two characteristic polynomials. No inverse of either diagonal block is required.

Exercise 5.S.13: Read one eigenvalue and solve the remaining block

The upper-left block is $[3]$ and the lower-right block is $B = \begin{bmatrix} 5 & -2 \\ -4 & 3 \end{bmatrix}$. Its characteristic polynomial is $(\lambda - 5)(\lambda - 3) - 8 = \lambda^2 - 8\lambda + 7 = (\lambda - 1)(\lambda - 7)$. By Exercise 5.S.12 the complete eigenvalue list is $3, 1, 7$, each with multiplicity one. The entries in the upper-right block have no effect on this list.

Exercise 5.S.14: Combine the spectra of two diagonal blocks

The blocks are $B = \begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} -7 & -4 \\ 3 & 1 \end{bmatrix}$. Their characteristic polynomials are

$$\det(\lambda I - B) = \lambda^2 - 5\lambda - 6 = (\lambda - 6)(\lambda + 1),$$

$$\det(\lambda I - C) = \lambda^2 + 6\lambda + 5 = (\lambda + 1)(\lambda + 5).$$

Multiply them by the block result. The eigenvalues are $6, -5$, each once, and -1 twice. Algebraic multiplicity is counted even when an eigenvalue occurs in both blocks.

Exercise 5.S.15: Split into the constant vector and coordinate-sum-zero space

Let $\mathbf{1} = (1, \ldots, 1)^T$. Then $J\mathbf{1} = n\mathbf{1}$, so $A\mathbf{1} = [a + (n-1)b]\mathbf{1}$. For z with coordinate sum zero, $Jz = 0$, and hence $Az = (a - b)z$. The coordinate-sum-zero subspace has dimension $n - 1$; together with $\mathbf{1}$ it gives a basis of $\mathbb{R}^n$. Thus the characteristic polynomial is

$$(\lambda - (a - b))^{n-1}(\lambda - (a + (n-1)b)).$$

When $b \neq 0$ and $n \geq 2$, the two eigenvalues are distinct, with multiplicities $n - 1$ and 1. If $b = 0$, they coincide at a and the combined multiplicity is n; for $n = 1$ only the value a is present. These qualifications prevent double-counting a coincident value.

Exercise 5.S.16: Apply the common-diagonal formula twice

For the 3×3 matrix, $a = 1, b = 2$. Thus $a - b = -1$ has multiplicity 2, and $a + 2b = 5$ has multiplicity 1. For the 5×5 matrix, $a = 7, b = 3$. Thus $a - b = 4$ has multiplicity 4, and $a + 4b = 19$ has multiplicity 1. These multiplicities add to the dimension in each case.

Exercise 5.S.17: The trace and determinant decide reality in dimension two

For a real matrix $A = [a_{ij}]$, expand

$$\det(\lambda I - A) = (\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

A real quadratic has both roots real exactly when its discriminant is nonnegative. Here that condition is $(\operatorname{tr} A)^2 - 4 \det A \geq 0$, equivalently

$$\det A \leq \left(\frac{\operatorname{tr} A}{2}\right)^2.$$

Equality includes a repeated real eigenvalue; it does not by itself assert diagonalizability.

Exercise 5.S.18: Compute the limiting spectral projection

The matrix has eigenvalues 1 and $3/5$, with eigenvectors $(-1, 2)^T$ and $(-3, 2)^T$. Distinct eigenvalues give a basis. The projection onto the 1-eigenspace along the other eigenspace is

$$E = \frac{A - (3/5)I}{1 - 3/5} = \begin{bmatrix} -1/2 & -3/4 \\ 1 & 3/2 \end{bmatrix}.$$

Indeed E acts as identity on the first eigenvector and as zero on the second, so on the whole eigenbasis

$$A^k = E + (3/5)^k(I - E).$$

The second term tends to zero. Therefore A^k tends to the stated matrix E, whose negative entries cause no contradiction: the original A already has a negative off-diagonal entry.

Exercise 5.S.19: A quadratic companion matrix

For $p(t) = 6 - 5t + t^2$, the companion matrix in the book's convention is

$$C_p = \begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix}.$$

Its characteristic polynomial is $\det(\lambda I - C_p) = \det \begin{bmatrix} \lambda & -1 \\ 6 & \lambda - 5 \end{bmatrix} = \lambda^2 - 5\lambda + 6 = p(\lambda)$. If using $\det(C_p - \lambda I)$ instead, the sign is the same in even dimension.

Exercise 5.S.20: A cubic companion matrix and its determinant

The coefficients $(-24, 26, -9, 1)$ give

$$C_p = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 24 & -26 & 9 \end{bmatrix}.$$

Expand $\det(\lambda I - C_p)$ along the first row:

$$\lambda \det \begin{bmatrix} \lambda & -1 \\ 26 & \lambda - 9 \end{bmatrix} - 24 = \lambda^3 - 9\lambda^2 + 26\lambda - 24.$$

This equals $(\lambda - 2)(\lambda - 3)(\lambda - 4) = p(\lambda)$. For the book's $\det(C_p - \lambda I)$ convention the polynomial is its negative, as the dimension is odd.

Exercise 5.S.21: Inductively compute the companion characteristic polynomial

For $n = 2$, direct expansion gives $\det \begin{bmatrix} -\lambda & 1 \\ -a_0 & -a_1 - \lambda \end{bmatrix} = \lambda^2 + a_1\lambda + a_0 = (-1)^2 p(\lambda)$. Suppose the identity holds in dimension $n - 1$. Expand $C_p - \lambda I$ down its first column. Only the top and bottom entries can contribute. The top contribution is $-\lambda$ times the determinant of the companion-form block associated with $q(t) = a_1 + a_2 t + \cdots + a_{n-1}t^{n-2} + t^{n-1}$. By induction this determinant is $(-1)^{n-1}q(\lambda)$. The bottom contribution is $(-a_0)(-1)^{n+1}$ times a determinant of 1; the latter minor is triangular with diagonal ones. Thus the total is

$$(-\lambda)(-1)^{n-1}q(\lambda) + (-1)^n a_0 = (-1)^n[\lambda q(\lambda) + a_0] = (-1)^n p(\lambda).$$

This completes the induction for all $n \geq 2$.

Exercise 5.S.22: A polynomial root supplies an explicit companion eigenvector

(a) The companion matrix is $C_p = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}$.

(b) Since $p(\lambda) = 0$, $\lambda^3 = -a_0 - a_1\lambda - a_2\lambda^2$. Therefore

$$C_p(1, \lambda, \lambda^2)^T = (\lambda, \lambda^2, -a_0 - a_1\lambda - a_2\lambda^2)^T = (\lambda, \lambda^2, \lambda^3)^T = \lambda(1, \lambda, \lambda^2)^T.$$

The first coordinate is 1, so the vector is nonzero even if $\lambda = 0$; it is genuinely an eigenvector.

Exercise 5.S.23: A Vandermonde matrix is an eigenvector matrix

Each column of V is the eigenvector from Exercise 5.S.22 for one root λ_j. Distinct roots give distinct eigenvalues, so the corresponding eigenvectors are independent. Thus V is invertible without needing to compute its inverse. The three eigenvector equations combine into $C_p V = V \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$. Multiply on the left by V^{-1} to obtain the requested diagonal matrix. Complex roots are allowed if matrices and independence are interpreted over $\mathbb{C}$.

Exercise 5.S.24: The companion-matrix idea behind a polynomial-root command

The basic method documented for MATLAB's `roots` is to construct a companion matrix and compute its eigenvalues. After removing leading zero coefficients and normalizing the leading nonzero coefficient, the polynomial is monic. Exercise 5.S.21 shows that the companion matrix's eigenvalues are exactly its roots. Numerical eigenvalue algorithms, typically using QR-based methods, then approximate those values. The companion matrix used by a program may be transposed or written in a different convention; its characteristic polynomial still supplies the same roots. The MATLAB reference is https://www.mathworks.com/help/matlab/ref/roots.html. This explains the algorithmic idea, not an assertion that every computed root has exact arithmetic accuracy: companion matrices can be sensitive to coefficient rounding, especially near repeated roots.

Exercise 5.S.25: A repeated eigenvalue prevents an exact diagonalization

For the supplied matrix, exact symbolic computation gives

$$\det(\lambda I - A) = (\lambda - 1)(\lambda - 2)^2.$$

Its eigenspaces are

$$E_1 = \text{Span}\{(1, -2, 2)^T\}, \qquad E_2 = \text{Span}\{(2, -5, 3)^T\}.$$

Each has dimension one, so there are only two independent eigenvectors for a three-dimensional matrix. Exact diagonalization is impossible. To see what happens to the requested checks, choose

$$D = \text{diag}(1, 2, 2), \quad P = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -5 & -5 \\ 2 & 3 & 3 \end{bmatrix}.$$

Then $AP - PD = 0$, but $\det P = 0$ because columns 2 and 3 coincide. Thus PDP^{-1} is undefined. A zero eigenvector-equation residual alone does not make P an invertible change of basis. Floating-point eigenvalue commands may split the repeated root slightly and return nearly parallel eigenvectors. An attempted inverse then amplifies rounding errors; exact ranks resolve the issue. The following executed symbolic checks reproduce the conclusion:

```
from sympy import Matrix, eye, symbols, factor
A = Matrix([[-3,-2,0],[14,7,-1],[-6,-3,1]])
z = symbols('z')
print(factor(A.charpoly(z).as_expr()))
print((A-eye(3)).nullspace())
print((A-2*eye(3)).nullspace())
P = Matrix([[1,2,2],[-2,-5,-5],[2,3,3]])
D = Matrix.diag(1,2,2)
assert A*P-P*D == Matrix.zeros(3)
assert P.det() == 0
```

Exercise 5.S.26: A nonzero nilpotent matrix cannot be diagonalized

For the given 4×4 matrix, exact calculation gives

$$\det(\lambda I - A) = \lambda^4, \qquad \text{Nul } A = \text{Span}\{(1, 2, 1, 0)^T\}.$$

The only eigenvalue is zero, with algebraic multiplicity four but only one independent eigenvector. Thus the matrix is not diagonalizable. In fact $A^4 = 0$ and $A^3 \neq 0$, another exact check of its nonzero nilpotent structure. An eigenvector matrix for $D = 0_4$ can only have columns proportional to $(1, 2, 1, 0)^T$. Such a P has rank at most one: $AP - PD = 0$ is possible, but P^{-1} is impossible. Were an invertible P to exist, $PDP^{-1} = 0$ would contradict the given nonzero A. A floating-point eigenvalue routine can again produce small nonzero numbers around the repeated zero root and unreliable eigenvectors. The exact calculation that settles the issue is:

```
A = Matrix([[-8,5,-2,0],[-5,2,1,-2],
            [10,-8,6,-3],[3,-2,1,0]])
assert A.charpoly().as_expr().expand() == A.charpoly().gen**4
assert len(A.nullspace()) == 1
assert A**4 == Matrix.zeros(4)
assert A**3 != Matrix.zeros(4)
```

No external data are needed, and the failure is a proved mathematical obstruction, not a software or setup failure.

6.1 Inner product, length, and orthogonality

For real columns u, v of equal length, $u \cdot v = u^T v$ is the sum of corresponding coordinate products. The norm is $\|v\| = \sqrt{v \cdot v}$, and distance is $\|u - v\|$. Two vectors are orthogonal precisely when their inner product is zero. Normalize a nonzero vector by dividing by its length, not by its squared length. The zero vector has no unit vector in its direction.

For a subspace W, its orthogonal complement $W^\perp$ contains the vectors orthogonal to *every* vector of W. Testing a spanning family suffices because the inner product distributes over linear combinations.

Practice problem 6.1.P1: A projection coefficient and its scaled vector

Compute $a \cdot b = (-2)(-3) + 1(1) = 7$ and $a \cdot a = (-2)^2 + 1^2 = 5$. Thus

$$\frac{a \cdot b}{a \cdot a} = \frac{7}{5}, \qquad \left(\frac{a \cdot b}{a \cdot a}\right) a = \begin{bmatrix} -14/5 \\ 7/5 \end{bmatrix}.$$

The first answer is a scalar, while the second is a vector.

Practice problem 6.1.P2: Normalize the three-coordinate vector

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{4}{3})^2 + (-1)^2 + (\frac{2}{3})^2} = \frac{\sqrt{29}}{3}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{4\sqrt{29}}{29} \\ -\frac{3\sqrt{29}}{29} \\ \frac{2\sqrt{29}}{29} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Practice problem 6.1.P3: Check orthogonality directly

The inner product is $5(4/3) + 6(-1) + (-1)(2/3) = 20/3 - 6 - 2/3 = 0$. Therefore d is orthogonal to c; checking the zero inner product is exactly the definition needed.

Practice problem 6.1.P4: Orthogonality survives normalization

The unit vector u from Practice 2 is $c/\|c\|$. By linearity in the second argument, $d \cdot u = (1/\|c\|)(d \cdot c) = 0$ using Practice 3.

Conclusion. Multiplying one of two orthogonal vectors by a scalar does not destroy orthogonality.

Exercise 6.1.1: Inner-product and norm calculations

Compute the two dot products from corresponding coordinates.

$$u \cdot u = (-1)(-1) + (2)(2) = 5,$$
$$v \cdot u = (4)(-1) + (6)(2) = 8.$$

Conclusion. The requested ratio is $(v \cdot u)/(u \cdot u) = \frac{8}{5}$.

Exercise 6.1.2: Inner-product and norm calculations

Compute the two dot products from corresponding coordinates.

$$w \cdot w = (3)(3) + (-1)(-1) + (-5)(-5) = 35,$$
$$x \cdot w = (6)(3) + (-2)(-1) + (3)(-5) = 5.$$

Conclusion. The requested ratio is $(x \cdot w)/(w \cdot w) = \frac{1}{7}$.

Exercise 6.1.3: Inner-product and norm calculations

The squared length is $w \cdot w = 35$. Multiply every coordinate by its reciprocal:

$$\frac{1}{w \cdot w} w = \frac{1}{35} \begin{bmatrix} 3 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} \frac{3}{35} \\ -\frac{1}{35} \\ -\frac{1}{7} \end{bmatrix}.$$

This is not generally a unit vector: normalization would divide by the square root of this denominator.

Exercise 6.1.4: Inner-product and norm calculations

The squared length is $u \cdot u = 5$. Multiply every coordinate by its reciprocal:

$$\frac{1}{u \cdot u} u = \frac{1}{5} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{5} \\ \frac{2}{5} \end{bmatrix}.$$

This is not generally a unit vector: normalization would divide by the square root of this denominator.

Exercise 6.1.5: Inner-product and norm calculations

The numerator is $u \cdot v = 8$ and the denominator is $v \cdot v = 52$. Therefore

$$\left(\frac{u \cdot v}{v \cdot v}\right) v = \frac{2}{13} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{8}{13} \\ \frac{12}{13} \end{bmatrix}.$$

The denominator is nonzero because the vector being projected onto is nonzero.

Exercise 6.1.6: Inner-product and norm calculations

The numerator is $w \cdot x = 5$ and the denominator is $x \cdot x = 49$. Therefore

$$\left(\frac{w \cdot x}{x \cdot x}\right) x = \frac{5}{49} \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{30}{49} \\ -\frac{10}{49} \\ \frac{15}{49} \end{bmatrix}.$$

The denominator is nonzero because the vector being projected onto is nonzero.

Exercise 6.1.7: Inner-product and norm calculations

By the norm definition,

$$\|w\| = \sqrt{(3)^2 + (-1)^2 + (-5)^2} = \sqrt{35} = \sqrt{35}.$$

The norm is the nonnegative square root, so there is no $\pm$ choice.

Exercise 6.1.8: Inner-product and norm calculations

By the norm definition,

$$\|x\| = \sqrt{(6)^2 + (-2)^2 + (3)^2} = \sqrt{49} = 7.$$

The norm is the nonnegative square root, so there is no $\pm$ choice.

Exercise 6.1.9: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(-30)^2 + (40)^2} = 50.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.10: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(-6)^2 + (4)^2 + (-3)^2} = \sqrt{61}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} -\frac{6\sqrt{61}}{61} \\ \frac{4\sqrt{61}}{61} \\ -\frac{3\sqrt{61}}{61} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.11: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{7}{4})^2 + (\frac{1}{2})^2 + (1)^2} = \frac{\sqrt{69}}{4}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{7\sqrt{69}}{69} \\ \frac{2\sqrt{69}}{69} \\ \frac{4\sqrt{69}}{69} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.12: Normalize without changing the direction

Compute the length, keeping the square root nonnegative.

$$\|v\| = \sqrt{(\frac{8}{3})^2 + (2)^2} = \frac{10}{3}.$$

Since $v \neq 0$, divide by this positive length rather than by its square:

$$u = \frac{v}{\|v\|} = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}.$$

The multiplier $1/\|v\|$ is positive, so the direction is unchanged; also $u \cdot u = (v \cdot v)/\|v\|^2 = 1$.

Conclusion. The displayed u is the unit vector in the given direction.

Exercise 6.1.13: Inner-product and norm calculations

Subtract the point coordinates first:

$$a - b = \begin{bmatrix} 10 \\ -3 \end{bmatrix} - \begin{bmatrix} -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 11 \\ 2 \end{bmatrix}.$$

Conclusion. The distance is $\|a - b\| = \sqrt{125} = 5\sqrt{5}$. Reversing the subtraction gives the same length.

Exercise 6.1.14: Inner-product and norm calculations

Subtract the point coordinates first:

$$a-b=\begin{bmatrix}0\\-5\\2\end{bmatrix}-\begin{bmatrix}-4\\-1\\8\end{bmatrix}=\begin{bmatrix}4\\-4\\-6\end{bmatrix}.$$

Conclusion. The distance is $\|a-b\|=\sqrt{68}=2\sqrt{17}$. Reversing the subtraction gives the same length.

Exercise 6.1.15: Inner-product and norm calculations

Compute the defining dot product:

$$a\cdot b=(8)(-2)+(-5)(-3)=-1.$$

Conclusion. The pair is not orthogonal because the product is nonzero.

Exercise 6.1.16: Inner-product and norm calculations

Compute the defining dot product:

$$a\cdot b=(12)(2)+(3)(-3)+(-5)(3)=0.$$

Conclusion. The pair is orthogonal because the product is zero.

Exercise 6.1.17: Inner-product and norm calculations

Compute the defining dot product:

$$a\cdot b=(3)(-4)+(2)(1)+(-5)(-2)+(0)(6)=0.$$

Conclusion. The pair is orthogonal because the product is zero.

Exercise 6.1.18: Inner-product and norm calculations

Compute the defining dot product:

$$a\cdot b=(-3)(1)+(7)(-8)+(4)(15)+(0)(-7)=1.$$

Conclusion. The pair is not orthogonal because the product is nonzero.

Exercise 6.1.19: Inner-product and norm calculations

(a) True. It is the squared-norm identity $\|v\|^2 = v \cdot v$.

(b) True. Scalar distributivity gives $u \cdot (cv) = c(u \cdot v)$.

(c) True. Squaring the equal distances and expanding gives $\|u\|^2 - 2u \cdot v + \|v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$. Hence $4u \cdot v = 0$, so the vectors are orthogonal.

(d) False. Take $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Its column space contains $(1, 0)^T$ and its null space contains $(-1, 1)^T$; their dot product is -1, not zero. The correct general pairing is *row space* with null space.

(e) True. For $w = \sum_j c_j v_j$ we have $x \cdot w = \sum_j c_j (x \cdot v_j) = 0$; thus $x \in W^\perp$.

Exercise 6.1.20: Inner-product and norm calculations

(a) True. Symmetry gives $u \cdot v = v \cdot u$, so their difference is zero.

(b) False for negative scalars. For $c = -1$ and $v \neq 0$, $\|cv\| = \|v\|$ but $c\|v\| = -\|v\|$. The correct rule uses $|c|$.

(c) True. Being orthogonal to every member of W is the definition of membership in $W^\perp$.

(d) True. Expand $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$. The given equality makes $2u \cdot v = 0$.

(e) True. $Ax = 0$ means every row has dot product zero with x. The same holds for every linear combination of rows, hence all of Row A.

Exercise 6.1.21: Inner-product and norm calculations

(b) Distribute using the transpose and matrix multiplication rules.

$$(u + v) \cdot w = (u + v)^T w = (u^T + v^T)w = u^T w + v^T w = u \cdot w + v \cdot w.$$

Here we used $(u + v)^T = u^T + v^T$ and distribution of matrix products across addition, both from Chapter 2.

(c) Move a scalar through the products.

The transpose rule $(cu)^T = cu^T$ and scalar associativity of products give

$$(cu) \cdot v = (cu)^T v = c(u^T v) = c(u \cdot v) = u^T (cv) = u \cdot (cv).$$

Each expression is a 1×1 product, identified with its real scalar value.

Exercise 6.1.22: Inner-product and norm calculations

We have $u \cdot u = u_1^2 + u_2^2 + u_3^2$. Every real square is nonnegative, so the sum is nonnegative. If the sum is zero, none of the nonnegative summands can be positive, so $u_i^2 = 0$ for every i. This forces all three entries to be zero. Conversely the zero vector has zero inner product with itself.

Conclusion. $u \cdot u \geq 0$, with equality if and only if $u = 0$.

Exercise 6.1.23: Inner-product and norm calculations

Compute directly rather than invoking the Pythagorean theorem:

$$u \cdot v = (2)(-7) + (-5)(-4) + (-1)(6) = 0,$$

$$\|u\|^2 = 30, \qquad \|v\|^2 = 101, \qquad u + v = \begin{bmatrix} -5 \\ -9 \\ 5 \end{bmatrix}.$$

$$\|u + v\|^2 = (-5)^2 + (-9)^2 + (5)^2 = 131 = 30 + 101.$$

The dot product is zero, so this example agrees with the orthogonal-sum identity, which has now been verified by arithmetic.

Exercise 6.1.24: Inner-product and norm calculations

Use bilinearity and symmetry to expand both squared norms:

$$\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2,$$

$$\|u - v\|^2 = \|u\|^2 - 2u \cdot v + \|v\|^2.$$

Adding cancels the two cross terms and gives $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$.

Conclusion. The parallelogram law holds for all $u, v \in \mathbb{R}^n$; orthogonality is not assumed.

Exercise 6.1.25: Inner-product and norm calculations

Orthogonality to $(a, b)^T$ is the single homogeneous equation $ax + by = 0$. If $a = b = 0$, every (x, y) works, so $H = \mathbb{R}^2$. If $(a, b) \neq (0, 0)$, the nonzero vector $(-b, a)^T$ is orthogonal to $(a, b)^T$ and its multiples solve the equation. Conversely, if $a \neq 0$ solve $x = -(b/a)y$; if $a = 0$ then $b \neq 0$ and the equation gives $y = 0$. Either way every solution is a multiple of $(-b, a)^T$.

Conclusion. For $v \neq 0$, H is the line through the origin perpendicular to v; for $v = 0$, it is the entire plane.

Exercise 6.1.26: Inner-product and norm calculations

We can write $W = \text{Nul}[5\ -6\ 7]$. The Chapter 4 theorem that a null space is a subspace therefore applies. Explicitly its equation is $5x_1 - 6x_2 + 7x_3 = 0$, one nonzero homogeneous linear constraint. Solving for x_1 leaves two freely chosen coordinates, so the geometric set is a plane through the origin. The normal direction is u because $u \cdot x = 0$ for every vector in the plane.

Conclusion. W is the plane perpendicular to $(5, -6, 7)^T$ through zero.

Exercise 6.1.27: Inner-product and norm calculations

By distributivity, $y \cdot (u + v) = y \cdot u + y \cdot v = 0 + 0 = 0$. The two zero values follow from the two given orthogonality assumptions.

Conclusion. y is orthogonal to $u + v$.

Exercise 6.1.28: Inner-product and norm calculations

Take any $w \in \text{Span}\{u, v\}$. There are real scalars c_1, c_2 with $w = c_1u + c_2v$. Then $y \cdot w = c_1(y \cdot u) + c_2(y \cdot v) = 0$. Because w was arbitrary, the claim holds for every vector in the span, not merely for its two generators.

Conclusion. $y \in (\text{Span}\{u, v\})^{\perp}$.

Exercise 6.1.29: Inner-product and norm calculations

An arbitrary $w \in W$ can be written $w = \sum_{j=1}^{p} c_j v_j$. Apply distributivity repeatedly:

$$x \cdot w = \sum_{j=1}^{p} c_j (x \cdot v_j) = \sum_{j=1}^{p} c_j 0 = 0.$$

Thus every w in W is orthogonal to x.

Conclusion. Testing orthogonality against a spanning family tests all of W.

Exercise 6.1.30: Inner-product and norm calculations

(a) Closure under scalar multiplication.

Let $z \in W^{\perp}$, $c \in \mathbb{R}$, and let u be any vector of W. Then $(cz) \cdot u = c(z \cdot u) = c0 = 0$. Since this holds for every $u \in W$, $cz \in W^{\perp}$.

(b) Closure under addition.

For $z_1, z_2 \in W^{\perp}$ and any $u \in W$, $(z_1 + z_2) \cdot u = z_1 \cdot u + z_2 \cdot u = 0 + 0 = 0$. Hence $z_1 + z_2 \in W^{\perp}$.

(c) Include zero and apply the subspace test.

The zero vector has zero dot product with every $u \in W$, so it lies in $W^{\perp}$. Together with (a) and (b), these are all three required subspace properties.

Conclusion. $W^{\perp}$ is a subspace of $\mathbb{R}^n$.

Exercise 6.1.31: Inner-product and norm calculations

If $x \in W$ and $x \in W^{\perp}$, then the latter membership makes x orthogonal to every member of W, including itself. Thus $x \cdot x = \|x\|^2 = 0$, which forces $x = 0$ by positive definiteness.

Conclusion. $W \cap W^{\perp} = \{0\}$.

Exercise 6.1.32: Inner-product and norm calculations

(a) Check all column lengths and mutual dot products.

The single exact calculation $A^T A = I_4$ states that every column has squared length 1 and all six off-diagonal pairwise products are zero. Thus the four lengths are $1, 1, 1, 1$ and the six requested dot products are 0.

(b)–(d) Perform three reproducible random experiments.

Using integer draws from $[-3, 3]$ with Python random seed 61032 gives the following three nonzero pairs. Listed equalities compare the original and transformed quantities.

$$u = \begin{bmatrix} 2 \\ 2 \\ 0 \\ -3 \end{bmatrix}, \quad v = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \end{bmatrix};$$

$$\|u\| = \|Au\| = \sqrt{17}, \quad \|v\| = \|Av\| = \sqrt{6},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = -\frac{\sqrt{102}}{17}.$$

$$u = \begin{bmatrix} -1 \\ 0 \\ 3 \\ 2 \end{bmatrix}, \quad v = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 1 \end{bmatrix};$$

$$\|u\| = \|Au\| = \sqrt{14}, \quad \|v\| = \|Av\| = \sqrt{14},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = \frac{5}{14}.$$

$$u = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \quad v = \begin{bmatrix} 0 \\ 2 \\ -1 \\ -1 \end{bmatrix};$$

$$\|u\| = \|Au\| = 2, \quad \|v\| = \|Av\| = \sqrt{6},$$

$$\cos\angle(u, v) = \cos\angle(Au, Av) = \frac{\sqrt{6}}{6}.$$

Prove the observed behavior for every pair.

For arbitrary u, v, $(Au)^T(Av) = u^T A^T Av = u^T v$. Taking $v = u$ proves norm preservation, and dividing by nonzero norms proves angle preservation for nonzero vectors.

Conclusion. A preserves lengths and angles; the numerical experiments are instances of this identity.

Exercise 6.1.33: Inner-product and norm calculations

Compute two actual random examples with one fixed nonzero direction.

Using seed 61033 and integer draws from $[-3, 3]$, take $v = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$, with $v \cdot v = 5$. Each scalar below multiplies this same explicitly given column.

$$x = \begin{bmatrix} 3 \\ -1 \\ 0 \\ 2 \end{bmatrix}, \quad y = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 1 \end{bmatrix},$$

$$T(x) = (\frac{8}{5})v, \quad T(y) = (-\frac{3}{5})v,$$
$$T(x+y) = (1)v, \quad T(10x) = (16)v.$$
$$x = \begin{bmatrix} -2 \\ 3 \\ 1 \\ 3 \end{bmatrix}, \quad y = \begin{bmatrix} -1 \\ 2 \\ 3 \\ -1 \end{bmatrix},$$
$$T(x) = (-\frac{1}{5})v, \quad T(y) = (-\frac{3}{5})v,$$
$$T(x+y) = (-\frac{4}{5})v, \quad T(10x) = (-2)v.$$

Verify the conjectured linearity algebraically.

For scalars a, b and any inputs x, y, distributivity gives

$$T(ax+by) = \frac{(ax+by)\cdot v}{v\cdot v}v = aT(x) + bT(y).$$

The fixed denominator is nonzero because $v \neq 0$.

Conclusion. Projection onto a fixed nonzero direction is linear; the displayed experiments satisfy additivity and scalar homogeneity exactly.

Exercise 6.1.34: Inner-product and norm calculations

Read the free variables from an exact row reduction.

$$\operatorname{rref}(A) = \begin{bmatrix} 1 & 0 & 5 & 0 & -\frac{1}{3} \\ 0 & 1 & 1 & 0 & -\frac{4}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are $1, 2, 4$, leaving free coordinates $3, 5$. A null-space basis, with the second vector scaled by 3 to avoid fractions, and a nonzero-row basis for the row space give

$$N = \begin{bmatrix} -5 & 1 \\ -1 & 4 \\ 1 & 0 \\ 0 & -1 \\ 0 & 3 \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 0 & 15 & 0 & -1 \\ 0 & 3 & 3 & 0 & -4 \\ 0 & 0 & 0 & 3 & 1 \end{bmatrix}.$$

The two columns of N have independent free-coordinate choices, and the three rows of R are scaled nonzero rows of RREF, so they are bases of the required spaces.

Multiply the two basis matrices.

$$RN = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Each entry is the dot product of one row-space basis vector with one null-space basis vector. All are zero, illustrating $(\text{Row } A)^{\perp} = \text{Nul } A$ from Theorem 3. The additional check $AN = 0$ verifies the null-space computation directly.

6.2 Orthogonal sets

An orthogonal set has pairwise zero inner products; it may include zero. An orthonormal set is orthogonal and has every length equal to 1, so its vectors are necessarily nonzero. For nonzero orthogonal columns u_j, a vector in their span has unique weights $c_j = (x \cdot u_j)/(u_j \cdot u_j)$.

For a line $L = \text{Span}\{u\}$, $u \neq 0$, the orthogonal projection of y is $\widehat{y} = ((y \cdot u)/(u \cdot u))u$. Its residual $y - \widehat{y}$ is perpendicular to the line; the distance to the line is the length of that residual. A rectangular matrix can have orthonormal columns, but the term *orthogonal matrix* is reserved here for a square matrix with $U^{-1} = U^T$.

Practice problem 6.2.P1: Two orthonormal vectors form a plane basis

Their dot product is $(-1/\sqrt{5})(2/\sqrt{5}) + (2/\sqrt{5})(1/\sqrt{5}) = 0$. Their squared lengths are $(1+4)/5 = 1$ and $(4+1)/5 = 1$. Thus they are orthonormal and in particular independent. Two independent vectors in $\mathbb{R}^2$ form a basis.

Conclusion. The proposed pair is an orthonormal basis of $\mathbb{R}^2$.

Practice problem 6.2.P2: The same projection using a different line generator

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 5 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{20}{5} = 4.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The projection is $(8, 4)^T$, the same point obtained with any nonzero generator of the same line.

Practice problem 6.2.P3: A rectangular isometry preserves inner products

The data from Example 6 are

$$U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}, \quad x = (\sqrt{2}, 3)^T, \quad y = (-3\sqrt{2}, 6)^T.$$

Multiplication gives $Ux = (3, -1, 1)^T$ and $Uy = (1, -7, 2)^T$. Consequently $(Ux)\cdot(Uy) = 3+7+2 = 12$, while $x \cdot y = (\sqrt{2})(-3\sqrt{2}) + 3(6) = -6 + 18 = 12$.

Conclusion. The two inner products agree exactly, although the outputs have three coordinates and the inputs only two.

Exercise 6.2.1: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 26 & 0 & 2 \\ 0 & 30 & 0 \\ 2 & 0 & 74 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.2: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 30 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

Exercise 6.2.3: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 54 & 0 & 0 \\ 0 & 126 & -30 \\ 0 & -30 & 11 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2), (1,3), (2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.4: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 38 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 56 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

The zero vector causes no difficulty for orthogonality, but it would prevent the set from being independent or orthonormal.

Exercise 6.2.5: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 23 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 122 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is orthogonal: all three off-diagonal products are zero.

Exercise 6.2.6: Orthogonality and orthogonal coordinates

Put the listed vectors in columns of V. The Gram matrix records every pairwise inner product:

$$V^TV = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 90 & -32 \\ 0 & -32 & 44 \end{bmatrix}.$$

The three products to test are the off-diagonal entries in positions $(1,2),(1,3),(2,3)$; diagonal entries instead record squared lengths.

Conclusion. The set is not orthogonal: at least one off-diagonal product is nonzero.

Exercise 6.2.7: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^TU = \begin{bmatrix} 13 & 0 \\ 0 & 52 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{39}{13} = 3.$$

$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{26}{52} = \frac{1}{2}.$$

Conclusion. $x = (3)u_1 + (\frac{1}{2})u_2$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.8: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 40 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{-15}{10} = -\frac{3}{2}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{30}{40} = \frac{3}{4}.$$

Conclusion. $x = (-\frac{3}{2})u_1 + (\frac{3}{4})u_2$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.9: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^T U = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{5}{2} = \frac{5}{2}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{-27}{18} = -\frac{3}{2}.$$
$$c_3 = \frac{x \cdot u_3}{u_3 \cdot u_3} = \frac{18}{9} = 2.$$

Conclusion. $x = (\frac{5}{2})u_1 + (-\frac{3}{2})u_2 + (2)u_3$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.10: Orthogonality and orthogonal coordinates

Verify the basis properties.

$$U^TU = \begin{bmatrix} 18 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 18 \end{bmatrix}.$$

All off-diagonal entries are zero and every diagonal entry is positive. Thus the vectors are nonzero and mutually orthogonal, hence independent. There are as many of them as the ambient dimension, so they form a basis.

Take inner products rather than solving a new linear system.

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{24}{18} = \frac{4}{3}.$$
$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{3}{9} = \frac{1}{3}.$$
$$c_3 = \frac{x \cdot u_3}{u_3 \cdot u_3} = \frac{6}{18} = \frac{1}{3}.$$

Conclusion. $x = (\frac{4}{3})u_1 + (\frac{1}{3})u_2 + (\frac{1}{3})u_3$. Multiplying the stated weights by the original columns returns the given x exactly.

Exercise 6.2.11: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 20 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{10}{20} = \frac{1}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -4 \\ 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.12: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 10 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-4}{10} = -\frac{2}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \begin{bmatrix} -\frac{2}{5} \end{bmatrix} = \begin{bmatrix} \frac{2}{5} \\ -\frac{6}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{3}{5} \\ \frac{1}{5} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.13: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 65 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-13}{65} = -\frac{1}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 4 \\ -7 \end{bmatrix} \begin{bmatrix} -\frac{1}{5} \end{bmatrix} = \begin{bmatrix} -\frac{4}{5} \\ \frac{7}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{14}{5} \\ \frac{8}{5} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.14: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 50 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{20}{50} = \frac{2}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} \begin{bmatrix} \frac{2}{5} \end{bmatrix} = \begin{bmatrix} \frac{14}{5} \\ \frac{2}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{4}{5} \\ \frac{28}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.2.15: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 100 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{100} = \frac{3}{10}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} \begin{bmatrix} \frac{3}{10} \end{bmatrix} = \begin{bmatrix} \frac{12}{5} \\ \frac{9}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{1} = 1$, not the projection length.

Exercise 6.2.16: Projection, residual, and distance to a line

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 5 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{15}{5} = 3.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -6 \\ 3 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{45} = 3\sqrt{5}$, not the projection length.

Exercise 6.2.17: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}.$$

The set is orthogonal but not normalized. Divide each column by its own positive length to get the following orthonormal columns:

$$Q = \begin{bmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{3}}{3} & 0 \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} \end{bmatrix}, \qquad Q^T Q = I.$$

Exercise 6.2.18: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Conclusion. The set is not orthogonal and hence not orthonormal. Merely normalizing the individual lengths cannot make a nonzero pairwise dot product become zero.

Exercise 6.2.19: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.20: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 \\ 0 & \frac{5}{9} \end{bmatrix}.$$

The set is orthogonal but not normalized. Divide each column by its own positive length to get the following orthonormal columns:

$$Q = \begin{bmatrix} -\frac{2}{3} & \frac{\sqrt{5}}{5} \\ \frac{1}{3} & \frac{2\sqrt{5}}{5} \\ \frac{2}{3} & 0 \end{bmatrix}, \qquad Q^T Q = I.$$

Exercise 6.2.21: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.22: Classify the set and normalize when appropriate

The pairwise products and squared lengths are recorded by

$$V^T V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Conclusion. The diagonal entries are all 1 and all off-diagonal entries are zero, so the set is already orthonormal.

Exercise 6.2.23: Orthogonality and orthogonal coordinates

(a) True. For example $(1, 0)^T, (1, 1)^T$ are independent but have dot product 1, so independence need not mean orthogonality.

(b) True. For a nonzero orthogonal family, take $y \cdot u_j$ to obtain $c_j = (y \cdot u_j)/(u_j \cdot u_j)$, without row reduction.

(c) False. Scaling two distinct vectors changes their dot product by a product of scalars; a zero product remains zero. Normalization uses nonzero scales and preserves orthogonality.

(d) False unless the matrix is square. The single column $(1, 0)^T$ is orthonormal but is a rectangular matrix, not an orthogonal matrix.

(e) False. The distance is $\|y - \widehat{y}\|$, not $\|\widehat{y}\|$. For nonzero $y \in L$, the distance is zero while $\|\widehat{y}\| = \|y\| > 0$.

Exercise 6.2.24: Orthogonality and orthogonal coordinates

(a) True. An orthogonal set can contain zero; $\{0, e_1\}$ is orthogonal but dependent. Nonzero orthogonal vectors are independent.

(b) False. The condition gives orthogonality only; the lengths may be different from one. For instance $\{2e_1, 3e_2\}$ meets it.

(c) True. $A^TA = I_n$ implies $\|Ax\|^2 = x^TA^TAx = x^Tx$. Taking nonnegative square roots gives length preservation.

(d) True. The factors of a nonzero c cancel in the projection formula, as proved explicitly in Exercise 31.

(e) True. Invertibility is part of the definition, and its inverse is its transpose.

Exercise 6.2.25: Orthogonality and orthogonal coordinates

Let U have orthonormal columns, so $U^TU = I$.

(b) Prove inner-product preservation first.

For all input vectors x, y, $(Ux) \cdot (Uy) = x^TU^TUy = x^Ty = x \cdot y$.

(a) Deduce preservation of lengths.

Set $y = x$ in the previous identity: $\|Ux\|^2 = \|x\|^2$. Since both norms are nonnegative, $\|Ux\| = \|x\|$.

(c) Deduce preservation of orthogonality in both directions.

Part (b) says that the two dot products are equal. Therefore

$$(Ux) \cdot (Uy) = 0 \quad \Longleftrightarrow \quad x \cdot y = 0.$$

This equivalence includes zero inputs, for which angle formulas would not be defined. As an additional consequence, for nonzero inputs the preserved dot products and norms also preserve the angle cosine.

Conclusion. All three conclusions of Theorem 7 follow from $U^TU = I$.

Exercise 6.2.26: Orthogonality and orthogonal coordinates

A nonzero orthogonal family is independent: in a zero combination, taking the dot product with each generator forces its coefficient to vanish. Thus the given n vectors are independent in $\mathbb{R}^n$. An independent set of exactly n vectors in this n-dimensional space is a basis, so its span is all of $\mathbb{R}^n$.

Conclusion. $W = \mathbb{R}^n$.

Exercise 6.2.27: Orthogonality and orthogonal coordinates

The orthonormal columns of U satisfy $U^TU = I_n$. Their nonzero orthogonality also makes them linearly independent. By the Invertible Matrix Theorem, a square matrix with independent columns is invertible. Multiplying $U^TU = I_n$ on the right by its inverse gives $U^T = U^{-1}$.

Conclusion. U is invertible and orthogonal. Squareness is essential to this inverse conclusion.

Exercise 6.2.28: Orthogonality and orthogonal coordinates

For an orthogonal square matrix, $U^{-1} = U^T$. Hence $UU^T = UU^{-1} = I_n$. The (i, j) entry of UU^T is the inner product of row i and row j. The identity matrix says these products are zero for distinct rows and one for each row with itself. Thus the rows are orthonormal. They are n nonzero orthogonal vectors in $\mathbb{R}^n$, so they form a basis.

Conclusion. The rows form an orthonormal basis, just as the columns do.

Exercise 6.2.29: Orthogonality and orthogonal coordinates

The product of invertible matrices is invertible and $(UV)^{-1} = V^{-1}U^{-1}$. Since U, V are orthogonal,

$$(UV)^{-1} = V^T U^T = (UV)^T,$$

using the transpose-of-product rule in the last equality.

Conclusion. UV is a square invertible matrix whose inverse is its transpose, so it is orthogonal.

Exercise 6.2.30: Orthogonality and orthogonal coordinates

A column permutation keeps the same column vectors, each of length one, with the same zero dot products between distinct columns. Thus V is still square with orthonormal columns. The result of Exercise 27 makes it invertible with inverse V^T. Equivalently write $V = UP$ for a permutation matrix P; then $V^T V = P^T U^T UP = P^T P = I$.

Conclusion. Every column permutation of an orthogonal matrix is orthogonal.

Exercise 6.2.31: Orthogonality and orthogonal coordinates

For $c \neq 0$, replace u by cu in the projection formula:

$$\frac{y \cdot (cu)}{(cu) \cdot (cu)}(cu) = \frac{c(y \cdot u)}{c^2(u \cdot u)}cu = \frac{y \cdot u}{u \cdot u}u.$$

The cancellation is valid because $c \neq 0$ and $u \neq 0$. Any two nonzero generators of a fixed line are related by just such a scale.

Conclusion. The projection depends on the line, not on the chosen nonzero generator or its orientation.

Exercise 6.2.32: Orthogonality and orthogonal coordinates

The two scaled vectors are nonzero because both original vectors and both scalars are nonzero. Their dot product is $(c_1 v_1) \cdot (c_2 v_2) = c_1 c_2 (v_1 \cdot v_2) = c_1 c_2 (0) = 0$. Thus they remain an orthogonal nonzero pair. Applying the same identity to each pair proves the assertion for a whole orthogonal family.

Conclusion. In particular normalization preserves orthogonality.

Exercise 6.2.33: Orthogonality and orthogonal coordinates

For any inputs x, y and scalars a, b,

$$\operatorname{proj}_L(ax+by) = \frac{(ax+by)\cdot u}{u\cdot u}u = a\frac{x\cdot u}{u\cdot u}u + b\frac{y\cdot u}{u\cdot u}u.$$

The right side is $a\operatorname{proj}_L x + b\operatorname{proj}_L y$, so both addition and scalar multiplication are preserved. The fixed nonzero u guarantees the denominator is positive.

Conclusion. Projection onto a fixed line through zero is linear.

Exercise 6.2.34: Orthogonality and orthogonal coordinates

Let $P(y) = \operatorname{proj}_L y$. Exercise 33 proves P linear; the identity map I is also linear. Reflection is $2P - I$. For scalars a, b,

$$(2P-I)(ay+bz) = 2aP(y) + 2bP(z) - ay - bz = a(2P(y)-y) + b(2P(z)-z).$$

Conclusion. Reflection in a fixed line through the origin is linear, with matrix $2uu^T/(u^Tu) - I$.

Exercise 6.2.35: Orthogonality and orthogonal coordinates

The appropriate calculation is the Gram matrix:

$$A^TA = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 100 & 0 & 0 \\ 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 100 \end{bmatrix}.$$

The six off-diagonal products are all zero, so the four columns are pairwise orthogonal. The diagonal entries also show that each column has length 10.

Conclusion. The columns are orthogonal, and their normalized matrix in the next exercise is $U = A/10$.

Exercise 6.2.36: Orthogonality and orthogonal coordinates

(a) Compute both products with the correct dimensions.

Normalization gives $U = A/10$, so

$$U^TU = I_4, \qquad UU^T = \frac{1}{100}\begin{bmatrix} 82 & 0 & -20 & 8 & 6 & 20 & 24 & 0 \\ 0 & 42 & 24 & 0 & -20 & 6 & 20 & -32 \\ -20 & 24 & 58 & 20 & 0 & 32 & 0 & 6 \\ 8 & 0 & 20 & 82 & 24 & -20 & 6 & 0 \\ 6 & -20 & 0 & 24 & 18 & 0 & -8 & 20 \\ 20 & 6 & 32 & -20 & 0 & 58 & 0 & 24 \\ 24 & 20 & 0 & 6 & -8 & 0 & 18 & -20 \\ 0 & -32 & 6 & 0 & 20 & 24 & -20 & 42 \end{bmatrix}.$$

The first product is 4×4 identity. The second is an 8×8 rank-four projection matrix, not I_8.

(b) Perform one reproducible random-vector calculation.

Integer draws from $[-3, 3]$ with seed 62036 give

$$y = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 0 \\ -1 \\ 2 \\ -3 \\ 0 \end{bmatrix}, \quad p = UU^T y = \begin{bmatrix} -1 \\ -\frac{47}{50} \\ \frac{1}{50} \\ -\frac{11}{10} \\ \frac{1}{5} \\ \frac{29}{50} \\ -\frac{9}{10} \\ \frac{57}{50} \end{bmatrix}, \quad z = y - p = \begin{bmatrix} 0 \\ -\frac{3}{50} \\ -\frac{51}{50} \\ \frac{11}{10} \\ -\frac{6}{5} \\ \frac{71}{50} \\ -\frac{21}{10} \\ -\frac{57}{50} \end{bmatrix}.$$

The formula $p = U(U^T y)$ explicitly writes p as a combination of the columns of U, so $p \in \operatorname{Col} U = \operatorname{Col} A$. Direct multiplication yields $z^T p = 0$.

(c) Check every column, not just p.

$$U^T z = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

These four entries are the four required column dot products.

(d) Extend the check to the whole column space.

If $w = Uc$, then $z \cdot w = z^T Uc = (U^T z)^T c = 0$. Thus $z \in (\operatorname{Col} A)^\perp$, and by construction $y = p + z$.

Conclusion. The displayed vectors give the requested orthogonal decomposition; the orthogonality also follows algebraically from $U^T(y - UU^T y) = 0$.

6.3 Orthogonal projections

For an orthogonal basis $u_1, \ldots, u_p$ of W, the orthogonal component of y in W is

$$\widehat{y} = \operatorname{proj}_W y = \sum_{j=1}^{p} \frac{y \cdot u_j}{u_j \cdot u_j} u_j.$$

The residual $z = y - \widehat{y}$ lies in $W^{\perp}$. Thus $y = \widehat{y} + z$ splits a vector into a part in the subspace and a perpendicular part. If the columns of U are *orthonormal*, this simplifies to $\widehat{y} = UU^T y$. The formula without the denominators is generally wrong for merely orthogonal columns of nonunit length.

For any $w \in W$, $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2$. Therefore $\widehat{y}$ is the unique closest point and the distance is $\|z\|$. Each numerical calculation below checks the residual against every generator, not just against the projection.

Practice problem 6.3.P1: Recognize a vector already in the subspace

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 66 & 0 \\ 0 & 6 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{88}{66} = \frac{4}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-2}{6} = -\frac{1}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -7 & -1 \\ 1 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix} = \begin{bmatrix} -9 \\ 1 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The projection equals $y = (-9, 1, 6)^T$ itself; the residual is zero.

Exercise 6.3.1: Compute and check an orthogonal projection

Use the orthogonal basis formula for all four coefficients:

$$c_j = \frac{x \cdot u_j}{u_j \cdot u_j}, \qquad (c_1, c_2, c_3, c_4) = (-\frac{8}{9}, -\frac{2}{9}, \frac{2}{3}, 2).$$

The fourth component and the sum of the first three components are

$$q = c_4u_4 = \begin{bmatrix} 10 \\ -6 \\ -2 \\ 2 \end{bmatrix}, \qquad p = x - q = \begin{bmatrix} 0 \\ -2 \\ 4 \\ -2 \end{bmatrix}.$$

Here $p = c_1u_1 + c_2u_2 + c_3u_3$ is in the first requested span and q is in the second. The basis orthogonality makes $p \cdot q = 0$.

Conclusion. $x = p + q$ with the two explicit columns above.

Exercise 6.3.2: Compute and check an orthogonal projection

For the four orthogonal generators, compute $c_j = (v \cdot u_j)/(u_j \cdot u_j)$. The exact coefficients are

$$(c_1, c_2, c_3, c_4) = (2, \frac{3}{7}, \frac{12}{7}, -\frac{8}{7}).$$

Thus the requested split is

$$p = c_1u_1 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 2 \end{bmatrix}, \qquad q = c_2u_2 + c_3u_3 + c_4u_4 = \begin{bmatrix} 2 \\ 1 \\ -5 \\ 1 \end{bmatrix}.$$

Adding the two columns gives the original v. The first is in $\text{Span}\{u_1\}$ and the second in $\text{Span}\{u_2, u_3, u_4\}$, and they are orthogonal.

Conclusion. $v = p + q$ as displayed.

Exercise 6.3.3: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{3}{2} = \frac{3}{2}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{5}{2} = \frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{2} \\ \frac{5}{2} \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.4: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{25} = \frac{6}{5}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-15}{25} = -\frac{3}{5}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & -4 \\ 4 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{6}{5} \\ -\frac{3}{5} \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.5: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 14 & 0 \\ 0 & 6 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{7}{14} = \frac{1}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-15}{6} = -\frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & 1 \\ -1 & -1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ -\frac{5}{2} \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.6: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 18 & 0 \\ 0 & 2 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-27}{18} = -\frac{3}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{5}{2} = \frac{5}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -4 & 0 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{3}{2} \\ \frac{5}{2} \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.7: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 14 & 0 \\ 0 & 42 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{0}{14} = 0.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{28}{42} = \frac{2}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 3 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{10}{3} \\ \frac{2}{3} \\ \frac{8}{3} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{7}{3} \\ \frac{7}{3} \\ \frac{7}{3} \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.8: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 \\ 0 & 14 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{3} = 2.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{7}{14} = \frac{1}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 \\ 1 & 3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{7}{2} \\ 1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -\frac{5}{2} \\ \frac{1}{2} \\ 2 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.9: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{3} = 2.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{10}{15} = \frac{2}{3}.$$
$$c_3 = \frac{y \cdot u_3}{u_3 \cdot u_3} = \frac{-2}{3} = -\frac{2}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \\ 0 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ -1 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.10: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{1}{3} = \frac{1}{3}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{14}{3} = \frac{14}{3}.$$
$$c_3 = \frac{y \cdot u_3}{u_3 \cdot u_3} = \frac{-5}{3} = -\frac{5}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{14}{3} \\ -\frac{5}{3} \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} -2 \\ 2 \\ 2 \\ 0 \end{bmatrix}.$$

The exact check $U^Tz = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Exercise 6.3.11: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^TU = \begin{bmatrix} 12 & 0 \\ 0 & 4 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{6}{12} = \frac{1}{2}.$$
$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{6}{4} = \frac{3}{2}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 3 & 1 \\ 1 & -1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ 2 \\ 4 \\ 2 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.12: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 26 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{10} = 3.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{26}{26} = 1.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -4 \\ -2 & 1 \\ -1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.13: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 15 & 0 \\ 0 & 3 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{10}{15} = \frac{2}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-7}{3} = -\frac{7}{3}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \\ -3 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ -\frac{7}{3} \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ -2 \\ 3 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ -4 \\ 4 \\ 0 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.14: Closest point and its perpendicular residual

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 14 & 0 \\ 0 & 49 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{7}{14} = \frac{1}{2}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{0}{49} = 0.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 2 & 5 \\ 0 & -2 \\ -1 & 4 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ -\frac{3}{2} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 1 \\ 4 \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition. For any other w in the subspace, $y - w = z + (\widehat{y} - w)$ is an orthogonal sum. Therefore $\|y - w\|^2 = \|z\|^2 + \|\widehat{y} - w\|^2 \geq \|z\|^2$, with equality only when $w = \widehat{y}$. This proves that the displayed projection is the unique closest point.

Exercise 6.3.15: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 35 & 0 \\ 0 & 14 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{35}{35} = 1.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-28}{14} = -2.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} -3 & -3 \\ -5 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \\ -1 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{40} = 2\sqrt{10}$, not the projection length.

Exercise 6.3.16: Decompose a vector and check each orthogonality condition

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 10 & 0 \\ 0 & 26 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{30}{10} = 3.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{26}{26} = 1.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & -4 \\ -2 & 1 \\ -1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -5 \\ -3 \\ 9 \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. The distance is the residual length $\|z\| = \sqrt{64} = 8$, not the projection length.

Exercise 6.3.17: Compute and check an orthogonal projection

(a) Compute the two distinct products.

$$U^T U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad UU^T = \begin{bmatrix} \frac{8}{9} & -\frac{2}{9} & \frac{2}{9} \\ -\frac{2}{9} & \frac{5}{9} & \frac{4}{9} \\ \frac{2}{9} & \frac{4}{9} & \frac{5}{9} \end{bmatrix}.$$

The first product checks orthonormal columns. The second acts on the ambient space and projects onto their span.

(b) Compute by the orthonormal projection formula.

$$\operatorname{proj}_W y = U(U^T y) = (UU^T)y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}.$$

Both methods are the same matrix product by associativity. Directly $U^T(y - \operatorname{proj}_W y) = 0$, checking the residual.

Exercise 6.3.18: Compute and check an orthogonal projection

(a) Compute the two distinct products.

$$U^T U = \begin{bmatrix} 1 \end{bmatrix}, \qquad UU^T = \begin{bmatrix} \frac{1}{10} & -\frac{3}{10} \\ -\frac{3}{10} & \frac{9}{10} \end{bmatrix}.$$

The first product checks orthonormal columns. The second acts on the ambient space and projects onto their span.

(b) Compute by the orthonormal projection formula.

$$\operatorname{proj}_W y = U(U^T y) = (UU^T)y = \begin{bmatrix} -2 \\ 6 \end{bmatrix}.$$

Both methods are the same matrix product by associativity. Directly $U^T(y - \operatorname{proj}_W y) = 0$, checking the residual.

Exercise 6.3.19: Compute and check an orthogonal projection

Plan. Remove the component of the given outside vector that lies in W; what remains is perpendicular to W.

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{-2}{6} = -\frac{1}{3}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{2}{30} = \frac{1}{15}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} \\ \frac{1}{15} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{2}{5} \\ \frac{4}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ \frac{2}{5} \\ \frac{1}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. Take $v = \begin{bmatrix} 0 \\ \frac{2}{5} \\ \frac{1}{5} \end{bmatrix}$. It is nonzero (its displayed entries are not all zero) and has zero dot product with u_1 and u_2.

Exercise 6.3.20: Compute and check an orthogonal projection

Plan. Remove the component of the given outside vector that lies in W; what remains is perpendicular to W.

Check the orthogonal directions and compute projection weights.

$$U^T U = \begin{bmatrix} 6 & 0 \\ 0 & 30 \end{bmatrix}.$$

The off-diagonal zeros verify pairwise orthogonality, and the positive diagonal entries are the squared lengths of the nonzero generators.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1} = \frac{1}{6} = \frac{1}{6}.$$

$$c_2 = \frac{y \cdot u_2}{u_2 \cdot u_2} = \frac{-1}{30} = -\frac{1}{30}.$$

Assemble the projection and subtract its residual.

$$\widehat{y} = \begin{bmatrix} 1 & 5 \\ 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{30} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{5} \\ -\frac{2}{5} \end{bmatrix}, \qquad z = y - \widehat{y} = \begin{bmatrix} 0 \\ \frac{4}{5} \\ \frac{2}{5} \end{bmatrix}.$$

The exact check $U^T z = 0$ means the residual is orthogonal to every generating column and hence to their entire span. Thus $y = \widehat{y} + z$ is the required orthogonal decomposition.

Conclusion. Take $v = \begin{bmatrix} 0 \\ \frac{4}{5} \\ \frac{2}{5} \end{bmatrix}$. It is nonzero (its displayed entries are not all zero) and has zero dot product with u_1 and u_2.

Exercise 6.3.21: Compute and check an orthogonal projection

(a) True. For any $w = c_1 u_1 + c_2 u_2$, $z \cdot w = c_1(z \cdot u_1) + c_2(z \cdot u_2) = 0$.

(b) True. This is the residual conclusion of the Orthogonal Decomposition Theorem.

(c) False. Orthogonal decomposition into W and $W^{\perp}$ is unique, so any orthogonal basis of the same W gives the same projection.

(d) True. For $y \in W$, the decomposition $y = y + 0$ already has its first part in W and its second in $W^{\perp}$, so uniqueness applies.

(e) True. With orthonormal columns, each projection coefficient is $u_j \cdot y$; collecting the coefficients gives $U(U^T y)$.

Exercise 6.3.22: Compute and check an orthogonal projection

(a) True. Such a vector is orthogonal to itself, so its squared length is zero and it must be zero.

(b) True. Each term $((y \cdot u_j)/(u_j \cdot u_j))u_j$ is the orthogonal projection onto the one-dimensional subspace $\text{Span}\{u_j\}$.

(c) True. The assumed sum is an orthogonal decomposition into W and $W^{\perp}$, and its uniqueness identifies the first part.

(d) False. The closest point is $\text{proj}_W y$; the stated vector is the residual. For a nonzero $y \in W$, the stated vector is zero while the unique best approximation is y.

(e) False. UU^T fixes only the vectors in $\text{Col}\, U$ in general. For $U = (1, 0)^T$ and $x = (0, 1)^T$, $UU^T x = 0 \neq x$.

Exercise 6.3.23: Compute and check an orthogonal projection

Apply orthogonal decomposition to the row space.

Theorem 3 identifies $(\text{Row}\, A)^{\perp} = \text{Nul}\, A$. Taking $W = \text{Row}\, A$ in the decomposition theorem gives $x = p + u$ with $p \in \text{Row}\, A$ and $u \in \text{Nul}\, A$.

Existence of a row-space solution for a consistent equation.

If $Ax = b$ has a solution x, decompose that solution as above. Since $Au = 0$, we obtain $Ap = A(x - u) = b$. Thus some row-space vector does solve the equation.

Uniqueness within the row space.

If p, p' are row-space solutions, then $A(p - p') = 0$. Their difference lies both in the row space and its perpendicular null space, so it is zero. Therefore $p = p'$.

Conclusion. Every vector has the required split, and every consistent system has exactly one solution belonging to the row space.

Exercise 6.3.24: Compute and check an orthogonal projection

(a) Check every type of pair.

Two distinct w_i are orthogonal by assumption, as are two distinct v_j. A mixed pair is orthogonal because $w_i \in W$ and $v_j \in W^\perp$. Thus the concatenated family is orthogonal. Its members are all nonzero because they belong to bases.

(b) Span an arbitrary ambient vector.

Every $x \in \mathbb{R}^n$ splits as $x = w + v$, where $w \in W$ and $v \in W^\perp$. Expand w in the w_i basis and v in the v_j basis. Adding the expansions expresses x using the concatenated family.

(c) Count the resulting basis.

A nonzero orthogonal family is independent, and (b) proves spanning. Hence this family is a basis of $\mathbb{R}^n$ with $p + q$ members.

Conclusion. $\dim W + \dim W^\perp = p + q = n$. This includes the zero-space boundary cases, where one of the bases is empty.

Exercise 6.3.25: Compute and check an orthogonal projection

Use the normalized matrix from Exercise 6.2.36.

That matrix is $U = A/10$, with the explicitly supplied 8×4 integer matrix A from Exercise 6.2.35. It satisfies $U^T U = I_4$.

$$\widehat{y} = U(U^T y) = \begin{bmatrix} 6 \\ 5 \\ 3 \\ 5 \\ 2 \\ 5 \\ 6 \\ 5 \\ 6 \\ 5 \\ 3 \\ 5 \\ 2 \\ 5 \\ 6 \\ 5 \\ 2 \\ 5 \\ 2 \\ 5 \\ 5 \end{bmatrix}, \qquad y - \widehat{y} = \begin{bmatrix} -\frac{1}{5} \\ \frac{3}{5} \\ -\frac{1}{5} \\ -\frac{1}{5} \\ \frac{1}{5} \\ -\frac{1}{5} \\ \frac{3}{5} \\ -\frac{1}{5} \\ \frac{3}{5} \end{bmatrix}.$$

Commands reproducing the exact calculation.

```
from sympy import Matrix, ones
A = Matrix([[-6,-3,6,1],[-1,2,1,-6],[3,6,3,-2],
            [6,-3,6,-1],[2,-1,2,3],[-3,6,3,2],
            [-2,-1,2,-3],[1,2,1,6]])
U = A/10
y = ones(8,1)
p = U*(U.T*y)
print(p)
assert U.T*(y-p) == Matrix.zeros(4,1)
```

The zero dot-product column $U^T(y - p) = 0$ checks that the residual is perpendicular to the subspace, proving that the displayed point is its projection and hence the unique closest point.

Exercise 6.3.26: Compute and check an orthogonal projection

Use the normalized matrix from Exercise 6.2.36.

That matrix is $U = A/10$, with the explicitly supplied 8×4 integer matrix A from Exercise 6.2.35. It satisfies $U^T U = I_4$.

$$\widehat{y} = U(U^T y) = \begin{bmatrix} \frac{1}{5} \\ \frac{23}{25} \\ \frac{11}{25} \\ 1 \\ -\frac{1}{5} \\ -\frac{11}{25} \\ \frac{3}{5} \\ -\frac{23}{25} \end{bmatrix}, \qquad y - \widehat{y} = \begin{bmatrix} \frac{4}{5} \\ \frac{2}{25} \\ \frac{14}{25} \\ 0 \\ -\frac{4}{5} \\ -\frac{14}{25} \\ -\frac{8}{5} \\ -\frac{2}{25} \end{bmatrix}.$$

Conclusion. The distance is $\|y - \widehat{y}\| = \sqrt{\frac{112}{25}} = \frac{4\sqrt{7}}{5}$. Here y denotes the exercise target b.

6.4 The Gram–Schmidt process

Gram–Schmidt replaces an independent ordered list by orthogonal directions with the same successive spans. At stage k, subtract the projections of x_k onto every earlier orthogonal direction. Independence guarantees the residual is nonzero; normalizing it gives the next unit direction.

For a full-column-rank matrix A, the normalized columns form Q with $Q^TQ = I$. The factor $R = Q^TA$ is square upper triangular and satisfies $A = QR$. Its diagonal can be made positive by using positive lengths in the normalizations. In the calculations below we verify both $Q^TQ = I$ and $QR = A$, not merely the shape of the proposed factors.

Practice problem 6.4.P1: Normalize a pair that is already orthogonal

The two given vectors satisfy $(1,1,1)\cdot(1/3,1/3,-2/3) = 0$. Their lengths are $\sqrt{3}$ and $\sqrt{6}/3$, respectively. Hence an orthonormal basis is

$$q_1 = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad q_2 = \frac{1}{\sqrt{6}}(1,1,-2)^T.$$

Each has squared length one, their dot product is zero, and their nonzero rescalings span the same space as the original pair.

Conclusion. The displayed q_1, q_2 are the desired orthonormal basis.

Exercise 6.4.1: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & 8 \\ 0 & 5 \\ -1 & -6 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (3).$$

Subtract all of those components:

$$v_2 = x_2 - (3)w_1 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 10 & 0 \\ 0 & 35 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.2: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 0 & 5 \\ 4 & 6 \\ 2 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (1).$$

Subtract all of those components:

$$v_2 = x_2 - (1)w_1 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 \\ 0 & 105 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.3: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 2 & 4 \\ -5 & -1 \\ 1 & 2 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}).$$

Subtract all of those components:

$$v_2 = x_2 - (\frac{1}{2})w_1 = \begin{bmatrix} 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$

Choose $w_2 = (\frac{2}{3})v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 30 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.4: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -3 \\ -4 & 14 \\ 5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 50 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.5: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 7 \\ -4 & -7 \\ 0 & -4 \\ 1 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (2).$$

Subtract all of those components:

$$v_2 = x_2 - (2)w_1 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 18 & 0 \\ 0 & 43 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.6: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -5 \\ -1 & 9 \\ 2 & -9 \\ -1 & 3 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-3).$$

Subtract all of those components:

$$v_2 = x_2 - (-3)w_1 = \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 15 & 0 \\ 0 & 61 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.7: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 2 & 4 \\ -5 & -1 \\ 1 & 2 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}).$$

Subtract all of those components:

$$v_2 = x_2 - (\frac{1}{2})w_1 = \begin{bmatrix} 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$

Choose $w_2 = (\frac{2}{3})v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 30 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{30}}{15} & \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{30}}{6} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{30}}{30} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Conclusion. The columns of Q are an orthonormal basis for the original column span.

Exercise 6.4.8: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -3 \\ -4 & 14 \\ 5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 50 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \dfrac{w_1}{\|w_1\|} & \cdots & \dfrac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{3\sqrt{2}}{10} & \frac{\sqrt{6}}{6} \\ -\frac{2\sqrt{2}}{5} & \frac{\sqrt{6}}{3} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Conclusion. The columns of Q are an orthonormal basis for the original column span.

Exercise 6.4.9: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{3}{2}, -\frac{1}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{3}{2})w_1 - (-\frac{1}{2})w_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}.$$

Choose $w_3 = v_3 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.10: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -1 & 6 & 6 \\ 3 & -8 & 3 \\ 1 & -2 & 6 \\ 1 & -4 & -3 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-3).$$

Subtract all of those components:

$$v_2 = x_2 - (-3)w_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}, \frac{5}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{1}{2})w_1 - (\frac{5}{2})w_2 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}.$$

Choose $w_3 = v_3 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.11: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (-1).$$

Subtract all of those components:

$$v_2 = x_2 - (-1)w_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (4, -1).$$

Subtract all of those components:

$$v_3 = x_3 - (4)w_1 - (-1)w_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{2})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.12: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (4).$$

Subtract all of those components:

$$v_2 = x_2 - (4)w_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (\frac{7}{2}, \frac{3}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{7}{2})w_1 - (\frac{3}{2})w_2 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{3})v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.13: Gram–Schmidt with every component removed

Use the known orthonormal factor to isolate R.

The supplied Q satisfies $Q^T Q = I_2$. Left multiplication of $A = QR$ by Q^T gives $R = Q^T A$, so

$$R = \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix}.$$

Verify the matrix product, not just triangularity.

$$Q = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ \frac{1}{6} & \frac{5}{6} \\ -\frac{1}{2} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{2} \end{bmatrix}, \qquad QR = \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix} = A.$$

Conclusion. The displayed upper triangular R is the required factor.

Exercise 6.4.14: Gram–Schmidt with every component removed

Use the known orthonormal factor to isolate R.

The supplied Q satisfies $Q^TQ = I_2$. Left multiplication of $A = QR$ by Q^T gives $R = Q^TA$, so

$$R = \begin{bmatrix} 7 & 7 \\ 0 & 7 \end{bmatrix}.$$

Verify the matrix product, not just triangularity.

$$Q = \begin{bmatrix} -\frac{2}{7} & \frac{5}{7} \\ \frac{5}{7} & \frac{2}{7} \\ \frac{2}{7} & -\frac{4}{7} \\ \frac{4}{7} & \frac{2}{7} \end{bmatrix}, \qquad QR = \begin{bmatrix} -2 & 3 \\ 5 & 7 \\ 2 & -2 \\ 4 & 6 \end{bmatrix} = A.$$

Conclusion. The displayed upper triangular R is the required factor.

Exercise 6.4.15: An exact QR factorization and its checks

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (-1).$$

Subtract all of those components:

$$v_2 = x_2 - (-1)w_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (4, -1).$$

Subtract all of those components:

$$v_3 = x_3 - (4)w_1 - (-1)w_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{2})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{5}}{5} & \frac{1}{2} & \frac{1}{2} \\ -\frac{\sqrt{5}}{5} & 0 & 0 \\ -\frac{\sqrt{5}}{5} & \frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{5}}{5} & -\frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{5}}{5} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

The exact check $Q^TQ = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^TA = \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.16: An exact QR factorization and its checks

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (4).$$

Subtract all of those components:

$$v_2 = x_2 - (4)w_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{7}{2}, \frac{3}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{7}{2})w_1 - (\frac{3}{2})w_2 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{3})v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{2}}{4} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{2}}{4} & \frac{1}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{2}}{4} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{2}}{4} & \frac{1}{2} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^T A = \begin{bmatrix} 2 & 8 & 7 \\ 0 & 2\sqrt{2} & 3\sqrt{2} \\ 0 & 0 & 6 \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.17: Gram–Schmidt with every component removed

(a) False as stated: $c = 0$ replaces a nonzero basis vector by zero, which makes the family dependent and unable to retain its old span. For $c \neq 0$ it would remain an orthogonal basis.

(b) True. Each residual is obtained by subtracting a combination of earlier directions, and the reverse equation reconstructs the new input. Induction proves equality of each pair of successive spans.

(c) True. Since $Q^T Q = I$, left multiplication of $A = QR$ gives $Q^T A = Q^T QR = R$.

Exercise 6.4.18: Gram–Schmidt with every component removed

(a) False without the requirement that all v_i are nonzero. In a three-dimensional W, the set $\{e_1, e_2, 0\}$ is orthogonal but is not a basis. If the three orthogonal vectors were all nonzero, they would be independent and would form a basis.

(b) True. If $x - \text{proj}_W x = 0$, then $x = \text{proj}_W x \in W$, contradicting $x \notin W$.

(c) True. Gram–Schmidt produces orthonormal columns of Q with exactly the same column span as A; they therefore form a basis of that column space.

Exercise 6.4.19: Gram–Schmidt with every component removed

Suppose $Rx = 0$. Multiply by Q to get $Ax = QRx = 0$. Independence of the columns of A means its homogeneous system has only the zero solution, so $x = 0$. Thus the square $n \times n$ matrix R has trivial null space. By the Invertible Matrix Theorem, it is invertible.

Conclusion. The conclusion requires neither guessing a determinant nor assuming R is already invertible; it follows from independence of A.

Exercise 6.4.20: Gram–Schmidt with every component removed

If $y \in \text{Col}\, A$, write $y = Ax = Q(Rx)$. This represents y as a combination of columns of Q, proving $\text{Col}\, A \subseteq \text{Col}\, Q$. Conversely, if $y \in \text{Col}\, Q$, write $y = Qz$ and set $x = R^{-1}z$. Then $Ax = QR(R^{-1}z) = Qz = y$, so $y \in \text{Col}\, A$.

Conclusion. The two inclusions give $\text{Col}\, A = \text{Col}\, Q$. Invertibility of R is needed for the reverse inclusion.

Exercise 6.4.21: Gram–Schmidt with every component removed

Start with the thin factorization $A = QR$, where Q has n orthonormal columns in $\mathbb{R}^m$ and R is invertible upper triangular. Extend those columns to an orthonormal basis of $\mathbb{R}^m$ by taking an orthonormal basis $Q_\perp$ of $(\operatorname{Col} Q)^\perp$; for example append independent standard coordinate directions and continue Gram–Schmidt. Set $Q_1 = [Q\ Q_\perp]$. Its m columns are orthonormal, so it is an orthogonal $m \times m$ matrix. Then

$$Q_1 \begin{bmatrix} R \\ 0_{(m-n)\times n} \end{bmatrix} = QR + Q_\perp 0 = A.$$

Conclusion. Adjoining the orthonormal complement produces the full QR factorization. The triangular factor in the top block remains the same R.

Exercise 6.4.22: Gram–Schmidt with every component removed

Using the orthogonal projection formula and nonzero basis vectors,

$$T(x) = \sum_{j=1}^{p} \frac{x \cdot u_j}{u_j \cdot u_j} u_j.$$

For any x, y and scalars a, b, substitute $ax+by$ and distribute each numerator. The fixed denominators give

$$T(ax + by) = \sum_j \frac{a(x \cdot u_j) + b(y \cdot u_j)}{u_j \cdot u_j} u_j = aT(x) + bT(y).$$

Conclusion. Projection onto a fixed subspace is linear.

Exercise 6.4.23: Gram–Schmidt with every component removed

Partition $Q = [Q_1\ Q_2]$ after its first p columns and partition the upper triangular matrix R compatibly:

$$R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix}.$$

The first p columns of QR are therefore $Q_1 R_{11}$, so $A_1 = Q_1 R_{11}$. A subset of the orthonormal columns of Q remains orthonormal, and R_{11} is upper triangular. Its diagonal entries are the first p nonzero diagonal entries of R, so it is invertible.

Conclusion. Retain the first p columns of Q and the leading $p \times p$ block of R to obtain a QR factorization of A_1.

Exercise 6.4.24: Four checked orthogonal directions in five coordinates

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -10 \\ 2 \\ -6 \\ 16 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} -5 \\ 1 \\ -3 \\ 8 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 3 \\ -3 \\ 0 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (-1, -4).$$

Subtract all of those components:

$$v_3 = x_3 - (-1)w_1 - (-4)w_2 = \begin{bmatrix} 6 \\ 0 \\ 6 \\ 6 \\ 0 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{6})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Column 4.

The component coefficients along the previous w_j are

$$\frac{x_4 \cdot w_j}{w_j \cdot w_j} : \quad (1, -3, -3).$$

Subtract all of those components:

$$v_4 = x_4 - (1)w_1 - (-3)w_2 - (-3)w_3 = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \\ -5 \end{bmatrix}.$$

Choose $w_4 = (\frac{1}{5})v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.25: QR factors for the five-by-four matrix

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -10 \\ 2 \\ -6 \\ 16 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} -5 \\ 1 \\ -3 \\ 8 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 3 \\ -3 \\ 0 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (-1, -4).$$

Subtract all of those components:

$$v_3 = x_3 - (-1)w_1 - (-4)w_2 = \begin{bmatrix} 6 \\ 0 \\ 6 \\ 6 \\ 0 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{6})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Column 4.

The component coefficients along the previous w_j are

$$\frac{x_4 \cdot w_j}{w_j \cdot w_j}: \quad (1, -3, -3).$$

Subtract all of those components:

$$v_4 = x_4 - (1)w_1 - (-3)w_2 - (-3)w_3 = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \\ -5 \end{bmatrix}.$$

Choose $w_4 = (\frac{1}{5})v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & \frac{\sqrt{2}}{2} \\ -\frac{3}{10} & -\frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{4}{5} & 0 & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^T A = \begin{bmatrix} 20 & -20 & -10 & 10 \\ 0 & 6 & -8 & -6 \\ 0 & 0 & 6\sqrt{3} & -3\sqrt{3} \\ 0 & 0 & 0 & 5\sqrt{2} \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.26: Gram–Schmidt with every component removed

Implement the specified orthonormal update in exact arithmetic.

```
from sympy import Matrix, zeros, sqrt, simplify
A = Matrix([[-10,13,7,-11],[2,1,-5,3],[-6,3,13,-3],
            [16,-16,-2,5],[2,1,-5,-7]])
Q = zeros(A.rows,0)
for k in range(A.cols):
    x = A[:,k]
    v = simplify(x-Q*(Q.T*x))
    q = simplify(v/sqrt(v.dot(v)))
    Q = Q.row_join(q)
R = simplify(Q.T*A)
assert simplify(Q.T*Q) == Matrix.eye(4)
assert simplify(Q*R) == A
print(Q)
print(R)
```

Record the actual computed factors.

$$Q = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & \frac{\sqrt{2}}{2} \\ -\frac{3}{10} & -\frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{4}{5} & 0 & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}, \qquad R = \begin{bmatrix} 20 & -20 & -10 & 10 \\ 0 & 6 & -8 & -6 \\ 0 & 0 & 6\sqrt{3} & -3\sqrt{3} \\ 0 & 0 & 0 & 5\sqrt{2} \end{bmatrix}.$$

The update subtracts all previous components at once through $Q(Q^T x)$. Nonzero residuals follow from independent input columns. These exact factors agree with Exercise 25; the two displayed assertions verify unit orthogonality and reconstruction.

Conclusion. The commands implement the requested recurrence and produce the explicit checked QR factorization above.

6.5 Least-squares problems

A least-squares solution $\widehat{x}$ minimizes the *output error* $\|b - Ax\|$, not $\|b - x\|$. The minimizing output $\widehat{b} = A\widehat{x}$ is the orthogonal projection of b onto $\operatorname{Col} A$. The residual $r = b - A\widehat{x}$ is orthogonal to every column, so

$$A^T r = 0 \quad \Longleftrightarrow \quad A^T A\widehat{x} = A^T b.$$

These are the normal equations. They always have solutions, but the coefficient vector need not be unique when A has dependent columns. The projection is still unique. With full column rank and a QR factorization $A = QR$, solve $R\widehat{x} = Q^T b$ by back substitution; do not explicitly form an inverse for numerical work.

Practice problem 6.5.P1: A least-squares problem can have zero error

For the given matrix, direct multiplication gives

$$A(2, -1, 0)^T = (5, -3, -5)^T = b.$$

Thus $\widehat{x} = (2, -1, 0)^T$ is already an exact solution and therefore a least-squares solution: its residual is zero, and no norm can be smaller than zero. Its associated least-squares error is 0. For a normal-equation check,

$$A^T A = \begin{bmatrix} 3 & 9 & 0 \\ 9 & 83 & 28 \\ 0 & 28 & 14 \end{bmatrix}, \quad A^T b = (-3, -65, -28)^T,$$

and the chosen vector satisfies that system as well.

Practice problem 6.5.P2: An orthogonal target has zero as its projection

If b is orthogonal to all columns, $A^T b = 0$. The projection of b onto $\operatorname{Col} A$ is zero, so $x = 0$ is a least-squares solution and the minimum error is $\|b\|$. More precisely all least-squares solutions satisfy $Ax = 0$; thus they form $\operatorname{Nul} A$. The coefficient vector is unique only when that null space is $\{0\}$.

Exercise 6.5.1: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} -4 \\ 11 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^TA \mid A^Tb] \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}.$$

Substituting into A^Tr gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.2: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}.$$

Taking column dot products gives

$$A^TA = \begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} -24 \\ -2 \end{bmatrix}.$$

The normal equation is $A^TA\widehat{x} = A^Tb$ with these entries.

(b) Solve that smaller system.

$$[A^TA \mid A^Tb] \sim \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 3 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Substituting into A^Tr gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.3: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 6 & 6 \\ 6 & 42 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 6 \\ -6 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^T A \mid A^T b] \sim \begin{bmatrix} 1 & 0 & \frac{4}{3} \\ 0 & 1 & -\frac{1}{3} \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}.$$

Substituting into $A^T r$ gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.4: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 6 \\ 14 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^T A \mid A^T b] \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}.$$

Substituting into $A^T r$ gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.5: Find every minimizing coefficient vector

The original data give

$$A^T A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}.$$

The reduced augmented normal matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Set the free variable $x_3 = t$. Reading the first two equations gives

$$\widehat{x} = \begin{bmatrix} 5 - t \\ t - 3 \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

All these coefficient vectors produce the same fitted output $A\widehat{x} = \begin{bmatrix} 2 \\ 2 \\ 5 \\ 5 \end{bmatrix}$. The residual $\begin{bmatrix} -1 \\ 1 \\ 3 \\ -3 \end{bmatrix}$ is orthogonal to all columns of A. Conversely every minimizer solves the normal equations, so this family contains all least-squares solutions, not just selected ones.

Exercise 6.5.6: Find every minimizing coefficient vector

The original data give

$$A^T A = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 3 & 0 \\ 3 & 0 & 3 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 27 \\ 12 \\ 15 \end{bmatrix}.$$

The reduced augmented normal matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Set the free variable $x_3 = t$. Reading the first two equations gives

$$\widehat{x} = \begin{bmatrix} 5 - t \\ t - 1 \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

All these coefficient vectors produce the same fitted output $A\widehat{x} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 5 \\ 5 \\ 5 \end{bmatrix}$. The residual $\begin{bmatrix} 3 \\ -2 \\ -1 \\ 1 \\ 0 \\ -1 \end{bmatrix}$ is orthogonal to all columns of A. Conversely every minimizer solves the normal equations, so this family contains all least-squares solutions, not just selected ones.

Exercise 6.5.7: Compute the minimum error as a residual norm

Using the least-squares solution in Exercise 3, the residual is $r = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$. Therefore

$$\|b - A\widehat{x}\| = \sqrt{(1)^2 + (3)^2 + (-3)^2 + (1)^2} = \boxed{2\sqrt{5}}.$$

The error is the norm, rather than its square. The sum of squared residuals is 20.

Exercise 6.5.8: Compute the minimum error as a residual norm

Using the least-squares solution in Exercise 4, the residual is $r = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$. Therefore

$$\|b - A\widehat{x}\| = \sqrt{(1)^2 + (1)^2 + (-2)^2} = \boxed{\sqrt{6}}.$$

The error is the norm, rather than its square. The sum of squared residuals is 6.

Exercise 6.5.9: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^T A = \begin{bmatrix} 14 & 0 \\ 0 & 42 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{2}{7} \\ \frac{1}{7} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} 3 \\ -3 \\ -3 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.10: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 24 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 9 \\ 12 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 4 \\ -1 \\ 4 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.11: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^TA = \begin{bmatrix} 54 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} 36 \\ 0 \\ 9 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 3 \\ 1 \\ 4 \\ -1 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{2}{3} \\ 0 \\ \frac{1}{3} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} 6 \\ -1 \\ -4 \\ 1 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.12: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^TA = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} 1 \\ 14 \\ -5 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{1}{3} \\ \frac{14}{3} \\ -\frac{5}{3} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} -3 \\ 3 \\ 3 \\ 0 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.13: Disqualify one candidate by finding a better one

Direct multiplication gives $Au = (11, -11, 11)^T$ and $Av = (7, -12, 7)^T$. With $b = (11, -9, 5)^T$, the residuals are $(0, 2, -6)^T$ and $(4, 3, -2)^T$, respectively. Their squared norms are 40 and 29. Since the candidate v has strictly smaller error than u, the vector u cannot minimize the error. This proves u is not a least-squares solution without solving the least-squares problem. It does not assert that v is a minimizer.

Exercise 6.5.14: Equal candidate errors do not prove minimality

Multiplication gives $Au = (3, 8, 2)^T$, $Av = (7, 2, 8)^T$. The two residuals from $b = (5, 4, 4)^T$ are $(2, -4, 2)^T$ and $(-2, 2, -4)^T$, both with squared norm 24. Now take the midpoint $z = (u + v)/2 = (5, -5)^T$. Linearity gives $Az = (5, 5, 5)^T$, whose residual is $(0, -1, -1)^T$, with squared norm $2 < 24$. Thus *neither* u nor v can be a least-squares solution. Exhibiting this one better candidate suffices; we have not computed the actual minimizer.

Exercise 6.5.15: Use the supplied QR factorization

The columns of Q are orthonormal, so solve $R\widehat{x} = Q^T b$. Here

$$Q^T b = \begin{bmatrix} (14 + 6 + 1)/3 \\ (-7 + 6 - 2)/3 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 5 \\ 0 & 1 \end{bmatrix}.$$

Back substitution first gives $\widehat{x}_2 = -1$, then $3\widehat{x}_1 + 5(-1) = 7$, so $\widehat{x}_1 = 4$.

Conclusion. $\widehat{x} = (4, -1)^T$. No explicit inverse of R is needed.

Exercise 6.5.16: Back substitute in another triangular system

The given columns of Q give

$$Q^T b = \begin{bmatrix} (-1 + 6 + 5 + 7)/2 \\ (1 + 6 - 5 + 7)/2 \end{bmatrix} = \begin{bmatrix} 17/2 \\ 9/2 \end{bmatrix}.$$

Thus $5\widehat{x}_2 = 9/2$ and $2\widehat{x}_1 + 3\widehat{x}_2 = 17/2$. These yield $\widehat{x}_2 = 9/10$, $\widehat{x}_1 = 29/10$.

Conclusion. $\widehat{x} = (2.9, 0.9)^T$, the unique least-squares solution.

Exercise 6.5.17: Five least-squares assertions

(a) **True.** This is the defining minimization problem in the Euclidean output norm.

(b) **True.** The nearest output in $\operatorname{Col} A$ is precisely $\widehat{b} = \operatorname{proj}_{\operatorname{Col} A} b$.

(c) **False.** The displayed inequality is reversed. A minimizer satisfies $\|b - A\widehat{x}\| \leq \|b - Ax\|$ for every x, not the other way around.

(d) **True.** The normal equations are equivalent to $A^T(b - A\widehat{x}) = 0$, the orthogonal-projection condition.

(e) **True.** Independent columns make A^TA invertible, so the always-consistent normal equations have exactly one solution.

Exercise 6.5.18: Distinguish input coefficients from fitted outputs

(a) **True.** An exact solution has zero error, the smallest possible.

(b) **False.** The point in $\operatorname{Col} A$ closest to b is $A\widehat{x}$, not generally $\widehat{x}$, which has n input coordinates rather than m output coordinates.

(c) **True.** The entries of $\widehat{x}$ are the weights whose column combination produces the projected output.

(d) **False without a rank assumption.** A^TA may be singular, so the displayed inverse may not exist. For the zero matrix every input is a minimizer, but A^TA is not invertible in positive dimension.

(e) **False as a numerical claim.** In exact arithmetic they characterize minimizers, but forming A^TA can magnify conditioning problems and roundoff. QR methods are often more reliable.

(f) **False as stated about the best computation.** With full column rank, the formula is algebraically valid, but solving $R\widehat{x} = Q^Tb$ by back substitution is preferable to explicitly forming R^{-1}. If full column rank is absent, the stated inverse may not exist at all.

Exercise 6.5.19: Prove equality of two null spaces

(a) Start with $Ax = 0$.

Multiplying by A^T gives $A^TAx = A^T0 = 0$.

(b) Start with $A^TAx = 0$.

Multiplying on the left by x^T gives $0 = x^TA^TAx = (Ax)^T(Ax) = \|Ax\|^2$. A sum of squares is zero only when every entry is zero, so $Ax = 0$.

Conclusion. Both inclusions hold, hence $\operatorname{Nul} A = \operatorname{Nul}(A^TA)$.

Exercise 6.5.20: Invertibility of the Gram matrix forces independent columns

If $Ax = 0$, Exercise 19 gives $A^TAx = 0$. Invertibility of A^TA forces $x = 0$, so the columns of A are independent. There is no step requiring A itself to be square or invertible.

Exercise 6.5.21: The converse and its rank consequences

(a) Independent columns give $\operatorname{Nul} A = \{0\}$. Exercise 19 makes $\operatorname{Nul}(A^TA) = \{0\}$ too. Since A^TA is square $n \times n$, the Invertible Matrix Theorem makes it invertible.

(b) There are n independent columns in $\mathbb{R}^m$, so $n \leq m$. Equivalently, n pivot columns require at least n rows.

(c) All n columns are pivot columns, so $\operatorname{rank} A = n$.

Exercise 6.5.22: The Gram matrix has the same rank

Both A and A^TA have n columns, so rank-nullity gives $\operatorname{rank} A = n - \dim \operatorname{Nul} A$ and $\operatorname{rank}(A^TA) = n - \dim \operatorname{Nul}(A^TA)$. The null spaces are equal by Exercise 19, so the subtracted dimensions are equal. Thus $\operatorname{rank}(A^TA) = \operatorname{rank} A$, even when the columns are dependent.

Exercise 6.5.23: Projection without an orthogonal column basis

Independent columns make A^TA invertible. Solving the normal equations gives $\widehat{x} = (A^TA)^{-1}A^Tb$. Multiply by A to get the fitted output:

$$\boxed{\widehat{b} = A(A^TA)^{-1}A^Tb}.$$

This is a formula for the projection in $\mathbb{R}^m$, not just its coefficient vector in $\mathbb{R}^n$. Orthogonality follows by computing $A^T(b - \widehat{b}) = A^Tb - A^TA(A^TA)^{-1}A^Tb = 0$.

Exercise 6.5.24: Orthonormal columns simplify the normal equations

If the columns are orthonormal, their Gram matrix is $A^TA = I_n$. Thus the normal equations reduce immediately to $\boxed{\widehat{x} = A^Tb}$. The corresponding projection is AA^Tb. Full column rank is automatic because no nontrivial combination of orthonormal columns can be zero.

Exercise 6.5.25: Fit two incompatible equations with a common left side

Let $s = x + y$. The squared error is

$$(2 - s)^2 + (4 - s)^2 = 2(s - 3)^2 + 2.$$

This is minimized exactly when $s = 3$. Hence all least-squares solutions are $(x, y) = (t, 3 - t)$, $t \in \mathbb{R}$. They all produce the projected output $(3, 3)^T$ and error $\sqrt{2}$. The fitted output is unique, while the two input weights are not.

Exercise 6.5.26: Design the approximate low-pass filter from the actual table

Use the table's rounded value $q = 0.7$, and define the periodic lists

$$y = (1, q, 0, -q, -1, -q, 0, q), \qquad w = (1, -q, 0, q, -1, q, 0, -q).$$

Indices are reduced modulo 8. Form an eight-row block $Y_k = (y_{k+2}, y_{k+1}, y_k)$ and another block $Z_k = (w_{k+2}, w_{k+1}, w_k)$ for $k = 0, \dots, 7$. Then the requested 16×3 system is explicitly

$$A = \begin{bmatrix} Y \\ Z \end{bmatrix}, \quad x = (a_0, a_1, a_2)^T, \quad b = (q, 0, -q, -1, -q, 0, q, 1, 0, 0, 0, 0, 0, 0, 0, 0)^T.$$

For clarity the two blocks are

$$Y = \begin{bmatrix} 0 & q & 1 \\ -q & 0 & q \\ -1 & -q & 0 \\ -q & -1 & -q \\ 0 & -q & -1 \\ q & 0 & -q \\ 1 & q & 0 \\ q & 1 & q \end{bmatrix}, \quad Z = \begin{bmatrix} 0 & -q & 1 \\ q & 0 & -q \\ -1 & q & 0 \\ q & -1 & q \\ 0 & q & -1 \\ -q & 0 & q \\ 1 & -q & 0 \\ -q & 1 & -q \end{bmatrix}.$$

Column dot products give $A^T A = (198/25) I_3$ and $A^T b = (14/5, 99/25, 14/5)^T$. The diagonal normal equations therefore yield

$$\boxed{a_0 = a_2 = 35/99 \approx 0.35353535, \qquad a_1 = 1/2}.$$

The normal-equation residual is exactly zero with these coefficients and the rational table entry $q = 7/10$. More generally the same computation for a table value q gives $a_0 = a_2 = q/(1 + 2q^2)$ and $a_1 = 1/2$. Putting $q = \sqrt{2}/2$ gives $a_0 = a_2 = \sqrt{2}/4$; using $q = 0.707$ gives a very close approximation. The rounded-table answer should not be silently replaced by the exact-trigonometric answer.

6.6 Applications to linear models

A model is linear in its *unknown parameters*, even if it contains x^2, $\sin x$, or $e^{-0.02x}$. For observations (x_i, y_i), write the predicted value as a row of a design matrix X times a parameter vector β. The observation equation is $y = X\beta + \varepsilon$. Least squares minimizes $\|\varepsilon\|$, and the fitted parameters solve $X^T X\widehat{\beta} = X^T y$. As a check, the fitted residual always satisfies $X^T(y - X\widehat{\beta}) = 0$. We retain unrounded coefficients in computations and round only displayed predictions.

Practice problem 6.6.P1: A seasonal term is still linear in its coefficient

For the proposed seasonal model, use

$$X = \begin{bmatrix} 1 & x_1 & \sin(2\pi x_1/12) \\ \vdots & \vdots & \vdots \\ 1 & x_n & \sin(2\pi x_n/12) \end{bmatrix}, \quad \beta = (\beta_0, \beta_1, \beta_2)^T, \quad y = (y_1, \ldots, y_n)^T.$$

The ith product row equals the predicted sales at month x_i. The sine values are known once the observation times are supplied; only their coefficients are unknown, so this is a linear least-squares model.

Exercise 6.6.1: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{9}{10} \\ \frac{2}{5} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{9}{10} + (\frac{2}{5})x}.$$

The residual $\begin{bmatrix} \frac{1}{10} \\ -\frac{3}{10} \\ \frac{3}{10} \\ -\frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{5}$.

Exercise 6.6.2: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 12 \\ 12 & 46 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 25 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} -\frac{3}{5} \\ \frac{7}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = -\frac{3}{5} + (\frac{7}{10})x}.$$

The residual $\begin{bmatrix} -\frac{1}{10} \\ \frac{1}{5} \\ -\frac{1}{5} \\ \frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{10}$.

Exercise 6.6.3: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 4 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{11}{10} \\ \frac{13}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{11}{10} + (\frac{13}{10})x}.$$

The residual $\begin{bmatrix} \frac{1}{5} \\ -\frac{1}{10} \\ -\frac{2}{5} \\ \frac{3}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{3}{10}$.

Exercise 6.6.4: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}, \qquad y = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 16 \\ 16 & 74 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{43}{10} \\ -\frac{7}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{43}{10} + (-\frac{7}{10})x}.$$

The residual $\begin{bmatrix} \frac{1}{10} \\ -\frac{1}{5} \\ \frac{1}{5} \\ -\frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{10}$.

Exercise 6.6.5: Exactly when does a line fit have unique parameters?

By Section 6.5, the normal equations have a unique solution exactly when the two columns of X are independent. They are the all-ones column and $(x_1, \ldots, x_n)^T$. If all x_i equal the same c, the second column is c times the first, so independence and uniqueness fail. Conversely, suppose $ax_i + b = 0$ at all data points and two abscissas differ. Subtracting the equations at those two points forces $a = 0$, then $b = 0$. Thus the columns are independent precisely when at least two distinct x-coordinates occur. Repeated observations at one x cannot identify both an intercept and a slope uniquely.

Exercise 6.6.6: Three distinct abscissas identify a quadratic fit

The three design columns are $1, x, x^2$. A dependence among them would give a polynomial $c_0+c_1t+c_2t^2$ that vanishes at every observed abscissa, in particular at three distinct ones. A nonzero polynomial of degree at most two cannot have three distinct roots. Hence all coefficients must be zero, proving independence of the columns. The full-column-rank least-squares theorem then gives a unique best-fitting parameter vector. The fitted polynomial may happen to have zero quadratic coefficient; uniqueness concerns the polynomial, not a requirement that its degree be exactly two.

Exercise 6.6.7: Fit a quadratic with no constant term

(a) Put the known powers into the design columns.

For the five observations, the model is

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix}, \quad y = (1.8, 2.7, 3.4, 3.8, 3.9)^T, \quad \beta = (\beta_1, \beta_2)^T.$$

There is no column of ones, because the proposed model has no intercept.

(b) Solve and state the curve.

The exact normal equations are

$$\begin{bmatrix} 55 & 225 \\ 225 & 979 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 521/10 \\ 403/2 \end{bmatrix}.$$

Their solution is $\widehat{\beta} = (14171/8050, -32/161)^T$. Thus the least-squares curve is

$$y = \frac{14171}{8050}x - \frac{32}{161}x^2 \approx 1.760373x - 0.198758x^2.$$

Substitution in the normal equations verifies both residual moments are zero. This certifies the fit with the supplied decimal data treated exactly.

Exercise 6.6.8: Fit a cubic cost model without fixed costs

(a) Record the general model.

Use $X_{i,:} = (x_i, x_i^2, x_i^3)$, $y = (y_1, \ldots, y_n)^T$, and $\beta = (\beta_1, \beta_2, \beta_3)^T$. Again no column of ones appears. The residual equation is $y = X\beta + \varepsilon$.

(b) Use the eight supplied observations.

Here $x_i = 4, 6, 8, 10, 12, 14, 16, 18$, and the observed costs are $1.58, 2.08, 2.50, 2.80, 3.10, 3.40, 3.80, 4.32$, in the stated thousand-unit scales. The normal equations are

$$\begin{bmatrix} 1136 & 16192 & 245312 \\ 16192 & 245312 & 3866368 \\ 245312 & 3866368 & 62617856 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 7254/25 \\ 100636/25 \\ 1501896/25 \end{bmatrix}.$$

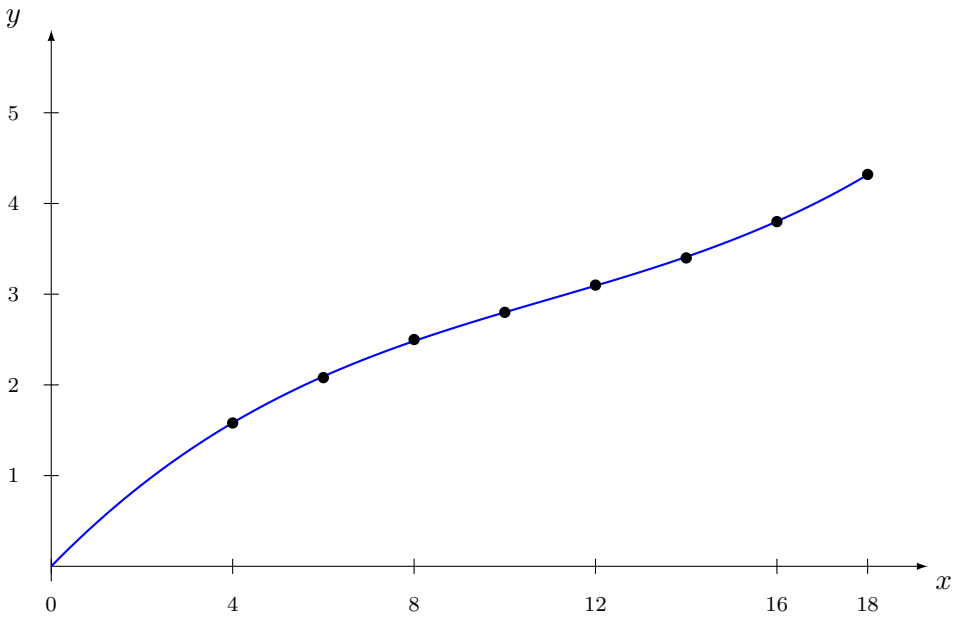
Solving with exact rational data gives

$$\widehat{\beta} = (8970503/17479000,\ -319181/9534000,\ 15221/14982000)^T.$$

The fitted curve is therefore

$$y \approx 0.513216031x - 0.0334781833x^2 + 0.00101595248x^3.$$

The exact residual satisfies $X^T(y - X\widehat{\beta}) = 0$. The graph below shows the fitted curve in blue and all eight observations as black points; both axes use the problem's thousand-unit scale.



Exercise 6.6.9: A trigonometric model linear in two coefficients

Use

$$X = \begin{bmatrix} \cos 1 & \sin 1 \\ \cos 2 & \sin 2 \\ \cos 3 & \sin 3 \end{bmatrix}, \quad y = (7.9, 5.4, -0.9)^T, \quad \beta = (A, B)^T.$$

Angles are in radians. Although the basis functions are trigonometric, the unknowns A, B enter linearly. Minimize $\|y - X\beta\|$ by solving $X^T X\widehat{\beta} = X^T y$, or use QR on X directly. A numerical solution gives $A \approx 2.34212062$, $B \approx 7.44750155$, so $y \approx 2.34212062\cos x + 7.44750155\sin x$. Using unrounded coefficients, the normal residual is below 3×10^{-14} in absolute value, a check on the computed fit.

Exercise 6.6.10: Estimate two initial radioactive masses

(a) Treat the decay functions as known columns.

At times $t_i = 10, 11, 12, 14, 15$, use

$$X_{i,:} = (e^{-0.02t_i}, e^{-0.07t_i}), \quad \beta = (M_A, M_B)^T,$$

with observation vector $(21.34, 20.68, 20.05, 18.87, 18.30)^T$. The decay constants are supplied, so the two masses, not the exponent rates, are the unknown linear parameters.

(b) Fit the two masses.

Numerical least squares gives $M_A \approx 19.94109495$ and $M_B \approx 10.10151413$ grams. Thus the fitted curve is

$$y(t) \approx 19.94109495e^{-0.02t} + 10.10151413e^{-0.07t}.$$

Both fitted masses are positive. The residual norm is approximately 0.0114745 grams, and the normal residual computed from unrounded coefficients is below 3×10^{-14}. The residual is not assumed to vanish, because the measured data are rounded.

Exercise 6.6.11: Linearize the comet equation in the correct parameters

The relation $r = \beta + e(r\cos\theta)$ is linear in β, e. For the given observations, use design rows $(1, r_i\cos\theta_i)$, not $(1, \cos\theta_i)$, and observation vector (r_i). Least squares gives $\beta \approx 1.45092922$ and $e \approx 0.81105251$; the fitted normal residual is below 2×10^{-14}. Since $0 < e < 1$, the predicted orbit is an ellipse. For a new angle, solve the original model for its new radius:

$$r = \frac{\beta}{1 - e\cos\theta}, \qquad r(4.6) \approx 1.32995445.$$

The denominator is positive here (indeed $e < 1$ makes it positive at every angle).

Conclusion. An elliptic orbit, with predicted radius about 1.330 in the same units as the observed radii.

Exercise 6.6.12: Regress pressure on logarithmic weight

Use design rows $(1, \ln w_i)$ and observation vector $(91, 98, 103, 110, 112)^T$. If the printed rounded logarithms $3.78, 4.11, 4.39, 4.73, 4.88$ are used, the fitted line is

$$p \approx 18.56421016 + 19.24070120\ln w.$$

At $w = 100$, using $\ln 100 \approx 4.605170186$, this predicts $p \approx 107.17091367$ mmHg. For clarity about precision: recomputing the logarithms directly from $w = 44, 61, 81, 113, 131$ instead of using their rounded table entries gives $p \approx 17.92434012 + 19.38499925\ln w$, and a prediction 107.19556074 mmHg. Both computations satisfy their respective normal equations to roundoff.

Conclusion. About 107 mmHg. The small difference between the two more detailed predictions comes from rounding the predictor data.

Exercise 6.6.13: Fit takeoff position and differentiate the fitted polynomial

(a) Build the cubic design matrix from the thirteen times.

For $t = 0, 1, \ldots, 12$, take row $(1, t, t^2, t^3)$ and the given position vector in feet. The normal equations have coefficient matrix

$$X^TX = \begin{bmatrix} 13 & 78 & 650 & 6084 \\ 78 & 650 & 6084 & 60710 \\ 650 & 6084 & 60710 & 630708 \\ 6084 & 60710 & 630708 & 6735950 \end{bmatrix},$$

with right side $X^Ty = (19001/5, 351277/10, 3480639/10, 35998009/10)^T$. Solving gives the exact coefficient vector

$$\widehat{\beta} = (-89/104,\ 75315/16016,\ 222437/40040,\ -313/11440)^T.$$

Thus, approximately,

$$y(t) = -0.855769231 + 4.702485015t + 5.555369630t^2 - 0.02736013986t^3.$$

The exact coefficients make $X^T(y - X\widehat{\beta}) = 0$; distinct times ensure the four design columns are independent, so the fit is unique.

(b) Differentiate before evaluating.

The fitted velocity is $y'(t) = \widehat{\beta}_1 + 2\widehat{\beta}_2 t + 3\widehat{\beta}_3 t^2$. At $t = 4.5$ this gives $\boxed{53.03868319 \text{ feet/second}}$. This estimates instantaneous velocity from the fitted curve, rather than dividing the fitted position by the elapsed time.

Exercise 6.6.14: The fitted line passes through the mean point

The residual $r = y - X\widehat{\beta}$ is orthogonal to each column of X, including the all-ones column. Thus its entries sum to zero: $\sum_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0$. Divide by $n > 0$ and use the definitions of the means to obtain

$$\overline{y} = \widehat{\beta}_0 + \widehat{\beta}_1 \overline{x}.$$

This says exactly that $(\overline{x}, \overline{y})$ lies on the fitted line.

Exercise 6.6.15: Derive the scalar normal equations

For design columns $\mathbf{1}$ and x, the four Gram entries are $\mathbf{1}^T\mathbf{1} = n$, $\mathbf{1}^T x = x^T\mathbf{1} = \sum x_i$, and $x^T x = \sum x_i^2$. Similarly, $X^T y = (\sum y_i, \sum x_i y_i)^T$. Multiplication of the resulting two-by-two system gives

$$n\widehat{\beta}_0 + \widehat{\beta}_1 \sum x_i = \sum y_i, \qquad \widehat{\beta}_0 \sum x_i + \widehat{\beta}_1 \sum x_i^2 = \sum x_i y_i,$$

which are precisely equations (7).

Exercise 6.6.16: Invert the two-by-two normal matrix

Write $S_x = \sum x_i$, $S_{xx} = \sum x_i^2$, $S_y = \sum y_i$, and $S_{xy} = \sum x_i y_i$. When at least two abscissas differ, $D = nS_{xx} - S_x^2 = n\sum_i (x_i - \overline{x})^2 > 0$. Thus the normal matrix inverse is

$$\frac{1}{D}\begin{bmatrix} S_{xx} & -S_x \\ -S_x & n \end{bmatrix}.$$

Multiplying it by $(S_y, S_{xy})^T$ gives

$$\boxed{\widehat{\beta}_0 = \frac{S_{xx}S_y - S_x S_{xy}}{D}, \qquad \widehat{\beta}_1 = \frac{nS_{xy} - S_x S_y}{D}}.$$

Equivalently $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$. If all abscissas coincide, $D = 0$ and this inverse formula is inapplicable, consistent with the failure of unique parameter identification.

Exercise 6.6.17: Centering the predictor makes the columns orthogonal

(a) Rewrite the actual data from Example 1.

Their mean abscissa is $(2+5+7+8)/4 = 5.5$. The transformed data are

$$(-3.5, 1), \quad (-0.5, 2), \quad (1.5, 3), \quad (2.5, 3).$$

The design matrix has columns $\mathbf{1}$ and $x^* = (-3.5, -0.5, 1.5, 2.5)^T$. Their dot product is the sum of x^*, which is zero, proving orthogonality.

(b) Solve the diagonal normal equations.

Here $X^TX = \operatorname{diag}(4, 21)$ and $X^Ty = (9, 7.5)^T$. Hence $\widehat{\beta}_0 = 9/4$ and $\widehat{\beta}_1 = 5/14$.

Conclusion. The centered fitted line is $y = 9/4 + (5/14)x^*$, with $x^* = x - 5.5$. Its uncentered form is $y = 2/7 + (5/14)x$.

Exercise 6.6.18: A general centered design has a diagonal Gram matrix

The two off-diagonal Gram entries are both $\sum_i x_i$, which is zero by the centered-coordinate assumption. Therefore

$$X^TX = \begin{bmatrix} n & 0 \\ 0 & \sum_i x_i^2 \end{bmatrix}.$$

If every centered abscissa is zero, the second diagonal entry vanishes; diagonal does not by itself mean invertible.

Exercise 6.6.19: Use an orthogonal decomposition of the observed vector

Write $y = X\widehat{\beta} + r$. The normal equations give $X^Tr = 0$, so r is orthogonal to $\operatorname{Col} X$, which contains $X\widehat{\beta}$. The Pythagorean theorem therefore applies:

$$\|y\|^2 = \|X\widehat{\beta}\|^2 + \|r\|^2.$$

With the problem's notation, this is $\mathrm{SS}(T) = \mathrm{SS}(R) + \mathrm{SS}(E)$. The zero-mean convention lets the total be interpreted as centered variation; the displayed norm identity itself follows from orthogonality.

Exercise 6.6.20: A computational formula for the fitted and residual sums

Using the normal equations in the middle step,

$$\|X\widehat{\beta}\|^2 = \widehat{\beta}^TX^TX\widehat{\beta} = \widehat{\beta}^TX^Ty.$$

Subtract this from the Pythagorean identity in Exercise 19 to obtain

$$\boxed{\mathrm{SS}(E) = y^Ty - \widehat{\beta}^TX^Ty}.$$

This avoids explicitly summing every residual square once the fit has been computed. In finite-precision arithmetic direct residual norms can be more stable when this subtraction nearly cancels two large numbers.

6.7 Inner product spaces

An inner product assigns real numbers $\langle u, v\rangle$ and satisfies symmetry, additivity, homogeneity, and positive definiteness. The associated norm is $\|u\| = \sqrt{\langle u, u\rangle}$; orthogonality means $\langle u, v\rangle = 0$. The formula depends on the space: a weighted coordinate product, evaluation of polynomials, and integration of functions are different inner products. For a nonzero vector p, projection onto its span is $\langle q, p\rangle p/\langle p, p\rangle$. For an orthogonal list, add these separate projections. Always use the inner product stipulated in the exercise rather than substituting the ordinary coordinate dot product.

Practice problem 6.7.P1: A zero vector has zero inner product with every vector

By homogeneity, $\langle 0, v\rangle = \langle 0u, v\rangle = 0\langle u, v\rangle = 0$, for any u in the space. Symmetry then gives $\langle v, 0\rangle = \langle 0, v\rangle = 0$. The choice of u does not matter, because its scalar multiple is zero.

Practice problem 6.7.P2: Additivity in the second argument

Use symmetry to move the sum into the first argument, where additivity is an axiom:

$$\langle u, v + w\rangle = \langle v + w, u\rangle = \langle v, u\rangle + \langle w, u\rangle = \langle u, v\rangle + \langle u, w\rangle.$$

The last equality applies symmetry to each term separately.

Exercise 6.7.1: A weighted norm and its orthogonal line

Example 1 uses $\langle u, v\rangle = 4u_1v_1 + 5u_2v_2$.

(a) Apply those weights to the stated vectors.

For $x = (1, 1)$ and $y = (5, -1)$, $\|x\|^2 = 4 + 5 = 9$, $\|y\|^2 = 4(25) + 5 = 105$, and $\langle x, y\rangle = 20 - 5 = 15$. Thus $\|x\| = 3$, $\|y\| = \sqrt{105}$, and $|\langle x, y\rangle|^2 = 225$.

(b) Solve the orthogonality equation.

A vector $z = (z_1, z_2)$ is orthogonal to y exactly when $20z_1 - 5z_2 = 0$, or $z_2 = 4z_1$.

Conclusion. All such vectors are $t(1, 4)$, $t \in \mathbb{R}$, including zero.

Exercise 6.7.2: Verify Cauchy–Schwarz with the weighted product

For $x = (3, -2)$ and $y = (-2, 1)$,

$$\langle x, y\rangle = 4(3)(-2) + 5(-2)(1) = -34,$$

$$\|x\|^2 = 4(9) + 5(4) = 56, \qquad \|y\|^2 = 4(4) + 5(1) = 21.$$

Therefore $|\langle x, y\rangle|^2 = 1156 \le 1176 = \|x\|^2\|y\|^2$. Both sides of $|\langle x, y\rangle| \le \|x\|\|y\|$ are nonnegative, so the squared comparison proves the desired inequality.

Evaluation convention for Exercises 3–8. The inner product is the dot product of the three value lists at $-1, 0, 1$, not of the polynomial coefficient lists. For Exercise 3, p and q have value vectors $(3, 4, 5)$ and $(1, 5, 1)$. For Exercise 4, their value vectors are $(-4, 0, 2)$ and $(5, 3, 5)$.

Exercise 6.7.3: Evaluate before multiplying

For $p(t) = 4 + t$ and $q(t) = 5 - 4t^2$,

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1) = 3(1) + 4(5) + 5(1) = \boxed{28}.$$

Each product uses values at the same input.

Exercise 6.7.4: A negative inner product is allowed

For $p(t) = 3t - t^2$ and $q(t) = 3 + 2t^2$,

$$\langle p, q\rangle = (-4)(5) + 0(3) + 2(5) = \boxed{-10}.$$

Only self-products must be nonnegative; products of different vectors can be negative, as this example shows.

Exercise 6.7.5: Norms from the first pair of value lists

Using the value vectors $(3, 4, 5)$ and $(1, 5, 1)$,

$$\|p\| = \sqrt{9 + 16 + 25} = \boxed{5\sqrt{2}}, \qquad \|q\| = \sqrt{1 + 25 + 1} = \boxed{3\sqrt{3}}.$$

The norm is the square root of the sum of squared values, not its square.

Exercise 6.7.6: Norms from the second pair

The value vectors are $(-4, 0, 2)$ and $(5, 3, 5)$, so

$$\|p\| = \sqrt{16 + 0 + 4} = \boxed{2\sqrt{5}}, \qquad \|q\| = \sqrt{25 + 9 + 25} = \boxed{\sqrt{59}}.$$

These are norms in the specified evaluation inner product.

Exercise 6.7.7: Project onto the first polynomial line

For the pair in Exercise 3, $\langle q, p\rangle = 28$ and $\langle p, p\rangle = 50$. Since $p \neq 0$, projection onto $\mathrm{Span}\{p\}$ is

$$\mathrm{proj}_{\mathrm{Span}\{p\}}\, q = \frac{28}{50}p = \boxed{\frac{14}{25}(4 + t)}.$$

A check is $\langle q - (14/25)p, p\rangle = 28 - (14/25)50 = 0$. Thus the residual is orthogonal to the whole one-dimensional target space.

Exercise 6.7.8: Project onto the second polynomial line

For the pair in Exercise 4, the coefficient is $\langle q, p\rangle/\langle p, p\rangle = -10/20 = -1/2$. Hence the projection is $\boxed{-\frac{1}{2}(3t - t^2) = (t^2 - 3t)/2}$. Its residual check is $-10 - (-1/2)20 = 0$. A negative projection coefficient is perfectly possible.

Exercise 6.7.9: Remove the lower-degree part of a quadratic

Use evaluation nodes $-3, -1, 1, 3$. The value lists for p_0, p_1, p_2 are $(1, 1, 1, 1)$, $(-3, -1, 1, 3)$, and $(9, 1, 1, 9)$.

(a) Project onto the first two orthogonal polynomials.

The first two lists have dot product zero. Also $\langle p_2, p_0\rangle = 20$, $\langle p_0, p_0\rangle = 4$, and $\langle p_2, p_1\rangle = -27 - 1 + 1 + 27 = 0$. Thus the projection is $5p_0 = 5$.

(b) Form and scale the orthogonal remainder.

The remainder $t^2 - 5$ has values $(4, -4, -4, 4)$. Divide by four to get $\boxed{q(t) = (t^2 - 5)/4}$, whose values are $(1, -1, -1, 1)$. It is orthogonal to both p_0, p_1 and is nonzero. The three polynomials have distinct increasing degrees, so they are independent and span the same space as $1, t, t^2$.

Exercise 6.7.10: Best quadratic approximation to an odd cubic

In the same four-node inner product, t^3 is orthogonal to the even polynomials p_0 and q: the products cancel at opposite nodes. Its only nonzero coefficient in the orthogonal list is on $p_1 = t$:

$$\frac{\langle t^3, t\rangle}{\langle t, t\rangle} = \frac{81 + 1 + 1 + 81}{9 + 1 + 1 + 9} = \frac{41}{5}.$$

Conclusion. The best approximation is $(41/5)t$. The best member of a quadratic-polynomial space need not have a nonzero quadratic term.

Exercise 6.7.11: A cubic projected onto a five-node quadratic space

Here evaluation is at $-2, -1, 0, 1, 2$, and the source's orthogonal polynomials are $p_0 = 1$, $p_1 = t$, $p_2 = t^2 - 2$. Odd-even cancellation makes the projection coefficients on p_0, p_2 zero. The coefficient on t is

$$\frac{\sum t_i^4}{\sum t_i^2} = \frac{16 + 1 + 0 + 1 + 16}{4 + 1 + 0 + 1 + 4} = \frac{17}{5}.$$

Conclusion. The orthogonal projection is $(17/5)t$.

Exercise 6.7.12: Choose a scaled orthogonal cubic

Subtract the projection from Exercise 11 to obtain $r(t) = t^3 - (17/5)t$. Its values at the five nodes are $(-6/5, 12/5, 0, -12/5, 6/5)$. Multiplying by $5/6$ yields

$$\boxed{p_3(t) = (5t^3 - 17t)/6},$$

with the requested values $(-1, 2, 0, -2, 1)$. The remainder is orthogonal to all of $\text{Span}\{p_0, p_1, p_2\}$. Its degree is three, so it is nonzero and adds a new direction. Thus the four polynomials are an orthogonal basis for $\mathcal{P}_3$.

Exercise 6.7.13: Transport the Euclidean product through an invertible map

Define $\langle u, v\rangle = (Au)^T(Av)$. Symmetry follows from symmetry of the dot product. For addition, $A(u+w) = Au + Aw$, so $\langle u+w, v\rangle = (Au+Aw)^T Av = \langle u, v\rangle + \langle w, v\rangle$. Similarly $A(cu) = cAu$ gives homogeneity. Finally $\langle u, u\rangle = \|Au\|^2 \geq 0$. If it is zero, $Au = 0$; invertibility gives $u = 0$. Conversely $u = 0$ clearly gives zero. All four axioms hold. The last step is why invertibility, rather than merely linearity of A, was required.

Exercise 6.7.14: The same construction for an injective transformation

Define $\langle u, v\rangle = T(u) \cdot T(v)$. Symmetry and the two linearity axioms follow exactly by applying linearity of T and the dot-product rules in $\mathbb{R}^n$. The self-product is $\|T(u)\|^2$, nonnegative and zero exactly when $T(u) = 0$. Because T is one-to-one and $T(0) = 0$, this happens only for $u = 0$. Thus positive definiteness holds, proving an inner product on V. Without injectivity, a nonzero kernel vector would have self-product zero and violate that axiom.

Exercise 6.7.15: Move a scalar out of the second argument

Symmetry and homogeneity in the first argument give

$$\langle u, cv\rangle = \langle cv, u\rangle = c\langle v, u\rangle = c\langle u, v\rangle.$$

These are real inner products, so no complex-conjugation convention is involved.

Exercise 6.7.16: The distance between two orthonormal vectors

Expand using bilinearity and symmetry:

$$\|u - v\|^2 = \langle u, u\rangle - 2\langle u, v\rangle + \langle v, v\rangle = 1 - 0 + 1 = 2.$$

Taking the nonnegative square root gives $\boxed{\|u - v\| = \sqrt{2}}$.

Exercise 6.7.17: Recover an inner product from two norms

Expand the two squared norms: $\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$ and $\|u-v\|^2 = \|u\|^2 - 2\langle u, v\rangle + \|v\|^2$. Subtracting cancels both self-products and leaves $\|u+v\|^2 - \|u-v\|^2 = 4\langle u, v\rangle$. Division by four proves the stated identity.

Exercise 6.7.18: The parallelogram identity

Add, instead of subtracting, the two expansions in Exercise 17. The mixed terms $2\langle u, v\rangle$ and $-2\langle u, v\rangle$ cancel, leaving

$$\boxed{\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2}.$$

The proof uses only inner-product axioms, so it is not restricted to coordinate vectors.

Exercise 6.7.19: Arithmetic mean dominates geometric mean

Use the ordinary dot product on the two supplied coordinate vectors. Their dot product is $2\sqrt{ab}$, while each squared norm is $a+b$. Cauchy–Schwarz gives $2\sqrt{ab} \le \sqrt{a+b}\sqrt{a+b} = a+b$, where nonnegativity of a, b makes the square roots defined and the left side nonnegative. Divide by two: $\boxed{\sqrt{ab} \le (a+b)/2}$. The case $a = b = 0$ is included.

Exercise 6.7.20: The square of a mean versus the mean of squares

For $u = (a, b)^T$ and $v = (1, 1)^T$, squared Cauchy–Schwarz gives $(a+b)^2 \le (a^2+b^2)(1+1) = 2(a^2+b^2)$. Divide by four to obtain

$$\boxed{\left(\frac{a+b}{2}\right)^2 \le \frac{a^2+b^2}{2}}.$$

This holds for all real a, b, with no nonnegativity assumption needed.

Exercise 6.7.21: An integral inner product that vanishes

The product expands as $(1-3t^2)(t-t^3) = t - 4t^3 + 3t^5$. Therefore

$$\langle f, g\rangle = \int_0^1 (t - 4t^3 + 3t^5)\,dt = \frac{1}{2} - 1 + \frac{1}{2} = \boxed{0}.$$

Both functions are continuous, so this is the specified inner product; they are orthogonal even though neither is zero.

Exercise 6.7.22: Another orthogonal pair of functions

Expanding $(5t-3)(t^3-t^2)$ gives $5t^4 - 8t^3 + 3t^2$. Thus $\langle f, g\rangle = \int_0^1 (5t^4 - 8t^3 + 3t^2)\,dt = 1 - 2 + 1 = \boxed{0}$. The integral of a product can cancel even when the functions have nonzero norms.

Exercise 6.7.23: An integral norm is a square root

For $f(t) = 1 - 3t^2$,

$$\|f\|^2 = \int_0^1 (1 - 6t^2 + 9t^4)\,dt = 1 - 2 + \frac{9}{5} = \frac{4}{5}.$$

Hence $\boxed{\|f\| = 2/\sqrt{5}}$. Nonnegativity determines the choice of square root.

Exercise 6.7.24: The norm of a cubic-minus-quadratic function

For $g(t) = t^3 - t^2$,

$$\|g\|^2 = \int_0^1 (t^6 - 2t^5 + t^4)\,dt = \frac{1}{7} - \frac{1}{3} + \frac{1}{5} = \frac{1}{105}.$$

Thus $\boxed{\|g\| = 1/\sqrt{105}}$.

Exercise 6.7.25: An orthogonal polynomial basis on a symmetric interval

On $[-1,1]$, oddness gives $\langle 1,t\rangle = 0$ and $\langle t,t^2\rangle = 0$. The projection of t^2 onto 1 has coefficient $(\int_{-1}^{1} t^2)/(\int_{-1}^{1} 1) = (2/3)/2 = 1/3$. Thus $t^2 - 1/3$ is orthogonal to both lower-degree polynomials. A conventional scaling gives the basis

$$\boxed{\{1,\ t,\ (3t^2-1)/2\}}.$$

The last polynomial has integral zero and its product with t is odd. All three are nonzero of distinct degrees, so they form a basis of $\mathrm{Span}\{1,t,t^2\}$. Their squared norms are $2, 2/3, 2/5$, respectively.

Exercise 6.7.26: Repeat the orthogonalization on a wider interval

On $[-2,2]$, the first two polynomials are again orthogonal by oddness. The coefficient to subtract from t^2 is $(\int_{-2}^{2} t^2)/(\int_{-2}^{2} 1) = (16/3)/4 = 4/3$. Scaling the remainder by 3 gives $\boxed{\{1,t,3t^2-4\}}$. The last polynomial has integral $3(16/3) - 4(4) = 0$, and its product with t is odd. Distinct degrees give independence and the same polynomial span. All squared norms are positive; the last is $256/5$.

Exercise 6.7.27: Complete the five-node orthogonal polynomial basis

Use the evaluation nodes $-2,-1,0,1,2$ and start with $p_0 = 1, p_1 = t, p_2 = t^2 - 2$. They are orthogonal. For t^3, only the projection onto t is nonzero, with coefficient $17/5$ as computed in Exercise 11. Set $r_3 = t^3 - (17/5)t$. For t^4, oddness makes the projections onto p_1 and r_3 zero. The constant projection coefficient is $34/5$, while

$$\frac{\langle t^4, p_2\rangle}{\langle p_2, p_2\rangle} = \frac{(\sum t_i^6) - 2(\sum t_i^4)}{14} = \frac{130-68}{14} = \frac{31}{7}.$$

Hence the final remainder is

$$r_4 = t^4 - \frac{34}{5} - \frac{31}{7}(t^2-2) = t^4 - \frac{31}{7}t^2 + \frac{72}{35}.$$

One convenient rescaled basis is

$$\boxed{\{1,t,t^2-2,5t^3-17t,35t^4-155t^2+72\}}.$$

Their distinct degrees prove independence, and Gram–Schmidt preserves the original span $\mathcal{P}_4$. Exact dot products of their five-node value vectors are zero for every distinct pair, verifying orthogonality.

Exercise 6.7.28: Orthogonalize powers of cosine by integration

Start with $q_0 = 1$ and $q_1 = \cos t$: their integral product over $[0, 2\pi]$ is zero. Since $\int \cos^2 t = \pi$ and $\int 1 = 2\pi$, the projection of $\cos^2 t$ onto the constant is $1/2$. Its projection onto $\cos t$ is zero because $\int_0^{2\pi} \cos^3 t\, dt = 0$. Thus $q_2 = \cos^2 t - 1/2$. For $\cos^3 t$, the constant and q_2 projections vanish by cancellation over a half-period. The remaining coefficient is

$$\frac{\int_0^{2\pi} \cos^4 t\, dt}{\int_0^{2\pi} \cos^2 t\, dt} = \frac{3\pi/4}{\pi} = \frac{3}{4}.$$

Therefore $q_3 = \cos^3 t - (3/4)\cos t$, giving the orthogonal basis

$$\boxed{\{1, \cos t, \cos^2 t - 1/2, \cos^3 t - (3/4)\cos t\}}.$$

The squared norms are $2\pi, \pi, \pi/4, \pi/16$, all nonzero. Equivalently the last two are $(1/2)\cos 2t$ and $(1/4)\cos 3t$, so the basis spans the same four-dimensional space as the original list.

6.8 Applications of inner product spaces

Two conventions matter. The book uses W to multiply *residuals*, so weighted least squares minimizes $\|W(y - X\beta)\|^2$. Therefore its normal matrix contains $W^TW = W^2$ when W is diagonal. Do not replace this by W unless the supplied weights have first been square-rooted. For Fourier approximations, the inner product is the ordinary integral $\langle f, g\rangle = \int_0^{2\pi} fg$; the constant coefficient is $a_0/2$, with $a_0 = (1/\pi)\int f$, while $a_k = (1/\pi)\int f\cos kt$ and $b_k = (1/\pi)\int f\sin kt$. These differing denominators arise because $\|1\|^2 = 2\pi$ and $\|\sin kt\|^2 = \|\cos kt\|^2 = \pi$ for positive integer k.

Practice problem 6.8.P1: Check the three integral products

For $q_1 = 1$, $q_2 = t$, $q_3 = 3t^2 - 4$ on $[-2, 2]$,

$$\langle q_1, q_2\rangle = \int_{-2}^{2} t\,dt = 0,$$

$$\langle q_1, q_3\rangle = [t^3 - 4t]_{-2}^{2} = 0,$$

$$\langle q_2, q_3\rangle = \int_{-2}^{2} (3t^3 - 4t)\,dt = 0.$$

The first and third vanish also by oddness. All three polynomials are nonzero, so the pairwise zero products prove an orthogonal set.

Practice problem 6.8.P2: Read approximations from a finite trigonometric expression

The function is already $f(t) = 3 - 2\sin t + 5\sin 2t - 6\cos 2t$. The first-order space retains only the constant and frequency-one terms, so its approximation is $3 - 2\sin t$. The third-order space contains all the displayed terms (there are no frequency-three terms to add), so its approximation is f itself. Orthogonality makes the discarded higher-frequency terms perpendicular to the lower-order space; this is why simply retaining terms gives the least-squares approximation here.

Exercise 6.8.1: Half-size residual multipliers at the two endpoints

Use design rows $(1, x_i)$ for $x_i = -2, -1, 0, 1, 2$ and $y = (0, 0, 2, 4, 4)^T$. Following the book's convention, take $W = \mathrm{diag}(1, 2, 2, 2, 1)$: the endpoint residuals receive half the multiplier of the interior ones. Scaling all multipliers by a common positive factor does not change the fit. The weighted normal equations are

$$X^TW^2X\widehat{\beta} = X^TW^2y, \qquad \begin{bmatrix} 14 & 0 \\ 0 & 16 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 28 \\ 24 \end{bmatrix}.$$

Thus $\widehat{\beta}_0 = 2$ and $\widehat{\beta}_1 = 3/2$.

Conclusion. The fitted line is $y = 2 + (3/2)x$. Here "half" refers to the residual multiplier used in the text, whose square enters the objective.

Exercise 6.8.2: A common rescaling leaves the minimizers unchanged

The second weighting matrix is $W_2 = 2W_1$, because every point's multiplier is doubled. For any parameter vector,

$$\|W_2(y - X\beta)\|^2 = 4\|W_1(y - X\beta)\|^2.$$

Multiplication by the positive constant 4 preserves every ordering of objective values, so the minimizing parameter vectors are exactly the same. The numerical minimum changes by that factor, but the fitted curve does not. This also holds when the minimizer is not unique.

Exercise 6.8.3: Add the cubic trend to the quadratic trend

Use nodes $-2, -1, 0, 1, 2$ and observations $g = (3, 5, 5, 4, 3)$. The orthogonal polynomials from Example 2 and the question are $p_0 = 1, p_1 = t, p_2 = t^2 - 2, p_3 = (5t^3 - 17t)/6$. Their inner products with the data and squared norms are

j	0	1	2	3
$\langle g, p_j \rangle$	20	-1	-7	2
$\langle p_j, p_j \rangle$	5	10	14	10

The cubic values are $(-1, 2, 0, -2, 1)$, so its coefficient is $2/10 = 1/5$. The fitted cubic trend is

$$\widehat{g} = 4p_0 - \tfrac{1}{10}p_1 - \tfrac{1}{2}p_2 + \tfrac{1}{5}p_3 = \boxed{5 - \tfrac{2}{3}t - \tfrac{1}{2}t^2 + \tfrac{1}{6}t^3}.$$

Because the basis is orthogonal, adding the cubic term does not require recomputing the earlier three coefficients.

Exercise 6.8.4: Orthogonal trends at six evenly spaced nodes

(a) Check the evaluation vectors.

At $-5, -3, -1, 1, 3, 5$, the proposed polynomials have lists

$$p_0 : (1, 1, 1, 1, 1, 1), \quad p_1 : (-5, -3, -1, 1, 3, 5),$$

$$p_2 : (5, -1, -4, -4, -1, 5).$$

The lists for p_1 and p_2 each sum to zero, so both are orthogonal to p_0. Their mutual product sums to zero by odd-even cancellation. Their squared norms are $6, 70, 84$, all positive. Thus they form the required orthogonal basis of the quadratic space.

(b) Project the actual data onto that basis.

For $g = (1, 1, 4, 4, 6, 8)$, the three inner products are $24, 50, 6$. Hence

$$\widehat{g} = \frac{24}{6}p_0 + \frac{50}{70}p_1 + \frac{6}{84}p_2 = 4 + \frac{5}{7}t + \frac{1}{14}\left(\frac{3}{8}t^2 - \frac{35}{8}\right).$$

Equivalently the quadratic is $\boxed{\widehat{g}(t) = 59/16 + (5/7)t + (3/112)t^2}$. The projection residual is orthogonal to all three basis lists, verifying the fit.

Exercise 6.8.5: Sines of distinct integer frequencies are orthogonal

For unequal positive integers m, n, use $2\sin mt\sin nt = \cos((m-n)t) - \cos((m+n)t)$. Both $m-n$ and $m+n$ are nonzero integers. Their cosine integrals on $[0, 2\pi]$ are zero, since $\int_0^{2\pi}\cos(kt)\,dt = [\sin(kt)/k]_0^{2\pi} = 0$. Thus the integral of the original product is zero, proving orthogonality.

Exercise 6.8.6: Sine and cosine terms are orthogonal even at equal frequency

Use $2\sin mt\cos nt = \sin((m+n)t) + \sin((m-n)t)$. Every nonzero integer-frequency sine integrates to zero over $[0, 2\pi]$. If $m = n$, the second term is identically $\sin 0 = 0$, so it also has zero integral without any division by zero. Thus $\langle \sin mt, \cos nt\rangle = 0$ for all the stated positive integers.

Exercise 6.8.7: The Fourier basis elements have squared norm pi

For positive integer k, the identities $\cos^2 kt = (1+\cos 2kt)/2$ and $\sin^2 kt = (1-\cos 2kt)/2$ give

$$\|\cos kt\|^2 = \tfrac{1}{2}(2\pi) + 0 = \pi, \qquad \|\sin kt\|^2 = \tfrac{1}{2}(2\pi) - 0 = \pi.$$

The Fourier setting uses integer frequencies. For an arbitrary noninteger real k, the boundary term $\sin(4\pi k)/(4k)$ need not vanish, so the integer-frequency condition is what justifies these equalities.

Exercise 6.8.8: A constant shift changes only the constant coefficient

For $f(t) = t - 1$, the mean coefficient is $(1/(2\pi))\int_0^{2\pi}(t-1)\,dt = \pi - 1$. For $k \geq 1$, integration by parts gives $a_k = 0$ and $b_k = -2/k$; the constant -1 has zero product with those sines and cosines.

Conclusion. The third-order approximation is $\pi - 1 - 2\sin t - \sin 2t - (2/3)\sin 3t$.

Exercise 6.8.9: Reverse the slope of the linear function

For $f(t) = 2\pi - t$, the constant coefficient is π. The nonconstant Fourier coefficients of t have their signs reversed: $a_k = 0$, $b_k = 2/k$. These also follow directly by integrating $(2\pi - t)\sin kt$ by parts.

Conclusion. The third-order approximation is $\pi + 2\sin t + \sin 2t + (2/3)\sin 3t$.

Exercise 6.8.10: Project an integrable square wave onto continuous trigonometric functions

The square wave in the exercise has a jump at π, so it is not literally a member of $C[0, 2\pi]$. Nevertheless it is bounded and Riemann integrable, as are all its products with the finite continuous trigonometric basis. The finite-dimensional least-squares problem $\min_g \int_0^{2\pi}(f-g)^2$ is therefore well defined on that continuous trigonometric subspace. Its normal equations use the same integrals; no assertion about an inner product on all pointwise piecewise-continuous functions is needed. Values assigned at isolated jump endpoints do not change any of these integrals. The mean is zero. For $k \geq 1$,

$$a_k = \frac{1}{\pi}\left(\int_0^{\pi}\cos kt\,dt - \int_{\pi}^{2\pi}\cos kt\,dt\right) = 0,$$

$$b_k = \frac{1}{\pi}\left(\int_0^{\pi} \sin kt\,dt - \int_{\pi}^{2\pi} \sin kt\,dt\right) = \frac{2(1-(-1)^k)}{\pi k}.$$

Thus even sine coefficients vanish and odd ones equal $4/(\pi k)$.

Conclusion. The third-order approximation is $(4/\pi)(\sin t + (1/3)\sin 3t)$.

Exercise 6.8.11: An exact finite trigonometric identity

The identity $\sin^2 t = (1-\cos 2t)/2$ already expresses the function in the third-order target subspace. It is therefore its own best approximation, with zero error. Its only nonzero coefficients are the constant $1/2$ and the cosine-two coefficient $-1/2$.

Conclusion. $1/2 - (1/2)\cos 2t$. No integration is necessary.

Exercise 6.8.12: Use the cosine triple-angle identity

From $\cos 3t = 4\cos^3 t - 3\cos t$, solve for the cube: $\cos^3 t = (3/4)\cos t + (1/4)\cos 3t$. This lies entirely in the third-order target space, so the best approximation is exact, again with zero error.

Conclusion. $(3/4)\cos t + (1/4)\cos 3t$.

Exercise 6.8.13: Each coefficient depends linearly on the function

For example,

$$a_k(f+g) = \frac{1}{\pi}\int_0^{2\pi} (f+g)\cos kt = a_k(f) + a_k(g)$$

by linearity of the integral. Replacing cosine by sine proves the same identity for b_k, and taking the constant term proves it for a_0 too. The fixed denominators are the same for both functions, so no extra normalization appears when adding coefficients.

Exercise 6.8.14: One additional correct Fourier coefficient cannot worsen the fit

The difference is $h - g = b_2 \sin 2t$. Since b_2 is the Fourier coefficient of f, the residual $f - h$ is orthogonal to $\sin 2t$. Therefore the Pythagorean theorem gives

$$\|f-g\|^2 = \|(f-h) + b_2 \sin 2t\|^2 = \|f-h\|^2 + \pi b_2^2.$$

Conclusion. h is at least as close as g. It is strictly closer if $b_2 \neq 0$; if $b_2 = 0$, the two functions and their errors are identical.

Exercise 6.8.15: A weighted cubic takeoff fit

Use the thirteen times and position observations from Exercise 6.6.13, with design row $(1, t, t^2, t^3)$ and the stated diagonal residual multipliers $1, 1, 1, 0.9, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1$. The normal equations must use $X^T W^2 X$, not $X^T W X$. For reproducibility their coefficient matrix and right side

are

$$\begin{bmatrix} 6.66 & 22.23 & 120.77 & 797.19 \\ 22.23 & 120.77 & 797.19 & 5956.13 \\ 120.77 & 797.19 & 5956.13 & 48490.23 \\ 797.19 & 5956.13 & 48490.23 & 420477.17 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 747.844 \\ 4815.438 \\ 35420.468 \\ 285262.44 \end{bmatrix}.$$

Solving with the supplied decimals treated as exact rational data gives, after rounding for display,

$$y(t) \approx -0.2685118943 + 3.6095481671t + 5.8575839604t^2 - 0.04770303371t^3.$$

The exact fitted residual satisfies $X^T W^2 (y - X\widehat{\beta}) = 0$, which checks that this curve minimizes the weighted objective. Differentiate and evaluate at 4.5 seconds:

$$y'(4.5) = \widehat{\beta}_1 + 9\widehat{\beta}_2 + 60.75\widehat{\beta}_3 \approx \boxed{53.42984451 \text{ feet/second}}.$$

The later observations receive less influence because their residuals are multiplied by smaller numbers before squaring.

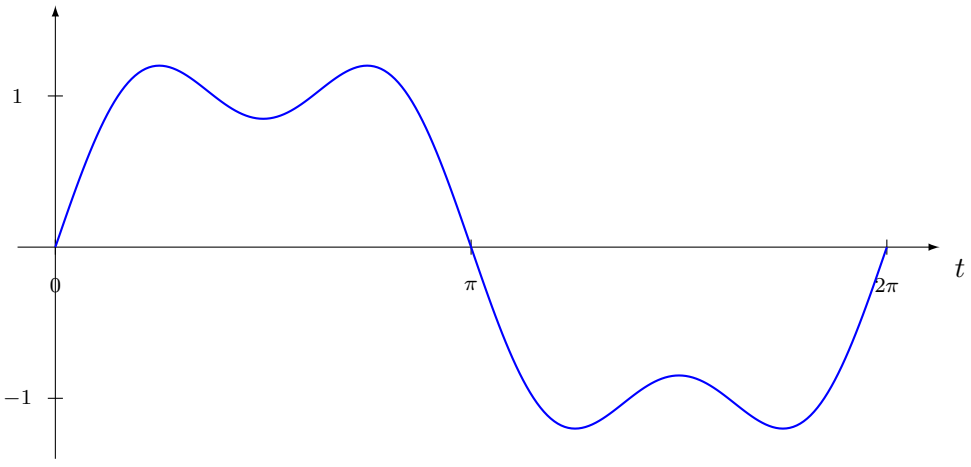
Exercise 6.8.16: Graph two Fourier orders and their periodic continuation

From Exercise 10, even sine coefficients vanish, so

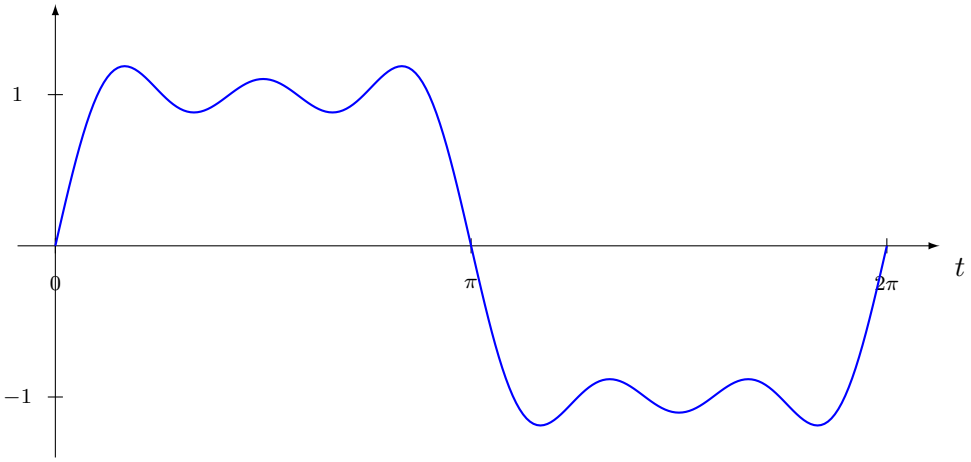
$$f_4(t) = \frac{4}{\pi}\left(\sin t + \frac{\sin 3t}{3}\right), \qquad f_5(t) = f_4(t) + \frac{4}{5\pi}\sin 5t.$$

Thus the fourth-order projection equals the third-order projection: its new frequency-four coefficients are zero. The fifth adds a nonzero frequency-five term. The requested three separate plots follow; the formulas determine the values exactly, while the curves are sampled densely for drawing. Both are 2π-periodic, explaining the continuation on the third plot.

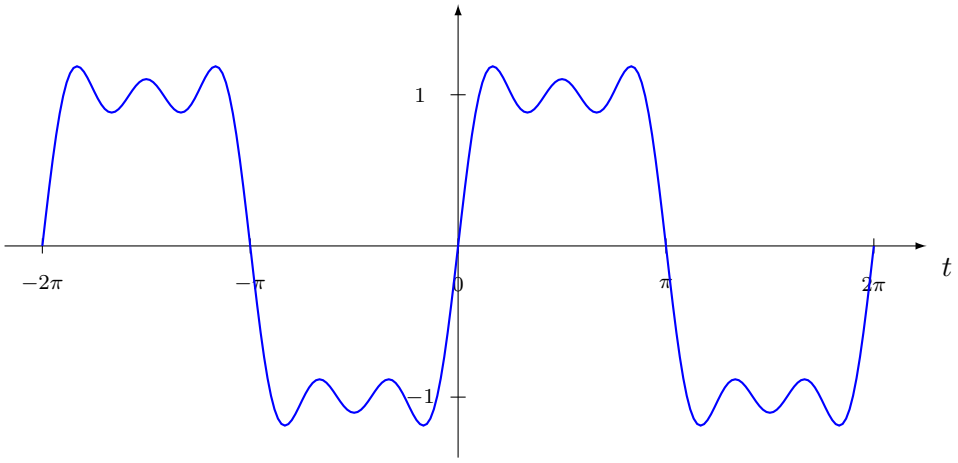
f_4 **on** $[0, 2\pi]$



f_5 **on** $[0, 2\pi]$



f_5 **on** $[-2\pi, 2\pi]$



Near the target's jumps the approximations overshoot; adding terms improves the integral fit without making these low-order continuous curves equal a discontinuous square wave. The finite trigonometric sums have value zero at the jump locations, the midpoint of the adjacent target levels.

6.S Chapter 6 supplementary exercises

For real vectors, $u \cdot v = u^T v$ and $\|u\| = \sqrt{u^T u}$. Orthogonal means perpendicular; orthonormal additionally requires unit length. The orthogonal decomposition $x = p + z$, with $p \in W$ and $z \in W^\perp$, has $p = \text{proj}_W x$. It yields the Pythagorean identity $\|x\|^2 = \|p\|^2 + \|z\|^2$. A least-squares solution is an *input* $\widehat{x}$ minimizing $\|b - Ax\|$; the corresponding closest *output* is $A\widehat{x}$. These two vectors can even have different numbers of coordinates.

Exercise 6.S.1: Nineteen true–false statements about orthogonality

(a) **False.** The zero vector has length zero. All lengths are nonnegative; only nonzero vectors have strictly positive length.

(b) **True.** $\| - v\| = | - 1|\|v\| = \|v\|$.

(c) **True.** This is the Euclidean distance definition.

(d) **False.** The rule is $\|rv\| = |r|\|v\|$. A negative r and nonzero v disprove the stated formula.

(e) **False.** The zero vector is orthogonal to every vector, but a set containing zero is dependent. Two *nonzero* orthogonal vectors are independent.

(f) **True.** $x \cdot (u - v) = x \cdot u - x \cdot v = 0 - 0 = 0$.

(g) **True.** Expanding gives $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$; comparison forces $u \cdot v = 0$.

(h) **True.** The analogous expansion with a minus sign gives $\|u - v\|^2 = \|u\|^2 - 2u \cdot v + \|v\|^2$, again forcing zero dot product.

(i) **False.** The projection onto nonzero u is $((y \cdot u)/(u \cdot u))u$, a multiple of u, not generally of y. For $y = (1, 1)$, $u = (1, 0)$ the projection $(1, 0)$ is not a multiple of y.

(j) **True.** A projection onto W belongs to W by its construction; equality with that projection places y in W.

(k) **True.** The set is the null space of the one-row matrix v^T for the fixed vector v, so is a subspace. This includes $v = 0$, when the set is all of $\mathbb{R}^n$.

(l) **False.** Their intersection is $\{0\}$, not empty. If x belongs to both, $x \cdot x = 0$, hence $x = 0$.

(m) **True.** For $i \neq j$, $(c_i v_i) \cdot (c_j v_j) = c_i c_j (v_i \cdot v_j) = 0$. Zero coefficients may introduce zero vectors, but pairwise orthogonality still holds.

(n) **False.** Orthonormal columns give $U^T U = I$. For $U = (1, 0)^T$, $UU^T = \text{diag}(1, 0) \neq I_2$. Squareness is needed to infer the reverse identity.

(o) **False.** Orthogonal columns need not have length one. The matrix $2I$ has orthogonal columns but $U^T U = 4I \neq I$.

(p) **True.** Square orthonormal-column U satisfies $U^T U = I$, hence $U^{-1} = U^T$ and $UU^T = I$. The latter identity says its rows are orthonormal.

(q) **True.** The projection and its residual are perpendicular, so the displayed identity is precisely Pythagoras.

(r) **False.** The minimizing input is $\widehat{x}$; $A\widehat{x}$ is the closest vector in $\operatorname{Col} A$. The inequality is valid, but it describes the output, not the least-squares solution itself.

(s) **False.** The normal equations are $A^TA\widehat{x} = A^Tb$. The inverse formula $\widehat{x} = (A^TA)^{-1}A^Tb$ requires independent columns. For $A = 0$, least-squares solutions exist but A^TA is not invertible.

Exercise 6.S.2: Inductively extend Pythagoras

For $p = 2$, orthonormality makes c_1v_1 perpendicular to c_2v_2; therefore $\|c_1v_1 + c_2v_2\|^2 = c_1^2\|v_1\|^2 + c_2^2\|v_2\|^2 = c_1^2 + c_2^2$. Assume the formula holds for $p-1$ terms. Let $w = \sum_{i=1}^{p-1} c_iv_i$. Since $w \cdot v_p = \sum_{i<p} c_i(v_i \cdot v_p) = 0$, Pythagoras gives

$$\|w + c_pv_p\|^2 = \|w\|^2 + c_p^2\|v_p\|^2 = \sum_{i=1}^{p-1} c_i^2 + c_p^2.$$

This completes induction. For real coefficients $c_i^2 = |c_i|^2$, so the result has exactly the form in the exercise.

Exercise 6.S.3: Bessel's inequality

Let $p = \sum_i (x \cdot v_i)v_i$, the projection of x onto the span of the orthonormal vectors. The residual $z = x - p$ is orthogonal to every v_i, because $z \cdot v_j = x \cdot v_j - (x \cdot v_j) = 0$. Thus $p \perp z$, and Exercise 2 yields

$$\|x\|^2 = \|p\|^2 + \|z\|^2 = \sum_i |x \cdot v_i|^2 + \|z\|^2 \geq \sum_i |x \cdot v_i|^2.$$

Nonnegativity of the residual's squared length is the entire source of the inequality. Equality holds precisely when x lies in the given span.

Exercise 6.S.4: An orthogonal map preserves an orthonormal basis

For each pair i, j,

$$(Uv_i) \cdot (Uv_j) = v_i^TU^TUv_j = v_i^Tv_j = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Thus the n transformed vectors are orthonormal. They are independent: a zero combination, dotted with each transformed vector, forces each coefficient to vanish. An independent set of n vectors in $\mathbb{R}^n$ is a basis, proving the full claim.

Exercise 6.S.5: Preserving all inner products characterizes orthogonality

Apply the given identity to $x = e_i$, $y = e_j$. The vectors Ue_i, Ue_j are columns i, j of U, so their dot product is $e_i \cdot e_j$, equal to 1 for $i = j$ and zero otherwise. Hence the columns of U are orthonormal: $U^TU = I$. Since U is square, this is exactly the condition that U is orthogonal.

Exercise 6.S.6: Real eigenvalues of an orthogonal matrix

Suppose $Uv = \lambda v$ with $v \neq 0$ and real λ. Orthogonal matrices preserve length, giving $\|v\| = \|Uv\| = \|\lambda v\| = |\lambda|\|v\|$. Divide by $\|v\| > 0$ to obtain $|\lambda| = 1$. For a real number this says $\lambda = 1$ or -1. It does not claim every eigenvalue is real: planar rotations can have nonreal eigenvalues of modulus one.

Exercise 6.S.7: A Householder matrix is orthogonal

Put $P = uu^T$ with $u^T u = 1$. Then $P^T = P$ and $P^2 = u(u^T u)u^T = P$. For $Q = I - 2P$, symmetry gives $Q^T = Q$ and multiplication gives $Q^2 = I - 4P + 4P^2 = I$. Consequently $Q^T Q = Q^2 = I$, so Q is orthogonal. This proves length preservation as well as its being its own inverse.

Exercise 6.S.8: A length-preserving linear map is orthogonal

(a) Recover dot products from lengths.

By expansion, $2a \cdot b = \|a+b\|^2 - \|a\|^2 - \|b\|^2$. For $a = T(x)$, $b = T(y)$, linearity and length preservation give

$$2T(x) \cdot T(y) = \|T(x+y)\|^2 - \|T(x)\|^2 - \|T(y)\|^2 = \|x+y\|^2 - \|x\|^2 - \|y\|^2 = 2x \cdot y.$$

In particular perpendicular inputs have perpendicular outputs.

(b) Apply this to the standard basis.

Each $T(e_i)$ has length one, and distinct $T(e_i), T(e_j)$ are perpendicular by part (a). These vectors are the columns of the square standard matrix, so that matrix has orthonormal columns and is orthogonal.

Exercise 6.S.9: Best approximation without orthogonalizing the generators

Set $A = [u\ v]$. The problem is to minimize $\|z - Ac\|$ for $c = (x_1, x_2)^T$. Solve the two normal equations

$$\begin{bmatrix} u \cdot u & u \cdot v \\ u \cdot v & v \cdot v \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} u \cdot z \\ v \cdot z \end{bmatrix}.$$

Their coefficient matrix is invertible because u, v are independent: for nonzero c, $c^T A^T A c = \|Ac\|^2 > 0$. The desired closest vector is $x_1 u + x_2 v$. These equations explicitly account for the nonzero cross term $u \cdot v$; using the separate orthogonal-projection coefficients would be wrong when the generators are not perpendicular.

Exercise 6.S.10: Scale the data, scale the least-squares solution

Independent columns imply $A^T A$ is invertible and the solution is unique. Thus replacing b by cb changes the solution to

$$(A^T A)^{-1} A^T (cb) = c(A^T A)^{-1} A^T b = c\widehat{x}.$$

One can also multiply the original normal equations by c and use uniqueness. The assertion even holds for $c = 0$, though the question only requests nonzero scalars.

Exercise 6.S.11: Fit three parallel inconsistent planes

Write $t = x - 2y + 5z$ and $\mu = (a + b + c)/3$. The squared residual is

$$(t-a)^2 + (t-b)^2 + (t-c)^2 = 3(t-\mu)^2 + (a-\mu)^2 + (b-\mu)^2 + (c-\mu)^2.$$

The equality follows by expanding; the mixed terms cancel because $(a-\mu) + (b-\mu) + (c-\mu) = 0$. The last three terms are constant, so the minimum occurs exactly at $t = \mu$. Every such t is attainable, for instance by taking $x = t, y = z = 0$. Therefore all least-squares solutions, and only those, form the plane

$$\boxed{x - 2y + 5z = (a + b + c)/3}.$$

The least-squares input is nonunique because this equation leaves two free variables, even though the best fitted output is unique.

Exercise 6.S.12: A Rayleigh quotient as a one-variable least-squares fit

Assume the proposed approximate eigenvector v is nonzero. Treat $v\lambda = Av$ as a least-squares system with one unknown. Its normal equation is $(v^T v)\lambda = v^T A v$. Since $v^T v = \|v\|^2 > 0$, its unique solution is

$$\boxed{\widehat{\lambda} = \frac{v^T A v}{v^T v}}.$$

This is the Rayleigh quotient. If $Av = \lambda_0 v$ exactly, the quotient returns λ_0. For $v = 0$ every scalar gives the same zero residual, so there would be no meaningful eigenvalue estimate; nonzeroness is essential.

Exercise 6.S.13: Identify the four fundamental subspaces

(a) If $u \in \operatorname{Nul} A$, each row r_i^T of A satisfies $r_i^T u = 0$. A linear combination of those rows is still perpendicular to u. Thus every member of $\operatorname{Row} A$ is perpendicular to every member of $\operatorname{Nul} A$, proving the stated containment.

(b) If $\operatorname{rank} A = r$, rank–nullity gives $\dim \operatorname{Nul} A = n - r$. A subspace of $\mathbb{R}^n$ and its orthogonal complement have dimensions adding to n, so $\dim(\operatorname{Nul} A)^\perp = r$. The row space also has dimension r. A contained subspace with the same finite dimension must be the whole space, so $\operatorname{Row} A = (\operatorname{Nul} A)^\perp$.

(c) Apply this result to A^T. Since $\operatorname{Row}(A^T) = \operatorname{Col} A$, it gives $\operatorname{Col} A = (\operatorname{Nul} A^T)^\perp$, now inside the ambient space $\mathbb{R}^m$.

Exercise 6.S.14: The consistency alternative in orthogonal-complement language

The system $Ax = b$ is consistent if and only if $b \in \operatorname{Col} A$. By Exercise 13(c), this is equivalent to $b \in (\operatorname{Nul} A^T)^\perp$, which says b is orthogonal to every solution of $A^T y = 0$. This proves both directions of the equivalence. Orthogonality to just one selected null vector would not in general be sufficient.

Exercise 6.S.15: A real triangular Schur factorization gives real eigenvalues

The factorization $A = URU^T$ with orthogonal U makes A similar to R. Similar matrices have the same characteristic polynomial. Since R is upper triangular, $\det(R - tI) = \prod_{i=1}^{n}(r_{ii} - t)$. Its n roots, counted with multiplicity, are the real diagonal entries r_{ii}. Hence A has n real eigenvalues counting multiplicities. The restriction to a real, genuinely triangular R matters; real Schur forms for matrices with nonreal eigenvalues use 2×2 blocks instead.

Exercise 6.S.16: The first step in constructing a real Schur factorization

(a) Since $Ue_1 = u_1$ and $Au_1 = \lambda_1 u_1$,

$$(U^T AU)e_1 = U^T Au_1 = \lambda_1 U^T u_1 = \lambda_1 e_1.$$

This is precisely the assertion about the first column. Here $U^T u_1 = e_1$ follows from the orthonormality of the columns.

(b) The block upper triangular form gives

$$\det(U^T AU - tI) = (\lambda_1 - t)\det(A_1 - tI).$$

Similarity makes this equal to $\det(A - tI)$, whose roots, with multiplicity, are the listed $\lambda_1, \ldots, \lambda_n$. Cancelling the polynomial factor $\lambda_1 - t$ leaves the roots $\lambda_2, \ldots, \lambda_n$ for A_1. This argument also handles repeated values of λ_1: only one copy is removed from the multiplicity list.

Exercise 6.S.17: Check a condition-number bound for perturbed data

Use the Euclidean norm throughout and $\kappa_2(A)$ as the condition number. Exact rational solution of the two systems gives

$$x = (3.94, .49)^T, \qquad \Delta x = (-1.04, 1.51)^T.$$

For example $A^{-1} = \begin{bmatrix} -110 & 310 \\ 160 & -450 \end{bmatrix}$, so multiplying by b and Δb verifies both vectors. The measured ratios and bound are

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .461796, \quad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .000154792, \quad \kappa_2(A) \approx 3362.9997.$$

Their product on the right is about .520567, which exceeds .461796. Thus the asserted inequality holds for this example, despite the large relative solution change.

Exercise 6.S.18: The same perturbation with a different original solution

The same inverse and the new right side give

$$x = (-491.17, 713.15)^T, \qquad \Delta x = (-1.04, 1.51)^T.$$

The perturbation is unchanged because A and Δb are unchanged. However the relative ratios use the new original vector:

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .00211738, \qquad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .00211778.$$

Multiplying the latter by 3362.9997 gives about 7.12211, much larger than .00211738. The bound is valid but far from attained. A condition number describes worst-case amplification, not the amplification of every particular perturbation.

Exercise 6.S.19: A small change along a relatively harmless direction

Exact solution of the two printed systems gives

$$x = (-8.642, -1730.054, 2368.716, -844.866)^T,$$

$$\Delta x = (.000030, .000140, -.000155, .000059)^T.$$

These terminating decimals are exact for the printed decimal data. Direct multiplication by the given integer A yields the stated b and Δb, respectively, providing an independent check. Numerically,

$$\kappa_2(A) \approx 23682.9279, \quad \frac{\|\Delta x\|_2}{\|x\|_2} \approx 7.17780 \cdot 10^{-8}, \quad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .000283204.$$

The right-hand bound is approximately 6.70710, far larger than the observed $7.18 \cdot 10^{-8}$. Therefore the inequality holds. The original solution is large, while the particular change in it is tiny.

Exercise 6.S.20: A perturbation close to worst-case amplification

With the new right side and perturbation, exact rational arithmetic gives

$$x = (2.541, 1.460, 1.174, -.101)^T, \qquad \Delta x = (-.002322, -.464848, .636450, -.227007)^T.$$

Multiplication checks $Ax = b$ and $A\Delta x = \Delta b$. The condition number is the same as in Exercise 19, since A is unchanged. The ratios are

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .259665, \qquad \frac{\|\Delta b\|_2}{\|b\|_2} \approx 1.09702 \cdot 10^{-5}.$$

The right-hand bound is .259806, slightly larger than .259665. Thus the inequality holds and is nearly sharp for this choice. All four experiments can be reproduced using exact SymPy solves for x and Δx, then NumPy's `linalg.cond(A)` and Euclidean vector norms. Matching the norm used for the condition number and the relative changes is essential.

7.1 Diagonalization of symmetric matrices

A real matrix is symmetric when $A^T = A$, and an orthogonal matrix satisfies $P^TP = I$, so $P^{-1} = P^T$. These are different definitions. The spectral theorem says that a real square matrix has an orthonormal eigenvector basis exactly when it is symmetric. To orthogonally diagonalize a symmetric matrix:

1. Find its real eigenvalues, with multiplicities.
2. For each λ, solve $(A - \lambda I)v = 0$ by row reduction.
3. Within a repeated eigenspace, use Gram–Schmidt if needed. Eigenvectors belonging to distinct eigenvalues are already perpendicular.
4. Normalize each nonzero vector v_i to $u_i = v_i/\|v_i\|$. Set $P = [u_1 \ \cdots \ u_n]$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ in the same order. Then $A = PDP^T$.

In the computations below, the columns of V are unnormalized orthogonal eigenvectors. A diagonal factor rescales the columns to form P. Every displayed diagonalization was checked exactly by $AP = PD$ and $P^TP = I$. Changing a column's sign or reordering columns together with D gives another equally valid answer.

Practice problem 7.1.P1: A square of a symmetric matrix is symmetric

Using the reverse-order transpose rule, $(A^2)^T = (AA)^T = A^TA^T = AA = A^2$, since $A^T = A$. Thus A^2 is symmetric. The reversal creates no difficulty here because the two factors are the same matrix.

Practice problem 7.1.P2: Square an orthogonal diagonalization

If $A = PDP^T$ with $P^TP = I$, then $A^2 = PDP^TPDP^T = PD^2P^T$. The same P is orthogonal and D^2 is diagonal, so this is an orthogonal diagonalization of A^2.

Exercise 7.1.1: Match entries across the diagonal

The two off-diagonal entries are both 5, so $\begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix}^T$ is the original matrix. It is symmetric.

Exercise 7.1.2: Opposite signs do not match

The upper-right entry is 5 and the lower-left entry is -5. They would have to be equal for symmetry. Therefore the matrix is not symmetric.

Exercise 7.1.3: One unequal pair is enough

Here $a_{12} = 2$ but $a_{21} = 4$. Thus $A^T \neq A$, so the matrix is not symmetric, despite the repeated entries within each row.

Exercise 7.1.4: Check all three off-diagonal pairs

The paired entries are $a_{12} = a_{21} = 8$, $a_{13} = a_{31} = 3$, and $a_{23} = a_{32} = -2$. The diagonal is unchanged by transpose. Thus this matrix is symmetric.

Exercise 7.1.5: An upper triangular matrix with nonzero off-diagonal entries

The entry $a_{12} = 2$ has counterpart $a_{21} = 0$. Thus the matrix is not symmetric. An upper triangular symmetric matrix would have to be diagonal, because its entries below and above the diagonal must both vanish.

Exercise 7.1.6: Symmetry requires a square matrix

The given array has three rows and four columns. Its transpose has four rows and three columns, so the two cannot be equal. It is not symmetric. The repeated numerical pattern does not overcome the shape mismatch.

Exercise 7.1.7: Orthonormal decimal columns

The two columns have squared lengths $.6^2 + .8^2 = 1$ and $.8^2 + (-.6)^2 = 1$. Their dot product is $.6(.8) + .8(-.6) = 0$. Thus A is orthogonal, and

$$A^{-1} = A^T = \begin{bmatrix} .6 & .8 \\ .8 & -.6 \end{bmatrix}.$$

Here the matrix happens to be symmetric too, so it is its own inverse.

Exercise 7.1.8: A rotation has its transpose as inverse

Each column has squared length $1/2 + 1/2 = 1$, and the dot product is $-1/2 + 1/2 = 0$. Hence the matrix is orthogonal and

$$A^{-1} = A^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Geometrically this reverses the 45° rotation of the original matrix.

Exercise 7.1.9: Perpendicular but not unit columns

The columns $(-5, 2)^T$ and $(2, 5)^T$ are perpendicular, but each has squared length $25 + 4 = 29$. Thus $A^T A = 29I$, not I. The matrix is not orthogonal. Having an inverse is a separate issue; orthogonality specifically requires the columns to be normalized.

Exercise 7.1.10: Another missing normalization

Each column has squared length $1 + 4 + 4 = 9$, and distinct columns have dot product $-2 - 2 + 4 = 0$. Therefore $A^T A = 9I$, not I. The given matrix is not orthogonal; dividing it by 3 would make it so.

Exercise 7.1.11: Verify a less obvious orthogonal matrix

Let the columns be c_1, c_2, c_3. Their squared lengths are

$$\|c_1\|^2 = 4/9 + 5/9 = 1, \quad \|c_2\|^2 = 4/9 + 1/5 + 16/45 = 1, \quad \|c_3\|^2 = 1/9 + 4/5 + 4/45 = 1.$$

Their pairwise dot products are $4/9 - 4/9 = 0$, $2/9 - 2/9 = 0$, and $2/9 - 2/5 + 8/45 = 0$. Thus all columns are orthonormal and the inverse is the transpose:

$$A^{-1} = \begin{bmatrix} 2/3 & 0 & \sqrt{5}/3 \\ 2/3 & 1/\sqrt{5} & -4/\sqrt{45} \\ 1/3 & -2/\sqrt{5} & -2/\sqrt{45} \end{bmatrix}.$$

Exercise 7.1.12: An orthogonal signed matrix

Write $A = H/2$, where H has the printed entries 1 or -1. Each column of H has squared length 4, and every pair of different columns has two agreeing signs and two disagreeing signs, so dot product zero. Thus $A^T A = I$, and

$$A^{-1} = A^T = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}.$$

Exercise 7.1.13: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t-4)(t-2)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = 2$: row reduction gives $x_1 = -x_2$. $\lambda = 4$: row reduction gives $x_1 = x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (2, 4).$$

Their squared lengths are $(2, 2)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{2}} V, \qquad D = \operatorname{diag}(2, 4).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.14: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t-6)(t+4)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = -4$: row reduction gives $x_1 = -x_2$. $\lambda = 6$: row reduction gives $x_1 = x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (-4, 6).$$

Their squared lengths are $(2, 2)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{2}} V, \qquad D = \operatorname{diag}(-4, 6).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.15: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $t(t-17)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = 0$: row reduction gives $x_1 = \frac{x_2}{4}$. $\lambda = 17$: row reduction gives $x_1 = -4x_2$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 4 \\ 4 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (0, 17).$$

Their squared lengths are $(17, 17)$. Hence an explicit answer is

$$P = \frac{1}{\sqrt{17}} V, \qquad D = \operatorname{diag}(0, 17).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.16: An orthogonal eigenvector basis

The characteristic polynomial $\det(A - tI)$ is $(t - 25)(t + 25)$. Its roots give the eigenvalues below.

Solve the eigenspace equations.

$\lambda = -25$: row reduction gives $x_1 = -\frac{4x_2}{3}$. $\lambda = 25$: row reduction gives $x_1 = \frac{3x_2}{4}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_2) = (-25, 25).$$

Their squared lengths are $(25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(-25, 25).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.17: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -2$: row reduction gives $x_1 = -x_3, \quad x_2 = 0$. $\lambda = 2$: row reduction gives $x_1 = x_3, \quad x_2 = -2x_3$. $\lambda = 5$: row reduction gives $x_1 = x_3, \quad x_2 = x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-2, 2, 5).$$

Their squared lengths are $(2, 6, 3)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{3}), \qquad D = \operatorname{diag}(-2, 2, 5).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.18: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -50$: row reduction gives $x_1 = \frac{3x_2}{4}, \quad x_3 = 0$. $\lambda = 3$: row reduction gives $x_1 = 0, \quad x_2 = 0$. $\lambda = 25$: row reduction gives $x_1 = -\frac{4x_2}{3}, \quad x_3 = 0$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 3 & 0 & 4 \\ 4 & 0 & -3 \\ 0 & 1 & 0 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-50, 3, 25).$$

Their squared lengths are $(25, 1, 25)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{1}{5}, 1, \frac{1}{5}), \qquad D = \operatorname{diag}(-50, 3, 25).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.19: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = -2$: row reduction gives $x_1 = -x_3, \quad x_2 = -\frac{x_3}{2}$. $\lambda = 7$: row reduction gives $x_1 = -\frac{x_2}{2} + x_3$. Within the 7-eigenspace, start with $(1, 0, 1)^T$ and $(-1/2, 1, 0)^T$. Subtracting the projection of the latter onto the former gives $(-1/4, 1, 1/4)^T$, a multiple of $(-1, 4, 1)^T$. This is the Gram–Schmidt step that produces the two orthogonal columns used below.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 4 \\ -2 & 1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (-2, 7, 7).$$

Their squared lengths are $(9, 2, 18)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{1}{3}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6}), \qquad D = \operatorname{diag}(-2, 7, 7).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.20: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = -2x_3, \quad x_2 = -2x_3$. $\lambda = 7$: row reduction gives $x_1 = -\frac{x_3}{2}, \quad x_2 = x_3$. $\lambda = 13$: row reduction gives $x_1 = x_3, \quad x_2 = -\frac{x_3}{2}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \\ -1 & -2 & 2 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (1, 7, 13).$$

Their squared lengths are $(9, 9, 9)$. Hence an explicit answer is

$$P = \frac{1}{3}V, \qquad D = \operatorname{diag}(1, 7, 13).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.21: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = -x_3, \quad x_2 = -x_4$. $\lambda = 5$: row reduction gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = -x_4$. $\lambda = 9$: row reduction gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ -1 & 0 & 1 & 1 \\ 0 & -1 & -1 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (1, 1, 5, 9).$$

Their squared lengths are $(2, 2, 4, 4)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, \frac{1}{2}), \qquad D = \operatorname{diag}(1, 1, 5, 9).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.22: An orthogonal eigenvector basis

Solve the eigenspace equations.

$\lambda = 0$: row reduction gives $x_1 = 0, \quad x_2 = -x_4, \quad x_3 = 0$. $\lambda = 2$: row reduction gives $x_2 = x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (0, 2, 2, 2).$$

Their squared lengths are $(2, 1, 1, 2)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, 1, 1, \frac{\sqrt{2}}{2}), \qquad D = \operatorname{diag}(0, 2, 2, 2).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.23: An orthogonal eigenvector basis

The printed vector $v = (1, 1, 1)^T$ satisfies $Av = 5v$, so its eigenvalue is 5. The exercise also asks to verify 2 is an eigenvalue; the vector $(1, -1, 0)^T$ satisfies $A(1, -1, 0)^T = 2(1, -1, 0)^T$. Thus the two assertions in the wording refer to different eigenvectors.

Solve the eigenspace equations.

$\lambda = 2$: row reduction gives $x_1 = -x_2 - x_3$. $\lambda = 5$: row reduction gives $x_1 = x_3, \quad x_2 = x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (2, 2, 5).$$

Their squared lengths are $(2, 6, 3)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{3}), \qquad D = \operatorname{diag}(2, 2, 5).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.24: An orthogonal eigenvector basis

Multiplication gives $Av_1 = 10v_1$ for the supplied $v_1 = (-2, 2, 1)^T$, and $Av_2 = v_2$ for $v_2 = (1, 1, 0)^T$. For the repeated eigenvalue 1, $(1, -1, 4)^T$ is a second eigenvector orthogonal to v_2.

Solve the eigenspace equations.

$\lambda = 1$: row reduction gives $x_1 = x_2 + \frac{x_3}{2}$. $\lambda = 10$: row reduction gives $x_1 = -2x_3, \quad x_2 = 2x_3$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & -2 \\ 0 & 4 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_3) = (1, 1, 10).$$

Their squared lengths are $(2, 18, 9)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6}, \frac{1}{3}), \qquad D = \operatorname{diag}(1, 1, 10).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.25: Symmetry, multiplicity, and a missing unit condition

(a) **True.** If $A = PDP^T$, then $A^T = (PDP^T)^T = PD^T P^T = PDP^T = A$.

(b) **True.** Symmetry gives $3u^T v = (Au)^T v = u^T Av = 4u^T v$, so $u^T v = 0$.

(c) **False.** I_n is symmetric but has only one distinct eigenvalue, namely 1 with multiplicity n.

(d) **False.** For $B = vv^T$, $B^2 = (v^T v)B$. A projection requires $B^2 = B$, which for nonzero v requires $\|v\| = 1$. For $v = 2e_1$, $B = 4e_1 e_1^T$ is not idempotent.

Exercise 7.1.26: Four consequences and nonconsequences of the spectral theorem

(a) **True.** This is the real symmetric spectral theorem.

(b) **True.** Taking the transpose of PDP^T gives the same matrix because D is real diagonal.

(c) **False.** A 90° rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is orthogonal but not symmetric, so it cannot be orthogonally diagonalized over $\mathbb{R}$.

(d) **True.** In the spectral theorem the orthogonal diagonalization provides one basis vector for each occurrence of an eigenvalue on the diagonal, giving equality of geometric and algebraic multiplicities.

Exercise 7.1.27: Three symmetric products

Use the reverse-order transpose rule and the hypothesis $A^T = A$:

$$(B^T AB)^T = B^T A^T B = B^T AB, \quad (B^T B)^T = B^T B, \quad (BB^T)^T = BB^T.$$

The products have square sizes $m \times m$, $m \times m$, and $n \times n$, respectively, so these equalities indeed establish symmetry in each case.

Exercise 7.1.28: Move a symmetric matrix across a dot product

For every x, y, $(Ax) \cdot y = (Ax)^T y = x^T A^T y = x^T Ay = x \cdot (Ay)$. The step using $A^T = A$ is exactly where symmetry matters. Without it the correct expression would contain A^T on the other side.

Exercise 7.1.29: The inverse retains the eigenvectors

Write $A = PDP^T$ with P orthogonal. Since A is invertible, none of the diagonal eigenvalues in D is zero, so D^{-1} exists and is diagonal. Then $A^{-1} = PD^{-1}P^T$, an orthogonal diagonalization with the same P. Its diagonal eigenvalues are the reciprocals of the original ones.

Exercise 7.1.30: A commuting product remains symmetric

Orthogonal diagonalizability makes both A and B symmetric. Therefore $(AB)^T = B^T A^T = BA = AB$, using the additional commutation hypothesis. Thus AB is symmetric and the spectral theorem makes it orthogonally diagonalizable. Without $AB = BA$, a product of two symmetric matrices need not be symmetric.

Exercise 7.1.31: Count a repeated eigenvalue's eigenspace dimension

Let $x = Py$. Since P is invertible, $Ax = \lambda x$ is equivalent to $(D - \lambda I)y = 0$. At positions where the diagonal of D differs from λ, this forces $y_i = 0$. At its k positions equal to λ, the corresponding y_i are arbitrary. Thus the solution space has exactly k independent coordinates. Mapping by P gives the eigenspace of A and preserves its dimension, so that dimension is k. Equivalently the k corresponding columns of P form its basis.

Exercise 7.1.32: A symmetric triangular matrix is diagonal

From $A = PRP^T$, solve for $R = P^T AP$. Taking transposes gives $R^T = P^T A^T P = P^T AP = R$, so R is symmetric. But R is also upper triangular. Entries below its diagonal are zero; symmetry forces their counterpart entries above the diagonal to be zero too. Thus all off-diagonal entries vanish and R is diagonal.

Exercise 7.1.33: A spectral decomposition for Example 2

Use the orthogonal eigenvectors $a = (-1, 1, 0)^T$, $b = (-1, -1, 2)^T$, $c = (1, 1, 1)^T$, with eigenvalues $8, 6, 3$ and squared lengths $2, 6, 3$. After normalization the spectral formula gives

$$\boxed{A = 4aa^T + bb^T + cc^T}.$$

Indeed the normalization divides each outer product by its squared length; thus the three coefficients are 8/2, 6/6, 3/3. Expanding these outer products gives $\begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$, the matrix from Example 2. This checks the decomposition entrywise.

Exercise 7.1.34: A spectral decomposition with a repeated eigenvalue

For Example 3, take $a = (1, 0, 1)^T$, $b = (-1, 4, 1)^T$, and $c = (-2, -1, 2)^T$. They are mutually perpendicular, with squared lengths $2, 18, 9$. The first two have eigenvalue 7, and the last has eigenvalue -2. Therefore

$$\boxed{A = \frac{7}{2}aa^T + \frac{7}{18}bb^T - \frac{2}{9}cc^T}.$$

For instance the $(1, 1)$ entry is $7/2 + 7/18 - 8/9 = 3$; expansion of all entries returns $\begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$. The two separate rank-one terms for eigenvalue 7 reflect its two-dimensional eigenspace.

Exercise 7.1.35: A rank-one projection from a unit vector

(a) $Bx = (uu^T)x = u(u^T x) = (x \cdot u)u$. Since $u \cdot u = 1$, this equals $((x \cdot u)/(u \cdot u))u$, the orthogonal projection formula.

(b) $B^T = (uu^T)^T = uu^T = B$ and $B^2 = u(u^T u)u^T = uu^T = B$.

(c) $Bu = u(u^T u) = u$. The unit vector is nonzero, so it is an eigenvector with eigenvalue 1.

Exercise 7.1.36: Every symmetric idempotent matrix is an orthogonal projection

(a) Show the proposed pieces are perpendicular.

Put $\widehat{y} = By$, $z = (I - B)y$. Symmetry and idempotence give

$$z^T \widehat{y} = y^T (I - B)^T By = y^T (B - B^2)y = 0.$$

(b) Verify the stronger orthogonality needed for a projection.

Clearly $\widehat{y} \in W = \operatorname{Col} B$. For any $w \in W$, write $w = Bv$; then $z^T w = y^T (I - B)Bv = 0$. Thus $z \in W^{\perp}$, not merely perpendicular to one particular vector $\widehat{y}$. The identity $y = By + (y - By)$ is therefore the orthogonal decomposition into a vector in W and one in $W^{\perp}$. Its uniqueness proves that $By = \operatorname{proj}_W y$.

Exercise 7.1.37: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -12$: row reduction gives $x_1 = x_4, \quad x_2 = -x_4, \quad x_3 = -x_4$. $\lambda = 4$: row reduction gives $x_1 = -x_4, \quad x_2 = -x_4, \quad x_3 = x_4$. $\lambda = 10$: row reduction gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = 18$: row reduction gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = -x_4$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (-12, 4, 10, 18).$$

Their squared lengths are $(4, 4, 4, 4)$. Hence an explicit answer is

$$P = \frac{1}{2}V, \qquad D = \operatorname{diag}(-12, 4, 10, 18).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.38: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = \frac{1}{4}$: row reduction gives $x_1 = 4x_4, \quad x_2 = 2x_4, \quad x_3 = 2x_4$. $\lambda = \frac{3}{10}$: row reduction gives $x_1 = -\frac{x_4}{4}, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = \frac{x_4}{2}$. $\lambda = \frac{11}{20}$: row reduction gives $x_1 = x_4, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = -2x_4$. $\lambda = \frac{3}{4}$: row reduction gives $x_1 = -x_4, \quad x_2 = 2x_4, \quad x_3 = -\frac{x_4}{2}$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 4 & 1 & 2 & 2 \\ 2 & 2 & -1 & -4 \\ 2 & -2 & -4 & 1 \\ 1 & -4 & 2 & -2 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (\frac{1}{4}, \frac{3}{10}, \frac{11}{20}, \frac{3}{4}).$$

Their squared lengths are $(25, 25, 25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(\frac{1}{4}, \frac{3}{10}, \frac{11}{20}, \frac{3}{4}).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.39: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -\frac{5}{4}$: row reduction gives $x_1 = -x_4, \quad x_2 = 2x_4, \quad x_3 = -\frac{x_4}{2}$. $\lambda = 0$: row reduction gives $x_1 = -x_4, \quad x_2 = -\frac{x_4}{2}, \quad x_3 = 2x_4$. $\lambda = \frac{3}{4}$: row reduction gives $x_1 = \frac{3x_3}{2} + x_4, \quad x_2 = x_3$. For the repeated eigenvalue $3/4$, the two chosen null vectors $(4, 2, 2, 1)^T$ and $(1, -2, -2, 4)^T$ have dot product $4 - 4 - 4 + 4 = 0$. Both have squared length 25, so dividing each by 5 normalizes them.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 2 & 2 & 4 & 1 \\ -4 & 1 & 2 & -2 \\ 1 & -4 & 2 & -2 \\ -2 & -2 & 1 & 4 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_4) = (-\frac{5}{4}, 0, \frac{3}{4}, \frac{3}{4}).$$

Their squared lengths are $(25, 25, 25, 25)$. Hence an explicit answer is

$$P = \frac{1}{5}V, \qquad D = \operatorname{diag}(-\frac{5}{4}, 0, \frac{3}{4}, \frac{3}{4}).$$

Dividing columns by these nonzero lengths gives $P^TP = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

Exercise 7.1.40: Compute eigenvalues, then row-reduce each eigenspace

The eigenvalues were computed from the characteristic polynomial. For the eigenvectors we solve null-space systems by row reduction, not by calling an eigenvector routine, as the exercise requests.

Solve the eigenspace equations.

$\lambda = -28$: row reduction gives $x_1 = -\frac{x_5}{4}, \quad x_2 = -\frac{x_5}{4}, \quad x_3 = -\frac{x_5}{4}, \quad x_4 = -\frac{x_5}{4}$. $\lambda = 8$: row reduction gives $x_1 = -x_2 - x_3, \quad x_4 = 0, \quad x_5 = 0$. $\lambda = 17$: row reduction gives $x_1 = x_5, \quad x_2 = x_5, \quad x_3 = x_5, \quad x_4 = x_5$. $\lambda = 32$: row reduction gives $x_1 = -\frac{x_4}{3}, \quad x_2 = -\frac{x_4}{3}, \quad x_3 = -\frac{x_4}{3}, \quad x_5 = 0$.

Normalize the orthogonal vectors and pair the same order with D.

The columns of

$$V = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 \\ 1 & 0 & -2 & 1 & 1 \\ 1 & 0 & 0 & 1 & -3 \\ -4 & 0 & 0 & 1 & 0 \end{bmatrix}$$

are mutually orthogonal eigenvectors in the order

$$(\lambda_1, \ldots, \lambda_5) = (-28, 8, 8, 17, 32).$$

Their squared lengths are $(20, 2, 6, 5, 12)$. Hence an explicit answer is

$$P = V \operatorname{diag}(\frac{\sqrt{5}}{10}, \frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{5}}{5}, \frac{\sqrt{3}}{6}), \qquad D = \operatorname{diag}(-28, 8, 8, 17, 32).$$

Dividing columns by these nonzero lengths gives $P^T P = I$, and the eigenspace equations give $AP = PD$. Multiplying the latter by P^T therefore verifies $A = PDP^T$.

7.2 Quadratic forms

For a symmetric matrix A, the expression $Q(x) = x^T Ax$ is a quadratic form. Diagonal entries are the coefficients of squares. Off-diagonal entries contribute twice: the coefficient of $x_i x_j$ for $i \neq j$ is $2a_{ij}$. Thus put *half* of each mixed-term coefficient in each of the two symmetric positions.

An orthogonal change $x = Py$ gives $Q(x) = y^T(P^T AP)y$. If P consists of orthonormal eigenvectors, then $P^T AP = D$ is diagonal and $Q = \sum_i \lambda_i y_i^2$. All eigenvalues positive means positive definite; all negative means negative definite. Nonnegative eigenvalues with a zero give positive semidefinite but not definite, and the analogous nonpositive case is negative semidefinite. Both signs give an indefinite form. A definite inequality is required only for $x \neq 0$, since every quadratic form has $Q(0) = 0$.

Practice problem 7.2.P1: Recognize positive semidefiniteness spectrally

A symmetric matrix is positive semidefinite exactly when all eigenvalues are nonnegative. In an orthonormal eigenbasis, $Q(x) = \sum_i \lambda_i y_i^2 \geq 0$ if every $\lambda_i \geq 0$. Conversely, if $\lambda_j < 0$, taking $y = e_j$ and $x = Pe_j$ gives $Q(x) = \lambda_j < 0$, contradicting semidefiniteness. This argument also includes zero eigenvalues and the zero matrix.

Exercise 7.2.1: Evaluate a two-variable form

(a) First multiply $Ax = (5x_1 + x_2/3, x_1/3 + x_2)^T$. Dotting with x gives $Q(x) = 5x_1^2 + (2/3)x_1x_2 + x_2^2$.

(b) For $x = (6, 1)^T$, the value is $5(36) + (2/3)6 + 1 = 185$.

(c) For $x = (1, 3)^T$, the value is $5 + (2/3)3 + 9 = 16$.

The two off-diagonal contributions both matter; omitting one changes the answer.

Exercise 7.2.2: Evaluate a three-variable form

(a) Expanding $x^T Ax$ and collecting the symmetric terms gives $Q(x) = 4x_1^2 + 2x_2^2 + x_3^2 + 6x_1x_2 + 2x_2x_3$. There is no x_1x_3 term because $a_{13} = a_{31} = 0$.

(b) At $x = (2, -1, 5)^T$, this is $16 + 2 + 25 - 12 - 10 = 21$.

(c) At $x = (1, 1, 1)^T/\sqrt{3}$, each square and each pair product is $1/3$. Thus $Q(x) = (4+2+1+6+2)/3 = 5$.

Exercise 7.2.3: Recover two symmetric coefficient matrices

The diagonal entries come from the square terms and each mixed coefficient is divided by two. Thus the matrices are

$$\text{(a)} \quad \boxed{\begin{bmatrix} 10 & -3 \\ -3 & -3 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 5 & 3/2 \\ 3/2 & 0 \end{bmatrix}}.$$

For example the first mixed coefficient is $-3 - 3 = -6$, and the second form has no x_2^2 term, so its bottom-right entry is zero.

Exercise 7.2.4: A large mixed coefficient and a pure mixed term

The same rule gives

$$\text{(a)} \quad \boxed{\begin{bmatrix} 20 & 15/2 \\ 15/2 & -10 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}}.$$

The two $1/2$ entries in part (b) sum to the required coefficient 1 of x_1x_2; neither diagonal entry is needed.

Exercise 7.2.5: Three-variable coefficient matrices

Record the three square coefficients on the diagonal, then place half of each of the three possible mixed coefficients in symmetric positions:

$$\text{(a)} \quad \boxed{\begin{bmatrix} 8 & -3 & 2 \\ -3 & 7 & -1 \\ 2 & -1 & -3 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}}.$$

In part (b) all square coefficients vanish, explaining the zero diagonal. Expanding either matrix as x^TAx reproduces every original coefficient.

Exercise 7.2.6: Mixed terms with odd coefficients

Fractions are necessary when a mixed coefficient is odd. The two matrices are

$$\text{(a)} \quad \boxed{\begin{bmatrix} 5 & 5/2 & -3/2 \\ 5/2 & -1 & 0 \\ -3/2 & 0 & 7 \end{bmatrix}}, \qquad \text{(b)} \quad \boxed{\begin{bmatrix} 0 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}}.$$

For instance $2(5/2) = 5$ and $2(-3/2) = -3$ give the two mixed coefficients in part (a); the absent x_2x_3 term gives a zero entry.

Exercise 7.2.7: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-4, 6)$ (counted with multiplicity). $\lambda = -4$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_2$. $\lambda = 6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 2)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{2}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-4, 6)$, so

$$Q(x) = \boxed{-4y_1^2 + 6y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.8: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 9 & -4 & 4 \\ -4 & 7 & 0 \\ 4 & 0 & 11 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(3, 9, 15)$ (counted with multiplicity). $\lambda = 3$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_3, \quad x_2 = -2x_3$. $\lambda = 9$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_3}{2}, \quad x_2 = x_3$. $\lambda = 15$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_3, \quad x_2 = -\frac{x_3}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \\ -1 & -2 & 2 \end{bmatrix}.$$

Their squared lengths are $(9, 9, 9)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{3}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(3, 9, 15)$, so

$$Q(x) = \boxed{3y_1^2 + 9y_2^2 + 15y_3^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.9: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(2, 7)$ (counted with multiplicity). $\lambda = 2$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 2x_2$. $\lambda = 7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(2, 7)$, so

$$Q(x) = \boxed{2y_1^2 + 7y_2^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.10: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 9 & -4 \\ -4 & 3 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(1, 11)$ (counted with multiplicity). $\lambda = 1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{x_2}{2}$. $\lambda = 11$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(1, 11)$, so

$$Q(x) = \boxed{y_1^2 + 11y_2^2}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.11: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 2 & 5 \\ 5 & 2 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-3, 7)$ (counted with multiplicity). $\lambda = -3$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_2$. $\lambda = 7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 2)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{2}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-3, 7)$, so

$$Q(x) = \boxed{-3y_1^2 + 7y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.12: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-6, -1)$ (counted with multiplicity). $\lambda = -6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -2x_2$. $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{x_2}{2}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}.$$

Their squared lengths are $(5, 5)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{5}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-6, -1)$, so

$$Q(x) = \boxed{-6y_1^2 - y_2^2}.$$

The form is **negative definite**. Every nonzero y has a nonzero squared coordinate with a negative coefficient, so $Q < 0$. Invertibility of P transfers this to nonzero x.

Exercise 7.2.13: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(0, 10)$ (counted with multiplicity). $\lambda = 0$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_2$. $\lambda = 10$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{3}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}.$$

Their squared lengths are $(10, 10)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{10}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(0, 10)$, so

$$Q(x) = \boxed{10y_2^2}.$$

The form is **positive semidefinite but not positive definite**. All terms are nonnegative, but a nonzero vector along a zero-eigenvalue coordinate gives $Q = 0$, preventing strict positive definiteness.

Exercise 7.2.14: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 8 & 3 \\ 3 & 0 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-1, 9)$ (counted with multiplicity). $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -\frac{x_2}{3}$. $\lambda = 9$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_2$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 3 \\ -3 & 1 \end{bmatrix}.$$

Their squared lengths are $(10, 10)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{\sqrt{10}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T AP = \operatorname{diag}(-1, 9)$, so

$$Q(x) = \boxed{-y_1^2 + 9y_2^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.15: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} -2 & 2 & 2 & 2 \\ 2 & -6 & 0 & 0 \\ 2 & 0 & -9 & 3 \\ 2 & 0 & 3 & -9 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-12, -8, -6, 0)$ (counted with multiplicity). $\lambda = -12$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = 0, \quad x_3 = -x_4$. $\lambda = -8$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = -6$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = -2x_4, \quad x_3 = x_4$. $\lambda = 0$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 3x_4, \quad x_2 = x_4, \quad x_3 = x_4$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & -1 & 2 & 1 \\ 1 & -1 & -1 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

Their squared lengths are $(2, 4, 6, 12)$ and their pairwise dot products vanish. Set

$$P = V \operatorname{diag}\left(\frac{\sqrt{2}}{2}, \frac{1}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{3}}{6}\right); \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-12, -8, -6, 0)$, so

$$Q(x) = \boxed{-12y_1^2 - 8y_2^2 - 6y_3^2}.$$

The form is **negative semidefinite but not negative definite**. All terms are nonpositive, but a nonzero vector along the zero-eigenvalue coordinate gives $Q = 0$, preventing strict negative definiteness.

Exercise 7.2.16: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 4 & \frac{3}{2} & 0 & -2 \\ \frac{3}{2} & 4 & 2 & 0 \\ 0 & 2 & 4 & \frac{3}{2} \\ -2 & 0 & \frac{3}{2} & 4 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(\frac{3}{2}, \frac{3}{2}, \frac{13}{2}, \frac{13}{2})$ (counted with multiplicity). $\lambda = \frac{3}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} + \frac{5x_4}{4}$, $\quad x_2 = -\frac{5x_3}{4} - \frac{3x_4}{4}$. $\lambda = \frac{13}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} - \frac{5x_4}{4}$, $\quad x_2 = \frac{5x_3}{4} - \frac{3x_4}{4}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 4 & 3 & 4 \\ -5 & 0 & 5 & 0 \\ 4 & -3 & 4 & -3 \\ 0 & 5 & 0 & -5 \end{bmatrix}.$$

Their squared lengths are $(50, 50, 50, 50)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{5\sqrt{2}}V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(\frac{3}{2}, \frac{3}{2}, \frac{13}{2}, \frac{13}{2})$, so

$$Q(x) = \boxed{\frac{3y_1^2}{2} + \frac{3y_2^2}{2} + \frac{13y_3^2}{2} + \frac{13y_4^2}{2}}.$$

The form is **positive definite**. Every nonzero y has a nonzero coordinate with a positive coefficient, so its value is strictly positive. Since P is invertible, the same holds for every nonzero x.

Exercise 7.2.17: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 1 & \frac{9}{2} & 0 & -6 \\ \frac{9}{2} & 1 & 6 & 0 \\ 0 & 6 & 1 & \frac{9}{2} \\ -6 & 0 & \frac{9}{2} & 1 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-\frac{13}{2}, -\frac{13}{2}, \frac{17}{2}, \frac{17}{2})$ (counted with multiplicity). $\lambda = -\frac{13}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} + \frac{5x_4}{4}, \quad x_2 = -\frac{5x_3}{4} - \frac{3x_4}{4}$. $\lambda = \frac{17}{2}$: reducing $(A - \lambda I)v = 0$ gives $x_1 = \frac{3x_3}{4} - \frac{5x_4}{4}, \quad x_2 = \frac{5x_3}{4} - \frac{3x_4}{4}$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 3 & 4 & 3 & 4 \\ -5 & 0 & 5 & 0 \\ 4 & -3 & 4 & -3 \\ 0 & 5 & 0 & -5 \end{bmatrix}.$$

Their squared lengths are $(50, 50, 50, 50)$ and their pairwise dot products vanish. Set

$$P = \frac{1}{5\sqrt{2}} V; \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^T A P = \operatorname{diag}(-\frac{13}{2}, -\frac{13}{2}, \frac{17}{2}, \frac{17}{2})$, so

$$Q(x) = \boxed{-\frac{13y_1^2}{2} - \frac{13y_2^2}{2} + \frac{17y_3^2}{2} + \frac{17y_4^2}{2}}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.18: Remove the mixed terms and identify the signs

The symmetric coefficient matrix is

$$A = \begin{bmatrix} 11 & -6 & -6 & -6 \\ -6 & -1 & 0 & 0 \\ -6 & 0 & 0 & -1 \\ -6 & 0 & -1 & 0 \end{bmatrix}.$$

Find orthogonal eigenvectors rather than guessing a change of variables.

The eigenvalues are $(-7, -1, 1, 17)$ (counted with multiplicity). $\lambda = -7$: reducing $(A - \lambda I)v = 0$ gives $x_1 = x_4, \quad x_2 = x_4, \quad x_3 = x_4$. $\lambda = -1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = -2x_4, \quad x_3 = x_4$. $\lambda = 1$: reducing $(A - \lambda I)v = 0$ gives $x_1 = 0, \quad x_2 = 0, \quad x_3 = -x_4$. $\lambda = 17$: reducing $(A - \lambda I)v = 0$ gives $x_1 = -3x_4, \quad x_2 = x_4, \quad x_3 = x_4$. Choosing free variables and, within a repeated eigenspace, orthogonalizing gives the eigenvectors as the columns of

$$V = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 1 & 2 & 0 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 \end{bmatrix}.$$

Their squared lengths are $(4, 6, 2, 12)$ and their pairwise dot products vanish. Set

$$P = V \operatorname{diag}(\frac{1}{2}, \frac{\sqrt{6}}{6}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{6}); \qquad x = Py.$$

Read the new form and its classification.

Exact multiplication gives $P^TAP = \operatorname{diag}(-7,-1,1,17)$, so

$$Q(x) = \boxed{-7y_1^2 - y_2^2 + y_3^2 + 17y_4^2}.$$

The form is **indefinite**. Taking y on a positive-eigenvalue coordinate axis makes $Q > 0$, while a negative-eigenvalue axis makes $Q < 0$. Thus both signs really occur.

Exercise 7.2.19: A unit-circle maximum

Since $x_1^2 + x_2^2 = 1$, $5x_1^2 + 8x_2^2 = 5 + 3x_2^2 \le 8$. Equality occurs at $x = (0,1)$ or $(0,-1)$. Thus the largest value is $\boxed{8}$. The calculation provides both an upper bound and a point attaining it.

Exercise 7.2.20: A negative coefficient lowers the maximum

Again use the constraint to write $5x_1^2 - 3x_2^2 = 5 - 8x_2^2 \le 5$. Equality occurs at $x = (1,0)$ or $(-1,0)$, so the largest value is $\boxed{5}$. Without the unit constraint, scaling x_1 would make the form arbitrarily large.

Exercise 7.2.21: Six true–false statements about quadratic forms

(a) **True.** The matrix of a form in this section means its canonical symmetric coefficient matrix, with mixed coefficients split equally. A nonsymmetric representing array can be replaced by its symmetric part.

(b) **True.** In a symmetric matrix the mixed coefficient is $2a_{ij}$, so all mixed coefficients vanish exactly when every off-diagonal entry vanishes.

(c) **True.** The principal-axis directions are given by an orthonormal eigenvector basis of the symmetric coefficient matrix.

(d) **False.** The condition is $Q(x) > 0$ for all $x \neq 0$. At $x = 0$ every quadratic form equals zero.

(e) **True.** In eigenvector coordinates the form is a sum of positive coefficients times squares, strictly positive for nonzero inputs.

(f) **True.** This is the stated form of a Cholesky factorization. Its existence is restricted to positive definite symmetric matrices; the statement describes the factorization, not a claim that every symmetric matrix admits one.

Exercise 7.2.22: Five further classification statements

(a) **True.** $\|x\|^2 = x^TIx$ is a quadratic form.

(b) **False.** Orthogonality of P alone is insufficient; its columns must be appropriate eigenvectors. For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $P = I$, the mixed term $2y_1y_2$ remains.

(c) **False.** Degenerate or empty level sets are also possible. For $A = I$, $c = -1$ the set is empty; for $A = \operatorname{diag}(1,0)$, $c = 0$ the level set is the line $x_1 = 0$, not one of the three listed curves.

(d) **False.** An indefinite form takes both positive and negative values. Semidefinite forms have only one weak sign. For example $x_1^2 - x_2^2$ is indefinite and neither kind of semidefinite.

(e) **True.** For each unit eigenvector u_i, $u_i^T A u_i = \lambda_i$. Strict negativity at every nonzero input therefore makes every eigenvalue negative.

Exercise 7.2.23: Compare characteristic-polynomial coefficients

For $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$,

$$\det(A - tI) = (a - t)(d - t) - b^2 = t^2 - (a + d)t + (ad - b^2).$$

On the other hand it is $(t - \lambda_1)(t - \lambda_2) = t^2 - (\lambda_1 + \lambda_2)t + \lambda_1\lambda_2$. Equality of polynomial coefficients gives

$$\boxed{\lambda_1 + \lambda_2 = a + d, \qquad \lambda_1\lambda_2 = \det A}.$$

The factor signs agree because there are two factors; in odd dimensions one must distinguish $\det(A - tI)$ from $\det(tI - A)$.

Exercise 7.2.24: Classify nonsingular two-variable forms without solving for roots

(a) If $\det A = ad - b^2 > 0$ and $a > 0$, then $ad > b^2 \geq 0$, so $d > 0$ and $a + d > 0$. The two real eigenvalues have positive product, so they have the same nonzero sign; their positive sum forces both positive. Thus the form is positive definite.

(b) If $\det A > 0$ and $a < 0$, then $ad > 0$ makes $d < 0$, so the eigenvalues have the same nonzero sign and negative sum. Both are negative, giving a negative definite form.

(c) If $\det A < 0$, the real eigenvalues have negative product, so one is positive and one negative. Their eigenvector directions give opposite values of the form, proving indefiniteness.

The hypothesis $\det A \neq 0$ excludes the semidefinite boundary cases.

Exercise 7.2.25: Gram matrices are positive semidefinite

The matrix $B^T B$ is symmetric. For any x, $x^T B^T B x = (Bx)^T (Bx) = \|Bx\|^2 \geq 0$, proving positive semidefiniteness. If B is square and invertible, $x \neq 0$ implies $Bx \neq 0$, making the last value strictly positive. Thus $B^T B$ is positive definite then. In fact independent columns suffice even for a rectangular B.

Exercise 7.2.26: Construct a positive definite square root

Write $A = PDP^T$ with orthogonal P and all $\lambda_i > 0$ on the diagonal of D. Let $C = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$, so $C^T C = D$, and set $B = PCP^T$. This B is symmetric and, for nonzero x with $y = P^T x \neq 0$, $x^T B x = \sum_i \sqrt{\lambda_i} y_i^2 > 0$, so B is positive definite. Finally

$$B^T B = PC^T P^T PCP^T = PC^T CP^T = PDP^T = A.$$

Both the factorization and the positive definiteness requested of B have now been verified.

Exercise 7.2.27: A sum of positive definite matrices

The sum $A+B$ is symmetric. For $x \neq 0$, $x^T(A+B)x = x^TAx + x^TBx > 0+0 = 0$, since each matrix is positive definite. Thus $A+B$ is positive definite, and its eigenvalues are all positive. The proof does not require A and B to commute or share eigenvectors; adding their eigenvalue lists would not be a valid general argument.

Exercise 7.2.28: The inverse of a positive definite matrix

An orthogonal diagonalization $A = PDP^T$ has positive diagonal entries λ_i. Its inverse is $A^{-1} = PD^{-1}P^T$, with positive eigenvalues $1/\lambda_i$. Hence its quadratic form is positive definite. For a direct check, given $x \neq 0$ let $y = A^{-1}x \neq 0$. Since A is symmetric, $x^TA^{-1}x = (Ay)^Ty = y^TAy > 0$. Thus both the eigenvalue argument and the defining inequality give the claim.

7.3 Constrained optimization

A quadratic form is an expression $Q(x) = x^T Ax$, where we take A to be symmetric. A coefficient of $x_i x_j$ with $i \neq j$ must be divided equally between a_{ij} and a_{ji}. For example, $6x_1x_2$ means $a_{12} = a_{21} = 3$, not 6.

The central argument. The spectral theorem gives an orthonormal eigenbasis $u_1, \ldots, u_n$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. Put $P = [u_1 \ \cdots \ u_n]$ and $x = Py$. Then $P^TP = I$, so

$$x^Tx = y^Ty, \qquad Q(x) = \sum_{j=1}^{n} \lambda_j y_j^2.$$

On the unit sphere, the nonnegative numbers y_j^2 add to one. Thus

$$\lambda_n \leq Q(x) \leq \lambda_1.$$

Taking $x = u_1$ attains the upper bound. If we also require $x^Tu_1 = 0$, then $y_1 = 0$, and the maximum becomes λ_2, attained at u_2. Eigenvalues here are listed *with multiplicities*; a repeated largest eigenvalue can remain the largest value after one direction is excluded. These observations explain the optimization steps below without requiring multivariable differentiation.

Practice problem 7.3.P1: Remove the mixed term using an orthonormal basis

What we must show. Diagonalize $3x_1^2 + 3x_2^2 + 2x_1x_2$ by $x = Py$. The symmetric matrix is $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Its characteristic polynomial is $(3-\lambda)^2 - 1$, with roots $4, 2$. For 4, solving $(A - 4I)v = 0$ gives $v_2 = v_1$; choose $(1,1)^T$. For 2, the equations give $v_2 = -v_1$; choose $(-1,1)^T$. Each vector has length $\sqrt{2}$, so use

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Its columns have length one and dot product zero. Therefore $P^TAP = \operatorname{diag}(4, 2)$, and $Q(Py) = 4y_1^2 + 2y_2^2$.

Conclusion. The displayed P gives the requested substitution. Normalizing the eigenvectors is essential to making P orthogonal.

Practice problem 7.3.P2: Translate the maximizing direction back to the original variables

For a unit x, the preceding change of variables gives $y_1^2 + y_2^2 = 1$. Consequently $Q(x) = 4y_1^2 + 2y_2^2 \leq 4$. Equality holds for $y = (1,0)^T$. But the requested vector is x, not y: multiplying by P gives $x = (1,1)^T/\sqrt{2}$. Its squared length is $(1+1)/2 = 1$.

Conclusion. The maximum is 4, attained at $(1,1)^T/\sqrt{2}$. Its negative also attains the same value.

Exercise 7.3.1: An orthogonal substitution in three variables

What we must show. Obtain the diagonal coefficients $9, 6, 3$ for the given quadratic form.

Build the matrix and find its eigenvectors.

The coefficients give

$$A = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 6 & -2 \\ 0 & -2 & 7 \end{bmatrix}.$$

Expanding $\det(\lambda I - A)$ gives $(\lambda - 9)(\lambda - 6)(\lambda - 3)$. Solving the corresponding homogeneous systems gives the convenient eigenvectors

$$v_9 = (-1, -2, 2)^T, \quad v_6 = (2, 1, 2)^T, \quad v_3 = (-2, 2, 1)^T.$$

For example, at $\lambda = 9$ the first equation is $-4a + 2b = 0$, so $b = 2a$; the last is $-2b - 2c = 0$, so $c = -b$. Taking $a = -1$ produces v_9. The other two vectors are checked by multiplying by A. Each has squared length 9, and their pairwise dot products are zero.

Normalize and put the columns in the requested order.

Set

$$P = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Then $P^T P = I$ and $AP = P\,\mathrm{diag}(9, 6, 3)$. Multiplying this last identity on the left by P^T gives $P^T AP = \mathrm{diag}(9, 6, 3)$.

Conclusion. With $x = Py$, the form is $9y_1^2 + 6y_2^2 + 3y_3^2$.

Exercise 7.3.2: Include the missing zero coefficient

What we must show. Obtain the stated diagonal form, keeping three coordinates. The symmetric matrix is

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}, \qquad \det(\lambda I - A) = \lambda(\lambda - 2)(\lambda - 5).$$

Corresponding eigenvectors are $(1, 1, 1)^T$, $(-2, 1, 1)^T$, and $(0, -1, 1)^T$, for $5, 2, 0$, respectively. Their squared lengths are $3, 6, 2$, and their pairwise dot products vanish. Thus take

$$P = \begin{bmatrix} 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix}.$$

Direct multiplication gives $P^T AP = \mathrm{diag}(5, 2, 0)$.

Conclusion. $x = Py$ transforms the form to $5y_1^2 + 2y_2^2 + 0y_3^2$. The zero coefficient does not remove a coordinate from the substitution.

Exercise 7.3.3: Exclude one principal direction

(a) Bound the form on all unit vectors.

Exercise 7.3.1 gives $Q(x) = 9y_1^2 + 6y_2^2 + 3y_3^2$ and $y_1^2 + y_2^2 + y_3^2 = 1$. Replacing 6 and 3 by 9 shows $Q(x) \le 9$.

(b) Give a maximizing unit vector in the original coordinates.

Take $u = (-1, -2, 2)^T/3$, the first column of P. Then $Au = 9u$ and $u^T u = 1$, so $Q(u) = u^T(9u) = 9$.

(c) Impose the orthogonality condition.

Because $x = Py$ and $u = Pe_1$, we have $x^T u = y^T e_1 = y_1$. Thus $x^T u = 0$ forces $y_1 = 0$. Now $Q(x) = 6y_2^2 + 3y_3^2 \le 6(y_2^2 + y_3^2) = 6$. Equality holds at the second column $(2, 1, 2)^T/3$ of P.

Conclusion. The answers are 9, $u = (-1, -2, 2)^T/3$, and 6.

Exercise 7.3.4: The same constraints with a null direction

Use the orthogonal coordinates of Exercise 7.3.2. On the unit sphere, $Q(x) = 5y_1^2 + 2y_2^2 \le 5(y_1^2 + y_2^2 + y_3^2) = 5$. The unit vector $u = (1, 1, 1)^T/\sqrt{3}$ has $Au = 5u$, so it attains 5. The additional equation $x^T u = 0$ is $y_1 = 0$. Therefore $Q(x) = 2y_2^2 \le 2$, with equality at $(-2, 1, 1)^T/\sqrt{6}$.

Conclusion. (a) 5; (b) $u = (1, 1, 1)^T/\sqrt{3}$; (c) 2.

Exercise 7.3.5: A negative mixed coefficient

The matrix is $A = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$. The equation $(5 - \lambda)^2 - 4 = 0$ gives eigenvalues $7, 3$. Their unit eigenvectors are $u = (-1, 1)^T/\sqrt{2}$ and $v = (1, 1)^T/\sqrt{2}$. They are perpendicular. Writing $x = au + bv$ gives $\|x\|^2 = a^2 + b^2$ and $Q(x) = 7a^2 + 3b^2$. Hence the unrestricted unit maximum is 7, attained at u. When $x^T u = 0$, we have $a = 0$ and $b^2 = 1$, so the value is 3.

Conclusion. (a) 7; (b) $(-1, 1)^T/\sqrt{2}$; (c) 3.

Exercise 7.3.6: Remember to halve an odd mixed coefficient

Here $A = \begin{bmatrix} 7 & 3/2 \\ 3/2 & 3 \end{bmatrix}$, since the mixed term is $3x_1x_2$. Its characteristic equation is

$$(7 - \lambda)(3 - \lambda) - \frac{9}{4} = 0,$$

whose roots are 15/2 and 5/2. At 15/2, the first row of $(A - \lambda I)v = 0$ gives $-v_1 + 3v_2 = 0$; choose $(3, 1)^T$. Its unit version is $u = (3, 1)^T/\sqrt{10}$. A perpendicular unit eigenvector is $v = (-1, 3)^T/\sqrt{10}$, for 5/2. For $x = au + bv$, the unit constraint gives $a^2 + b^2 = 1$, and $Q(x) = (15/2)a^2 + (5/2)b^2$. Requiring $x^T u = 0$ sets $a = 0$.

Conclusion. (a) 15/2; (b) $(3, 1)^T/\sqrt{10}$; (c) 5/2.

Exercise 7.3.7: Normalize the largest-eigenvalue vector

The given coefficients give

$$A = \begin{bmatrix} -2 & 2 & 0 \\ 2 & -1 & 2 \\ 0 & 2 & 0 \end{bmatrix}.$$

Its eigenvalues are $2, -1, -4$, so the maximum on the unit sphere is 2. To find a maximizing vector, solve $(A - 2I)v = 0$. The first row gives $-4v_1 + 2v_2 = 0$, hence $v_2 = 2v_1$. The last gives $2v_2 - 2v_3 = 0$, hence $v_3 = v_2$. Taking $v_1 = 1$ gives $v = (1, 2, 2)^T$, with length 3. Consequently $x = v/3$ is unit, and $Q(x) = x^T(2x) = 2$.

Conclusion. One required vector is $(1, 2, 2)^T/3$.

Exercise 7.3.8: A repeated largest eigenvalue

The matrix is

$$A = \begin{bmatrix} 7 & -4 & -2 \\ -4 & 1 & -4 \\ -2 & -4 & 7 \end{bmatrix}.$$

Its eigenvalues are $9, 9, -3$. At 9, every row equation reduces to $v_1 + 2v_2 + v_3 = 0$. Choose $v = (-1, 0, 1)^T$ in this plane and normalize: $x = (-1, 0, 1)^T/\sqrt{2}$. The squared length is one, $Ax = 9x$, and $Q(x) = 9$. No unit vector can exceed the largest eigenvalue.

Conclusion. $(-1, 0, 1)^T/\sqrt{2}$ is a maximizing unit vector; the maximum is 9.

Exercise 7.3.9: Find the maximum without finding a vector

The mixed coefficient -2 gives $A = \begin{bmatrix} 7 & -1 \\ -1 & 3 \end{bmatrix}$. Its characteristic equation is

$$(7 - \lambda)(3 - \lambda) - 1 = \lambda^2 - 10\lambda + 20 = 0.$$

The quadratic formula gives $\lambda = 5 \pm \sqrt{5}$. By the unit-sphere bound established at the start, the maximum is the larger root. No eigenvector is needed for the question asked.

Conclusion. The maximum is $5 + \sqrt{5}$.

Exercise 7.3.10: A form with both positive and negative values

Now $A = \begin{bmatrix} -3 & -1 \\ -1 & 5 \end{bmatrix}$. The equation is

$$(-3 - \lambda)(5 - \lambda) - 1 = \lambda^2 - 2\lambda - 16 = 0,$$

with roots $1 \pm \sqrt{17}$. The constraint is exactly the unit-sphere constraint, so the larger eigenvalue gives the maximum.

Conclusion. The maximum is $1 + \sqrt{17}$.

Exercise 7.3.11: Evaluate a form at a unit eigenvector

If $Ax = 3x$ and $x^T x = 1$, substitute the eigenvector equation first:

$$x^T Ax = x^T(3x) = 3(x^T x) = 3.$$

This particular calculation does not require symmetry of A; the two stated equations are sufficient.

Conclusion. The value is 3.

Exercise 7.3.12: Every eigenvalue lies between the extreme unit values

Let λ be an eigenvalue of the real symmetric matrix A. The spectral theorem supplies a nonzero real eigenvector v. Normalize it: $x = v/\|v\|$. Then $x^T x = 1$ and $Ax = \lambda x$, so $x^T Ax = \lambda x^T x = \lambda$. By definition, m is a lower bound and M is an upper bound for the values of the form at all unit vectors. They therefore bound this particular value.

Conclusion. $m \le \lambda \le M$. Normalization places the eigenvector on the very set over which the extrema are defined.

Exercise 7.3.13: Every intermediate value is attained

What we must show. Given $m \le t \le M$, construct a unit x with $x^T Ax = t$.

First separate the constant case.

If $M = m$, then $t = m$ and any unit eigenvector for m works. The proposed expression can use $\alpha = 0$, giving $x = u_n$. This also handles the one-dimensional case without demanding two perpendicular vectors in a one-dimensional space.

For $M > m$, choose the interpolation weight explicitly.

Put $\alpha = (t - m)/(M - m)$. The inequalities on t imply $0 \le \alpha \le 1$, and rearrangement gives $t = (1 - \alpha)m + \alpha M$. Let $Au_1 = Mu_1$ and $Au_n = mu_n$, where the vectors are unit. Distinct eigenvalues of a symmetric matrix have orthogonal eigenvectors: indeed $u_1^T Au_n = mu_1^T u_n$ and, by symmetry, the same expression equals $(Au_1)^T u_n = Mu_1^T u_n$. Since $M - m \neq 0$, $u_1^T u_n = 0$.

Check both the constraint and the desired value.

Define $x = \sqrt{1 - \alpha}\, u_n + \sqrt{\alpha}\, u_1$. Expanding gives

$$x^T x = (1 - \alpha)\underbrace{u_n^T u_n}_{1} + \alpha \underbrace{u_1^T u_1}_{1} + 2\sqrt{\alpha(1 - \alpha)}\underbrace{u_n^T u_1}_{0} = 1.$$

The mixed terms in $x^T Ax$ also vanish, because, for example, $u_n^T Au_1 = Mu_n^T u_1 = 0$. The remaining terms give

$$x^T Ax = (1 - \alpha)m + \alpha M = t.$$

Conclusion. Every number in $[m, M]$ occurs. The square roots in x make the squared coefficients, rather than the coefficients, add to one.

Exercise 7.3.14: An exact computer-assisted four-dimensional example

Convert the mixed terms to a symmetric matrix.

$$A = \frac{1}{2}\begin{bmatrix} 0 & 1 & 3 & 30 \\ 1 & 0 & 30 & 3 \\ 3 & 30 & 0 & 1 \\ 30 & 3 & 1 & 0 \end{bmatrix}.$$

To check its spectrum without trusting rounded output, use the four pairwise perpendicular vectors

$$(1,1,1,1)^T, \quad (1,-1,-1,1)^T, \quad (-1,1,-1,1)^T, \quad (-1,-1,1,1)^T.$$

Multiplication by A gives these same vectors times $17, 13, -14, -16$, respectively. Each has length 2; dividing them by 2 gives an orthonormal eigenbasis and verifies the entire spectrum.

Answer the three optimization questions.

The maximum on unit vectors is 17, attained at $u = (1,1,1,1)^T/2$. Orthogonality to u removes its coordinate, leaving the next largest eigenvalue, 13. It is attained at $(1,-1,-1,1)^T/2$.

Conclusion. (a) 17; (b) $(1,1,1,1)^T/2$; (c) 13.

Exercise 7.3.15: A negative constrained maximum

$$A = \frac{1}{2}\begin{bmatrix} 0 & 3 & 5 & 7 \\ 3 & 0 & 7 & 5 \\ 5 & 7 & 0 & 3 \\ 7 & 5 & 3 & 0 \end{bmatrix}.$$

The same four orthogonal sign vectors from Exercise 7.3.14 are eigenvectors, now with eigenvalues $15/2, -1/2, -5/2, -9/2$, in that order. For instance, the first row sum is $(3+5+7)/2 = 15/2$, verifying the first coordinate of the all-ones eigenvector equation; the other rows have the same sum. Checking the other three sign vectors verifies the remaining eigenvalues. Their unit versions form a basis, so no eigenvalue is omitted. Thus the unit-sphere maximum is $15/2$, at $u = (1,1,1,1)^T/2$. On $u^\perp$, the largest remaining eigenvalue is $-1/2$, attained at $(1,-1,-1,1)^T/2$. A maximum need not be positive.

Conclusion. (a) $15/2$; (b) $(1,1,1,1)^T/2$; (c) $-1/2$.

Exercise 7.3.16: One diagonal term and six mixed terms

The symmetric matrix and a factored characteristic polynomial are

$$A = \begin{bmatrix} 4 & -3 & -5 & -5 \\ -3 & 0 & -3 & -3 \\ -5 & -3 & 0 & -1 \\ -5 & -3 & -1 & 0 \end{bmatrix},$$

$$\det(\lambda I - A) = (\lambda - 9)(\lambda - 3)(\lambda - 1)(\lambda + 9).$$

For an exact check, vectors $(-2,0,1,1)^T$, $(1,-3,1,1)^T$, $(0,0,-1,1)^T$, and $(1,1,1,1)^T$ have eigenvalues $9, 3, 1, -9$, respectively. Each assertion follows by one matrix–vector multiplication, and the vectors

are nonzero and pairwise perpendicular. The first has length $\sqrt{6}$, so choose $u = (-2, 0, 1, 1)^T/\sqrt{6}$. Then $Q(u) = 9$. With $x \perp u$, the maximum becomes 3, attained at $(1, -3, 1, 1)^T/\sqrt{12}$.

Conclusion. (a) 9; (b) $(-2, 0, 1, 1)^T/\sqrt{6}$; (c) 3.

Exercise 7.3.17: The largest eigenvalue can itself be negative

Here

$$A = \begin{bmatrix} -6 & -2 & -2 & -2 \\ -2 & -10 & 0 & 0 \\ -2 & 0 & -13 & 3 \\ -2 & 0 & 3 & -13 \end{bmatrix}.$$

Its characteristic polynomial is $(\lambda + 4)(\lambda + 10)(\lambda + 12)(\lambda + 16)$. Nonzero eigenvectors for $-4, -10, -12, -16$, respectively, are

$$(-3, 1, 1, 1)^T, \quad (0, -2, 1, 1)^T, \quad (1, 1, 1, 1)^T, \quad (0, 0, -1, 1)^T.$$

They are pairwise perpendicular. Checking $Av = \lambda v$ for each one verifies the spectrum independently of an eigenvalue routine. The largest eigenvalue is -4, since $-4 > -10 > -12 > -16$; one does *not* choose the largest absolute value. The first vector has squared length 12. After excluding its unit direction, -10 is the largest remaining eigenvalue, attained at $(0, -2, 1, 1)^T/\sqrt{6}$.

Conclusion. (a) -4; (b) $(-3, 1, 1, 1)^T/\sqrt{12}$; (c) -10.

Reproducing the computer checks in 14–17. Enter each displayed matrix using exact fractions, not rounded decimals. For a symmetric matrix, the following Python/SymPy commands return an exact spectrum and check a chosen vector. Replace `entries`, `vector`, and `eigenvalue` with the data displayed in the corresponding solution:

```
from sympy import Matrix, Rational, eye
A = Matrix(entries)
print(A.eigenvals())
v = Matrix(vector)
lam = eigenvalue
assert A == A.T
assert A*v == lam*v
u = v / (v.dot(v))**Rational(1, 2)
assert (u.T*u)[0] == 1
```

The exact eigenvector certificates above also give a complete verification without installing software. The matrices, results, and certificates are all specified; the computer problems do not require an external dataset.

7.4 The singular value decomposition

For an $m \times n$ real matrix, a full singular value decomposition (SVD) is $A = U\Sigma V^T$, where U and V are square orthogonal matrices and Σ is rectangular diagonal with nonnegative entries in decreasing order. The singular values are the square roots of the eigenvalues of A^TA. Positive singular values count rank. The distinction between V and the displayed factor V^T is essential when reading singular vectors.

Right singular vectors satisfy $A^TAv_i = \sigma_i^2 v_i$; when $\sigma_i > 0$, set $u_i = Av_i/\sigma_i$. Complete the left columns orthonormally if needed. This method is useful for understanding the theory; for ill-conditioned large matrices a direct numerical SVD avoids squaring the condition number by forming A^TA.

Practice problem 7.4.P1: Transpose an SVD and track zero conventions

Transposing the factorization gives

$$A^T = (U\Sigma V^T)^T = V\Sigma^T U^T.$$

Both outer matrices are orthogonal, and Σ^T is rectangular diagonal with the same nonnegative diagonal entries. Thus it is an SVD of the transpose. The positive singular values of A and A^T coincide, including multiplicities. If singular values are listed as all square roots of the eigenvalues of A^TA, the number of additional zero entries can differ for a rectangular matrix because A^TA and AA^T have different sizes. There is no disagreement about the nonzero singular values.

Exercise 7.4.1: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}.$$

Its eigenvalues are $9,\ 1$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(3,\ 1).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.2: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 25 & 0 \\ 0 & 0 \end{bmatrix}.$$

Its eigenvalues are $25,\ 0$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(5,\ 0).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.3: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 6 & \sqrt{6} \\ \sqrt{6} & 7 \end{bmatrix}.$$

Its eigenvalues are $9,\ 4$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(3,\ 2).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.4: Nonnegative singular values

Compute the positive-semidefinite matrix

$$A^TA = \begin{bmatrix} 3 & 2\sqrt{3} \\ 2\sqrt{3} & 7 \end{bmatrix}.$$

Its eigenvalues are $9,\ 1$. Singular values are their nonnegative square roots, listed in decreasing order:

$$(3,\ 1).$$

A negative diagonal entry of A does not make a negative singular value: lengths and square roots here are nonnegative.

Exercise 7.4.5: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} -3 & 0 \\ 0 & 0 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 9 & 0 \\ 0 & 0 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $9,\ 0$. Their nonnegative square roots are $3,\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3$ gives

$$u_1 = \frac{Av_1}{3} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^2$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.6: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $4,\ 1$. Their nonnegative square roots are $2,\ 1$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 2$ gives

$$u_1 = \frac{Av_1}{2} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 1$ gives

$$u_2 = \frac{Av_2}{1} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.7: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $9,\ 4$. Their nonnegative square roots are $3,\ 2$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{2\sqrt{5}}{5} & -\frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3$ gives

$$u_1 = \frac{Av_1}{3} = \begin{bmatrix} \frac{\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 2$ gives

$$u_2 = \frac{Av_2}{2} = \begin{bmatrix} -\frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{5}}{5} & -\frac{2\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ -\frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.8: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^TA = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $16,\ 1$. Their nonnegative square roots are $4,\ 1$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{\sqrt{5}}{5} & -\frac{2\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^TV = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 4$ gives

$$u_1 = \frac{Av_1}{4} = \begin{bmatrix} \frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 1$ gives

$$u_2 = \frac{Av_2}{1} = \begin{bmatrix} -\frac{\sqrt{5}}{5} \\ \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{2\sqrt{5}}{5} & -\frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \\ -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 2×2, 2×2, and 2×2, respectively. Exact multiplication gives $U^TU = I$, $V^TV = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.9: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 7 & 1 \\ 0 & 0 \\ 5 & 5 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 74 & 32 \\ 32 & 26 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $90,\ 10$. Their nonnegative square roots are $3\sqrt{10},\ \sqrt{10}$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{2\sqrt{5}}{5} & -\frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3\sqrt{10}$ gives

$$u_1 = \frac{Av_1}{3\sqrt{10}} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = \sqrt{10}$ gives

$$u_2 = \frac{Av_2}{\sqrt{10}} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3\sqrt{10} & 0 \\ 0 & \sqrt{10} \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ -\frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.10: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 4 & -2 \\ 2 & -1 \\ 0 & 0 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 20 & -10 \\ -10 & 5 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $25,\ 0$. Their nonnegative square roots are $5,\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 5$ gives

$$u_1 = \frac{Av_1}{5} = \begin{bmatrix} -\frac{2\sqrt{5}}{5} \\ -\frac{\sqrt{5}}{5} \\ 0 \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{5} & -\frac{2\sqrt{5}}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 5 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} -\frac{2\sqrt{5}}{5} & \frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{2\sqrt{5}}{5} \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.11: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $90,\ 0$. Their nonnegative square roots are $3\sqrt{10},\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} -\frac{3\sqrt{10}}{10} & \frac{\sqrt{10}}{10} \\ \frac{\sqrt{10}}{10} & \frac{3\sqrt{10}}{10} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 3\sqrt{10}$ gives

$$u_1 = \frac{Av_1}{3\sqrt{10}} = \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{1}{3} & \frac{2\sqrt{2}}{3} & 0 \\ -\frac{2}{3} & \frac{\sqrt{2}}{6} & \frac{\sqrt{2}}{2} \\ -\frac{2}{3} & \frac{\sqrt{2}}{6} & -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 3\sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} -\frac{3\sqrt{10}}{10} & \frac{\sqrt{10}}{10} \\ \frac{\sqrt{10}}{10} & \frac{3\sqrt{10}}{10} \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.12: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $3,\ 2$. Their nonnegative square roots are $\sqrt{3},\ \sqrt{2}$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = \sqrt{3}$ gives

$$u_1 = \frac{Av_1}{\sqrt{3}} = \begin{bmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = \sqrt{2}$ gives

$$u_2 = \frac{Av_2}{\sqrt{2}} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

Complete these orthonormal columns to an orthonormal basis of $\mathbb{R}^3$; the additional columns do not come from dividing by a zero singular value. They can be chosen in $\operatorname{Nul} A^T$.

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{3}}{3} & 0 & -\frac{\sqrt{6}}{3} \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The dimensions are 3×3, 3×2, and 2×2, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.13: Build and verify all three SVD factors

What we must show. Find a full singular value decomposition of

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}.$$

Find right singular vectors from a symmetric matrix.

Compute

$$G = A^T A = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}.$$

The eigenvalues of G in decreasing order, including zeros, are $25,\ 9,\ 0$. Their nonnegative square roots are $5,\ 3,\ 0$. Choose orthonormal eigenvectors in the same order:

$$V = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{6} & -\frac{2}{3} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{6} & \frac{2}{3} \\ 0 & \frac{2\sqrt{2}}{3} & \frac{1}{3} \end{bmatrix}.$$

These columns are found by solving $(G - \lambda I)v = 0$ and normalizing; if an eigenspace has dimension greater than one, Gram–Schmidt is used within it. The exact checks are $V^T V = I$ and $Gv_i = \sigma_i^2 v_i$.

Find left singular vectors, dividing only by positive singular values.

For $i = 1$, $\sigma_i = 5$ gives

$$u_1 = \frac{Av_1}{5} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}.$$

For $i = 2$, $\sigma_i = 3$ gives

$$u_2 = \frac{Av_2}{3} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

Assemble and check the factors.

A valid choice is

$$U = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}, \qquad V^T = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{6} & -\frac{\sqrt{2}}{6} & \frac{2\sqrt{2}}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}.$$

The dimensions are 2×2, 2×3, and 3×3, respectively. Exact multiplication gives $U^T U = I$, $V^T V = I$, and $U\Sigma V^T = A$, checking orthogonality and every entry of the reconstruction. Nonzero sign changes in matching columns of U, V can produce different-looking but equivalent answers.

Exercise 7.4.14: A unit vector attaining the largest stretch

For any unit vector x, expand it in the orthonormal right singular basis: $x = c_1v_1 + c_2v_2$, where $c_1^2 + c_2^2 = 1$. Exercise 7 gives singular values 3 and 2, so

$$\|Ax\|^2 = 9c_1^2 + 4c_2^2 \le 9(c_1^2 + c_2^2) = 9.$$

The largest possible length is therefore 3, attained at the first right singular vector

$$x = \begin{bmatrix} \frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \end{bmatrix}.$$

It is a unit vector, and direct multiplication gives $\|Ax\| = 3$. Its negative is another maximizer.

Exercise 7.4.15: Read rank and subspace bases without reconstruction

(a) Count positive singular values.

The displayed diagonal factor has positive entries 7.10 and 3.10 and then zero, so $\operatorname{rank} A = 2$.

(b) Read the appropriate columns of the two orthogonal factors.

A basis of $\operatorname{Col} A$ is given by the first two columns of U:

$$\left\{ \begin{bmatrix} .40 \\ .37 \\ -.84 \end{bmatrix}, \begin{bmatrix} -.78 \\ -.33 \\ -.52 \end{bmatrix} \right\}.$$

For the null space take column 3 of V, which is *row 3 of the printed V^T factor*:

$$\operatorname{Nul} A = \operatorname{Span} \left\{ \begin{bmatrix} .58 \\ -.58 \\ .58 \end{bmatrix} \right\}.$$

The entries are rounded descriptions of the true singular vectors; they need not give exact orthogonality or exact zero residuals when interpreted as exact terminating decimals. The requested structural information comes from the stated SVD, not from recomputing rank after rounding.

Exercise 7.4.16: A rectangular SVD with a two-dimensional null space

The diagonal factor has two positive entries, 12.48 and 6.34, so $\operatorname{rank} A = 2$. Since A has four columns, its nullity is $4 - 2 = 2$. A basis of $\operatorname{Col} A$ is supplied by the first two U columns:

$$\left\{ \begin{bmatrix} -.86 \\ .31 \\ .41 \end{bmatrix}, \begin{bmatrix} -.11 \\ .68 \\ -.73 \end{bmatrix} \right\}.$$

The last two columns of V are the last two rows of the printed right-hand factor V^T. Thus a basis of $\operatorname{Nul} A$ is

$$\left\{ \begin{bmatrix} .65 \\ .08 \\ -.16 \\ -.73 \end{bmatrix}, \begin{bmatrix} -.34 \\ .42 \\ -.84 \\ -.08 \end{bmatrix} \right\}.$$

As in Exercise 15, the printed decimal columns are approximations to the exact basis vectors associated with the given SVD.

Exercise 7.4.17: SVD consequences and proofs

Invertibility makes every singular value positive. Reverse the factor order to obtain

$$A^{-1} = V\Sigma^{-1}U^T.$$

This has orthogonal outer factors and positive diagonal entries $1/\sigma_1, \ldots, 1/\sigma_n$. If an SVD is required to list singular values in decreasing order, reverse this order: let J be the permutation matrix reversing the standard basis. Then

$$A^{-1} = (VJ)(J^T\Sigma^{-1}J)(UJ)^T.$$

Its diagonal is $1/\sigma_n \geq \cdots \geq 1/\sigma_1$, as required. In particular the largest singular value of the inverse is the reciprocal of the smallest singular value of A.

Exercise 7.4.18: SVD consequences and proofs

Taking determinants of $A = U\Sigma V^T$ gives $\det A = (\det U)(\det \Sigma)(\det V^T)$. Orthogonal matrices have determinants ± 1, while $\det \Sigma = \sigma_1 \cdots \sigma_n \geq 0$. Taking absolute values proves $|\det A| = \prod_{j=1}^{n} \sigma_j$. If A is singular, a singular value is zero and both sides are zero.

Exercise 7.4.19: SVD consequences and proofs

Use orthogonality in the products:

$$A^TA = V\Sigma^TU^TU\Sigma V^T = V(\Sigma^T\Sigma)V^T,$$

$$AA^T = U\Sigma V^TV\Sigma^TU^T = U(\Sigma\Sigma^T)U^T.$$

The middle matrices are diagonal. Their positive entries are $\sigma_1^2, \ldots, \sigma_r^2$, with zeros in the remaining positions. Thus the columns of V and U are the respective eigenvectors, with those squared eigenvalues; their nonnegative square roots are the singular values. For a rectangular matrix, the numbers of structural zeros on the two sides can differ.

Exercise 7.4.20: SVD consequences and proofs

A real positive definite matrix in this section is symmetric. Its orthogonal diagonalization $A = PDP^T$ has strictly positive eigenvalues on the diagonal of D. Thus the factors already satisfy the SVD requirements, with $U = V = P$ and $\Sigma = D$. If necessary reorder the eigenvectors and diagonal entries together so that the positive diagonal is decreasing. Positive definiteness is what avoids any sign adjustments to make singular values nonnegative.

Exercise 7.4.21: SVD consequences and proofs

Because $P^TP = I_m$,

$$(PA)^T(PA) = A^TP^TPA = A^TA.$$

The two Gram matrices are identical, not merely similar. Hence their eigenvalues and their nonnegative square roots agree, proving the singular values agree. Geometrically an orthogonal transformation of the output does not change any length.

Exercise 7.4.22: SVD consequences and proofs

Assume there is a second right singular direction, so $n \geq 2$. For a unit $x \perp v_1$, expand $x = \sum_{j=2}^{n} c_j v_j$ with $\sum_{j=2}^{n} c_j^2 = 1$. Orthogonality of the left singular vectors gives

$$\|Ax\|^2 = \sum_{j=2}^{n} \sigma_j^2 c_j^2 \leq \sigma_2^2 \sum_{j=2}^{n} c_j^2 = \sigma_2^2.$$

The bound is attained at $x = v_2$. Thus the maximum length is σ_2, including the cases of repeated singular values or $\sigma_2 = 0$. The order $\sigma_1 \geq \sigma_2 \geq \cdots$ is used in the inequality.

Exercise 7.4.23: SVD consequences and proofs

The product $U\Sigma$ has first r columns $\sigma_j u_j$ and all other columns zero. Row j of V^T is v_j^T. By the column-times-row expansion of a matrix product,

$$U\Sigma V^T = \sum_{j=1}^{r} (\sigma_j u_j) v_j^T = \sum_{j=1}^{r} \sigma_j u_j v_j^T.$$

Every term is an $m \times n$ rank-one matrix. The dimensions match, and zero-singular-value terms contribute nothing.

Exercise 7.4.24: SVD consequences and proofs

Transpose the SVD and use $U^T u_j = e_j$:

$$A^T u_j = V\Sigma^T U^T u_j = V\Sigma^T e_j = V(\sigma_j e_j) = \sigma_j v_j \qquad (1 \leq j \leq r).$$

Thus multiplication by A^T sends a positive left singular direction to its matching right singular direction, with the same scale.

Exercise 7.4.25: SVD consequences and proofs

Find the standard matrix A of T and compute a full SVD $A = U\Sigma V^T$. Let $\mathcal{B}$ be the ordered columns of V in the domain, and let $\mathcal{C}$ be the ordered columns of U in the codomain. They are bases because the matrices are orthogonal. The coordinate matrix is

$$[T]_{\mathcal{C} \leftarrow \mathcal{B}} = U^{-1} A V = U^T A V = \Sigma.$$

This is the requested rectangular diagonal matrix, including zero rows or columns where dimensions and rank require them. Using different bases on input and output is what permits this for every rectangular linear map.

Exercise 7.4.26: Numerical SVD with measured reconstruction checks

What we must show. Compute an SVD of

$$A = \begin{bmatrix} -18 & 13 & -4 & 4 \\ 2 & 19 & -4 & 12 \\ -14 & 11 & -12 & 8 \\ -2 & 21 & 4 & 8 \end{bmatrix}.$$

Following the construction in the examples, compute eigenvectors of A^TA in decreasing eigenvalue order. Take the positive square roots to obtain the singular values, then set $u_i = Av_i/\sigma_i$. Exact row reduction confirms rank 3, so exactly that many singular values are positive. The full-precision calculation gives factors whose entries, rounded to two decimals as requested, are

$$U \approx \begin{bmatrix} 0.50 & -0.50 & -0.50 & -0.50 \\ 0.50 & 0.50 & 0.50 & -0.50 \\ 0.50 & -0.50 & 0.50 & 0.50 \\ 0.50 & 0.50 & -0.50 & 0.50 \end{bmatrix}.$$

$$\Sigma \approx \begin{bmatrix} 40.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 20.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 10.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.00 \end{bmatrix}.$$

$$V^T \approx \begin{bmatrix} -0.40 & 0.80 & -0.20 & 0.40 \\ 0.80 & 0.40 & 0.40 & 0.20 \\ 0.40 & -0.20 & -0.80 & 0.40 \\ -0.20 & -0.40 & 0.40 & 0.80 \end{bmatrix}.$$

Before rounding, the Frobenius reconstruction error $\|U\Sigma V^T - A\|_F$ was 2.35×10^{-14}, and the orthogonality errors for U, V were 2.48×10^{-15} and 8.2×10^{-16}. Rounding the displayed entries increases residuals and need not retain exact orthogonality; the decomposition is specified numerically to two decimals, not as an exact factorization of the rounded factors. A reproducible implementation of the stated construction is:

```
import numpy as np
lam,V = np.linalg.eigh(A.T@A)
idx = np.argsort(lam)[::-1]
lam,V = lam[idx],V[:,idx]
r = 3 if A.shape[1] == 4 else 4  # exact ranks here
s = np.sqrt(lam[:r])
U = (A@V[:,:r])/s
if r < A.shape[0]:
    Q,_ = np.linalg.qr(U,mode='complete')
    U = np.column_stack((U,Q[:,r:]))
Sigma = np.zeros(A.shape)
Sigma[np.arange(r),np.arange(r)] = s
print(U,Sigma,V.T)
```

Use floating-point arrays for the displayed input. Eigenvector signs may differ between libraries; changing the matching left and right signs does not change the product. A direct numerical SVD was also used to check the computed singular values.

Exercise 7.4.27: Numerical SVD with measured reconstruction checks

What we must show. Compute an SVD of

$$A = \begin{bmatrix} 6 & -8 & -4 & 5 & -4 \\ 2 & 7 & -5 & -6 & 4 \\ 0 & -1 & -8 & 2 & 2 \\ -1 & -2 & 4 & 4 & -8 \end{bmatrix}.$$

Following the construction in the examples, compute eigenvectors of A^TA in decreasing eigenvalue order. Take the positive square roots to obtain the singular values, then set $u_i = Av_i/\sigma_i$. Exact row reduction confirms rank 4, so exactly that many singular values are positive. The full-precision calculation gives factors whose entries, rounded to two decimals as requested, are

$$U \approx \begin{bmatrix} -0.57 & 0.65 & 0.42 & 0.27 \\ 0.63 & 0.24 & 0.68 & -0.29 \\ 0.07 & 0.63 & -0.53 & -0.56 \\ -0.51 & -0.34 & 0.29 & -0.73 \end{bmatrix}.$$

$$\Sigma \approx \begin{bmatrix} 16.46 & 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 12.16 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 4.87 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 4.31 & 0.00 \end{bmatrix}.$$

$$V^T \approx \begin{bmatrix} -0.10 & 0.61 & -0.21 & -0.52 & 0.55 \\ 0.39 & -0.29 & -0.84 & 0.14 & 0.19 \\ 0.74 & 0.27 & 0.07 & -0.38 & -0.49 \\ 0.41 & -0.50 & 0.45 & -0.23 & 0.58 \\ -0.36 & -0.48 & -0.19 & -0.72 & -0.29 \end{bmatrix}.$$

Before rounding, the Frobenius reconstruction error $\|U\Sigma V^T - A\|_F$ was 1.13×10^{-14}, and the orthogonality errors for U, V were 2.99×10^{-15} and 1.04×10^{-15}. Rounding the displayed entries increases residuals and need not retain exact orthogonality; the decomposition is specified numerically to two decimals, not as an exact factorization of the rounded factors. A reproducible implementation of the stated construction is:

```
import numpy as np
lam,V = np.linalg.eigh(A.T@A)
idx = np.argsort(lam)[::-1]
lam,V = lam[idx],V[:,idx]
r = 3 if A.shape[1] == 4 else 4  # exact ranks here
s = np.sqrt(lam[:r])
U = (A@V[:,:r])/s
if r < A.shape[0]:
    Q,_ = np.linalg.qr(U,mode='complete')
    U = np.column_stack((U,Q[:,r:]))
Sigma = np.zeros(A.shape)
Sigma[np.arange(r),np.arange(r)] = s
print(U,Sigma,V.T)
```

Use floating-point arrays for the displayed input. Eigenvector signs may differ between libraries; changing the matching left and right signs does not change the product. A direct numerical SVD was also used to check the computed singular values.

Exercise 7.4.28: Singular values and a condition number

The matrix referred to from Section 2.3 is

$$A = \begin{bmatrix} 4 & 0 & -3 & -7 \\ -6 & 9 & 9 & 9 \\ 7 & -5 & 10 & 19 \\ -1 & 2 & 4 & -1 \end{bmatrix}.$$

A stable direct SVD, rather than forming the potentially ill-conditioned Gram matrix numerically, gives the singular values

$$(25.96092922,\ 14.45655782,\ 4.669971173,\ 2.286801047).$$

All are positive (also $\det A = -4008 \neq 0$ in exact arithmetic). The spectral condition number is therefore

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_4} \approx 11.3525089.$$

The computation is reproducible with

```
s = np.linalg.svd(A,compute_uv=False)
print(s,s[0]/s[-1])
```

The ratio measures the greatest stretching relative to the least stretching. A large ratio explains sensitivity to perturbations even though the matrix is exactly invertible.

Exercise 7.4.29: Singular values and a condition number

The matrix referred to from Section 2.3 is

$$A = \begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 6 & 4 & 2 & 8 & -8 \\ 7 & 5 & 3 & 10 & 9 \\ 9 & 6 & 4 & -9 & -5 \\ 8 & 5 & 2 & 11 & 4 \end{bmatrix}.$$

A stable direct SVD, rather than forming the potentially ill-conditioned Gram matrix numerically, gives the singular values

$$(25.9343222,\ 16.7554367,\ 11.29172511,\ 1.078525439,\ 0.0003779276369).$$

All are positive (also $\det A = 2 \neq 0$ in exact arithmetic). The spectral condition number is therefore

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_5} \approx 68622.45484.$$

The computation is reproducible with

```
s = np.linalg.svd(A,compute_uv=False)
print(s,s[0]/s[-1])
```

The ratio measures the greatest stretching relative to the least stretching. A large ratio explains sensitivity to perturbations even though the matrix is exactly invertible.

7.5 Applications to image processing and statistics

A data matrix has one observation in each *column* and one measured variable in each row. Its mean column is $M = N^{-1}\sum_{k=1}^{N} X_k$. Subtract M from every column to form B; then the sample covariance matrix is $S = BB^T/(N-1)$, for $N > 1$.

For a unit coefficient vector c, the scalar index $y = c^T X$ has variance $c^T S c$. The greatest possible variance is the largest eigenvalue of S, attained at a corresponding unit eigenvector. Principal component scores normally use the centered observation $X - M$. Centering changes an index by a constant and hence does not change its variance. Signs of principal component directions are arbitrary: reversing a unit eigenvector reverses every score but preserves its explained variance.

Practice problem 7.5.P1: Center the weight-height data and calculate covariance

The five-column mean is $M = (130, 65)^T$. Subtracting it from every observation gives

$$B = \begin{bmatrix} -10 & -5 & -5 & 5 & 15 \\ -4 & -5 & -1 & 3 & 7 \end{bmatrix}.$$

Each row sums to zero, checking the centering step. The cross-product is

$$BB^T = \begin{bmatrix} 400 & 190 \\ 190 & 100 \end{bmatrix}.$$

Since the sample has $N = 5$, divide by $N - 1 = 4$:

$$S = \frac{1}{4}BB^T = \begin{bmatrix} 100 & \frac{95}{2} \\ \frac{95}{2} & 25 \end{bmatrix}.$$

Its diagonal entries are the weight and height sample variances; its off-diagonal entry is their sample covariance.

Practice problem 7.5.P2: A one-number size index

Diagonalizing the covariance matrix gives variances approximately 123.01859218 and 1.98140782. A unit direction for the larger one is

$$c \approx \begin{bmatrix} 0.899901 \\ 0.436094 \end{bmatrix}.$$

Thus a centered size score is $y \approx 0.899901(w - 130) + 0.436094(h - 65)$. Its variance is the larger eigenvalue. The total variance is $\operatorname{tr} S = 125$, so the fraction retained is 0.98414874, or about 98.4149%. This index uses the original units specified by the question; standardizing the two variables to unit variance would define a different PCA problem.

Exercise 7.5.1: Center all observations and form the sample covariance

Compute the mean of each row using all six observations.

$$M = \begin{bmatrix} 12 \\ 10 \end{bmatrix}.$$

Subtract that column from every observation.

$$B = \begin{bmatrix} 7 & 10 & -6 & -9 & -10 & 8 \\ 2 & -4 & -1 & 5 & 3 & -5 \end{bmatrix}.$$

Each row sums to zero. This checks that no observation was omitted and that the rows and columns were not reversed.

Form the sample covariance with denominator $6 - 1$.

$$BB^T = \begin{bmatrix} 430 & -135 \\ -135 & 80 \end{bmatrix}, \qquad S = \frac{1}{5} BB^T = \begin{bmatrix} 86 & -27 \\ -27 & 16 \end{bmatrix}.$$

For example, the off-diagonal entry is the sum of six products of paired centered measurements, divided by 5. The two diagonal entries are sums of squares divided by 5, so they are nonnegative as variances must be.

Exercise 7.5.2: Center all observations and form the sample covariance

Compute the mean of each row using all six observations.

$$M = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

Subtract that column from every observation.

$$B = \begin{bmatrix} -3 & 1 & -2 & 2 & 3 & -1 \\ -6 & 2 & -3 & -1 & 6 & 2 \end{bmatrix}.$$

Each row sums to zero. This checks that no observation was omitted and that the rows and columns were not reversed.

Form the sample covariance with denominator $6 - 1$.

$$BB^T = \begin{bmatrix} 28 & 40 \\ 40 & 90 \end{bmatrix}, \qquad S = \frac{1}{5} BB^T = \begin{bmatrix} \frac{28}{5} & 8 \\ 8 & 18 \end{bmatrix}.$$

For example, the off-diagonal entry is the sum of six products of paired centered measurements, divided by 5. The two diagonal entries are sums of squares divided by 5, so they are nonnegative as variances must be.

Exercise 7.5.3: Find both principal component directions

Use the exact covariance matrix from Exercise 1:

$$S = \begin{bmatrix} 86 & -27 \\ -27 & 16 \end{bmatrix}.$$

Its characteristic equation is $t^2 - 102t + 647 = 0$. In decreasing order its eigenvalues are approximately

$$\lambda_1 = 95.20407221, \qquad \lambda_2 = 6.79592779.$$

Choose the following orthonormal eigenvector columns, in that same order:

$$P \approx \begin{bmatrix} 0.946515 & 0.322659 \\ -0.322659 & 0.946515 \end{bmatrix}.$$

The two principal component scores are the entries of $Y = P^T(X - M)$, with mean

$$M = \begin{bmatrix} 12 \\ 10 \end{bmatrix}.$$

Their covariance matrix is diagonal, $P^T S P = \operatorname{diag}(\lambda_1, \lambda_2)$, so they are uncorrelated. The first score explains about 93.33733% of total variance; the second explains the remainder. A sign-reversed column gives an equivalent principal component.

Exercise 7.5.4: Find both principal component directions

Use the exact covariance matrix from Exercise 2:

$$S = \begin{bmatrix} \frac{28}{5} & 8 \\ 8 & 18 \end{bmatrix}.$$

Its characteristic equation is $\frac{5t^2 - 118t + 184}{5} = 0$. In decreasing order its eigenvalues are approximately

$$\lambda_1 = 21.92126474, \qquad \lambda_2 = 1.67873526.$$

Choose the following orthonormal eigenvector columns, in that same order:

$$P \approx \begin{bmatrix} 0.440130 & 0.897934 \\ 0.897934 & -0.440130 \end{bmatrix}.$$

The two principal component scores are the entries of $Y = P^T(X - M)$, with mean

$$M = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

Their covariance matrix is diagonal, $P^T S P = \operatorname{diag}(\lambda_1, \lambda_2)$, so they are uncorrelated. The first score explains about 92.88672% of total variance; the second explains the remainder. A sign-reversed column gives an equivalent principal component.

Exercise 7.5.5: A leading principal component from a supplied covariance matrix

The full covariance matrix needed for this calculation is supplied in the exercise:

$$S = \begin{bmatrix} \frac{4103}{25} & \frac{3273}{100} & \frac{2026}{25} \\ \frac{3273}{100} & \frac{13486}{25} & \frac{24913}{100} \\ \frac{2026}{25} & \frac{24913}{100} & \frac{18911}{100} \end{bmatrix}.$$

No original satellite image is required to diagonalize this matrix. Compute its symmetric eigenproblem and normalize the eigenvector of largest eigenvalue. The eigenvalues, in decreasing order, are approximately

$$(677.49712437,\ 181.58112062,\ 33.59175501).$$

A maximizing unit direction is

$$c \approx \begin{bmatrix} 0.129554 \\ 0.874423 \\ 0.467547 \end{bmatrix}.$$

Thus $y_1 = c_1\widehat{x}_1 + c_2\widehat{x}_2 + c_3\widehat{x}_3$, where the hats denote mean-deviation variables, is the first principal component. The uncentered combination has the same variance. The maximum variance is approximately 677.49712437, and total variance is $\operatorname{tr} S = \frac{89267}{100}$. Therefore the explained fraction is

$$\frac{\lambda_1}{\operatorname{tr} S} \approx 0.75895586, \quad \text{or } 75.89559\%.$$

The actual computation used the real symmetric eigenvalue routine:

```
eigenvalues,P = np.linalg.eigh(S)
c = P[:,-1]
print(c,eigenvalues[-1],
      100*eigenvalues[-1]/np.trace(S))
```

The maximum eigenvalue is the last one because `eigh` returns increasing order. A reversed eigenvector sign is equivalent. Orthogonality and the residual $Sc - \lambda_1 c$ were checked at full precision before rounding.

Exercise 7.5.6: A leading principal component from a supplied covariance matrix

The full covariance matrix needed for this calculation is supplied in the exercise:

$$S = \begin{bmatrix} \frac{741}{25} & \frac{919}{50} & 5 \\ \frac{919}{50} & \frac{1041}{50} & \frac{703}{50} \\ 5 & \frac{703}{50} & \frac{2921}{100} \end{bmatrix}.$$

No original satellite image is required to diagonalize this matrix. Compute its symmetric eigenproblem and normalize the eigenvector of largest eigenvalue. The eigenvalues, in decreasing order, are approximately

$$(51.69566979,\ 24.53510921,\ 3.43922100).$$

A maximizing unit direction is

$$c \approx \begin{bmatrix} 0.615525 \\ 0.599424 \\ 0.511683 \end{bmatrix}.$$

Thus $y_1 = c_1\widehat{x}_1 + c_2\widehat{x}_2 + c_3\widehat{x}_3$, where the hats denote mean-deviation variables, is the first principal component. The uncentered combination has the same variance. The maximum variance is approximately 51.69566979, and total variance is $\operatorname{tr} S = \frac{7967}{100}$. Therefore the explained fraction is

$$\frac{\lambda_1}{\operatorname{tr} S} \approx 0.64887247, \quad \text{or } 64.88725\%.$$

The actual computation used the real symmetric eigenvalue routine:

```
eigenvalues,P = np.linalg.eigh(S)
c = P[:,-1]
print(c,eigenvalues[-1],
      100*eigenvalues[-1]/np.trace(S))
```

The maximum eigenvalue is the last one because `eigh` returns increasing order. A reversed eigenvector sign is equivalent. Orthogonality and the residual $Sc - \lambda_1 c$ were checked at full precision before rounding.

Exercise 7.5.7: A maximal-variance scalar combination

For the data in Exercise 1, the covariance matrix is

$$S = \begin{bmatrix} 86 & -27 \\ -27 & 16 \end{bmatrix}.$$

Choose the unit coefficient direction

$$c \approx \begin{bmatrix} 0.946515 \\ -0.322659 \end{bmatrix}.$$

The requested uncentered variable is $y_1 \approx (0.946515)x_1 + (-0.322659)x_2$. Its variance is $c^T S c = \lambda_1 \approx 95.20407221$, the maximum by the quadratic-form theorem. Using $x_1 - M_1, x_2 - M_2$ instead would simply subtract its mean and leave that variance unchanged. The total variance is $\operatorname{tr} S = 102$, so the explained percentage is approximately 93.33733%. Both the unit-length constraint and the covariance eigenvector equation were checked before rounding the coefficients.

Exercise 7.5.8: A maximal-variance scalar combination

For the data in Exercise 2, the covariance matrix is

$$S = \begin{bmatrix} \frac{28}{5} & 8 \\ 8 & 18 \end{bmatrix}.$$

Choose the unit coefficient direction

$$c \approx \begin{bmatrix} 0.440130 \\ 0.897934 \end{bmatrix}.$$

The requested uncentered variable is $y_1 \approx (0.440130)x_1 + (0.897934)x_2$. Its variance is $c^T S c = \lambda_1 \approx 21.92126474$, the maximum by the quadratic-form theorem. Using $x_1 - M_1, x_2 - M_2$ instead would simply subtract its mean and leave that variance unchanged. The total variance is $\operatorname{tr} S = \frac{118}{5}$, so the explained percentage is approximately 92.88672%. Both the unit-length constraint and the covariance eigenvector equation were checked before rounding the coefficients.

Exercise 7.5.9: A unit eigenvector maximizes the index variance

The sample variance of $y = c^T X$ is $c^T S c$, with $c^T c = 1$. The characteristic roots of the supplied symmetric matrix are 9, 6, 3, with multiplicity. Its largest eigenvalue is 9. Solving $(S - (9)I)c = 0$ and normalizing gives

$$c = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}.$$

The exact checks are $c^T c = 1$ and $Sc = (9)c$. Hence the index

$$y = (\frac{1}{3})x_1 + (\frac{2}{3})x_2 + (\frac{2}{3})x_3$$

has the largest possible variance, 9. This accounts for the fraction $\frac{1}{2}$ of the total variance 18.

Exercise 7.5.10: A unit eigenvector maximizes the index variance

The sample variance of $y = c^T X$ is $c^T S c$, with $c^T c = 1$. The characteristic roots of the supplied symmetric matrix are 15, 3, 3, with multiplicity. Its largest eigenvalue is 15. Solving $(S - (15)I)c = 0$ and normalizing gives

$$c = \begin{bmatrix} \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact checks are $c^T c = 1$ and $Sc = (15)c$. Hence the index

$$y = (\frac{\sqrt{6}}{6})x_1 + (\frac{\sqrt{6}}{3})x_2 + (\frac{\sqrt{6}}{6})x_3$$

has the largest possible variance, 15. This accounts for the fraction $\frac{5}{7}$ of the total variance 21.

Exercise 7.5.11: Covariance identities and their dimensions

(a) The transformed mean remains zero.

Write $B = [X_1 \ \cdots \ X_N]$ and $C = [Y_1 \ \cdots \ Y_N] = P^T B$. Since the original mean is zero, $\sum_k X_k = 0$. Thus

$$\frac{1}{N}\sum_k Y_k = P^T \left(\frac{1}{N} \sum_k X_k \right) = 0.$$

Equivalently, $B\mathbf{1} = 0$ implies $C\mathbf{1} = P^T B\mathbf{1} = 0$. No orthogonality of P is required.

(b) Compute the new covariance using part (a).

Because both matrices are already centered,

$$S_Y = \frac{1}{N-1} CC^T = \frac{1}{N-1} P^T BB^T P = P^T SP.$$

The transpose on C^T reverses factors and gives $C^T = B^T P$, which explains the rightmost P.

Exercise 7.5.12: Covariance identities and their dimensions

For orthogonal P, the relation $X = PY$ is equivalent to $Y = P^T X$. Centering commutes with that linear change of variables, and Exercise 11 gives covariance $S_Y = P^T SP$. Total variance is the trace. The cyclic trace identity yields

$$\operatorname{tr}(S_Y) = \operatorname{tr}(P^T SP) = \operatorname{tr}(SPP^T) = \operatorname{tr}(SI) = \operatorname{tr}(S).$$

Thus orthogonal changes redistribute variance among new coordinate directions without creating or destroying its total. This proof also covers data not originally centered, because covariance first subtracts the transformed mean.

Exercise 7.5.13: Covariance identities and their dimensions

Let $M = N^{-1} \sum_{k=1}^{N} X_k$ and $B = [X_1 - M \ \cdots \ X_N - M]$. Multiply the column partition of B by the row partition of B^T:

$$BB^T = \sum_{k=1}^{N} (X_k - M)(X_k - M)^T.$$

Hence the sample covariance is

$$\boxed{S = \frac{1}{N-1} \sum_{k=1}^{N} (X_k - M)(X_k - M)^T.}$$

Every summand is a $p \times p$ outer-product matrix. Its (i, j) entry is the product of the centered measurements of variables i and j in observation k. When $p = 1$, the outer product is just the square $(t_k - m)^2$, giving exactly the usual scalar sample-variance formula. The sum has N terms, one per observation; the lowercase n printed as an upper bound in the displayed scalar formula should be read as that sample size.

7.S Chapter 7 supplementary exercises

These exercises connect symmetric spectral decompositions, quadratic-form positivity, Gram matrices, singular value decompositions, and minimum-length solutions. A symmetric matrix is positive definite when $x^TAx > 0$ for every $x \neq 0$, not merely for one vector. A pseudoinverse provides a least-squares solution even when an exact solution does not exist; that distinction is important in the final two printed exercises.

Exercise 7.S.1: Seventeen true–false statements

(a) **True.** If $A = PDP^T$ with P orthogonal and D diagonal, then $A^T = PD^TP^T = A$.

(b) **False.** The 90° rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is orthogonal but is not symmetric.

(c) **True.** $\|Ax\|^2 = x^TA^TAx = x^Tx$ for orthogonal A. Taking nonnegative square roots gives the length equality.

(d) **False.** Principal axes require mutually perpendicular eigenvector directions. For $A = I_2$, every invertible P diagonalizes A, but $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has nonorthogonal columns, so its columns do not give orthogonal principal axes.

(e) **False.** Orthogonal columns need not be unit columns. For $P = 2I$, $P^T = 2I \neq I/2 = P^{-1}$.

(f) **False.** All coefficients of $x_1^2 + x_2^2 + 4x_1x_2$ are positive, but its value at $(1, -1)$ is -2.

(g) **False.** $x_1^2 - x_2^2$ is positive at e_1 and negative at e_2, so positivity at one input does not give definiteness.

(h) **True.** The symmetric coefficient matrix admits an orthonormal eigenbasis, and its orthogonal change of variables removes all mixed terms by the Principal Axes Theorem.

(i) **False.** For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, the largest diagonal entry is 1, but the unit vector $(1, 1)^T/\sqrt{2}$ gives $x^TAx = 2$, the largest eigenvalue.

(j) **False without a constraint.** A positive definite form is unbounded above: $Q(tx) = t^2Q(x) \to \infty$ for fixed $x \neq 0$. The greatest eigenvalue is the maximum only when $\|x\| = 1$ is imposed.

(k) **False.** If P is orthogonal, nonzero u gives $x = Pu \neq 0$, so $(Pu)^TA(Pu) > 0$. An invertible variable change preserves positive definiteness rather than turning it negative.

(l) **False.** An indefinite form has a symmetric coefficient matrix with both positive and negative eigenvalues. “Not definite eigenvalues” is not a mathematical alternative definition; semidefinite forms are not indefinite merely because an eigenvalue is zero.

(m) **True.** Substitution gives $(Pu)^TA(Pu) = u^TP^TAPu = u^T(P^{-1}AP)u$.

(n) **False.** Orthonormal columns are required for UU^T to be the projection formula. For $U = 2e_1$ and $x = e_1$, $UU^Tx = 4e_1$, whereas the projection onto its column line is e_1.

(o) **True.** An SVD gives $\|Bx\|^2 = \sum_i \sigma_i^2 y_i^2 \le \sigma_1^2 \sum_i y_i^2 = \sigma_1^2$ for $y = V^T x$ and $\|x\| = 1$.

(p) **True as an existence statement.** From an ordinary SVD $B = U\Sigma V^T$, take $P = U$ and $Q = V^T$. The transpose of an orthogonal matrix is orthogonal, so $B = P\Sigma Q$ has exactly the asserted form. This does not mean one can omit a transpose while keeping the same right singular-vector matrix V; the new symbol Q denotes its transpose.

(q) **False.** The singular values of $A^T A$ are the squares of those of A. For $A = 2I$, they are 4 versus 2, so the lists differ.

Exercise 7.S.2: Build a symmetric matrix with prescribed eigenvectors

(a) Each outer product satisfies $(u_i u_i^T)^T = u_i u_i^T$. A real linear combination of symmetric matrices is symmetric, so $A^T = A$.

(b) For each j, orthonormality gives

$$Au_j = \sum_i \lambda_i u_i (u_i^T u_j) = \lambda_j u_j.$$

Every u_j is nonzero, so it is an eigenvector with eigenvalue λ_j. Because these n vectors are a basis, the list contains all eigenvalues, including their multiplicities.

Exercise 7.S.3: Rank counts the nonzero spectral summands

Write $A = PDP^T$ with orthogonal P. Multiplication by invertible matrices preserves rank, so $\operatorname{rank} A = \operatorname{rank} D$, the number of nonzero diagonal eigenvalues. Thus exactly r terms in $A = \sum_i \lambda_i u_i u_i^T$ are nonzero. For each such term, every column is a multiple of u_i, and at least one column is nonzero because $u_i \neq 0$ and $\lambda_i \neq 0$. Each nonzero term therefore has rank one. Discarding zero terms represents A as a sum of r rank-one matrices, as claimed. When $r = 0$, this is the empty sum representing zero.

Exercise 7.S.4: An orthogonal column/null decomposition for a symmetric matrix

(a) A vector z is perpendicular to every column of A precisely when $A^T z = 0$. Since $A^T = A$, this says $z \in \operatorname{Nul} A$. Therefore $(\operatorname{Col} A)^\perp = \operatorname{Nul} A$.

(b) Orthogonally decompose y relative to $W = \operatorname{Col} A$: $y = \widehat{y} + z$, where $\widehat{y} \in \operatorname{Col} A$ and $z \in (\operatorname{Col} A)^\perp = \operatorname{Nul} A$. Such a decomposition exists by the Orthogonal Decomposition Theorem and is unique.

Exercise 7.S.5: A nonzero-eigenvalue eigenvector lies in the column space

If $Av = \lambda v$ with $\lambda \neq 0$, then $v = A(v/\lambda)$. This explicitly writes v as an output of A, so $v \in \operatorname{Col} A$. The nonzero-eigenvalue condition permits division; a zero-eigenvalue eigenvector need not belong to the column space.

Exercise 7.S.6: Construct the decomposition using an eigenbasis

Choose an orthonormal eigenbasis u_i and write $y = \sum_i c_i u_i$. Separate the sum into

$$\widehat{y} = \sum_{\lambda_i \neq 0} c_i u_i, \qquad z = \sum_{\lambda_i = 0} c_i u_i.$$

Every summand of $\widehat{y}$ belongs to $\operatorname{Col} A$ by Exercise 5, so $\widehat{y} \in \operatorname{Col} A$. Every summand of z is annihilated by A, so $z \in \operatorname{Nul} A$. Their sum is y, giving the requested second proof. The two sums are perpendicular because they use disjoint parts of an orthonormal basis.

Exercise 7.S.7: Prove the Cholesky characterization

A Cholesky factorization implies positive definiteness.

If $A = R^T R$ with invertible real R, it is symmetric and $x^T A x = \|Rx\|^2 > 0$ whenever $x \neq 0$. Thus A is positive definite.

A positive definite matrix supplies such a factorization.

Exercise 7.2.26 gives an invertible positive definite square root B with $A = B^T B$. Take a QR factorization $B = QR$, choosing the signs of columns of Q and rows of R so the nonzero diagonal entries of R are positive. Here Q is square orthogonal and R is upper triangular. Then $A = B^T B = R^T Q^T Q R = R^T R$. The full rank of B makes R invertible, and the sign choices give the required positive diagonal. This proves both directions of the equivalence.

Exercise 7.S.8: Convert Cholesky into an LU factorization

From Exercise 7, $A = R^T R$ with upper triangular R and positive r_{ii}. Let $D = \operatorname{diag}(r_{11}, \ldots, r_{nn})$. Define $L = R^T D^{-1}$, which is lower triangular with diagonal entries one, and $U = DR$, which is upper triangular with diagonal entries $r_{ii}^2 > 0$. Then $LU = R^T D^{-1} DR = R^T R = A$, as required. The parenthetical converse in the text is understood for symmetric matrices and the usual unit-lower-triangular LU convention. Without symmetry, an arbitrary upper triangular matrix with positive diagonal need not be a positive definite *symmetric* matrix.

Exercise 7.S.9: A Gram matrix has the same rank as its original matrix

$G = A^T A$ is symmetric and $x^T G x = \|Ax\|^2 \geq 0$, so it is positive semidefinite. Moreover, $Gx = 0$ implies $0 = x^T G x = \|Ax\|^2$, hence $Ax = 0$. Conversely $Ax = 0$ immediately implies $Gx = A^T 0 = 0$. Thus $\operatorname{Nul} G = \operatorname{Nul} A$. Both matrices have the same number n of columns, so rank–nullity gives $\operatorname{rank} G = n - \dim \operatorname{Nul} G = n - \dim \operatorname{Nul} A = \operatorname{rank} A$. This also proves why forming a Gram matrix does not create new independent column directions in exact arithmetic.

Exercise 7.S.10: Construct a rank-revealing Gram factor

A symmetric positive semidefinite G has r positive eigenvalues $\lambda_1, \ldots, \lambda_r$ and the rest zero. Let U_r contain corresponding orthonormal eigenvectors, and $D_r = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$. Then $G = U_r D_r U_r^T$.

Choose the $r \times n$ matrix

$$\boxed{A = D_r^{1/2} U_r^T}.$$

Its Gram matrix is $A^T A = U_r D_r U_r^T = G$. Equivalently the hinted factor $B = U_r D_r^{1/2}$ satisfies $G = BB^T$ and $A = B^T$. When $r = 0$, $G = 0$ and the $0 \times n$ empty matrix has zero Gram matrix under the empty-sum convention.

Exercise 7.S.11: A polar decomposition, including singular matrices

Take a square SVD $A = U\Sigma V^T$, with U, V orthogonal and Σ diagonal with nonnegative entries. Set

$$P = U\Sigma U^T, \qquad Q = UV^T.$$

Then P is symmetric positive semidefinite by its orthogonal diagonalization, and $\operatorname{rank} P = \operatorname{rank} \Sigma = \operatorname{rank} A$. Also $Q^T Q = VU^T UV^T = I$, so Q is orthogonal. Finally $PQ = U\Sigma U^T UV^T = U\Sigma V^T = A$. No inverse singular values were used, so the construction works for singular A as well. The orthogonal factor need not be unique in that case.

Exercise 7.S.12: The pseudoinverse produces two orthogonal projections

Use the reduced SVD $A = U_r D V_r^T$ and $A^+ = V_r D^{-1} U_r^T$, where the r singular values in D are positive.

(a) Since $V_r^T V_r = I_r$, $AA^+ = U_r D D^{-1} U_r^T = U_r U_r^T$. The columns of U_r are an orthonormal basis of $\operatorname{Col} A$; therefore $AA^+ y$ is the orthogonal projection of y onto that space.

(b) Similarly $A^+ A = V_r V_r^T$, the projection onto $\operatorname{Col} V_r = \operatorname{Row} A$.

(c) Multiplying the formulas and using orthonormality gives

$$AA^+ A = U_r U_r^T U_r D V_r^T = A, \qquad A^+ AA^+ = V_r V_r^T V_r D^{-1} U_r^T = A^+.$$

Thus both requested triple-product identities hold.

Exercise 7.S.13: A consistent system's unique minimum-length solution

(a) Since $b = Ax$ for some x by consistency, $x^+ = A^+ b = A^+ Ax$ is the projection of x onto $\operatorname{Row} A$ by Exercise 12(b). Hence $x^+ \in \operatorname{Row} A$.

(b) Since $b \in \operatorname{Col} A$, its projection there is itself, so $Ax^+ = AA^+ b = b$ by Exercise 12(a). Thus x^+ is indeed a solution.

(c) For any other solution u, $A(u - x^+) = b - b = 0$. Thus $u - x^+ \in \operatorname{Nul} A$, perpendicular to $x^+ \in \operatorname{Row} A = (\operatorname{Nul} A)^\perp$. Pythagoras yields $\|u\|^2 = \|x^+\|^2 + \|u - x^+\|^2 \geq \|x^+\|^2$. Equality holds only when $u = x^+$. Taking square roots proves the minimum length and its uniqueness.

Exercise 7.S.14: The minimum-length least-squares solution for arbitrary data

Let $\widehat{b} = AA^+ b$, the projection of b onto $\operatorname{Col} A$. A vector x is a least-squares solution of $Ax = b$ exactly when $Ax = \widehat{b}$, since $\widehat{b}$ is the unique closest output. This new equation is consistent. By Exercise

13 its unique minimum-length solution is $A^+\widehat{b}$. The triple-product identity from Exercise 12 gives $A^+\widehat{b} = A^+AA^+b = A^+b$. Thus $\boxed{A^+b}$ is the unique minimum-length least-squares solution, even when the original equation itself has no solution.

Exercise 7.S.15: A pseudoinverse and an inconsistent printed right side

Check consistency before claiming to solve the original system.

Rows 3 and 4 of the printed A are identical, but the prescribed $b = (6, -1, -4, 6)^T$ has unequal entries 3 and 4. Hence $Ax = b$ has *no exact solution*. The requested pseudoinverse calculation is nevertheless well defined: it returns the minimum-length least-squares solution from Exercise 14, not an exact solution from Exercise 13. We retain the printed data and make that distinction explicit.

Compute from an SVD, then verify the exact rational matrix.

An executed numerical SVD has nonzero singular values approximately 9.84443449, 2.62465501, and 1.09466692. Its fourth singular value is at roundoff level, representing an exact zero. In NumPy the construction can be reproduced by

```
U,d,Vt = np.linalg.svd(A,full_matrices=False)
r = np.count_nonzero(d > 1e-12)
Ap = (Vt[:r,:].T/d[:r])@U[:,:r].T
```

The exact rational result, verified by the four pseudoinverse identities, is

$$A^+ = \frac{1}{40}\begin{bmatrix} -2 & -14 & 13 & 13 \\ -2 & -14 & 13 & 13 \\ -2 & 6 & -7 & -7 \\ 2 & -6 & 7 & 7 \\ 4 & -12 & -6 & -6 \end{bmatrix}.$$

Here the verification is $AA^+A = A$, $A^+AA^+ = A^+$, and symmetry of AA^+ and A^+A; all products were checked with exact rational arithmetic. Multiplying by the printed right side gives

$$\widehat{x} = A^+b = (7/10, 7/10, -4/5, 4/5, 3/5)^T, \qquad A\widehat{x} = (6, -1, 1, 1)^T.$$

The residual is $(0, 0, -5, 5)^T$ and is perpendicular to every column of A, since its two nonzero components multiply identical rows with opposite weights. Thus $A^T(b - A\widehat{x}) = 0$, independently checking the least-squares claim.

Check row-space membership and the strict norm inequality.

An explicit witness is $\widehat{x} = A^T(7/100, -91/100, 129/100, 0)^T$, so $\widehat{x} \in \operatorname{Row} A$. Choose $u = (-1, 1, 0, 0, 0)^T$. Its first two columns cancel, so $Au = 0$; also $\widehat{x} \cdot u = 0$. Finally $\|\widehat{x}\|^2 = 131/50$ and $\|\widehat{x} + u\|^2 = 231/50$, proving $\|\widehat{x}\| < \|\widehat{x} + u\|$.

Exercise 7.S.16: A second minimum-length least-squares computation

Again distinguish least squares from exact consistency.

Here row 4 is three times row 3, whereas the prescribed right side has $6 \neq 3(-4)$. Thus the printed system is also inconsistent. Its pseudoinverse solution must be interpreted via Exercise 14.

Construct and verify the pseudoinverse.

The executed SVD's nonzero singular values are approximately $12.9536237, 1.44552724, .337763285$. Using the SVD formula in Exercise 15 and verifying exact products gives

$$A^+ = \frac{1}{20}\begin{bmatrix} 10 & 0 & -1 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 40 & 10 & 30 \\ 10 & -20 & -7 & -21 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

All four pseudoinverse identities hold exactly. The resulting input and fitted output are

$$\widehat{x} = (23/10, 0, 5, -9/10, 0)^T, \qquad A\widehat{x} = (6, -1, 7/5, 21/5)^T.$$

The residual is $(0, 0, -27/5, 9/5)^T$. Its row-weight combination vanishes because row 4 equals three times row 3: $(-27/5)\,\mathrm{row}_3\, A + (9/5)\,\mathrm{row}_4\, A = 0$. Hence the normal equations hold and this is a least-squares solution.

Verify the two geometric checks.

The identity $\widehat{x} = A^T(7/10, 109/10, 27, 0)^T$ places it in the row space. Take the nonzero null vector $u = e_2 = (0, 1, 0, 0, 0)^T$, since column 2 is zero. It is perpendicular to $\widehat{x}$, whose second coordinate is zero. The squared norms are $\|\widehat{x}\|^2 = 311/10$ and $\|\widehat{x} + u\|^2 = 321/10$, giving the requested strict inequality. These checks establish the minimum-length property without falsely claiming that the inconsistent original equation has been solved exactly.

8.1 Affine combinations

An affine combination is a linear combination whose coefficients sum to 1. They may be negative: nonnegativity is an additional condition for a *convex* combination, studied later. The affine hull aff S is the set of all affine combinations from S.

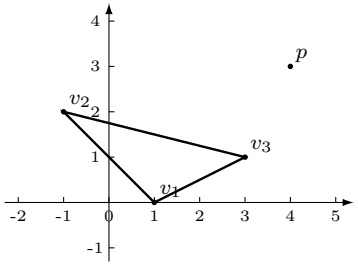
For nonempty $S = \{v_1, \dots, v_k\}$,

$$\operatorname{aff} S = v_1 + \operatorname{Span}\{v_2 - v_1, \dots, v_k - v_1\}.$$

An equivalent computational method appends a coordinate 1 to each point. Solving $[\widetilde{v}_1 \ \cdots \ \widetilde{v}_k]c = \widetilde{p}$ then enforces both the point equation and $\sum c_j = 1$. An affine set is closed under affine combinations; it may be empty under the usual closure convention. A nonempty affine set is a translated subspace, or flat.

Practice problem 8.1.P1: Three noncollinear points fill the plane affinely

The edge differences from v_1 are $v_2 - v_1 = (-2, 2)$ and $v_3 - v_1 = (2, 1)$. Their determinant is $(-2)1 - 2 \cdot 2 = -6 \neq 0$, so they span $\mathbb{R}^2$. Hence the affine hull of the three points is all of $\mathbb{R}^2$, and it must contain p.



For explicit weights solve $p - v_1 = c_2(v_2 - v_1) + c_3(v_3 - v_1)$:

$$-2c_2 + 2c_3 = 3, \qquad 2c_2 + c_3 = 3.$$

Adding suitable multiples gives $c_2 = 1/2$, $c_3 = 2$, and then $c_1 = 1 - c_2 - c_3 = -3/2$. Thus $p = -\frac{3}{2}v_1 + \frac{1}{2}v_2 + 2v_3$. These weights sum to 1 and reproduce $(4, 3)$; a negative weight is permitted for an affine combination.

Exercise 8.1.1: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & -2 & 0 & 3 \\ 2 & 2 & 4 & 7 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & -\frac{3}{2} & -\frac{3}{2} \\ 0 & 0 & 1 & \frac{5}{2} & \frac{1}{2} \end{bmatrix}.$$

There is a free parameter; setting it to zero gives one valid choice. Other choices are also allowed, since an affine representation need not be unique for a dependent set. One affine coefficient column is

$$c = \begin{bmatrix} 2 \\ -\frac{3}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.2: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 2 & 2 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} -5 \\ 2 \\ 4 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.3: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} -3 & 0 & 4 \\ 1 & 4 & -2 \\ 1 & -2 & 6 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 17 \\ 1 \\ 5 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 17 \\ 1 \\ 5 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.4: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & 2 & 4 \\ 2 & -6 & 3 \\ 0 & 7 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} -3 \\ 4 \\ -4 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & \frac{13}{5} \\ 0 & 1 & 0 & -\frac{2}{5} \\ 0 & 0 & 1 & -\frac{6}{5} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} \frac{13}{5} \\ -\frac{2}{5} \\ -\frac{6}{5} \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} -3 \\ 4 \\ -4 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.5: Use orthogonality before checking the affine condition

The given orthogonal basis has squared lengths $6, 5, 30$. Orthogonal expansion therefore gives the unique linear coefficients

$$c_1 = \frac{p \cdot b_1}{6}, \qquad c_2 = \frac{p \cdot b_2}{5}, \qquad c_3 = \frac{p \cdot b_3}{30}.$$

Because the basis is linearly independent, these are the only possible coefficients. The representation is affine exactly when their sum is 1.

(a) The point $p = \begin{bmatrix} 3 \\ 8 \\ 4 \end{bmatrix}$.

The dot products with b_1, b_2, b_3 are $18,\ -5,\ -30$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (3)b_1 + (-1)b_2 + (-1)b_3$ is an affine combination.

(b) The point $p = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $12,\ 0,\ 30$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 3.$$

The unique linear coefficients do not sum to 1, so no affine combination of these basis points equals p.

(c) The point $p = \begin{bmatrix} 0 \\ -1 \\ -5 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $-6,\ 10,\ 0$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (-1)b_1 + (2)b_2 + (0)b_3$ is an affine combination.

Exercise 8.1.6: Use orthogonality before checking the affine condition

The given orthogonal basis has squared lengths $6, 5, 30$. Orthogonal expansion therefore gives the unique linear coefficients

$$c_1 = \frac{p \cdot b_1}{6}, \qquad c_2 = \frac{p \cdot b_2}{5}, \qquad c_3 = \frac{p \cdot b_3}{30}.$$

Because the basis is linearly independent, these are the only possible coefficients. The representation is affine exactly when their sum is 1.

(a) The point $p = \begin{bmatrix} 0 \\ -19 \\ -5 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $-24,\ 10,\ 90$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} -4 \\ 2 \\ 3 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (-4)b_1 + (2)b_2 + (3)b_3$ is an affine combination.

(b) The point $p = \begin{bmatrix} \frac{3}{2} \\ -\frac{13}{10} \\ -\frac{1}{2} \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $\frac{6}{5},\ \frac{5}{2},\ 9$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} \frac{1}{5} \\ \frac{1}{2} \\ \frac{3}{10} \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (\frac{1}{5})b_1 + (\frac{1}{2})b_2 + (\frac{3}{10})b_3$ is an affine combination.

(c) The point $p = \begin{bmatrix} 5 \\ -4 \\ 0 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $6,\ 5,\ 30$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 3.$$

The unique linear coefficients do not sum to 1, so no affine combination of these basis points equals p.

Exercise 8.1.7: Distinguish a linear span from an affine hull

Place the three given independent vectors in the columns of

$$V = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 3 & 0 & 1 \\ 0 & 4 & 1 \end{bmatrix}.$$

A target is in $\operatorname{Span} S$ if $Vc = p$ is consistent; it is in $\operatorname{aff} S$ only if its coefficients also sum to 1. Independence makes a consistent coefficient column unique.

(a) Test p_1**.**

$$\operatorname{rref}[V \mid p_1] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, so the point is in $\operatorname{Span} S$. The sum is 2, so the point is not in $\operatorname{aff} S$.

(b) Test p_2**.**

$$\operatorname{rref}[V \mid p_2] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} 2 \\ -4 \\ 3 \end{bmatrix}$, so the point is in $\operatorname{Span} S$. The sum is 1, so the point is also in $\operatorname{aff} S$.

(c) Test p_3.

$$\operatorname{rref}[V \mid p_3] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The augmented last column has a pivot, so the system is inconsistent. The point is not in $\operatorname{Span} S$ and therefore cannot be in the smaller set $\operatorname{aff} S$.

Exercise 8.1.8: Distinguish a linear span from an affine hull

Place the three given independent vectors in the columns of

$$V = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 3 & 6 & 12 \\ -2 & -5 & -6 \end{bmatrix}.$$

A target is in $\operatorname{Span} S$ if $Vc = p$ is consistent; it is in $\operatorname{aff} S$ only if its coefficients also sum to 1. Independence makes a consistent coefficient column unique.

(a) Test p_1.

$$\operatorname{rref}[V \mid p_1] = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$, so the point is in $\operatorname{Span} S$. The sum is 3, so the point is not in $\operatorname{aff} S$.

(b) Test p_2.

$$\operatorname{rref}[V \mid p_2] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The augmented last column has a pivot, so the system is inconsistent. The point is not in $\operatorname{Span} S$ and therefore cannot be in the smaller set $\operatorname{aff} S$.

(c) Test p_3.

$$\operatorname{rref}[V \mid p_3] = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} -2 \\ 6 \\ -3 \end{bmatrix}$, so the point is in $\operatorname{Span} S$. The sum is 1, so the point is also in $\operatorname{aff} S$.

Exercise 8.1.9: Represent a translated line by two points

Write the free scalar as t to avoid confusing it with a vector coordinate. Choose

$$v_1 = p = \begin{bmatrix} -3 \\ 0 \end{bmatrix}, \qquad v_2 = p + u = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Then $p + tu = (1 - t)v_1 + tv_2$, whose two weights sum to 1. Conversely any affine combination of v_1, v_2 has precisely this form. Thus the entire solution set is $\operatorname{aff}\{v_1, v_2\}$, as required.

Exercise 8.1.10: Represent a translated line by two points

Write the free scalar as t to avoid confusing it with a vector coordinate. Choose

$$v_1 = p = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}, \qquad v_2 = p + u = \begin{bmatrix} 6 \\ -2 \\ 2 \end{bmatrix}.$$

Then $p + tu = (1 - t)v_1 + tv_2$, whose two weights sum to 1. Conversely any affine combination of v_1, v_2 has precisely this form. Thus the entire solution set is $\operatorname{aff}\{v_1, v_2\}$, as required.

Exercise 8.1.11: Affine closure, flats, and hull identities

(a) **True.** This is the definition of the affine hull.

(b) **False.** For $S = \{e_1, e_2\} \subset \mathbb{R}^2$, zero is a linear combination but not an affine combination: its unique linear coefficients are $(0, 0)$, whose sum is 0, not 1.

(c) **True.** The hull is $v_1 + \operatorname{Span}\{v_2 - v_1\}$, a one-dimensional flat when $v_2 \neq v_1$.

(d) **False.** A flat is a translated subspace and need not contain zero. The line $\{(x, 1) : x \in \mathbb{R}\}$ is a flat but not a subspace.

(e) **True.** A plane in $\mathbb{R}^3$ has dimension $2 = 3 - 1$, which is exactly the hyperplane dimension.

Exercise 8.1.12: Affine closure, flats, and hull identities

(a) **False.** The only affine combinations of one point x use total coefficient 1, so $\operatorname{aff}\{x\} = \{x\}$.

(b) **True.** If a set contains its affine hull, it contains every affine combination of its points, which is the closure condition defining an affine set. Conversely an affine set contains all those combinations. Every set is contained in its own affine hull, so containment here is actually equality.

(c) **True.** A one-dimensional flat is a line.

(d) **False in general.** A hyperplane in $\mathbb{R}^n$ has dimension $n-1$. A two-dimensional flat is a hyperplane in $\mathbb{R}^3$, but not in $\mathbb{R}^4$.

(e) **True.** Write the flat as $p + W$ with W a subspace. If $0 = p + w$ for some $w \in W$, then $p = -w \in W$, so $p + W = W$ is a subspace.

Exercise 8.1.13: Affine closure, flats, and hull identities

Suppose $c_2(v_2 - v_1) + c_3(v_3 - v_1) = 0$. Regrouping gives $(-c_2 - c_3)v_1 + c_2v_2 + c_3v_3 = 0$. Since v_1, v_2, v_3 form a basis, all three coefficients are zero, in particular $c_2 = c_3 = 0$. Thus the two differences are independent. Their span therefore has dimension 2 and is a plane through zero in $\mathbb{R}^3$.

Exercise 8.1.14: Affine closure, flats, and hull identities

The affine-hull identity gives

$$\operatorname{aff}\{v_1, v_2, v_3\} = v_1 + \operatorname{Span}\{v_2 - v_1, v_3 - v_1\}.$$

By Exercise 13 the difference span is two-dimensional, so this is a plane. It contains v_1, $v_2 = v_1 + (v_2 - v_1)$, and $v_3 = v_1 + (v_3 - v_1)$. Conversely any plane containing those three points must contain every affine combination of them, so this is precisely the plane through them. The independence of the differences rules out a line instead.

Exercise 8.1.15: Affine closure, flats, and hull identities

Take solutions $x_1, \dots, x_k$ and coefficients c_j with $\sum_j c_j = 1$. Matrix linearity gives

$$A\left(\sum_j c_j x_j\right) = \sum_j c_j A x_j = \sum_j c_j b = \left(\sum_j c_j\right) b = b.$$

Thus every affine combination is another solution. This proves the solution set is affine. If the system is inconsistent, its solution set is empty and the closure assertion holds vacuously; if it has a solution p, the more concrete form is $p + \operatorname{Nul} A$.

Exercise 8.1.16: Affine closure, flats, and hull identities

For points x_j satisfying $x_j \cdot v = k$ and coefficients summing to 1,

$$\left(\sum_j c_j x_j\right) \cdot v = \sum_j c_j (x_j \cdot v) = \sum_j c_j k = k.$$

Hence the set is closed under affine combinations. If $v = 0$, the set is all of $\mathbb{R}^n$ when $k = 0$, and empty otherwise; both satisfy that closure convention.

Exercise 8.1.17: Affine closure, flats, and hull identities

Take $S = \{(0, 0, 5), (1, 0, 5), (0, 1, 5)\}$. Their differences from the first point are e_1, e_2, which span all horizontal directions. Thus

$$\operatorname{aff} S = (0, 0, 5) + \operatorname{Span}\{e_1, e_2\} = \{(x_1, x_2, 5) : x_1, x_2 \in \mathbb{R}\}.$$

This is exactly the required plane, not merely a subset of it.

Exercise 8.1.18: Affine closure, flats, and hull identities

Choose $p = (6, 0, 0)$, $u = (-1, 2, 0)$, $v = (3, 0, 2)$, and the four distinct points

$$S = \{p, p + u, p + v, p + u + v\} = \{(6, 0, 0), (5, 2, 0), (9, 0, 2), (8, 2, 2)\}.$$

Each satisfies $2x_1 + x_2 - 3x_3 = 12$. The two directions u, v are independent, so the affine hull is the plane $p + \operatorname{Span}\{u, v\}$. Conversely, a point on the required plane has the expression

$$p + \frac{x_2}{2}u + \frac{x_3}{2}v = \left(6 - \frac{x_2}{2} + \frac{3x_3}{2}, x_2, x_3\right),$$

and the first coordinate equals x_1 by its plane equation. Thus no points of the desired plane are missing from the hull.

Exercise 8.1.19: Affine closure, flats, and hull identities

Take finitely many points $y_j = f(x_j)$ in $f(S)$ with $x_j \in S$, and coefficients summing to 1. Since S is affine, $x = \sum_j c_j x_j$ is in S. Linearity gives $\sum_j c_j y_j = f(x)$, which is in $f(S)$. Thus $f(S)$ is closed under affine combinations and is affine. If S is empty, its image is empty and the same closure statement is vacuous.

Exercise 8.1.20: Affine closure, flats, and hull identities

For points $x_j \in S$ we know $f(x_j) \in T$. If $\sum_j c_j = 1$, linearity gives $f(\sum_j c_j x_j) = \sum_j c_j f(x_j)$. Because T is affine, this image belongs to T, hence the combined input belongs to S. This verifies affine closure of the preimage, including the empty-preimage case.

Exercise 8.1.21: Affine closure, flats, and hull identities

Every point of aff A is an affine combination of finitely many points of A. They also lie in B because $A \subseteq B$. Since B is affine, the combination lies in B. Therefore aff $A \subseteq B$.

Exercise 8.1.22: Affine closure, flats, and hull identities

The affine set aff B contains B and hence contains A. Applying Exercise 21 with that affine set gives aff $A \subseteq$ aff B.

Exercise 8.1.23: Affine closure, flats, and hull identities

Both $A \subseteq A \cup B$ and $B \subseteq A \cup B$. Exercise 22 therefore gives $\operatorname{aff} A \subseteq \operatorname{aff}(A \cup B)$ and $\operatorname{aff} B \subseteq \operatorname{aff}(A \cup B)$. Taking their union proves the requested inclusion.

Exercise 8.1.24: Affine closure, flats, and hull identities

Take $A = \{(0,0)\}$ and $B = \{(1,0)\}$. Each singleton is its own affine hull, so their hull union is just two points. In contrast, $\operatorname{aff}(A \cup B) = \{(t,0) : t \in \mathbb{R}\}$ is an entire line. For instance $(1/2, 0)$ lies in the latter but not the former, proving strict inclusion.

Exercise 8.1.25: Affine closure, flats, and hull identities

Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, monotonicity of affine hull gives $\operatorname{aff}(A \cap B) \subseteq \operatorname{aff} A$ and $\operatorname{aff}(A \cap B) \subseteq \operatorname{aff} B$. Membership in both is exactly membership in their intersection, yielding the asserted inclusion.

Exercise 8.1.26: Affine closure, flats, and hull identities

Let $A = \{(0,0), (1,0), (0,1)\}$ and $B = \{(0,0), (2,0), (0,2)\}$. Their intersection is the singleton $\{(0,0)\}$, whose affine hull is itself. Each original set contains three noncollinear points and hence has affine hull $\mathbb{R}^2$. Thus $\operatorname{aff} A \cap \operatorname{aff} B = \mathbb{R}^2$, strictly larger than $\operatorname{aff}(A \cap B) = \{(0,0)\}$. This example does not rely on a convention for the affine hull of the empty set.

8.2 Affine independence

An indexed set of points is affinely dependent if there are coefficients, not all zero, whose sum is 0 and whose weighted point sum is 0. It is affinely independent otherwise. Appending a final coordinate 1 converts this definition into the ordinary null-space test for linear dependence.

An affinely independent set gives *unique* affine coefficients, called barycentric coordinates, for every point in its affine hull. These coefficients sum to 1, but may be negative outside the convex region between the points. For a triangle, the coordinate belonging to a vertex is zero on the line through the other two vertices, positive on the vertex's side of that line, and negative on the opposite side. This geometric sign rule is particularly useful in Exercises 15 and 16.

Practice problem 8.2.P1: A fast collinearity test

For points v_1, v_2, v_3, compute the two differences $u = v_2 - v_1$ and $w = v_3 - v_1$. They are collinear exactly when u, w are linearly dependent, which can be checked by a two-column row reduction. In $\mathbb{R}^2$ one can simply test $\det[u\ w] = 0$. Repeated points also satisfy the test. If the differences are independent, the three points are not on one line and are affinely independent.

Practice problem 8.2.P2: A nontrivial relation with coefficient sum zero

Append a final 1 to each point and form

$$H = \begin{bmatrix} 4 & 1 & 5 & 1 \\ 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & \frac{5}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} 4 \\ -5 \\ -3 \\ 4 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(4)v_1 + (-5)v_2 + (-3)v_3 + (4)v_4 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.1: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 3 & 0 & 2 \\ -3 & 6 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} -2 \\ -1 \\ 3 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(-2)v_1 + (-1)v_2 + (3)v_3 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.2: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 2 & 5 & -3 \\ 1 & 4 & -2 \\ 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column is a pivot column, so $Hc = 0$ has only the zero solution. Hence the points are affinely independent and no affine dependence relation exists.

Exercise 8.2.3: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 1 & -2 & 2 & 0 \\ 2 & -4 & -1 & 15 \\ -1 & 8 & 11 & -9 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every column is a pivot column, so $Hc = 0$ has only the zero solution. Hence the points are affinely independent and no affine dependence relation exists.

Exercise 8.2.4: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} -2 & 0 & 1 & -2 \\ 5 & -3 & -2 & 7 \\ 3 & 7 & -6 & -3 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & \frac{6}{5} \\ 0 & 1 & 0 & -\frac{3}{5} \\ 0 & 0 & 1 & \frac{2}{5} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} -6 \\ 3 \\ -2 \\ 5 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(-6)v_1 + (3)v_2 + (-2)v_3 + (5)v_4 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.5: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 5 & 5 \\ -2 & 1 & 1 & -3 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & \frac{4}{3} \\ 0 & 1 & 0 & -\frac{5}{3} \\ 0 & 0 & 1 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} -4 \\ 5 \\ -4 \\ 3 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(-4)v_1 + (5)v_2 + (-4)v_3 + (3)v_4 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.6: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 3 & -1 & 5 & 5 \\ 1 & -2 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every column is a pivot column, so $Hc = 0$ has only the zero solution. Hence the points are affinely independent and no affine dependence relation exists.

Exercise 8.2.7: Unique barycentric coordinates in a higher-dimensional affine hull

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 2 \\ 2 & 0 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 5 \\ 4 \\ -2 \\ 2 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} -2 \\ 4 \\ -1 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 5 \\ 4 \\ -2 \\ 2 \end{bmatrix}$. Thus the requested affine combination exists. Since the points are affinely independent, the coefficient column is unique and is precisely the requested barycentric coordinate column.

Exercise 8.2.8: Unique barycentric coordinates in a higher-dimensional affine hull

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 4 \\ -2 & 0 & -6 \\ 1 & 2 & 5 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} -1 \\ 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} -1 \\ 1 \\ -4 \\ 0 \end{bmatrix}$. Thus the requested affine combination exists. Since the points are affinely independent, the coefficient column is unique and is precisely the requested barycentric coordinate column.

Exercise 8.2.9: Affine independence and barycentric meaning

(a) **True.** The differences are based at v_2, not v_1, but the same criterion works for any base point. If $\sum_{j\neq 2} c_j(v_j - v_2) = 0$ is nontrivial, give v_2 coefficient $-\sum_{j\neq 2} c_j$. The resulting nontrivial point relation has coefficient sum zero.

(b) **False.** Independent homogeneous columns mean the original points are affinely *independent*, not dependent.

(c) **False.** A dependence requires coefficient sum 0. The distinct points $v_1 = (1, 0)$, $v_2 = (2, 0)$ are affinely independent, but $2v_1 - v_2 = 0$ has coefficient sum 1, satisfying the proposed incorrect test.

(d) **False.** A negative coordinate can occur for a point in the affine hull, but outside the convex hull. For the two points $(0, 0), (1, 0)$, the point $(2, 0)$ has barycentric coordinates $(-1, 2)$ and is still in their affine line.

(e) **True for the stated triangle.** A point in a closed triangular region is a convex combination of the vertices, so its barycentric coordinates are nonnegative; an edge or vertex can give zeros. The ray is merely a way to locate that point and does not change this condition.

Exercise 8.2.10: Affine independence and barycentric meaning

(a) **False.** A nontrivial affine dependence gives a nontrivial linear dependence of the homogeneous forms with the very same coefficients. They cannot be independent.

(b) **True.** Three independent differences based at v_1 imply affine independence of the four original points by the difference criterion.

(c) **False without affine independence.** For $S = \{(1, 0), (2, 0), (3, 0)\}$, the middle point is both itself and $\frac{1}{2}(1, 0) + \frac{1}{2}(3, 0)$. These are distinct affine representations.

(d) **True as an affine color rule.** Use the three barycentric weights for each color component at the vertices. Inside the triangle this interpolates the colors; outside, negative weights give affine extrapolation and may leave a physical color range.

(e) **True.** A point on an edge has coefficient zero for the opposite vertex. Hence the three coefficients are not all strictly positive, including the vertex endpoints of the edge.

Exercise 8.2.11: Affine independence and barycentric meaning

Append a 1 to every point of $\mathbb{R}^3$, producing vectors in $\mathbb{R}^4$. Five or more vectors in a four-dimensional vector space are linearly dependent. Therefore some nontrivial coefficient column has both weighted point sum zero and coefficient sum zero. That is an affine dependence of the original indexed points.

Exercise 8.2.12: Affine independence and barycentric meaning

The homogeneous forms lie in $\mathbb{R}^{n+1}$. If $p \geq n + 2$, there are more columns than the dimension, so their homogeneous system has a free variable and a nonzero solution c. Its first n equations give $\sum_j c_j v_j = 0$, and its last gives $\sum_j c_j = 0$. This is precisely the affine-dependence definition.

Exercise 8.2.13: Affine independence and barycentric meaning

Suppose $c_1v_1 + c_2v_2 = 0$, $c_1 + c_2 = 0$, and the coefficient pair is nonzero. Then $c_2 = -c_1$, with $c_1 \neq 0$. The point equation becomes $c_1(v_1 - v_2) = 0$, so division by c_1 gives $v_1 = v_2$. Conversely, if $v_1 = v_2$, take $(c_1, c_2) = (1, -1)$. Their sum is zero and their weighted point sum is zero, so the indexed pair is affinely dependent. Both directions use only the definition.

Exercise 8.2.14: Affine independence and barycentric meaning

For $S_1 = \{(1,0), (2,0), (3,0)\}$, the relation $v_1 - 2v_2 + v_3 = 0$ has nonzero coefficients summing to zero, so this set is both affinely and linearly dependent.

For $S_2 = \{(1,0), (2,0)\}$, the relation $2v_1 - v_2 = 0$ proves linear dependence, but the two points are distinct and hence affinely independent by Exercise 13. All points in both examples are nonzero, as requested. Linear dependence does not force its coefficients to sum to zero.

Exercise 8.2.15: Barycentric coordinates and signs in the printed figure

(a) Check affine independence.

The homogeneous matrix has determinant $-8 \neq 0$, so its three columns are independent and the original points are affinely independent.

$$H = \begin{bmatrix} -1 & 0 & 2 \\ 2 & 4 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) Solve for all five ordered coordinate triples.

The inverse mapping, obtained by row reduction, is

$$H^{-1} = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{4} & 1 \\ \frac{1}{4} & \frac{3}{8} & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{8} & \frac{1}{2} \end{bmatrix}.$$

Thus multiply $H^{-1}(x, y, 1)^T$ for each given point. The exact results are

point	r	s	t
p_1	$-\frac{3}{4}$	$\frac{9}{8}$	$\frac{5}{8}$
p_2	0	$\frac{1}{2}$	$\frac{1}{2}$
p_3	$\frac{7}{4}$	$-\frac{5}{8}$	$-\frac{1}{8}$
p_4	$\frac{3}{4}$	$-\frac{5}{8}$	$\frac{7}{8}$
p_5	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{5}{8}$

Each row sums to 1 and reconstructs its point, supplying independent checks on the arithmetic.

(c) Use the positions in Figure 8 rather than inventing coordinates for its unlabeled points.

A coordinate is zero on the opposite edge line, positive on its own vertex's side, and negative on the other side. Reading the three side tests in the figure gives

$$p_6 : (-,-,+), \qquad p_7 : (0,+,-), \qquad p_8 : (+,+,-).$$

The zero for p_7 matters: the dot is on the *extended line* through v_2, v_3, beyond v_2, not in an adjacent open region. Its second weight is positive and third negative along that extension. Point p_6 lies across both edge lines opposite v_1 and v_2; point p_8 is opposite v_3 but on the other two vertices' positive sides.

Exercise 8.2.16: A plotted triangle with extended sides and boundary points

(a) Verify independence of the triangle vertices.

The homogeneous matrix

$$H = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

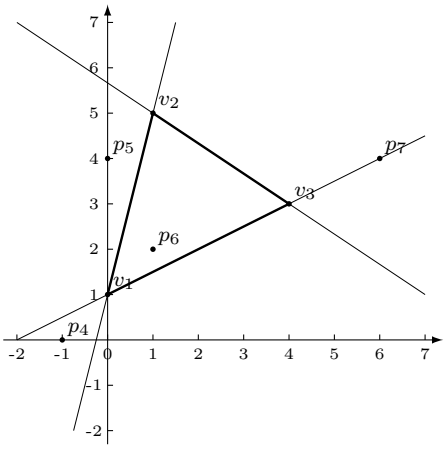
has determinant $-14 \neq 0$, proving affine independence.

(b) Compute the first three coordinate triples.

Solve $Hc = (p_x, p_y, 1)^T$. Row reduction gives

point	r	s	t
p_1	$-\frac{2}{7}$	$\frac{5}{7}$	$\frac{4}{7}$
p_2	$\frac{2}{7}$	$-\frac{5}{7}$	$\frac{10}{7}$
p_3	$\frac{2}{7}$	$\frac{2}{7}$	$\frac{3}{7}$

(c) Plot the remaining points and test which side of each line they occupy.



The thin lines extend the three sides; the bold triangle joins the actual vertices. For a weight belonging to v_j, compare the point with v_j across the line through the other two vertices. These side tests give

$$p_4 : (+,-,-), \qquad p_5 : (+,+,-), \qquad p_6 : (+,+,+), \qquad p_7 : (-,0,+).$$

Here p_6 lies inside the triangle, while $p_7 = (6, 4)$ lies exactly on the extension of the side through $v_1 = (0, 1)$ and $v_3 = (4, 3)$, so its v_2 coordinate is zero. This boundary check prevents treating every exterior point as having three nonzero coordinates. No numerical barycentric weights are needed for these sign conclusions.

Exercise 8.2.17: Barycentric proofs and geometric consequences

What we must show. Prove both existence and uniqueness in Theorem 6. Existence is the definition of $p \in \operatorname{aff} S$: some coefficients c_j sum to 1 and satisfy $p = \sum_j c_j v_j$. For uniqueness suppose also $p = \sum_j d_j v_j$ with $\sum_j d_j = 1$. Subtract the two point equations and the two coefficient sums:

$$\sum_j (c_j - d_j) v_j = 0, \qquad \sum_j (c_j - d_j) = 0.$$

Affine independence says the only coefficient column satisfying both zero equations is zero. Thus $c_j - d_j = 0$ for every j, so all weights agree. This is uniqueness of the entire ordered coordinate column, not merely uniqueness of the represented point.

Exercise 8.2.18: Barycentric proofs and geometric consequences

The ordered vertices are $ae_1, be_2, ce_3, 0$, with $a, b, c > 0$ in the stated standard position. If $p = (p_1, p_2, p_3)$ has barycentric coordinates (r, s, t, u), coordinate comparison gives $ar = p_1$, $bs = p_2$, and $ct = p_3$. The sum condition then determines the last coordinate. Hence

$$(r, s, t, u) = \left(\frac{p_1}{a}, \frac{p_2}{b}, \frac{p_3}{c}, 1 - \frac{p_1}{a} - \frac{p_2}{b} - \frac{p_3}{c} \right).$$

These formulas work for every point of $\mathbb{R}^3$, not only interior points. The point belongs to the closed tetrahedron exactly when all four coordinates are nonnegative. The three edge vectors ae_1, be_2, ce_3 are independent, so the coordinate column is unique.

Exercise 8.2.19: Barycentric proofs and geometric consequences

Choose a nontrivial affine-dependence coefficient triple c_1, c_2, c_3. By assumption $\sum_j c_j p_j = 0$ and $\sum_j c_j = 0$. Linearity gives

$$\sum_j c_j f(p_j) = f\left(\sum_j c_j p_j \right) = f(0) = 0.$$

The coefficients have not changed and are still not all zero, so the indexed image points satisfy an affine dependence. The map may identify some points; this does not destroy the displayed indexed relation.

Exercise 8.2.20: Barycentric proofs and geometric consequences

Suppose the translated points have a coefficient relation with $\sum_j c_j = 0$ and $\sum_j c_j (p_j + q) = 0$. Expanding yields

$$0 = \sum_j c_j p_j + \left(\sum_j c_j \right) q = \sum_j c_j p_j.$$

Affine independence of the original points now forces every $c_j = 0$. Therefore no nontrivial affine dependence exists among the translated points. Translation changes location but not affine independence.

Exercise 8.2.21: Barycentric proofs and geometric consequences

The homogeneous-column determinant is

$$D = \det \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ 1 & 1 & 1 \end{bmatrix}.$$

Subtract its first column from columns 2 and 3, then expand across the final row. This gives $D = \det[b-a\ c-a]$. The parallelogram on those two edges has area $|D|$, and the triangle has half that area. The stipulated vertex order makes $D > 0$, so the triangle area is $D/2$ without absolute-value bars. For a different order one would need $|D|/2$.

Exercise 8.2.22: Barycentric proofs and geometric consequences

A point on the line is $p = (1-t)a + tb$ for some real t. Because these two weights sum to 1, its homogeneous form satisfies $\widetilde{p} = (1-t)\widetilde{a} + t\widetilde{b}$. Therefore the three homogeneous columns are linearly dependent, and $\det[\widetilde{a}\ \widetilde{b}\ \widetilde{p}] = 0$. This also follows by expanding linearly in the third column, where both resulting determinants have repeated columns.

Exercise 8.2.23: Barycentric proofs and geometric consequences

Write $\widetilde{p} = r\widetilde{a} + s\widetilde{b} + t\widetilde{c}$ and let $D = \det[\widetilde{a}\ \widetilde{b}\ \widetilde{c}] > 0$. Expand a determinant linearly in its replaced column. Repeated-column terms vanish, so

$$\det[\widetilde{p}\ \widetilde{b}\ \widetilde{c}] = rD, \quad \det[\widetilde{a}\ \widetilde{p}\ \widetilde{c}] = sD, \quad \det[\widetilde{a}\ \widetilde{b}\ \widetilde{p}] = tD.$$

Since p is interior, it lies on the same side of each opposite edge as the replaced vertex; each displayed subtriangle has the same positive orientation as the original triangle. Thus its ordinary area is one half of its displayed determinant. Divide those area expressions by $D/2$, the area of the full triangle, to obtain

$$r = \frac{\text{area}(\triangle pbc)}{\text{area}(\triangle abc)}, \qquad s = \frac{\text{area}(\triangle apc)}{\text{area}(\triangle abc)},$$

$$t = \frac{\text{area}(\triangle abp)}{\text{area}(\triangle abc)}.$$

These are positive ratios for an interior point. Signed determinants, rather than unsigned areas alone, are needed if extending the formulas outside the triangle.

Exercise 8.2.24: Barycentric proofs and geometric consequences

Because q lies on the segment bc, it has the form $q = (1-t)b + tc$ for some $0 \le t \le 1$. The given line point therefore has affine representation

$$p = xa + (1-x)(1-t)b + (1-x)tc.$$

Its three weights sum to 1. Appending final coordinates and expanding the determinant linearly in its first column gives

$$\det[\widetilde{p}\ \widetilde{b}\ \widetilde{c}] = x \det[\widetilde{a}\ \widetilde{b}\ \widetilde{c}],$$

because both terms involving $\widetilde{b}$ or $\widetilde{c}$ in the first column have repeated columns. The original triangle is nondegenerate, so its vertices are affinely independent. Uniqueness of barycentric coordinates makes x exactly the first coordinate of p.

Also $p - q = x(a - q)$, so for $0 \le x \le 1$, $\|p - q\|/\|a - q\| = x$. For other real x, it is an *oriented* relative position; the ordinary distance ratio is $|x|$. A vertex color and the color at q combine as $xC_a + (1 - x)C_q$, varying linearly and continuously as the point moves. This is the stated smooth interpolation along the segment.

8.3 Convex combinations

A linear combination allows arbitrary coefficients. An affine combination requires that their sum be 1. A convex combination requires both sum 1 and *nonnegative* coefficients. Thus $\operatorname{conv} S \subseteq \operatorname{aff} S \subseteq \operatorname{Span} S$. For an affinely independent list, its affine coordinates are unique: a negative coordinate rules out convex membership, all positive coordinates put the point in the relative interior of its simplex, and zero coordinates locate it on a face. A convex hull consists of finite convex combinations; it need not include limit points not obtained by those combinations.

Practice problem 8.3.P1: Test a triangle in three-dimensional space

Use the supplied vertices $v_1 = (6, 2, 2)$, $v_2 = (7, 1, 5)$, $v_3 = (-2, 4, -1)$. To find affine coordinates solve $p - v_1 = a(v_2 - v_1) + b(v_3 - v_1)$, then use coefficients $(1 - a - b, a, b)$. For $p_1 = (1, 3, 1)$ the solution is $a = 1/3, b = 2/3$; checking gives $p_1 = 0v_1 + (1/3)v_2 + (2/3)v_3$. All coefficients are nonnegative, so $p_1 \in \operatorname{conv} S$. For $p_2 = (3, 2, 1)$ the first two coordinate equations give $a = 1, b = 1/2$, but the third would require $3a - 3b = -1$, whereas it equals $3/2$. Thus p_2 is not even in the affine hull and cannot be in the convex hull.

Practice problem 8.3.P2: The hull of the positive reciprocal curve

What we must show. Describe the hull of $S = \{(x, 1/x) : x > 0\}$ exactly. No convex combination has $x \le 0$. Moreover convexity of $1/x$ on positive inputs gives $\sum_i \lambda_i/x_i \ge 1/(\sum_i \lambda_i x_i)$. For example this follows directly from Cauchy–Schwarz applied to $\sqrt{\lambda_i x_i}$ and $\sqrt{\lambda_i/x_i}$. Hence the hull lies in $\{(x, y) : x > 0, y \ge 1/x\}$. Conversely, for a point (x, y) in that set draw the line of slope -1 through it, $u + v = x + y$. Its intersections with $uv = 1$ have positive x-coordinates given by roots of $u^2 - (x + y)u + 1 = 0$. The polynomial at $u = x$ is $1 - xy \le 0$, so x lies between the two roots. Thus (x, y) lies on the line segment joining these two curve points, or is itself the single tangency point. It is a convex combination of them. Therefore the hull is exactly the region on and above the curve.

Exercise 8.3.1: A triangle with a missing upper edge

Every convex combination of vertical-segment points and $(2, 0)$ can be written $(1 - \lambda)(0, a) + \lambda(2, 0)$ with $0 \le \lambda \le 1$ and $0 \le a < 1$. (Combine all the vertical points into their weighted average first.) Thus the hull is

$$\{(x, y) : 0 \le x < 2,\ 0 \le y < 1 - x/2\} \ \cup\ \{(2, 0)\}.$$

Conversely any point in the first set uses $\lambda = x/2$ and $a = y/(1 - x/2)$, which satisfy the required inequalities. The lower edge and right vertex are included, but the sloping upper edge and the point $(0, 1)$ are excluded. Replacing the strict inequality by a weak one would add points not actually in the convex hull.

Exercise 8.3.2: Three convex hulls of graphs

(a) The reciprocal graph starting at $x = 1/2$.

The exact hull is

$$\{(1/2, 2)\} \ \cup\ \{(x, y) : x > 1/2,\ 1/x \le y < 2\}.$$

Every combination has $x \geq 1/2$, lies on or above the reciprocal curve by convexity, and has $y \leq 2$. Equality $y = 2$ forces every positive-weight input to be the endpoint $(1/2, 2)$, so it occurs only there. For the converse fix $x > 1/2$ and $1/x \leq y < 2$. Put $b = (2x-1)/(2-y)$. The lower bound on y gives $b \geq x$. The chord joining $(1/2, 2)$ and $(b, 1/b)$ has height $2 - (2x-1)/b = y$ at horizontal coordinate x. Because $1/2 < x \leq b$, the point lies on that segment and is in the hull.

(b) The sine graph on the whole real line.

The hull is the closed strip $\{(x, y) : -1 \leq y \leq 1\}$. The range of sine bounds every combination within the strip. Conversely its maximum points $(\pi/2 + 2j\pi, 1)$ extend in both horizontal directions. For any fixed real x, two such points bracket x, so their segment contains $(x, 1)$. The minimum points similarly give $(x, -1)$. The segment between these two hull points contains every (x, y) in the strip.

(c) The square-root graph.

The hull is

$$\{(0,0)\} \ \cup \ \{(x,y) : x > 0, \ 0 < y \leq \sqrt{x}\}.$$

Concavity of square root bounds the height of a convex combination by $\sqrt{x}$; nonnegativity is automatic. If $x > 0$, some positive-weight input has positive x and therefore positive height, so height cannot be zero. Conversely, for $x > 0$ and $0 < y \leq \sqrt{x}$, take $a = (x/y)^2$ and $\lambda = y^2/x \in (0, 1]$. Then $\lambda(a, \sqrt{a}) + (1-\lambda)(0,0) = (x, y)$. This proves the formula and explains why the positive horizontal axis is not included.

Exercise 8.3.3: Use the unique coordinates from Section 8.1, Exercise 5

The three orthogonal generators are $b_1 = (2, 1, 1)$, $b_2 = (1, 0, -2)$, $b_3 = (2, -5, 1)$. Their squared lengths are $6, 5, 30$, so the unique linear coefficients of p are $c_j = (p \cdot b_j)/(b_j \cdot b_j)$. For the three supplied points the results are

p	(c_1, c_2, c_3)	sum
$(3, 8, 4)$	$(3, -1, -1)$	1
$(6, -3, 3)$	$(2, 0, 1)$	3
$(0, -1, -5)$	$(-1, 2, 0)$	1

The first and third have negative coordinates; the second does not even have coefficient sum one. Since the linear representation in a basis is unique, no alternative nonnegative sum-one representation exists.

Conclusion. None of the three points is in $\operatorname{conv} S$.

Exercise 8.3.4: A second list with the same orthogonal generators

The same dot-product calculation gives

p	(c_1, c_2, c_3)	sum
$(0, -19, -5)$	$(-4, 2, 3)$	1
$(3/2, -13/10, -1/2)$	$(1/5, 1/2, 3/10)$	1
$(5, -4, 0)$	$(1, 1, 1)$	3

Only the middle row has all coefficients nonnegative and sum one. Thus $p_2 \in \operatorname{conv} S$, while p_1, p_3 are not.

Exercise 8.3.5: Convex coordinates in a tetrahedron

Append a final coordinate 1 to each of the four supplied v-vectors. Solving the resulting 4×4 systems for the two points gives

$$[p_1]_S = (-1/6, 1/3, 2/3, 1/6), \qquad [p_2]_S = (1/3, 1/3, 1/6, 1/6).$$

Here coordinates refer to affine/barycentric coordinates. The appended matrix is invertible, so these representations are unique. For instance $(1/3)v_1 + (1/3)v_2 + (1/6)v_3 + (1/6)v_4 = (0, -2, 2) = p_2$. The first representation has a negative coefficient and therefore is not convex; the second has all coefficients positive and sum one.

Conclusion. p_1 is outside the convex hull; p_2 is in it.

Exercise 8.3.6: Distinguish the linear, affine, and convex hulls

The supplied orthogonal vectors are $v_1 = (2, 0, -1, 2)$, $v_2 = (0, -2, 2, 1)$, $v_3 = (-2, 1, 0, 2)$; each has squared length 9. For a proposed p, first compute $c_j = (p \cdot v_j)/9$ and check whether $\sum c_j v_j = p$. Only after this check do sum-one and sign tests apply.

(a) For $p_1 = (-1, 2, -3/2, 5/2)$, the coefficients are $(1/2, -1/2, 1)$ and reconstruct p_1. They sum to one but include a negative coefficient. Thus p_1 is in the span and affine hull, but not the convex hull.

(b) For $p_2 = (-1/2, 0, 1/4, 7/4)$, the coefficients are $(1/4, 1/4, 1/2)$. They reconstruct p_2, sum to one, and are nonnegative. Thus p_2 is in all three hulls.

(c) For $p_3 = (6, -4, 1, -1)$, the coefficients are $(1, 1, -2)$ and reconstruct p_3, but their sum is zero. Hence p_3 is in the span only, not the affine or convex hull.

(d) For $p_4 = (-1, -2, 0, 4)$, the dot-product coefficients are $(2/3, 8/9, 8/9)$, but the reconstruction error is $p_4 - \sum c_j v_j = (-5/9, -10/9, -10/9, 0) \neq 0$. Thus p_4 is not in the span and consequently in neither smaller hull.

Exercise 8.3.7: Barycentric coordinates locate four planar points

(a) Solve the affine-coordinate systems.

With $T = \{(-1, 0), (2, 3), (4, 1)\}$, solve

$$\begin{pmatrix} -1 & 2 & 4 \\ 0 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix} c = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

The four answers, in the listed order, are

$$(1/3, 1/6, 1/2), \quad (0, 1/2, 1/2), \quad (1/2, -1/4, 3/4), \quad (1/2, 3/4, -1/4).$$

(b) Use signs and zero entries.

p_1 has all positive entries and is inside the triangle. p_2 has one zero and the other entries positive, so it is on the edge joining vertices two and three. p_3, p_4 have negative entries and are outside.

Exercise 8.3.8: Another barycentric-coordinate table

For $T = \{(2,0),(0,5),(-1,1)\}$, use the appended-coordinate system

$$\begin{pmatrix} 2 & 0 & -1 \\ 0 & 5 & 1 \\ 1 & 1 & 1 \end{pmatrix} c = (x, y, 1)^T.$$

For p_1, p_2, p_3, p_4 its solutions are respectively

$$(12/13, 3/13, -2/13), \quad (8/13, 2/13, 3/13), \quad (2/3, 0, 1/3), \quad (9/13, -1/13, 5/13).$$

Thus p_1 is outside, p_2 inside, p_3 on the edge joining vertices one and three, and p_4 outside. The p_3 supplied in the exercise is $(1, 1/3)$, not $(1, 1)$; preserving this fraction is essential.

Exercise 8.3.9: Read positions from tetrahedral coordinates

A point is inside precisely when all four barycentric coordinates are positive. Nonnegative coordinates with a zero put it on a surface; two zeros put it on an edge. A negative coordinate puts it outside. Therefore $p_1 = (2, 0, 0, -1)$ is outside; $p_2 = (0, 1/2, 1/4, 1/4)$ is on the face opposite vertex one, but not an edge; $p_3 = (1/2, 0, 3/2, -1)$ is outside; $p_4 = (1/3, 1/4, 1/4, 1/6)$ is inside; and $p_5 = (1/3, 0, 2/3, 0)$ is on the edge joining vertices one and three.

Exercise 8.3.10: A second tetrahedral list

Using the same criteria, $q_1 = (1/8, 1/4, 1/8, 1/2)$ is inside; $q_2 = (3/4, -1/4, 0, 1/2)$ is outside; $q_3 = (0, 3/4, 1/4, 0)$ is on the edge joining vertices two and three; $q_4 = (0, -2, 0, 3)$ is outside; and $q_5 = (1/3, 1/3, 1/3, 0)$ is on the face opposite vertex four, not an edge.

Exercise 8.3.11: Three statements about convexity

(a) **False.** Sum one makes a combination affine, but negative coefficients are still possible. Convexity also requires each coefficient nonnegative.

(b) **False.** A singleton $S = \{v\}$ has $\operatorname{conv} S = S$. More generally this equality holds for any convex S.

(c) **False.** Two singleton sets are convex but their union $\{(0,0),(2,0)\}$ misses the midpoint $(1,0)$.

Exercise 8.3.12: Three more statements about hulls

(a) **True.** Containing every segment between two of its points is exactly the definition of convexity.

(b) **True.** If two points belong to both convex sets, their whole joining segment belongs to both, hence to the intersection. The empty intersection is also convex under the usual vacuous definition.

(c) **False.** Carathéodory's theorem says *at most* six points in $\mathbb{R}^5$, not necessarily six distinct available points. A singleton S is an immediate counterexample to the stated requirement.

Exercise 8.3.13: A linear image preserves convexity

For $u = f(x), v = f(y)$ in $f(S)$ and $0 \le t \le 1$, linearity gives $(1-t)u + tv = f((1-t)x + ty)$. Convexity of S puts the input $(1-t)x + ty$ in S, so this output is in $f(S)$. Hence $f(S)$ is convex. Empty sets cause no exception.

Exercise 8.3.14: A linear preimage preserves convexity

For $x, y \in S$, the images $f(x), f(y)$ lie in T. Their combination $(1-t)f(x) + tf(y)$ lies in T for $0 \le t \le 1$ by convexity. This equals $f((1-t)x + ty)$ by linearity, so the input $(1-t)x + ty$ lies in S. Thus the whole segment is in S.

Exercise 8.3.15: Eliminate a vertex in two different ways

Direct coordinate addition checks both supplied identities: $p = (1/3)v_1 + (1/3)v_2 + (1/6)v_3 + (1/6)v_4 = (2, 1)$ and $v_1 - v_2 + v_3 - v_4 = 0$. Add s times the zero relation to the coefficients. They become

$$(1/3 + s,\ 1/3 - s,\ 1/6 + s,\ 1/6 - s).$$

Their sum stays one and their combination stays p. For $s = 1/6$ and $s = -1/6$ all entries are nonnegative and one is zero, giving respectively

$$p = \tfrac{1}{2}v_1 + \tfrac{1}{6}v_2 + \tfrac{1}{3}v_3, \qquad p = \tfrac{1}{6}v_1 + \tfrac{1}{2}v_2 + \tfrac{1}{3}v_4.$$

This is the coefficient-elimination step in the proof of Carathéodory's theorem.

Exercise 8.3.16: A second coefficient-elimination example

The printed instruction says "Repeat Exercise 9," but the supplied convex combination and zero relation clearly call for the procedure of Exercise 15. We carry out that intended task explicitly. The given coefficients are $(1, 72, 37, 11)/121$, whose sum is one; coordinate substitution gives $p = (1, 2)$. The relation $10v_1 - 6v_2 + 7v_3 - 11v_4 = 0$ also has coefficient sum zero. Add $s(10, -6, 7, -11)$ to the original coefficients. For $s = 1/121$, the last coefficient becomes zero and we obtain

$$p = \tfrac{1}{11}v_1 + \tfrac{6}{11}v_2 + \tfrac{4}{11}v_3.$$

For $s = -1/1210$, the first becomes zero and the result is

$$p = \tfrac{3}{5}v_2 + \tfrac{3}{10}v_3 + \tfrac{1}{10}v_4.$$

Each list is nonnegative, sums to one, and reconstructs p.

Exercise 8.3.17: A containing convex set contains the hull

If $A \subseteq B$ and B is convex, then any finite convex combination of points in A is a convex combination of points in B and therefore belongs to B. Every point of $\operatorname{conv} A$ is such a combination, so $\operatorname{conv} A \subseteq B$.

Exercise 8.3.18: Convex hull is monotone under inclusion

If $A \subseteq B$, every convex combination of points in A is also one of points in B. Therefore $\operatorname{conv} A \subseteq \operatorname{conv} B$. This also follows from the preceding exercise because $\operatorname{conv} B$ is convex and contains A.

Exercise 8.3.19: Union of hulls need not fill the combined hull

(a) Prove the inclusion.

Since $A \subseteq A \cup B$ and $B \subseteq A \cup B$, monotonicity gives both $\operatorname{conv} A \subseteq \operatorname{conv}(A \cup B)$ and the corresponding inclusion for B. Taking their union proves the claim.

(b) Show strict inclusion can occur.

Let $A = \{(0,0)\}$ and $B = \{(2,0)\}$. The union of their hulls has just two points, whereas the combined hull is the whole segment and includes $(1,0)$.

Exercise 8.3.20: Intersection of hulls can be larger

(a) Apply monotonicity twice.

Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, its hull is contained in each hull, hence in their intersection.

(b) Give an example with a new intersection point.

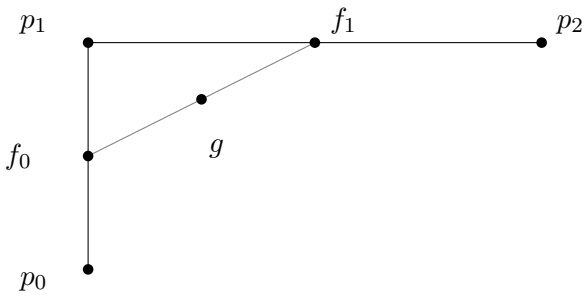
Take $A = \{(-1,0),(1,0)\}$ and $B = \{(0,-1),(0,1)\}$. The sets have empty intersection, so its hull is empty. But their two segments intersect at zero, showing that the claimed inclusion can be strict.

Exercise 8.3.21: The midpoint construction for a quadratic curve

At $t = 1/2$,

$$f_0 = \tfrac{1}{2}(p_0 + p_1), \quad f_1 = \tfrac{1}{2}(p_1 + p_2), \quad g = \tfrac{1}{2}(f_0 + f_1) = \tfrac{1}{4}p_0 + \tfrac{1}{2}p_1 + \tfrac{1}{4}p_2.$$

Thus first mark the midpoints of p_0p_1 and p_1p_2, join them, and mark the midpoint of that new segment. The original diagram below uses the same right-angle configuration as the source, with convenient schematic coordinates $p_0 = (0,0), p_1 = (0,2), p_2 = (4,2)$.

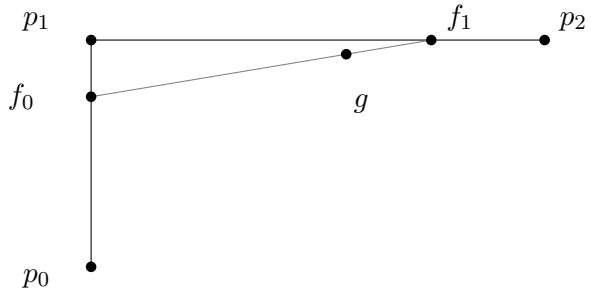


Exercise 8.3.22: A three-quarter construction

Now $f_0 = (1/4)p_0 + (3/4)p_1$ and $f_1 = (1/4)p_1 + (3/4)p_2$. Place each point three quarters of the way along its directed segment. Then place g three quarters of the way from f_0 to f_1:

$$g = \tfrac{1}{16}p_0 + \tfrac{3}{8}p_1 + \tfrac{9}{16}p_2.$$

The same schematic coordinates give $f_0 = (0, 3/2)$, $f_1 = (3, 2)$, and $g = (9/4, 15/8)$, as shown.



Exercise 8.3.23: Expand the quadratic Bézier formula

Substitute both definitions into $g = (1 - t)f_0 + tf_1$ and distribute:

$$g(t) = (1 - t)^2 p_0 + 2t(1 - t)p_1 + t^2 p_2.$$

For $0 \le t \le 1$, the three coefficients are nonnegative and their sum is $(1 - t)^2 + 2t(1 - t) + t^2 = ((1 - t) + t)^2 = 1$. Thus this formula is a convex combination of the control points and lies in their convex hull for the entire parameter interval.

Exercise 8.3.24: The cubic curve also stays inside its controls

Apply the preceding formula to the two quadratic curves and collect terms:

$$h(t) = (1 - t)^3 p_0 + 3t(1 - t)^2 p_1 + 3t^2(1 - t)p_2 + t^3 p_3.$$

The four coefficients are nonnegative on $[0, 1]$ and add to $((1 - t) + t)^3 = 1$ by the binomial theorem. Hence every point of the cubic curve is a convex combination of the four controls and lies in their convex hull. Equivalently, both quadratic points already lie in that convex hull, which is itself convex.

8.4 Hyperplanes

A linear functional has scalar output: $f : \mathbb{R}^n \to \mathbb{R}$. It can be written $f(x) = n^T x$ for a fixed vector n. When $n \neq 0$, the level set $[f : d] = \{x : n^T x = d\}$ is a hyperplane. Its two open sides have $f(x) < d$ and $f(x) > d$. To separate sets weakly, put them in opposite closed sides; to separate strictly, put them in opposite open sides. Here "open" and "closed" refer to the ambient $\mathbb{R}^n$, not just the set's own line or plane. In Euclidean space compact means closed and bounded.

Practice problem 8.4.P1: An explicit line where two planes intersect

The normal equations are $x_1 + x_2 - 2x_3 = -3$ and $-2x_1 + x_2 + 3x_3 = 7$, using $n_1 \cdot p_1 = -3$ and $n_2 \cdot p_2 = 7$. Subtracting the second equation from the first after elimination gives the reduced equations $x_1 - (5/3)x_3 = -10/3$, $x_2 - (1/3)x_3 = 1/3$. Thus their intersection is

$$x = (-10/3, 1/3, 0) + t(5/3, 1/3, 1), \qquad t \in \mathbb{R}.$$

Substitution verifies both plane equations for every t, and the single free variable shows that every point in the intersection has this form.

Exercise 8.4.1: An implicit equation for a line

The direction from $(-1, 4)$ to $(3, 1)$ is $(4, -3)$. Choose the perpendicular normal $n = (3, 4)$, since $3(4) + 4(-3) = 0$. Then $d = n \cdot (-1, 4) = -3 + 16 = 13$, so take $f(x, y) = 3x + 4y$ and $d = 13$. Both points satisfy this equation, and its nonzero normal makes it the unique line through those two distinct points.

Exercise 8.4.2: A second line functional

The direction between $(1, 4)$ and $(-2, -1)$ is $(-3, -5)$. A perpendicular normal is $(5, -3)$; its dot product with the direction is $-15 + 15 = 0$. At the first point it has value $5 - 12 = -7$. Thus $f(x, y) = 5x - 3y$ and $d = -7$ give the required line $[f : d]$.

Exercise 8.4.3: Open, closed, or neither

(a) $y > 0$ is **open**, not closed. Every point has a ball staying above the axis; $(0, 1/k)$ tends to the excluded origin.

(b) The segment $x = 2$, $1 \leq y \leq 3$ is **closed**, not open. It includes its endpoints and is the intersection of closed constraints. Every disk about a point on it contains points with $x \neq 2$.

(c) The segment $x = 2$, $1 < y < 3$ is **neither**. It has no ambient open disk, and $(2, 1 + 1/k)$ tends to an excluded endpoint.

(d) The positive branch $xy = 1$, $x > 0$ is **closed**, not open. For a convergent sequence on it, the limit satisfies $xy = 1$ by continuity; its x cannot be zero or negative, so the limit stays on this branch. No disk about a curve point lies entirely on the curve.

(e) The region $xy \geq 1$, $x > 0$ is **closed**, not open. For a finite convergent sequence its limit has $x \geq 0$ and $xy \geq 1$, so $x = 0$ is impossible and the limit remains in the set. Boundary points with $xy = 1$ prevent openness.

Exercise 8.4.4: Five curved planar sets

(a) The circle $x^2 + y^2 = 1$ is **closed**, not open, by continuity of $x^2 + y^2$ and the absence of any disk contained in a circle.

(b) The region $x^2 + y^2 > 1$ is **open**, not closed: continuity preserves its strict inequality locally, but $(1 + 1/k, 0)$ tends to the excluded circle.

(c) The region $x^2 + y^2 \leq 1$, $y > 0$ is **neither**. The included arc point $(0, 1)$ is not interior; $(0, 1/k)$ tends to the excluded origin on the diameter.

(d) The region $y \geq x^2$ is **closed**, not open. It is defined by the nonnegative value of the continuous function $y - x^2$, and equality points have nearby points below the boundary.

(e) The region $y < x^2$ is **open**, not closed. It is a strict continuous inequality; for example $(0, -1/k)$ tends to the excluded origin.

Exercise 8.4.5: Compactness and convexity for the sets of Exercise 3

(a) The upper open half-plane is unbounded, hence not compact, but it is convex because averages of positive y-coordinates remain positive.

(b) The closed vertical segment is closed and bounded, hence compact. A segment is convex.

(c) The open vertical segment is not closed, hence not compact. It is convex: averaging two coordinates strictly between 1 and 3 keeps that coordinate strictly between the same bounds.

(d) The positive reciprocal curve is unbounded, hence not compact. It is not convex: the midpoint of $(1, 1)$ and $(2, 1/2)$ is $(3/2, 3/4)$, whose coordinate product is $9/8 \neq 1$.

(e) The region on and above the reciprocal curve is unbounded, hence not compact. It is convex: for positive x_1, x_2 and $0 \leq t \leq 1$,

$$\frac{1}{(1-t)x_1 + tx_2} \leq \frac{1-t}{x_1} + \frac{t}{x_2}.$$

This reciprocal convexity inequality, together with $y_i \geq 1/x_i$, puts each segment point on or above the same curve. Its horizontal coordinate stays positive. The inequality also follows by expanding the nonnegative quantity $t(1-t)(x_1 - x_2)^2$ after clearing positive denominators.

Exercise 8.4.6: Compactness and convexity for the sets of Exercise 4

(a) The circle is compact (closed and bounded) but not convex: $(1, 0)$ and $(-1, 0)$ have midpoint zero, off the circle.

(b) The exterior of the disk is unbounded and not compact. It is not convex: $(2, 0), (-2, 0)$ are in it but their midpoint is not.

(c) The upper half-disk with the diameter excluded is not closed, hence not compact. It is convex as the intersection of the closed unit disk and the convex open half-plane $y > 0$.

(d) The epigraph $y \geq x^2$ is unbounded, hence not compact. It is convex because $((1-t)x_1 + tx_2)^2 \leq (1-t)x_1^2 + tx_2^2 \leq (1-t)y_1 + ty_2$. The first inequality follows by subtracting the left side to get $t(1-t)(x_1 - x_2)^2 \geq 0$.

(e) The region $y < x^2$ is unbounded and not compact. It is not convex: $(-1, 1/2), (1, 1/2)$ are in it, but their midpoint $(0, 1/2)$ is not.

Exercise 8.4.7: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (0, 2, 3)^T, \qquad n^T p_1 = 11.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 11$. The equation $n^T x = 11$ contains all the points. The difference vectors have rank 2, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.8: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (-4, -3, 6)^T, \qquad n^T p_1 = 8.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 8$. The equation $n^T x = 8$ contains all the points. The difference vectors have rank 2, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.9: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (3, -1, 2, 1)^T, \qquad n^T p_1 = 5.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 5$. The equation $n^T x = 5$ contains all the points. The difference vectors have rank 3, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.10: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (-2, 3, -5, 1)^T, \qquad n^T p_1 = 4.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 4$. The equation $n^T x = 4$ contains all the points. The difference vectors have rank 3, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.11: Use the sign of the functional relative to its level

Here $f(x) = 2x_1 + x_2 + 5x_3 - x_4$ and $d = f(p) = 2 - 3 + 5 - 2 = 2$. At the origin $f(0) - d = -2 < 0$. For the three proposed points the functional values are $f(v_1) = 5$, $f(v_2) = -2$, and $f(v_3) = 2$. Thus v_2 is on the same open side as the origin, v_1 is on the opposite side, and v_3 lies in the hyperplane itself, on neither open side.

Exercise 8.4.12: Weak separation without strict parallel separation

With the prescribed normal $n = (3, 1, -2)$, use $f(x) = 3x_1 + x_2 - 2x_3$. Evaluation at the three points of A gives $-5, 4, 3$; at those of B it gives $7, 4, 6$. Hence $H = [f : 4]$ separates the sets weakly: $f(A) \le 4 \le f(B)$. But $\max f(A) = 4 = \min f(B)$, so there is no real number strictly between these two bounds. No hyperplane parallel to H strictly separates them. This only answers the prescribed-normal question; it does not assert that no differently oriented separating hyperplane could exist.

Exercise 8.4.13: The intersection of two hyperplanes in four dimensions

The two constants are $n_1 \cdot p_1 = 4$ and $n_2 \cdot p_2 = 22$. Thus solve

$$x_1 + 2x_2 + 4x_3 + 2x_4 = 4, \qquad 2x_1 + 3x_2 + x_3 + 5x_4 = 22.$$

Elimination gives $x_1 - 10x_3 + 4x_4 = 32$ and $x_2 + 7x_3 - x_4 = -14$. Take $x_3 = s, x_4 = t$ freely. The explicit answer is

$$x = (32, -14, 0, 0) + s(10, -7, 1, 0) + t(-4, 1, 0, 1), \qquad s, t \in \mathbb{R}.$$

Substitution checks both equations. The two direction vectors are independent because their third and fourth coordinates isolate s, t. Every solution was obtained through the two free variables, so this is the entire intersection, a two-dimensional flat.

Exercise 8.4.14: Possible dimensions of an intersection of flats

Choose $p \in F_1 \cap F_2$ and translate it to zero. Then $F_1 = p + U$, $F_2 = p + V$ with $\dim U = \dim V = 4$, and $F_1 \cap F_2 = p + (U \cap V)$. The dimension formula gives

$$\dim(U \cap V) = \dim U + \dim V - \dim(U + V) \ge 8 - 6 = 2,$$

and the intersection has dimension at most four. Each possibility occurs: take $U = \text{Span}\{e_1, e_2, e_3, e_4\}$ in $\mathbb{R}^6$, and take V respectively as $\text{Span}\{e_1, e_2, e_5, e_6\}$, $\text{Span}\{e_1, e_2, e_3, e_5\}$, or U. The intersection dimensions are $2, 3, 4$.

Exercise 8.4.15: An already implicit hyperplane equation

The equation $Ax = b$ is $x_1 - 3x_2 + 4x_3 - 2x_4 = 5$. Thus $f(x) = x_1 - 3x_2 + 4x_3 - 2x_4$ and $d = 5$. The coefficient vector is nonzero, so its level set is a hyperplane in $\mathbb{R}^4$.

Exercise 8.4.16: A null space of codimension one

Set $f(x) = 2x_1 + 5x_2 - 3x_3 + 6x_5$ and $d = 0$. Then $[f : 0]$ is exactly the given null space in $\mathbb{R}^5$. The fourth coordinate is free because its coefficient is zero; the functional itself is nonzero, so this is a hyperplane rather than the whole space.

Exercise 8.4.17: Describe a row plane by its perpendicular direction

Solve $Bn = 0$ for $B = \left(\begin{smallmatrix} 1 & 3 & 5 \\ 0 & 2 & 4 \end{smallmatrix}\right)$. The second equation gives $n_2 = -2n_3$, the first then gives $n_1 = n_3$. Take $n = (1, -2, 1)$. Both independent rows are perpendicular to n, so their two-dimensional span equals $\{x : n^T x = 0\}$. Thus $f(x) = x_1 - 2x_2 + x_3$, $d = 0$.

Exercise 8.4.18: A second row plane

For $B = \left(\begin{smallmatrix} 1 & 4 & -5 \\ 0 & -2 & 8 \end{smallmatrix}\right)$, solving $Bn = 0$ gives $n_2 = 4n_3$ and $n_1 = -11n_3$. Choose $n = (-11, 4, 1)$. Since the two rows of B are independent, their span is the plane $-11x_1 + 4x_2 + x_3 = 0$. Take $f(x) = -11x_1 + 4x_2 + x_3$ and $d = 0$.

Exercise 8.4.19: A column plane and a transpose null space

A normal must be perpendicular to both columns, so solve $B^T n = 0$: $n_1 + 4n_2 - 7n_3 = 0$, $2n_2 - 6n_3 = 0$. These give $n_2 = 3n_3$, $n_1 = -5n_3$. Thus $n = (-5, 3, 1)$ is a normal. The columns are independent, so the column space is the full plane $-5x_1 + 3x_2 + x_3 = 0$. Take this expression as f and take $d = 0$.

Exercise 8.4.20: A second column-plane equation

The normal equations are $n_1 + 5n_2 - 4n_3 = 0$ and $2n_2 - 4n_3 = 0$. Choose $n_3 = 1$, giving $n = (-6, 2, 1)$. The two original columns are independent, so $H = \text{Col}\, B$ equals $\{x : -6x_1 + 2x_2 + x_3 = 0\}$. Hence $f(x) = -6x_1 + 2x_2 + x_3$, $d = 0$.

Exercise 8.4.21: Four assertions about functionals and separation

(a) **False.** A linear functional maps $\mathbb{R}^n$ *to* $\mathbb{R}$, not generally from $\mathbb{R}$ to $\mathbb{R}^n$.

(b) **False.** The correct general formula is $f(x) = n^T x$, a dot product with a fixed vector, not scalar multiplication kx, which has vector output when $n > 1$. For example $f(x_1, x_2) = x_1 + x_2$ is a functional.

(c) **True.** A point in both sets would have its functional value simultaneously strictly below and strictly above the same level d. That is impossible.

(d) **False.** Two disjoint closed convex sets need not have a positive separating gap if neither is compact. For example $A = \{(x, y) : y \geq e^x\}$ and $B = \{(x, 0) : x \in \mathbb{R}\}$ are closed, convex, and disjoint. Any functional bounded on one side along the entire line B must have zero x-coefficient. Separation would then have to be horizontal, but A has positive y values arbitrarily close to zero, while B has $y = 0$. No horizontal level lies strictly between the sets.

Exercise 8.4.22: Four further assertions about separating hyperplanes

(a) **True.** A nonzero functional has a nonzero coefficient vector, so each of its real level sets is a hyperplane.

(b) **False.** The normal must be nonzero. If $n = 0, d = 0$, the set is all of $\mathbb{R}^n$; if $n = 0, d \neq 0$, it is empty. Neither is a hyperplane.

(c) **False.** Convexity is missing. Take $A = \{0\}$ and B the unit circle. They are disjoint, with A compact and B closed, but the center cannot be strictly separated from all points of the surrounding circle. If $b \in B$, both b and $-b$ belong to it, so no functional can put both strictly on the same side of a level separating them from zero.

(d) **False.** Failure of one chosen hyperplane does not imply failure of every choice. Take $A = \{(0, 0)\}$, $B = \{(2, 0)\}$, and the line $y = 100$, which does not strictly separate them. Their convex hulls are still disjoint, and the line $x = 1$ does strictly separate them.

Exercise 8.4.23: Separate a point from a triangle by an edge normal

Take $f(x, y) = 3x - 2y$. Its values at the three listed triangle vertices are $1, 9, 9$, whereas $f(p) = 10$ for $p = (4, 1)$. Every convex combination of the vertices therefore has value at most 9. Thus the line $[f : 19/2]$, or $3x - 2y = 9.5$, strictly separates the triangle from p: all its hull values are below 9.5, and the point's value is above.

Exercise 8.4.24: A second point outside a triangle

Take $f(x, y) = -2x + 3y$. At $(1, 2), (5, 1), (4, 4)$ the values are $4, -7, 4$; at $p = (2, 3)$ the value is 5. Thus $[f : 9/2]$ strictly separates p from the triangle, since its entire convex hull has functional value at most $4 < 4.5 < 5$.

Exercise 8.4.25: Separate two open balls using their center direction

Let $p = (4, 1)$, $f(x) = p \cdot x = 4x_1 + x_2$, and $v = (3/4)p = (3, 3/4)$. Set $d = f(v) = 51/4$. For $x \in B(0, 3)$, Cauchy–Schwarz gives $f(x) < 3\sqrt{17} < 51/4$. For $x \in B(p, 1)$, it gives $f(x) = 17 + p \cdot (x - p) > 17 - \sqrt{17} > 51/4$. Thus $[f : 51/4]$ strictly separates the balls. The hinted point v lies in neither

ball because $\|v\| = 3\sqrt{17}/4 > 3$ and $\|v - p\| = \sqrt{17}/4 > 1$. The strict inequalities are justified by $\sqrt{17} < 17/4$ and $3\sqrt{17} < 51/4$, which follow on squaring positive quantities.

Exercise 8.4.26: Translate the two-ball construction

The center difference is $n = p - q = (4, -2)$, with $\|n\| = \sqrt{20}$. Take $f(x) = 4x_1 - 2x_2$, so $f(q) = 2$, $f(p) = 22$. The point $v = q + (3/4)(p - q) = (5, 3/2)$ has $f(v) = 17$. For $x \in B(q, 3)$, $f(x) < 2 + 3\sqrt{20} < 17$; for $x \in B(p, 1)$, $f(x) > 22 - \sqrt{20} > 17$. Therefore $[f : 17]$ strictly separates the balls. The inequalities follow from $\sqrt{20} < 5$, and they leave a positive gap between the sets.

Exercise 8.4.27: A closed set whose convex hull loses closedness

Let

$$S = \{(n, 1/n), (-n, 1/n) : n = 1, 2, 3, \ldots\}.$$

This set is closed: any convergent sequence in it is bounded, so its integer first coordinates range through only finitely many possibilities; its limit is in that finite subset of S. But the midpoint of $(n, 1/n)$ and $(-n, 1/n)$ is $(0, 1/n)$, so these points lie in $\operatorname{conv} S$ and tend to $(0, 0)$. No convex combination of points in S has height zero, because it is a finite weighted average of strictly positive heights. Thus $(0, 0)$ is not in the convex hull, which is consequently not closed.

Exercise 8.4.28: Disjoint hulls need not give a strict separating gap

Take the closed set $B = S$ of Exercise 8.4.27 and the compact set $A = \{(0, 0)\}$. Every point in $\operatorname{conv} B$ has positive height, so the two convex hulls are disjoint. Suppose a hyperplane strictly separated A and B. After changing the functional's sign if necessary, its level would satisfy $0 = f(0) < d < f(b)$ for every $b \in B$. Taking the midpoint of the two points with first coordinates $\pm n$ would give $f(0, 1/n) > d$. Letting $n \to \infty$ contradicts $d > 0$, since $f(0, 1/n) \to 0$. Thus strict separation is impossible. The compact-set separation theorem cannot be applied to $\operatorname{conv} B$ as a closed set here: Exercise 8.4.27 shows that it is not closed.

Exercise 8.4.29: An open ball is convex

For $x, y \in B(p, \delta)$ and $0 \le t \le 1$, the triangle inequality gives

$$\|(1 - t)x + ty - p\| = \|(1 - t)(x - p) + t(y - p)\| \le (1 - t)\|x - p\| + t\|y - p\| < \delta.$$

At $t = 0$ or 1 this is the original strict inequality; between them it is a weighted average of two numbers strictly below δ. Thus the whole segment remains inside the open ball.

Exercise 8.4.30: Convex hull preserves boundedness

Since S is bounded, choose R with $\|s\| \le R$ for every $s \in S$. For a convex combination $y = \sum_i \lambda_i s_i$, use the triangle inequality, nonnegativity, and sum-one property to obtain

$$\|y\| \le \sum_i \lambda_i \|s_i\| \le \sum_i \lambda_i R = R.$$

Hence the same ball bounds $\operatorname{conv} S$. If S is empty its hull is empty and bounded as well.

8.5 Polytopes

A polytope is the convex hull of finitely many points. Its minimal representation consists exactly of its extreme points, or vertices. For a linear functional, the value at a convex combination is the same combination of the values at its generators. Hence its maximum is the largest generator value; the full maximizing set is the convex hull of *all* generators tying for that largest value. The analogous statement holds for minima. A k-face has dimension k; vertices have dimension zero, edges dimension one, and facets of an n-polytope dimension $n-1$.

Practice problem 8.5.P1: Turn five inequalities into a vertex list

In the first quadrant the constraints are $x+3y\le 12$, $x+2y\le 9$, $2x+y\le 12$. The axis vertices are $(0,0),(6,0),(0,4)$, obtained by imposing the most restrictive intercept on each axis. Pairwise boundary intersections are $(3,3),(5,2),(24/5,12/5)$. The last violates $x+2y\le 9$, because $24/5+24/5=48/5>9$. The first two pass the remaining inequalities. Thus the minimal representation is

$$P=\operatorname{conv}\{(0,0),(6,0),(5,2),(3,3),(0,4)\}.$$

Each retained point is a genuine corner of the feasible polygon; no retained point is a convex combination of the others. The printed final sentence calls these "three" vertices while listing five; there are five.

Exercise 8.5.1: All maximizers of three functionals on a triangle

At $p_1=(1,0), p_2=(2,3), p_3=(-1,2)$, the functional values are

	p_1	p_2	p_3	max	all maximizing points
$x-y$	1	-1	-3	1	p_1
$x+y$	1	5	1	5	p_2
$-3x+y$	-3	-3	5	5	p_3

These respectively answer (a), (b), (c). To justify completeness, write $x=\sum_i \lambda_i p_i$ with nonnegative coefficients of sum one. If a positive coefficient were attached to a vertex with value below the maximum, their weighted average would be strictly below that maximum. Here each maximum occurs at just one vertex, so no other point can attain it.

Exercise 8.5.2: Two maximizing edges

For $p_1=(0,-1), p_2=(2,1), p_3=(1,2)$, evaluation gives

	p_1	p_2	p_3	max	all maximizing points
$x+y$	-1	3	3	3	$\operatorname{conv}\{p_2,p_3\}$
$x-y$	1	1	-1	1	$\operatorname{conv}\{p_1,p_2\}$
$-2x+y$	-1	-3	0	0	p_3

In each tied case every point on the listed edge has the same value by linearity. A convex combination involving the lower-value vertex with positive weight has smaller value, so the whole maximizing set is exactly that edge.

Exercise 8.5.3: Minima for the first triangle

Use the values from Exercise 8.5.1 and select the smallest in each row. For (a), the minimum of $x - y$ is -3, attained only at p_3. For (b), the minimum of $x + y$ is 1, attained at every point of the edge $\text{conv}\{p_1, p_3\}$ and nowhere else. For (c), the minimum of $-3x + y$ is -3, attained exactly on $\text{conv}\{p_1, p_2\}$. The weighted-average argument used for maxima also proves these sets contain all minimizers.

Exercise 8.5.4: Minima for the second triangle

From the values in Exercise 8.5.2: (a) $x + y$ has minimum -1 only at p_1; (b) $x - y$ has minimum -1 only at p_3; (c) $-2x + y$ has minimum -3 only at p_2. No other point can attain one of these strict vertex minima, because a combination using another vertex would increase the weighted average.

Exercise 8.5.5: A polygon bounded by two sloping lines

The inequalities are $x + 2y \leq 10$, $3x + y \leq 15$, $x, y \geq 0$. The axis endpoints are $(5, 0)$ and $(0, 5)$. Solve both boundary equations: substitute $y = 15 - 3x$ into the first, obtaining $x + 30 - 6x = 10$, so $x = 4, y = 3$. This intersection is in the first quadrant. Together with the origin these are all corners:

$$P = \text{conv}\{(0, 0), (5, 0), (4, 3), (0, 5)\}.$$

Each corner is the meeting point of two boundary lines of the feasible polygon, and each lies on its boundary with a change in direction, so none is redundant in the minimal representation.

Exercise 8.5.6: Another two-constraint polygon

Here $2x + 3y \leq 18$, $4x + y \leq 16$, $x, y \geq 0$. The axis endpoints are $(4, 0), (0, 6)$. The two sloping boundaries intersect where $y = 16 - 4x$ and $2x + 3(16 - 4x) = 18$, giving $x = 3, y = 4$. Thus

$$P = \text{conv}\{(0, 0), (4, 0), (3, 4), (0, 6)\}.$$

All listed points satisfy all constraints and are the successive corners of the feasible polygon, hence give its minimal representation.

Exercise 8.5.7: Three boundaries meet at one corner

The constraints are $x + 3y \leq 18$, $x + y \leq 10$, $4x + y \leq 28$, with nonnegative coordinates. The restrictive axis intercepts are $(7, 0)$ and $(0, 6)$. Solving any pair of the sloping equalities gives the same point $(6, 4)$, which satisfies all three with equality. Hence there is only one non-axis corner, not three:

$$P = \text{conv}\{(0, 0), (7, 0), (6, 4), (0, 6)\}.$$

The inequality $x + y \leq 10$ touches this corner but introduces no new vertex.

Exercise 8.5.8: A redundant inequality

The constraints are $2x + y \leq 8$, $x + y \leq 6$, $x + 2y \leq 7$, with $x, y \geq 0$. Their restrictive axis endpoints are $(4, 0), (0, 7/2)$. The first and third boundaries meet at $(3, 2)$, which satisfies $x + y \leq 6$. The intersections with the middle boundary are $(2, 4)$ and $(5, 1)$; they violate respectively $x + 2y \leq 7$ and $2x + y \leq 8$ and must be discarded. Therefore

$$P = \text{conv}\{(0, 0), (4, 0), (3, 2), (0, 7/2)\}.$$

The discarded points are intersections of boundary lines, but they are not feasible and therefore cannot be vertices of this polytope.

Exercise 8.5.9: An extreme point that is not an exposed vertex

Let $D = \{(x, y) : x^2 + (y - 1)^2 \leq 1\}$ and $K = \text{conv}(D \cup \{(3, 0)\})$. Every point of K has $y \geq 0$, and its points with $y = 0$ form precisely the segment $[(0, 0), (3, 0)]$. Indeed the disk itself touches $y = 0$ only at the origin. If the origin were a strict convex combination of two distinct points in K, their nonnegative heights would both have to be zero. Their nonnegative horizontal coordinates on that segment could then average to zero only if both points were the origin. Thus the origin is extreme. However, the only supporting line to the disk at the origin is its tangent $y = 0$. Any supporting line to K there must also support the contained disk, so it must be that same line. It intersects K in the whole segment to $(3, 0)$, not in a singleton.

Conclusion. The origin is extreme but is not a vertex in the section's supporting-hyperplane (zero-dimensional exposed-face) sense.

Exercise 8.5.10: A closed unbounded counterexample to profile generation

Take the ray $S = \{(x, 0) : x \geq 0\}$. It is closed and convex. Its origin is extreme, while every point with $x > 0$ is the midpoint of two other ray points. Thus its profile is $\{(0, 0)\}$. The convex hull of that profile is just the singleton, not the whole ray. This shows why closedness without boundedness is insufficient.

Exercise 8.5.11: A bounded but nonclosed counterexample

Take the half-open segment $S = \{(x, 0) : 0 \leq x < 1\}$. It is bounded and convex. Its profile is again just the origin, since any other point has two distinct nearby points on either side inside the segment. The hull of the nonempty profile is a singleton, not S. This time the missing hypothesis is closedness.

Exercise 8.5.12: Face counts for simplices

(a) Count the faces of the five-simplex.

A five-simplex has six vertices; each k-face is determined by choosing $k + 1$ of them. Thus its counts for $k = 0, 1, 2, 3, 4$ are $6, 15, 20, 15, 6$. Their alternating sum is $6 - 15 + 20 - 15 + 6 = 2$, equal to $1 + (-1)^4$, as Euler's formula requires.

(b) Display the pattern and explain it.

$n\backslash k$	0	1	2	3	4
1	2	–	–	–	–
2	3	3	–	–	–
3	4	6	4	–	–
4	5	10	10	5	–
5	6	15	20	15	6

The general formula is $f_k(S^n) = \binom{n+1}{k+1}$ for $0 \le k \le n-1$. Affine independence ensures that each chosen $k+1$ vertices defines a k-simplex face and that different subsets give different faces. Dashes denote dimensions not belonging to the list of proper faces.

Exercise 8.5.13: Face counts for hypercubes

(a) Count a five-cube and check Euler.

Choose k coordinates allowed to vary and fix each remaining coordinate at either endpoint. Thus $f_k(C^n) = 2^{n-k}\binom{n}{k}$. For $n = 5$ the counts are $32, 80, 80, 40, 10$; $32 - 80 + 80 - 40 + 10 = 2 = 1 + (-1)^4$.

(b) The requested chart.

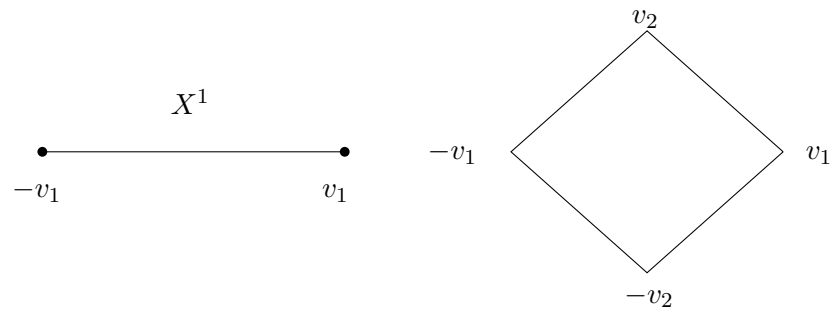
$n\backslash k$	0	1	2	3	4
1	2	–	–	–	–
2	4	4	–	–	–
3	8	12	6	–	–
4	16	32	24	8	–
5	32	80	80	40	10

The coordinate-choice argument proves the formula, rather than only predicting it from these few values.

Exercise 8.5.14: Crosspolytopes in increasing dimensions

(a) Draw the first two cases.

X^1 is the segment joining $-v_1$ to v_1. X^2 is the parallelogram with successive vertices $v_1, v_2, -v_1, -v_2$. Below are representatives using standard-coordinate generators; an invertible linear change gives the analogous pictures for any independent generators.



(b) The three-dimensional case.

There are 6 vertices, 12 edges, and 8 triangular facets. The standard-coordinate version is a regular octahedron; for arbitrary independent generators it is a linearly distorted octahedron with the same face counts.

(c) The four-dimensional case.

Its counts are $8, 24, 32, 16$. The alternating sum is $8 - 24 + 32 - 16 = 0 = 1 + (-1)^3$, as required.

(d) Explain the general formula.

A proper face selects $k+1$ of the n opposite vertex pairs, and selects one sign from each selected pair; both opposite vertices cannot belong to one proper face. These choices give

$$f_k(X^n) = 2^{k+1}\binom{n}{k+1}, \qquad 0 \le k \le n-1.$$

Every such selection is a simplex face: for the standard crosspolytope a supporting functional assigning the selected signed coordinates value one and all other coordinate coefficients zero exposes exactly that face. Linear equivalence preserves the face structure for general independent generators.

Exercise 8.5.15: Face counts for a pyramid

Let the $(n-1)$-dimensional base be Q and the apex lie outside its affine hull.

(a) Vertices.

All base vertices remain vertices and the apex adds one, so $f_0(P^n) = f_0(Q) + 1$.

(b) Intermediate dimensions.

A k-face not containing the apex is a k-face of Q. A k-face containing the apex is the pyramid over a $(k-1)$-face of Q. Thus $f_k(P^n) = f_k(Q) + f_{k-1}(Q)$ for $1 \le k \le n-2$.

(c) Facets.

There is one base facet Q; each of its $(n-2)$-faces joined to the apex gives a side facet. Therefore $f_{n-1}(P^n) = 1 + f_{n-2}(Q)$, for $n \ge 2$. These types are disjoint and cover every face of a pyramid.

Exercise 8.5.16: Four statements about profiles and simplices

(a) **True.** This is the definition of a polytope.

(b) **True.** An extreme point cannot lie strictly between two other points of the convex set. If $p \in uv$ and $p \neq u$, it must equal v.

(c) **False.** Compactness is necessary in the profile theorem; the ray in Exercise 8.5.10 is a counterexample.

(d) **True.** A four-simplex has five vertices; omit one at a time to obtain its five facets, each a three-simplex, or tetrahedron.

Exercise 8.5.17: Four more statements about faces

(a) **False.** A three-dimensional cube has six square facets.

(b) **True.** For polytopes the minimal-representation theorem identifies extreme points with vertices. The distinction seen for a nonpolytope in Exercise 8.5.9 does not occur for polytopes.

(c) **False.** A maximizing point need not itself be extreme. On the square $[-1,1]^2$, $f(x,y)=x$ is maximal at $(1,0)$, the midpoint of the edge joining $(1,-1)$ and $(1,1)$.

(d) **True.** A two-dimensional polytope is a polygon; each edge joins successive vertices around its boundary, giving equally many vertices and edges.

Exercise 8.5.18: Removing an extreme point preserves convexity

Suppose v is extreme and $x,y \in S \setminus \{v\}$. Their segment lies in S by convexity and cannot pass through v in its interior, by extremality; neither endpoint is v. Thus the whole segment lies in $S \setminus \{v\}$, proving that set convex. Conversely, if $S \setminus \{v\}$ is convex and v were strictly between some distinct $x,y \in S$, neither could equal v. Their segment would lie in the set with v removed, contradicting its containing v. Hence v is extreme. Singleton cases work because the empty set is convex.

Exercise 8.5.19: Add two positive dilations of one convex set

An arbitrary element of $cS+dS$ is $cx+dy$ with $x,y \in S$. Since $c,d>0$,

$$cx+dy=(c+d)\left(\frac{c}{c+d}x+\frac{d}{c+d}y\right)\in(c+d)S$$

by convexity. This proves one inclusion. Conversely, for each $s \in S$, $(c+d)s=cs+ds \in cS+dS$, proving the reverse inclusion. Thus $cS+dS=(c+d)S$.

Exercise 8.5.20: Convexity cannot be omitted

Take $S=\{(0,0),(1,0)\}$ and $c=d=1$. Then $S+S=\{(0,0),(1,0),(2,0)\}$, whereas $2S=\{(0,0),(2,0)\}$. The point $(1,0)$ occurs only in the sum. Thus the equality from Exercise 8.5.19 can fail for a nonconvex set.

Exercise 8.5.21: A Minkowski sum of convex sets is convex

Take two points $x=a_1+b_1$ and $y=a_2+b_2$ in $A+B$. For $0 \le t \le 1$,

$$(1-t)x+ty=((1-t)a_1+ta_2)+((1-t)b_1+tb_2).$$

The first parenthesis belongs to A and the second to B by their convexity. Thus their sum is in $A+B$. This proves the entire segment is included, as required.

Exercise 8.5.22: Why the list of regular polyhedra has five entries

(a) Count incidences in two ways.

Every one of the r facets has k edges, giving kr facet-edge incidences. Each edge belongs to two facets, giving $2e$; hence $kr = 2e$. Likewise every one of the v vertices meets s edges, while every edge has two endpoints, so $sv = 2e$.

(b) Substitute those counts into Euler's formula.

From $v - e + r = 2$, substitute $v = 2e/s$ and $r = 2e/k$, then divide by $2e$:

$$\frac{1}{s} + \frac{1}{k} = \frac{1}{2} + \frac{1}{e}.$$

(c) Exhaust the possible integer pairs.

A face has at least three sides, so $k \geq 3$; at least three edges meet at any vertex, so $s \geq 3$. The equation makes $1/s + 1/k > 1/2$. If $k \geq 6$, its left side is at most $1/3 + 1/6 = 1/2$, impossible. For $k = 3$, the strict inequality gives $s < 6$, so $s = 3, 4, 5$. For $k = 4$, it gives $s < 4$, so $s = 3$. For $k = 5$, it gives $s < 10/3$, again $s = 3$. The counts follow from $e = (1/s + 1/k - 1/2)^{-1}$ and the incidence formulas:

k	s	v	e	r	polyhedron
3	3	4	6	4	tetrahedron
4	3	8	12	6	cube
3	4	6	12	8	octahedron
5	3	20	30	12	dodecahedron
3	5	12	30	20	icosahedron

These are the five realized regular convex polyhedra named in the exercise, and the integer argument rules out every other possibility.

8.6 Curves and surfaces

A cubic Bézier curve has four controls and formula

$$B(t) = (1-t)^3 p_0 + 3t(1-t)^2 p_1 + 3t^2(1-t)p_2 + t^3 p_3, \qquad 0 \le t \le 1.$$

The coefficients are nonnegative and sum to one, so the curve lies in the convex hull of its controls. A uniform cubic B-spline segment is a *different* curve, with weights

$$S(t) = \tfrac{1}{6}\big[(1-t)^3 p_0 + (3t^3 - 6t^2 + 4)p_1 + (-3t^3 + 3t^2 + 3t + 1)p_2 + t^3 p_3\big].$$

We use this formula, identical to the Practice Problems' equations (14)–(15), for the B-spline calculations below. Exercise 2 prints an inconsistent coefficient; its solution explicitly diagnoses the discrepancy before correcting it. The numbered exercises here concern curve segments and subdivision; the section's surface discussion uses these same one-variable operations in both parameter directions.

Practice problem 8.6.P1: The initial endpoint of a B-spline segment

Substitute $t = 0$ into $S(t)$ to obtain

$$S(0) = \tfrac{1}{6}p_0 + \tfrac{2}{3}p_1 + \tfrac{1}{6}p_2.$$

The coefficients are nonnegative and sum to one, so this point is in $\operatorname{conv}\{p_0, p_1, p_2\}$. Under the stated affine-independence assumption its unique affine coordinates are $(1/6, 2/3, 1/6)$. They differ from $(1, 0, 0)$, the coordinates of p_0, so it is not p_0 under that assumption. Without independence the formula still holds, but a degenerate choice such as all controls equal can make the endpoint coincide with a control.

Practice problem 8.6.P2: The final endpoint

At $t = 1$ the coefficients give

$$S(1) = \tfrac{1}{6}p_1 + \tfrac{2}{3}p_2 + \tfrac{1}{6}p_3.$$

Hence it is in $\operatorname{conv}\{p_1, p_2, p_3\}$. When those points are affinely independent, its affine coordinates are $(1/6, 2/3, 1/6)$, not $(0, 0, 1)$, so it differs from p_3. This proves the claimed endpoint behavior with its required nondegeneracy condition.

Exercise 8.6.1: Translate every control by the same vector

Write the four Bézier weights as $b_j(t)$. Their sum is $((1-t)+t)^3 = 1$. Therefore

$$B(t) + b = \sum_{j=0}^{3} b_j(t)p_j + \left(\sum_{j=0}^{3} b_j(t)\right) b = \sum_{j=0}^{3} b_j(t)(p_j + b).$$

This is a cubic Bézier curve with translated controls $p_j + b$. Translation is affine rather than linear, and the sum-one identity is exactly what makes the same control transformation work.

Exercise 8.6.2: Diagnose the printed coefficient before proving the B-spline properties

The exercise prints $3t(1-t) - 3t + 4$ as the coefficient of p_1 before the factor $1/6$, whereas the preceding Practice Problem equation (15) has $3t(1-t)^2 - 3t + 4$. The missing square changes the mathematics. With the printed unsquared expression, the four weights sum to

$$1 + \tfrac{1}{2}t^2(1-t),$$

not one. At $t = 1/2$ they sum to $17/16$. For example, if all four controls are one fixed nonzero vector p, the printed curve gives $(17/16)p$, which is outside their singleton convex hull. Thus part (a) is false literally as printed. We now use the consistent B-spline formula (14)–(15), with that square restored, for this exercise and later references to it.

(a) Prove convex-hull containment for the intended formula.

The first and fourth weights are $(1-t)^3/6$ and $t^3/6$, hence nonnegative. For the second use $3t(1-t)^2 - 3t + 4 \geq 4 - 3t \geq 1$ on $[0,1]$. For the third use $3t^2(1-t) + 3t + 1 \geq 1$. Expanding the four numerators makes their sum 6, so their weights sum to one. Thus $S(t)$ is a convex combination of the controls.

(b) Translate using the same sum-one property.

If the four weights are $s_j(t)$, then $S(t) + b = \sum_j s_j(t)(p_j + b)$. This is a B-spline with controls translated by b. The literal unsquared formula would not have this affine-invariance property: its translated-control expression adds $[1 + \frac{1}{2}t^2(1-t)]b$, not b.

Exercise 8.6.3: First and second derivatives of a cubic Bézier curve

(a) Differentiate the polynomial weights and regroup by differences.

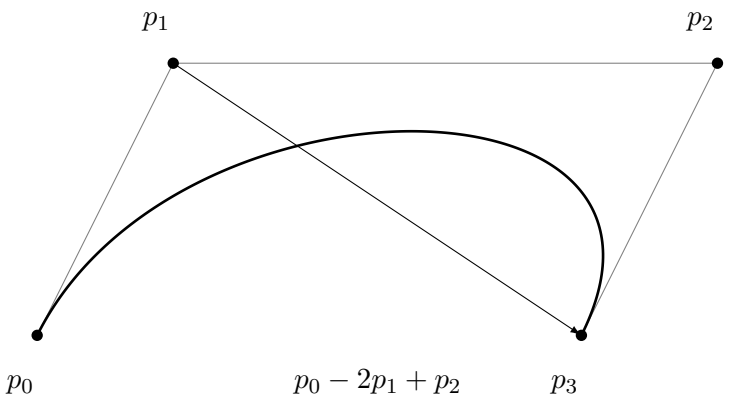
$$B'(t) = 3\big[(1-t)^2(p_1 - p_0) + 2t(1-t)(p_2 - p_1) + t^2(p_3 - p_2)\big].$$

In particular $B'(0) = 3(p_1 - p_0)$ and $B'(1) = 3(p_3 - p_2)$. The initial tangent direction goes from p_0 toward p_1; the final one goes from p_2 toward p_3, provided those differences are nonzero. Yes, $B'(1) = 0$ is possible: it occurs exactly when $p_3 = p_2$. At such an endpoint the zero tangent vector does not specify a direction.

(b) Differentiate once more.

$$B''(t) = 6\big[(1-t)(p_0 - 2p_1 + p_2) + t(p_1 - 2p_2 + p_3)\big].$$

Thus $B''(0) = 6(p_0 - 2p_1 + p_2)$ and $B''(1) = 6(p_1 - 2p_2 + p_3)$. With p_1 as origin, let $a = p_0 - p_1$, $c = p_2 - p_1$. Their vector sum $a + c = p_0 - 2p_1 + p_2$ has the direction of $B''(0)$. Construct the parallelogram on these two vectors; its diagonal gives the requested direction (unless the vector is zero). The schematic plot below uses controls $(0,0), (1,2), (5,2), (4,0)$ and shows that diagonal starting at p_1.



Exercise 8.6.4: Derivatives of the consistent cubic B-spline

(a) Compute the tangent and its endpoint values.

Differentiating the corrected formula gives

$$S'(t) = \tfrac{1}{2}\big[-(1-t)^2 p_0 + (3t^2 - 4t)p_1 + (-3t^2 + 2t + 1)p_2 + t^2 p_3\big].$$

Hence $S'(0) = (p_2 - p_0)/2$ and $S'(1) = (p_3 - p_1)/2$. The initial tangent is parallel to the segment from p_0 to p_2, and the final tangent to the segment from p_1 to p_3, when nonzero. If both vanish, then $p_2 = p_0 = a$ and $p_3 = p_1 = b$. Substitution gives

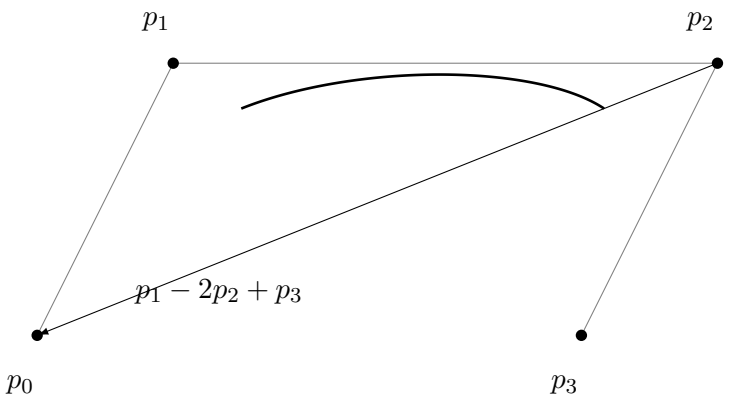
$$S(t) = \big(\tfrac{1}{3} + t^2 - \tfrac{2}{3}t^3\big)a + \big(\tfrac{2}{3} - t^2 + \tfrac{2}{3}t^3\big)b, \quad S'(t) = 2t(1-t)(a-b).$$

If $a \neq b$, this traverses the straight segment from $(a + 2b)/3$ to $(2a + b)/3$, with zero speeds at both endpoints but positive speed inside. It need not be constant. It is constant only if $a = b$.

(b) Compute the second derivative and its geometry.

$$S''(t) = (1-t)(p_0 - 2p_1 + p_2) + t(p_1 - 2p_2 + p_3).$$

In particular $S''(0) = p_0 - 2p_1 + p_2$ and $S''(1) = p_1 - 2p_2 + p_3$. Taking p_2 as origin, add the vectors $p_1 - p_2$ and $p_3 - p_2$; their parallelogram diagonal has the direction of $S''(1)$. The plot uses the same schematic controls as Exercise 8.6.3 but the B-spline curve and this second diagonal.



Exercise 8.6.5: Joining two cubic Bézier segments with matching velocities

(a) Equal derivatives give the midpoint condition.

At their shared control p_3, the first segment has derivative $3(p_3 - p_2)$ and the second has derivative $3(p_4 - p_3)$. Using the same unit parameter length for both pieces, C^1 continuity therefore requires and is guaranteed by

$$p_3 - p_2 = p_4 - p_3, \quad \text{or} \quad p_4 = 2p_3 - p_2.$$

Thus p_3 is the midpoint of p_2p_4.

(b) The zero-derivative case.

Both derivatives vanish exactly when $p_2 = p_3 = p_4$. The joined parametrization is still C^1, because its one-sided derivatives agree and are zero. But a zero velocity supplies no regular tangent direction; one cannot infer a smooth nonstationary join merely from parallel tangents. The remaining controls can give different departing and arriving directions at higher order, so C^2 continuity is not automatic.

Exercise 8.6.6: Adjacent B-spline segments automatically join

The first segment has controls p_0, p_1, p_2, p_3; the next has p_1, p_2, p_3, p_4.

(a) Check the endpoints.

By the endpoint formulas, $x(1) = (p_1 + 4p_2 + p_3)/6 = y(0)$, proving G^0 continuity.

(b) Check the derivatives with their actual scale factors.

The first segment has $x'(1) = (p_3 - p_1)/2$; the next has $y'(0) = (p_3 - p_1)/2$. They agree, so the combined unit-parameter curve is C^1 at the join. The last new control p_4 does not enter these equalities.

Exercise 8.6.7: A second derivative condition determines the next control

The C^1 condition from Exercise 8.6.5 gives $p_4 = 2p_3 - p_2$. The C^2 condition equates

$$x''(1) = 6(p_3 - 2p_2 + p_1), \qquad y''(0) = 6(p_3 - 2p_4 + p_5).$$

Cancel 6 and p_3, obtaining $p_5 = p_1 - 2p_2 + 2p_4$. Substitute the already determined p_4 to get

$$p_5 = p_1 - 4p_2 + 4p_3.$$

Thus p_4 and p_5 are fixed by the original controls and C^2 continuity; only the new segment's final control p_6 is free.

Exercise 8.6.8: Adjacent B-spline segments also match accelerations

For controls p_0, p_1, p_2, p_3, the final second derivative is $x''(1) = p_1 - 2p_2 + p_3$. For the next segment, with controls shifted forward by one, the initial second derivative is $y''(0) = p_1 - 2p_2 + p_3$. They are equal. Together with the endpoint and first-derivative equalities proved in Exercise 8.6.6, this establishes C^2 continuity at the join.

Exercise 8.6.9: The quartic Bézier basis matrix

Let $G = [\,p_0\ p_1\ p_2\ p_3\ p_4\,]$ and $u(t) = (1, t, t^2, t^3, t^4)^T$. Expand each of the five quartic weights in powers of t and put its coefficients in a *row* of M_B:

$$M_B = \begin{pmatrix} 1 & -4 & 6 & -4 & 1 \\ 0 & 4 & -12 & 12 & -4 \\ 0 & 0 & 6 & -12 & 6 \\ 0 & 0 & 0 & 4 & -4 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

For example its second row produces $4t - 12t^2 + 12t^3 - 4t^4 = 4t(1-t)^3$, the weight of p_1. Thus $x(t) = GM_Bu(t)$ reproduces the stated quartic curve.

Exercise 8.6.10: The uniform cubic B-spline basis matrix

Using $G = [\,p_0\ p_1\ p_2\ p_3\,]$ and $u(t) = (1, t, t^2, t^3)^T$, read the four consistent B-spline weights into rows:

$$M_S = \tfrac{1}{6}\begin{pmatrix} 1 & -3 & 3 & -1 \\ 4 & 0 & -6 & 3 \\ 1 & 3 & 3 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then $GM_Su(t)$ gives the B-spline formula from the Practice Problems. The factor $1/6$ is essential for normalization and cannot be dropped. The printed missing-square error in Exercise 2 is not used in this matrix.

Exercise 8.6.11: Three statements about Bézier curves

(a) **True.** A cubic Bézier curve uses four controls.

(b) **False.** For a quadratic curve the initial derivative is $2(p_1 - p_0)$, not $p_1 - p_0$. The direction is the same when nonzero, but the tangent vector has the extra factor two.

(c) **True.** The endpoint derivatives are $2(p_2 - p_1)$ and $2(p_3 - p_2)$, which agree exactly when $p_2 = (p_1 + p_3)/2$. The two pieces already share their endpoint, so this ensures C^1 continuity.

Exercise 8.6.12: Three statements about invariance and matrices

(a) **False.** Translations preserve the Bézier construction too, because the weights sum to one, as Exercise 8.6.1 proves.

(b) **True.** Equality of the ending and starting points is exactly G^0 continuity; matching derivatives requires an additional condition.

(c) **False.** The control columns form the geometry matrix G. The basis matrix contains fixed polynomial coefficients, independent of the controls.

Exercise 8.6.13: The first half's subdivided control points

(a) Use the endpoint derivative equation.

Equation (12) says $3(q_1 - q_0) = \frac{3}{2}(p_1 - p_0)$. Since $q_0 = p_0$, divide by 3 and rearrange: $q_1 = (p_0 + p_1)/2$.

(b) Rearrange the midpoint derivative equation.

Equation (13) gives $q_3 - q_2 = (-p_0 - p_1 + p_2 + p_3)/8$. Multiplying by eight and solving for q_2 gives $8q_2 = 8q_3 + p_0 + p_1 - p_2 - p_3$.

(c) Substitute the actual curve midpoint.

Use $q_3 = (p_0 + 3p_1 + 3p_2 + p_3)/8$ from equation (8). Then $8q_2 = 2p_0 + 4p_1 + 2p_2$, so

$$q_2 = (p_0 + 2p_1 + p_2)/4 = \tfrac{1}{2}[q_1 + (p_1 + p_2)/2].$$

Thus it is exactly the requested midpoint of two already constructed midpoints.

Exercise 8.6.14: The second half's subdivided controls

(a) Justify the endpoint derivative chain.

The equality $3(r_3 - r_2) = z'(1)$ is the Bézier endpoint formula. The equality $z'(1) = \frac{1}{2}x'(1)$ is the chain rule for $z(t) = x(1/2 + t/2)$. Finally $\frac{1}{2}x'(1) = \frac{3}{2}(p_3 - p_2)$ is the original endpoint formula.

(b) Solve for r_2.

Since $r_3 = p_3$, the equalities give $r_2 = (p_2 + p_3)/2$.

(c) Justify the initial derivative chain.

The identity $3(r_1 - r_0) = z'(0)$ is the initial Bézier derivative formula; $z'(0) = \frac{1}{2}x'(1/2)$ follows from the same reparametrization and chain rule.

(d) Use the original midpoint derivative.

Since $x'(1/2) = \frac{3}{4}(-p_0 - p_1 + p_2 + p_3)$, $3(r_1 - r_0) = \frac{3}{8}(-p_0 - p_1 + p_2 + p_3)$. Divide by 3 and rearrange to obtain $8r_1 = -p_0 - p_1 + p_2 + p_3 + 8r_0$.

(e) Substitute $r_0 = x(1/2)$.

The midpoint formula yields $8r_1 = 2p_1 + 4p_2 + 2p_3$. Consequently

$$r_1 = (p_1 + 2p_2 + p_3)/4 = \tfrac{1}{2}\big[r_2 + (p_1 + p_2)/2\big].$$

Exercise 8.6.15: A subdivision algorithm using only midpoint operations

(a) Apply the chain rule at the common midpoint.

With $y(t) = x(t/2)$ and $z(t) = x(1/2 + t/2)$, $y'(1) = \frac{1}{2}x'(1/2) = z'(0)$.

(b) Translate that equality into a control-point midpoint.

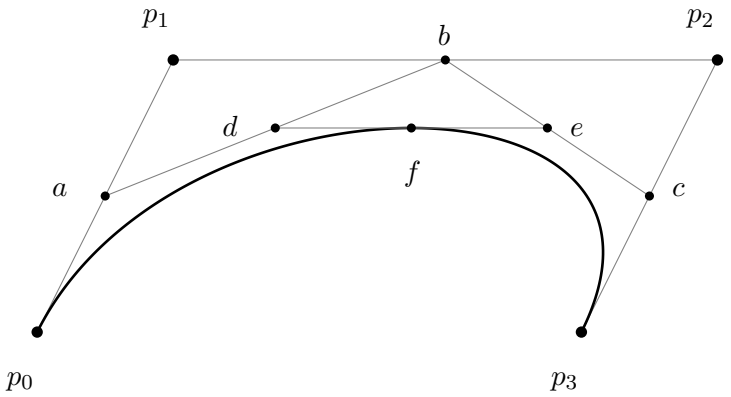
The endpoint formulas give $3(q_3 - q_2) = 3(r_1 - r_0)$. Using $q_3 = r_0$ and rearranging gives $2q_3 = q_2 + r_1$, so q_3 is the midpoint of $q_2 r_1$.

(c) State the full reusable algorithm.

Compute three first-level midpoints

$$a = (p_0 + p_1)/2, \quad b = (p_1 + p_2)/2, \quad c = (p_2 + p_3)/2;$$

then two second-level midpoints $d = (a + b)/2$, $e = (b + c)/2$; then the last midpoint $f = (d + e)/2$. The left controls are (p_0, a, d, f) and the right controls are (f, e, c, p_3). The formulas in Exercises 8.6.13–14 show $a = q_1, d = q_2, f = q_3 = r_0$, $e = r_1, c = r_2$. Thus the algorithm computes all required controls once, shares the repeated midpoint, and uses only sums and division by two.



Exercise 8.6.16: Endpoint positions and velocities determine the curve

The endpoint formulas recover all controls from the four specified vectors:

$$p_0 = x(0), \quad p_3 = x(1), \quad p_1 = x(0) + x'(0)/3, \quad p_2 = x(1) - x'(1)/3.$$

These equations determine one and only one control list, and substituting that list into the cubic Bézier formula determines one and only one curve. The derivative scale factor 3 is needed; just choosing any control points on the tangent lines would not determine the parameterized curve uniquely.

Exercise 8.6.17: Elevate a quadratic curve to cubic degree without changing it

(a) Match both endpoint positions and both endpoint velocities.

For $w(t) = (1-t)^2p_0+2t(1-t)p_1+t^2p_2$, $w(0) = p_0$, $w(1) = p_2$, $w'(0) = 2(p_1-p_0)$, and $w'(1) = 2(p_2-p_1)$. The preceding recovery formulas give cubic controls

$$r_0 = p_0, \quad r_1 = \tfrac{1}{3}p_0 + \tfrac{2}{3}p_1, \quad r_2 = \tfrac{2}{3}p_1 + \tfrac{1}{3}p_2, \quad r_3 = p_2.$$

Their endpoint derivatives are $3(r_1 - r_0) = w'(0)$ and $3(r_3 - r_2) = w'(1)$, so all four requirements are met.

(b) Check equality for every parameter.

Insert these controls into the cubic formula and collect the original p_0, p_1, p_2 coefficients. They become respectively

$$(1-t)^3 + t(1-t)^2 = (1-t)^2,$$
$$2t(1-t)^2 + 2t^2(1-t) = 2t(1-t), \qquad t^2(1-t) + t^3 = t^2.$$

Thus $x(t) = w(t)$ identically, including at both endpoints. The curve's polynomial degree representation has changed, but its geometry and parametrization have not.

Exercise 8.6.18: Multiply the scalar blocks by the control vectors

Treat the displayed p_j as vector blocks. Row-by-row block multiplication gives

$$\begin{pmatrix} p_0 \\ -3p_0 + 3p_1 \\ 3p_0 - 6p_1 + 3p_2 \\ -p_0 + 3p_1 - 3p_2 + p_3 \end{pmatrix}.$$

For controls in $\mathbb{R}^n$, this is a column of four vector blocks, or $4n$ scalar entries when expanded. These are the constant, linear, quadratic, and cubic vector coefficients of the curve; they agree with direct expansion of its four Bernstein weights.